

Intelligent Security API

Developer Guide

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1 Reading Guide

Chapter	Description
Overview	Includes the ISAPI overview, applicable products, terms and definitions, abbreviations, and update history.
ISAPI Framework	Read the chapter to take a quick look at the ISAPI framework and basic functions.
Quick Start Guide	Read the chapter to quickly understand the programming process of basic functions such as authentication, message parsing, real-time live view, playback, and event uploading.
API Reference	Start programming according to API definitions.
How-To Video Guidance	How-to videos demonstrate detailed steps of different integration tasks.

2 Overview

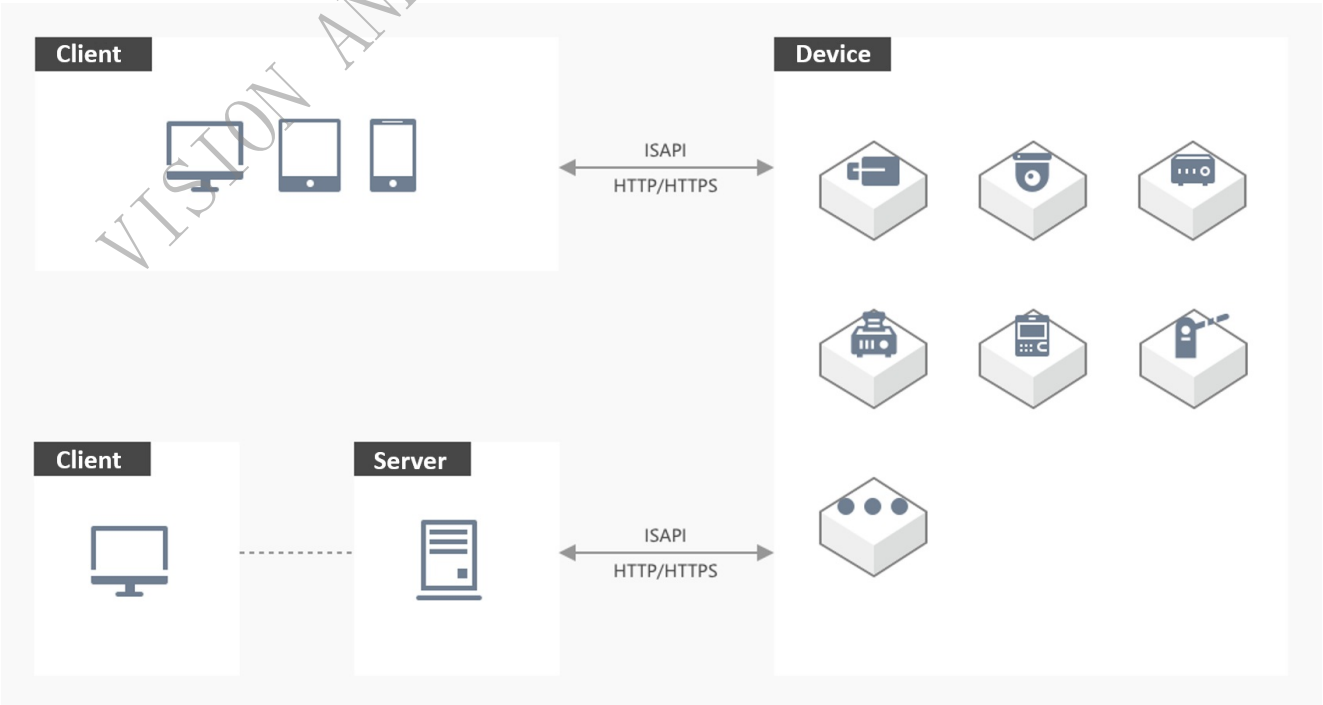
2.1 Introduction

Intelligent Security API (hereinafter referred to as ISAPI) is an application layer protocol based on HTTP (Hypertext Transfer Protocol) and adopts the REST (Representational State Transfer) architecture for communication between security devices (cameras, DVRs, NVRs, etc.) and the platform or client software.

Since established in 2013, ISAPI has included more than 11,000 APIs for different functions, including device management, vehicle recognition, parking lot management, intelligent facial application, access control management, interrogation management, and recording management. It is applicable to industries such as traffic, fire protection, education, and security inspection.

2.1.1 Application Scenario

When you integrate devices via ISAPI, the device acts as the server to listen on the fixed port and the user's application acts as the client to actively log in to the device for communication. To achieve the above goals, the device should be configured with a fixed IP address and the requests from the client can reach the server.



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2.1.2 Layers in the Network Model

ISAPI is an application layer protocol based on HTTP, thereby it inherits all specifications and properties from HTTP.

Protocols frequently used along with ISAPI include SADP (Search Active Device Protocol) based on multicast for discovering and activating devices, RTSP (Real-Time Streaming Protocol) based on TCP/UDP for live view and video playback of devices, etc.

Application Layer	Device Discovery SADP	Signaling Interaction ISAPI (HTTP)	Media Stream RTSP/RTP
Transport Layer	MCAST	TCP	TCP/UDP
Network Layer	IP		
Network Interface	Hardware Drive Interface		

2.2 Product Scope

- DeepinMind AI Products
  - DeepinMind Central AI

    DS-IC0116-F, DS-IE1016-03U/BA, DS-IE1016-03U/BA(07), DS-IE1016-03U/BA(A7), DS-IE1016-A7U/BA(H10), DS-IE1024-03U/BA, DS-IE1032-03U/BA, DS-IE1032-03U/BA(07), DS-IE1032-03U/BA(A7), DS-IE1032-A7U/BA(H10), DS-IE1064-03U/BA, DS-IE1064-03U/BA(07), DS-IE1064-03U/BA(A7), DS-IE1064-A7U/BA(H10), DS-IE1080-03U/BA, DS-IE2016-SU/BA-K2, DS-IE2032-SU/BA-K4, DS-IE2064-SU/BA-K6, DS-IE6308-E/FA, DS-IE6316-E/FA, DS-IE6316-EL/FA, DS-IE6332-E/FA, DS-IE6332-EL/FA, DS-IE6364-E/FA, DS-IF0100-AI, DS-IF0206-P4I, DS-IF1032-03U/X, DS-IF1032-03U/X(07), DS-IF1032-03U/X(A7), DS-IF1064-03U/X, DS-IF1064-03U/X(07), DS-IF1064-03U/X(A7), DS-IF2003-A3H/NF, DS-IF2004-C1U/X, DS-IF2006-A3H/NF, DS-IF2006-C1U/X, DS-IS5300, DS-IX2001-A3U/X, DS-IX2001-A7U/LX, DS-IX2001-A7U/LX(B), DS-IX2002-A1U/X, DS-IX2002-A3U/X, DS-IX2002-A7U/LX, DS-IX2002-A7U/LX(B), DS-IX2003-A3U/X, DS-IX2003-A7U/LX(B), DS-IX2004-A1U/X, DS-IX2128-C1U/XP, DS-IX2256-C1U/XP, DS-VD22D-B/HW37, DS-VD22D-B/HW38, DS-VD22D-B/HW42

2.3 Terms And Definitions

2.3.1 IoTChannel

IoTChannel refers to channels of IoT devices.

2.3.2 Event

Event refers to the information uploaded by the device actively. Event needs to be uploaded by the device in real time for the immediate response from the platform. If the device is offline, the event can be stored in the cache first and then be uploaded again when the connection is restored.

2.3.3 Arming

Arming means that the client establishes connection with the device so that events can be uploaded to the client via the connection. The client can subscribe to some event types, and the device will upload the specified events only, otherwise the device will upload all types of events to the client.

2.4 Symbols And Acronyms

IOT: Internet of Things

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OSD: On-screen display

ISAPI: Intelligent Security API.

ISUP: Intelligent Security Uplink Protocol.

HTTP: Hypertext Transfer Protocol.

NVR: Network Video Recorder

IPC: IP Camera

FDLib: Face Picture Library

FDID: Face Picture Library ID

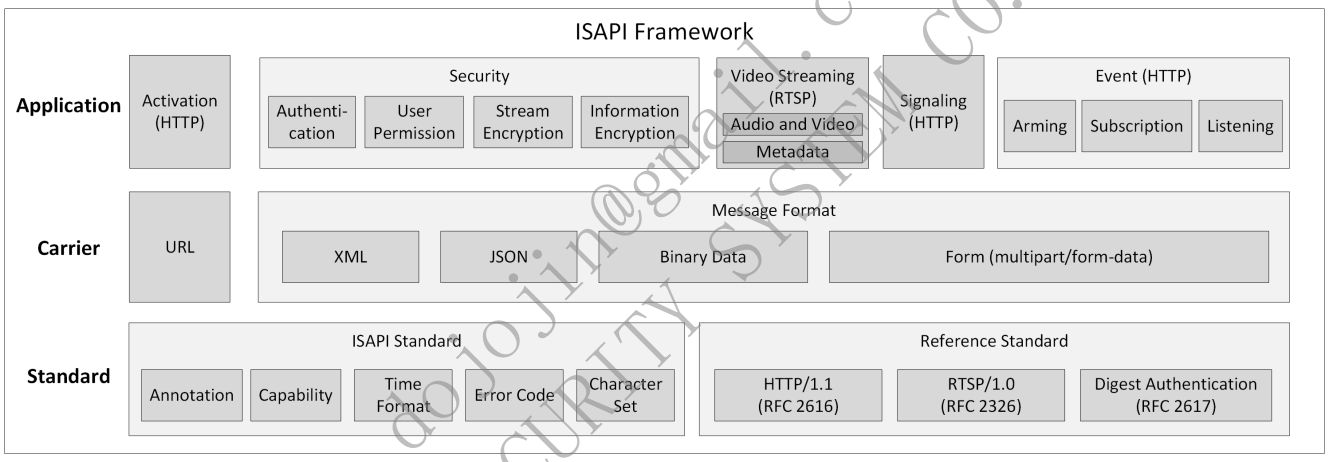
PID: Face Picture ID in the Face Picture Library

2.5 Update History

No update record

3 ISAPI Framework

3.1 Overview



Notes:

In general, ISAPI refers to the communication protocol based on the HTTP standard. As ISAPI is usually used along with RTSP (Real-Time Streaming Protocol), the RTSP standard is brought into the ISAPI system.

The metadata scheme for transmitting additional information of the stream is extended based on the RTSP standard to transmit the video stream and the structured intelligent information of the stream simultaneously. It is compatible with the RTSP standard.

3.2 Activation

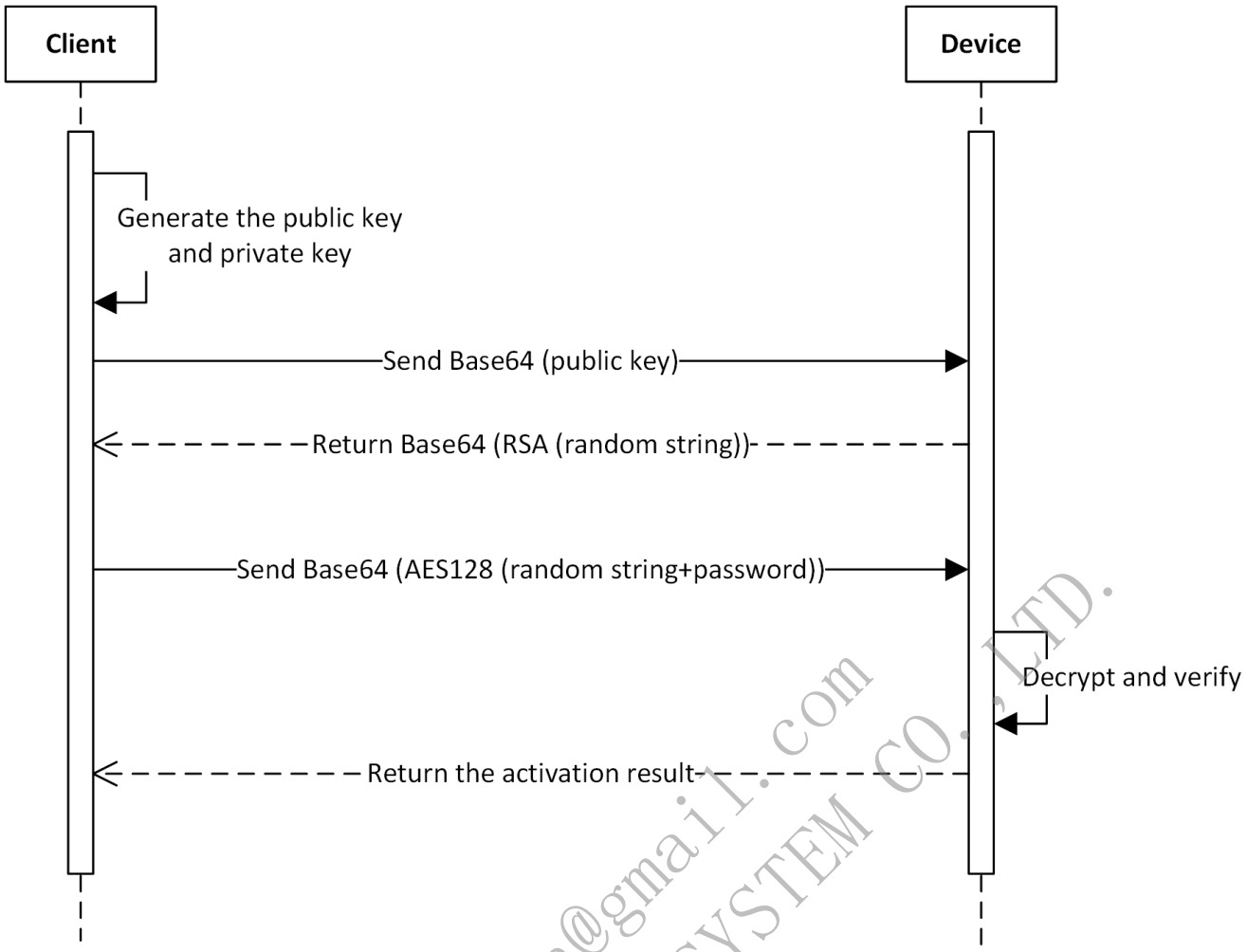
The purpose of activation is to ensure that the user can set the password for the device and the password meets the security requirement. After the device is activated, you can use the related functions.

ISAPI is a communication protocol running on the application layer. When activating the device via ISAPI, you should know the device's IP address and make sure that the device is connected to the client.

The web application built in the device supports activating the device via ISAPI. When you enter the device's IP address in the address bar of the web browser on the PC, you can activate the device according to the activation guide.

If you want to activate the device on your own application, you need to integrate the activation function via ISAPI. The API calling flow and related APIs are shown below.

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Firstly, two operations are defined:

- bytesToHexstring: it is used to convert a byte array (the length is N) to a hexadecimal string (the length is 2N). For example, `127,10,23` -> `7f0a17`
- hexStringToBytes: it is used to convert a hexadecimal string (the length is 2N) to a byte array (the length is N). For example, `7f0a17` -> `127,10,23`

1. The client generates a public and private key pair (1024 bits), and gets the 128-byte modulus in the public key (hereinafter referred to as public key modulus). If the length is longer than 128, the leading 0 needs to be removed.
2. The client converts the public key modulus to a 256-byte public key string via bytesToHexstring and sends the public key string to the device in XML message (related URI: `POST /ISAPI/Security/challenge`) after being encoded by Base64.
3. The device parses the request to obtain a 256-byte public key string decoded by Base64 and converts it to a 128-byte public key modulus via hexStringToBytes. The complete public key is the combination of obtained public key modulus and public exponent (the default value is `'010001'`).
4. The device generates a 32-byte hexadecimal random string, calls the RSA API to encrypt the random string with the private key, converts the encrypted data to a string via bytesToHexstring, encodes the string by Base64, and then sends it to the client.
5. The client decodes the string from the device by Base64, converts it via hexStringToBytes to get the encrypted data, decrypts the encrypted data with the private key via RSA to obtain a 32-byte hexadecimal random string, converts the obtained string via hexStringToBytes to get a 16-byte AES key. Then the client uses the AES key to encrypt the `"string consisting of the first 16 characters of the random string and the real password"` by AES128 ECB mode (with zero-padding method) to get a ciphertext, and converts the ciphertext via bytesToHexstring, encodes it by Base64, and sends it to the device in XML message (related URI: `PUT /ISAPI/System/activate`). Note: If the first 16 characters of the random string are `aaaabbbbccccdddd` and the real password is `Abc12345`, the data before encryption is `aaaabbbbccccddddAbc12345`. This can ensure that the client uses the random string as the key for encryption.
6. The device decodes the string by Base64, converts it via hexStringToBytes to get the ciphertext, uses the AES key to decrypt the ciphertext by AES128 ECB mode, and gets the real password via removing the first 16 characters.

7. The device verifies the password and returns the activation result.

Notes:

- You can get the device's activation status by calling the URI `GET /SDK/activateStatus` which requires no authentication.
- Devices also support to be activated via SADP (Search Active Device Protocol) which is based on the communication protocol of the data link layer. With SADP, you do not have to know the IP address of the device but need to ensure that the device and the application running SADP are connected to the same router. SADP also supports discovering devices in the LAN, changing the password of the devices, and so on. The HCSadpSDK is provided for SADP integration, including the developer guide, plug-in, and sample demo which can be used as a simple SADP tool.

3.3 Security Mechanism

3.3.1 Authentication

When the client applications send requests to devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Currently, all mainstream request class libraries of HTTP have encapsulated digest authentication. See details in Authentication of Quick Start Guide.

3.3.2 User Permission

There are three kinds of users with different permissions for access control and management.

Administrator: Has the permission to access all supported resources and should keep activated all the time. It is also known as "admin".

Operator: Has the permission to access general resources and a part of advanced resources.

Normal User: Only has the permission to access general resources.

3.3.3 Information Encryption

During ISAPI integration, the HTTPS service of devices is enabled by default. When the client applications communicate with devices via HTTPS, the information can be transmitted securely.

3.4 Video Streaming

3.4.1 Audio and Video Stream

ISAPI supports getting and setting stream media parameters of the device, such as video resolution, encoding format, and stream.

Cameras support standard RTSP (Real-Time Streaming Protocol, see details in RFC 7826). Client applications can get the stream from devices via RTSP.

For details about real-time streaming and video playback, refer to **Real-Time Live View** and **Playback** in Quick Start Guide.

3.4.2 Metadata

The metadata is the structured intelligent information generated by intelligent devices. When the client applications get the audio and/or video stream from devices via RTSP, the metadata will be returned by the device at the same time. For example, to display the face target frame, face information, vehicle target frame, license plate number, vehicle information, and other information on the video stream, the client applications can overlay the above information on the video image.

Before using the metadata, you need to enable the metadata function of the device and then get the stream from the device via RTSP. Some devices support subscribing to the metadata by type. For details about the process of integrating the metadata function, refer to **Metadata Management**.

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4 Quick Start Guide

4.1 Authentication

When the client applications send requests to the devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Client applications only need to call APIs of the class library to implement the digest authentication. The sample code is shown below.

4.1.1 C/C++ (libcurl)

```
// #include <curl/curl.h>
// Callback Function
static size_t OnWriteData(void* buffer, size_t size, size_t nmemb, void* lpVoid)
{
    std::string* str = dynamic_cast<std::string*>((std::string *)lpVoid);
    if( NULL == str || NULL == buffer )
    {
        return -1;
    }

    char* pData = (char*)buffer;
    str->append(pData, size * nmemb);
    return nmemb;
}

std::string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
std::string strResponseData;
CURL *pCurlHandle = curl_easy_init();
curl_easy_setopt(pCurlHandle, CURLOPT_CUSTOMREQUEST, "GET");
curl_easy_setopt(pCurlHandle, CURLOPT_URL, strUrl.c_str());
// Set the user name and password
curl_easy_setopt(pCurlHandle, CURLOPT_USERPWD, "admin:admin12345");
// Set the authentication method to the digest authentication
curl_easy_setopt(pCurlHandle, CURLOPT_HTTPAUTH, CURLAUTH_DIGEST);
// Set the callback function
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEFUNCTION, OnWriteData);
// Set the parameters of the callback function to get the returned information
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEDATA, &strResponseData);
// Timeout settings for receiving the data. If receiving data is not completed within 5 seconds, the application will exit directly
curl_easy_setopt(pCurlHandle, CURLOPT_TIMEOUT, 5);
// Set the redirection times to avoid too many redirections
curl_easy_setopt(pCurlHandle, CURLOPT_MAXREDIRS, 1);
// Connection timeout duration. If the duration is too short, the client application will be disconnected before the data request sent by the application
reaches the device
curl_easy_setopt(pCurlHandle, CURLOPT_CONNECTTIMEOUT, 5);
CURLcode nRet = curl_easy_perform(pCurlHandle);
if (0 == nRet)
{
    // Output the received message
    std::cout << strResponseData << std::endl;
}
curl_easy_cleanup(pCurlHandle);
```

4.1.2 C# (WebClient)

```
// using System.Net;
// using System.Net.Security;
try
{
    string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
    WebClient client = new WebClient();
    // Set the user name and password
    client.Credentials = new NetworkCredential("admin", "admin12345");
    byte[] responseData = client.DownloadData(strUrl);
    string strResponseData = Encoding.UTF8.GetString(responseData);
    // Output received information
    Console.WriteLine(strResponseData);
}
catch (Exception ex)
{
    Console.WriteLine(ex.Message);
}
```

4.1.3 Java (HttpClient)

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```java
// import org.apache.commons.httpclient.HttpClient;
String url = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
HttpClient client = new HttpClient();
// Set the user name and password
UsernamePasswordCredentials creds = new UsernamePasswordCredentials("admin", "admin12345");
client.getState().setCredentials(AuthScope.ANY, creds);
GetMethod method = new GetMethod(url);
method.setDoAuthentication(true);
int statusCode = client.executeMethod(method);
byte[] responseData = method.getResponseBodyAsString().getBytes(method.getResponseCharSet());
String strResponseData = new String(responseData, "utf-8");
method.releaseConnection();
// Output received information
System.out.println(strResponseData);
```

4.1.4 Python (requests)

```python
# - *- coding: utf-8 -*-

import requests
request_url = 'http://192.168.18.84:80/ISAPI/System/deviceInfo'
# Set the authentication information
auth = requests.auth.HTTPDigestAuth('admin', 'admin12345')
# Send the request and receive response
response = requests.get(request_url, auth=auth)
# Output response content
print(response.text)
```

4.2 Message Parsing

4.2.1 Message Format

During the process of communication and interaction via ISAPI, the request and response messages are often text data in XML or JSON format. Besides that, the data of firmware packages and configuration files is in binary format. A request can also be in form format with multiple formats of data (multipart/form-data).

4.2.1.1 XML

Generally, the `Content-Type` in the headers of the HTTP request is `application/xml; charset="UTF-8"`.

Request and response messages in XML format are all encoded with UTF-8 standards in ISAPI.

The namespace `http://www.isapi.org/ver20/XMLSchema` and ISAPI version number `2.0` of XML messages are configured by default, see the example below.

```xml
<?xml version="1.0" encoding="UTF-8"?>
<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <Node>
    <id>1</id>
    <enabled>true</enabled>
    <nodeName>nodeName</nodeName>
    <level>level1</level>
  </Node>
</NodeList>
```

4.2.1.2 JSON

The `Content-Type` in the headers of the HTTP request is often `application/json`.

To distinguish between APIs with XML messages and those with JSON messages, ISAPI adds the query parameter format=json to all request URLs with JSON messages, e.g., `http://192.168.1.1:80/ISAPI/System/Sensor/thermometrySensor?format=json`. Messages of request URLs without the query parameter format=json are usually in XML format. However, there may be some exceptions, and the message format is subject to the API definition.

Request and response messages in JSON format are all encoded by UTF-8 in ISAPI.

4.2.1.3 Binary Data

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For the firmware and configuration files, the `Content-Type` in the header of an HTTP request is often `application/octet-stream`.

4.2.1.4 Form (multipart/form-data)

When multiple pieces of data are submitted at the same time in an ISAPI request (e.g., the person information and face picture need to be submitted at the same time when a face record is added to the face picture library), the `Content-Type` in the header of the corresponding HTTP request is usually `multipart/form-data, boundary=AaB03x`, where the boundary is a variable used to separate the entire HTTP body into multiple units and each unit is a piece of data with its own headers and body. In `Content-Disposition` of form unit headers, the `name` property refers to the form unit name, which is required for all form units; the `filename` property refers to the file name of form unit body, which is required only when the form unit body is a file. In headers of form units, `Content-Length` refers to the body length, which starts after CRLF(`\r\n`) and ends before two hyphens (`--`) of next form. There should be a CRLF used as the delimiter of two form units before two hyphens (`--`), and the `Content-Length` of previous form unit does not include the CRLF length. For the detailed format description, refer to RFC 1867 (Form-Based File Upload in HTML). Pay attention to two hyphens (`--`) before and after the boundary.

Notes

- In RFC specifications, it is strongly recommended to contain the field `Content-Length` in the entity header, and there is no requirement that the field `Content-Length` should be contained in the header of each form element. The absence of field `Content-Length` in the header should be considered when the client and device programs parse the form data.
- To avoid the conflict between message content and boundary value, it is recommended to use a longer and more complex string as the boundary value.

The example of ISAPI form data submitted by a client to a device is as follows.

```
POST /ISAPI/Intelligent/FDLib/pictureUpload
Content-Type: multipart/form-data; boundary=e5c2f8c5461142aea117791dade6414d
Content-Length: 56789

--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="PictureUploadData";
Content-Type: application/xml
Content-Length: 1234

<PictureUploadData/>
--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="face_picture"; filename="face_picture.jpg";
Content-Type: image/jpeg
Content-Length: 34567

Picture Data
--e5c2f8c5461142aea117791dade6414d--
```

The example of ISAPI form data responded from a device to a client is as follows.

In ISAPI messages, when there are multiple form units, three nodes (`pid`, `contentid`, and `filename`) are used for linking form units. The corresponding relations are as follows:

Node Name	Form Field	Description
pid	name	pid in XML/JSON messages corresponds to the name property of Content-Disposition in form headers.
contentid	Content-ID	contentid in XML/JSON messages corresponds to Content-ID in form headers.
filename	filename	filename in XML/JSON messages corresponds to filename property of Content-Disposition in form headers.

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```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=136a73438ecc4618834b999409d05bb9
Content-Length: 56789

--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="mixedTargetDetection";
Content-Type: application/json
Content-Length: 811

{
    "ipAddress": "172.6.64.7",
    "macAddress": "01:17:24:45:D9:F4",
    "channelID": 1,
    "dateTime": "2009-11-14T15:27+08:00",
    "eventType": "mixedTargetDetection",
    "eventDescription": "Mixed target detection",
    "deviceID": "123456789",
    "CaptureResult": [{
        "targetID": 1,
        "Human": {
            "Rect": {
                "height": 1.0,
                "width": 1.0,
                "x": 0.0,
                "y": 0.0
            },
            "contentID1": "humanImage", /*human body thumbnail*/
            "contentID2": "humanBackgroundImage", /*human body background picture*/
            "pId1": "9d48a26f7b8b4f2390c16808f93f3534", /*human body thumbnail ID */
            "pId2": "5EE7078E07BB47CF860DE8E4E9A85F28" /*ID of human body background picture*/
        }
    }]
}
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="9d48a26f7b8b4f2390c16808f93f3534"; filename="humanImage.jpg";
Content-Type: image/jpeg
Content-Length: 34567
Content-ID: humanImage

Picture Data
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="5EE7078E07BB47CF860DE8E4E9A85F28"; filename="humanBackgroundImage.jpg";
Content-Type: image/jpeg
Content-Length: 345678
Content-ID: humanBackgroundImage

Picture Data
--136a73438ecc4618834b999409d05bb9--
```

4.2.2 Annotation

The field descriptions of ISAPI request and response messages are marked as annotations in the example messages as shown below.

```
<?xml version="1.0" encoding="UTF-8"?>

<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, node list, attr:version{req, string, version No., range:[,]}-->
  <Node>
    <!--ro, opt, object, node information-->
    <id>
      <!--ro, req, int, node No., range:[,], step:, unit:, unitType:-->1
    </id>
    <enabled>
      <!--ro, opt, bool, whether to enable-->true
    </enabled>
    <nodeName>
      <!--ro, req, string, node name, range:[1,32]-->test
    </nodeName>
    <level>
      <!--ro, opt, enum, level, subType:string,
      [level1#Level 1,level2#Level 2,level3#Level 3]-->level1
    </level>
  </Node>
</NodeList>
```

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```
{
    "name":  "test",
    /*ro, req, string, name, range:[1,32]*/
    "type":  "type1",
    /*ro, req, enum, type, subType:string, [type1#type 1,type2#type 2]*/
    "enabled":  true,
    /*ro, opt, bool, enable or not, desc:xxxxxxx*/
    "NodeList": {
    /*opt, object, node list, dep:and,{$.enabled,eq,true}*/
        "scene":  1,
        /*req, enum, scene, subType:int, [1#scene 1; 2#scene 2; 3#scene 3]*/
        "ID":  1
        /*req, int, No., range:[1,8], step:, unit:, unitType:*/
    }
}
```

Key annotations are shown in the table below.

Annotation	Description	Remark
ro	Attribute: Read-Only	This field can only be obtained and cannot be edited.
wo	Attribute: Write-Only	This field can only be edited and cannot be obtained.
req	Attribute: Required	This field is required for request messages sent to the device and response messages returned from the device.
opt	Attribute: Optional	This field is optional for request messages sent to the device and response messages returned from the device.
dep	Attribute: Dependent	This field is valid and required when specific conditions are satisfied.
object	Field Type: Object	The field of type object contains multiple sub-fields.
list	Field Type: List	The subType following it refers to the data type of sub-items in the list.
subType	Field Type: String	The range following it refers to the maximum and the minimum string size of the field.
int	Field Type: Int	The range following it refers to the maximum and the minimum value of the field.
float	Field Type: Float	The range following it refers to the maximum and the minimum value of the field.
bool	Field Type: Boolean	The value can be true or false.
enum	Field Type: Enumeration	The subType following it indicates that the enumerators are of type string or int. The [] following the subType contains all enumerators.
subType	Sub-Type of Field	When the type of field is list or enum, the value of subType is the data type of each sub-object.
desc	Field Description	The detailed description of the field.

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4.2.3 Capability Set

ISAPI has designed capability sets for almost all functions, APIs, and fields. URLs for getting the capability set end with `/capabilities`. Some URLs may contain query parameters in the format: `/capabilities?format=json&type=xxx`.

There are two types of fields in the capability message of ISAPI: whether the device supports a function and the value range of a field in an API.

Whether the device supports a function: it is often in the format `isSupportXxxxxxxx`, which indicates that whether the device supports a function and a set of APIs for implementing this function.

The capability message example in JSON format is shown below:

```
{
    "isSupportMap":  true,
    /*ro, opt, bool, whether it supports the e-map function, desc:/ISAPI/SDT/Management/map/capabilities?format=json*/
    "isSupportAlgTrainResourceInfo":  true,
    /*ro, opt, bool, whether it supports only getting the resource information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/ResourceInfo?format=json*/
    "isSupportAlgTrainAuthInfo":  true,
    /*ro, opt, bool, whether it supports ony getting the authorization information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/SoftLock/AuthInfo?format=json*/
    "isSupportAlgTrainNodeList":  true,
    /*ro, opt, bool, whether it supports only getting the node information of the algorithm training platform, desc:/ISAPI/SDT/algorithmTraining/NodeList?
format=json*/
    "isSupportNAS":  true
    /*ro, opt, bool, whether it supports mounting and unmounting NAS, desc:/ISAPI/SDT/Management/NAS/capabilities?format=json*/
}
```

The capability message example in XML format is shown below:

```
<isSupportNetworkStatus>
    <!--ro, opt, bool, whether it supports searching the network status, desc: related API (/ISAPI/System/Network/status?format=json)-->true
</isSupportNetworkStatus>
```

The value range of the field: the maximum value, minimum value, the maximum size, the minimum size, options, and so on of each field of the API.

The example of JSON format is shown below:

```
{
    "boolType": {
        /*req, object, example of the capability of type bool*/
        "@opt": [true, false]
        /*req, array, options, subType: bool*/
    },
    "integerType": {
        /*req, object, example of the capability of type integer*/
        "@min": 0,
        /*ro, req, int, the minimum value*/
        "@max": 100
        /*ro, req, int, the maximum value*/
    },
    "stringType": {
        /*req, object, example of the capability of type string*/
        "@min": 0,
        /*ro, req, int, the minimum string size*/
        "@max": 32
        /*ro, req, int, the maximum string size*/
    },
    "enumType": {
        /*req, object, capability example of type enum*/
        "@opt": ["enum1", "enum2", "enum3"]
        /*req, array, options, subType: string*/
    }
}
```

The example of XML format is shown below:

```
<boolType opt="true,false" def="true">
    <!--ro, opt, bool, example of the capability of type bool-->true
</boolType>
<integerType min="0" max="100">
    <!--ro, opt, int, example of the capability of type int-->50
</integerType>
<stringType min="0" max="64">
    <!--ro, opt, string, example of the capability of type string-->test
</stringType>
<enumType opt="red,white,black" def="red">
    <!--ro, opt, string, example of the capability of type enum-->white
</enumType>
```

Note: For the same capability set, devices of different models and versions may return different results. The values shown in this document are only examples for reference. The capability set actually returned by the device takes precedence.

4.2.4 Time Format

ISAPI adopts ISO 8601 Standard Time Format, which is the same as W3C Standard Date and Time Formats.

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Format: `YYYY-MM-DDThh:mm:ss.sTZD`

```
YYYY = the year consisting of four decimal digits
MM = the month consisting of two decimal digits (01-January, 02-February, and so forth)
DD = the day consisting of two decimal digits (01 to 31)
hh = the hour consisting of two decimal digits (00 to 23, a.m. and p.m. are not allowed)
mm = the minute consisting of two decimal digits (00 to 59)
ss = the second consisting of two decimal digits (00 to 59)
s = one or more digits representing the fractional part of a second
TZD = time zone identifier (Z or +hh:mm or -hh:mm)
```

Example: 2017-08-16T20:17:06.123+08:00 refers to 20:17:06.123 on August 16, 2017 (local time which is 8 hours ahead of UTC). The plus sign (+) indicates that the local time is ahead of UTC, and the minus sign (-) means that the local time is behind UTC.

After the DST is enabled, the local time and time difference will change compared with UTC, and the values of related fields also need to be changed. Disabling the DST will bring into the opposite effect.

Example: In 1986, the DST was in effect from May 4 at 2:00 a.m. (GMT+8). During the DST period, the clocks were moved one hour ahead, which means that there was one less hour on that day. When the DST ends at 2:00 a.m. on September 14, 1986, the clocks were moved one hour back and there was an extra hour on that day. The changes of the time are as follows:

- DST Starts: 1986-05-04T02:00:00+08:00 --> 1986-05-04T03:00:00+09:00
- DST Ends: 1986-09-14T02:00:00+09:00 --> 1986-09-14T01:00:00+08:00

Notes:

- The time difference cannot be simply used to determine the time zone. Because when the DST starts, the time difference will change and it cannot represent the actual time zone.
- Both TZ (UTC time, e.g., 1986-05-03T18:00:00Z) and TD (local time and time difference, e.g., 1986-05-04T02:00:00+08:00) meet the time format standards of ISO 8601. In ISAPI, the TD format is recommended to be used in messages sent from the user applications and the devices.
- A few old-version devices will return the time in TZ format. For representing the time difference information and forward compatibility, an extra field `timeDiff` is added as shown in the example below. User applications need to support both TD format and TZ format when parsing the time in the messages returned by devices.

```
{
    "dateTime": "1986-05-03T18:00:00Z", /*device time. The value in TZ format is the UTC time and the value in TD format is the time difference between the device's local time and UTC*/
    "timeDiff": "+08:00" /*optional, time difference between the local time and UTC time. If this field does not exist, the user application will convert the dateTime into the local time for use*/
}
```

4.2.5 Character Set

To prevent characters not commonly used from resulting in exceptions in device programs and user applications, ISAPI limits the valid field values of type string to a specific range of characters. Character sets allowed to be used in the fields of type string in ISAPI are listed below.

- Single-byte character set: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and special characters (see details in the table below).
- Multi-byte character set: language characters based on Unicode and encoded by UTF-8 (UTF-8 encoding is a transformation format of Unicode character set. For details, refer to RFC 2044).

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No.	Name	Special Character	No.	Name	Special Character
1	Open Parenthesis	(	18	Dollar Sign	$
2	Close Parenthesis	)	19	Percent Sign	%
3	Plus Sign	+	20	Ampersand	&
4	Comma	,	21	Close Single Quotation Mark	'
5	Minus Sign	-	22	Asterisk	*
6	Period	.	23	Slash	/
7	Semicolon	;	24	Smaller Than	<
8	Equal Sign	=	25	Greater Than	>
9	At Sign	@	26	Question Mark	?
10	Open Square Bracket	[	27	Caret	^
11	Close Square Bracket	]	28	Open Single Quotation Mark	'
12	Underscore	_	29	Vertical Bar	|
13	Open Brace	{	30	Tilde	~
14	Close Brace	}	31	Double Quotation Marks	"
15	Space		32	Colon	:
16	Exclamation Mark	!	33	Backslash	|
17	Octothorpe	#			

The valid characters that can be used in some special fields are listed below.

- User name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 30 in the special character table.
- Password: User Name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 33 in the special character table.
- Names displayed on the UI (device name, person name, face picture library name, etc.): lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters.
- Normal fields of type string support lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters by default.

4.2.6 Error Processing

When requesting via ISAPI failed (the HTTP status code is not 200), the device will return the HTTP status code and ISAPI error code. For HTTP status codes, refer to 10 Status Code Definitions in RFC 2616. For ISAPI error codes, refer to Error Code Dictionary.

Message Example:

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```
HTTP/1.1 403 Forbidden
Content-Type: application/json; charset="UTF-8"
Date: Thu, 15 Jul 2021 20:43:30 GMT
Content-Length: 229
Connection: Keep-Alive

{
    "requestURL": "/ISAPI/Event/triggers/notifications/channels/whiteLightAlarm",
    "statusCode": 4,
    "statusString": "Invalid Operation",
    "subStatusCode": "notSupport",
    "errorCode": 1073741825,
    "errorMsg": "notSupport"
}
```

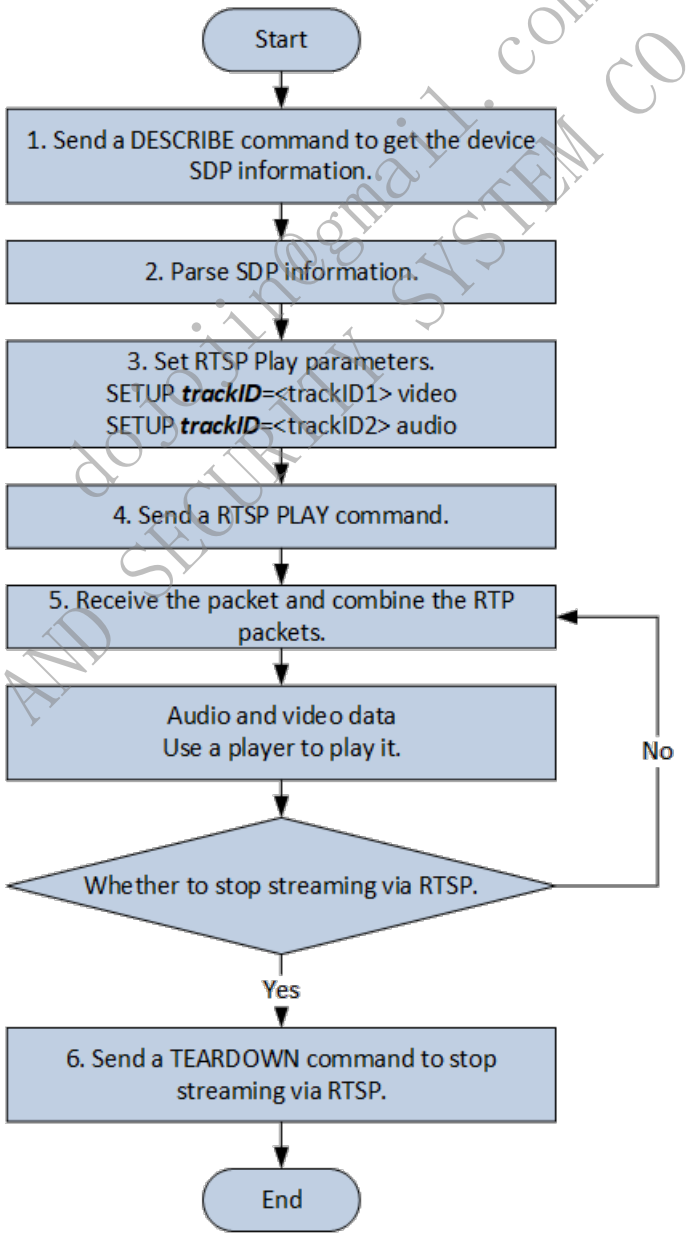
4.3 Real-Time Live View

4.3.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

4.3.2 API Calling Flow



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1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`. Digest authentication with devices is required before this step. 2. The client parses the media SDP information returned by the device. 3. Set RTSP play parameters, that is to set the track ID parsed from SDP information via SETUP commands. For

example, trackID=1 indicates videos while trackID=2 indicates audios. 4. The client sends an RTSP PLAY command, and the device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`. 5. The client receives the RTP packet sent by the device. Divided RTP packets should be assembled on the client before being parsed. 6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address; `[:port]` is optional, and 554 by default; `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, the IP address of the target device is `172.7.203.11`, and the streaming address of main stream for channel 17 will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2-=value2…)`, such as `/Streaming/Channels/101?transportmode=unicast`.

4.3.3 Example

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

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```
RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.22.126
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:6154
a=range:npt=now-
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=4D0014;packetization-mode=0
a=control:trackID=1
b=AS:6144
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=2
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000020000011171011040lF000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP requests, and the server responds to them.

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 2
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Session:1127293610;timeout=60
Transport:RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=433122ab
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

6. The client sends an RTSP PLAY request.

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

7. The server sends audio and video stream data.

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```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. .....d1.w....c....".T....g....).i.........a....7.S..~J.............X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

4.4 Event Uploading

When the rules configured on the device are triggered, the device will generate event messages (e.g., motion detection, etc.) and actively upload them to the client. ISAPI supports three methods to receive event messages uploaded by the device, that is, in arming mode, in listening mode, and via subscription.

4.4.1 Arming

The client establishes a HTTP persistent connection with the device to receive event messages from the device.

There are two methods (arming with subscription and arming without subscription) to receive events from the device. The arming without subscription is to get all event messages from the device via HTTP GET method, while the arming with subscription is to get messages of subscribed events via HTTP POST method.

Notes

- ISAPI arming (with or without subscription) uses the HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send event messages continuously, while it's not supported for clients to send any message to the device via the connection.
- When the heartbeat timed out and no message is received from the device, you should terminate the arming connection and try establishing a new one.

4.4.1.1 Arming without Subscription

1. Establish the connection of arming without subscription: `GET /ISAPI/Event/notification/alertStream` and keep the connection alive via configuring `Connection: keep-alive` in HTTP headers on the client.
2. Receive events sent by the device. The event message will be separated and parsed by boundary. For parsing details, see **Event Message Parsing** below.
3. Terminate the arming connection when no event message needs to be received.

Event Message Parsing:

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```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

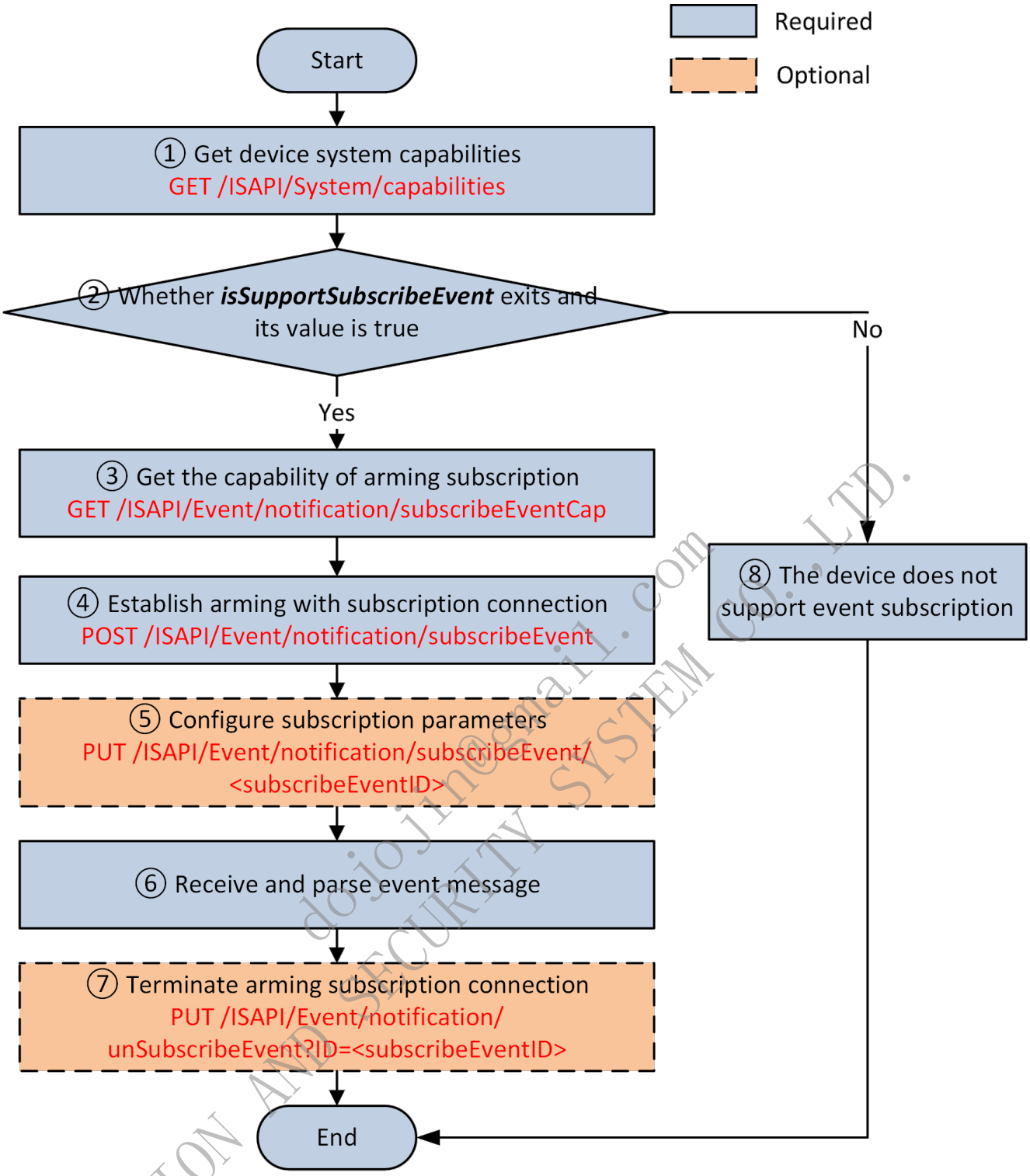
<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

Note: `<data_gateway_ip>` and `<frontier>` are variables, `[Picture Data]` indicates the raw data of a picture.

4.4.1.2 Arming with Subscription

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: `GET /ISAPI/Event/notification/subscribeEventCap`.
4. Establish a connection of arming with subscription: `POST /ISAPI/Event/notification/subscribeEvent`. You need to set `Connection: keep-alive` in HTTP headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: `GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`. Then, edit the parameters based on the existing subscription configurations: `PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`.
6. Receive events sent by the device. The event messages will be separated and parsed by the boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: `PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>`. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note：

Three types of data will be transmitted on the arming link: `<SubscribeEventResponse/>`, `<EventNotificationAlert/>`, and picture data. `<SubscribeEventResponse/>` is the data of first form sent by the device after arming established, see the response parameters of URL (`POST /ISAPI/Event/notification/subscribeEvent`) for details; and `<EventNotificationAlert/>` is the event content or heartbeat, you can identify the event type via field `eventType`, e.g., for heartbeat, the value of `eventType` is `heartBeat`.

Event Messages Parsing:

Client Creates Arming with Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

Server Responds to Request and Push Event Message

When the client sends a request to the device for establishing an arming connection, the device sends data in HTTP form format (multipart/form-data). In HTTP request headers, the `Content-Type` is `multipart/form-data, boundary=AaB03x`, of which the boundary is a variable used to divide the HTTP body into multiple units, and each unit has its headers and body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Please note two hyphens `--` before and after boundary. In normal conditions, devices will not close the arming link actively, and the end mark `--AaB03x--` will not be sent on the arming link.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Picture Data
--AaB03x--
```

The description of some keywords are as follows:

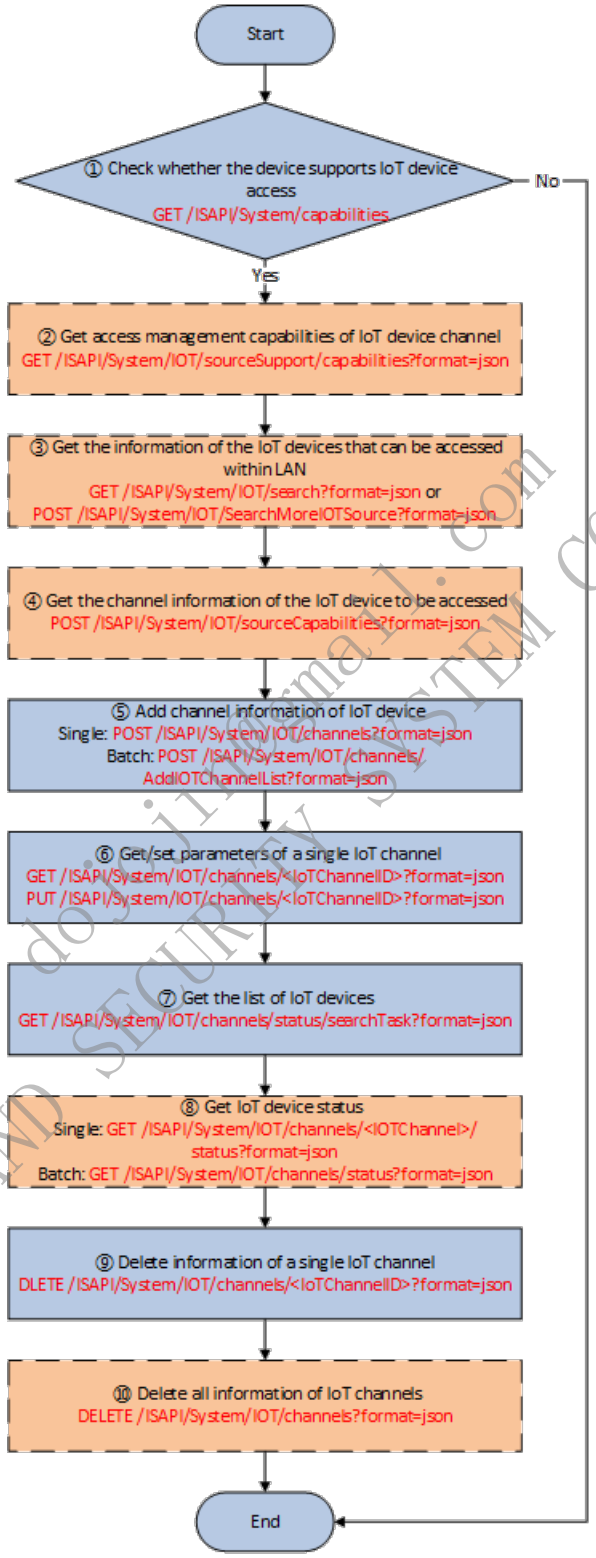
Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. A form message which starts with `--boundary` and ends with `--boundary--`.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description. `form-data` is a piece of form data.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next `\r\n` to the next `--boundary`.

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5 Device Management (General)

5.1 4 IoT Channel Management

5.1.1 IoT Device



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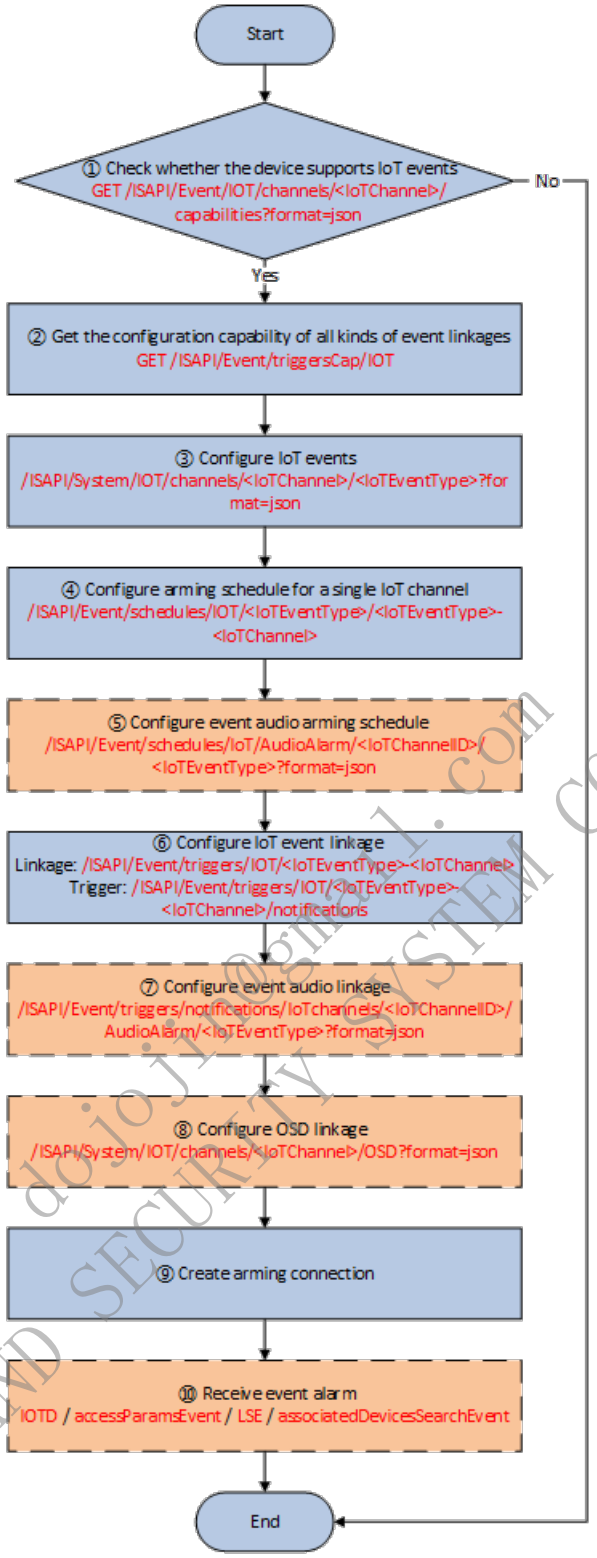
1. Check whether the device supports IoT device access: `GET /ISAPI/System/capabilities`. If the node `IOTCap` exists in the response message, it means that the device supports IoT device access. The content of `IOTCap` is the capability set of IoT device access.
2. (Optional) Get access management capabilities of IoT device channel: `GET /ISAPI/System/IOT/sourceSupport/capabilities?format=json`. Its function is similar to that of the previous step, so it is optional.
3. Get the information of the IoT devices that can be accessed within LAN.

- All results: `GET /ISAPI/System/IOT/search?format=json`
- Results with search conditions: `POST /ISAPI/System/IOT/SearchMoreIOTSource?format=json`

4. Get the channel information of the IoT device to be accessed: `POST /ISAPI/System/IOT/sourceCapabilities?format=json`
5. Add channel information of IoT device.
   - Single: `POST /ISAPI/System/IOT/channels?format=json`
   - Batch: `POST /ISAPI/System/IOT/channels/AddIOTChannelList?format=json`
6. Get/set parameters of a single IoT channel.
   - Get: `GET /ISAPI/System/IOT/channels/<IoTChannelID>?format=json`
   - Set: `PUT /ISAPI/System/IOT/channels/<IoTChannelID>?format=json`
7. Get the list of IoT devices: `GET /ISAPI/System/IOT/channels/status/searchTask?format=json`
8. (Optional) Get IoT device status. Its function is similar to that of the previous step, so it is optional. It is recommended to get device status via the device list in the previous step.
   - Single: `GET /ISAPI/System/IOT/channels/<IOTChannel>/status?format=json`
   - Batch: `GET /ISAPI/System/IOT/channels/status?format=json`
9. Delete information of a single IoT channel: `DLETE /ISAPI/System/IOT/channels/<IoTChannelID>?format=json`
10. Delete all information of IoT channels: `DELETE /ISAPI/System/IOT/channels?format=json`

5.1.2 IoT Event

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doi.org/10.1109/...

1. Get the event capabilities supported by the IoT channel: `GET /ISAPI/Event/IOT/channels/<IoTChannel>/capabilities?format=json`
2. Get the configuration capability of all kinds of event linkages: `GET /ISAPI/Event/triggersCap/IOT`
3. Configure IoT events.
   - Get IoT event configuration capability: `GET /ISAPI/System/IOT/channels/<IoTChannel>/<IoTEventType>/capabilities?format=json`
   - Set IoT event parameters: `PUT /ISAPI/System/IOT/channels/<IoTChannel>/<IoTEventType>?format=json`
   - Get IoT event parameters: `GET /ISAPI/System/IOT/channels/<IoTChannel>/<IoTEventType>?format=json`
4. Configure arming schedule for a single IoT channel.
   - Get arming schedule of a single IoT channel: `GET /ISAPI/Event/schedules/IOT/<IoTEventType>/<IoTEventType>-<IoTChannel>`
   - Set arming schedule of a single IoT channel: `PUT`

`/ISAPI/Event/schedules/IOT/<IoTEventType>/<IoTEventType>-<IoTChannel>`

5. (Optional) Configure event audio arming schedule. If it is not configured, the device will play audio alarm whenever a linkage alarm is triggered. If it is configured, the device will only play the audio alarm within the arming schedule, and will upload the alarm without playing audio for the rest of the time.
   - Get event audio linkage configuration capability: `GET /ISAPI/Event/triggers/notifications/IoTChannels/<IoTChannelID>/AudioAlarm/<IoTEventType>/capabilities?format=json`
   - Set event audio arming schedule parameters: `PUT /ISAPI/Event/schedules/IoT/AudioAlarm/<IoTChannelID>/<IoTEventType>?format=json`
   - Get event audio arming schedule parameters: `GET /ISAPI/Event/schedules/IoT/AudioAlarm/<IoTChannelID>/<IoTEventType>?format=json`
6. Configure IoT event linkage, which consists of two parts: linkage method configuration and trigger method configuration.
   - Set IoT event linkage method: `PUT /ISAPI/Event/triggers/IOT/<IoTEventType>-<IoTChannel>`
   - Get IoT event linkage method: `GET /ISAPI/Event/triggers/IOT/<IoTEventType>-<IoTChannel>`
   - Set IoT event trigger method: `PUT /ISAPI/Event/triggers/IOT/<IoTEventType>-<IoTChannel>/notifications`
   - Get IoT event trigger method: `GET /ISAPI/Event/triggers/IOT/<IoTEventType>-<IoTChannel>/notifications`
7. (Optional) Configure event audio linkage.
   - Get event audio linkage configuration capability: `GET /ISAPI/Event/triggers/notifications/IoTChannels/<IoTChannelID>/AudioAlarm/<IoTEventType>/capabilities?format=json`
   - Set event audio linkage parameters: `PUT /ISAPI/Event/triggers/notifications/IoTChannels/<IoTChannelID>/AudioAlarm/<IoTEventType>?format=json`
   - Get event audio linkage parameters: `GET /ISAPI/Event/triggers/notifications/IoTchannels/<IoTChannelID>/AudioAlarm/<IoTEventType>?format=json`
8. Configure IoT channel OSD linkage.
   - Get OSD linkage configuration capability: `GET /ISAPI/System/IOT/channels/<IoTChannel>/OSD/capabilities?format=json`
   - Set OSD linkage parameters: `PUT /ISAPI/System/IOT/channels/<IoTChannel>/OSD?format=json`
   - Get OSD linkage parameters: `GET /ISAPI/System/IOT/channels/<IoTChannel>/OSD?format=json`
9. Upload alarm events. There are four types of IoT events:
   - IoT device events: `IOTD`
   - Getting access parameters: `accessParamsEvent`
   - Logistics scanning event: `LSE`
   - Getting the list of linked devices: `associatedDevicesSearchEvent`

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5.2 Arming and Subscription

5.2.1 Introduction to the Function

With arming and subscription, the client can establish HTTP persistent connection with the device, and continuously receive the event messages from the device.

For ISAPI event arming, the client can receive all types of events by GET method, or receive the subscribed events only by POST method.

5.2.2 API Calling Flow

5.2.2.1 Without Subscription

1. Establish a connection for arming: `GET /ISAPI/Event/notification/alertStream`. You need to set `Connection: keep-alive` in HTTP Headers.
2. When receiving events sent by the device, the event messages can be separated and parsed by boundary. See "Parsing Event Messages" below for details.
3. Disable the arming connection when you do not need to receive event messages.

5.2.2.1.2 Syntax

```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
uri="/ISAPI/Event/notification/alertStream", cnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/form-data; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!-- Some alarms are in JSON format, so the upper layer should parse based on the Content-Type field -->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/jpeg
Content-Length: image_length

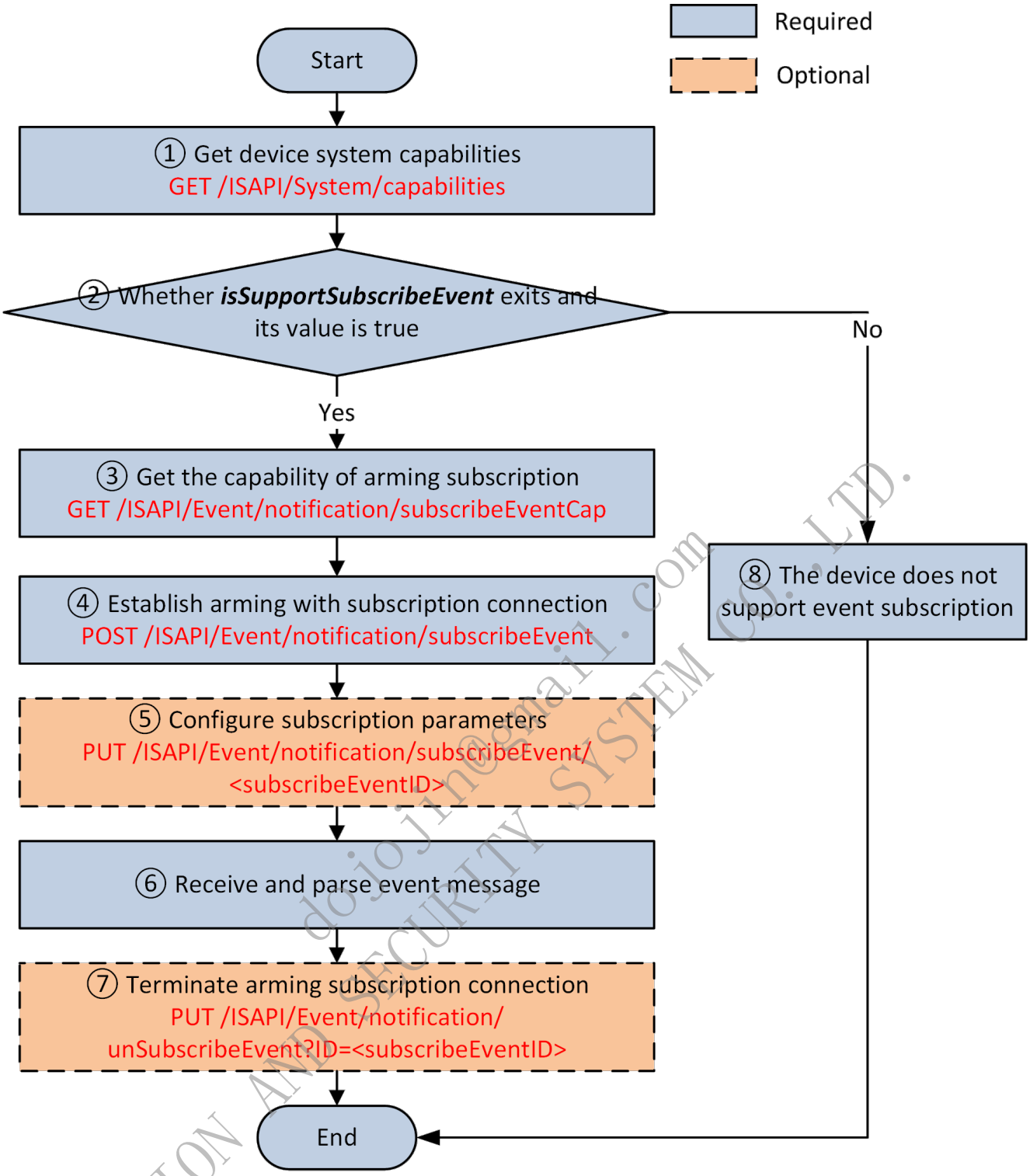
[Image Data]
--<frontier>
```

Note: `<data_gateway_ip>` and `<frontier>` are variables, and `[Image Data]` is an abbreviated representation indicating the raw data of an image at this location.

5.2.2.2 Subscription

5.2.2.2.1 API Calling Flow

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: `GET /ISAPI/Event/notification/subscribeEventCap`.
4. Establish a connection of arming with subscription: `POST /ISAPI/Event/notification/subscribeEvent`. You need to set `Connection: keep-alive` in HTTP Headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: `GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`. Then, edit the parameters based on the existing subscription configurations: `PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`.
6. Receive events sent by the device. The event messages will be separated and parsed by boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: `PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>`. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note:

Three types of data will be transmitted on the arming link: `<SubscribeEventResponse/>`, `<EventNotificationAlert/>`, and picture data. `<SubscribeEventResponse/>` is the data of first form sent by the device after arming established, see the response parameters of URL (`POST /ISAPI/Event/notification/subscribeEvent`) for details; and `<EventNotificationAlert/>` is the event content or heartbeat, you can identify the event type via field `eventType`, e.g., for heartbeat, the value of `eventType` is `heartBeat`.

5.2.2.2.2 Example

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Host: device_ip
Accept-Language: zh-cn
Date: YourDate
Content-Type: application/xml;
Content-Length: text_length
Connection: Keep-Alive

<SubscribeEvent/>
HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

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5.2.2.3 Event Messages Parsing

After the arming connection with the device is established, the data sent by the device is in HTTP form format (multipart/form-data). In an HTTP request, `Content-Type` in Headers is usually `multipart/form-data, boundary=AaB03x`, and boundary is a variable used to divide HTTP Body into multiple units, each being a set of data and has its own Headers and Body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Note the dash `--` before and after boundary. Under normal circumstances, the device will not actively close the arming connection, so the device will not send the form format end symbol `--AaB03x--` on the arming connection.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Image Data
--AaB03x--
```

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. --boundary is the start of a form. --boundary-- is the end of the whole HTTP form message.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next \r\n to the next --boundary.

5.2.3 Restriction Description

Note that ISAPI arming (with or without subscription) uses HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send out event messages continuously, while you cannot send any message to the device via the connection.

After the heartbeat time, if you do not receive any message from the device, you should disable the arming connection and try establishing a new one.

5.2.4 Sample Messages

5.2.4.1 Establish Arming Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

5.2.4.2 The Device Responses and Uploads an Event Message

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```
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <ipAddress>10.17.133.46</ipAddress>
    <portNo>80</portNo>
    <protocol>HTTP</protocol>
    <macAddress>44:19:b6:6d:24:85</macAddress>
    <channelID>1</channelID>
    <dateTime>2017-05-04T11:20:02+08:00</dateTime>
    <activePostCount>0</activePostCount>
    <eventType>heartBeat</eventType>
    <eventState>active</eventState>
    <eventDescription>heartBeat</eventDescription>
</EventNotificationAlert>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length
Content-ID: image_ID

[Picture Data]
--<frontier>
```

5.3 Calling Flow of Device Packet Capture

5.3.1 Function Introduction

When problems arise after a device is deployed on site, interaction message between the device and the external network is necessary to help developers for troubleshooting. Packet capture can be stored on the local device, and packet capture files can be exported after capture is complete. Also, packet capture files can be uploaded to cloud storage, and packet capture data can be obtained in real-time even if the device does not have the storage space.

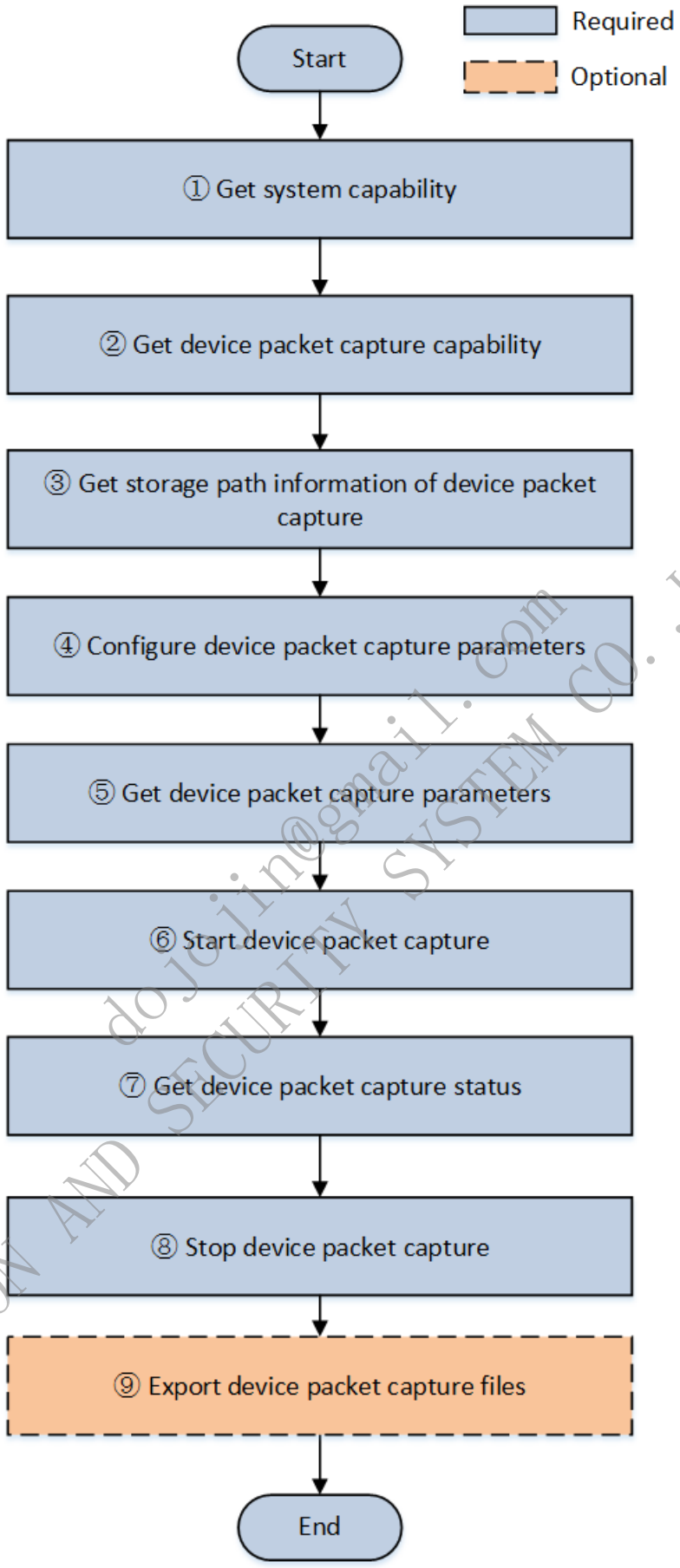
1. Device packet capture: save packet capture files on the local device, and export the files after capture is complete. Also, uploading packet capture files to cloud storage after capture is complete is supported. Then the client can obtain the storage URL and download packet capture files from the cloud storage.

2. Device real-time packet capture: after it is enabled, the device returns an URI for downloading packet capture data. The client can submit this URI to the browser to download the packet data. The device transmits packet data via HTTP Chunked, and users can store the packet data through the browser.

5.3.2 API Calling Flow

5.3.2.1 Device Packet Capture

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`. Get to know if the device supports packet capture by the field: `<isSupportNetworkCapture>true</isSupportNetworkCapture>`.

2. Check if the device supports packet captures: `GET /ISAPI/System/networkCapture/capabilities?format=json`. If `isSupportManualControl` is true, the device supports packet capture. If `isSupportManualControlAsyn` is true, the device supports asynchronous packet capture.

3. Get storage path information of device packet capture: `GET /ISAPI/System/networkCapture/StoragePathInfo?format=json`.

4. Configure device packet capture parameters such as capture duration, storage path, port, and address: `PUT /ISAPI/System/networkCapture/captureParams?format=json`.

5. Get device packet capture parameters such as capture duration, storage path, port, and address: `GET /ISAPI/System/networkCapture/captureParams?format=json`.

6. Start device packet capture: depending on the parameters, packet capture files can be saved on the local device for export after capture is complete, or packet capture files can be uploaded to cloud storage and downloaded via the storage URL, or the packet capture data can be returned in real-time.

Start device packet capture: `PUT /ISAPI/System/networkCapture/manualStart?format=json&asyn=<asyn>&realTime=<realTime>`.

Note:

- If the API does not contain URL parameters, the packet capture file is saved on the local device.
- If the API contains `asyn=true`, the packet capture file is automatically uploaded to cloud storage after capture is complete.

7. After starting capture, you can repeatedly get capture status, including whether the capture is ongoing, the size of the packet capture data, and the progress and storage URL for uploading the data to cloud storage.

Get status of device packet capture: `GET /ISAPI/System/networkCapture/manualStatus?format=json`.

8. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

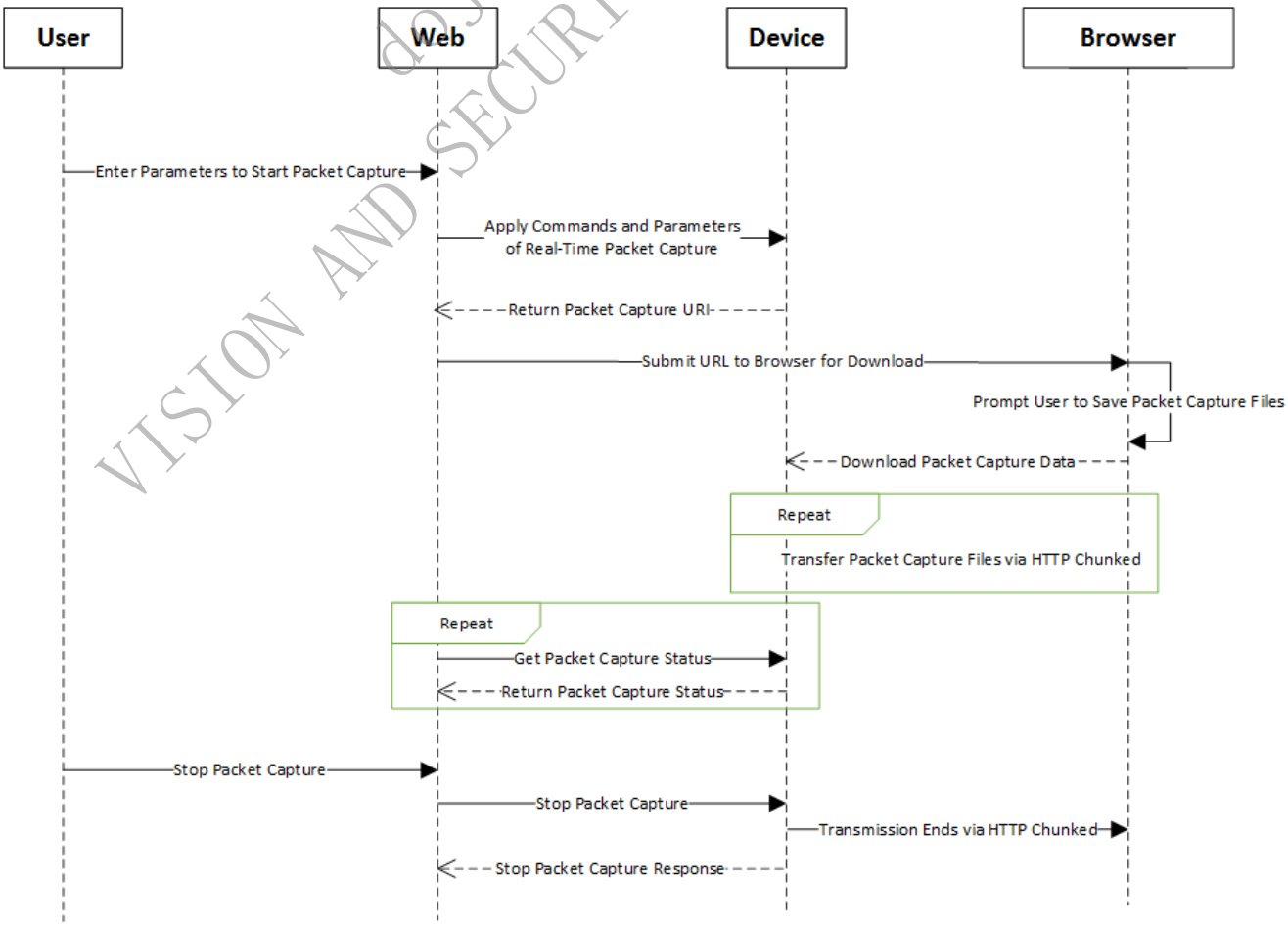
Stop device packet capture: `PUT /ISAPI/System/networkCapture/manualStop?format=json`.

9. (Optional) If packet capture data is stored on the local device, packet capture files need to be exported. If packet capture files are saved to cloud storage or captured in real-time, there is no need to export packet capture files.

Export device packet capture files: `GET /ISAPI/System/networkCapture/exportFile?format=json`.

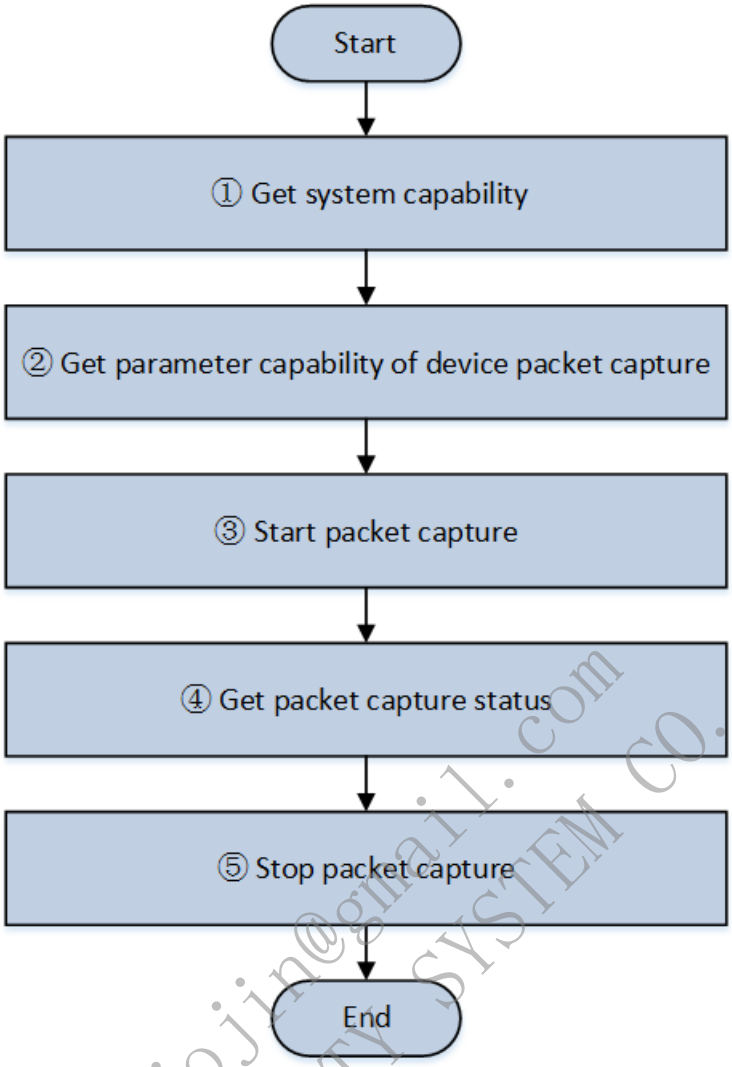
5.3.2.2 Device Real-Time Packet Capture

The process of real-time packet capture is shown as the following:



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See the following figure for the calling flow:



1. Get device system capabilities: `GET /ISAPI/System/capabilities`. `<isSupportStartNetworkCapture>true</isSupportStartNetworkCapture>` indicates the device supports starting packet capture. `<isSupportStopNetworkCapture>true</isSupportStopNetworkCapture>` indicates the device supports stopping packet capture. `<isSupportGetNetworkCaptureStatus>true</isSupportGetNetworkCaptureStatus>` indicates the device supports getting packet capture status.

2. Get capabilities of packet capture parameters: `GET /ISAPI/System/NetworkCaptureParams/capabilities?format=json`. The `realTimeEnabled` field indicates whether the device supports real-time packet capture.

3. Set the field realTimeEnabled as true in the parameters applied to the device to start device packet capture. The device returns an URI, and the client can download the real-time packet capture data from the device through a browser.

Start packet capture: `POST /ISAPI/System/StartNetworkCapture?format=json&security=<security>&iv=<iv>`.

Note:

- If `realTimeEnabled=true` is contained when starting device packet capture, it indicates packet capture data is uploaded in real-time by HTTP Chunked. The URL for downloading the returned packet capture data is valid for 30 seconds by default. If the download is attempted after this time, the device should return an HTTP 404 status code.

4. After starting packet capture, you can repeatedly get packet capture status, including whether packet capture is ongoing and the size of the packet capture data.

Get packet capture status: `GET /ISAPI/System/GetNetworkCaptureStatus?format=json`.

5. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

Stop device packet capture: `POST /ISAPI/System/StopNetworkCapture?format=json`.

5.4 Device Peripherals Upgrade

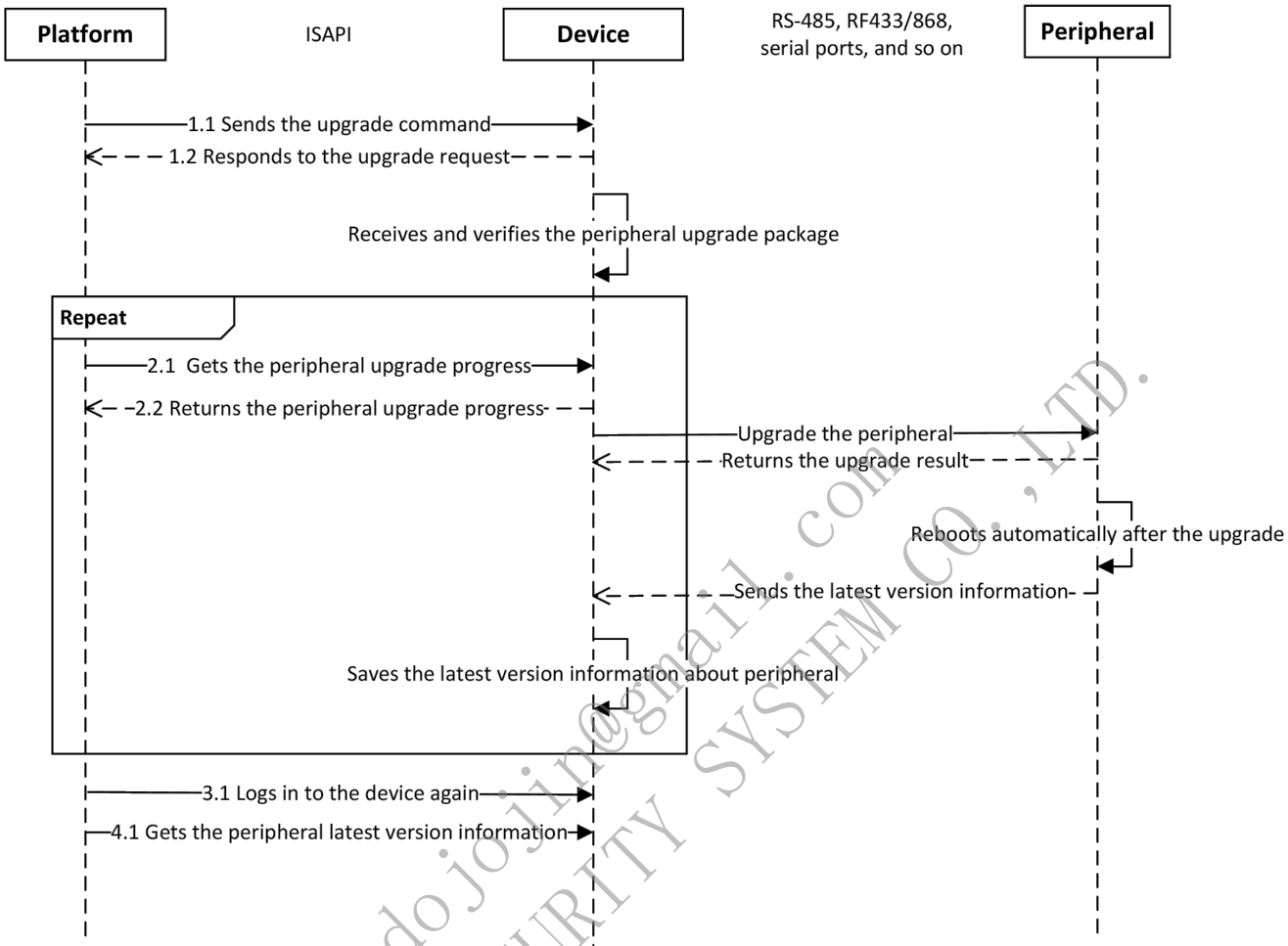
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5.4.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades device peripherals via ISAPI.

5.4.2 API Calling Flow

The sequence diagram of upgrading device peripherals by the platform is shown below.



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1. Get the device system capability `GET /ISAPI/System/capabilities` and check whether the device supports upgrading peripherals. If the field `isSupportAcsUpdate` is returned and its value is true, it indicates that the device supports this function, otherwise, the device does not support this function.
2. Get the capability of upgrading the peripherals module `GET /ISAPI/System/AcsUpdate/capabilities`, and get the types and IDs of peripherals that support upgrading.
3. The platform sends the upgrade command `POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>`. In the URL `type` refers to the peripheral type, `moduleAddress` refers to the peripheral module address, and `indexID` refers to the ID of peripheral to be upgraded. The platform will apply the upgrade peripheral package to the device.
4. Get the peripheral upgrade progress `GET /ISAPI/System/upgradeStatus?type=<Type>`.
5. Log in to the device again.
6. Get the peripheral latest version information.

5.5 Device Time Sync

5.5.1 Introduction to the Function

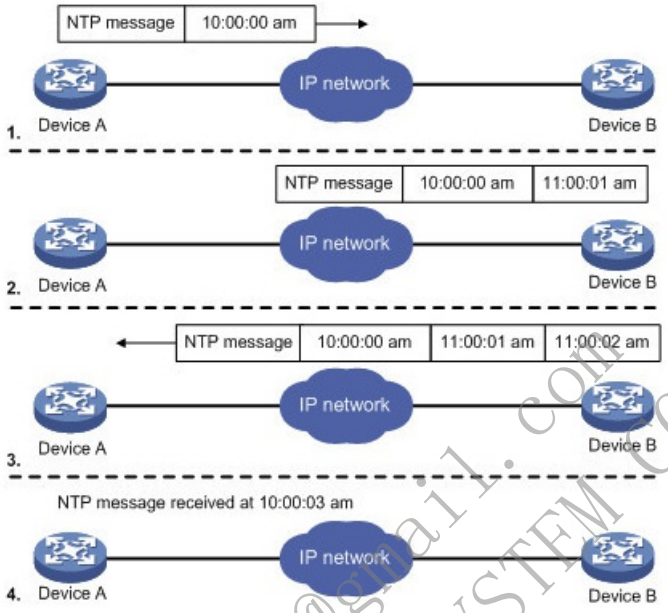
Time sync is a method to synchronize the time of all devices connecting to the NTP server, so that all devices can share the same clock time for providing related functions based on time. Supported time sync types: NTP time sync, manual sync, satellite time sync, platform time synchronization, etc. The following describes the method of NTP time sync.

5.5.1.1 NTP Time Sync

The local system of running NTP can receive sync from other clock sources (self as client), other clocks can sync from the local system (self as server), and sync with other devices.

The basic working principle of NTP is shown in the picture. Device A and Device B are connected via the network, and their systems follow their own independent system time. To auto sync their time, you can set device time auto sync via NTP. For example:

Before time sync between Device A and Device B, the time of Device A is 10:00:00 am, and that of Device B is 11:00:00 am. Device B is set as the server of NTP server, so that the time of Device A should be synchronized with that of Device B. The time of NTP message transmitted between Device A and Device B is 1 second.



The working process of system clock synchronization is as follows:

Device A sends an NTP message to Device B with a timestamp of 10:00:00 am (T1) that is when it leaves Device A.

When the NTP message reaches Device B. Device B will add its own timestamp, which is 11:00:01 am (T2).

Then the NTP message leaves Device B with Device B's timestamp, which is 11:00:02 am (T3).

Device A receives the response message, and the local time of Device A is 10:00:03 am (T4).

Above all, Device A can calculate two important parameters:

Round-trip delay of NTP message: Delay = (T4-T1) - (T3-T2) = 2 seconds.

Time difference between Device A and Device B: offset = ((T2-T1)+(T3-T4))/2=1 h.

Device A can sync its own time with that of Device B according to calculation results.

5.5.2 API Calling Flow

5.5.2.1 Time Sync Configuration

1. Get the Capability of Device Time synchronization Management

You can call this API to get the time sync types currently supported by the device, such as NTP time sync, manual time sync, satellite time sync, EZ platform time sync.

Get the capability: `GET /ISAPI/System/time/capabilities`.

2. Set device time synchronization management parameters

You can configure the time synchronization mode as follows：

Get device time synchronization management parameters: `GET /ISAPI/System/time`;

Set device time synchronization management parameters: `PUT /ISAPI/System/time`;

NTP time synchronization: See 4.2.2 NTP Time Sync (Client).

Manual time synchronization: Set the value of `timeMode` to `manual`, and set the device local time in nodes `localTime`、`timeZone`.

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Satellite time synchronization: Set the value of `timeMode` to `satellite`, and set the device local time in nodes `satelliteInterval`.

Platform time synchronization: Set the value of `timeMode` to `platform`.

Note： For manual time synchronization (time offset including time zone offset): Set manual time synchronization: localTime refers to the local time on device (time offset excluded, in format like 2019-02-28T10:50:44); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00); Get manual time synchronization: localTime refers to the local time on device (time offset included, in format like 2019-02-28T10:50:44+8:30); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00).

5.5.2.2 Time Zone Configuration

1. Get device time zone configuration capability

Call `GET /ISAPI/System/capabilities` to get the system capability. When `isSupportTimeZone` is returned, the time zone configuration is supported by the device.

2. Configure time zone parameters

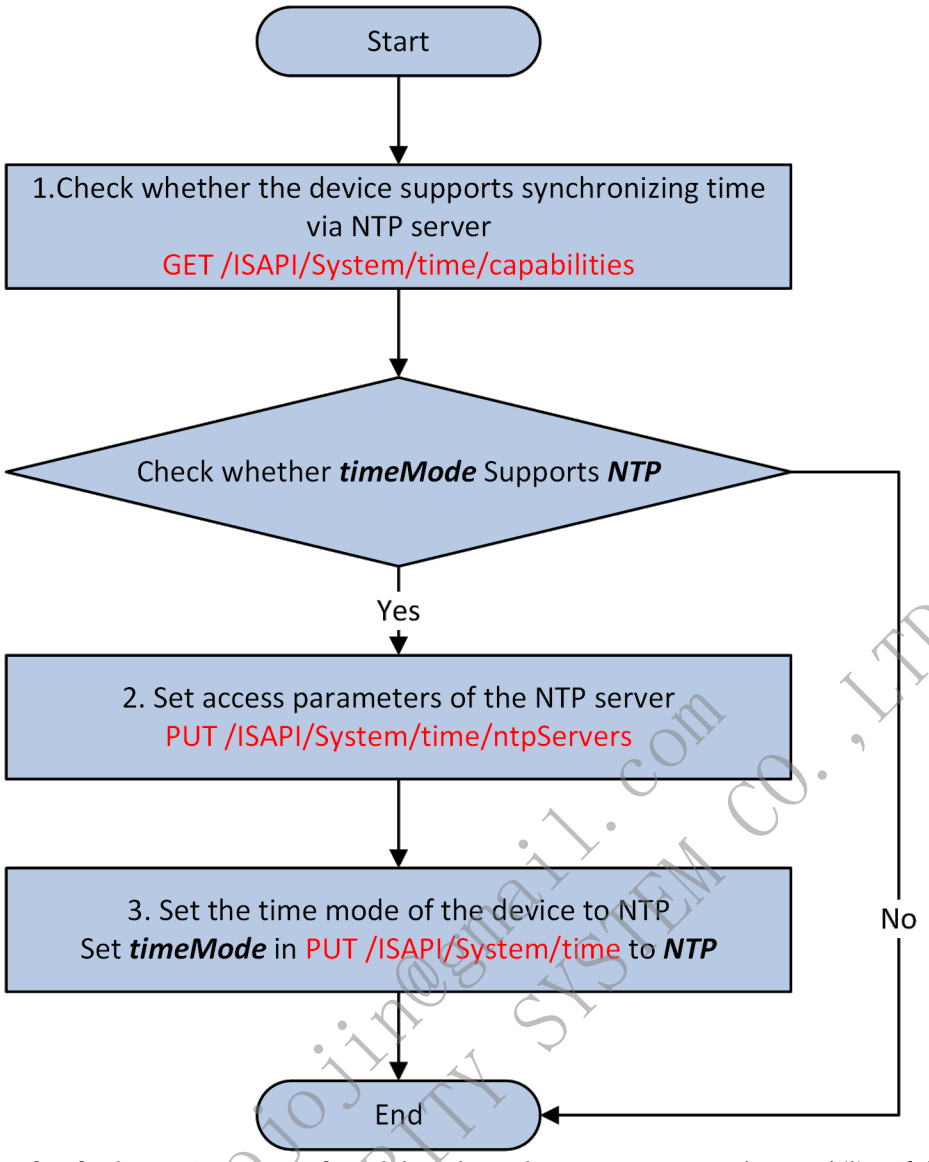
Get the device's time zone parameters: `GET /ISAPI/System/time/timeZone`. Set the device's time zone parameters: `PUT /ISAPI/System/time/timeZone`.

If DST (Daylight Saving Time) is disabled, the example of returned time zone parameters is: CST-8:00:00. It refers to UTC+8, and -8:00:00 is the UTC local time. If DST (Daylight Saving Time) is enabled, the example of returned time zone parameters is: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.2/02:00:00. It refers to UTC+8, the DST time is 30 minutes ahead of local time, the DST starts at 02:00:00 on the first Sunday of April and ends at 02:00:00 on the fifth Tuesday of October. MX.Y.Z: X is the month, Y is the week number in the month, Z is the day of a week (0-Sunday, 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday, 6-Saturday).

5.5.2.3 NTP Time Sync (Client)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as client):

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1. Check whether the device supports synchronizing time via NTP server Get the capability of the device: `GET /ISAPI/System/time/capabilities`, and check whether `timeMode` supports `NTP`.

2. Set access parameters of the NTP server

Supports accessing the NTP server by IP address to synchronize the device time.

Get the access parameter capability of the NTP server: `GET /ISAPI/System/time/ntpServers/capabilities`

Set access parameters of the NTP server: `PUT /ISAPI/System/time/ntpServers`

Get access parameters of the NTP server: `GET /ISAPI/System/time/ntpServers`

3. Set the time mode of the device to NTP

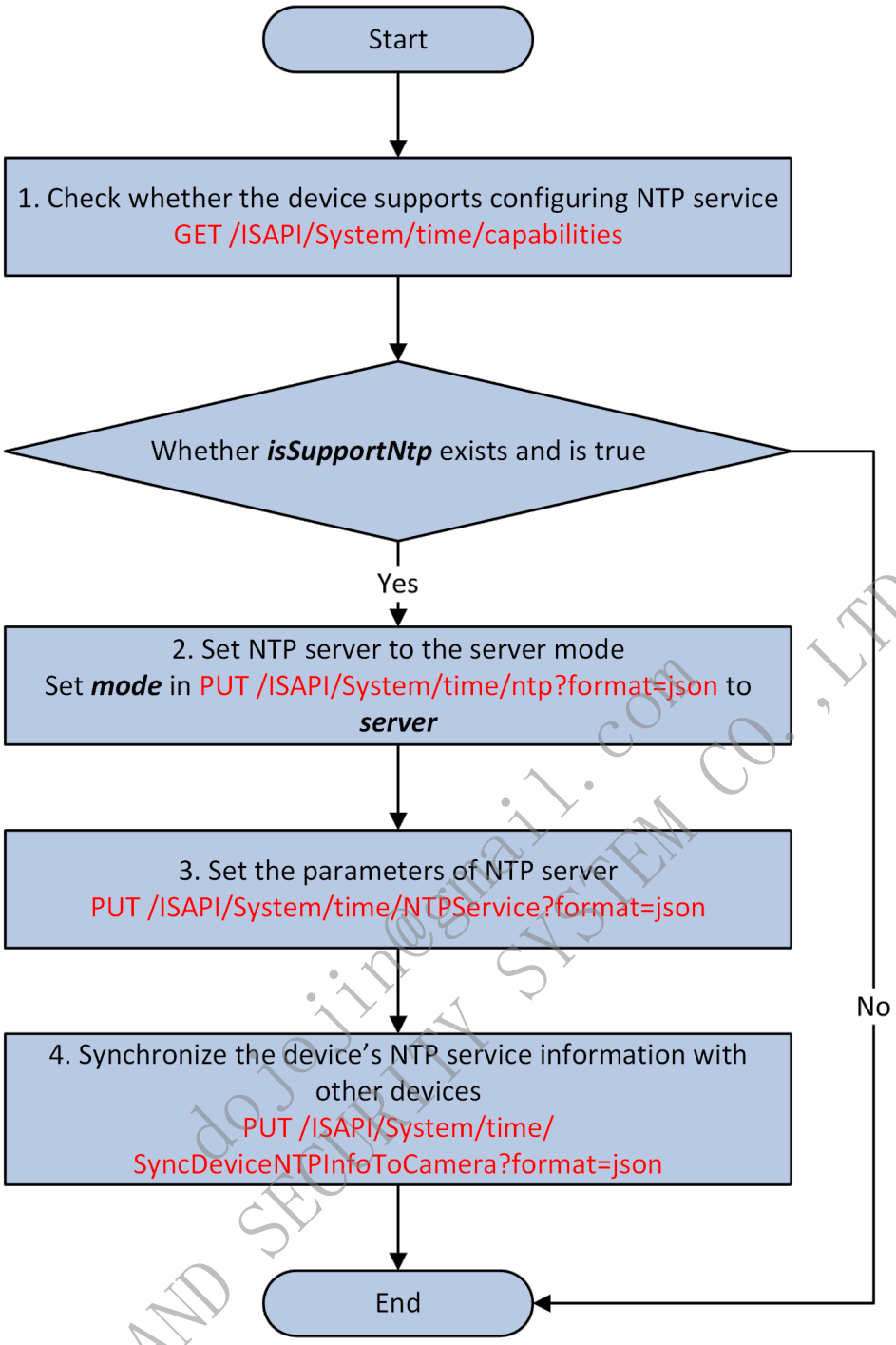
Supports setting the value of `timeMode` to `NTP`.

Get device time synchronization management parameters: `GET /ISAPI/System/time` Set device time synchronization management parameters: `PUT /ISAPI/System/time`

5.5.2.4 NTP Time Sync (Server Mode)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as server):

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1. Check whether the device supports configuring NTP service Get the capability of device time synchronization management: `GET /ISAPI/System/time/capabilities`; If `isSupportNtp` is returned, it indicates that the device supports time synchronization management.

2. Set NTP server to the server mode

Supports setting the value of `mode` to `server`.

Get the capability of server mode: `GET /ISAPI/System/time/ntp/capabilities?format=json`

Set NTP to server mode: `PUT /ISAPI/System/time/ntp?format=json`

Get parameters of NTP server mode: `GET /ISAPI/System/time/ntp?format=json`

3. Set the parameters of NTP server

Supports setting the IP address of the NTP server.

Get the capability of NTP server: `GET /ISAPI/System/time/NTPService/capabilitis?format=json`

Set the NTP server parameters: `PUT /ISAPI/System/time/NTPService?format=json`

Get the parameters of the NTP server: `GET /ISAPI/System/time/NTPService?format=json`

4. Synchronize the device's NTP service information with other devices

Supports synchronizing the time information to the camera.

Get the capability set of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json`

Synchronize device's NTP service information with the camera: `PUT /ISAPI/System/time/SyncDeviceNTPInfoToCamera?format=json`

Get the progress of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/Progress?format=json`

Search for the results of synchronizing device's NTP service information with the camera: `POST /ISAPI/System/time/SyncDeviceNTPInfoToCamera/SearchResult?format=json`

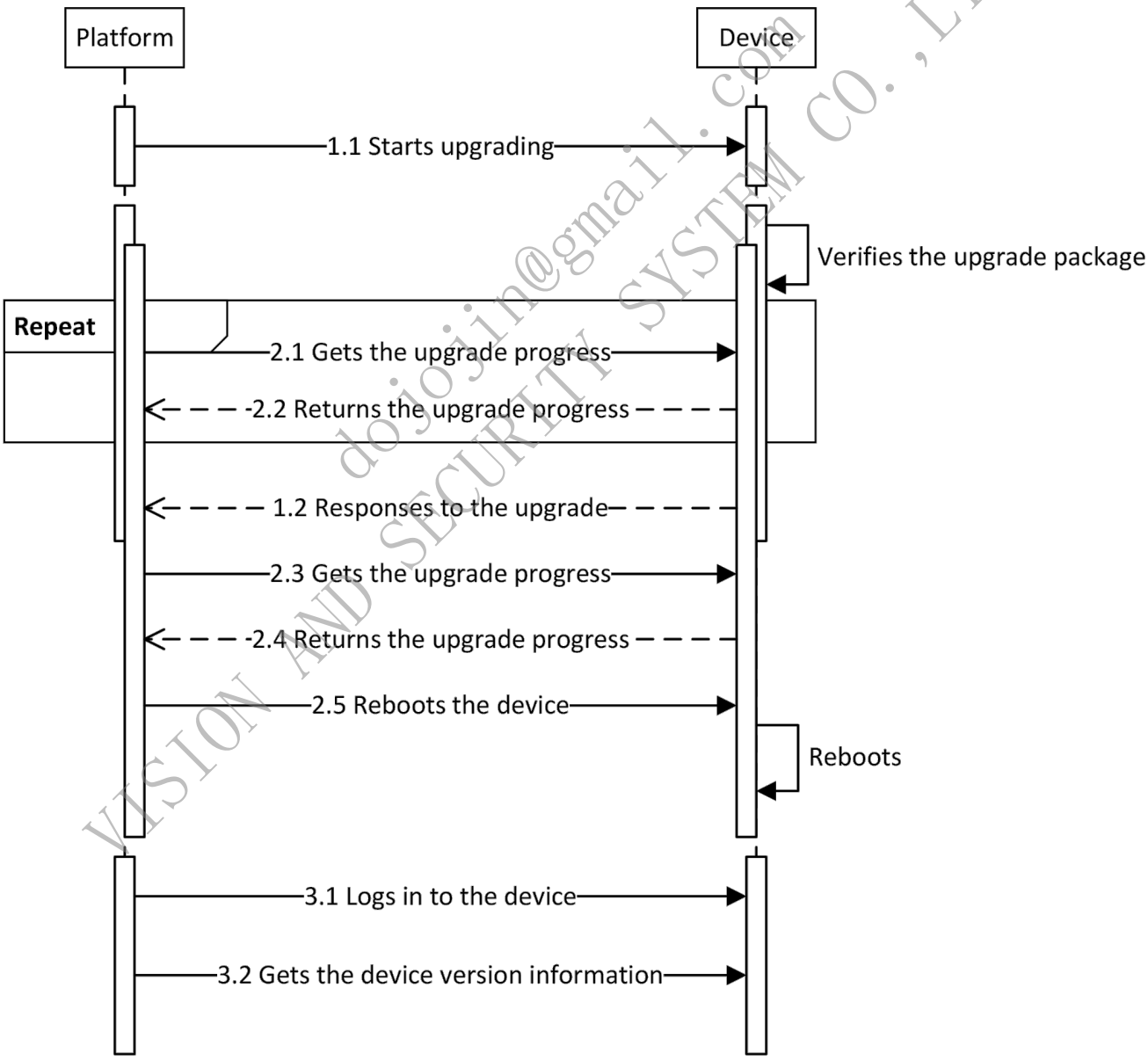
5.6 Device Upgrade

5.6.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades devices via ISAPI.

5.6.2 API Calling Flow

The sequence diagram of upgrading devices by the platform is shown below.



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1. **Upgrade devices.**

   Upgrade the device firmware: `POST /ISAPI/System/updateFirmware`.

2. **Get the device upgrade progress.**

Get the device upgrade progress: `GET /ISAPI/System/upgradeStatus`.

3. **Reboot devices.**

   Reboot devices: `PUT /ISAPI/System/reboot`.

5.7 Mutually Exclusive Functions

5.7.1 Introduction to the Function

Some functions are mutually exclusive due to the device performance (for example, function A and function B cannot run at the same time, i.e, only one of them is allowed at one time).

5.7.2 API Calling Flow

The following three APIs are available for the integration of mutually exclusive functions:

1. Get the information of mutually exclusive functions: `GET /ISAPI/System/mutexFunction/capabilities?format=json`. Call this URL to get the list of existing mutually exclusive functions supported by the device. Note: NVR devices only support setting exlusive function "perimeter" (perimeter protection), and do not support "linedetection" (line crossing detection), "fielddetection" (intrusion detection), "regionEntrance" (region entrance), or "regionExiting" (region exiting).

2. Search for the functions that are mutually exclusive with a specified function: `POST /ISAPI/System/mutexFunction?format=json`. Based on the list of mutually exclusive functions returned by `GET /ISAPI/System/mutexFunction/capabilities?format=json`, you can search for the mutual exclusion status of a specified function and see whether to change the settings and disbale the mutually exclusive function.

3. Get the mutual exclusion information when device function exception occurs: `GET /ISAPI/System/mutexFunctionErrorMsg`. After getting the error code, you can call this API to get the current mutually exclusive functions.

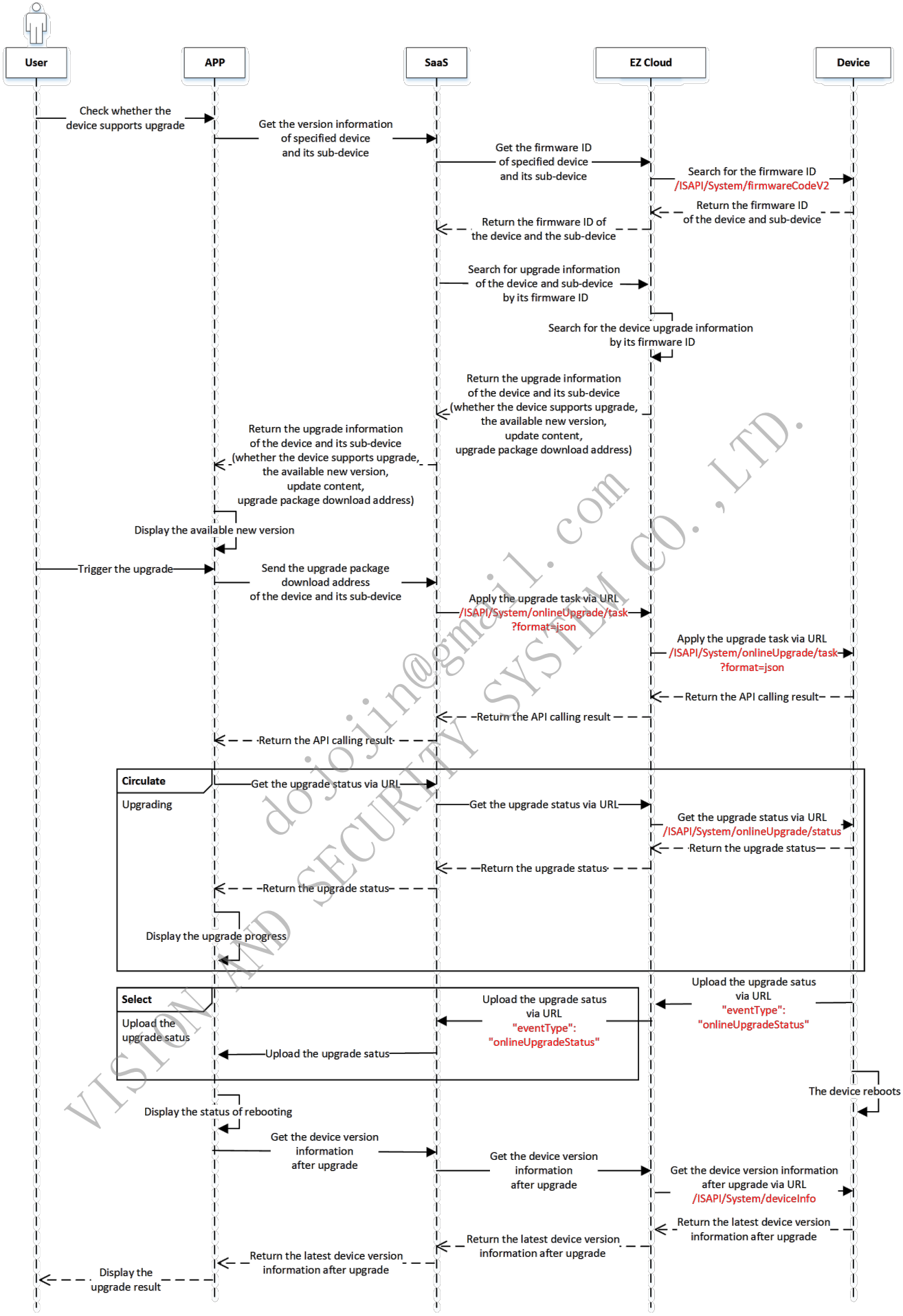
5.8 Remote Online Upgrade Based on The EZ2.0

5.8.1 Introduction to the Function

Considering that the device is connected to EZ via EZ 2.0 Protocol, EZ OTA upgrade solution is not supported. This solution applies to specified version upgrade.

5.8.2 API Calling Flow

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1. Search for the device firmware ID: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information by its firmware ID. If there is an available new version, apply the upgrade task to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade devices (e.g., NVR) or sub-devices (e.g., IPC) via the node

device (e.g., NVR) or sub-device (e.g., IPC) via the node `<channel>`.

3. During the device upgrade, you can get the real-time upgrade progress on the platform: `GET /ISAPI/System/onlineUpgrade/status`.
4. After the device upgrade is finished, the device will upload the `"onlineUpgradeStatus"` and reboot automatically.
5. After both upgrade and reboot are finished, you can get the latest device version information on the platform.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

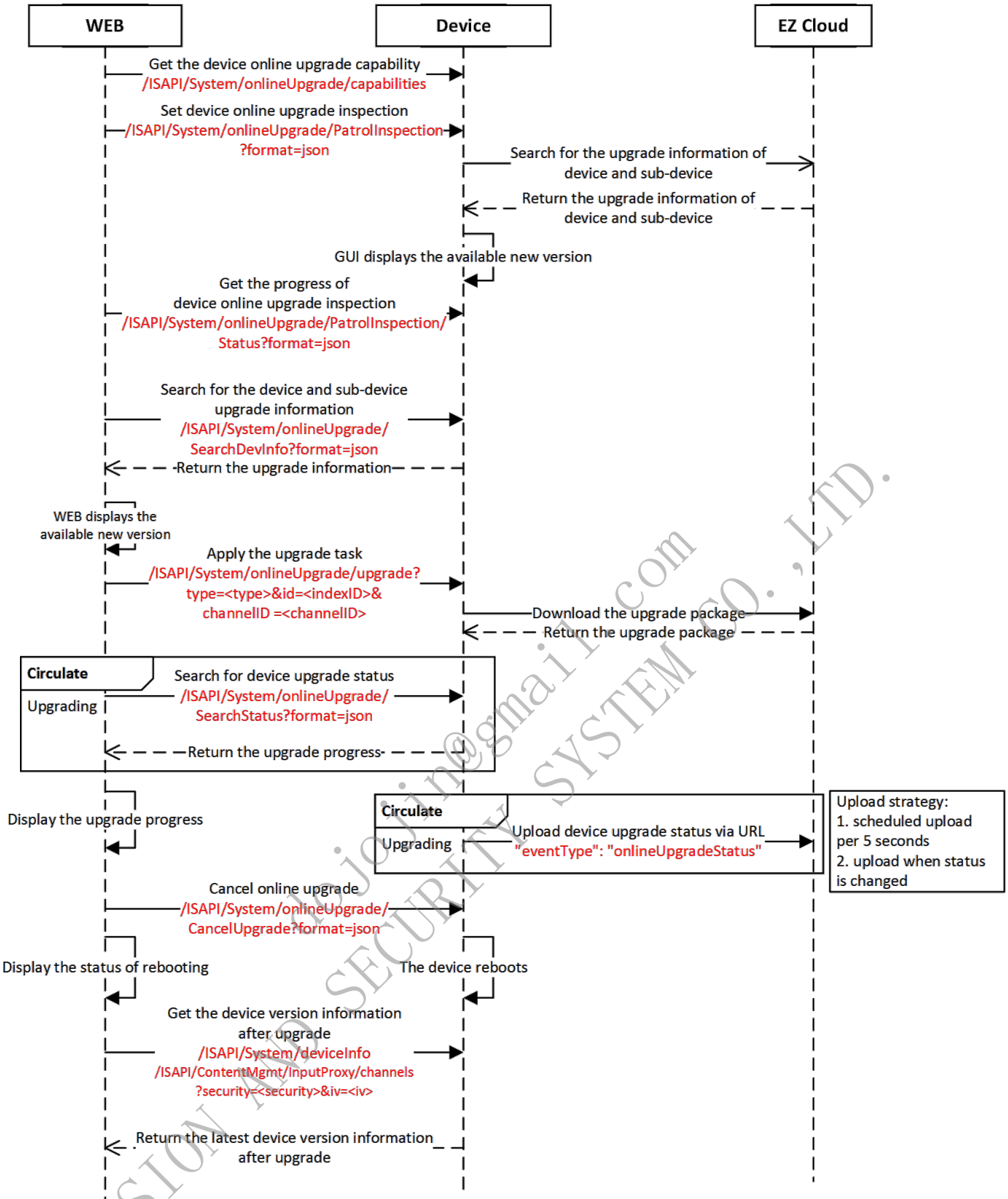
5.9 Remote Online Upgrade Based on the WEB Client

5.9.1 Introduction to the Function

This solution applies to remote online upgrade of the devices and sub-devices connected to WEB client.

5.9.2 API Calling Flow

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1. Get device online upgrade capability: `GET /ISAPI/System/onlineUpgrade/capabilities`.
2. Manually trigger the device online upgrade inspection: `POST /ISAPI/System/onlineUpgrade/PatrolInspection?format=json`. After it is enabled, the device will search for the available new version.
3. During the inspection, the WEB client can get the real-time inspection progress: `GET /ISAPI/System/onlineUpgrade/PatrolInspection/Status?format=json`.
4. After the device inspection is finished, search for the device and sub-device upgrade information: `GET /ISAPI/System/onlineUpgrade/SearchDevInfo?format=json`.
5. Apply the upgrade task: `PUT /ISAPI/System/onlineUpgrade/upgrade?type=<type>&id=<indexID>&channelID =<channelID>`. You can upgrade a specified device via the node `<channelID>` in the URL. Specify sub-device upgrade by configuring `<type>`.

6. You can search for online upgrade status during the upgrade: `GET /ISAPI/System/onlineUpgrade/SearchStatus?format=json`.
7. You can cancel the upgrade during the upgrade: `PUT /ISAPI/System/onlineUpgrade/CancelUpgrade?format=json`.
8. After the upgrade and reboot are finished, you can get the latest device version information.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

5.10 Serial Port Accessed External Device Management

5.10.1 Introduction to the Function

Information management of device accessed the serial are as follows: 1. Configure manufacturer, type, and model information of the specific serial port access device. 2. Search for the device type or model supported by the specific serial port.

5.10.2 API Calling Flow

1. Check whether the device supports information management of serial port. Get the capability of the device serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportDeviceInfo>` is returned, the device supports information configuration of devices access the serial port.

2. Set the information of serial port:

- Get the capability of device information parameters of a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Get device information parameters access single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Set device information parameter of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.

3. Check whether the device supports linking information of devices access the serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportSearchDeviceInfoRelations>` is returned, it indicates that the device supports searching for linked information od devices access the serial port.

4. Search for linked information of devices access the serial port.

- Get the capability of searching for linked parameters of information of devices access a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations/capabilities?format=json`.
- Search for linked parameters of information of devices access a single serial port:`POST /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations?format=json`.

5.11 Serial Port Data Transparent Transmission

5.11.1 Introduction to the Function

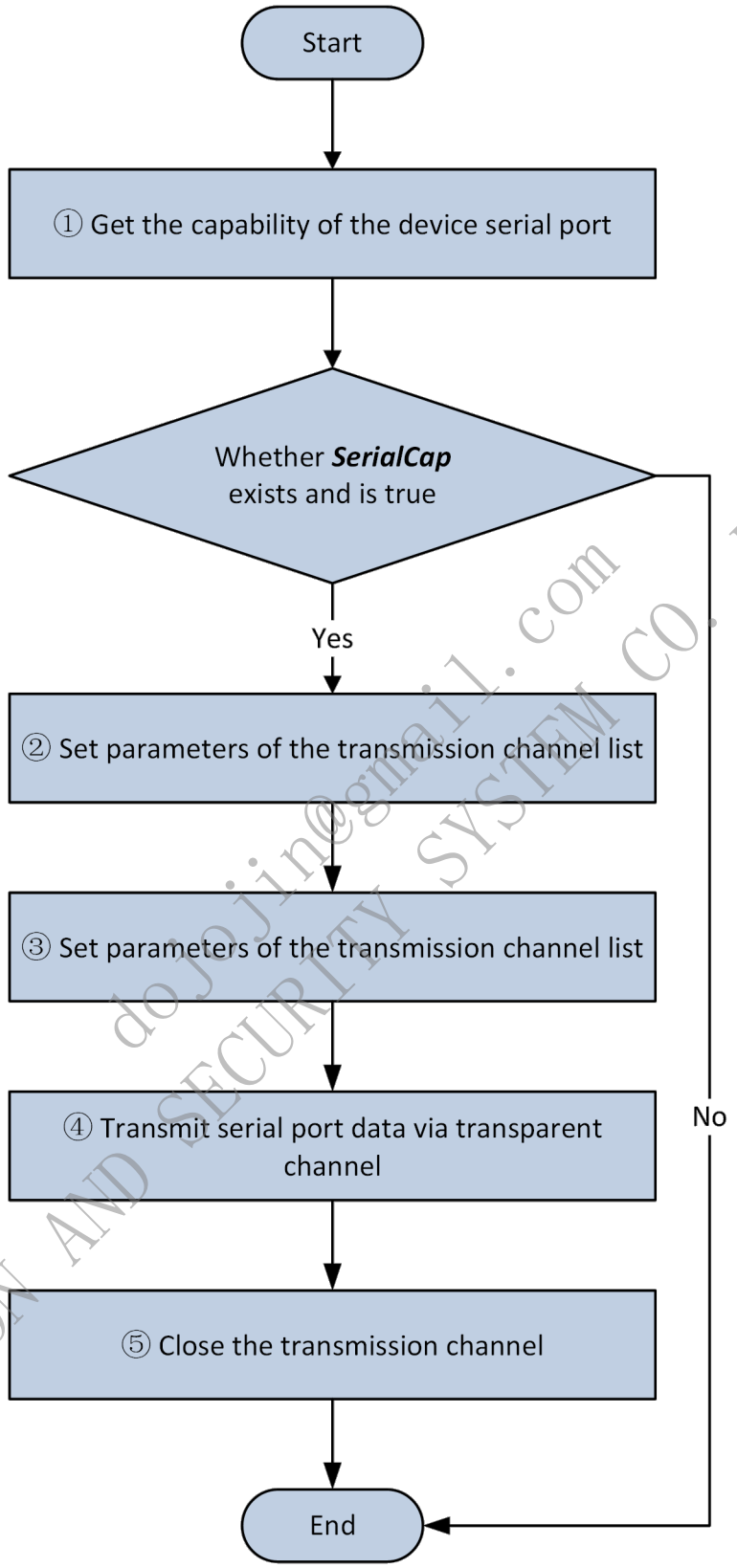
RS485, RS422 and RS232 serial ports external to the device are used as transparent channels to transmit serial port data. Supports the client sending serial data to the device, which then forwards the data to the serial port; conversely, when the serial port sends data to the device, which transparently transmits the data to the client. Note that the device's transparent transmission of serial data is half-duplex communication, meaning that bidirectional communication can be implemented between the device and the client, but the communication data is not necessarily in a question-and-answer format, and the client must match the request and response relationship on its own.

Since most standard HTTP request libraries do not support the persistent connection for receiving and sending data in real time, it is recommended that the client uses two TCP clients to implement the sending and receiving of

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transparently transmitted serial data.

5.11.2 API Calling Flow



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1. Check whether the device supports serial port data transmission.

Get the capability of the device serial port: `GET /ISAPI/System/capabilities`. If `SerialCap` is returned and the value is true, it indicates that the device supports the functions of the serial port.

2. Set parameters of the transmission channel list.

- Get parameters of the specific transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.
- Configure parameters of the specific transmission channel: `GET`

`/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.

3. Open the transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/open`;

4. Transmit serial port data via transparent channel.

- Receive data uploaded by device serial port through transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData`.
- Send data to device serial port through transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData`.

5. Close the transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/close`.

5.12 Serial Port Parameter Configuration

5.12.1 Introduction to the Function

Serial port parameter configuration.

5.12.2 API Calling Flow

1. Check whether the device supports configuring serial port parameters.

Get the capability of device serial port: `GET /ISAPI/System/capabilities. If is returned and its value is true, the device supports functions of serial port.

3. Get parameters of all serial ports.

- Get the capability of all serial ports: `GET /ISAPI/System/Serial/capabilities`.
- Get control parameters of all serial ports: `GET /ISAPI/System/Serial/ports?permissionController=<indexID>`.

4. Set control parameters of a single serial port.

- Get control parameters of single serial port: `GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>`.
- Configure control parameters of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>`.

5. Get the status of single serial port:`GET /ISAPI/System/Serial/ports/<portID>/status`.

5.13 Sub-device Batch Upgrade

5.13.1 Introduction to the Function

Sub-device Batch Upgrade (Single Task): applicable to situations of upgrading multiple sub-devices using one upgrade package. Application scenarios include: when the UWB positioning anchor connects to the web, upgrading multiple tags (sub-devices) through the positioning engine (gateway), with the same upgrade package.

Management of Sub-device Batch Upgrade Tasks: There are many types of sub-devices, such as LoRa nodes, which have a slow upgrade process and different models of LoRa nodes in the field also require different upgrade packages. Therefore, it is necessary to support the creation of multiple upgrade tasks at once, entrusting the upgrade to the devices, to enhance the practicality of batch upgrading sub-devices.

5.13.2 API Calling Flow

Sub-device Batch Upgrade (Single Task)

Get the device capability: `/ISAPI/System/capabilities`.

If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that the device support the function.

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If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that the device support getting the progress of batch upgraading sub-devices.

Sub-device batch upgrade: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.

Batch sub-device upgrade progress search: `POST /ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json`.

Sub-device Batch Upgrade Task Management

Get device system capabilities `/ISAPI/System/capabilities`, If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that batch upgrade of sub devices is supported,

If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that progress of batch upgrading sub devices can be searched,

If the node `isSupportSearchBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading devices can be searched,

If the node `isSupportModifyBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading sub devices can be edited,

If the node `isSupportDeleteBulkUpgradeChildDeviceTask` exists and is true, it indicates that task of batch upgrading sub devices can be deleted.

Get the capability of batch upgrading sub-device: `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`.

Batch upgrade sub devices: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.****

Search for tasks of batch sub-device upgrade: `POST /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`.

Edit tasks of batch sub-device upgrade: **`PUT /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`**.****

Delete tasks of batch sub-device upgrade: `PUT /ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json`.

Note: 1. The two methods for batch upgrading child devices are the same API: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`. If the `BulkUpgradeChildDeviceListCap.taskID` is returned in `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`, it indicates that the task method is supported in the sub device upgrade. 2. To determine which sub devices can be upgraded, you can use the API `POST /ISAPI/IoTGateway/Childmanage/SearchChild?format=json`. Sub devices with the `tag` `isUpgradable` and `value` `true` indicate that the sub devices support upgrades.

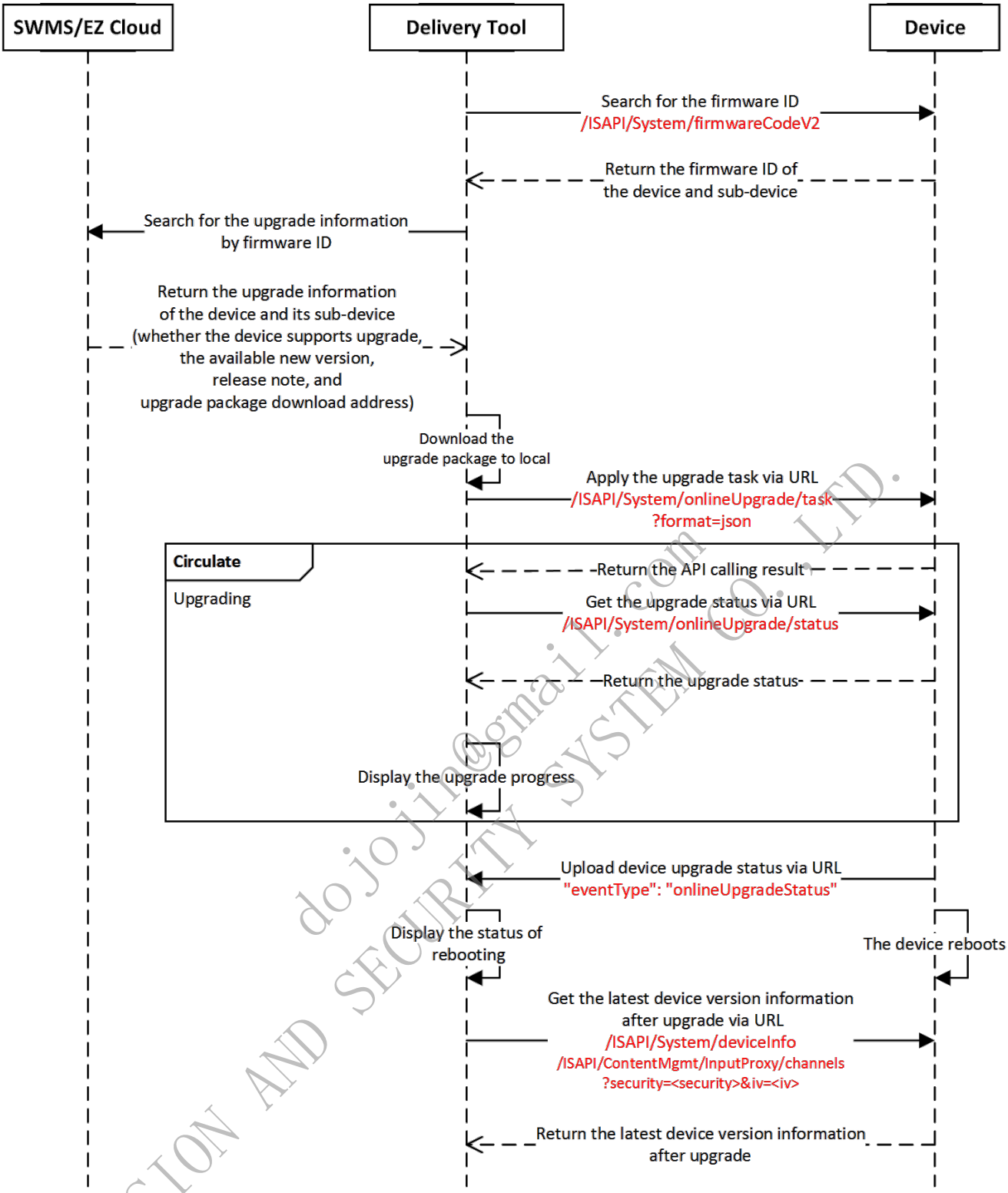
5.14 Upgrade Based on Interactive Tools Local Deployment

5.14.1 Introduction to the Function

This solution applies to the upgrade of devices and sub-devices connected via ISAPI. If the device is connected via ISAPI protocols, it is required to set up HTTP(s) service to store device upgrade package and generate URL for downloading upgrade package. If the device is connected via HCNetSDK, you can implement the device upgrade via transmitting ISAPI.

5.14.2 API Calling Flow

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1. Search for the device firmware ID: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information by firmware ID on the SWMS/EZ Cloud, download upgrade package to local storage, and apply the upgrade task to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade device (e.g., NVR) or sub-device (e.g., camera) via the node `<channel>`.
3. During the upgrade, you can get the real-time upgrade progress: `GET /ISAPI/System/onlineUpgrade/status`.
4. After upgrade is completed, the device will upload the `"onlineUpgradeStatus"` (upgrade status), and reboot automatically.
5. After the device rebooting, you can get the latest device version information.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=`

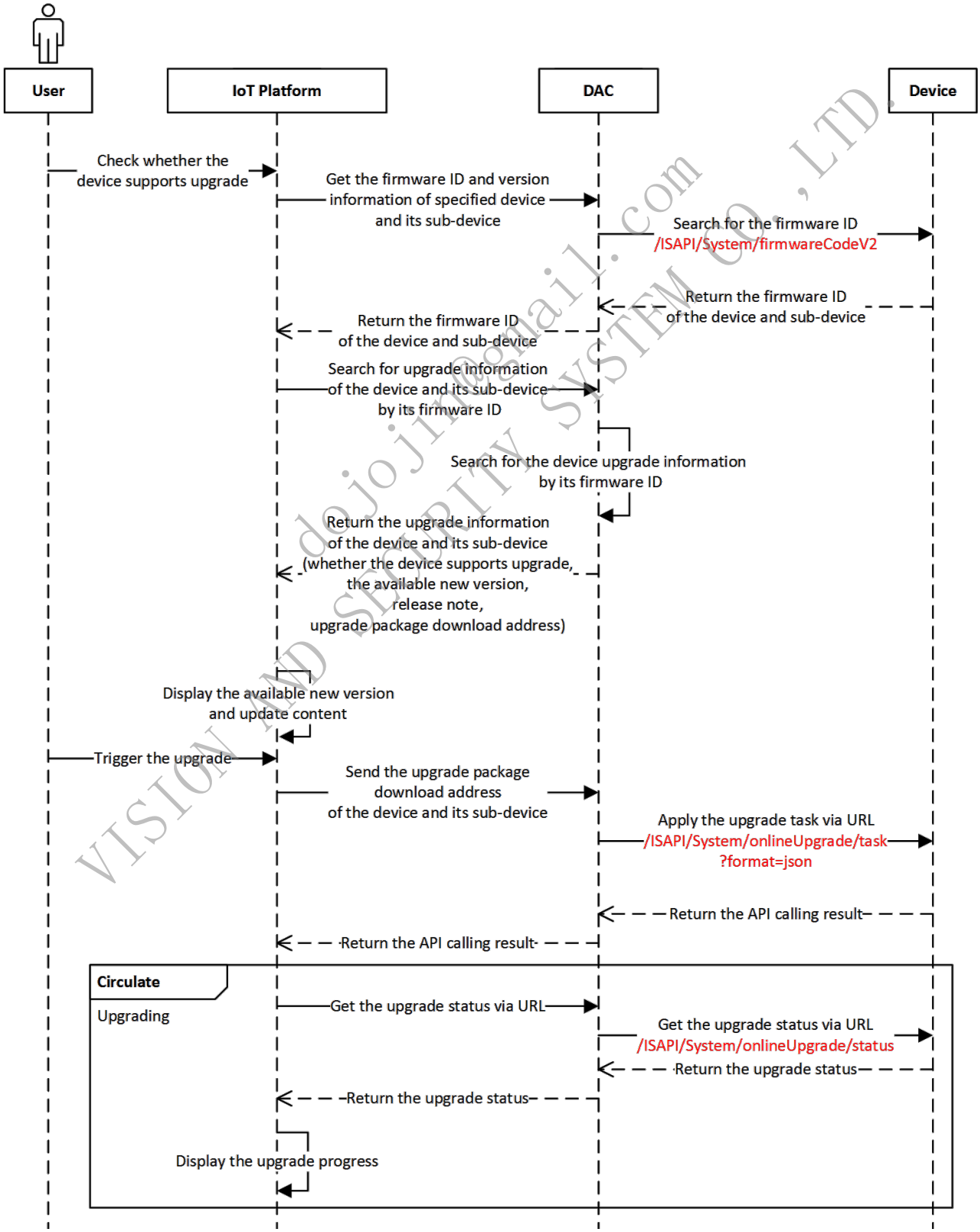
`<security>&iv=<iv>`.

5.15 Upgrade Based on Platform Local Deployment

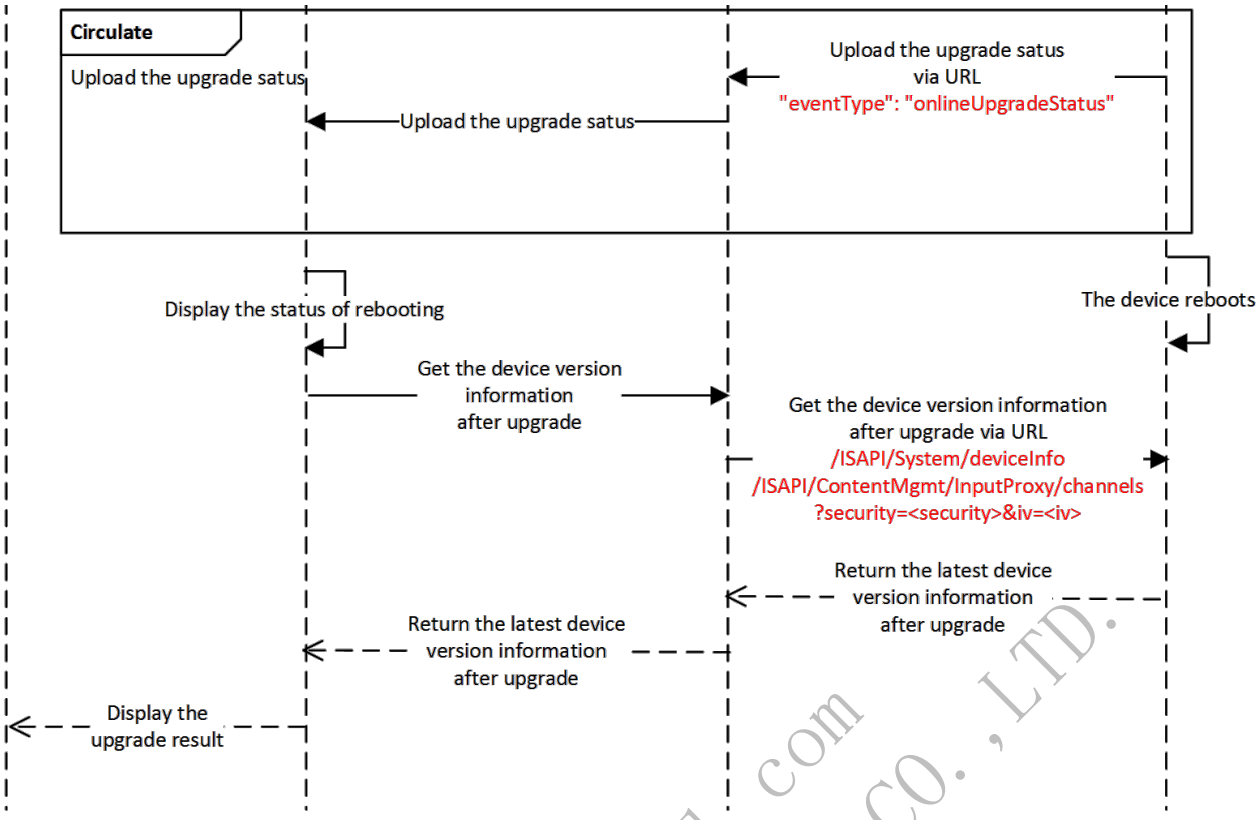
5.15.1 Introduction to the Function

This solution applies to the upgrade of devices and sub-devices connected via ISAPI and ISUP. If via ISAPI, it is required to set up HTTP(s) service or transfer binary upgrade package files so as to store device upgrade package and generate URL for downloading upgrade package. If via ISUP, it is required to set up FTP service to store device upgrade package and generate URL for downloading upgrade package. If the device is connected via HCNetSDK protocols, you can realize device upgrade via transmitting ISAPI.

5.15.2 API Calling Flow



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1. Search for device firmware ID on the platform: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information in DAC by device firmware ID on the platform. If there is an available new version, DAC will trigger an upgrade task and apply it to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade device (e.g., NVR) or sub-device (e.g., IPC) via the node `<channel>`.
3. During the upgrade, you can get the real-time upgrade progress on the platform: `GET /ISAPI/System/onlineUpgrade/status`.
4. After the device upgrade is finished, the device will upload the `"onlineUpgradeStatus"` (upgrade status), and will reboot automatically.
5. After the upgrade and reboot are finished, you can get the latest device version information on the platform.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

6 Video (General)

6.1 Async Capturing Manually

6.1.1 Introduction to the Function

The client sends a capture command to the device, and the device immediately responds by returning the URL of the storage server where the captured picture is stored. The client can request image data from the storage server based on the received URL. It is suitable for the limited network bandwidth between the client and the device or the client centrally manages a large number of devices. The client can choose to use the image URL address carried in the alarm content to request pictures from the storage server when the network conditions improve or there is no concurrency of a large number of requests, so as to reduce the probability of image download failure and the risk of network congestion.

6.1.2 API Calling Flow

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1. The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportSnapshotAsync` is returned and its value is `true`, async manual capturing shall be supported.
2. (Optional) The client gets the async manual capturing capability of a specified channel: `GET /ISAPI/Streaming/channels/<channelID>/picture/async/capabilities?format=json`.
3. The client controls the specified channel to manually and asynchronously capture pictures: `GET /ISAPI/Streaming/channels/<channelID>/picture/async?format=json&imageType=<imageType>&URLType=<URLType>`, and the device returns the URL of the storage server for capturing pictures.

Remarks:

- If the step 2 returns a failure during the integration， it is recommended that you check whether the channel supports async manual capturing according to steps 1 and 3.

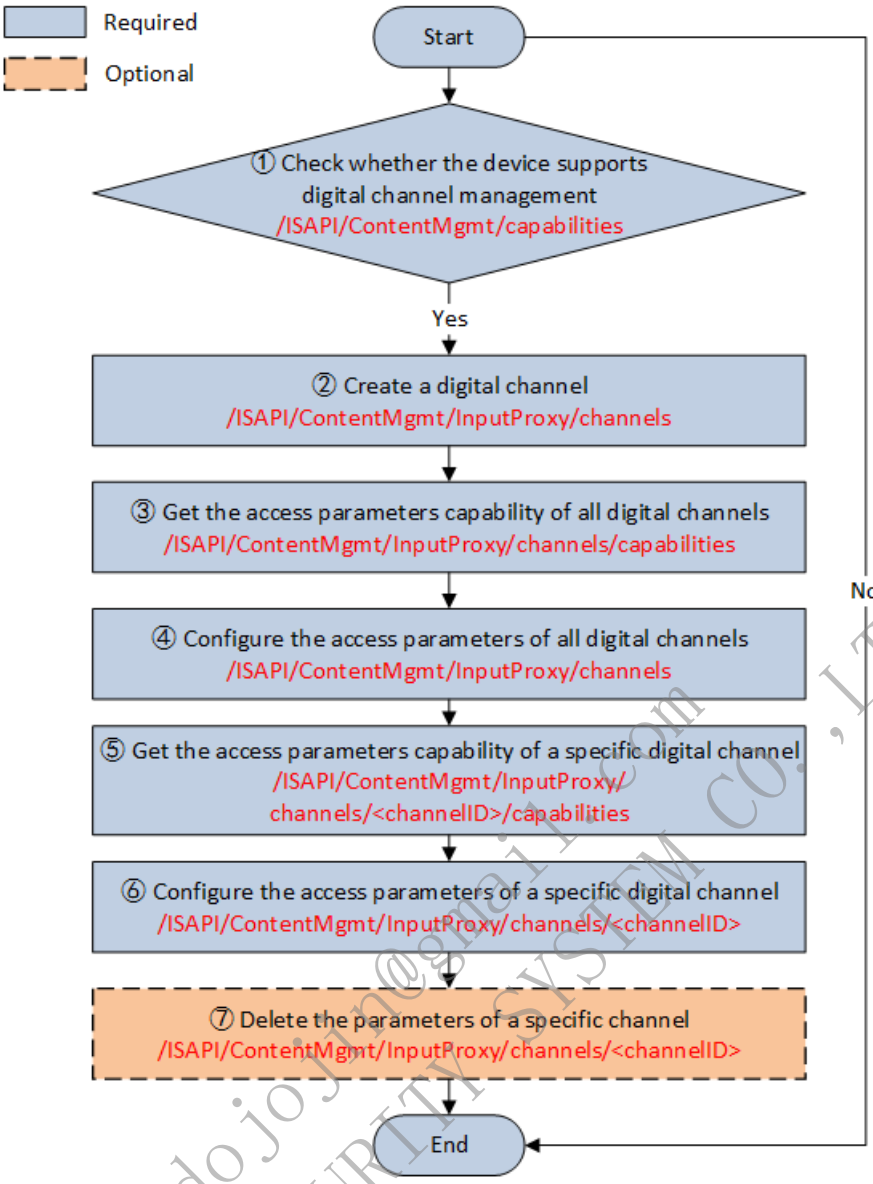
6.2 Digital Channel Access Management

6.2.1 Introduction to the Function

- Video channel includes analog channel and digital channel. The service involves channel access protocols, streaming protocols, address information, version No. of channel access firmware, and so on.
- Analog channel refers to the local video channels of the device.
- Digital channel refers to the video channels accessed by the device by IP address or POE port (e.g., NVR accessing the video channel of network camera).
- Analog channel No. and digital channel No. are unique. Devices that do not support digital channels will only return analog channel No. For example, analog channel No. ranges from 1 to 8 (determined by the number of analog channel ports of the device), digital channel No. ranges from 9 to 32 (within the maximum number of digital channels, determined by the number of added network cameras).

6.2.2 API Calling Flow

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1. Get the capability node to check whether the device supports digital channel management: `GET /ISAPI/ContentMgmt/capabilities`. If the value of the node `iSptInputProxyChanCap` is true, it indicates that the device supports digital channel management.
2. Add digital channels: `POST /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.
3. Get the access parameters capability of all digital channels: `GET /ISAPI/ContentMgmt/InputProxy/channels/capabilities?security=<security>&iv=<iv>`.
4. Configure the access parameters of all digital channels:

   Get: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

   Configure: `PUT /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.
5. Get the capability of configuring the access parameters of a specific digital channel: `GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/capabilities`.
6. Configure the access parameters of a specific digital channel:

   Get: `GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>?security=<security>&iv=<iv>`.

   Configure: `PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>?security=<security>&iv=<iv>`.
7. Delete the parameters of a specific digital channel: `DELETE /ISAPI/ContentMgmt/InputProxy/channels/<channelID>`.

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6.2.3 Video Channel Application

1.Get the list of video channels, which is the sum of digital channels and analog channels. Digital channels and analog channels need to be obtained separately:

- 1) Call "Get all video input channel parameters" to get all analog channels via GET /ISAPI/System/Video/inputs/channels and VideoInputChannelList parameter is returned. The id represents the analog channel No.
- 2) Call the "Get all digital channel access parameters" to get all digital channels via GET /ISAPI/ContentMgmt/InputProxy/channels?security=&iv= and InputProxyChannelList parameter is returned. The id represents the digital channel No.

2. Video channel function configuration supports unified configuration of analog and digital channels. Both the analog channel interface starting with `/ISAPI/System/Video/inputs/channels/<channelID>/` and the digital channel interface starting with `/ISAPI/ContentMgmt/InputProxy/channels/<channelID>/` are supported. **It is recommended to use the channel configuration interface starting with `/ISAPI/System/Video/inputs/channels/<channelID>/`, where <channelID> represents all channel No.s supported by the device, including analog channels and digital channels.**

> For example, configure text overlay parameters for a channel: Both `GET/PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays` and `GET/PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/overlays` are supported, and it is recommended to use `/ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

6.3 Live View

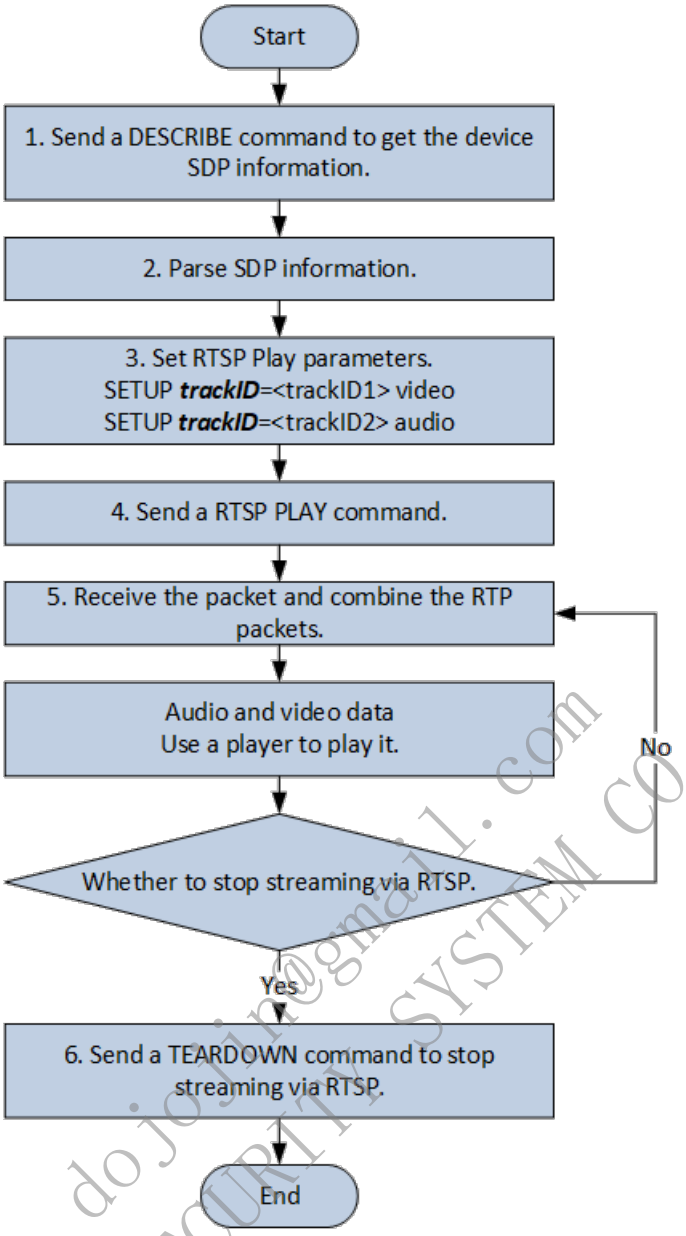
6.3.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

6.3.2 API Calling Flow

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1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`; Digest authentication with devices is required.
2. The client parses the media SDP information returned by the device.
3. Set RTSP play parameters, set the track ID parsed from SDP information via SETUP commands. For example, trackID=1 indicates videos while trackID=2 indicates audios.
4. The client sends an RTSP PLAY command, and device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`.
5. The client receives the RTP package sent by device. Divided RTP packets should be assembled on the client before being parsed.
6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address, `[:port]` is optional, and 554 by default, `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream); for example, the IP address of the target device is `172.7.203.11`, and if the channel number is 172, the streaming address will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2-=value2…)`, such as `/Streaming/Channels/101?transportmode=unicast`.

6.3.3. Example

1. A client sends an RTSP DESCRIBE request.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

```
RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s=Media Server V4.22.126
i=Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:6154
a=range:npt=now-
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=4D0014;packetization-mode=0
a=control:trackID=1
b=AS:6144
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=2
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000020000011171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP requests, and the server responds to them.

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 2
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

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```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Session:1127293610;timeout=60
Transport:RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=433122ab
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

6. The client sends an RTSP PLAY request.

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

7. The server sends audio and video stream data.

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. .....d1.w....c....".T....g....).i.........a....7.S..~J.............X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

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6.4 Manual Capturing Events

6.4.1 Introduction to the Function

According to the preset information and capture interval configured by the client, the device manually captures images at the specified preset and reports the image information in the form of events. It is used for the client to identify emergency situations and perform an emergency patrol on some presets for analysis. It has a higher priority than the daily scheduled patrol capture. Common usage scenarios: according to the scheduled capturing event (eventType: imageCapture) rules, the store camera regularly audits in a number of key areas. If in the event of an emergency, and a worker wants to immediately re-inspect each area, you can set a manual capture rule to complete an independent audit. After the audit is completed, the camera will continue to audit according to the scheduled capturing rules.

6.4.2 API Calling Flow

1. The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportManualImageCapture` returns `true`, manual capturing shall be supported.
2. The client gets manual capturing capability for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture/capabilities?format=json`.
3. The client sets the parameters of manual capture events for a specified channel: Get parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture?format=json`. Set parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture?format=json`.
4. Receive the event uploaded by the device in arming or listening mode: `eventType:manualImageCapture`.

Remarks:

- Each time the client sets manual capture event parameters, the channel immediately takes a patrol capture of the relevant presets. When the last preset capture is finished, the manual capture event ends.

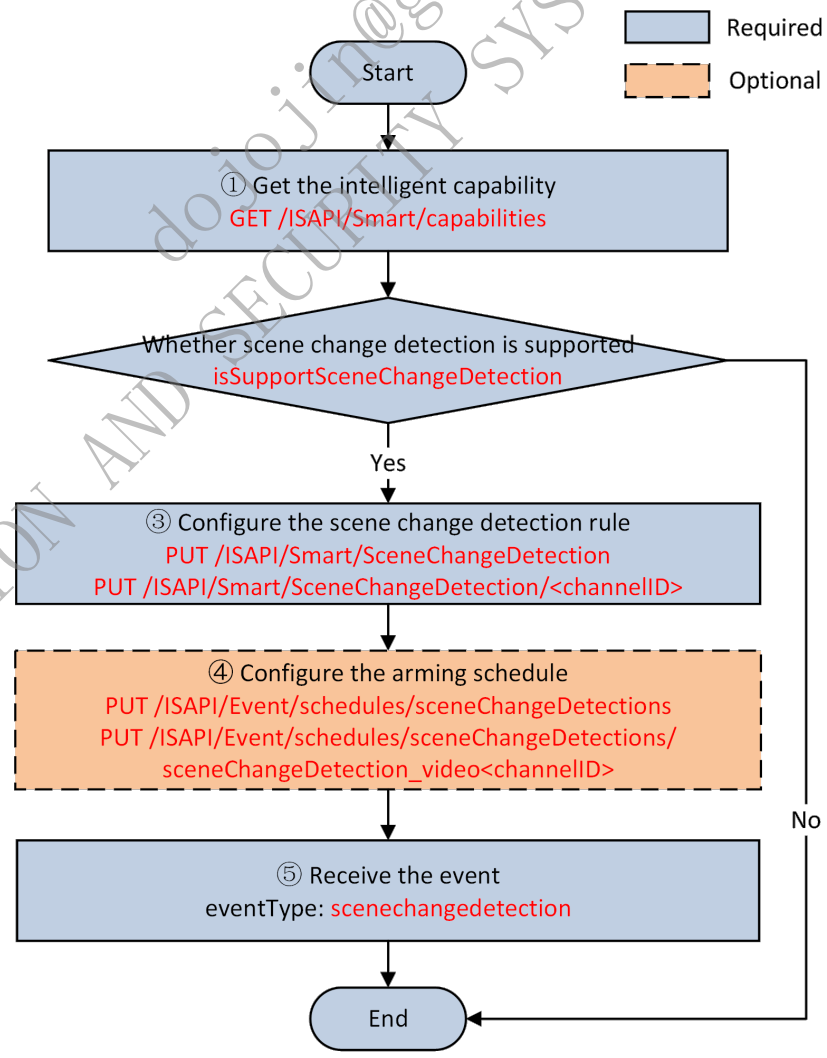
6.5 Scene Change Detection

6.5.1 Function Introduction

The scene change detection function identifies whether a scene has changed based on configured rules. Once a scene change is detected, an alarm is triggered.

This function is commonly used to monitor whether external forces have caused changes in camera's viewing angle. When the viewing angle is changed by external interference, or the monitoring scene is tampered, affecting normal monitoring, relevant personnel can be notified promptly to take actions.

6.5.2 API Calling Flow



1. Get the device's intelligent capability.

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`GET /ISAPI/Smart/capabilities`. If the response returns `isSupportSceneChangeDetection` as `true`, it indicates that the device supports the scene change detection.

2. Configure the detection rules.
   - Get the scene change detection capability: `GET /ISAPI/Smart/SceneChangeDetection/<channelID>/capabilities`.
   - Get scene change detection parameters: `GET /ISAPI/Smart/SceneChangeDetection`.
   - Set scene change detection parameters: `PUT /ISAPI/Smart/SceneChangeDetection`.
   - Get scene change detection parameters for a specific channel: `GET /ISAPI/Smart/SceneChangeDetection/<channelID>`.
   - Set scene change detection parameters for a specific channel: `PUT /ISAPI/Smart/SceneChangeDetection/<channelID>`.

   Note: A higher **sensitivityLevel** makes the camera more sensitive to scene changes. Adjust it based on actual environment.

3. (Optional) Configure the arming schedule.
   - Get schedule parameters of scene change detection for all channels: `GET /ISAPI/Event/schedules/sceneChangeDetections`.
   - Set schedule parameters of scene change detection for all channels: `PUT /ISAPI/Event/schedules/sceneChangeDetections`.
   - Get schedule parameters of scene change detection for a specific channel: `GET /ISAPI/Event/schedules/sceneChangeDetections/sceneChangeDetection_video<channelID>`.
   - Set schedule parameters of scene change detection for a specific channel: `PUT /ISAPI/Event/schedules/sceneChangeDetections/sceneChangeDetection_video<channelID>`.

   Note:
   - Up to 8 time periods can be configured per day.
   - Time periods must not overlap.

4. Receive the event (`"eventType":"scenechangedetection"`).

6.6 Sub-device's Transparent Transmission

6.6.1 Introduction to the Function

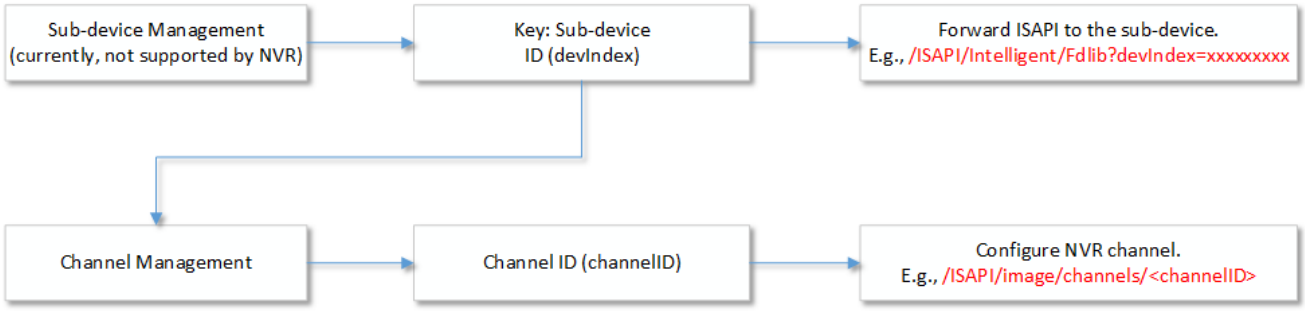
In actual application scenarios, the NVR being used is generally not the latest version, if an IPC with latest functions is added to the system, the NVR will fail to recognize the IPC's new functions due to version incompatibility, so the platform (including device access platforms, client softwares, device's web plug-ins, etc, hereinafter referred to as the platform) will fail to operate new functions of the IPC.

Besides, in video channel resource management, the platform can only get information about digital channels (also known as IP channels) added to the NVR, but not the device that the channels belong to. For example, two video channels (channel 13 and channel 18) of a dual-lens people counting camera are added to the NVR. The platform can only get that digital channel 13 and 18 are added to the NVR, but not that the two channels belong to one camera. Such information loss may lead to repeated operations, which will further cause problems in actual practices. For example, users may fail to upgrade IPC via digital channel No. if not knowing the relationship between camera and digital channel.

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To settle above problems, the sub-device's transmission function is introduced to the video channel resources management. In this way, users can directly access to the device of the digital channels via the devIndex (sub-device index) allocated to the device by NVR.

Notably, despite the introduction of the sub-device's transparent transmission function, the NVR does not function as an access gateway, and so it does not function with sub-device management on gateway. Refer to the Sub-device Management of Device Management for details.

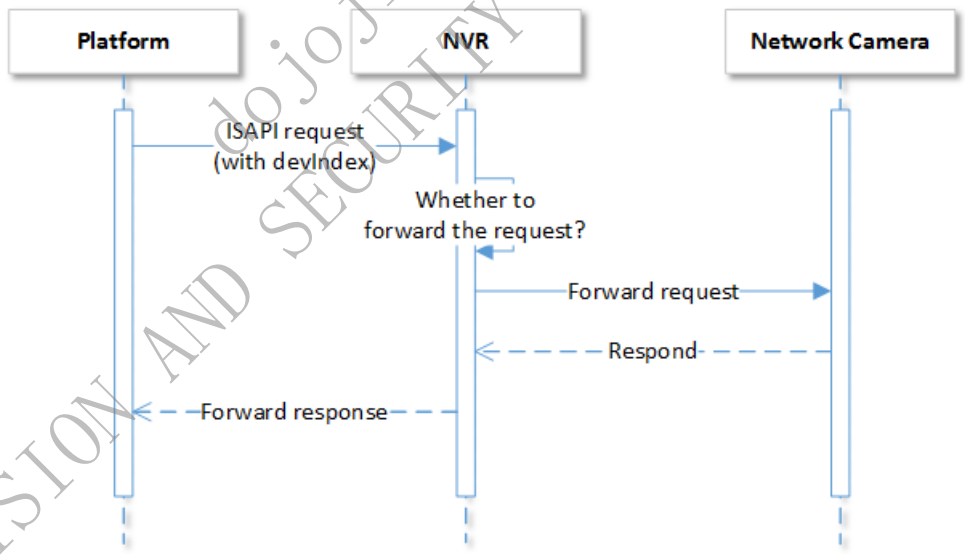


Steps are as follows:

1. First, the platform will check whether the NVR supports sub-device's transparent transmission. If it is supported, follow the next step. Second, add devIndex to the request URL.
2. For the ISAPI request sent to the NVR by the platform, if it contains devIndex, it will be forwarded to the specific sub-device; otherwise, the NVR will respond to the request.
3. When the request is forwarded to the sub-device by the NVR, delete the devIndex in the request URL so as to realize ISAPI interaction.

For example:

1. Send the request URL (`GET /ISAPI/System/deviceInfo`) to get the information of the platform's NVR device.
2. Send the request URL (`GET /ISAPI/System/deviceInfo?devIndex=<devIndex>`) to get the information of a specific sub-device of the NVR.

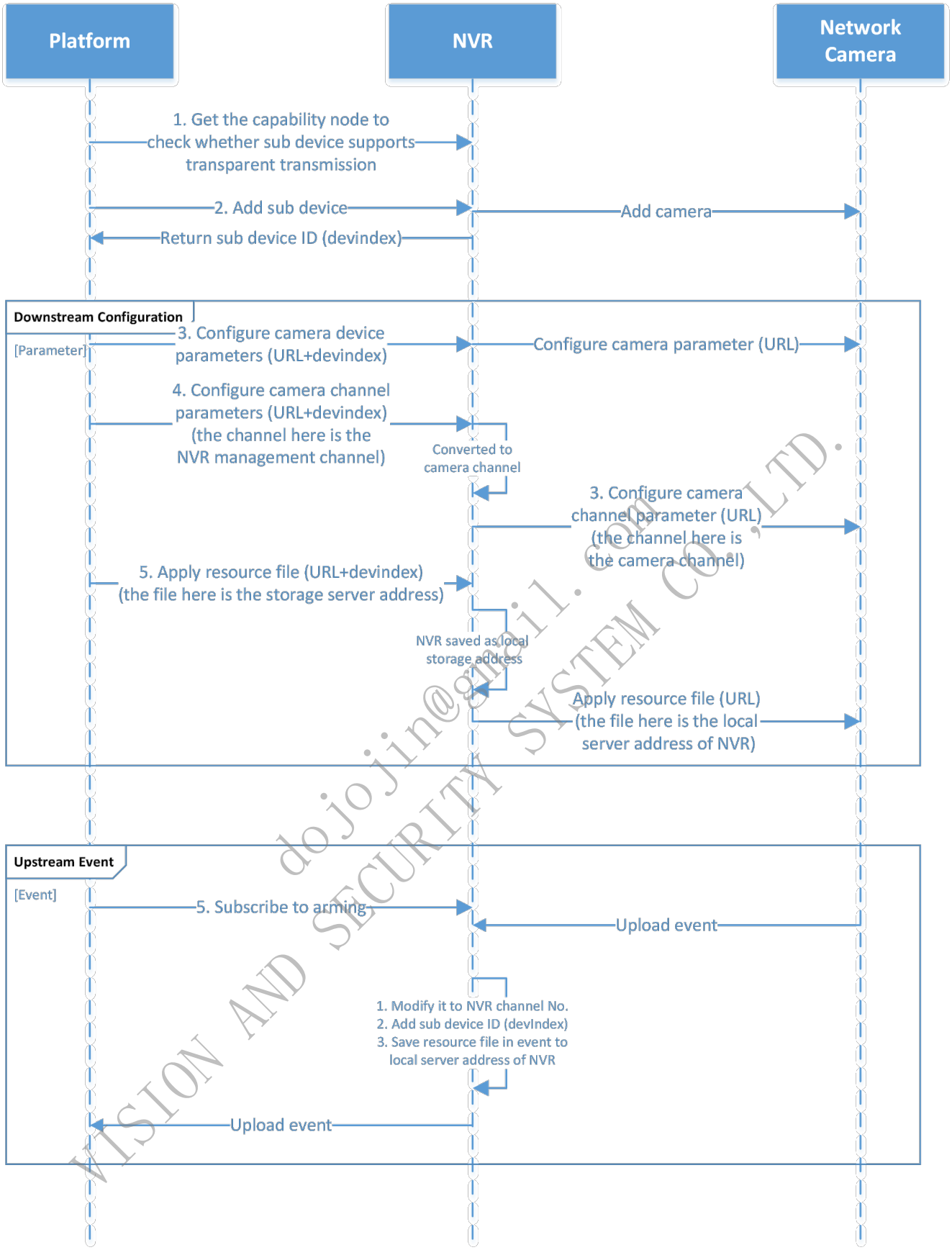


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Note:

1. This function is applicable to transitory connections, while inapplicable to persistent connections (e.g., arming subscription, two-way audio).
2. It is required to delete the devIndex in the URL before NVR forwards the ISAPI request.
3. For the VCA function of NVR, the transmission solution of sub-devices is not adopted. You can switch the analysis unit by the following URI: /ISAPI/Smart/analysisUnitSwitch/channels/`<channelID>`/event/`<EventType>`.

6.6.2 API Calling Flow



1. Get the capability node to check whether the device supports transparent transmission. If the value of the node `isSupportDevIndex` is true, it indicates that the device supports transparent transmission.
2. Create digital channels: `POST /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`. The sub device ID `devindex` will be returned.
3. Get access parameters of all digital channels: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.
   Each digital channel information contains one parameter `devIndex`, which is the same for digital channels of a same physical device.

4. When splicing ISAPI request, if the URL contains `devIndex=<devIndex>`, it represents a request for the sub-device. For example, you can send `12345678` (the request URL) to get the information of the sub-device `GET /ISAPI/System/deviceInfo?devIndex=12345678` of the target device. Note: 1) For the downstream configuration of a video channel, the platform will apply the channel managed by NVR, and before applying, the NVR will transform channel into the video channel. 2) When the platform applies the storage address of resource files, the camera cannot access the storage service to get resource files because of the network isolation between the camera and platform, so the storage address must be saved as a local storage address by the NVR before being applied.

6.7 Sync Capturing Manually

6.7.1 Introduction to the Function

The client sends a capture command to the device, and the device immediately returns the captured picture data in respond. It is often used in scenarios with high network bandwidth between clients and devices to facilitate the rapid transmission of image data, such as a LAN built on a company campus.

6.7.2 API Calling Flow

1. (Optional) The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportManualSnapPicture` returns `true`, sync manual capturing shall be supported.
2. (Optional) The client gets sync manual capturing capability for a specified channel: `GET /ISAPI/Streaming/channels/<trackStreamID>/picture/capabilities?format=json`.
3. The client controls the specified channel to perform sync manual capturing: `GET /ISAPI/Streaming/channels/<trackStreamID>/picture?snapShotImageType=<snapShotImageType>&videoResolutionWidth=<videoResolutionWidth>&videoResolutionHeight=<videoResolutionHeight>&imageQuality=<imageQuality>&x=<x>&y=<y>&hight=<hight>&width=<width>`, and the device returns the captured binary picture data.

Remarks:

- If you only need to capture a JEPG picture, you can jump to step 3.
- If you need to capture pictures other than JEPG or capture in the designated area of the video, you need to perform steps 1 and 2 to determine whether the device supports it. If it does, go to step 3.
- In step 3, the value ranges of `videoResolutionWidth` and `videoResolutionHeight` are the same as `<videoResolutionWidth> and <videoResolutionHeight>` returned by `GET /ISAPI/Streaming/channels/<trackStreamID>/capabilities`.

7 People Counting

7.1 People Counting Statistics Search

7.1.1 Introduction to the Function

For devices with local storage function, the people counting data will be stored in the local storage after the people counting function is enabled. You can search the local data to get multiple types of reports by setting different conditions.

7.1.2 API Calling Flow

7.1.2.1 Single Channel

1. Get the capability of searching for people counting data of a single channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/search/capabilities`.
2. Search for people counting data of a single channel: `POST /ISAPI/System/Video/inputs/channels/<channelID>/counting/search`.

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7.1.2.2 Multiple Channels

1. Get video capability: `GET /ISAPI/System/Video/capabilities`. If the node `isSupportMultiChannelCounting` is returend and its value is `true`, it indicates that searching for people counting data of multiple channels is supported.
2. Get the capability of searching for people counting data of multiple channels: `GET /ISAPI/System/Video/inputs/channels/counting/search/capabilities`.
3. Search for the people counting data of multiple channels: `POST /ISAPI/System/Video/inputs/channels/counting/search`.

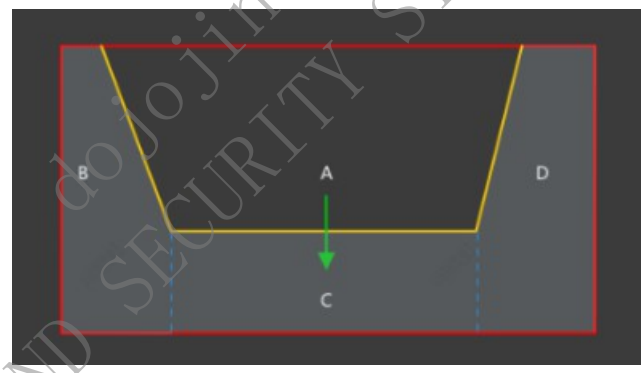
7.2 People Flow Direction Analysis

7.2.1 Introduction to the Function

People flow direction analysis is used for counting the people who crosses detection lines and enters the detection area. The flow analysis is valid only when the detection area is quadrilateral and the detection line and the detection area are overlapped only in the start and end points.

The requirements for drawing the rule/detection area of flow direction analysis are as follows:

1. The detection line (in yellow) and the detection area (in red) should have but two overlapped points.
2. The detection line should be perpendicular to the detection direction (along the direction of green arrow).
3. The four points on the detection line respectively are: point 1, 2, 3, 4.
4. Along the detection direction (along the green arrow), draw two parallel lines from point 2 and 3 to intersect the detection area (in red). The area between the two parallel lines is the counting region of area 2 (area C in the figure), and the rest two are area 1 (area B in the figure) and area 3 (area D in the figure).



7.2.2 API Calling Flow

The process of people flow direction analysis is the same with that of the real-time people counting. You need to enable the flow direction analysis to receive and parse the flow statistics data.

1. Get the capability of people counting: `GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/capabilities`. If `flowDirectionAnalysis` is returned, it indicates that the device supports flow analysis.
2. Configure people counting parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting`. Configure detection lines parameters (`CountingLineItemList`) and rule parameters (`CountingRegionList.RuleRegionCoordinatesList`), select scene mode (`countingSceneMode`), enable the flow direction analysis (`flowDirectionAnalysis` is `true`).
3. Receive the event uploaded by devices in arming or listening mode: `PeopleCounting`. Parse `peopleCounting.FlowStatisticsList` to get the flow data.

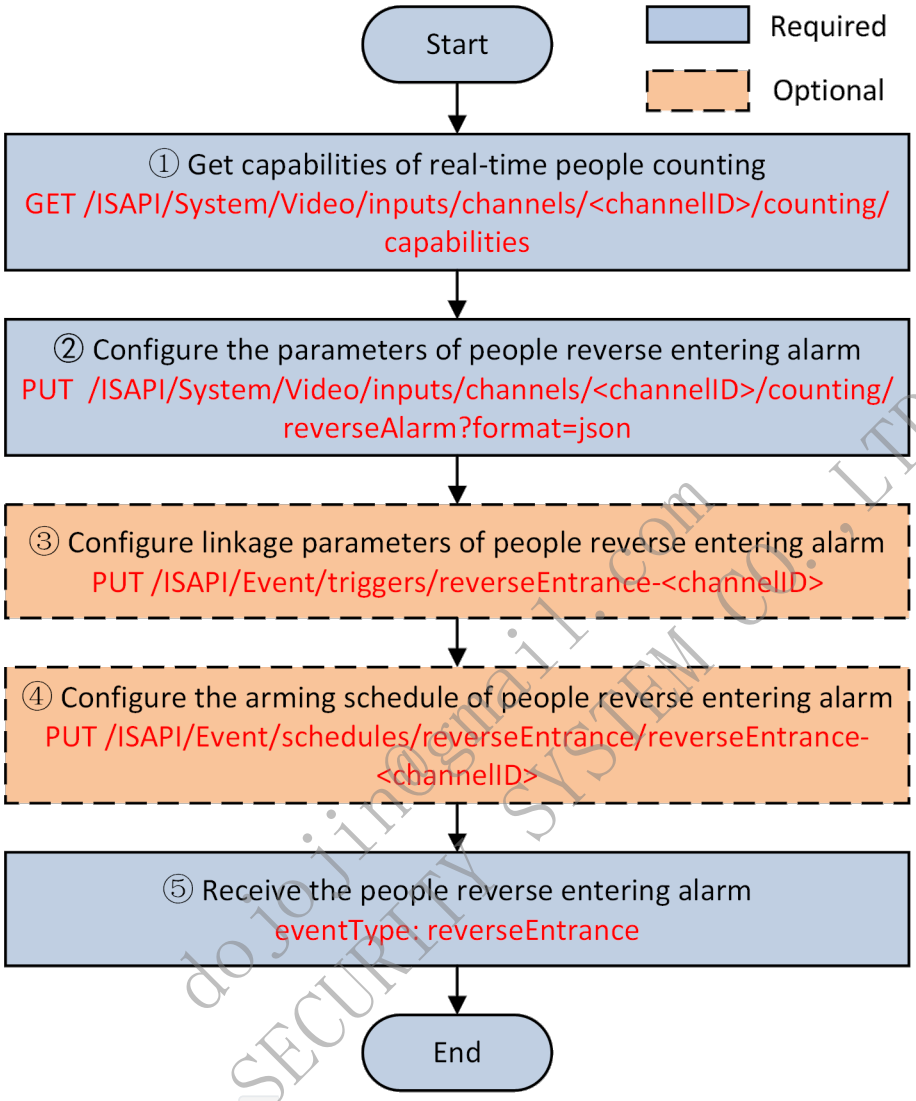
7.3 People Reverse Entering Alarm

7.3.1 Introduction to the Function

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The people reverse entering alarm will be triggered if a person crosses the detection line from the opposite direction within a specified period. It applies to the circumstance when only single direction passing is allowed within a specified period. For example, if someone exits the workshop within working time, an alarm will be uploaded for notification.

7.3.2 API Calling Flow



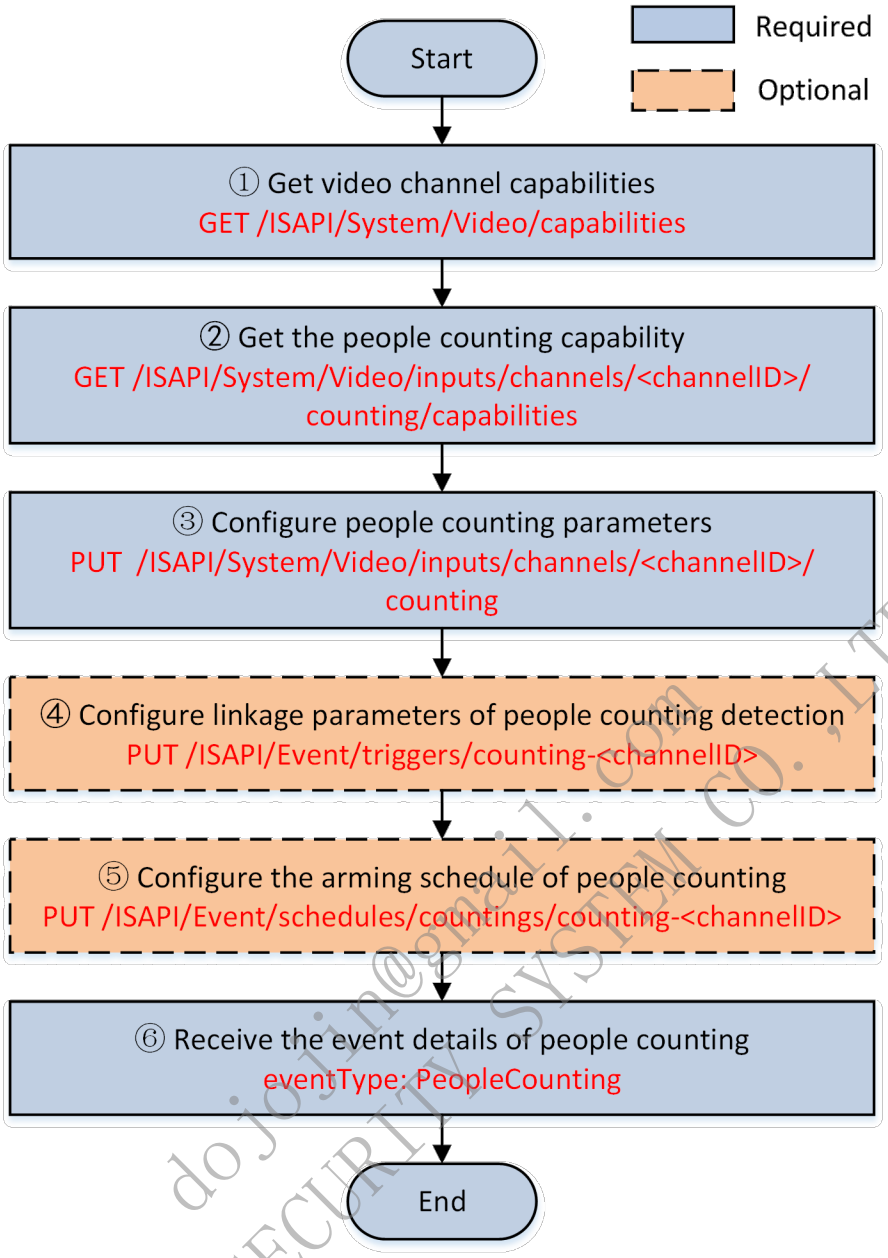
1. Get the people counting capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/capabilities`. If the node `isSupportReverseAlarm` is returned and its value is `true`, it indicates that people reverse entering alarm is supported.
2. Configure parameters of the people reverse entering alarm: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting/reverseAlarm?format=json`.
3. (Optional) Configure linkage parameters of the people reverse entering alarm: `PUT /ISAPI/Event/triggers/reverseEntrance-<channelID>`.
4. (Optional) Configure the arming schedule for the people reverse entering alarm: `PUT /ISAPI/Event/schedules/reverseEntrance/reverseEntrance-<channelID>`.
5. Receive events reported by device via arming or listening: `reverseEntrance`.

7.4 Real-time People Counting

7.4.1 Introduction to the Function

The function is to count line crossing persons or objects in a specified time period. It is commonly applied in the entrance and exit of the scenic spots, malls, stores, etc.

7.4.2 API Calling Flow



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1. Get video channel capabilities: `GET /ISAPI/System/Video/capabilities`. If the nodes `isSupportCounting` and `isSupportMultiChannelCounting` are returned and the value is true, it indicates that the device supports real-time people counting.
2. Get the people counting capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/capabilities`.
3. Configure people counting parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting`.
4. (Optional) Configure linkage rules for people counting detection: `PUT /ISAPI/Event/triggers/counting-<channelID>`.
5. (Optional) Configure the arming schedule for people counting: `PUT /ISAPI/Event/schedules/countings/counting-<channelID>`.
6. Receive events uploaded by device in listening or arming mode: `PeopleCounting`.

Notes:

- During people counting, considering the height difference between children and adults, you can enable the children detection mode so as to improve the accuracy. a. Get the capability of people counting. If `ChildFilter` is returned, it indicates that the device supports counting the number of children. b. Enable the children counting during the configuration of people counting parameters (set `enabled` of `ChildFilter` to true). c. Other operations are same as above steps.
- The people counting detection event (`PeopleCounting`) supports the following data types: a. Real-time statistics data (`statisticalMethods` is `realTime`): after enabled, persons' entry and exit will trigger the event data

uploading, and the data is the real-time people counting statistics from 12:00 A.M. to the current clock on the day. b. Statistics data by period (`statisticalMethods` is `timeRange`): the device will upload the statistics data according to the scheduled uploading interval (`dataUploadCycle`) within a specified period. c. Statistics data triggered by alarm input (`statisticalMethods` is `signalTrigger`): the device is connected to the alarm input signal, and the statistics data will be accumulated and uploaded every time the signal is triggered.

- The dimensions of statistics data are as follows: the number of people entering (`enter`), the number of people exiting (`exit`), the number of people repeating passing (`duplicatePeople`, which is only valid for people entering, and deduplication is invalid for people exiting), the number of people according to different face attributes, including face expression (`FaceExpressionList`), gender (`GenderList`), mask wearing (`MaskList`), etc. See details in the event message.
- If the client software fails to receive the people counting event uploaded by devices due to exceptions, or the connected devices have operated for a while, you can get the history people counting data via URI (POST /ISAPI/System/Video/inputs/channels/counting/search).

8 Abnormal Event Detection

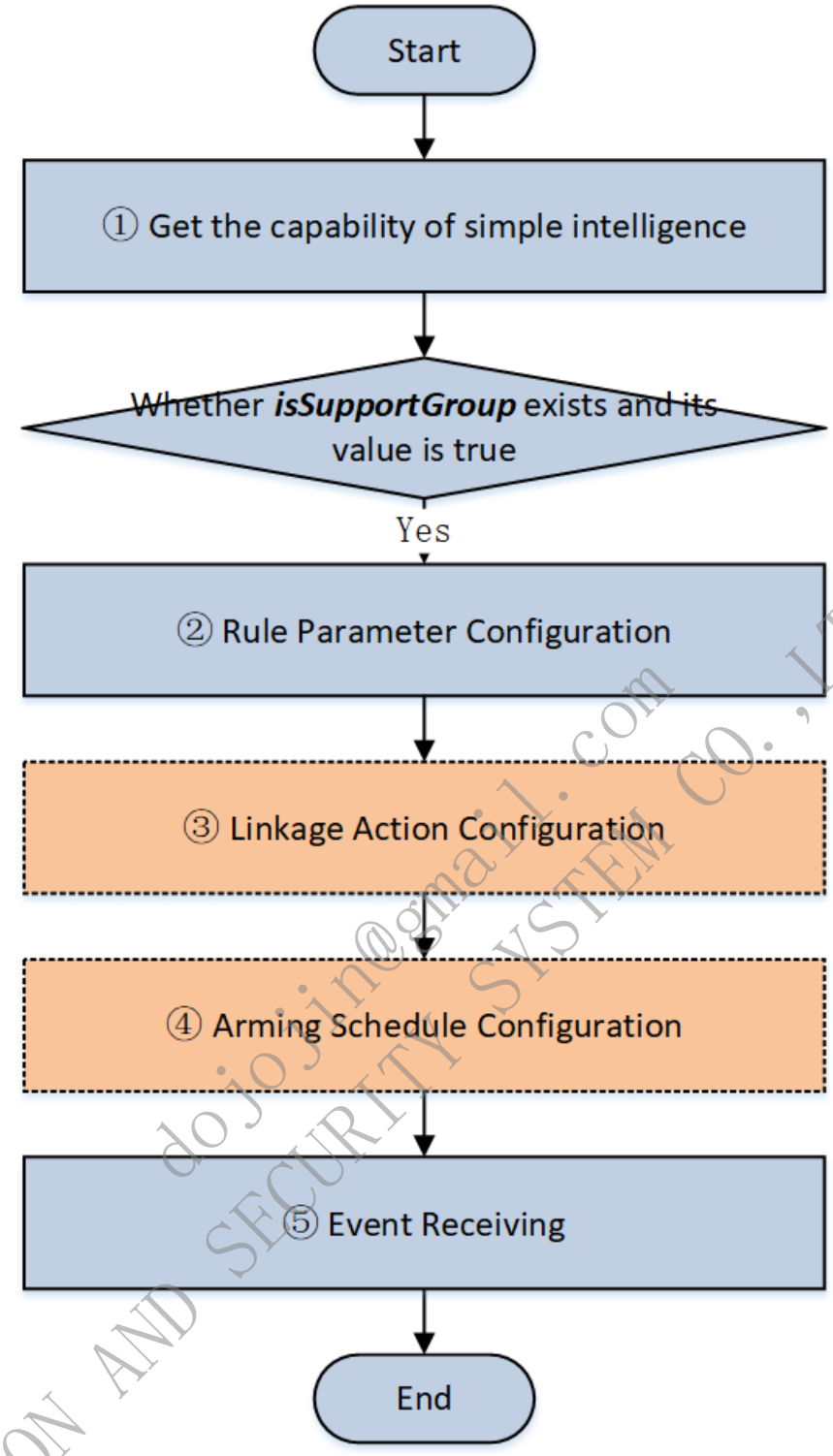
8.1 People Gathering

8.1.1 Introduction to the Function

The people gathering function is used for triggering the alarm in the fastest and best way when the people density in the set area exceeds the set threshold. The people gathering function is often used for performing the real-time monitoring of the abnormal behavior of people in the monitoring area, effectively preventing accidents caused by negligence, so as to avoid potential safety accidents and improve the supervision efficiency.

8.1.2 API Calling Flow

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1. Get the apability of AI intelligent search.

   Get the `isSupportGroup` in the capability of simple intelligence (related URI: `GET /ISAPI/Smart/capabilities`).

2. Configure rule parameters.

   Configure All Channels

- Get people gathering parameters (related URI: `GET /ISAPI/Smart/group`).
- Configure people gathering parameters (related URI: `PUT /ISAPI/Smart/group`).

  Configure Specified Channel

- Get the capability of people gathering parameters (related URI: `GET /ISAPI/Smart/group/<channelID>/capabilities`).

- Get the people gathering parameters of a specified channel (related URI: `GET /ISAPI/Smart/group/<channelID>`).
- Configure the people gathering parameters of a specified channel (related URI: `PUT /ISAPI/Smart/group/<channelID>`).

Configure All Areas of Specified Channel

- Get the capability of the parameters of all areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions/capabilities`).
- Get the parameters of all areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions`).
- Configure the parameters of all areas for people gathering (related URI: `PUT /ISAPI/Smart/group/<channelID>/regions`).

Configure Specified Channel in Specified Channel

- Get the capability of the parameters of specified areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions/<regionID>/capabilities`).
- Get the parameters of specified areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions/<regionID>`).
- Configure the parameters of specified areas for people gathering (related URI: `PUT /ISAPI/Smart/group/<channelID>/regions/<regionID>`). Note: The percentage refers to the percentage of people number accounts for the alarming area. An alarm will be triggered in the system when the percentage is higher than the set percentage. Otherwise, no alarm will be triggered.

3. (Optional) Configure arming linkage parameters.

- Get the capability of event linkage configuration (related URI: `GET /ISAPI/Event/triggersCap`), whether the message `EventTriggersCap` contains the node `GroupDetectionTriggerCap`.
- Get the linkage parameters of people gathering (related URI: `GET /ISAPI/Event/triggers/group-<channelID>`).
- Configure linkage parameters of people gathering (related URI: `PUT /ISAPI/Event/triggers/group-<channelID>`).

4. (Optional) Configure arming schedule.

- Get the arming schedule of alarm outputs for all channels (related URI: `GET /ISAPI/Event/schedules/groups`).
- Configure the arming schedule of people gathering for all channels (related URI: `PUT /ISAPI/Event/schedules/groups`).
- Get the arming schedule of people gathering for a specified channel (related URI: `GET /ISAPI/Event/schedules/groups/group-<channelID>`).
- Configure the arming schedule of people gathering for a specified channel (related URI: `PUT /ISAPI/Event/schedules/groups/group-<channelID>`). Note: 1. Supports setting up to 8 time periods in one day on the timeline. 2. Overlapped time between any two time periods is not allowed when setting the arming schedule.

5. Receive event: `"eventType":"group"`, including the following. `detectionPictureTransType` refers to the transmission method of the pictures. The field value is bkgUrl for the picture URL or the picture encoding data via base64 when the value of detectionPictureTransType is "url" or "base64". The field value is the picture Content-ID when the value of detectionPictureTransType is "binary".

9 Facial Recognition

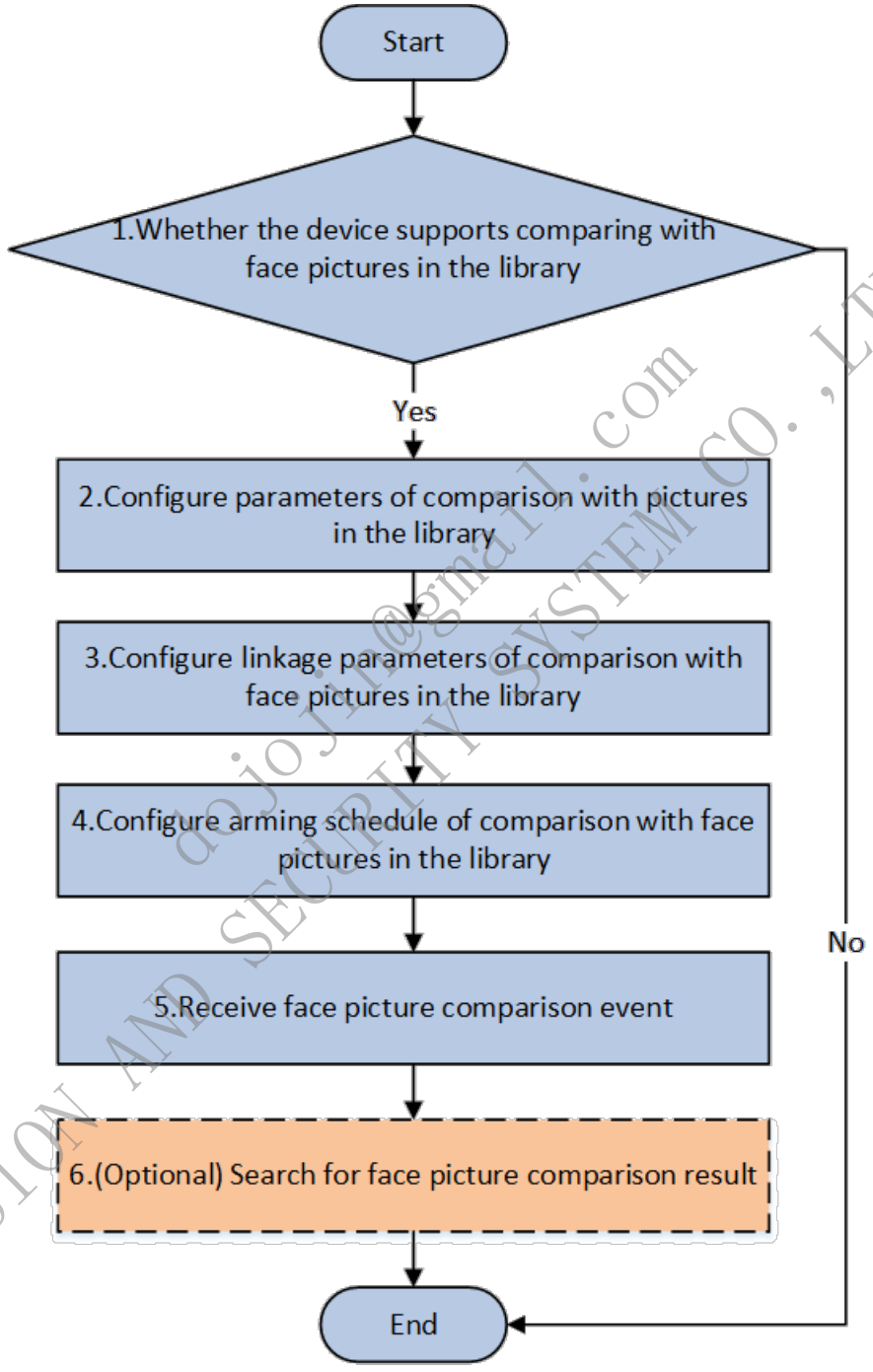
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9.1 . Comparison with Pictures in the Library

9.1.1 Introduction to the Function

Note: This feature depends on the face picture library management (refer to Face Picture Library Management Field for details). Analyze and model the face captured in the camera, then compare the face picture with pictures in the library, and lastly return the comparison similarity.

9.1.2 API Calling Flow



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1. **Whether the device supports comparing with face pictures in the library:**

   Get the capability node to evaluate whether the device supports comparing pictures with face pictures in the library, `GET /ISAPI/System/capabilities`; If the node `<isSupportFaceContrast>` is returned and its value is `true`, it indicates that the device supports comparing pictures with face pictures in the library.

2. **The parameters of comparison with pictures in the library including:**

   Get the capability of face picture comparison, `GET /ISAPI/Intelligent/channels/<channelID>/faceContrast/capabilities`. Configure the parameters of face

picture comparison, `PUT /ISAPI/Intelligent/channels/<channelID>/faceContrast`; Get the configuration of face picture comparison, `GET /ISAPI/Intelligent/channels/<channelID>/faceContrast`; **Note: The rule parameters support configuring the triggering threshold of face similarity event and specifying information including face picture library ID.**

3. **The linkage parameters of comparison with face pictures in the library:**

   Two sets of configurations linked to business systems are provided due to the front-end and back-end device differences:

   Linkage action of face picture library: for front-end devices (e.g., Deepinview), a business system of single channel and multiple face picture libraries. Therefore it supports configuring different arming linkage parameters via each compared face picture library. Call `GET /ISAPI/Event/triggersCap` to evaluate whether it supports the event linkage configuration capability, it does if `<FaceLibTriggerCap>` is returned, and the relevant configuration APIs are as follows: Configure the linkage action parameters, `PUT /ISAPI/Event/triggers/faceLib-<channelID>/<FDID>`; if users have added a customized face picture library ID when creating one, they can also call `PUT /ISAPI/Event/triggers/faceLib-<channelID>/<customFaceLibID>?FDType=custom` to configure via `customFaceLibID`; get the linkage action parameters, `GET /ISAPI/Event/triggers/faceLib-<channelID>/<FDID>`; Customizing the face picture library ID follows the same flow.

   Linkage action of channel: for back-end devices (e.g., DeepinMind), a business system of multiple channels and single face picture library. Therefore it supports configuring different arming linkage parameters via each video channel. Call `GET /ISAPI/Event/triggersCap` to evaluate whether it supports the event linkage configuration capability, it does if `<FaceContrastTriggerCap>` is returned, and the relevant configuration APIs are as follows: Configure the linkage action parameters of face picture comparison, `PUT /ISAPI/Event/triggers/faceContrast-<channelID>`; get the linkage action parameters of face picture comparison, `GET /ISAPI/Event/triggers/faceContrast-<channelID>`;

4. **Arming schedule of comparison with face pictures in the library**:

   Two sets of arming schedule configurations linked to business systems are provided due to the front-end and back-end device differences: Arming schedule of face picture library: Configure the single-channel arming schedule of face picture library, `PUT /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>`; if users have added a customized face picture library ID when creating one, they can also call `PUT /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>/<customFaceLibID>?FDType=custom` to configure via `customFaceLibID`; get the single-channel arming schedule of face picture library, `GET /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>`; Customizing the face picture library ID follows the same flow. Arming schedule of channel: Configure the arming schedule of face picture comparison, `PUT /ISAPI/Event/schedules/faceContrast-<channelID>`; get the arming schedule of face picture comparison, `GET /ISAPI/Event/schedules/faceContrast-<channelID>`;

5. **Face picture comparison event**:

   When the face captured on the camera is compared with pictures in the library, and the similarity has reached the rule parameters including the threshold, the device will upload the face picture comparison event; the event information mainly includes comparison similarity, captured face and its attributes, and the corresponding registered face picture and its attributes, `"eventType": "alarmResult"`.

6. **Search for face picture comparison result:**

   If supported by device, the face picture comparison result will be saved to the device local storage, and you can search for comparison history via the following API: Search for face picture comparison information, `GET /ISAPI/Intelligent/FDLib/FCSearch`.

9.2 . Search Via Picture Comparison In Face Capture Library

9.2.1 Introduction to the Function

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You can search for matched face pictures in face capture library via comparison with the uploaded local face picture. The general application scenario: compare the target face picture with face pictures in the face capture library to track the target according to the comparison result.

9.2.2 Search via Picture Comparison Synchronously

9.2.1.1 Introduction to the Function

After the request of searching via picture comparison synchronously is applied, the search result will be returned directly in the response message.

9.2.1.2 API Calling Flow

1. **Check whether the device supports searching via picture comparison synchronously.**

   Get the capability of searching via picture comparison: `GET /ISAPI/SDT/Face/searchByPic/capabilities`. If the node `isSupportSyncSearch` is returned and its value is `true`, it indicates that searching via picture comparison synchronously is supported.

   Note: For some devices supporting searching via picture comparison synchronously, the API for getting the capability is invalid, you can call `POST /ISAPI/SDT/Face/searchByPic?supportSync=true` directly.

2. **Submit the task of synchronously searching via picture comparison.**

   Synchronously search for pictures via picture comparison: `POST /ISAPI/SDT/Face/searchByPic?supportSync=true`.

   Note: Searching via picture comparison synchronously supports three types of pictures (`dataType`): picture URL (`faceURL`), picture model data (`targetModelData`), which should be generated in advance via `POST /ISAPI/SDT/Face/pictureAnalysis`, and binary picture data, which is in form format and follows the JSON message.

   The Request Message Example of Binary Picture Data

```
POST /ISAPI/SDT/Face/searchByPic?supportSync=true
Host: device_ip:port
Accept-Language: zh-cn
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive

--<frontier>
Content-Disposition: form-data; name="Event_Type"
Content-Type: application/json

[JSON Message]
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[Picture Data]
--<frontier>--
```

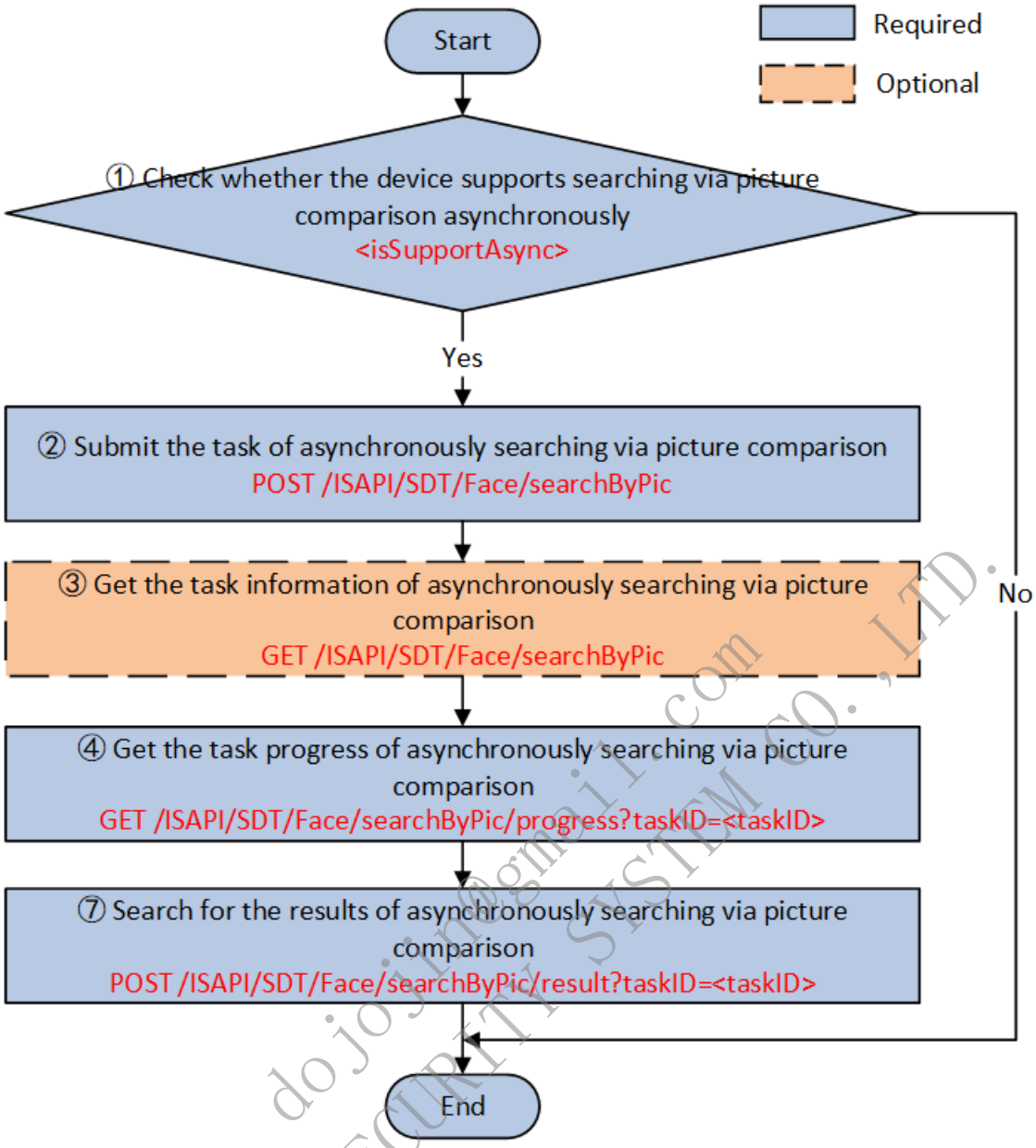
9.2.2 Search via Picture Comparison Asynchronously

9.2.2.1 Introduction to the Function

Submit the task of asynchronously searching via picture comparison and search for results by the returned task ID.

9.2.2.2 API Calling Flow

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1. **Check whether the device supports searching via picture comparison asynchronously.**

   Get capabilities of searching via picture comparison: `GET /ISAPI/SDT/Face/searchByPic/capabilities`. If the node `isSupportAsync` is returned and its value is `true`, it indicates that searching via picture comparison asynchronously is supported.

2. **Submit the task of asynchronously searching via picture comparison.**

   Asynchronously search via picture comparison: `POST /ISAPI/SDT/Face/searchByPic`. The task ID (`taskID`) is returned and will be used in the APIs below.

   Delete the task of asynchronously searching via picture comparison: `DELETE /ISAPI/SDT/Face/searchByPic?format=json&taskID=<taskID>`.

3. **(Optional) Get the task information of asynchronously searching via picture comparison.**

   `GET /ISAPI/SDT/Face/searchByPic`.

4. **Get the task progress of asynchronously searching via picture comparison.**

   Get the task progress: `GET /ISAPI/SDT/Face/searchByPic/progress?taskID=<taskID>`. When the progress (`progress`) is 100%, you can proceed to the following step.

5. **Search for the results of asynchronously searching via picture comparison.**

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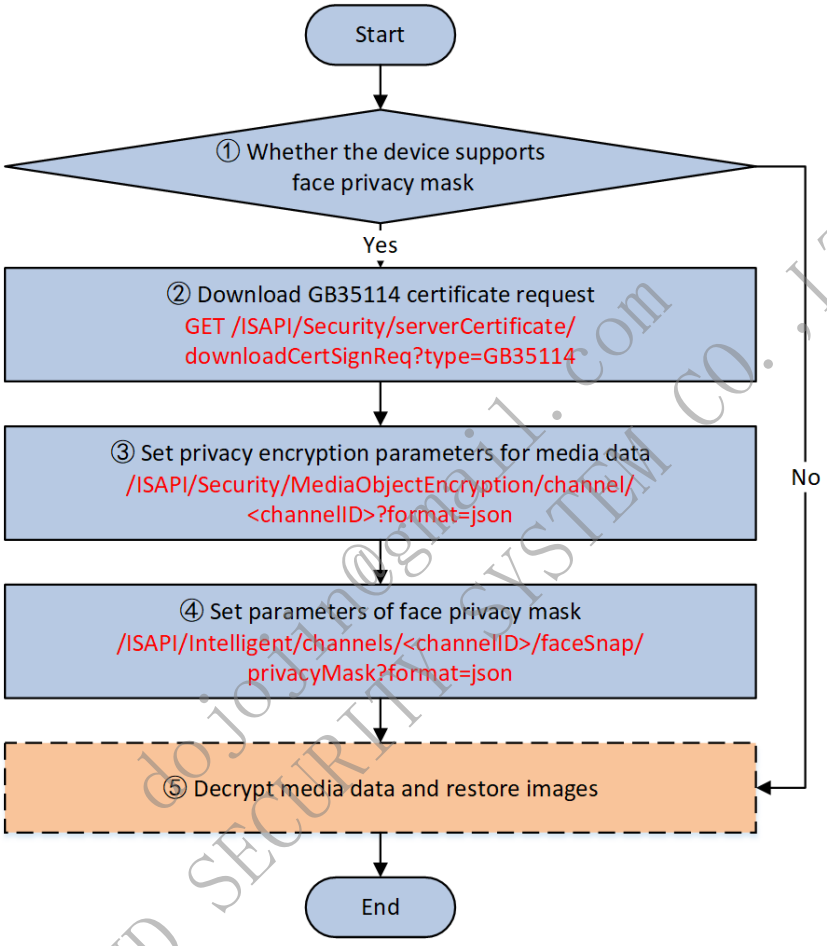
Search for results of asynchronously searching via picture comparison: `POST /ISAPI/SDT/Face/searchByPic/result?taskID=<taskID>`.

9.3 Face Capture

9.3.1 Introduction to the Function

Face capture is to detect whether face pictures show up in images. If set conditions are met and there are faces showing up, the face pictures will be captured and uploaded as alarms.

9.3.2 API Calling Flow



1. **Whether the device supports face capture:**

   Get the system capabilities: `GET /ISAPI/System/capabilities`. If the node (its parent node is) is returned as true, it indicates that the device supports face capture.

2. **Whether the device supports multi-scene face capture:**

   Get the intelligent capabilities of device: `GET /ISAPI/Intelligent/channels/<channelID>/capabilities`. If the node is returned and its value is true, it indicates that the device supports face capture.

3. **Set face picture capturing rules of multiple scenes:**

   If multi-scene capture of a channel is required, you can set the face picture capturing rules for multiple scenes.

   Set the face capture rules of a specified scene: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/<SID>`;

   Get the face capture rules of a specified scene: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/<SID>`;

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Set the face picture capturing rules of multiple scenes for a specified channel: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene`;

Get the face picture capturing rules of multiple scenes for a specified channel: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene`.

4. **Set face picture capturing rules of a specific scene.**

   If specific-scene capture of a channel is required, you can set the face picture capturing rules.

   Get the capabilities of captured face picture rules `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/capabilities`;

   Set the face capture rule parameters: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule`;

   Get the face capture rule parameters: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule`;

   Get the default parameters of face capture rule: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule?defualt=<defualt>`.

5. **Set linkage mode of face picture capture.**

   Set the linkage parameters of face capture: `PUT /ISAPI/Event/triggers/faceSnap-<channelID>`;

   Get the linkage parameters of face capture: `GET /ISAPI/Event/triggers/faceSnap-<channelID>`;

   Delete the linkage of face capture: `DELETE /ISAPI/Event/triggers/faceSnap-<channelID>`.

6. **Configure the arming schedule of face capture.**

   Set the arming schedule of face capture for a specified channel: `PUT /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>`;

   Get the arming schedule of face capture for a specified channel: `GET /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>`;

   Set the arming schedule of face capture: `PUT /ISAPI/Event/schedules/faceSnap`;

   Get the arming schedule of face capture: `GET /ISAPI/Event/schedules/faceSnap`.

7. **Receive face capture and recognition events.**

   If the wanted face shows up in the set area, face recognition event will be reported. The event information includes face pictures and attribute information, etc. `"eventType": "faceCapture"`.

9.4 Face Modeling and Face Picture Comparison

9.4.1 Face Picture Analysis

9.4.1.1 Introduction to the Function

Face picture analysis includes face detection and face modeling. After analyzing facial features (including age, gender, glasses, smiling, face angle, face frame, etc) of specified pictures, the platform performs modeling and returns model data. To perform face picture comparison in the name list, users need to submit the task of face picture comparison. For details, refer to the process of synchronous/asynchronous face picture comparison.

9.4.1.2 API Calling Flow

1. Get the capabilities of analyzing face pictures: `GET /ISAPI/SDT/Face/pictureAnalysis/capabilities`. If calling succeeded, it indicates that the device supports face picture analysis function.

2. Analyze face pictures (modeling): `POST /ISAPI/SDT/Face/pictureAnalysis`. The API will return the analysis result to the caller synchronously.

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to the caller synchronously.

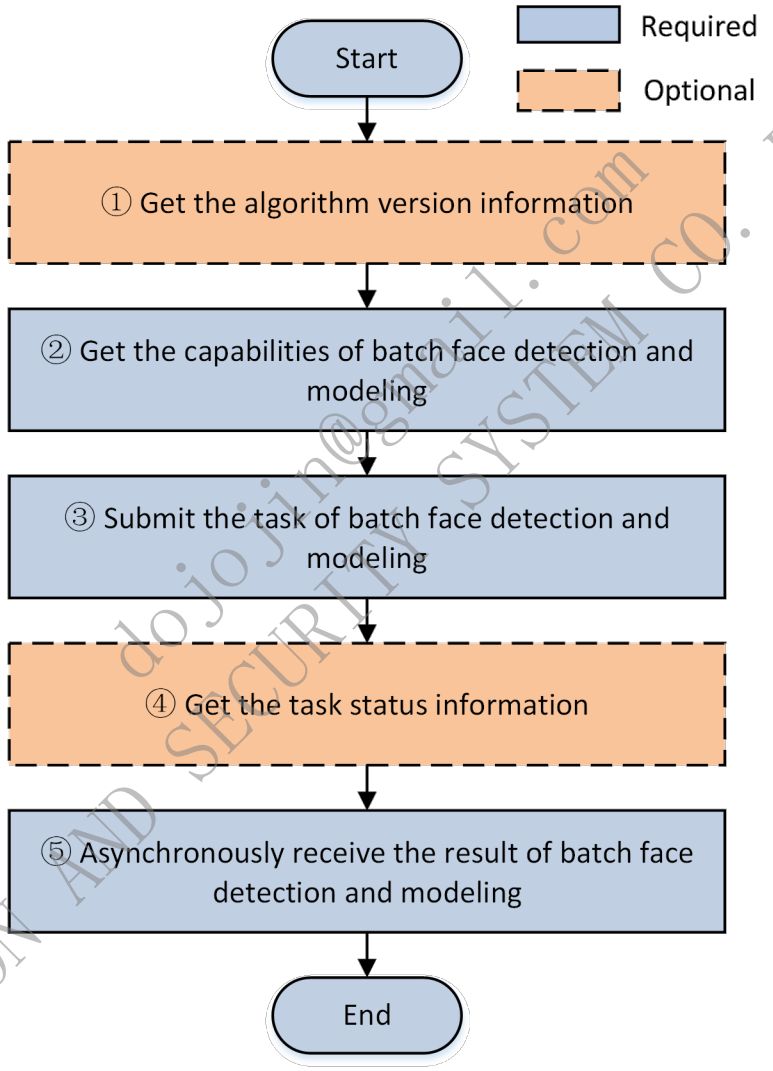
9.4.2 Batch Face Picture Analysis

9.4.2.1 Introduction to the Function

Batch face picture analysis supports detecting faces in multiple pictures, modeling and analyzing face thumbnails, uploading the analysis result via event type, and linking to corresponding tasks via task ID. Make sure the algorithm resources allocation is completed before performing batch face picture modeling detection. To perform face picture comparison in the name list, users need to submit the task of face picture comparison. For details, refer to the process of synchronous/asynchronous face picture comparison.

Batch picture analysis allows users to specify whether to import pictures into the face picture library. If importing pictures into library is supported, users can search for the status of the pictures imported to the library in a batch.

9.4.2.2 API Calling Flow



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1. Get the version information of intelligent algorithm scheduling supported by the device: `GET /ISAPI/Intelligent/algorithmInfos?format=json`.
2. Get the capabilities of batch face detection and modeling: `GET /ISAPI/SDT/Face/pictureAnalysis/batch/capabilities`. Check whether the device supports batch face detection and modeling.
3. Submit the task of batch face detection and modeling: `POST /ISAPI/SDT/Face/pictureAnalysis/batch`. It is required to upload the pictures to server or cloud storage considering only importing the face pictures by URL is allowed.
4. Get the task status information: `GET /ISAPI/SDT/Management/Task/status`.

5. Asynchronously receive the result of batch face picture analysis mode: `eventType: analysisResult`.

9.4.3 Synchronous Face Picture Comparison

9.4.3.1 Introduction to the Function

Synchronous face picture comparison analyzes the facial features in a single picture first, and then compares the picture with pictures in the face picture library, and returns matched face records. It is commonly used for the analysis of a single picture. Users can obtain the face modeling data via face picture modeling function. For details, refer to the process of (batch) face picture analysis.

9.4.3.2 API Calling Flow

1. Get the capabilities of synchronous face picture comparison task: `GET /ISAPI/SDT/Management/Task/facePicture/sync/capabilities?format=json`.
2. Submit the task of synchronous face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture/sync?format=json`. The API will return the analysis result to the caller synchronously.

9.4.3.3 Exception Handling

9.4.3.3.1 Error Code

statusCode	statusString	subStatusCode	Error Code	Description	Remarks
4	Invalid Operation	noMoreTasksCanBeAdded	0x4000118A	No more tasks can be added.	No more tasks can be added. Delete the completed or unneeded tasks and try again.
4	Invalid Operation	userScheOffline	0x40001040	Task scheduler server is offline.	Please contact the technical support.
4	Invalid Operation	clusterNodeOffline	0x40001075	Cluster node is offline.	Cluster node is offline. Check the cluster status.
4	Invalid Operation	destinationaddressinvalid	0x400011c6	Invalid third-party address.	Invalid third-party address. Check again.
4	Invalid Operation	cloudAnalysisRequestInformationBuildFailed	0x4000110C	Creating request message of cloud analysis failed.	Please provide related log and packet capture information and contact the

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				failed.	the technical support.
4	Invalid Operation	cloudAnalysisResponseInformationParseFailed	0x4000110D	Parsing response message of cloud analysis failed.	Please provide related log and packet capture information and contact the technical support.
4	Invalid Operation	cloudAnalysisServerReturnFailed	0x400010FC	No response is returned by cloud analytic service.	Please provide related log and packet capture information and contact the technical support.
4	Invalid Operation	numberReachedLlimit	0x40001119	The number reached limit.	No more configured items can be added. Delete unneeded items and try again.
4	Invalid Operation	tsModelError	0x40001254	Forwarding service module error.	Incorrect task receiving address. Check again.
4	Invalid Operation	anModelError	0x40001255	Computing node error.	Please contact the technical support.
4	Invalid Operation	schedulingdatabaseisfull	0x400011c4	Database disk is full.	Database disk is full. Try again later or cancel related tasks and try again.
					Scheduling module

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4	Invalid Operation	ssModelError	0x4000124c	Scheduling module error.	error. Please contact the technical support.
6	Invalid Content	requestMemoryNULL	0x6000003F	Request memory is null.	Please contact the technical support.
2	Device Busy	noMemory	0x20000001	Insufficient memory.	Confirm the remaining memory by checking the value of the free field via the directory path of the linux system. Please provide related log information and contact the technical support.

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9.4.4 Asynchronous Face Picture Comparison

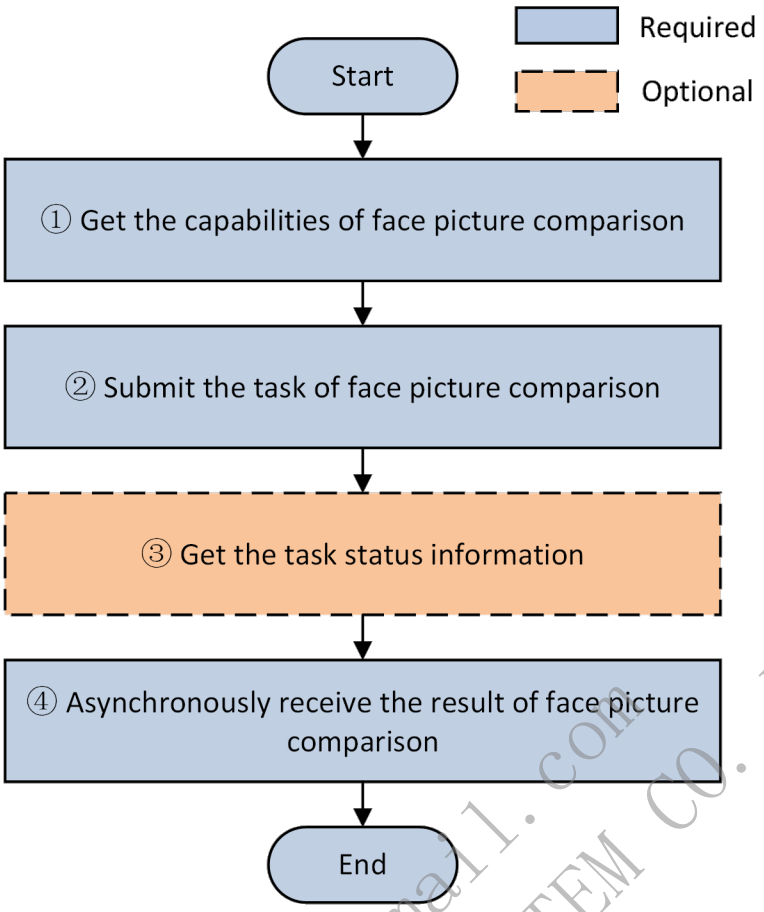
9.4.4.1 Introduction to the Function

Asynchronous face picture comparison analyzes face information in multiple pictures and compares with the faces in the face picture library, and then returns the matched face records. It is commonly used for analyzing multiple pictures. Users can obtain the face modeling data via face picture modeling function. For details, refer to the process of (batch) face picture analysis.

Asynchronous face picture comparison supports single and batch task.

9.4.4.2 API Calling Flow

9.4.4.2.1 Submit the Task of Face Picture Comparison



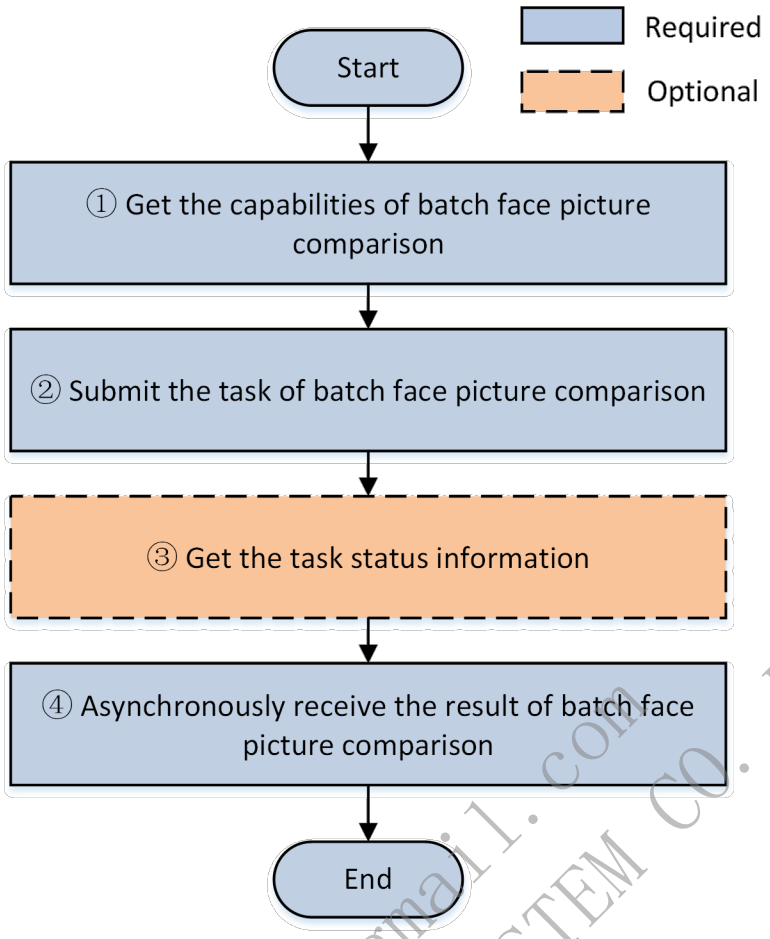
1. Get the capability set of the face picture comparison task: `GET /ISAPI/SDT/Management/Task/facePicture/capabilities`.
2. Submit the face picture comparison task: `POST /ISAPI/SDT/Management/Task/facePicture`.

   Notes: It can be applied in scenes including blocklist alarm, stranger detection alarm, frequently appeared person alarm, and one person one archive. Users can distinguish various scenes according to the node `type` of the imported parameters `relateGroup`.
3. Get the task status information: `GET /ISAPI/SDT/Management/Task/status`.
4. Asynchronously receive the result of face picture comparison: `eventType: alarmResult`. The corresponding task is linked via the task ID.

Notes: Users can get the supported maximum number of pictures to be analyzed via the `maxPictureNum` of `GET /ISAPI/SDT/Management/Task/facePicture/capabilities`.

9.4.4.2.2 Submit the Task of Batch Face Picture Comparison

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1. Get the capabilities of the task of batch face picture comparison: `GET /ISAPI/SDT/Management/Task/facePicture/batch/capabilities`.
2. Submit the task of batch face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture/batch`.

   Notes: It can be applied in scenes including blocklist alarm, stranger detection alarm, frequently appeared person alarm, and one person one archive. Users can distinguish various scenes according to the node `type` of the imported parameters `relateGroup`.
3. Get the task status information: `GET /ISAPI/SDT/Management/Task/status`.
4. Asynchronously receive the result of face picture comparison: `eventType: alarmResult`. The corresponding task is linked via the task ID.

Notes: Users can get the supported maximum number of pictures to be analyzed via the `maxPictureNum` of `GET /ISAPI/SDT/Management/Task/facePicture/batch/capabilities`.

9.4.5 Transmitted Parameters

9.4.5.1 Introduction to the Function

For data center products, transmitted parameters (targetAttrs) are imported when the platform applies tasks. The transmitted parameters will not be parsed and will be returned when the analysis result is uploaded to link to corresponding alarms. Transmitted content is agreed between the platform and device.

Notes: The message length of the transmitted parameters is restricted, which depends on the corresponding capability set of the task API returned by the device.

9.4.5.2 Sample Message

9.4.5.2.1 The Task of Batch Face Picture Comparison

The task of batch face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture/batch`. The sample message of the transmitted parameters targetAttrs are as follows:

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```
"targetAttrs": {
/*opt, object, the analysis device will not parse the transmitted content*/
    "deviceId": "1",
    /*opt, string, device ID, range:[,]*/
    "deviceChannel": 1,
    /*opt, int, device channel No., range:[,], step:, unit:, unitType:*/
    "deviceName": "shebei",
    /*opt, string, device name, range:[,]*/
    "faceTime": "2016-09-10T01:01:00.000Z",
    /*opt, string, capture time, range:[,]*/
    "rect": {
        /*opt, object, background picture frame information of face*/
        "height": 1.0,
        /*opt, float, height, range:[,], step:, unit:, unitType:*/
        "width": 1.0,
        /*opt, float, width, range:[,], step:, unit:, unitType:*/
        "x": 0.0,
        /*opt, float, X coordinate, range:[,], step:, unit:, unitType:*/
        "y": 0.0
        /*opt, float, Y coordinate, range:[,], step:, unit:, unitType:*/
    },
    "bkgUrl": "http://absddsdfsdfsdk.jpg",
    /*opt, string, background picture, range:[,]*/
    "cameraIp": "10.11.41.26",
    /*req, string, camera IP address, range:[,]*/
    "cameraIndexCode": "3301023820382193",
    /*req, string, camera No., range:[,]*/
    "longitude": 116.39737,
    /*opt, float, latitude, range:[,], step:, unit:, unitType:*/
    "latitude": 116.39737,
    /*opt, float, latitude, range:[,], step:, unit:, unitType:*/
    "name": "Jack",
    /*opt, string, name, range:[,]*/
    "registerGender": "male",
    /*opt, string, gender, range:[,]*/
    "nation": 1,
    /*opt, int, nation, range:[,], step:, unit:, unitType:*/
    "bornTime ": "2016-01-30",
    /*opt, string, birthday, range:[,]*/
    "address": "New York, the United States",
    /*opt, string, address, range:[,]*/
    "certificateNumber": "111111111111111111",
    /*opt, string, ID card, range:[,]*/
    "credImgUrl": "http://XXX",
    /*opt, string, picture URL, range:[,]*/
    "matchResult": 0,
    /*opt, int, the number of matched results, range:[,], step:, unit:, unitType:*/
    "linkFaceBodyId": "XXX",
    /*opt, string, linkage ID between face and human body picture, range:[,]*/
    "linkFaceVehicleId": "XXX",
    /*opt, string, linkage ID between face and vehicle, range:[,]*/
    "linkFaceMacId": "XXX",
    /*opt, string, linkage ID between face and MAC, range:[,]*/
    "Coordinates": [
        /*req, array, coordinate, subType:object, range:[,]*/
        {
            "timeStamp": "test",
            /*req, string, timestamp, range:[,]*/
            "VT": {
                /*req, object, VT*/
                "h": 1.0,
                /*opt, float, height, range:[,], step:, unit:, unitType:*/
                "w": 1.0,
                /*opt, float, width, range:[,], step:, unit:, unitType:*/
                "x": 0.0,
                /*opt, float, X coordinate, range:[,], step:, unit:, unitType:*/
                "y": 0.0
                /*opt, float, Y coordinate, range:[,], step:, unit:, unitType:*/
            },
            "isPTValid": true,
            /*req, bool, whether the PT is valid*/
            "PT": {
                /*req, object, PT*/
                "x": 0.0,
                /*opt, float, X coordinate, range:[,], step:, unit:, unitType:*/
                "y": 0.0,
                /*opt, float, Y coordinate, range:[,], step:, unit:, unitType:*/
                "z": 0.0
                /*opt, float, Z coordinate, range:[,], step:, unit:, unitType:*/
            },
            "humanBodyClarityType": 1,
            /*ro, opt, enum, type of body completeness, subType:int, (1#totally covered,2#upper body not covered,3#whole body not covered)*/
            "posture": "sitting",
            /*ro, opt, enum, gesture, subType:string, ("sitting", "standingUp")*/
            "motion": "yes",
            /*ro, opt, enum, whether the person is in motion, subType:string, ("yes", "no")*/
            "height": 1,
            /*ro, opt, int, height, range:[0,1000], step:1, unit:cm, unitType:length*/
            "humanRect": {
            /*ro, opt, object, coordinates of the human body in the video, desc:coordinates in the video (x,y,width,height); float (accurate to 3 decimal
places). The origin is the upper-left corner of the screen */
                "h": 1.000,
```

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```
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "w": 1.000,
                /*ro, req, float, width, range:[0.000,1.000], step:, unit:, unitType:*/
                "x": 0.000,
                /*ro, req, float, X coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
                "y": 0.000
                /*ro, req, float, Y coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
            },
            "faceRect": {
            /*ro, opt, object, coordinates of the human face in the video, desc: coordinates in the video (x,y,width,height); float (accurate to 3 decimal
places). The origin is the upper-left corner of the screen */
                "h": 1.000,
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "w": 1.000,
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "x": 0.000,
                /*ro, req, float, X coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
                "y": 0.000
                /*ro, req, float, Y coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
            },
            "hsRect": {
            /*ro, opt, object, coordinates of the head and shoulder frame in the video, desc: coordinates in the video (x,y,width,height); float (accurate
to 3 decimal places). The origin is the upper-left corner of the screen */
                "h": 1.000,
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "w": 1.000,
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "x": 0.000,
                /*ro, req, float, X coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
                "y": 0.000
                /*ro, req, float, Y coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
            }
        }
    ],
    "targetID": "1",
    /*req, string, the tracking ID of a single camera, range:[,]*/
    "traComplete": true
    /*req, bool, boll*/
}
```


Notes: The length limit of the transmittance node `targetAttrs` depends on the `targetAttrsMaxLen` of the `GET /ISAPI/SDT/Management/Task/facePicture/batch/capabilities` (API of getting the capabilities of batch face picture comparison).

9.4.6 Result Recipient List

9.4.6.1 Introduction to the Function

Users can specify address of the analysis result recipient by configuring the `destinationType` and `destinationUrl` of the `destination` node. Related APIs:

1. The task of face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture`.
2. The task of batch face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture/batch`.

9.4.6.2 Corresponding Relationship List

The relationship between `destinationType` and `destinationUrl` is as follows:

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destinationType	destinationType Description	destinationUrl Format	Rema
ServerKafka	Kafka address.	{"topic":"test","url":"10.17.139.184:2181"}	
faceCaptureLib	Captured face picture library of big data.	{"topic":"SNAP_IMAGE_INFO_TOPIC","url":"172.10.201.156:2181"}	
faceAlarmUpload	Upload list alarms by ISAPI.	http://IP:PORT/****(****represents custom string description and empty is allowed)	
AlarmRMQ	AlarmRMQ.	{"topic":"test_face_alarm:test_face","url":"10.15.117.65:6005"}	
KafkaFaceLabel	Fusion big data tagging.	" {"topic":"LABEL_SNAP_IMAGE_INFO_TOPIC","url":"10.66.106.149:2181"}"	
faceLabelUpload	Face picture tagging.	{"topic":"SNAP_IMAGE_INFO_TOPIC","url":"172.10.201.156:2181"}	Its UR same of the faceC

9.5 Face Picture Library Management

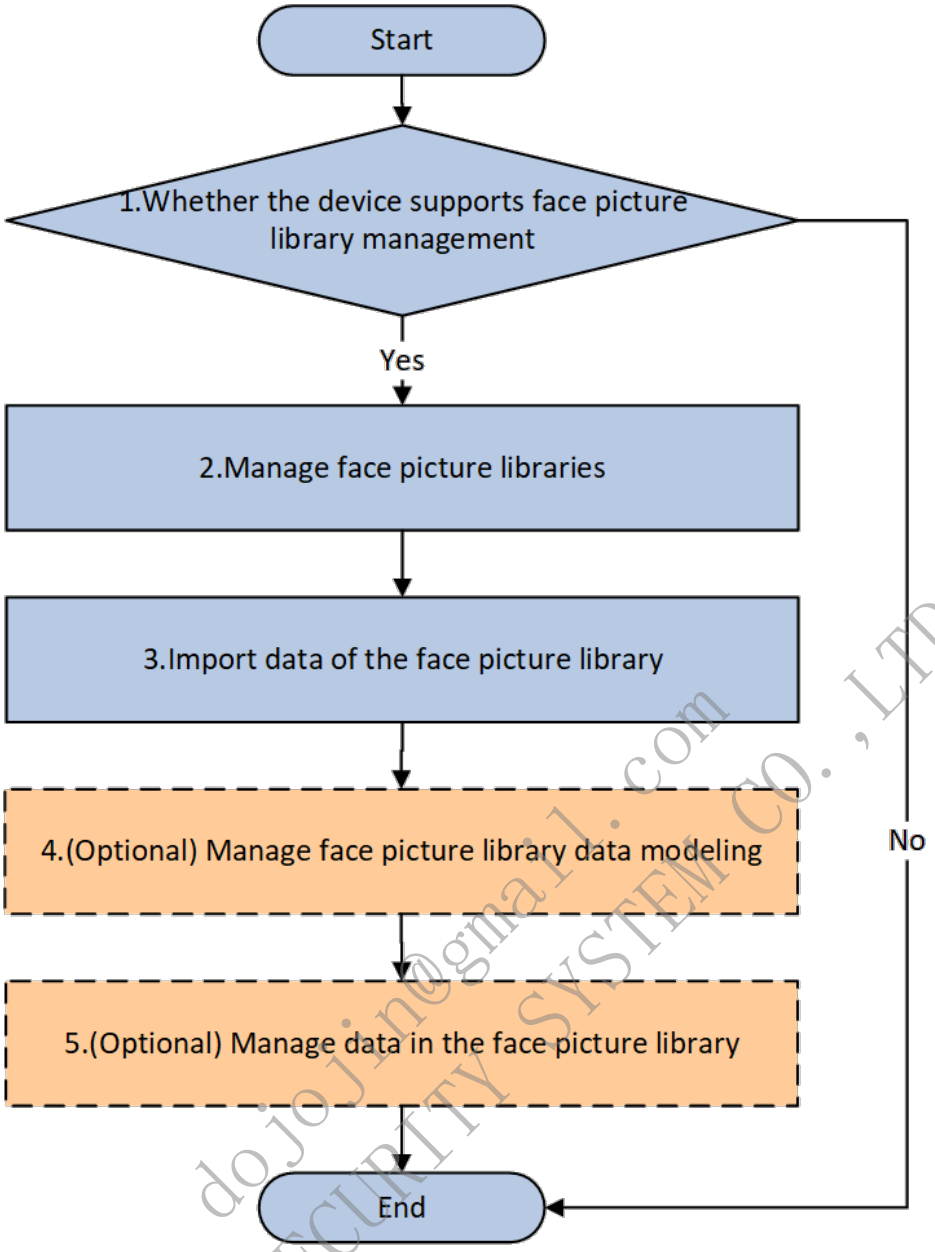
9.5.1 Face Picture Library Management

9.5.1.1 Introduction to the Function

This function is mainly used to manage operations on face picture libraries on the device, such as adding, deleting, editing, or searching for the face picture libraries, importing face pictures and additional person information to the libraries, editing, deleting, or searching for face pictures and additional person information in the libraries, and face picture modeling. This function is also the prerequisite of other face picture library functions such as face picture library comparison.

9.5.1.2 API Calling Flow

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1. Get the capability of managing face picture libraries for checking functions supported by the device: `GET /ISAPI/Intelligent/FDLib/capabilities`.
2. Manage face picture libraries.
   i. Create a face picture library: `POST /ISAPI/Intelligent/FDLib`. After the face picture library is created, the device will automatically generate the face picture library ID (that is, `FDID` is returned in the message `<FDLibInfoList>`). If you need to manage the face picture library ID by yourself, you can configure the custom face picture library ID (`customFaceLibID`) in the request message `<CreateFDLibList>` as needed. Afterwards, you can perform operations on the face picture library by the custom ID instead of `FDID`.
   ii. Edit parameters of the face picture library: `PUT /ISAPI/Intelligent/FDLib/<FDID>`. If you have added the custom face picture library ID in the above steps, you can also call `PUT /ISAPI/Intelligent/FDLib/<customFaceLibID>?FDType=custom` to configure parameters using `customFaceLibID`.
   iii. Get parameters of the face picture library: `GET /ISAPI/Intelligent/FDLib/<FDID>`. Operations involved with the custom face picture library ID are the same as those mentioned above.
   iv. Delete the face picture library: `DELETE /ISAPI/Intelligent/FDLib/<FDID>`. Operations involved with the custom face picture library ID are the same as those mentioned above.
3. Import data of the face picture library.

i. Import face picture data and automatically perform modeling, `POST /ISAPI/Intelligent/FDLib/pictureUpload`. By repeatedly applying the previous API, you can import the face picture data (including the face picture and additional face picture information `FaceAppendData`) to the corresponding face picture library; after it is successfully imported, the device will automatically model on the picture which could take some time, then the a face picture ID will be automatically generated (i.e., `PID` is returned in response parameters `MaskInfo`) by device after modeling succeeds; if you want to manually manage the face picture ID, you can add customized face picture ID (`customHumanID`) as needed when applying the request parameter `PictureUploadData`. `PID` can be replaced with customized face picture ID for subsequent face-picture-related operations.

ii. Import face picture library data in concurrent operation mode, `POST /ISAPI/Intelligent/FDLib/pictureUpload?type=concurrent`. If you want to improve the import efficiency, you can apply concurrent operation that the device will only store the picture not model on the picture. You will need to manually model on the picture (refer to Face Picture Library Data Modeling Management Flow for details). Note: Back-end DeepinMind devices will automatically perform modeling after data is imported even though under concurrent operation circumstances; if you demand to manually perform modeling, you can still apply modeling management.

4. (Optional) Manage face picture library data modeling. If you imported face picture library data in concurrent operation mode, you can select a time for device data modeling management after importing completed:
   i. Modeling on registered face pictures, `GET /ISAPI/Intelligent/FDLib/manualModeling`;
   ii. Search for modeling control results, `GET /ISAPI/Intelligent/FDLib/manualModelingRangeTask`;
   iii. Get modeling progress status, `GET /ISAPI/Intelligent/FDLib/manualModeling/progressState`.

5. (Optional) Manage data in the face picture library. You can search, edit, and delete data in the face picture library.
   i. Search for face picture data: `POST /ISAPI/Intelligent/FDLib/FDSearch`;
   ii. Edit face picture data: `PUT /ISAPI/Intelligent/FDLib/FDModify?format=json`, it is for editing data of imported pictures in binary format;
   iii. Edit face picture data in the library: `PUT /ISAPI/Intelligent/FDLib/FDSearch?format=json&FDID=<FDID>&FPID=<FPID>&faceLibType=<faceLibType>`, it is for editing data of imported pictures in URL format;
   iv. Delete face picture data in the library: `PUT /ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&FDID=<FDID>&faceLibType=<FDType>`.

9.5.1.3 Exception Handling

9.5.1.3.1 Error Code

statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Rem
1	OK	ok	0x00000001			
2	Device Busy	noMemory	0x20000001		Not enough memory.	
4	Invalid Operation	notSupport	0x40000001		Not supported.	
4	Invalid Operation	faceLibDiskFull	0x400020F4		Face picture library storage media is full.	
4	Invalid Operation	faceLibraryIDNotExis	0x400020FD		The face picture library does not exist.	
					The face	

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4	Invalid Operation	faceLibInitializing	0x40002136		picture library is initializing. Please wait.	
4	Invalid Operation	notMatchFacelib	0x4000a004		Face picture library is not matched.	
4	Invalid Operation	existSameNameFacelib	0x4000a00a		Face picture library with duplicate name exists.	
4	Invalid Operation	delFacelibFailed	0x4000a00b		Deleting the face picture library failed.	
4	Invalid Operation	modifyFacelibFailed	0x4000a00c		Editing the face picture library failed.	
4	Invalid Operation	createFacelibFailed	0x4000a00d		Creating a face picture library failed.	
4	Invalid Operation	facelibAlreadyUsed	0x4000a010		The face picture library has been used.	
6	Invalid Content	faceLibraryDatabaseError	0x60000032		Database version of face picture library mismatch.	
6	Invalid Content	faceLibraryError	0x60000034		Face picture library error.	
6	Invalid Content	multipleFaceObjectError	0x60000035		Multiple faces are recognized in the pictures to be imported into the face picture library.	
6	Invalid Content	faceLibPicImportingError	0x6000003A		Importing picture into library...	
6	Invalid Content	faceLibraryNoData	0x60000069		No data found in face picture	

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					library.	
6	Invalid Content	NoDataForModelingError	0x6000006B		There is no data to be modeled in the face picture library.	
6	Invalid Content	customFaceLibIDRepeat	0x6000006C		Duplicate ID of custom face picture library.	
6	Invalid Content	customFaceDataDeleteing	0x600000A1		Deleting original customized face picture library data... The upper level cannot duplicate the customized face picture library of this ID.	
6	Invalid Content	analysisEnginesFacelibImporting	0x60001005		Importing face picture library to analysis engine...	
6	Invalid Content	facelibDataProcessing	0x6000100b		Processing face picture library data.	
6	Invalid Content	personInfoExtendValueIsTooLong	0x60001083		Customized tag for face picture library exceeded limit.	
6	Invalid Content	personInfoExtendValueContainsInvalidChar	0x60001084		Customized tag for face picture library contains invalid character(s).	

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9.5.1.4 Message Example

Import face pictures to face picture library in binary format and automatically perform modeling, `POST /ISAPI/Intelligent/FDLib/pictureUpload`:

```
POST http://10.16.96.29/ISAPI/Intelligent/FDLib/pictureUpload HTTP/1.1
If-Modified-Since: 0
Accept: */*
X-Requested-With: XMLHttpRequest
Content-Type: multipart/form-data; boundary=----------------------------7e27010d1dba
Referer: http://10.16.96.29/doc/page/config.asp
Accept-Language: zh-CN
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/5.0 (compatible; MSIE 10.0; Windows NT 6.1; WOW64; Trident/6.0)
Connection: Keep-Alive
Content-Length: 903330
DNT: 1
Host: 10.16.96.29
Pragma: no-cache
Cookie: language=zh; WebSession=48b693ca2b7301e3119d; sdMarkMenu=14%3AfaceSnapLib; sdMarkTab_1_0=0%3AsettingBasic

----------------------------7e27010d1dba
Content-Disposition: form-data; name="PictureUploadData"

<?xml version='1.0' encoding='UTF-8'?
><PictureUploadData><FDID>55869AFEA1E840958120C97054720372</FDID><FaceAppendData><name>20171222103007395_2</name></FaceAppendData></PictureUploadData>
----------------------------7e27010d1dba
Content-Disposition: form-data; name="importImage"; filename="20171222103007395_2.jpeg"
Content-Type: image/jpeg

【Picture data in binary format】
----------------------------7e27010d1dba--
```

9.6 Frequently Appeared Person Detection

9.6.1 Introduction to the Function

Alarm will be triggered when the person who repeatedly appears for a certain number of times within the specified time period (in-house personnel filtered out). When users cannot provide photos of the person who is in the arming status, such as habitual thieves and highly risky personnel, it enables a complete personnel control by identifying the personnel in unusual behavior, providing help for decision making of users to determine the suspicious personnel.
Applicable scenarios:

A. Company sites/finance agency outlets/hospital registration / queuing for ticket at scenic spots, etc.

Pre-alarm will be triggered for personnel who frequently come to stay and linger.

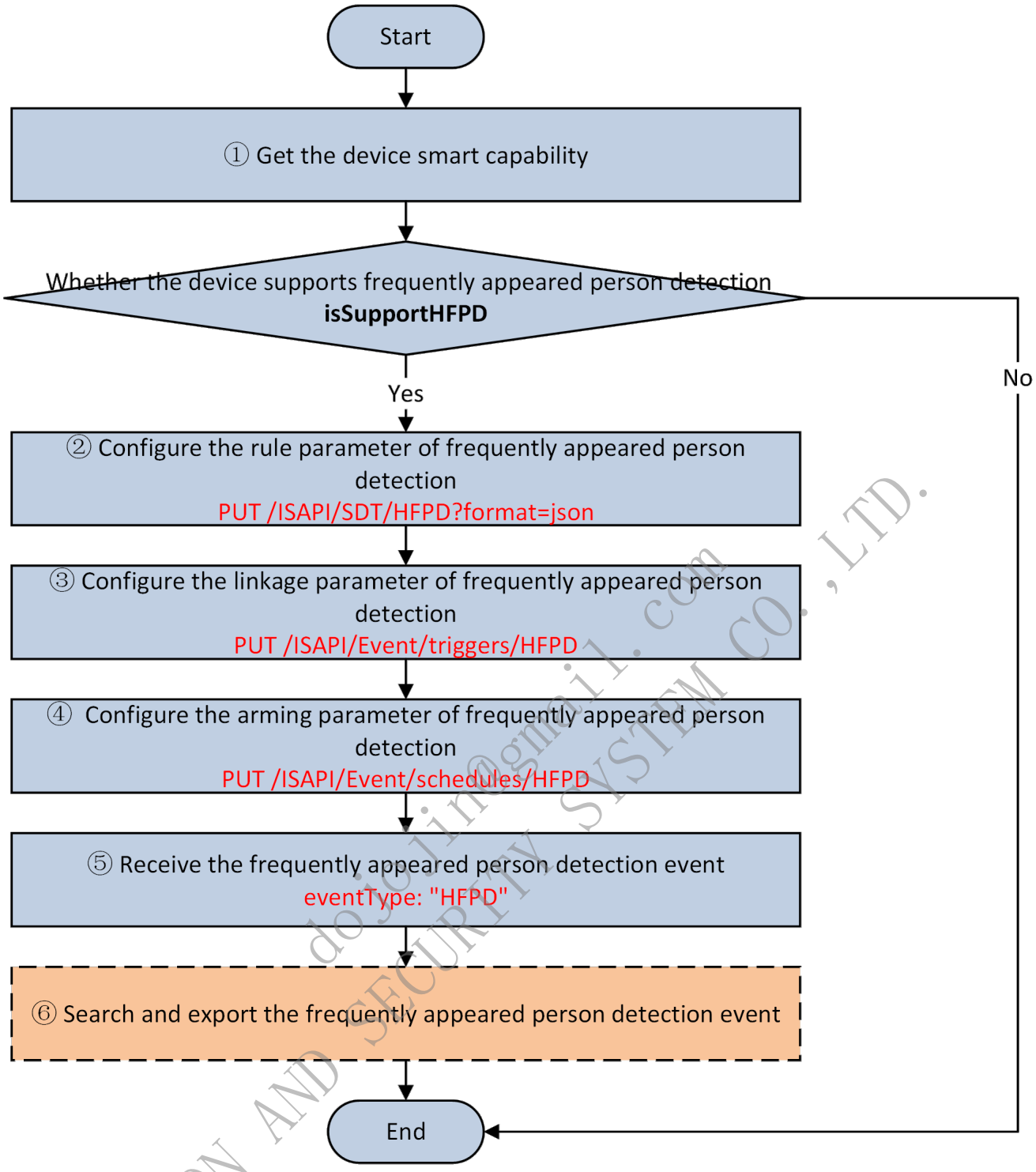
B. Identify frequent petitioners at the door of the Letters and Complaints Bureau.

C. Identify the regular customers during the 4S car shops activities.

D. Identify the personnel who often come to stay in the selected primary spots at the gate of the court.

9.6.2 API Calling Flow

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1. Check whether the frequently appeared person detection is supported by device:**

Get the frequently appeared person detection capability supported by the device `GET /ISAPI/Smart/capabilities`. If the node `<isSupportHFPD>` is returned and its value is `true`, it indicates that the frequently appeared person detection function is supported by the device.

2. Parameters of the frequently appeared person detection rule, including the following parameters:

Get the rule parameters capability of frequently appeared person detection `GET /ISAPI/SDT/HFPD/capabilities?format=json`.

Configure the rule parameters of frequently appeared person detection `PUT /ISAPI/SDT/HFPD?format=json`.

Get the rule parameters of frequently appeared person detection `GET /ISAPI/SDT/HFPD?format=json`.

3. Rule parameters of the frequently appeared person detection, including the following parameters:

Get the linkage parameters capability of frequently appeared person detection `GET /ISAPI/Event/triggersCap`.

Configure the linkage parameters of frequently appeared person detection `PUT /ISAPI/Event/triggers/HFPD`.

Get the linkage parameters of frequently appeared person detection `GET /ISAPI/Event/triggers/HFPD`.

4. Parameters of the frequently appeared person detection rule, including the following parameters:

Configure the linkage parameters of frequently appeared person detection `PUT /ISAPI/Event/schedules/HFPD`.

Get the linkage parameters of frequently appeared person detection `GET /ISAPI/Event/schedules/HFPD`.

5. Frequently appeared person detection event:

The device will report the frequently appeared person detection event when the appearance of arming face pictures reaches a specified number of times. The event information mainly includes face attributes, face pictures, and matched face picture library.

6. Search for the frequently appeared person detection event:

Check whether searching for the frequently appeared person detection event is supported by device: `GET /ISAPI/SDT/HFPD/capabilities?format=json`.

Search the asynchronous data:

Request for searching for data: `POST /ISAPI/SDT/HFPD/searchTask/eventRecordData?format=json`.

Get the searching progress: `GET /ISAPI/SDT/Management/Task/progress/<taskID>?format=json`.

Search data: `POST /ISAPI/SDT/HFPD/searchTask/result?format=json`.

Search the synchronous data:

Request for searching data: `POST /ISAPI/SDT/HFPD/searchTask/eventRecordData/sync?format=json`.

Get data details: `GET /ISAPI/SDT/HFPD/searchTask/result/recordData/<ID>?format=json`.

7. Export the searched results of the frequently appeared person detection event:

Check whether the frequently appeared person detection is supported by device: `GET /ISAPI/SDT/HFPD/capabilities?format=json`.

Request for exporting data: `POST /ISAPI/SDT/HFPD/search/dataPackage?format=json`.

Get the data exporting progress: `GET /ISAPI/SDT/Management/Task/progress/<taskID>?format=json;`.

Get the exported data: `GET /ISAPI/SDT/HFPD/search/dataPackage/<taskID>?format=json`.

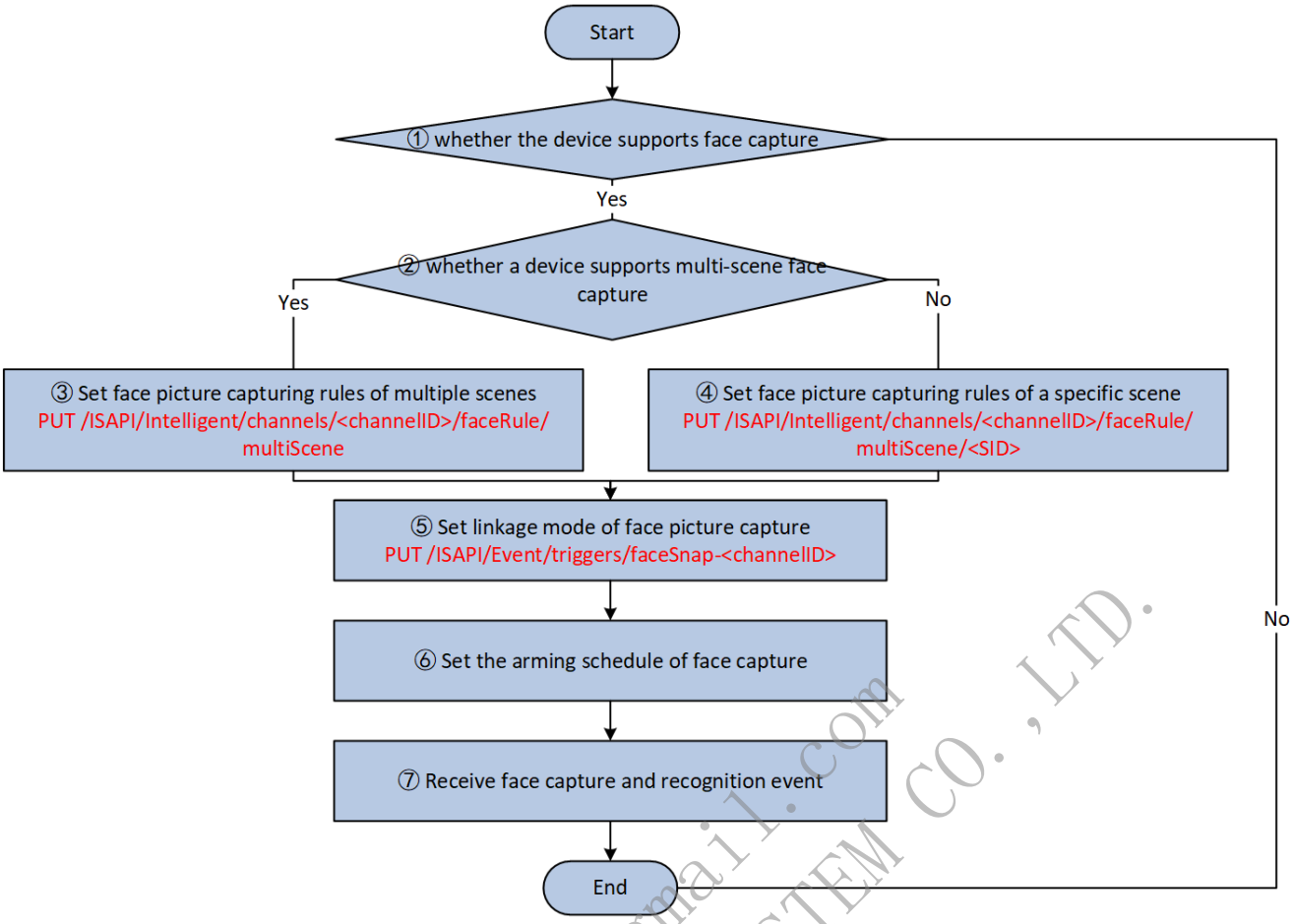
9.7 Privacy Mask on Faces

9.7.1 Introduction to the Function

Privacy mask on faces is to encrypt privacy contents for face capture and media data (see "Video Privacy Mask" API calling flow); it is the process of privacy masking for all faces of cameras and the encrypted images can be restored. First of all, all faces need to be identified, the original media data of faces in the images and masking block data will be replaced, the original media data of faces will be added to the private data of the live view stream and encrypted, then reported to the user synchronously; and the user can decrypt the private data to get the original media data and restore the face image. **Note: The current image restoration only supports private streams, which needs to be restored by integrating the Playback Library.** The following figure shows the effect of face privacy mask by mosaic processing.

9.7.2 API Calling Flow

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1. **Whether the video channel supports face privacy mask.** Get the channel intelligence capabilities: `GET /ISAPI/Intelligent/channels/<channelID>/capabilities`, if the node `<isSupportFacePrivacyMask>` is returned as true, it indicates that the device supports face privacy masking.

2. **Set privacy encryption parameters in media data.** Users apply the encryption parameters including secondary key `secretKey` and the update time of the primary key `secretKeyUpdateFrequency` to devices, etc. Get the encryption capabilities of media data privacy: `GET /ISAPI/Security/MediaObjectEncryption/channel/<channelID>/capabilities?format=json`. Set the encryption parameters of media data privacy `PUT /ISAPI/Security/MediaObjectEncryption/channel/<channelID>?format=json`.Get the encryption parameters of media data privacy: `GET /ISAPI/Security/MediaObjectEncryption/channel/<channelID>?format=json`.

3. **Set parameters of face privacy mask.** Users can enable face privacy mask for a channel `enabled`. Get the capability of face privacy masking: `GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask/capabilities?format=json`;
Set the parameters of face privacy mask: `PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json`; Get the parameters of face privacy mask: `GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json`;

4. **(Optional) Decrypt media data and restore images**. The device generates the primary key and updates the primary key periodically according to the primary key update time in the media data privacy encryption parameters, then replaces the original media data with the masking block in the specified sensitive area in the live view screen, after which the images in the sensitive area are encrypted by the primary key and added to the private data of the live view stream, and at the same time the primary key is encrypted by the secondary key and added to the private data of the preview stream. The key is also encrypted by the secondary key and added to the private data of the live view stream. The user can parse the private data in the live view stream and decrypt it with the secondary key to get the primary key, then decrypt the private data with the primary key to get the original media data and restore the image.

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Note: At present, image restoration is only supported by private streams. You need to get the private data in the live view stream by integrating Playback Library, and get the primary key by decrypting the secondary key, then decrypt the private data by the primary key to get the original media data; and finally call the Playback Library API to restore the image. You can be refer to Playback library interface integration manual for playback library related APIs.

10 Human Body Recognition

10.1 Search by Human Body Picture

10.1.1 Function Introduction

Search by human body picture is the feature-based image search. It allows users to submit a human body picture as an input, and the system retrieves the most relevant images from a large-scale image records based on the input.

10.1.2 API Calling Flow

Capability Check

- Get device system capabilities: `GET /ISAPI/System/capabilities`. If `isSupportHumanRecognition` returns true, this function is supported.
- Get human body detection capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`.

Human Detection and Modeling

- Human detection: `POST /ISAPI/Intelligent/humanRecognition?format=json`.
- Human modeling: `POST /ISAPI/Intelligent/humanModel?format=json`.

Synchronous Search

- Get synchronous search capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`. If `isSupportSync` returns true, the synchronous search is supported.
- Submit a synchronous search request: `POST /ISAPI/Intelligent/humanRecognition/searchByPic/sync?format=json`.

Asynchronous Search

- Get asynchronous search capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`. If `isSupportAsync` returns true, the asynchronous search is supported.
- Submit an asynchronous search task: `POST /ISAPI/Intelligent/humanRecognition/searchByPic?format=json`.
- Get the asynchronous search task progress: `GET /ISAPI/Intelligent/humanRecognition/searchByPic/progress?format=json&taskID=<taskID>`.
- Get the search task result: `POST /ISAPI/Intelligent/humanRecognition/searchByPic/result?format=json`.
- Delete the search task: `DELETE /ISAPI/Intelligent/humanRecognition/searchByPic?format=json&taskID=<taskID>`.
- Search for task information.
  - Get human detection capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`. If `isSupportTaskInfo` returns true, searching for task information is supported.
  - Get task search capabilities: `GET /ISAPI/Intelligent/humanRecognition/searchByPic/taskInfo/capabilities?format=json`.
  - Search for task information: `POST /ISAPI/Intelligent/humanRecognition/searchByPic/taskInfo?format=json`.

Export Image Search Results

- Get human detection capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`. If the node `isSupportExport` returns true, exporting image search results is supported.
- Submit the exporting task: `POST /ISAPI/Intelligent/humanRecognition/searchByPic/result/export?format=json`.
- Get the progress of exporting task: `GET`

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`/ISAPI/Intelligent/humanRecognition/searchByPic/result/export/progress?format=json&taskID=<taskID>`.

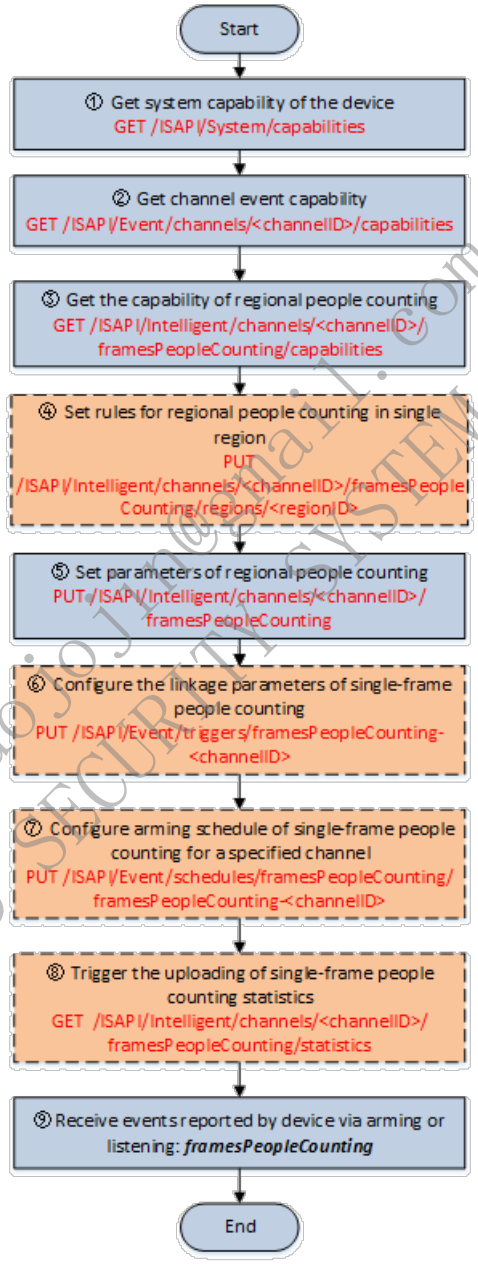
- Stop the exporting task: `PUT /ISAPI/Intelligent/humanRecognition/searchByPic/result/export/stop?format=json`.

11 Crowd Trend Detection

11.1 1 Introduction to the Function

The people counting statistics is calculated by frame and reported via alarm.

11.1.2 API Calling Flow



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1. Get system capability of the device: `GET /ISAPI/System/capabilities`. If the node `isSupportFramesPeopleCounting` is returned, it indicates that regional people counting is supported;
2. Get channel event capability: `GET /ISAPI/Event/channels/<channelID>/capabilities`. if the node `framesPeopleCounting` is returned, it indicates that regional people counting is supported by the corresponding channel;
3. Get the capability of regional people counting and get the supported parameters: `GET /ISAPI/Intelligent/channels/<channelID>/framesPeopleCounting/capabilities`;
4. (Optional) Set rules for regional people counting in single region: `PUT /ISAPI/Intelligent/channels/<channelID>/framesPeopleCounting/regions/<regionID>`;

5. Set parameters of regional people counting: `PUT /ISAPI/Intelligent/channels/<channelID>/framesPeopleCounting`;
6. (Optional) Configure the linkage parameters of single-frame people counting: `PUT /ISAPI/Event/triggers/framesPeopleCounting-<channelID>`;
7. (Optional) Configure arming schedule of single-frame people counting for a specified channel: `PUT /ISAPI/Event/schedules/framesPeopleCounting/framesPeopleCounting-<channelID>`;
8. (Optional) Trigger the uploading of single-frame people counting statistics: `GET ISAPI/Intelligent/channels/<channelID>/framesPeopleCounting/statistics`;
9. Receive events reported by device via arming or listening: `framesPeopleCounting`.

Regional People Counting Statistics Search

11.1.1 Introduction to the Function

Get the history or the latest statistics of regional people counting by channel or device.

11.1.2 API Calling Flow

11.1.2.1 Search for Statistics by Time and Channel

1. Evaluate whether the device channel supports regional people counting, i.e., the Regional people counting capability flow;
2. Search for regional people counting statistics by time and channel: `GET /ISAPI/Intelligent/channels/<channelID>/framesPeopleCounting/statistics/search?startTime=<startTime>&endTime=<endTime>`.

11.1.2.2 Search for Statistics by Time

1. Get intelligent capability: `GET /ISAPI/Intelligent/capabilities`. If the node `isSupportFramesPeopleCountingSearch` is returned, it indicates that regional people counting is supported;
2. Get the search capability: `GET /ISAPI/Intelligent/framesPeopleCounting/search/capabilities?format=json`;
3. Search for device statistics within a specified time period: `POST /ISAPI/Intelligent/framesPeopleCounting/search?format=json`.

11.1.2.3 Get the Latest Statistics

1. Evaluate whether the device channel supports regional people counting: `GET /ISAPI/Intelligent/channels/<channelID>/framesPeopleCounting/capabilities`. If the node `isSupportSerachCountingResult` is returned, it indicates that getting the latest regional people counting statistics is supported.
2. Get the latest regional people counting alarm result: `GET /ISAPI/Intelligent/channels/<channelID>/framesPeopleCounting/result`.

12 Video Recognition

12.1 Face Picture Library Management

12.1.1. Search via Picture Comparison in Face Capture Library

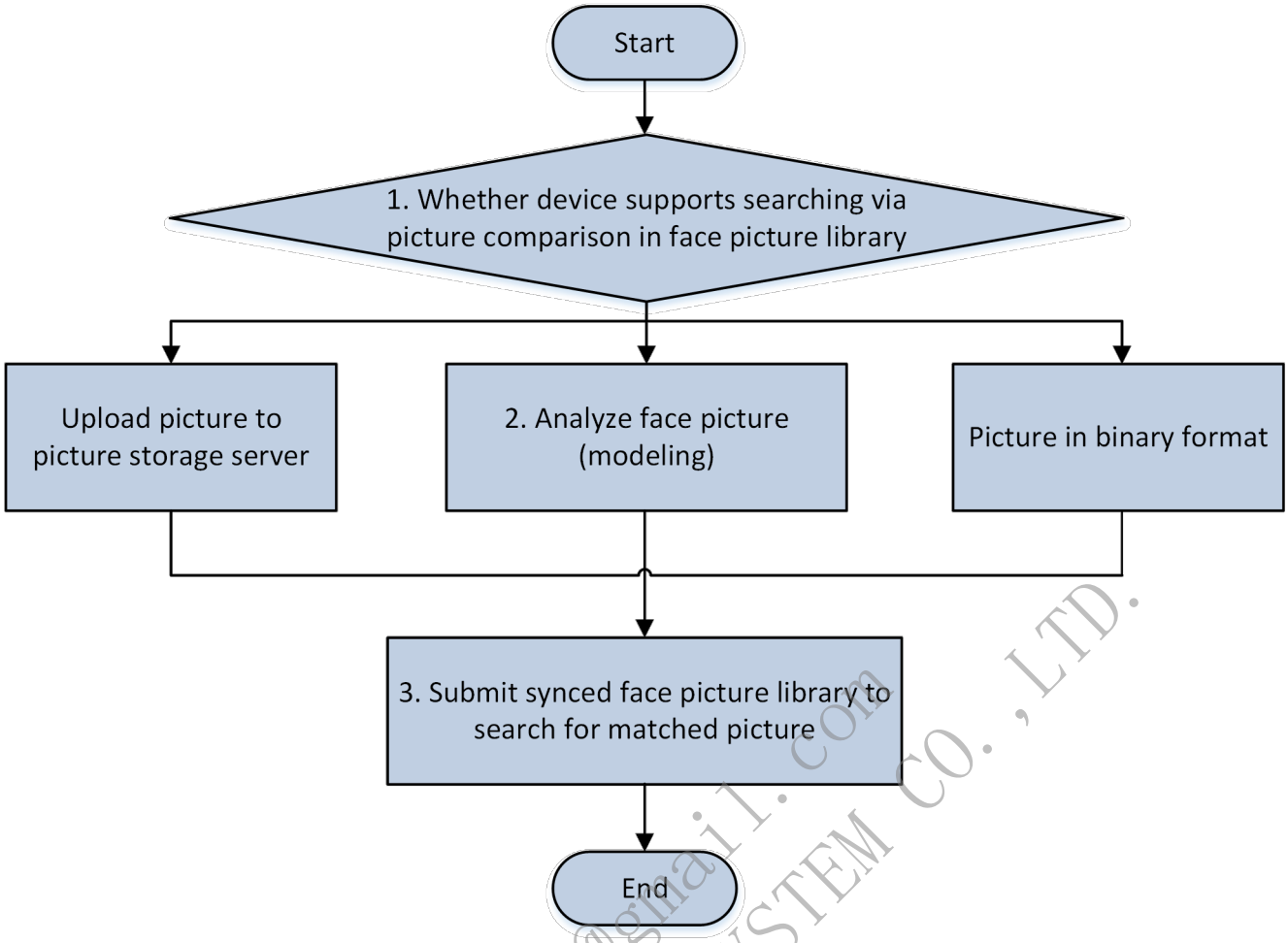
12.1.1.1 Introduction to the Function

Picture comparison with captured picture library

Scene application: Search for similar person in face picture library, which can be used for identity confirmation.

12.1.1.2 API Calling Flow

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1. **Check whether the device supports searching by picture synchronously.**

   Get the capability of searching via picture comparison: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`. If the node `isSuportFSSearchByPic` is returned and its value is `true`, it indicates that searching via picture comparison synchronously is supported.

   Note: For some devices supporting searching via picture comparison synchronously, the API for getting the capability is invalid, you can call `POST /ISAPI/Intelligent/FDLib/searchByPic?format=json` directly.

2. **Three methods are supported when searching picture.**

   By URL:

   Based on the picture storage server, picture can be uploaded and you can get the picture URL to download the picture file via the URL.

   By model data:

   Get face picture capability: `GET /ISAPI/SDT/Face/pictureAnalysis/capabilities`;

   Analyze (model) face pictures: `POSTT /ISAPI/SDT/Face/pictureAnalysis`; Return picture modeling data: `targetModelData`.

   By binary picture:

   Read binary picture files.

3. **Submit synced face picture library to use picture to search picture**

   Search via picture comparison in face capture library: `POST /ISAPI/Intelligent/FDLib/searchByPic?format=json`;

Note: Input parameter node `dataType` include three modes. Picture URL corresponds to `faceURL`, model data corresponds to `targetModelData`, binary data mode is in form of form, and its JSON data is followed by binary picture data.

The request message example of binary picture data is as follows:

```
POST /ISAPI/Intelligent/FDLib/searchByPic?format=json
Host: device_ip:port
Accept-Language: zh-cn
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive

--<frontier>
Content-Disposition: form-data; name=""
Content-Type: application/json

[JSON message]
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[Picture data]
--<frontier>--
```

12.2 Professional Smart Abnormal Event Detection

12.2.1 Introduction to the Function

Professional Smart Abnormal Event Detection includes the following features:

- Line Crossing Detection
- Intrusion Detection
- Region Entrance Detection
- Region Exiting Detection
- Loitering Detection
- People Gathering Detection
- Rapid Movement Detection
- Parking Detection
- Unattended Baggage Detection
- Object Removal Detection
- Running Detection
- Abnormal Number of People Detection
- Person Spacing Change Detection
- Violent Motion Detection
- On/Off Duty Detection
- Falling Down Detection
- Lingering Detection
- Person Density Detection
- Climbing Detection
- Trend Analysis
- In-Toilet Overtime Detection
- Sudden Change of Sound Intensity Detection
- Student Standing Up Detection
- Getting Up from Bed Detection

Professional smart abnormal event detection is designed for specific scenarios: workplace activity detection and pedestrian activity detection.

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Workplace activity detection includes but not limited to:

- Using a Mobile Phone
- Number of People Changes
- On/Off Duty
- Person Lingering

Pedestrian activity detection includes but not limited to:

- Running Detection
- People Gathering Detection
- Violent Motion Detection
- Person Falling Detection

You can configure workplace activity detection and pedestrian activity detection rules and alarm linkage actions to suit your needs.

Abnormal event detection can be categorized into two types: Basic smart abnormal event detection and professional smart abnormal event detection. Their differences are as follows:

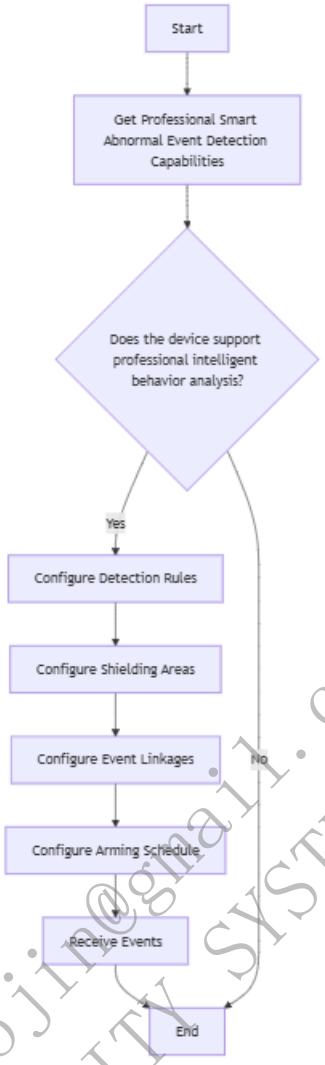
1. **Algorithms**: Traditional algorithms are used for basic smart abnormal event detection, while deep learning algorithms are used for professional smart abnormal event detection.
2. **Detection Effect**: Basic smart abnormal event detection can only analyze targets as people or vehicles. Professional smart abnormal event detection can not only distinguish between targets as people or vehicles but also identify specific behaviors, such as the number of people and human behaviors such as using a mobile phone, violent motion, and running.
3. **Supported Events**: Smart Basic Intelligent Abnormal Behavior Detection supports events such as:
    - Line Crossing Detection
    - Intrusion Detection
    - Region Entrance Detection
    - Region Exiting Detection
    - Loitering Detection
    - People Gathering Detection
    - Rapid Movement Detection
    - Parking Detection
    - Unattended Baggage Detection
    - Object Removal Detection

Professional Smart Abnormal Event Detection supports all the events mentioned above and additional events including:

- Running Detection
- Abnormal Number of People Detection
- Person Spacing Change Detection
- Violent Motion Detection
- On/Off Duty Detection
- Falling Down Detection
- Lingering Detection
- Person Density Detection
- Climbing Detection
- Trend Analysis
- In-Toilet Overtime Detection
- Sudden Change of Sound Intensity Detection
- Student Standing Up Detection
- Getting Up from Bed Detection

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12.2.2 API Calling Flow



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Before taking the following steps, ensure that this smart detection has been enabled for devices.

1. **Get Professional Smart Abnormal Event Detection Capabilities**
   - Call `GET /ISAPI/Intelligent/capabilities` to get professional smart abnormal event detetection capabilities and the field `isBehaviorSupport` is returned as `true` in the response message.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/capabilities` to get professional smart abnormal event detetection capabilities for a specified channel and the field `isBehaviorSupport` is returned as `true` in the response message.

   Note: `isBehaviorSupport`as `true` indicates support for professional smart abnormal event detetection. The fields of supported events are as follows:
   - `isLineDetectionSupport` indicates support for **Line Crossing Detection**.
   - `isFieldDetectionSupport` indicates support for **Intrusion Detection**.
   - `isRegionEntranceSupport` indicates support for **Region Entrance Detection**.
   - `isRegionExitingSupport` indicates support for **Region Exiting Detection**.
   - `isLoiteringSupport` indicates support for **Loitering Detection**.
   - `isGroupSupport` indicates support for **People Gathering Detection**.
   - `isRapidMoveSupport` indicates support for **Rapid Movement Detection**.
   - `isParkingSupport` indicates support for **Parking Detection**.
   - `isUnattendedBaggageSupport` indicates support for **Unattended Baggage Detection**.
   - `isAttendedBaggageSupport` indicates support for **Object Removal Detection**.
   - `isRunningSupport` indicates support for **Running Detection**.

- `isPeopleNumChange` indicates support for **Abnormal Number of People Detection**.
- `isSpacingChangeSupport` indicates support for **Person Spacing Change Detection**.
- `isViolentMotionSupport` indicates support for **Violent Movement Detection**.
- `isLeavePositionSupport` indicates support for **On/Off Duty Detection**.
- `isFailDownSupport` indicates support for ** Falling Down Detection**.
- `isRetentionParamSupport` indicates support for **Lingering Detection**.
- `isSupportPersonDensityDetection` indicates support for **Person Density Detection**.
- `isVcaEventAdvReachHeightSupport` indicates support for **Climbing Detection**.
- `isSupportSituationAnalysis` indicates support for **Trend Analysis**.
- `isSupportToiletTarry` indicates support for **In-Toilet Overtime Detection**.
- `isSupportAudioAbnormal` indicates support for **Sudden Change of Sound Intensity Detection**.
- `isSupportStandUp` indicates support for **Student Standing Up Detection**.
- `isSupportGetUp` indicates support for **Getting Up from Bed Detection**.

2. **Configure Detection Rules**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/capabilities` to get professional smart abnormal event detection capabilities.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to get detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to configure detection rules.
   - Call `DELETE /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to delete detection rules.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>` to get all detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>` to configure all detection rules.

   Note:
   - The `eventType` field in the rules indicates the corresponding event type.
   - The `ruleType` field indicates the event support drawing a detection line or a polygon detection area. Some events (sudden change of sound intensity) do not support drawing a box or line, some events (line crossing, climbing) support drawing a line, and some events (except sudden change of sound intensity, line crossing, and climbing) support for polygon detection areas.

3. **(Optional) Configure Shielding Areas**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/Shield/capabilities` to get professional smart area shielding capabilities.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/Shield` to get shielding areas for all events.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/Shield` to configure shielding areas for all events.

4. **(Optional) Configure Event Linkages**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/notifications` to get event linkages.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/notifications` to configure event linkages.

5. **(Optional) Configure Arming Schedule**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/schedules` to get the schedule to disable the professional smart abnormal event detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/schedules` to configure the schedule to disable the professional smart abnormal event detection rules.

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Note:

- Up to eight time periods can be configured per day.
- Each two time periods cannot overlap.

6. **Receive Events**
   - Line Crossing Detection: `"eventType":"linedetection"`
   - Intrusion Detection: `"eventType":"fielddetection"`
   - Region Entrance Detection: `"eventType":"regionEntrance"`
   - Region Exiting Detection: `"eventType":"regionExiting"`
   - Loitering Detection: `"eventType":"loitering"`
   - People Gathering Detection: `"eventType":"group"`
   - Rapid Movement Detection: `"eventType":"rapidMove"`
   - Parking Detection: `"eventType":"parking"`
   - Unattended Baggage Detection: `"eventType":"unattendedBaggage"`
   - Object Removal Detection: `"eventType":"attendedBaggage"`
   - Running Detection: `"eventType":"running"`
   - Abnormal Number of People Detection: `"eventType":"peopleNumChange"`
   - Person Spacing Change Detection: `"eventType":"spacingChange"`
   - Violent Movement Detection: `"eventType":"violentMotion"`
   - On/Off Duty Detection: `"eventType":"leavePosition"`
   - Falling Down Detection: `"eventType":"failDown"`
   - Lingering Detection: `"eventType":"retention"`
   - Person Density Event: `"eventType":"personDensityDetection"`
   - Climbing Detection: `"eventType":"advReachHeight"`
   - Trend Analysis: `"eventType":"situationAnalysis"`
   - In-Toilet Overtime Detection: `"eventType":"toiletTarry"`
   - Sudden Change of Sound Intensity Detection: `"eventType":"audioAbnormal"`
   - Student Standing Up Detection: `"eventType":"standUp"`
   - Getting Up from Bed Detection: `"eventType":"getUp"`
   - Sitting Still Detection: `"eventType":"sitQuietly"`
   - Key Target Standing Up Detection: `"eventType":"keyPersonGetUp"`
   - Using Mobile Phone Detection: `"eventType":"playCellphone"`
   - Yard Lingering: `"eventType":"yardTarry"`
   - Key Target Standing Up Detection: `"eventType":"keyPersonGetUp"`
   - Sitting Still Detection: `"eventType":"sitQuietly"`
   - Yard Lingering: `"eventType":"yardTarry"`
   - Using Mobile Phone Detection: `"eventType":"playCellphone"`

Note:

- The content of `targetAttrs` is in JSON format. When the event format is XML, use CDATA (Character Data) to include this data in the event and transmit it to the platform.
- CDATA (Character Data) is used to include large amounts of text data in XML documents. This text may contain special characters (such as `<`, `>`, `&`), which have special meanings in XML and typically need to be escaped. CDATA blocks allow these special characters to be included directly in the document without escaping, such as including JavaScript scripts or CSS styles in XML documents.

12.3 Upgrading the Deep Learning Model for Perimeter Intrusion Detection

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12.3.1 Introduction to the Function

You can upgrade the deep learning model for perimeter intrusion detection. You can also upgrade both the local training model for NVRs and the training model for network camera channels.

12.3.2 API Calling Flow

1. Determine if the device supports upgrading the deep learning model for perimeter intrusion detection: Call `GET /ISAPI/Intelligent/capabilities`. If the field `isSupportUpgradeDepthPerimeterModel` is returned as `true` in the response message, this function is supported.
2. Get the deep learning model upgrade capabilities: `GET /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/capabilities?format=json`
3. Set the model version: `PUT /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/VersionOperation?format=json`
    - You cannot specify a version to upgrade. Only rolling back to the previous version and restoring to the factory model version are currently supported.
4. Get the model upgrade progress: `GET /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/process?format=json`
5. Set the model version for a channel: `PUT /ISAPI/Intelligent/depthPerimeter/channels/<channelID>/upgradeTrainModel/VersionOperation?format=json`
6. Get the model upgrade progress for a channel: `GET /ISAPI/Intelligent/depthPerimeter/channels/<channelID>/upgradeTrainModel/process?format=json`

13 Image Intelligent Search Based on Large Mode

13.1 Text-to-Image Search

13.1.1 Function Introduction

Use a text to search for the target information, including capture channel, capture time, target type, target similarity, capture image URL, target image URL, and target rectangle frame coordinates.

13.1.2 API Calling Flow

1. Check if the device supports the text-to-image search function.

Request:

```
GET /ISAPI/ContentMgmt/capabilities
```

Response:

```
<isSupportSearchTextTaskStatus>true</isSupportSearchTextTaskStatus>
```

Only the field `<isSupportSearchTextTaskStatus>` is returned with `true` value, the device supports this function.

2. Check if the text-to-image search is enabled. Request:

```
GET /ISAPI/ContentMgmt/PicSearchByText?format=json
```

Response:

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```
{"enabled": true}
```

If it is disabled, enable it via following URL. Request:

```
PUT /ISAPI/ContentMgmt/PicSearchByText?format=json

{
    "enabled": true,
    "channelList": [1]
}
```

3. Execute the text-to-image search.

Inputs: Search text, specified channel, and similarity threshold. Outputs: Capture channel, capture time, target type, similarity, capture picture URLs, and rectangle frame coordinates. This is an asynchronous API, which gets the search task ID (`taskID`) when a search request is submitted, and then queries the results according to search task ID (`taskID`).

Search Request:

```
POST /ISAPI/ContentMgmt/AsynSearchByTextTask?format=json

{
    "startTime": "2025-05-12T00:00:00+08:00",
    "endTime": "2025-05-12T23:59:59+08:00",
    "channelList": [1, 2],
    "text": "Person in black top",
    "similarity": 60
}
```

Search Response:

```
{"taskID": "ffc9bb4a-bb4f-4216-bcc7-1d6efb2"}
```

Result Query Request:

```
POST /ISAPI/ContentMgmt/SearchTextTaskStatus?format=json

{
    "taskID": "ffc9bb4a-bb4f-4216-bcc7-1d6efb2",
    "searchResultPosition": 1,
    "maxResults": 18,
    "similarity": 60
}
```

Result Query Response:

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```
{
    "taskID": "ffc9bb4a-bb4f-4216-bcc7-1d6efb2",
    "runStatus": "completed",
    "numOfMatches": 18,
    "totalMatches": 200,
    "progress": 100,
    "matchResults": [
        {
            "id": 1,
            "time": "2025-05-12T09:20:13+08:00",
            "channelID": 1,
            "targetType": "human",
            "similarity": 85,
            "imageUUID": "5862098914022649856",
            "image": {
                "filePathType": "URL",
                "filePath": "http://10.15.97.210:80/picture/Streaming/tracks/103/?
name=ch00001_0e0083a003b8c7f4800041f400004ba58010515a5be72e85f000&size=270144"
            },
            "targetImage": {
                "filePathType": "URL",
                "filePath": "http://10.15.97.210:80/picture/Streaming/tracks/103/?
name=ch00001_0e0083a003b909e8800009a1800009b18010515a5be72e85f000&size=39448"
            },
            "targetRect": {
                "height": 0.70709,
                "width": 0.157127,
                "x": 0.50076,
                "y": 0.0001
            }
        }
    ]
}
```


14 Audio (General)

14.1 Audio Recording Integration Description

14.1.1 Introduction to the Function

Audio Recording includes functions such as manual audio recording, searching, downloading, and deleting audio.

14.1.2 API Calling Flow

1. **Get the capability:**

   Get device system capability: `GET /ISAPI/System/capabilities`.

   Get device storage capability: `GET /ISAPI/ContentMgmt/capabilities`.

2. **Set audio name** (placed after the default name):

   Get parameters of setting a custom audio name of a specific audio channel: `GET /ISAPI/ContentMgmt/recordAudio/<audioID>/customFileName?format=json`.

   Set parameters of setting a custom audio name of a specific audio channel: `PUT /ISAPI/ContentMgmt/recordAudio/<audioID>/customFileName?format=json`.

3. **Record Audio:**

   Get device storage capability: `GET /ISAPI/ContentMgmt/capabilities`. If the node `isSupportRecordAudio` is returned and its value is `true`, the device supports audio recording.

   Get audio recording parameter capability of a specific audio channel: `GET /ISAPI/ContentMgmt/recordAudio/<audioID>/capabilities?format=json`.

   Get audio recording parameters of a specified channel: `GET /ISAPI/ContentMgmt/recordAudio/<audioID>?format=json`.

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Set audio recording parameters of a specified channel: `PUT /ISAPI/ContentMgmt/recordAudio/<audioID>?format=json`.

Note: In the parameters, if the value of `enabled` is `true`, the recording is on; if the value of `enabled` is `false`, the recording is off.

4. **Search the audio:**

   Search audio information by month (check whether there is any audio every day in a specific month):

   Get device storage capability: `GET /ISAPI/ContentMgmt/capabilities`. If `isSupportAudioDailyDistribution` is returned and its value is `true`, the device supports searching audio information by month.

   Get monthly audio information of a specific audio channel: `POST /ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json`.

   Search audio information (search by date):

   Get device storage capability: `GET /ISAPI/ContentMgmt/capabilities`. If `isSupportSearchAudio` is returned and its value is `true`, the device supports searching audio information.

   Get the capability of searching audio: `GET /ISAPI/ContentMgmt/recordAudio/search/capabilities?format=json`.

   Search audio information: `POST /ISAPI/ContentMgmt/recordAudio/search?format=json`.

5. **Download the audio:**

   Get device storage capability: `GET /ISAPI/ContentMgmt/capabilities`. If `isSupportDownloadAudio` is returned and its value is `true`, the device supports downloading audio.

   Get the capability of exporting an audio file with its custom name: `GET /ISAPI/ContentMgmt/recordAudio/fileNameDownload/capabilities?format=json`.

   Export an audio file with the custom name (binary): `POST /ISAPI/ContentMgmt/recordAudio/fileNameDownload/binary?format=json`.

6. **Delete the audio:**

   Get device storage capability: `GET /ISAPI/ContentMgmt/capabilities`. If `isSupportDeleteAudio` is returned and its value is `true`, the device supports deleting audio by name.

   Delete specific audio file: `PUT /ISAPI/ContentMgmt/recordAudio/deleteFile?format=json`.

   Note: For the file name in the input, see the returned results in `/ISAPI/ContentMgmt/recordAudio/search?format=json`.

7. **Audio playback:**

1. The client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 4
Authorization: Digest username="", realm="323852ae0234c718f2d4198b", nonce="34e30476d", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="883546f4c19dd156fb3a490266c99715"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
```

2. The server responds that an authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 5
WWW-Authenticate: Digest realm="323852ae0234c718f2d4198b", nonce="55e5895b9", algorithm="MD5"
```

3. The client resends an RTSP DESCRIBE command with the authentication information.

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```
DESCRIBE rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 7
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="cf33e4dc6b86a2fdd2e5b26d25e7b99d"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Accept: Accept:application/sdp
```

4. The server returns the SDP information.

```
RTSP/1.0 200 OK
CSeq: 7
Content-Type: application/sdp
Content-Length: 546
Date: Wed, 18 Aug 2021 07:29:25 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.60.000
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:4106
a=range:clock=20210818T151815Z-20210818T151908Z
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H265/90000
a=control:trackID=video
b=AS:4096
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=audio
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000040005001171010401F000000FA0000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends the RTSP SETUP command twice, and the server responds.

```
SETUP rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=video
RTSP/1.0
CSeq: 8
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61048-61049
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61048-61049;server_port=62000-62001;ssrc=d6d3ed5
CSeq: 8
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

```
SETUP rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=audio
RTSP/1.0
CSeq: 9
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61050-61051
Session: 225263317
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61050-61051;server_port=62002-62003;ssrc=d6d3ed6
CSeq: 9
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

6. The client sends an RTSP PLAY command, and the server starts pushing playback data stream.

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```
PLAY rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 10
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="4d8b7ceb301c06b40a9ae411189a909f"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
Range: npt=0.000-
```

```
RTSP/1.0 200 OK
Range: clock=20210818T151815Z-20210818T151908Z
Session: 225263317
CSeq: 10
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

7. The client sends an RTSP TEARDOWN command to stop playback.

```
TEARDOWN rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 11
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="1d3f6f8d07d7087d341560b125445456"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
```

```
RTSP/1.0 200 OK
CSeq: 11
Date: Wed, 18 Aug 2021 07:29:38 GMT
Session: 225263317
Connection: close
```

15 API Reference

15.1 设备通用

15.1.1 事件订阅

15.1.1.1 Establish arming connection

Request URL

GET /ISAPI/Event/notification/alertStream?deployID=<deployID>

Query Parameter

Parameter Name	Parameter Type	Description
deployID	string	0 (client-side arming), 1 (real-time arming). Defaults to the client-side arming when delpoyID does not exist.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, event massage, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPV4 address of the device that triggers an alarm-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPV6 address of the device that triggers an alarm-->test
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the device that triggers an alarm, range:[0,65535]-->0
  </portNo>
  <protocol>
    <!--ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address, range:[0,17]-->00-16-EA-AE-3C-40
  </macAddress>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers an alarm, range:[0,65535]-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm triggering time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, req, int, the uploading times of the same alarm, range:[0,65535]-->0
  </activePostCount>
  <eventType>
    <!--ro, req, string, heartbeat type-->test
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:"active" (triggered), "inactive" (not triggered, the heartbeat data)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, range:[0,128]-->test
  </eventDescription>
</EventNotificationAlert>
```

15.1.1.2 Set alarm/event subscription parameters

Request URL

PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all"(upload all alarms/events), "list"(upload specified alarm/event)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, "all"-upload all alarms/events,"list"-upload specified alarm/event, subType:string, desc:"all"(upload all alarms/events), "list"(upload
specified alarm/event)-->list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--opt, object, uploading mode of specified alarm/event-->
      <type>
        <!--req, enum, alarm/event types, subType:string, desc:see details in event types: "ADAS"(advanced driving assistance system), "ADASAlarm"(advanced
driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent", "CDsStatus", "DBD"
(driving behavior detection) "GPSUpload", "HFPD"(frequently appeared person), "IO"(I/O Alarm), "IOTD", "LES", "LFPD"(low frequency person detection),
"PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-
alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage",
"audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch",
"changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk
unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling",
"facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection", "fireEscapeDetection",
"flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"
(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature
alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person
getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection), "listSyncException"(list synchronization exception),
"loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event), "mixedTargetDetection"(multi-target-type detection),
```

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```
"modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"
(staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm),
"personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection),
"personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception),
"policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection",
"radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient
recording period), "recordException", "regionEntrance", "regionExiting", "retention"(people overstay detection), "rollover", "running"(people running),
"safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"(angular acceleration alarm), "severeHDFailure"(HDD major fault
detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO",
"spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection),
"targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime
detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list
alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException", "videoloss", "violationAlarm", "violentMotion"(violent motion
detection), "yardTarry"(playground overstay detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code
event of access control), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent",
"ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching consumption set meals),
"ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison"-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, operation sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, event sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)--
>cloudStorageURL
      </pictureURLType>
      <channels>
        <!--opt, string, channel information linked to event, desc:event linked channel information,and multiple channels can be linked,each channel is
separated by comma,e.g.,"1,2,3,4..."-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get
all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--opt, enum, event sub-type, subType:string, desc:"success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--opt, array, subType:object-->
        <childDevID>
          <!--opt, string, range:[1,128]-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--opt, string-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--opt, string-->1,2,3,4
      </audioChannels>
    </Event>
  </EventList>
  <channels>
    <!--opt, string, event linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4...", desc:event
linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4..."-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--opt, enum, alarm picture format, subType:string, desc:"binary"(binary), "localURL"(device local URL), "cloudStorageURL"(cloud storage URL)--
>cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:5 minute (default). If GUID is not reconnected in the internal, a new GUID
will be generated since the device starts a new arm period-->300
    </interval>
    <StatusSub>
      <!--opt, object, status description-->
      <all>
        <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--opt, bool, status of subscribed channels, desc:reporting is not required if all is true-->true
```

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```
      </channel>
      <hd>
        <!--opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
      </hd>
      <capability>
        <!--opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <level>
    <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
  </level>
  <middleLevelPictureEnabeld>
    <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--opt, array, subType:object-->
    <childDevID>
      <!--opt, string, range:[1,128]-->test
    </childDevID>
  </childDevIDList>
  <radarChannels>
    <!--opt, string-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object-->
    <defaultLevel>
      <!--ro, req, enum, subType:int-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, subType:object-->
      <event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, string, range:[0,64]-->test
        </eventType>
        <level>
          <!--ro, req, enum, subType:int-->1
        </level>
        <channels>
          <!--ro, opt, string, range:[0,256]-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
  <audioChannels>
    <!--opt, string-->1,2,3,4
  </audioChannels>
</SubscribeEvent>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

15.1.1.3 Get alarm/event subscription parameters

Request URL

GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	When an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), the (subscription ID) will be returned in message SubscribeEventResponse. When an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), and if subscribeEventID is applied, it can be used to replace the in SubscribeEventResponse.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--ro, opt, int, heartbeat interval time, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--ro, opt, enum, channel subscription mode, subType:string, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel
list)-->list
  </channelMode>
  <eventMode>
    <!--ro, req, enum, event subscription mode, subType:string, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)--
>list
  </eventMode>
  <EventList>
    <!--ro, opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--ro, opt, object, subscription of a specified alarm/event-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:alarm/event types, which are obtained from the capability, refer to Alarm/Event Types
(eventType)-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--ro, opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
```

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```
        <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
      <channels>
        <!--ro, opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--ro, opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--ro, opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--ro, opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--ro, opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--ro, opt, enum, event sub-type, subType:string, desc:"success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--ro, opt, array, subType:object-->
        <childDevID>
          <!--ro, opt, string, range:[1,128]-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--ro, opt, string-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--ro, opt, string-->1,2,3,4
      </audioChannels>
      <childDeviceFormat>
        <!--ro, opt, bool-->true
      </childDeviceFormat>
    </Event>
  </EventList>
  <channels>
    <!--ro, opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is all, this node should not be applied-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--ro, opt, enum, format unified configuration of alarm pictures, subType:string, desc:"binary", "localURL" (device local URL), "cloudStorageURL" (cloud storage URL). The node indicates the upload mode of all event pictures. If the node is applied, the <pictureURLType> of the <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--ro, opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--ro, opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--ro, opt, object, status description-->
      <all>
        <!--ro, opt, bool, whether to subscribe to all channels, HDDs, and capability set changes-->false
      </all>
      <channel>
        <!--ro, opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
      </channel>
      <hd>
        <!--ro, opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
      </hd>
      <capability>
        <!--ro, opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the value of <all> is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
```

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```
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--ro, opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--ro, opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--ro, opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <level>
    <!--ro, opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
  </level>
  <middleLevelPictureEnabeld>
    <!--ro, opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--ro, opt, array, subType:object-->
    <childDevID>
      <!--ro, opt, string, range:[1,128]-->test
    </childDevID>
  </childDevIDList>
  <radarChannels>
    <!--ro, opt, string-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object-->
    <defaultLevel>
      <!--ro, req, enum, subType:int-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, subType:object-->
      <event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, string, range:[0,64]-->test
        </eventType>
        <level>
          <!--ro, req, enum, subType:int-->1
        </level>
        <channels>
          <!--ro, opt, string, range:[0,256]-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
  <audioChannels>
    <!--ro, opt, string-->1,2,3,4
  </audioChannels>
  <retransmissionID>
    <!--ro, opt, string, range:[1,64]-->123456
  </retransmissionID>
  <retransmissionTimeoutDuration>
    <!--ro, opt, int, range:[1,168], unit:h-->1
  </retransmissionTimeoutDuration>
  <SubscribeISAPIMessage>
    <!--ro, opt, object-->
    <EventTypeList>
      <!--ro, req, array, subType:object-->
      <eventType>
        <!--ro, req, enum, subType:string-->AIOP_Video
      </eventType>
      <uploadPicEnabled>
        <!--ro, opt, bool-->false
      </uploadPicEnabled>
    </EventTypeList>
    <EventList>
      <!--ro, req, array, subType:object, range:[0,200]-->
      <Event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, enum, subType:string-->AIOP_Video
        </eventType>
        <uploadPicEnabled>
          <!--ro, opt, bool-->false
        </uploadPicEnabled>
      </Event>
    </EventList>
  </SubscribeISAPIMessage>
</SubscribeEvent>
```

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15.1.1.4 Subscribe to events/alarms in arming mode

Request URL

POST /ISAPI/Event/notification/subscribeEvent?deployID=<deployID>

Query Parameter

Parameter Name	Parameter Type	Description
deployID	enum	0 (device local arming), 1 (real-time arming by platform). If the deployID is absent, it defaults to device local arming.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval time, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel
list)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, event subscription mode, subType:string, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)--
>list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--opt, object, subscription of a specified alarm/event-->
      <type>
        <!--req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
(video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
"TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
"alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioException", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm",
"faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection",
"fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event),
"highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
conflicts), "keyPersonGetUp"(key person getting up detection), "LeavePosition"(absence detection), "linedetection"(line crossing detection),
"listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event),
"mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
"nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
"peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
"personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
"radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system),
"takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
"unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException",
"videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "CertificateCaptureEvent"(person ID
capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event),
"TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status),
"certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison), "regionTargetNumberCounting" (regional target statistics)--
>mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, minor operation type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, minor event type, desc:this node is required when type is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)--
>cloudStorageURL
      </pictureURLType>
      <channels>
        <!--opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
```

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```
          <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get
all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel
check-in.
3. When this node is not applied, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--opt, array, (sub-device local identifier) device serial No. list, subType:object, desc:arm sub device (sensor)-->
        <childDevID>
          <!--opt, string, (sub-device local identifier) device serial No. list, range:[1,128], desc:arm sub device (sensor)-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--opt, string, arm by radar channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--opt, string, arm by audio channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </audioChannels>
      <childDeviceFormat>
        <!--opt, bool, upload by sub device, desc:upload by sub device-->true
      </childDeviceFormat>
      <unitChannels>
        <!--opt, string, arm by optical fiber channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </unitChannels>
    </Event>
  </EventList>
  <channels>
    <!--opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is
all, this node should not be applied-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--opt, enum, format unified configuration of alarm pictures, subType:string, desc:"binary", "localURL" (device local URL), "cloudStorageURL" (cloud
storage URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is
not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end
devices is local URL of the device-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client
software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--opt, object, status information-->
      <all>
        <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--opt, bool, status of subscribed channels, desc:it is not required if the value of <all> is true-->true
      </channel>
      <hd>
        <!--opt, bool, subscribe to HDD status, desc:it is not required if the value of <all> is true-->true
      </hd>
      <capability>
        <!--opt, bool, subscribe to changing status of capability set, desc:it is not required if the value of <all> is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When
this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform
focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-
->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will
be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm--
>true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no
related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
```

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```
<identityKey>
  <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
</identityKey>
<ConferenceIDList>
  <!--opt, array, video conference ID list, subType:object-->
  <ConferenceID>
    <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
  </ConferenceID>
</ConferenceIDList>
<subscribeEventID>
  <!--opt, string, custom subscription ID. The platform should guarantee the uniqueness of ID when applying it. The automatic network replenishment of events will be enabled by default after the node is applied, range:[1,128], desc:when an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), if this node is applied, it can be used to replace <id> (subscription ID) returned in message SubscribeEventResponse-->test
</subscribeEventID>
<level>
  <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
</level>
<middleLevelPictureEnabeld>
  <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
</middleLevelPictureEnabeld>
<childDevIDList>
  <!--opt, array, (sub-device local identifier) device serial No. list, subType:object, desc:arm sub device (sensor)-->
  <childDevID>
    <!--opt, string, (sub-device local identifier) device serial No. list, range:[1,128], desc:arm sub device (sensor)-->test
  </childDevID>
</childDevIDList>
<SubscribeISAPIMessage>
  <!--opt, object-->
  <EventTypeList>
    <!--opt, array, event list, subType:object, desc:event list-->
    <eventType>
      <!--req, enum, event type, subType:string, desc:event type-->AIOP_Video
    </eventType>
    <uploadPicEnabled>
      <!--opt, bool, whether to upload the event picture, desc:whether to upload the event picture-->false
    </uploadPicEnabled>
  </EventTypeList>
  <EventList>
    <!--req, array, event list, subType:object, range:[0,200]-->
    <Event>
      <!--opt, object, a single event-->
      <eventType>
        <!--req, enum, event type, subType:string, desc:event type-->AIOP_Video
      </eventType>
      <uploadPicEnabled>
        <!--opt, bool, whether to upload the event picture, desc:whether to upload the event picture-->false
      </uploadPicEnabled>
    </Event>
  </EventList>
</SubscribeISAPIMessage>
<radarChannels>
  <!--opt, string, batch arming by radar channel, desc:if this field is applied, radarChannels in Event will not take effect. To arm all radar channels, this field should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
</radarChannels>
<qosLevel>
  <!--ro, opt, object, subscription events qos level, desc:subscription events qos level-->
  <defaultLevel>
    <!--ro, req, enum, default qos level, subType:int, desc:default qos level-->0
  </defaultLevel>
  <eventList>
    <!--ro, opt, array, custom event qos level list, subType:object-->
    <event>
      <!--ro, opt, object, custom event qos level-->
      <eventType>
        <!--ro, req, string, event type, range:[0,64], desc:its values refer to event types of different domains-->test
      </eventType>
      <level>
        <!--ro, req, enum, qos level of this event type, subType:int, desc:qos level of this event type-->1
      </level>
      <channels>
        <!--ro, opt, string, video resource channel No., range:[0,256], desc:no such node means it applies to all channels; otherwise it applies to the channels listed in the array, and channels not listed use default qos level-->0,1,2
      </channels>
    </event>
  </eventList>
</qosLevel>
<audioChannels>
  <!--opt, string, arm by audio channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
</audioChannels>
<unitChannels>
  <!--opt, string, batch arming by optical fiber channel, desc:if this field is applied, unitChannels in Event will not take effect. To arm all fiber channels, this node should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
</unitChannels>
<retransmissionID>
  <!--opt, string, ANR arming ID, range:[1,64], desc:UUID (an unique ID generated by the client). If this field is applied during establishing the event arming/subscription connection, the ANR of subscribed event will be enabled by default; when the network reconnected, the platform can apply the same UUID to tell the device to re-upload saved event information during network disconnection period-->123456
</retransmissionID>
<retransmissionTimeoutDuration>
  <!--opt, int, data transmission timeout for ANR, range:[1,168], unit:h, desc:data transmission timeout for ANR-->1
</retransmissionTimeoutDuration>
<subscribeWithoutPic>
  <!--opt, bool-->false
```

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```
  </subscribeWithoutPic>
</SubscribeEvent>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEventResponse xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, subscription result, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, subscription ID, which is returned by the device. The device should guarantee the uniqueness of ID, range:[1,64]-->1
  </id>
  <FailedEventList>
    <!--ro, opt, array, list of events, of which subscription failed, subType:object, desc:when subscription of one or multiple events failed, this node
should be returned. The client software can check if all events are subscribed to by checking whether the node <FailedEventList> is returned-->
    <Event>
      <!--ro, opt, object, event, of which the subscription failed. The event/alarm type name can be obtained via the capability set-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:event type-->ADAS
      </type>
      <minorAlarm>
        <!--ro, opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL).-->binary
      </pictureURLType>
      <channels>
        <!--ro, opt, string, channel No., desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <subStatusCode>
        <!--ro, req, enum, error code, subType:string, desc:refer to the error code dictionary-->activeNumMax
      </subStatusCode>
      <minorType>
        <!--ro, opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel
check-in.
3. When this node is not returned, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
      </minorType>
    </Event>
  </FailedEventList>
</SubscribeEventResponse>
```

15.1.1.5 Get the alarm/event subscription capability

Request URL

GET /ISAPI/Event/notification/subscribeEventCap

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEventCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <format opt="xml,json">
    <!--ro, opt, string, supported message data format, attr:opt{req, string}-->xml
  </format>
  <heartbeat min="1" max="180">
    <!--ro, opt, int, heartbeat interval time, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->1
  </heartbeat>
  <channelMode opt="all,list">
    <!--ro, opt, enum, channel subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all channels), "list" (subscribe to
channels according to channel list)-->list
  </channelMode>
```

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```
  <eventMode opt="all,list,allAndChild">
    <!--ro, req, enum, event subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all alarms/events), "list" (subscribe to
specified alarms/events)-->list
  </eventMode>
  <EventList>
    <!--ro, opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--ro, opt, object, subscription of a specified alarm/event-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
(video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
"TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
"alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm",
"faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection",
"fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event),
"highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
conflicts), "keyPersonGetUp"(key person getting up detection), "LeavePosition"(absence detection), "linedetection"(line crossing detection),
"listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event),
"mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
"nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
"peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
"personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
"radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system),
"takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
"unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException",
"videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent",
"IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "certificateCaptureEvent"(person ID capture comparison
event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event),
"TransactionRecordEvent", "HealthInfoSyncQuery" (health information search event), "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"
(searching the consumption status), "certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison),
"regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
      </type>
      <minorAlarm opt="0x400,0x401,0x402,0x403">
        <!--ro, opt, string, minor alarm type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"--
>0x400,0x401
      </minorAlarm>
      <minorException opt="0x400,0x401,0x402,0x403">
        <!--ro, opt, string, minor exception type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401
      </minorException>
      <minorOperation opt="0x400,0x401,0x402,0x403">
        <!--ro, opt, string, minor operation type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401
      </minorOperation>
      <minorEvent opt="0x01,0x02,0x03,0x04">
        <!--ro, opt, string, minor event type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"--
>0x400,0x401
      </minorEvent>
      <pictureURLType opt="binary,localURL,cloudStorageURL,multipart" def="cloudStorageURL">
        <!--ro, opt, enum, alarm picture format, subType:string, attr:opt{req, string},def{req, string}, desc:"binary" (binary), "localURL" (device local
URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
      <channels>
        <!--ro, opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <ConferenceIDList size="1">
        <!--ro, opt, array, video conference ID list, subType:object, attr:size{req, int}, desc:when this node does not exist, it indicates to subscribe to
all conferences-->
        <ConferenceID min="1" max="32">
          <!--ro, opt, string, ID of subscribed video conference, range:[1,32], attr:min{req, int},max{req, int}-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget opt="true,false">
        <!--ro, opt, bool, whether to upload all detected targets, attr:opt{req, string}, desc:the default value is false. In perimeter protection, this
node is used for NVR to get all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData opt="true,false">
        <!--ro, opt, bool, whether to upload the electronic fence data related to the event, attr:opt{req, string}, desc:true by default-->true
      </uploadMEFData>
      <minorType opt="success,failed">
        <!--ro, opt, string, event sub-type, attr:opt{req, string},
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel
check-in.-->success
      </minorType>
      <childDevIDList min="1" max="8">
        <!--ro, opt, array, (sub-device local identifier) device serial No. list, subType:object, attr:min{req, int},max{req, int}, desc:arm sub device
(sensor)-->
        <childDevID min="1" max="128">
          <!--ro, opt, string, (sub-device local identifier) device serial No. list, range:[1,128], attr:min{req, int},max{req, int}, desc:arm sub device
(sensor)-->test
```

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```
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--ro, opt, string, radar channel arming capability, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--ro, opt, string, audio channel arming capability, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </audioChannels>
      <childDeviceFormat opt="true,false">
        <!--ro, opt, bool, upload event by sub device, attr:opt{req, string}-->true
      </childDeviceFormat>
      <unitChannels>
        <!--ro, opt, string, optical fiber channel arming capability of device, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </unitChannels>
    </Event>
  </EventList>
  <channels>
    <!--ro, opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is all, this node should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
  </channels>
  <pictureURLType opt="binary,localURL,cloudStorageURL,multipart" def="cloudStorageURL">
    <!--ro, opt, enum, format unified configuration of alarm pictures, subType:string, attr:opt{req, string},def{req, string}, desc:"binary", "localURL" (device local URL), "cloudStorageURL" (cloud storage URL). The node indicates the upload mode of all event pictures. If the node is applied, the <pictureURLType> of the <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--ro, opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--ro, opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--ro, opt, bool, status information-->true
      <all>
        <!--ro, opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--ro, opt, bool, status of subscribed channels, desc:reporting is not required if all is true-->true
      </channel>
      <hd>
        <!--ro, opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
      </hd>
      <capability>
        <!--ro, opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online opt="true,false">
          <!--ro, opt, bool, whether the channel online status is subscribed, attr:opt{req, string}, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </online>
        <record opt="true,false">
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, attr:opt{req, string}, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </record>
        <recordStatus opt="true,false">
          <!--ro, opt, bool, whether the recording status is subscribed, attr:opt{req, string}, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </recordStatus>
        <signal opt="true,false">
          <!--ro, opt, bool, whether the signal status is subscribed, attr:opt{req, string}, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
        <arming opt="true,false">
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, attr:opt{req, string}, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey max="64">
    <!--ro, opt, string, subscription connection interaction key, range:[1,64], attr:max{req, int}, desc:N/A-->test
  </identityKey>
  <ConferenceIDList size="1">
    <!--ro, opt, array, video conference ID list, subType:object, attr:size{req, int}-->
    <ConferenceID min="1" max="32">
      <!--ro, opt, string, ID of subscribed video conference, range:[1,32], attr:min{req, int},max{req, int}-->test
    </ConferenceID>
  </ConferenceIDList>
  <isSupportModifySubscribeEvent>
    <!--ro, opt, bool, whether the device supports the management of arming subscription, desc:related API: /ISAPI/Event/notification/subscribeEvent/<ID>-->true
  </isSupportModifySubscribeEvent>
  <subscribeEventID min="1" max="128">
    <!--ro, opt, string, custom subscription ID. The platform should guarantee the uniqueness of ID when applying it. The automatic network replenishment of events will be enabled by default after the node is applied, range:[1,128], attr:min{req, int},max{req, int}, desc:when an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), if this node is applied, it can be used to replace <id> (subscription ID) returned in message SubscribeEventResponse-->test
  </subscribeEventID>
  <level opt="high,middle,low">
    <!--ro, opt, enum, arming level, subType:string, attr:opt{req, string}, desc:"high" (default value), "middle", "low"-->high
```

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```
    <!--ro, opt, enum, arming level, subType:string, attr:opt{req, string}, desc:high (default value), middle, low-->high
  </level>
  <middleLevelPictureEnabeld opt="true,false">
    <!--ro, opt, bool, whether to enable picture uploading for middle-level arming, attr:opt{req, string}, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <deployID opt="1,2">
    <!--ro, opt, enum, arming type, subType:int, attr:opt{req, string}, desc:0 (by client software), 1 (real-time)-->0
  </deployID>
  <childDevIDList min="1" max="8">
    <!--ro, opt, array, (sub-device local identifier) device serial No. list, subType:object, attr:min{req, int},max{req, int}, desc:arm sub device
(sensor)-->
    <childDevID min="1" max="128">
      <!--ro, opt, string, (sub-device local identifier) device serial No. list, range:[1,128], attr:min{req, int},max{req, int}, desc:arm sub device
(sensor)-->test
    </childDevID>
  </childDevIDList>
  <SubscribeISAPIMessage>
    <!--ro, opt, object-->
    <eventType opt="AIOP_Video,AIOP_Polling_Video,AIOP_Picture,AIOP_Polling_Snap,TPS">
      <!--ro, req, enum, subType:string, attr:opt{req, string}-->AIOP_Video
    </eventType>
    <eventPicUploadType opt="AIOP_Picture">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->AIOP_Picture
    </eventPicUploadType>
    <EventList>
      <!--ro, opt, object, event list-->
      <Event>
        <!--ro, opt, object, a single event-->
        <eventType opt="AIOP_Video,AIOP_Polling_Video,AIOP_Picture,AIOP_Polling_Snap,TPS,audioexception">
          <!--ro, req, enum, subType:string, attr:opt{req, string}-->AIOP_Video
        </eventType>
        <eventPicUploadType opt="AIOP_Picture">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->AIOP_Picture
        </eventPicUploadType>
      </Event>
    </EventList>
  </SubscribeISAPIMessage>
  <radarChannels>
    <!--ro, opt, string, if this field is applied, radarChannels in Event will not take effect. To arm all radar channels, this field should not be applied.
To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas, desc:if this field is applied, radarChannels in
Event will not take effect. To arm all radar channels, this field should not be applied. To arm a part of channels, you can list the channel numbers.
Multiple channels are separated by commas-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object, subscription events qos level, desc:subscription events qos level-->
    <defaultLevel opt="0,1,2">
      <!--ro, req, enum, default qos level, subType:int, attr:opt{req, string}, desc:default qos level-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, custom event qos level list, subType:object-->
      <event>
        <!--ro, opt, object, custom event qos level-->
        <eventType>
          <!--ro, req, string, event type, range:[0,64], desc:its values refer to event types of different domains-->test
        </eventType>
        <level opt="0,1,2">
          <!--ro, req, enum, qos level of this event type, subType:int, attr:opt{req, string}, desc:qos level of this event type-->1
        </level>
        <channels>
          <!--ro, opt, string, video resource channel No., range:[0,256], desc:no such node means it applies to all channels; otherwise it applies to the
channels listed in the array, and channels not listed use default qos level-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
  <isSupportUnSubscribeEvent>
    <!--ro, opt, bool, whether the device supports canceling arming subscription-->true
  </isSupportUnSubscribeEvent>
  <audioChannels>
    <!--ro, opt, string, audio channel arming capability, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
  </audioChannels>
  <unitChannels>
    <!--ro, opt, string, batch arming capability of optical fiber channel, desc:if this field is applied, unitChannels in Event will not take effect. To arm
all fiber channels, this node should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas
-->1,2,3,4
  </unitChannels>
  <retransmissionID min="1" max="64">
    <!--ro, opt, string, ANR arming ID, range:[1,64], attr:min{req, int},max{req, int}, desc:UUID, which is unique and generated by the platform-->123456
  </retransmissionID>
  <retransmissionTimeoutDuration min="1" max="168" def="168">
    <!--ro, opt, int, data transmission timeout for ANR, unit:h, attr:min{req, int},max{req, int},def{req, int}, desc:data transmission timeout for ANR-->0
  </retransmissionTimeoutDuration>
  <isSupportRetransmissionStatus>
    <!--ro, opt, bool-->true
  </isSupportRetransmissionStatus>
  <isSupportCancelRetransmission>
    <!--ro, opt, bool-->true
  </isSupportCancelRetransmission>
  <isSupportSubscribeWithoutPic>
    <!--ro, opt, bool-->true
  </isSupportSubscribeWithoutPic>
</SubscribeEventCap>
```

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15.1.1.6 Cancel subscribing alarm/event

Request URL

PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	Subscription event ID, which should match the id field in the SubscribeEventResponse of the response message from the POST /ISAPI/Event/notification/subscribeEvent?deployID=.This field is optional. When the protocol is /ISAPI/Event/notification/unSubscribeEvent, it indicates that all subscription links are to be unsubscribed. When the protocol is /ISAPI/Event/notification/unSubscribeEvent?ID=XXX, it indicates that a specific subscription link is to be unsubscribed.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

15.1.2 异常检测

15.1.2.1 HDD error

EventType:diskerror

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"diskerror" (HDD Error)-->diskerror
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:HDD Error-->diskerror
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <diskNo>
    <!--ro, opt, int, HDD number-->1
  </diskNo>
  <HDDList>
    <!--ro, opt, array, HDD list, subType:object-->
    <HDD>
      <!--ro, opt, object, HDD-->
      <id>
        <!--ro, req, int, index, desc:it starts from 1-->1
      </id>
      <diskNumber>
        <!--ro, req, int, HDD No.-->1
      </diskNumber>
    </HDD>
  </HDDList>
</EventNotificationAlert>
```

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15.1.2.2 Invalid access

EventType:illaccess

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event parameters, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the alarm device-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the alarm device-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the alarm device-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, transfer protocol type, range:[0,64], desc:transfer protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZVIZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, No, of the channel that triggers an alarm, desc:No. of the video channel that triggers an alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, the uploading times of the same alarm, desc:event frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"illaccess" (invalid access)-->illaccess
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for persistent events: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:illaccess-invalid access-->illaccess
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <username>
    <!--ro, opt, string, locked user name, range:[0,32], desc:the number of attempts by current user name has reached limit. The user name and IPv4 address may be locked at the same time-->test
  </username>
  <ipV4Address>
    <!--ro, opt, string, locked IPv4 address, desc:the number of attempts by current IPv4 address has reached limit. The user name and Ipv4 address may be locked at the same time-->test
  </ipV4Address>
  <lockStatus>
    <!--ro, opt, enum, locking status, subType:string, desc:; “unlocked”, “locked”-->unlocked
  </lockStatus>
  <retryLoginTime>
    <!--ro, opt, int, remaining attempts, range:[0,65535], desc:it is valid when the value of lockStatus is “unlocked”-->0
  </retryLoginTime>
  <lockedTime>
    <!--ro, opt, int, locking duration (minutes), range:[0,65535], unit:秒, desc:it is valid when the value of lockStatus is “locked”-->0
  </lockedTime>
  <accessInfoList>
    <!--ro, opt, array, invalid access information list, subType:object-->
    <accessInfo>
      <!--ro, opt, object, invalid access information-->
      <ipV4Address>
        <!--ro, opt, string, Invalid access IP address-->test
      </ipV4Address>
      <dateTime>
        <!--ro, opt, datetime, invalid access time-->2004-05-03T17:30:08+08:00
      </dateTime>
    </accessInfo>
  </accessInfoList>
</EventNotificationAlert>
```

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15.1.2.3 Network disconnected

EventType:nicbroken

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers an alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers an alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the device that triggers an alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, communication protocol types, range:[0,64], desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome" (the value
should be "HTTP" when ISAPI protocol is transmitted via EZ protocol, and the value should be "EHome" when ISAPI protocol is transmitted via ISUP)-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers an alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, the uploading times of the same alarm, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, string, event type, desc:"nicbroken" (network disconnected)-->nicbroken
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:"nicbroken" (network disconnected)-->nicbroken
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DevicePositionInfo>
    <!--ro, opt, object, device location, desc:the current location of the device when triggering the event, which is mainly for on-board device-->
    <divisionEW>
      <!--ro, opt, enum, eastern/western hemisphere, subType:string, desc:"E" (eastern hemisphere), "W" (western hemisphere)-->E
    </divisionEW>
    <longitude>
      <!--ro, opt, int, longitude, desc:degree*3600*100 + minute*60*100 + second*100-->100
    </longitude>
    <divisionNS>
      <!--ro, opt, enum, southern/northern hemisphere, subType:string, desc:"S" (southern hemisphere), "N" (northern hemisphere)-->N
    </divisionNS>
    <latitude>
      <!--ro, opt, int, latitude, desc:degree*3600*100 + minute*60*100 + second*100-->100
    </latitude>
    <direction>
      <!--ro, opt, int, vehicle direction, desc:actual direction (degree) (due north as 0 degree, calculated clockwise)-->100
    </direction>
    <speed>
      <!--ro, opt, int, speed, desc:cm/h-->100
    </speed>
    <height>
      <!--ro, opt, int, height, unit:cm, desc:cm-->1
    </height>
  </DevicePositionInfo>
</EventNotificationAlert>
```

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15.1.3 非视频通道管理(S)

15.1.3.1 Get the event search capability

Request URL

GET /ISAPI/SDT/Management/EventSearch/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message


```
{
    "startTime":  "1970-01-01T00:00:00+08:00",
    /*ro, req, datetime, start time, desc:start time,ISO8601_time*/
    "endTime":  "1970-01-01T00:00:00+08:00",
    /*ro, req, datetime, end time, desc:end time,ISO8601_time*/
    "resultMaxNum":  100,
    /*ro, req, int, supported maximum number of searching*/
    "channelIDLen": {
    /*ro, opt, object, channel ID length*/
        "min":  0,
        /*ro, req, int, minimum value, step:1*/
        "max":  32
        /*ro, req, int, maximum value, step:1*/
    },
    "eventType": {
    /*ro, opt, object, event type, desc:studentStoodUp-student stands up, accessController-access controller, videoIntercom-video intercom, Luminite-
Luminite security control panel, GJD-GJD security control panel, cameraDetector-detector, securityControlPanel-security control panel. Multiple selections
are separated by comma*/
        "@opt":  "studentStoodUp, accessController,videoIntercom, OPTEX,Luminite,GJD, cameraDetector, securityControlPanel",
        /*ro, req, string, range*/
        "#text":  "studentStoodUp"
        /*ro, opt, string*/
    },
    "type": {
    /*ro, opt, object, channel type, desc:video-video channel, IOT-IOT channel; if this node does not exist, it indicates video channel*/
        "@opt":  "video,IOT",
        /*ro, req, string*/
        "#text":  "video"
        /*ro, opt, string*/
    },
    "AccessController": {
    /*ro, opt, object, access control event*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        },
        "nameLen": {
        /*ro, opt, object, name length*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  32
            /*ro, req, int, maximum value, step:1*/
        },
        "cardNoLen": {
        /*ro, opt, object, card No. length*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  32
            /*ro, req, int, maximum value, step:1*/
        }
    },
    "VideoIntercom": {
    /*ro, opt, object, Video Intercom Event*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        },
        "nameLen": {
        /*ro, opt, object, name length*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  32
            /*ro, req, int, maximum value, step:1*/
        },
        "cardNoLen": {
        /*ro, opt, object, card No. length*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  32
            /*ro, req, int, maximum value, step:1*/
        }
    },
    "OPTEX": {
    /*ro, opt, object*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
```


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```
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        }
    },
    "Luminite": {
    /*ro, opt, object*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        }
    },
    "GJD": {
    /*ro, opt, object*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        }
    },
    "CameraDetector": {
    /*ro, opt, object*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        }
    },
    "SecurityControlPanel": {
    /*ro, opt, object, security control panel event*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test"
            /*ro, req, string, range*/
        },
        "eventMainType": {
        /*ro, opt, object*/
            "@opt": ["alarm", "fault", "operation", "others"]
            /*ro, req, array, subType:string*/
        },
        "zoneNo": {
        /*ro, req, object, zone No.*/
            "@size":  1,
            /*ro, req, int, step:1*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  16,
            /*ro, req, int, maximum value, step:1*/
            "#text":  1
            /*ro, opt, int*/
        },
        "ipAddress": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, minimum value, range:[0,128], step:1*/
            "@max":  128
            /*ro, req, int, maximum value, range:[0,128], step:1*/
        },
        "devIndex": {
        /*ro, opt, object, maximum value*/
            "@min":  0,
            /*ro, req, int, minimum value, range:[0,256], step:1*/
            "@max":  256
            /*ro, req, int, maximum value, range:[0,256], step:1*/
        },
        "code": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, minimum value, range:[0,5000], step:1*/
            "@max":  5000
            /*ro, req, int, maximum value, range:[0,5000], step:1*/
        },
        "devName": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int, range:[1,48], step:1*/
            "@max":  48
```


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```
            /*ro, req, int, range:[1,48], step:1*/
        }
    },
    "isSupportEventExport":  true
    /*ro, opt, bool*/
}
```

15.1.3.2 Search for event information

Request URL

POST /ISAPI/SDT/Management/EventSearch?format=json

Query Parameter

None

Request Message

```
{
    "searchID":  "test",
    /*req, string, search ID, desc:this node is used to check whether the search is the same one. Due to device performance limitation, you can search
asynchronously by applying the search conditions multiple times and getting the searching progress (the searchID should be the same)*/
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults":  100,
    /*req, int, the maximum number of search results this time*/
    "startTime":  "1970-01-01T08:00:00+08:00",
    /*req, datetime, start time, desc:start time,ISO8601_time*/
    "endTime":  "1970-01-01T09:00:00+08:00",
    /*req, datetime, end time, desc:end time,ISO8601_time*/
    "choiceChannel": [
    /*opt, array, camera information, subType:object*/
        {
            "channelID":  "26c92893d2a94bc1840451b78444e0f9"
            /*opt, string, camera No.*/
        }
    ],
    "eventType":  "accessController",
    /*req, enum, event type, subType:string, desc:"studentStoodUp" (student stands up), "accessController" (access controller), "videoIntercom" (video
intercom), "OPTEX" (OPTEX security control panel), "Luminite" (Luminite security control panel), "GJD" (GJD security control panel), "cameraDetector"
(detector), "securityControlPanel" (security control panel), multiple selections are separated with commas*/
    "type":  "video",
    /*opt, enum, channel type, subType:string, desc:"video" (video channel), "IOT" (IOT channel), if this node does not exist, it indicates video channel*/
    "AccessController": {
    /*opt, object*/
        "eventType":  "test",
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
        "name":  "test",
        /*opt, string, name*/
        "cardNo":  "test"
        /*opt, string, card No.*/
    },
    "VideoIntercom": {
    /*opt, object*/
        "eventType":  "test",
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
        "name":  "test",
        /*opt, string, name*/
        "cardNo":  "test"
        /*opt, string, card No.*/
    },
    "OPTEX": {
    /*opt, object, OPTEX security control panel*/
        "eventType":  "test"
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
    },
    "Luminite": {
    /*opt, object, Luminite security control panel*/
        "eventType":  "test"
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
    },
    "GJD": {
    /*opt, object, GJD security control panel*/
        "eventType":  "test"
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
    },
    "CameraDetector": {
```

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```
    /*opt, object*/
        "eventType":  "test"
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
    },
    "SecurityControlPanel": {
    /*opt, object, security control panel*/
        "eventType":  "test",
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
        "eventMainType":  "alarm",
        /*opt, enum, subType:string*/
        "zoneNo": [1, 2],
        /*opt, array, zone No., subType:int*/
        "ipAddress":  "10.1.200.1",
        /*opt, string, device's IPv4 address, range:[0,128]*/
        "devIndex":  "test",
        /*opt, string, device No., range:[0,256]*/
        "code":  0,
        /*opt, int, range:[0,5000]*/
        "devName":  "test"
        /*opt, string, range:[1,48]*/
    }
}
```

Response Message

```
{
    "responseStatusStrg":  "MORE",
    /*ro, req, enum, searching status, subType:string, desc:"OK" (searching ended), "MORE" (search again for more results), "NO MATCH" (no matched data
found)*/
    "numOfMatches":  1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches":  1,
    /*ro, opt, int, total number of matched results*/
    "eventInfo": [
    /*ro, opt, array, event information, subType:object*/
        {
            "captureTime":  "1970-01-01T08:00:00+08:00",
            /*ro, req, datetime, capture time*/
            "channelID":  "test",
            /*ro, opt, string, camera No.*/
            "channelName":  "test",
            /*ro, opt, string, camera name*/
            "picUrl":  "test",
            /*ro, opt, string, picture URL*/
            "subPicUrl":  "test",
            /*ro, opt, string, thumbnail URL*/
            "eventType":  "accessController",
            /*ro, req, enum, event type, subType:string, desc:"studentStoodUp" (student stands up), "accessController" (access controller), "videoIntercom"
(video intercom), "OPTEX" (OPTEX security control panel), "Luminite" (Luminite security control panel), "GJD" (GJD security control panel), "cameraDetector"
(detector), "securityControlPanel" (security control panel), multiple selections are separated with commas*/
            "studentStoodUp": {
            /*ro, opt, object, student stand up event information*/
                "studentsStoodUp_number":  4
                /*ro, opt, int, the number of students who are standing up*/
            },
            "AccessController": {
            /*ro, req, object, event type*/
                "eventType":  "test",
                /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
                "name":  "test",
                /*ro, opt, string, name*/
                "cardNo":  "test",
                /*ro, opt, string, card No.*/
                "cardType":  "invalid",
                /*ro, opt, enum, card type, subType:string, desc:"invalid" (invalid card), "ordinary" (normal card), "disabled" (card for the disabled),
"blocklist" (card in blocklist), "patrol" (patrol card), "super" (super card), "guest" (visitor card), "remove" (removed card)*/
                "videoChannelInfo": [
                /*ro, opt, array, video channel information, subType:object*/
                    {
                        "videoChannel":  1,
                        /*ro, req, int, video channel No.*/
                        "picUrl":  "test"
                        /*ro, opt, string, picture URL*/
                    }
                ]
            },
            "VideoIntercom": {
            /*ro, opt, object, video intercom*/
                "eventType":  "test",
                /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
                "name":  "test",
                /*ro, opt, string, name*/
                "cardNo":  "test",
                /*ro, opt, string, card No.*/
                "cardType":  "invalid",
                /*ro, opt, enum, card type, subType:string, desc:"invalid" (invalid card), "ordinary" (normal card), "disabled" (card for the disabled),
"blocklist" (card in blocklist), "patrol" (patrol card), "super" (super card), "guest" (visitor card), "remove" (removed card)*/
```

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```
            "videoChannelInfo": [
            /*ro, opt, array, video channel information, subType:object*/
                {
                    "videoChannel":  1,
                    /*ro, req, int, video channel No.*/
                    "picUrl":  "test"
                    /*ro, opt, string, picture URL*/
                }
            ]
        },
        "OPTEX": {
        /*ro, opt, object, OPTEX security control panel*/
            "eventType":  "test",
            /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
            "videoChannelInfo": [
            /*ro, opt, array, video channel information, subType:object*/
                {
                    "videoChannel":  1,
                    /*ro, req, int, video channel No.*/
                    "picUrl":  "test"
                    /*ro, opt, string, picture URL*/
                }
            ]
        },
        "Luminite": {
        /*ro, opt, object, Luminite security control panel*/
            "eventType":  "test",
            /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
            "videoChannelInfo": [
            /*ro, opt, array, video channel information, subType:object*/
                {
                    "videoChannel":  1,
                    /*ro, req, int, video channel No.*/
                    "picUrl":  "test"
                    /*ro, opt, string, picture URL*/
                }
            ]
        },
        "GJD": {
        /*ro, opt, object, GJD security control panel*/
            "eventType":  "test",
            /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
            "videoChannelInfo": [
            /*ro, opt, array, video channel information, subType:object*/
                {
                    "videoChannel":  1,
                    /*ro, req, int, video channel No.*/
                    "picUrl":  "test"
                    /*ro, opt, string, picture URL*/
                }
            ]
        },
        "CameraDetector": {
        /*ro, opt, object*/
            "eventType":  "test",
            /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
            "videoChannelInfo": [
            /*ro, opt, array, video channel information, subType:object*/
                {
                    "videoChannel":  1,
                    /*ro, req, int, video channel No.*/
                    "picUrl":  "test"
                    /*ro, opt, string, picture URL*/
                }
            ]
        },
        "SecurityControlPanel": {
        /*ro, opt, object*/
            "eventType":  "test",
            /*ro, opt, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
            "eventMainType":  "alarm",
            /*ro, opt, enum, subType:string*/
            "videoChannelInfo": [
            /*ro, opt, array, video channel information, subType:object*/
                {
                    "videoChannel":  1,
                    /*ro, req, int, video channel No.*/
                    "picUrl":  "test"
                    /*ro, opt, string, picture URL*/
                }
            ],
            "zoneNo":  1,
            /*ro, opt, int, zone No., desc:zone No.,it is valid when eventType values "zone"*/
            "code":  1103,
            /*ro, opt, int, CID event No., desc:CID event No.,it is valid when eventType values "zone"*/
            "zoneName":  "test",
            /*ro, opt, string, zone name*/
            "ipAddress":  "test",
            /*ro, opt, string, device's IPv4 address, range:[0,128]*/
            "devIndex":  "test",
            /*ro, opt, string, device No., range:[0,256]*/
            "subSysNo":  1,
            /*ro, opt, int*/
```

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```
                "recordID":  "test",
                /*ro, opt, string*/
                "deviceName":  "test"
                /*ro, opt, string, range:[1,48]*/
            }
        }
    ]
}
```

15.1.3.3 Get access management capability of IoT device channel

Request URL

GET /ISAPI/System/IOT/sourceSupport/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "IOTSourceSupport": {
    /*ro, req, object, supported source information*/
        "sourceNum":  10,
        /*ro, req, int, the number of protocols*/
        "IOTSourceDescriptions": [
        /*ro, opt, array, protocol information, subType:object*/
            {
                "addressingFormatType": {
                /*ro, opt, object, address type*/
                    "@opt":  "ipaddress,hostname",
                    /*ro, req, string, range*/
                    "#text":  "ipaddress"
                    /*ro, opt, string, sample value*/
                },
                "protocol": {
                /*ro, opt, object, protocol type*/
                    "@opt":  "tcp,udp,muticast",
                    /*ro, req, string, range*/
                    "#text":  "tcp"
                    /*ro, opt, string, sample value*/
                },
                "portNo": {
                /*ro, opt, object, device port No.*/
                    "@min":  0,
                    /*ro, req, int, minimum value*/
                    "@max":  65535,
                    /*ro, req, int, maximum value*/
                    "#text":  80
                    /*ro, opt, int, sample value*/
                },
                "userName": {
                /*ro, opt, object, user name*/
                    "@min":  0,
                    /*ro, req, int, minimum value*/
                    "@max":  32,
                    /*ro, req, int, maximum value*/
                    "#text":  "admin"
                    /*ro, opt, string, sample value*/
                },
                "password": {
                /*ro, opt, object, password*/
                    "@min":  0,
                    /*ro, req, int, minimum value*/
                    "@max":  16,
                    /*ro, req, int, maximum value*/
                    "#text":  "12345"
                    /*ro, opt, string, sample value*/
                },
                "enableTiming": {
                /*ro, opt, object, whether time sync on the device is supported*/
                    "@opt": [true, false]
                    /*ro, opt, array, optional value, subType:bool*/
                }
            }
        ],
        "DeviceInductiveType": {
        /*ro, opt, object, device inductive type, desc:Multiple types of devices might need to be put into one device group for management, so inductive
type is introduced to sort out the devices.*/
            "inductiveType1": ["accessController", "videoIntercom"],
            /*ro, opt, array, inductive type 1, subType:string, desc:"accessController" (access controller), "videoIntercom" (video intercom)*/
            "inductiveType2": ["OPTEX", "Luminite", "GJD", "securityControlPanel", "Adam", "Arteco", "FM2466"]
            /*ro, opt, array, inductive type 2, subType:string, desc:"OPTEX"(OPTEX security control panel protocol), "Luminite" (Luminite security control
panel), "GJD" (GJD security control panel), "securityControlPanel" (security control panel), "Adam" (Adam module protocol), "Arteco" (Arteco module
protocol), "FM2466" (FM2466 protocol)*/
        }
    }
}
```

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15.1.4 在线升级管理

15.1.4.1 Get the device online upgrade capability

Request URL

GET /ISAPI/System/onlineUpgrade/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, online upgrade, attr:version{req, string, protocolVersion}-->
  <firmwareNum max="10">
    <!--ro, req, int, number of supported online upgrade packages, attr:max{req, int}-->1
  </firmwareNum>
  <firmwareCode max="256">
    <!--ro, req, int, maximum length of firmware ID, attr:max{req, int}-->1
  </firmwareCode>
  <firmwareVersion max="64">
    <!--ro, req, int, maximum length of version, attr:max{req, int}-->1
  </firmwareVersion>
  <firmwareCodeNumOnce max="10">
    <!--ro, req, int, maximum number of obtained firmware IDs, attr:max{req, int}-->1
  </firmwareCodeNumOnce>
  <upgradePercent min="0" max="100">
    <!--ro, req, int, upgrade progress capability in percentage, attr:min{req, int},max{req, int}-->1
  </upgradePercent>
  <Version>
    <!--ro, opt, object, upgrade package version information-->
    <newVersion max="64">
      <!--ro, req, string, version information of the new upgrade package, attr:max{req, int}-->test
    </newVersion>
    <changeLog max="64">
      <!--ro, req, string, log of the new upgrade package, attr:max{req, int}-->test
    </changeLog>
  </Version>
  <rebootAfterUpgrade>
    <!--ro, opt, enum, upgrade options after device reboot, subType:string, desc:"auto" (auto upgrade), "manual" (manual upgrade)-->auto
  </rebootAfterUpgrade>
  <DeviceParameter>
    <!--ro, opt, object, online upgrade parameters-->
    <isSupportAutoDownloadPackage>
      <!--ro, opt, bool, whether it supports automatic download of upgrade package-->true
    </isSupportAutoDownloadPackage>
    <notSupportAutoUpgrade>
      <!--ro, opt, bool, whether it no longer supports automatic download of upgrade package and automatic upgrade, desc:corresponding URL: PUT
/ISAPI/System/onlineUpgrade/upgrade-->true
    </notSupportAutoUpgrade>
    <isSupportTimingUpgrade>
      <!--ro, opt, bool, whether it supports scheduled upgrade-->true
    </isSupportTimingUpgrade>
  </DeviceParameter>
  <ManualDownloadPackage>
    <!--ro, opt, object, manual download of upgrade package-->
    <supportOperation opt="start,cancel,pause,resume">
      <!--ro, opt, enum, supported operations, subType:string, attr:opt{req, string}, desc:"start", "cancel", "pause", "resume"-->start
    </supportOperation>
  </ManualDownloadPackage>
  <isSupportIgnoreCurrentVersion>
    <!--ro, opt, bool, whether it supports ignoring the current version-->true
  </isSupportIgnoreCurrentVersion>
  <UpgradePackageTask>
    <!--ro, opt, object, whether the device supports applying upgrade tasks, desc:if this node is returned, it indicates that the following URLs are also
supported: /ISAPI/System/onlineUpgrade/task?format=json, /ISAPI/System/onlineUpgrade/status/<ID>-->
    <isSupportUpgradeModules>
      <!--ro, opt, bool, whether the device supports upgrading sub modules-->true
    </isSupportUpgradeModules>
    <isSupportUpgradeChannels>
      <!--ro, opt, bool, whether the device supports upgrading sub channels-->true
    </isSupportUpgradeChannels>
    <isSupportDownloadStrategy>
      <!--ro, opt, bool, whether the device supports strategy parameters of downloading upgrade package, desc:corresponding filed is "downloadStrategy" of
URL (/ISAPI/System/onlineUpgrade/task?format=json)-->true
    </isSupportDownloadStrategy>
    <isSupportDelayedUpgradeByDateTime>
      <!--ro, opt, bool, whether the device supports delaying the upgrade by time, desc:corresponding filed is "delayedUpgradeDateTime" of URL
(/ISAPI/System/onlineUpgrade/task?format=json)-->true
    </isSupportDelayedUpgradeByDateTime>
    <upgradePackageAddressType opt="URL, ftpIPAddress, ftpDomainAddress">
      <!--ro, opt, string, attr:opt{req, string}-->URL
    </upgradePackageAddressType>
  </UpgradePackageTask>
  <isSupportPatrolInspection>
    <!--ro, opt, bool, whether the device supports patrol inspection-->true
  </isSupportPatrolInspection>
  <isSupportBatchUpgrade>
    <!--ro, opt, bool, whether the device supports batch online upgrades-->true
  </isSupportBatchUpgrade>
  <isSupportCancelUpgrade>
    <!--ro, opt, bool, whether the device supports canceling online upgrade-->true
  </isSupportCancelUpgrade>
  <isSupportPackagePause>
    <!--ro, opt, bool, whether the device supports pausing upgrade package download, desc:/ISAPI/System/onlineUpgrade/PackagePause?format=json-->true
  </isSupportPackagePause>
  <isSupportPackageResume>
    <!--ro, opt, bool, whether the device supports resuming upgrade package download, desc:/ISAPI/System/onlineUpgrade/PackageResume?format=json-->true
  </isSupportPackageResume>
  <isSupportCancelUpgradeByTaskID>
    <!--ro, opt, bool, whether the device supports canceling online upgrade by task ID, desc:corresponding to the field assignByTaskID of operationType in
URL PUT  /ISAPI/System/onlineUpgrade/CancelUpgrade?format=json-->true
  </isSupportCancelUpgradeByTaskID>
```

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```
</OnlineUpgradeCap>
```

15.1.4.2 Set the online device upgrade parameters

Request URL

PUT /ISAPI/System/onlineUpgrade/deviceParameter?format=json

Query Parameter

None

Request Message

```
{
    "OnlineUpgradeParameter": {
    /*req, object, online upgrade parameters*/
        "autoDownloadPackage":  true,
        /*req, bool, whether to enable automatic download of upgrade package*/
        "timingUpgrade":  true,
        /*opt, bool, whether to enable scheduled upgrade*/
        "upgradeTime":  "00:00:00"
        /*opt, time, upgrade time, dep:and,{$.OnlineUpgradeParameter.timingUpgrade,eq,true}*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode when statusCode is not 1*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

15.1.4.3 Get the device online upgrading parameters

Request URL

GET /ISAPI/System/onlineUpgrade/deviceParameter?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "OnlineUpgradeParameter": {
    /*ro, req, object, online upgrade parameters*/
        "autoDownloadPackage":  true,
        /*ro, req, bool, whether to enable automatic downloading of upgrade package*/
        "timingUpgrade":  true,
        /*ro, opt, bool, whether to enable scheduled upgrade*/
        "upgradeTime":  "00:00:00"
        /*ro, opt, time, upgrade time, dep:and,{$.OnlineUpgradeParameter.timingUpgrade,eq,trur}*/
    }
}
```

15.1.4.4 Get the online upgrade package downloading progress

Request URL

GET /ISAPI/System/onlineUpgrade/downloadPackage/status?format=json

Query Parameter

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None

Request Message

None

Response Message

```
{
    "DownloadPackageStatus": {
    /*ro, req, object, upgrade package downloading status*/
        "status":  "notDownload",
        /*ro, req, enum, download status, subType:string, desc:"notDownload", "downloading", "downloadFailed", "pause", "finish", "incorrectPackage"
(incorrect upgrade package format), "hdOperationFailed" (HD operation failed)*/
        "total":  0.0,
        /*ro, opt, float, upgrade package total size, desc:unit: MB, corrects to one decimal place*/
        "remain":  0.0,
        /*ro, opt, float, remaining space, desc:unit: MB, corrects to one decimal place*/
        "speed":  0.0,
        /*ro, opt, float, download speed, desc:unit: MB, corrects to one decimal place*/
        "remainTime":  0,
        /*ro, opt, int, estimated remaining time, unit:s*/
        "progress":  0
        /*ro, req, int, progress, range:[0,100]*/
    }
}
```

15.1.4.5 Start to download upgrade package to device

Request URL

PUT /ISAPI/System/onlineUpgrade/downloadPackage?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

15.1.4.6 Get the online upgrade progress of device

Request URL

GET /ISAPI/System/onlineUpgrade/status

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, online upgrade status, attr:version{req, string, protocolVersion}-->
  <status>
    <!--ro, req, enum, online upgrade status, subType:string, desc:"notUpgrade" (not upgraded), "upgrading", "successful" (upgraded), "LanguageMismatch" (language mismatch), "writeFlashError" (failed to write to flash), "packageTypeMismatch" (upgrade package type mismatch), "packageVersionMismatch" (upgrade version mismatch), "netUnreachable" (network disconnected), "unknownError" (unknown error), "engineVersionMismatch" (mismatch of upgrade package engine version)-->notUpgrade
  </status>
  <percent>
    <!--ro, req, int, current upgrade progress in percentage, range:[0,100]-->1
  </percent>
  <onlineUpdateDevInfoList>
    <!--ro, opt, object, online upgrade device information, desc:containing the online upgrade status of sub devices, sub modules, or sub channels-->
    <onlineUpdateDevInfo>
      <!--ro, opt, object, online upgrade device information-->
      <moduleType>
        <!--ro, req, enum, module type, subType:string, desc:"keypad", "detector", "cardReader" (card reader), "siren" (sounder), "relay", "transmitter" (transmitter), "remoteCtrl" (remote control),"repeater" (repeater), "SSD", "extensionModule" (extension module)-->keypad
      </moduleType>
      <moduleID>
        <!--ro, req, int, module ID, range:[1,128], step:1-->1
      </moduleID>
      <model>
        <!--ro, opt, string, model, range:[1,32]-->test
      </model>
      <status>
        <!--ro, opt, enum, upgrade status, subType:string, desc:"allowance" (upgradeable, the upgrade command is not triggered), "ready" (waiting for upgrade, the upgrade command has been triggered), "upgriding" (upgrading), "success" (upgraded), "failed" (upgrade failed)-->allowance
      </status>
      <percent>
        <!--ro, opt, int, current upgrade percentage, range:[0,100]-->1
      </percent>
      <updateTime>
        <!--ro, opt, datetime, upgrade completion time-->2004-05-03T17:30:08+08:00
      </updateTime>
    </onlineUpdateDevInfo>
  </onlineUpdateDevInfoList>
</OnlineUpgradeStatus>
```

15.1.4.7 Get the new version information

Request URL

GET /ISAPI/System/onlineUpgrade/version?check=<check>

Query Parameter

Parameter Name	Parameter Type	Description
check	string	When check is false (default), the device does not need to interact with the server and only needs to return the results of periodic checks. When check is true, the device needs to interact with the server and return the latest results.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeVersion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, new version information, attr:version{req, string, protocolVersion}-->
  <newVersionAvailable>
    <!--ro, req, bool, whether the new version already is available-->true
  </newVersionAvailable>
  <newVersion>
    <!--ro, opt, string, version information of the new upgrade package-->test
  </newVersion>
  <changeLog>
    <!--ro, opt, string, updating logs of the new upgrade package version-->test
  </changeLog>
  <changeLevel>
    <!--ro, opt, enum, subType:int-->2
  </changeLevel>
</OnlineUpgradeVersion>
```

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15.1.4.8 Upload online upgrade status event

EventType:onlineUpgradeStatus

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:the event triggering frequency. The pulse event is an event which is triggered continuously according to the detection frequency of the device, e.g., the motion detection. The instant event is the event triggered when object recognition occurs, and the recognition of each object will trigger the event once, e.g., the face capture. When the pulse events are triggered, this node is used to tell whether the events are triggered by the same source. For example, if it is motion detection, the alarms will be uploaded continuously according to the device detection frequency; if it changes to another source, then the counting will start again. It can be regarded as the event triggering frequency*/
    "eventType":  "onlineUpgradeStatus",
    /*ro, req, string, event type*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid event or event starts), "inactive" (invalid event or the event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "Online Upgrade Status",
    /*ro, req, string, event description, desc:online upgrade status event*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID, desc:PUID. This field must be returned when ISAPI event information is transmitted via ISUP. It is the same as deviceID of URI (/ISAPI/System/Network/Ehome)*/
    "serialNumber":  "test",
    /*ro, opt, string, device short serial No., range:[1,9]*/
    "status":  "failure",
    /*ro, req, enum, upgrade status, subType:string, desc:"failure" (upgrade failed), "success" (upgrade succeeded), "upgrading"*/
    "failedReason":  "languageMismatch",
    /*ro, opt, enum, cause of failure, subType:string, desc:"languageMismatch" (language mismatch), "writeFlashError" (failed to write to flash), "packageTypeMismatch" (upgrade package type mismatch), "packageVersionMismatch" (upgrade version mismatch), "netUnreachable" (network disconnected), "unknownError" (unknown error)*/
    "moduleType":  "cardReader",
    /*ro, opt, enum, device component type, subType:string, desc:"cardReader" (card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm programma of card reader), "dockStationHost" (dock station host), "dockStationScreen" (dock station screen)*/
    "moduleID":  1,
    /*ro, opt, int, device component No., desc:it depends on the node moduleType*/
    "channel":  1,
    /*ro, opt, int, channel No., desc:it is used for IP channel (to upgrade digital channel managed by NVR)*/
    "taskID":  "test",
    /*ro, opt, string, task ID, range:[1,64], desc:generated when creating upgrade tasks, related URI: /ISAPI/System/onlineUpgrade/task?format=json*/
    "process":  1
    /*ro, opt, int, upgrade progress, range:[1,100], step:1, unit:%, dep:or,{$.status,eq,upgrading}, desc:progress percentage*/
}
```

15.1.5 串口配置

15.1.5.1 Get serial port authentication service capability

Request URL

GET /ISAPI/System/Serial/authentication/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "AuthenticationCap": {
    /*ro, opt, object, the serial port authentication service*/
        "enabled": {
        /*ro, req, object, whether to enable serial port authentication service*/
            "@opt": [true, false]
            /*ro, opt, array, whether to enable serial port authentication service, subType:bool*/
        },
        "serviceClosurePeriod": {
        /*ro, opt, object, service life, desc:unit: day it is required when enabled is "false"*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  30,
            /*ro, opt, int, the maximum value*/
            "@def":  7
            /*ro, opt, int, the default value*/
        }
    }
}
```

15.1.5.2 Set parameters of serial authentication service

Request URL

PUT /ISAPI/System/Serial/authentication?format=json

Query Parameter

None

Request Message

```
{
    "Authentication": {
    /*opt, object*/
        "enabled":  false,
        /*req, bool, whether to enable serial authentication service*/
        "serviceClosurePeriod":  7
        /*opt, int, service life, unit:d, desc:it is required when enabled is "false"*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

15.1.5.3 Get parameters of serial authentication service

Request URL

GET /ISAPI/System/Serial/authentication?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "Authentication": {
    /*ro, opt, object*/
        "enabled":  false,
        /*ro, req, bool, whether to enable serial authentication service*/
        "serviceClosurePeriod":  7
        /*ro, opt, int, service life, unit:d, desc:it is required when enabled is "false"”*/
    }
}
```

15.1.5.4 Get the serial port capability of the device

Request URL

GET /ISAPI/System/Serial/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SerialCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, range of RS-485 serial port numbers supported by the device, attr:version{req, string, protocolVersion}-->
  <rs485PortNums>
    <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->0
  </rs485PortNums>
  <supportRS232Config>
    <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
  </supportRS232Config>
  <rs422PortNums>
    <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->0
  </rs422PortNums>
  <rs232PortNums>
    <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->0
  </rs232PortNums>
  <rs485WorkMode opt="Led, CaptureTrigger,transparent">
    <!--ro, opt, string, range of RS-485 serial port No. supported by the device, attr:opt{req, string}-->test
  </rs485WorkMode>
  <rs232SerialNumber opt="1,2,3">
    <!--ro, opt, int, range of RS-232 serial port numbers supported by the device, attr:opt{req, string}-->1
  </rs232SerialNumber>
  <rs485SerialNumber opt="4,5,6">
    <!--ro, opt, int, range of RS-485 serial port numbers supported by the device, attr:opt{req, string}-->1
  </rs485SerialNumber>
  <isSupportAuthenticationService>
    <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
  </isSupportAuthenticationService>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports configuring the information about the serial port device-->true
  </isSupportDeviceInfo>
  <isSupportSearchDeviceInfoRelations>
    <!--ro, opt, bool, whether the device supports searching the linkage information of the serial port device-->true
  </isSupportSearchDeviceInfoRelations>
</SerialCap>
```

15.1.5.5 Get the capability of a specific serial port

Request URL

GET /ISAPI/System/Serial/ports/<portID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

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```
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port No., attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, ID-->0
  </id>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <encryptKey min="0" max="10">
    <!--ro, opt, string, key, attr:min{req, int},max{req, int}, desc:key-->test
  </encryptKey>
  <serialNumber opt="1,2,3,4,5">
    <!--ro, opt, int, serial No., attr:opt{req, string}-->0
  </serialNumber>
  <serialPortType opt="RS485,RS422,RS232">
    <!--ro, opt, string, serial port type, attr:opt{req, string}-->RS485
  </serialPortType>
  <serialAddress min="0" max="10">
    <!--ro, opt, int, serial port address, attr:min{req, int},max{req, int}, desc:serial port address-->1
  </serialAddress>
  <duplexMode opt="half,full">
    <!--ro, opt, string, duplex mode of the serial port, attr:opt{req, string}-->half
  </duplexMode>
  <direction opt="monodirectional,bdirectional">
    <!--ro, opt, string, attr:opt{req, string}-->monodirectional
  </direction>
  <baudRate opt="1200,2400,4800,9600,19200,38400,57600,115200">
    <!--ro, opt, int, attr:opt{req, string}-->1200
  </baudRate>
  <dataBits opt="6,7,8">
    <!--ro, opt, int, attr:opt{req, string}-->6
  </dataBits>
  <parityType opt="none,even,odd,mark,space">
    <!--ro, opt, string, attr:opt{req, string}-->none
  </parityType>
  <stopBits opt="1,1.5,2">
    <!--ro, opt, string, stop bit, attr:opt{req, string}-->1
  </stopBits>
  <workMode opt="console,transparent,audiomixe,screenCtrl,ptzCtrl,keyboard,matrix,audioMixers">
    <!--ro, opt, string, working mode, attr:opt{req, string}-->console
  </workMode>
  <flowCtrl opt="none,software,hardware">
    <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
  </flowCtrl>
  <deviceName min="0" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </deviceName>
  <deviceProtocol min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </deviceProtocol>
  <mode opt="readerMode,clientMode,externMode,stairsControl,accessControlHost,disabled,custom,cardReceiver,QRCodeReader">
    <!--ro, opt, string, working mode, attr:opt{req, string}-->readerMode
  </mode>
  <modeChangeReboot>
    <!--ro, opt, bool-->true
  </modeChangeReboot>
  <outputDataType opt="cardNo,employeeNo,auto">
    <!--ro, opt, string, output data type, attr:opt{req, string}, desc:data type output from the door station to the elevator controller: floorNumber (floor
No.,default),cardNo (card No.)-->cardNo
  </outputDataType>
  <copyToAll opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </copyToAll>
  <isVariable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isVariable>
  <PTZInfo>
    <!--ro, opt, object-->
    <bindChannelNo min="0" max="10">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->0
    </bindChannelNo>
    <PTZProtocolList>
      <!--ro, req, object-->
      <PTZProtocol>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, protocol index-->0
        </id>
        <protoDesc>
          <!--ro, req, string, protocol description-->test
        </protoDesc>
      </PTZProtocol>
    </PTZProtocolList>
  </PTZInfo>
  <ScreenCtrlInfo>
    <!--ro, opt, object, ScreenCtrlInfo, desc:this field is valid when the value of workMode is "screenCtrl"-->
    <ScreenCtrlProtoList>
      <!--ro, req, object, ScreenCtrlProtoList-->
      <ScreenCtrlProto>
        <!--ro, opt, object, ScreenCtrlProto-->
        <id>
          <!--ro, req, int, protocol index-->0
        </id>
```

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```
        <protoDesc>
          <!--ro, req, string, protocol description-->test
        </protoDesc>
      </ScreenCtrlProto>
    </ScreenCtrlProtoList>
  </ScreenCtrlInfo>
  <MatrixCtrlInfo>
    <!--ro, opt, object, MatrixCtrlInfo-->
    <MatrixCtrlProtoList>
      <!--ro, req, object, MatrixCtrlProtoList-->
      <MatrixCtrlProto>
        <!--ro, opt, object, MatrixCtrlProto-->
        <id>
          <!--ro, req, int, protocol index-->0
        </id>
        <protoDesc>
          <!--ro, req, string, protocol description-->test
        </protoDesc>
      </MatrixCtrlProto>
    </MatrixCtrlProtoList>
  </MatrixCtrlInfo>
  <KeyBoardCtrlInfo>
    <!--ro, opt, object, KeyBoardCtrlInfo-->
    <KeyBoardCtrlProtoList>
      <!--ro, req, object, KeyBoardCtrlProtoList-->
      <KeyBoardCtrlProto>
        <!--ro, opt, object, KeyBoardCtrlProto-->
        <id>
          <!--ro, req, int, protocol index-->0
        </id>
        <protoDesc>
          <!--ro, req, string, protocol description-->test
        </protoDesc>
      </KeyBoardCtrlProto>
    </KeyBoardCtrlProtoList>
  </KeyBoardCtrlInfo>
  <ControlAddress>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <address min="0" max="255">
      <!--ro, req, int, IP address, attr:min{req, int},max{req, int}-->0
    </address>
  </ControlAddress>
  <rs485WorkMode
opt="Led,sensor,CaptureTrigger,transparent,GPS,ModbusRTU,supplementLight,audibleVisualAlarm,columnSpeaker,commModeUsing433MHz,signalLampControl">
    <!--ro, opt, string, working mode of RS-485 serial port, dep:and,{$.SerialPortList.SerialPort.serialPortType,eq,RS485}, attr:opt{req, string},
desc:working mode of RS-485 serial port,which is used for LED display or triggering transmission of captured pictures. This node is valid only when
<serialPortType> is "RS485"-->sensor
  </rs485WorkMode>
  <SensorCtrlList>
    <!--ro, opt, object-->
    <Sensor>
      <!--ro, opt, object-->
      <id min="1" max="3">
        <!--ro, req, int, index, attr:min{req, int},max{req, int}-->1
      </id>
      <name min="1" max="32">
        <!--ro, req, string, attr:min{req, int},max{req, int}-->visibilitySensor
      </name>
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable, attr:opt{req, string}, desc:whether to enable the serial port: "true,false"-->false
      </enabled>
    </Sensor>
  </SensorCtrlList>
  <stairsControl>
    <!--ro, opt, object, elevator control parameters-->
    <outputDataType opt="floorNumber,cardNo">
      <!--ro, opt, enum, data type output from the door station to the elevator controller, subType:string, attr:opt{req, string}, desc:floorNumber (floor
No.,default),cardNo (card No.)-->floorNumber
    </outputDataType>
  </stairsControl>
  <comMode opt="active,passive">
    <!--ro, opt, enum, communication mode, subType:string, attr:opt{req, string}, desc:communication mode-->active
  </comMode>
  <ModbusRTU>
    <!--ro, opt, object-->
    <ModbusRTUBaudRate opt="1200,2400,4800,9600,19200" def="9600">
      <!--ro, opt, int, attr:opt{req, string},def{req, string}-->9600
    </ModbusRTUBaudRate>
    <ModbusRTUDataBits opt="5,6,7,8" def="8">
      <!--ro, opt, int, attr:opt{req, string},def{req, string}-->6
    </ModbusRTUDataBits>
    <ModbusRTUParityType opt="none,even,odd" def="none">
      <!--ro, opt, string, attr:opt{req, string},def{req, string}-->none
    </ModbusRTUParityType>
    <ModbusRTUStopBits opt="1,2" def="1">
      <!--ro, opt, string, stop bit, attr:opt{req, string},def{req, string}-->1
    </ModbusRTUStopBits>
    <ModbusRTUFlowCtrl opt="none,software,hardware">
      <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
    </ModbusRTUFlowCtrl>
    <decodeAddress min="0" max="255" def="0">
```

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```
    <decoderAddress min="0" max="255" def="0">
      <!--ro, opt, int, decoder IP address, attr:min{req, int},max{req, int},def{req, int}-->1
    </decoderAddress>
    <online opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </online>
  </ModbusRTU>
  <accessName min="1" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </accessName>
  <serialPortUse opt="acquAndCtrl,swipe,strongElectric">
    <!--ro, opt, string, attr:opt{req, string}-->acquisition
  </serialPortUse>
  <acquisitionProtocol opt="modbus,custom">
    <!--ro, opt, string, dep:and,{$.SerialPort.serialPortUse,eq,acquisition}, attr:opt{req, string}-->modbus
  </acquisitionProtocol>
  <serialProtocol opt="FECbus,GB26875">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->GB26875
  </serialProtocol>
  <SupplementLightList size="5">
    <!--ro, opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}, attr:size{req, int}-->
    <SupplementLight>
      <!--ro, opt, object-->
      <SupplementLightID min="1" max="5">
        <!--ro, req, int, range:[1,5], attr:min{req, int},max{req, int}-->1
      </SupplementLightID>
    </SupplementLight>
  </SupplementLightList>
  <permissionController opt="0,1">
    <!--ro, opt, enum, subType:int, attr:opt{req, string}-->1
  </permissionController>
  <deviceModel min="0" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </deviceModel>
  <deviceSoftwareVersion min="0" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </deviceSoftwareVersion>
  <audibleVisualAlarmType opt="CIRCLE,AFE80,AFE82,XY">
    <!--ro, opt, enum, CIRCLE; AFE80; AFE82; XY, subType:string, attr:opt{req, string}, desc:CIRCLE; AFE80; AFE82; XY-->CIRCLE
  </audibleVisualAlarmType>
  <cardReaderType opt="K1108AM">
    <!--ro, opt, string, attr:opt{req, string}-->test
  </cardReaderType>
  <strongElectricType opt="19M04-RO/T,19M08-RO/T">
    <!--ro, opt, string, attr:opt{req, string}-->test
  </strongElectricType>
  <signalLampControl>
    <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,signalLampControl}-->
    <protocolType opt="GB920,Hisense,LESER,JERE">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->GB920
    </protocolType>
  </signalLampControl>
</SerialPort>
```

15.1.5.6 Set the parameters of a specific serial port supported by the device

Request URL

PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	--
childDevID	string	--
deviceIndex	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, serial port ID-->0
  </id>
  <enabled>
    <!--opt, bool, whether to enable the function-->true
  </enabled>
  <encryptKey>
```

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```
    <!--opt, string-->test
  </encryptKey>
  <serialNumber>
    <!--opt, int, serial No.-->0
  </serialNumber>
  <serialPortType>
    <!--opt, enum, serial port type: "RS485","RS422","RS232", subType:string, desc:serial port type: "RS485","RS422","RS232"-->RS485
  </serialPortType>
  <serialAddress>
    <!--opt, int-->1
  </serialAddress>
  <duplexMode>
    <!--opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"”-->half
  </duplexMode>
  <direction>
    <!--opt, enum, "monodirectional,bdirectional", subType:string, desc:"monodirectional, bdirectional”-->monodirectional
  </direction>
  <baudRate>
    <!--opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--opt, int-->6
  </dataBits>
  <parityType>
    <!--opt, enum, parity type, subType:string, desc:"none, even, odd, mark, space”-->none
  </parityType>
  <stopBits>
    <!--opt, string, stop bit: "1,1.5,2"-->1
  </stopBits>
  <workMode>
    <!--opt, enum, work mode, subType:string, desc:working mode: "console","transparent","audiomixer","stairsControl"-elevator control,"cardReader"-card
reader,"disabled","custom". This node is required only when <serialPortType> is set to "RS232"-->console
  </workMode>
  <flowCtrl>
    <!--opt, enum, "none,software,hardware", subType:string, desc:"none, software, hardware”-->none
  </flowCtrl>
  <rs485WorkMode>
    <!--opt, enum, working mode of RS-485 serial port, which is used for LED display or triggering transmission of captured pictures: "Led, CaptureTrigger".
This node is valid only when <serialPortType> is "RS485", subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}, desc:working mode of RS-485 serial
port,which is used for LED display or triggering transmission of captured pictures: "Led,CaptureTrigger". This node is valid only when <serialPortType> is
"RS485"-->sensor
  </rs485WorkMode>
  <copyToAll>
    <!--opt, bool-->true
  </copyToAll>
  <deviceName>
    <!--ro, opt, string-->test
  </deviceName>
  <deviceProtocol>
    <!--ro, opt, int-->1
  </deviceProtocol>
  <mode>
    <!--opt, enum, work mode, subType:string, desc:deq,working mode: "readerMode,clientMode,externMode,accessControlHost,disabled",this node is valid only
when <serialPortType> is "RS485"-->readerMode
  </mode>
  <outputDataType>
    <!--opt, enum, output data type, subType:string, dep:and,{$.SerialPort.mode,eq,accessControlHost}, desc:"cardNo,employeeNo", this node is valid when
<mode>is "accessControlHost”-->cardNo
  </outputDataType>
  <isVariable>
    <!--opt, bool-->true
  </isVariable>
  <PTZInfo>
    <!--opt, object, PTZ information-->
    <bindChannelNo>
      <!--req, int-->0
    </bindChannelNo>
    <PTZProtocolList>
      <!--req, object-->
      <PTZProtocol>
        <!--opt, object-->
        <id>
          <!--req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--req, string-->test
        </protoDesc>
      </PTZProtocol>
    </PTZProtocolList>
  </PTZInfo>
  <ScreenCtrlInfo>
    <!--opt, object-->
    <ScreenCtrlProtoList>
      <!--req, object-->
      <ScreenCtrlProto>
        <!--opt, object-->
        <id>
          <!--req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--req, string-->test
        </protoDesc>
      </ScreenCtrlProto>
```

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```
    </ScreenCtrlProtoList>
  </ScreenCtrlInfo>
  <MatrixCtrlInfo>
    <!--opt, object-->
    <MatrixCtrlProtoList>
      <!--req, object-->
      <MatrixCtrlProto>
        <!--opt, object-->
        <id>
          <!--req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--req, string-->test
        </protoDesc>
      </MatrixCtrlProto>
    </MatrixCtrlProtoList>
  </MatrixCtrlInfo>
  <KeyBoardCtrlInfo>
    <!--opt, object-->
    <KeyBoardCtrlProtoList>
      <!--req, object-->
      <KeyBoardCtrlProto>
        <!--opt, object-->
        <id>
          <!--req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--req, string-->test
        </protoDesc>
      </KeyBoardCtrlProto>
    </KeyBoardCtrlProtoList>
  </KeyBoardCtrlInfo>
  <ControlAddress>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <address>
      <!--req, int, address, range:[0,255]-->1
    </address>
  </ControlAddress>
  <SensorCtrlList>
    <!--opt, object-->
    <Sensor>
      <!--opt, object-->
      <id>
        <!--req, int, ID, range:[1,3]-->1
      </id>
      <name>
        <!--req, string, sensor name, range:[1,32]-->visibilitySensor
      </name>
      <enabled>
        <!--req, bool, whether to enable the function, desc:whether to enable the serial port: "true,false"-->false
      </enabled>
    </Sensor>
  </SensorCtrlList>
  <stairsControl>
    <!--opt, object, elevator control parameters, this node is valid when the value of the node <mode> is stairsControl, dep:and,
{$.SerialPort.mode,eq,stairsControl}-->
    <outputDataType>
      <!--opt, enum, output data type, subType:string, desc:"cardNo,employeeNo", this node is valid when <mode>is "accessControlHost"-->floorNumber
    </outputDataType>
  </stairsControl>
  <comMode>
    <!--opt, enum, subType:string-->active
  </comMode>
  <ModbusRTU>
    <!--opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
    <ModbusRTUBaudRate>
      <!--opt, enum, subType:string-->9600
    </ModbusRTUBaudRate>
    <ModbusRTUDataBits>
      <!--opt, enum, subType:int-->6
    </ModbusRTUDataBits>
    <ModbusRTUParityType>
      <!--opt, enum, subType:string-->none
    </ModbusRTUParityType>
    <ModbusRTUStopBits>
      <!--opt, enum, subType:string-->1
    </ModbusRTUStopBits>
    <ModbusRTUFlowCtrl>
      <!--opt, enum, subType:string-->none
    </ModbusRTUFlowCtrl>
    <decoderAddress>
      <!--opt, int, range:[0,255]-->0
    </decoderAddress>
  </ModbusRTU>
  <accessName>
    <!--opt, string, range:[1,32]-->test
  </accessName>
  <serialPortUse>
    <!--opt, enum, subType:string-->acquisition
  </serialPortUse>
  <acquisitionProtocol>
```

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```
    <!--opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,acquisition}-->modbus
  </acquisitionProtocol>
  <serialProtocol>
    <!--opt, enum, subType:string-->GB26875
  </serialProtocol>
  <SupplementLightList>
    <!--opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}-->
    <SupplementLight>
      <!--opt, object-->
      <SupplementLightID>
        <!--req, int, range:[1,5]-->1
      </SupplementLightID>
    </SupplementLight>
  </SupplementLightList>
  <cardReaderType>
    <!--opt, enum, subType:string-->K1108AM
  </cardReaderType>
</SerialPort>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error reason description in detail, desc:error reason description in detail-->OK
  </subStatusCode>
</ResponseStatus>
```

15.1.5.7 Get the parameters of a specific port supported by the device

Request URL

GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	0 (main permission controller), 1 (sub permission controller) When this parameter does not exist, it indicates getting the serial port information of the main permission controller (i.e., the device serial port information).
childDevID	string	--
deviceIndex	string	For C60S devices connected to multiple LED sending cards, the URL needs to include deviceIndex= to distinguish which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, serial port ID-->0
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable the serial port-->true
  </enabled>
  <serialPortType>
    <!--ro, opt, enum, serial port type, subType:string, desc:"RS485", "RS422", "RS232"-->RS485
  </serialPortType>
  <duplexMode>
    <!--ro, opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"-->half
  </duplexMode>
  <direction>
    <!--ro, opt, enum, "monodirectional,bdirectional", subType:string, desc:"monodirectional", "bdirectional"-->monodirectional
  </direction>
  <baudRate>
    <!--ro, opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--ro, opt, int-->6
  </dataBits>
  <parityType>
    <!--ro, opt, enum, parity type, subType:string, desc:parity type: "none,even,odd,mark,space"-->none
  </parityType>
  <stopBits>
    <!--ro, opt, enum, stop bit, subType:string, desc:stop bit-->1
  </stopBits>
  <workMode>
    <!--ro, opt, enum, working mode, subType:string, desc:"console", "transparent", "audiomixer", "screenCtrl", "ptzCtrl", "keyboard", "matrix",
"audioMixers"-->console
  </workMode>
  <flowCtrl>
    <!--ro, opt, enum, "none,software,hardware", subType:string, desc:"none", "software", "hardware"-->none
  </flowCtrl>
</SerialPort>
```

15.1.5.8 Get the list of serial ports supported by the device

Request URL

GET /ISAPI/System/Serial/ports?permissionController=<indexID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	0 (main permission controller), 1 (sub permission controller) If this parameter does not exist, it indicates getting the serial port information of main permission controller, that is, the serial port information of the device.
deviceIndex	string	For C60S device connecting multiple LED sending cards, the URL needs to include the parameter deviceIndex= to specify which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPortList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <SerialPort>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->0
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <serialPortType>
      <!--ro, opt, enum, subType:string-->RS485
    </serialPortType>
    <duplexMode>
      <!--ro, opt, enum, subType:string-->half
    </duplexMode>
    <direction>
      <!--ro, opt, enum, subType:string-->monodirectional
    </direction>
    <baudRate>
      <!--ro, opt, enum, subType:int-->2400
    </baudRate>
    <dataBits>
      <!--ro, opt, int-->6
    </dataBits>
    <parityType>
      <!--ro, opt, enum, subType:string-->none
    </parityType>
    <stopBits>
      <!--ro, opt, enum, subType:string-->1
    </stopBits>
    <workMode>
      <!--ro, opt, enum, subType:string-->console
    </workMode>
    <flowCtrl>
      <!--ro, opt, enum, subType:string-->none
    </flowCtrl>
  </SerialPort>
</SerialPortList>
```

15.1.6 系统维护

15.1.6.1 Get the storage capability of the device

Request URL

GET /ISAPI/ContentMgmt/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<RacmCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device storage capability, attr:version{req, string, protocolVersion}-->
  <isSupportZeroChan>
    <!--ro, opt, bool, whether the device supports channel-zero, desc:related URI: /ISAPI/ContentMgmt/ZeroVideo/channels-->true
  </isSupportZeroChan>
  <inputProxyNums>
    <!--ro, opt, int, supported number of digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->0
  </inputProxyNums>
  <eSATANums>
    <!--ro, opt, int, supported number of eSATA interfaces-->0
  </eSATANums>
  <miniSASNums>
    <!--ro, opt, int, supported number of miniSAS interfaces-->0
  </miniSASNums>
  <nasNums>
    <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->0
  </nasNums>
  <ipSanNums>
    <!--ro, opt, int, supported number of IPSANs-->0
  </ipSanNums>
  <isSupportRaid>
    <!--ro, opt, bool, whether the device supports RAID-->true
  </isSupportRaid>
  <isSupportExtHdCfg>
    <!--ro, opt, bool, whether the device supports advanced HDD management-->true
  </isSupportExtHdCfg>
```

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```
  <isSupportTransCode>
    <!--ro, opt, bool, whether the device supports live view transcoding-->true
  </isSupportTransCode>
  <isSupportIpcImport>
    <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/ipcConfig-->true
  </isSupportIpcImport>
  <isSupportIpcStreamType>
    <!--ro, opt, bool, whether the device supports the stream type of the network camera-->true
  </isSupportIpcStreamType>
  <isSupportIOInputProxy>
    <!--ro, opt, bool, whether the device supports dynamic input channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/inputs-->true
  </isSupportIOInputProxy>
  <isSupportIOOutputProxy>
    <!--ro, opt, bool, whether the device supports dynamic output channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/outputs-->true
  </isSupportIOOutputProxy>
  <isSupportPTZRs485Proxy>
    <!--ro, opt, bool, whether the device supports dynamic PTZ RS-485 channels, desc:related URI: /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities-->true
  </isSupportPTZRs485Proxy>
  <isSupportSMARTTest>
    <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
  </isSupportSMARTTest>
  <recordSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofN
on-MotorVehicle,
illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin
,temperatureIntervalMeasurement,AIOP,securityChecking,scheduledRecordUpload,variableDataEvent,smokeAndFireDetection,smokeDetection,AIOP_Video,AIOP_Polling_S
nap,BatteryCarInfoEvent,indoorPasswayBlock,detectorTemperatureAlarm,fireNoRegulation,detectorMotionAlarm,detectorShelterAlarm,detectorRunningStatus,faceCapt
ure,UVPhotonDetection,DevPowerStatusReport,ChannelClearEvent,channelOccupy,ChannelPassingEvent,personQueueCounting,personQueueTime,leavePosition,localButton
Record,callRecord,partialDischarge,gasLeakage,smokeAndFireSample,passback,linkedTracking,CallRecordAnswered,CallRecordNotAnswered,lensMotorLocking">
    <!--ro, opt, enum, video search type, subType:string, attr:opt{req, string}, desc:"AllEvent" (all events), "CMR" (capture by schedule), "MOTION" (motion
detection), "ALARM", "EDR", "ALARMANDMOTION", "pir", "wlsensor", "callhelp", "facedetection", "FieldDetection", "scenechangedetection", "LineDetection",
"regionEntrance", "regionExiting", "loitering", "group", "rapidMove", "parking", "unattendedBaggage", "attendedBaggage", "vehicleDetection", "manual",
"manualSnapShot", "playSnapShot", "safetyHelmet",  "personDensityDetection", "smokeDetection", "ATMPanel", "ATMSurround", "ATMFace", "ATMSafetyCabin",
"temperatureIntervalMeasurement", "AIOP", "securityChecking", "scheduledRecordUpload", "thermalVehicleDetection", "waterQualityDetection",
"waterLevelDetection", "methaneConcentrationException", "variableDataEvent"-->AllEvent
  </recordSearchType>
  <isSupportActivateIpc>
    <!--ro, opt, bool, whether the device supports activating the network camera-->true
  </isSupportActivateIpc>
  <isSupportCheckIpcSecurity>
    <!--ro, opt, object, whether the device supports IPC security verification-->
    <isSupportCheckPassword>
      <!--ro, opt, bool, whether the device supports password detection-->true
    </isSupportCheckPassword>
  </isSupportCheckIpcSecurity>
  <isSupportMainAndSubRecord>
    <!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
  </isSupportMainAndSubRecord>
  <isSupportSyncIPCPassword>
    <!--ro, opt, bool, whether the device supports synchronizing the password of the network camera-->true
  </isSupportSyncIPCPassword>
  <isSupportTransferIPC>
    <!--ro, opt, bool, whether the device supports transparent transmission of the network camera-->true
  </isSupportTransferIPC>
  <isSupportSmartSearch>
    <!--ro, opt, bool, whether the device supports video dual-VCA, desc:related URI: /ISAPI/ContentMgmt/SmartSearch/capabilities-->true
  </isSupportSmartSearch>
  <isSupportPOS>
    <!--ro, opt, bool, whether the device supports POS-->true
  </isSupportPOS>
  <isSupportLinkNum>
    <!--ro, opt, bool, whether the device supports getting the number of links, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/linkNum-->true
  </isSupportLinkNum>
  <isSupportStorageExtraInfo>
    <!--ro, opt, bool, whether the device supports configuring additional storage information, desc:related URI:
/ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities-->true
  </isSupportStorageExtraInfo>
  <diskGroupNums>
    <!--ro, opt, int, supported number of disk groups, desc:related URI: /ISAPI/ContentMgmt/Storage/diskGroup-->0
  </diskGroupNums>
  <ExternalDevice>
    <!--ro, opt, object, external device storage-->
    <USB>
      <!--ro, opt, object, USB information-->
      <debugLogOutput>
        <!--ro, opt, bool, whether to enable the debugging log, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
      </debugLogOutput>
      <isSupportCapturePackage>
        <!--ro, opt, bool, whether the device supports configuring capture and filter function of a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/capturePackage/capabilities?format=json-->true
      </isSupportCapturePackage>
      <isSupportStreamStorage>
        <!--ro, opt, bool, whether the device supports saving stream to a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/streamStorage/capabilities?format=json-->true
      </isSupportStreamStorage>
    </USB>
  </ExternalDevice>
  <customProtocolNums>
    <!--ro, opt, int, number of custom protocols for accessing network cameras, desc:related URI: /ISAPI/ContentMgmt/InputProxy/customProtocols-->0
  </customProtocolNums>
```

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```
  <isSupportIPCTiming>
    <!--ro, opt, bool, whether the device supports synchronizing the time of the network camera, desc:relate to enableTiming on URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>-->true
  </isSupportIPCTiming>
  <SecurityLog>
    <!--ro, opt, object, security log, desc:security log-->
    <isSupportSecurityLog>
      <!--ro, opt, bool, whether the device supports the security log-->true
    </isSupportSecurityLog>
    <isSupportLogServer>
      <!--ro, opt, bool, whether it supports security log server configuration-->true
    </isSupportLogServer>
  </SecurityLog>
  <certificateValidationEnabled>
    <!--ro, opt, bool, whether to enable the certification validation, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <certificateValidationEnabled> node supports-->true
  </certificateValidationEnabled>
  <defaultAdminPortEnabled>
    <!--ro, opt, bool, whether to enable the default communication port, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <defaultAdminPortEnabled> node supports-->true
  </defaultAdminPortEnabled>
  <isSupportLogDataPackage>
    <!--ro, opt, bool, whether the device supports exporting the log files, desc:related URI: /ISAPI/ContentMgmt/logSearch/dataPackage-->true
  </isSupportLogDataPackage>
  <CloudStorageServerCap>
    <!--ro, opt, object, capability of the cloud storage service-->
    <isSupportCloudStorageParameter>
      <!--ro, opt, bool, whether the device supports configuring the cloud storage parameters, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/capabilities?format=json-->true
    </isSupportCloudStorageParameter>
  </CloudStorageServerCap>
  <isSupportWebPrivatePlaybackByUTC>
    <!--ro, opt, bool, whether it supports playback (via private protocol) according to UTC-->true
  </isSupportWebPrivatePlaybackByUTC>
  <isSupportFindCommonFileByUTC>
    <!--ro, opt, bool, whether it supports searching for ordinary files (private protocols) by UTC time-->true
  </isSupportFindCommonFileByUTC>
  <isSupportFindEventFileByUTC>
    <!--ro, opt, bool, whether it supports searching for event related videos (via private protocol) according to UTC-->true
  </isSupportFindEventFileByUTC>
  <isSupportSmartSearchRecordByUTC>
    <!--ro, opt, bool, whether it supports VCA search by UTC time-->true
  </isSupportSmartSearchRecordByUTC>
  <isSupportMRDSearchByTimeZone>
    <!--ro, opt, bool, whether it supports searching for files (private protocols) by monthly calendar by UTC time-->true
  </isSupportMRDSearchByTimeZone>
  <isSupportSearchRecordLabelByUTC>
    <!--ro, opt, bool, whether it supports searching for video tags by UTC time-->true
  </isSupportSearchRecordLabelByUTC>
  <isSupportSearchPictureByUTC>
    <!--ro, opt, bool, whether it supports searching for pictures by UTC time-->true
  </isSupportSearchPictureByUTC>
  <isSupportSmartSearchPictureByUTC>
    <!--ro, opt, bool, whether it supports smart pictures search by UTC time-->true
  </isSupportSmartSearchPictureByUTC>
  <isSupportFindLogByUTC>
    <!--ro, opt, bool, whether it supports searching log by UTC time-->true
  </isSupportFindLogByUTC>
  <isSupportUploadRecordByUTC>
    <!--ro, opt, bool, whether it supports uploading files to cloud storage by UTC time-->true
  </isSupportUploadRecordByUTC>
  <isSupportPlaybackByUTC>
    <!--ro, opt, bool, whether it supports playback and locating by UTC time-->true
  </isSupportPlaybackByUTC>
  <isSupportFaceDetecttionAlarmByTimeZone>
    <!--ro, opt, bool, whether it supports reporting face detection by UTC time, desc:the time should be before isSupportCountingSearchByUTC and after
isSupportPlaybackByUTC-->true
  </isSupportFaceDetecttionAlarmByTimeZone>
  <isSupportCountingSearchByUTC>
    <!--ro, opt, bool, whether it supports searching people counting results by UTC time-->true
  </isSupportCountingSearchByUTC>
  <eventRecordSearch>
    <!--ro, opt, object, whether the device supports searching videos by event, desc:related URI: /ISAPI/ContentMgmt/eventRecordSearch?format=json-->
    <eventType
opt="all,batch,alarmResult,faceSnap,vehicleMatchResult,hawkResult,VMD,alarmIn,inquestAlarm,behavior,posAlarm,accessControllerEvent,IOTD,trialAlarm,rapidMove
,defocus,scenechangedetection,PIR,fireSmartFireDetect,shipsDetection,TMA,TDA,framesPeopleCounting,personQueueTime,peopleDetection,safetyHelmetDetection,mixe
dTargetDetection,TMPA,smokeDetection,smokeAndFireDetection,abnormalReboot,audioexception,ANPR,objectsThrownDetection,audioAnalysis,HFPD,shelteralarm,cityMan
agement,hideStart,hideStop,sleepOnduty,PerimeterTargetDetEvent">
      <!--ro, opt, enum, event type, subType:string, attr:opt{req, string}, desc:“vehicleMatchResult” (vehicle comparison), “hawkResult” (video analysis
task), “faceContrast” (face picture comparison), “faceSnap” (face capture)-->alarmResult
    </eventType>
    <alarmResultAuxEventType opt="all,alarmResultFailure,alarmResultSuccess,whiteList">
      <!--ro, opt, enum, sub-type of face picture comparison alarm, subType:string, dep:and,{$.RacmCap.eventRecordSearch.eventType,eq,alarmResult},
attr:opt{req, string}, desc:"alarmResultFailure" (face picture comparison failed), "alarmResultSuccess" (face picture comparison succeeded), "whiteList"
(allowlist alarm, i.e., strangers alarm)-->alarmResultFailure
    </alarmResultAuxEventType>
    <behavior>
      <!--ro, opt, object, abnormal event detection, dep:and,{$.EventSearchDescription.eventType,eq,behavior}-->
      <behaviorEventType
opt="any,linedetection,regionEntrance,regionExiting,fielddetection,loitering,parking,violentMotion,group,unattendedBaggage,attendedBaggage,audioAbnormal,lea
vePosition,peopleNumChange,retention,failDown,running,spacingChange,playCellphone,objectPlacedOrTaken,tailing,advReachHeight">
        <!--ro, req, enum, abnormal event detection event type, subType:string, attr:opt{req, string}, desc:"any", "linedetection" (line crossing
detection), "regionEntrance", "regionExiting", "fielddetection" (intrusion detection), "loitering" (loitering detection), "parking" (parking detection),
```

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```
"violentMotion", "group" (people gathering), "unattendedBaggage", "attendedBaggage" (object removal), "audioAbnormal" (sudden change of sound intensity detection)-->any
      </behaviorEventType>
    </behavior>
    <cityManagement>
      <!--ro, opt, object, City Management, dep:and,{$.RacmCap.eventRecordSearch.eventType,eq,cityManagement}-->
      <subEventType opt="unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad">
        <!--ro, opt, string, sub event type, attr:opt{req, string}-->unregisteredStreetVendor
      </subEventType>
    </cityManagement>
    <ANPRSearch>
      <!--ro, opt, object-->
      <licensePlate min="1" max="32">
        <!--ro, opt, string, license plate number, attr:min{req, int},max{req, int}-->A283KY77
      </licensePlate>
      <province opt="0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34">
        <!--ro, opt, string, N/A, attr:opt{req, string}-->0
      </province>
    </ANPRSearch>
  </eventRecordSearch>
  <isSupportTimeSearch>
    <!--ro, opt, bool, whether it supports searching for start and end time of recording by channel, desc:/ISAPI/ContentMgmt/time/search?format=json-->true
  </isSupportTimeSearch>
  <isSupportSSDSMARTTest>
    <!--ro, opt, bool, whether the device supports S.M.A.R.T. detection on SSD, desc:related URI: /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/start-->true
  </isSupportSSDSMARTTest>
  <isSupportChanStatusPage>
    <!--ro, opt, string, whether it supports searching the digital channel by page, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/statusPage/capabilities-->test
  </isSupportChanStatusPage>
  <isSupportRecordSearchByTargetType>
    <!--ro, opt, bool, whether it supports searching for event recordings by target types (pedestrians and vehicles)-->true
  </isSupportRecordSearchByTargetType>
  <isSupportLogSearch>
    <!--ro, opt, bool, whether it supports log search-->true
  </isSupportLogSearch>
  <isSupportSpare>
    <!--ro, opt, bool, whether the device supports hot spare configuration, desc:related capability set: /ISAPI/ContentMgmt/spare/capabilities?format=json-->true
  </isSupportSpare>
  <isSupportPTEventTableTemplate>
    <!--ro, opt, bool-->true
  </isSupportPTEventTableTemplate>
  <isSupportPTEventTableFile>
    <!--ro, opt, bool-->true
  </isSupportPTEventTableFile>
  <isSupportAudioDailyDistribution>
    <!--ro, opt, bool, whether it supports searching for audio file information by monthly calendar, desc:related URI: /ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json-->true
  </isSupportAudioDailyDistribution>
  <isSupportDownloadAudio>
    <!--ro, opt, bool, whether it supports downloading audio files by file name, desc:related URI: /ISAPI/ContentMgmt/recordAudio/fileNameDownload/capabilities?format=json-->true
  </isSupportDownloadAudio>
  <isSupportAudioChanCap>
    <!--ro, opt, bool, whether the device supports managing audio channel resources, desc:related URL: /ISAPI/ContentMgmt/audioChannel/capabilities?format=json-->true
  </isSupportAudioChanCap>
  <isSupportRedirectStorageUrl>
    <!--ro, opt, bool-->true
  </isSupportRedirectStorageUrl>
  <isNotSupportHDDBadSectorsTestStart>
    <!--ro, opt, bool-->false
  </isNotSupportHDDBadSectorsTestStart>
  <isSupportSearchRAIDStoragePerformance>
    <!--ro, opt, bool-->true
  </isSupportSearchRAIDStoragePerformance>
  <isSupportSearchDiskConfigurationSpecificationList>
    <!--ro, opt, bool-->true
  </isSupportSearchDiskConfigurationSpecificationList>
  <isSupportPicSearchByText>
    <!--ro, opt, bool-->true
  </isSupportPicSearchByText>
  <isSupportAsynSearchByTextTask>
    <!--ro, opt, bool-->true
  </isSupportAsynSearchByTextTask>
  <isSupportSearchTextTaskStatus>
    <!--ro, opt, bool-->true
  </isSupportSearchTextTaskStatus>
  <isSupportDeleteAsynSearchByTextTask>
    <!--ro, opt, bool-->true
  </isSupportDeleteAsynSearchByTextTask>
</RacmCap>
```

15.1.6.2 Get device system capability

Request URL

GET /ISAPI/System/capabilities?type=<all>

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Query Parameter

Parameter Name	Parameter Type	Description
all	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device system capability, attr:version{opt, string, protocolVersion}-->
  <isSupportPreview>
    <!--ro, opt, bool, whether the device supports live view-->true
  </isSupportPreview>
  <isSupportSystemMaintain>
    <!--ro, opt, bool, whether the device supports system maintenance, desc:for traffic devices, this node is required and must be true-->true
  </isSupportSystemMaintain>
  <isSupportReboot>
    <!--ro, opt, bool, whether the device supports rebooting, desc:for traffic devices, this node is required and must be true.-->true
  </isSupportReboot>
  <isSupportFactoryReset>
    <!--ro, opt, bool, whether the device supports restoring to default settings, desc:for traffic devices, this node is required and must be true-->true
  </isSupportFactoryReset>
  <isSupportConfigurationData>
    <!--ro, opt, bool, whether the device supports importing and exporting the configuration files safely, desc:for traffic devices, this node is required
and must be true-->true
  </isSupportConfigurationData>
  <isSupportUpdatefirmware>
    <!--ro, opt, bool, whether the device supports upgrading, desc:for traffic devices, this node is required and must be true-->true
  </isSupportUpdatefirmware>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports getting the device information, desc:for traffic devices, this node is required and must be true-->true
  </isSupportDeviceInfo>
  <isSupportTime>
    <!--ro, opt, bool, whether the device supports time configuration, desc:for traffic device, this node is required and must be true. For other devices,
the isSupportTimeCap is used to configure time-->true
  </isSupportTime>
  <SysCap>
    <!--ro, opt, object, system capability-->
    <isSupportDst>
      <!--ro, opt, bool, whether the device supports DST (Daylight Saving Time)-->true
    </isSupportDst>
    <NetworkCap>
      <!--ro, opt, object, network capability, desc:related URI: /ISAPI/System/Network/capabilities-->
      <isSupportWireless>
        <!--ro, req, bool, whether the device supports wireless network-->true
      </isSupportWireless>
      <isSupportWired>
        <!--ro, opt, bool, whether it supports wired network-->true
      </isSupportWired>
      <isSupportWAN>
        <!--ro, opt, bool, whether the device supports WAN (Wide Area Network)-->true
      </isSupportWAN>
      <isSupportPPPoE>
        <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
      </isSupportPPPoE>
      <isSupportBond>
        <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
      </isSupportBond>
      <isSupport802_1x>
        <!--ro, req, bool, whether the device supports 802.1x protocol-->true
      </isSupport802_1x>
      <isSupportNtp>
        <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
      </isSupportNtp>
      <isSupportFtp>
        <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
      </isSupportFtp>
      <isSupportUpnp>
        <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play ) protocol-->true
      </isSupportUpnp>
      <isSupportPNP>
        <!--ro, opt, bool, whether the device supports plug-and-play protocol-->true
      </isSupportPNP>
      <isSupportDdns>
        <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
      </isSupportDdns>
      <isSupportHttps>
        <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
      </isSupportHttps>
      <isSupport28181>
        <!--ro, opt, bool, N/A-->true
      </isSupport28181>
      <SnmpCap>
        <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
```

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    <isSupport>
      <!--ro, req, bool, whether the device supports SNMP-->true
    </isSupport>
  </SnmpCap>
  <isSupportExtNetCfg>
    <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
  </isSupportExtNetCfg>
  <isSupportIPFilter>
    <!--ro, opt, bool, whether the device supports IP filtering-->true
  </isSupportIPFilter>
  <isSupportSSH>
    <!--ro, opt, bool, whether the device supports SSH-->true
  </isSupportSSH>
  <isSupportNetPreviewStrategy>
    <!--ro, opt, bool, whether the device supports network preview strategy-->true
  </isSupportNetPreviewStrategy>
  <isSupportEZVIZ>
    <!--ro, opt, bool, whether the device supports EZ protocol-->true
  </isSupportEZVIZ>
  <isSupportEhome>
    <!--ro, opt, bool, whether the device supports ISUP server configuration-->true
  </isSupportEhome>
  <isSupportWirelessDial>
    <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
  </isSupportWirelessDial>
  <isSupportWirelessServer>
    <!--ro, opt, bool, whether the device supports wireless server-->true
  </isSupportWirelessServer>
  <isSupportWPS>
    <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
  </isSupportWPS>
  <isWirelessMutexWithWirelessServer>
    <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
  </isWirelessMutexWithWirelessServer>
  <isSupportMACFilter>
    <!--ro, opt, bool, whether the device supports physical address filtering-->true
  </isSupportMACFilter>
  <GB28181Cap>
    <!--ro, opt, object, N/A-->
    <isSupportGB28181Service>
      <!--ro, opt, bool, N/A-->true
    </isSupportGB28181Service>
  </GB28181Cap>
  <WPS>
    <!--ro, opt, object, Wi-Fi Protected Setup-->
    <NetworkInterfaceList size="2">
      <!--ro, opt, array, list of network nodes, subType:object, attr:size{opt, string}-->
      <NetworkInterface>
        <!--ro, opt, object, network node-->
        <id>
          <!--ro, req, string, NIC index-->1
        </id>
        <enabled>
          <!--ro, req, bool, whether to enable or not-->true
        </enabled>
        <isSupportAutoConnect>
          <!--ro, opt, bool, whether the device supports automatic connection-->true
        </isSupportAutoConnect>
        <isSupportDevicePinCode>
          <!--ro, opt, bool, whether the device supports PIN code-->true
        </isSupportDevicePinCode>
        <isSupportDevicePinCodeUpdate>
          <!--ro, opt, bool, whether the device supports updating the PIN code-->true
        </isSupportDevicePinCodeUpdate>
        <ApPinCode>
          <!--ro, opt, object, PIN code of the access point-->
          <ssid min="1" max="1">
            <!--ro, opt, string, SSID (Service Set Identifier), attr:min{opt, string},max{opt, string}-->test
          </ssid>
          <pinCode min="1" max="1">
            <!--ro, opt, string, PIN code, attr:min{opt, string},max{opt, string}-->test
          </pinCode>
        </ApPinCode>
      </NetworkInterface>
    </NetworkInterfaceList>
  </WPS>
  <isSupportRFIDData>
    <!--ro, opt, bool, whether the device supports RFID collection configuration-->true
  </isSupportRFIDData>
  <isSupportwifiProbeSSID>
    <!--ro, opt, bool, whether the device supports configuration of Wi-Fi signal detection-->true
  </isSupportwifiProbeSSID>
  <isSupportWifiProbe>
    <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
  </isSupportWifiProbe>
  <verificationCode min="1" max="1">
    <!--ro, opt, string, verification code, attr:min{opt, string},max{opt, string}-->test
  </verificationCode>
  <WPSCap>
    <!--ro, opt, object, capability of configuring Wi-Fi Protected Setup-->
    <isSupport>
      <!--ro, req, bool, whether the device supports Wi-Fi Protected Setup-->true
    </isSupport>
    <isSupportAutoConnect>
```

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```
        <!--ro, req, bool, whether the device supports automatic connection-->true
      </isSupportAutoConnect>
    </WPSCap>
    <NetWorkMode>
      <!--ro, opt, object, network mode configuration-->
      <workMode>
        <!--ro, opt, string, network mode-->true
      </workMode>
    </NetWorkMode>
    <VerificationCodeModification>
      <!--ro, opt, object, device verification code configuration-->
      <verificationCodeType opt="normal,empty">
        <!--ro, opt, string, verification code type, attr:opt{opt, string}-->empty
      </verificationCodeType>
      <isSupportDeclarationURL>
        <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
      </isSupportDeclarationURL>
      <isSupportPrivacyPolicyURL>
        <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
      </isSupportPrivacyPolicyURL>
      <verificationCodeModify opt="true,false">
        <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{opt, string}-->true
      </verificationCodeModify>
      <Hyperlinks>
        <!--ro, opt, object, hyperlink configuration-->
        <declarationURL>
          <!--ro, opt, string, hyperlink to the Terms of Service-->test
        </declarationURL>
        <privacyPolicyURL>
          <!--ro, opt, string, hyperlink to the Privacy Policy-->test
        </privacyPolicyURL>
      </Hyperlinks>
      <isSupportVerificationCodeCheck>
        <!--ro, opt, bool, whether the device supports verifying the verification code-->true
      </isSupportVerificationCodeCheck>
      <isSupportOldVerificationCode>
        <!--ro, opt, bool, whether the device supports the EZ password configuration of old version (the password of old version consists of six
characters)-->true
      </isSupportOldVerificationCode>
      <maintenanceVerificationCodeType opt="normal,empty">
        <!--ro, opt, string, maintenance verification code type, attr:opt{opt, string}, desc:used to identify the verificationCode type in
ISAPI/System/Network/EZVIZ?security=<security>&iv=<iv>&moduleType=<moduleType>, valid only when the moduleType in the URL is maintenanceModule-->empty
      </maintenanceVerificationCodeType>
      <maintenanceVerificationCodeModify opt="true,false">
        <!--ro, opt, bool, whether the device maintenance verification code has been modified, attr:opt{opt, string}, desc:used to identify the
verificationCode type in ISAPI/System/Network/EZVIZ?security=<security>&iv=<iv>&moduleType=<moduleType>, valid only when the moduleType in the URL is
maintenanceModule-->true
      </maintenanceVerificationCodeModify>
    </VerificationCodeModification>
    <isSupportIntegrate>
      <!--ro, opt, bool, whether the device supports access protocol configuration-->true
    </isSupportIntegrate>
    <isSupportPlatformAccess>
      <!--ro, opt, bool, whether the device supports platform access, desc:GET /ISAPI/System/Network/platformAccessFilter/capabilities (platform access IP
address filtering configuration (Ehome and GB28181 allowlist configuration))-->true
    </isSupportPlatformAccess>
    <isSupportIntelligentBoost>
      <!--ro, opt, bool, whether the device supports intelligent acceleration-->true
    </isSupportIntelligentBoost>
    <isSupportResourceStatistics>
      <!--ro, opt, bool, whether the device supports getting the network resource information-->true
    </isSupportResourceStatistics>
    <isSupportBandwidthLimit>
      <!--ro, opt, bool, whether the device supports bandwidth limit-->true
    </isSupportBandwidthLimit>
    <isSupportPOEPortsDisableAdaptiveServer>
      <!--ro, opt, bool, whether the device supports the POE protocol-->true
    </isSupportPOEPortsDisableAdaptiveServer>
    <isSupportPOEConfiguration>
      <!--ro, opt, bool, whether the device supports configuring the POE NIC parameters-->true
    </isSupportPOEConfiguration>
    <isSupportGetLinkSocketIP>
      <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
    </isSupportGetLinkSocketIP>
    <isSupportWebSocket>
      <!--ro, opt, bool, whether the device supports WebSocket-->true
    </isSupportWebSocket>
    <isSupportWebSocketS>
      <!--ro, opt, bool, whether the device supports WebSocketS-->true
    </isSupportWebSocketS>
    <Adaption>
      <!--ro, opt, object, network self-adaption configuration-->
      <streamType opt="0,1,2,3,4,5,7,8,9,10">
        <!--ro, opt, string, stream type that supports network self-adaption during live view, attr:opt{opt, string}, desc:0 (main stream), 1 (sub-
stream), 2 (third stream), 3 (fourth stream/virtual stream), 4 (fifth stream), 5 (sixth stream), 7 (seventh stream), 8 (eighth stream), 9 (ninth stream), 10
(tenth stream), and so forth. The index value here is the same as that in the live view structure NET_DVR_PREVIEWINFO of Device Network SDK. Note that 6 is
an invalid value, and from the seventh stream (index value: 7), the index value is the same as that for live view via RTSP-->test
      </streamType>
      <isSupportPlayback>
        <!--ro, opt, bool, whether the device supports playback-->true
      </isSupportPlayback>
    </Adaption>
    <isSupportVideoImgDB>
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        <!--ro, opt, bool, whether the device supports image and video library configuration-->true
      </isSupportVideoImgDB>
      <isSupportVideoImgLibService>
        <!--ro, opt, bool, whether the device supports image and video library service-->true
      </isSupportVideoImgLibService>
      <isSupportWirelessWifi>
        <!--ro, opt, bool, whether the device supports Wi-Fi status-->true
      </isSupportWirelessWifi>
      <isSupportPrivatePlatom>
        <!--ro, opt, bool, whether the device supports accessing third party platform-->true
      </isSupportPrivatePlatom>
      <isSupportGantryETCServer>
        <!--ro, opt, bool-->true
      </isSupportGantryETCServer>
      <isSupportEventDataOverWebSocket opt="true,false">
        <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{opt, string}-->true
      </isSupportEventDataOverWebSocket>
      <isSupportDynamicHostName>
        <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) configuration-->true
      </isSupportDynamicHostName>
      <isSupportTCPAcceleration>
        <!--ro, opt, bool, whether the device supports TCP acceleration-->true
      </isSupportTCPAcceleration>
      <isSupportFlowControl>
        <!--ro, opt, bool, whether the device supports traffic control-->true
      </isSupportFlowControl>
      <isSupportEZVIZTiming>
        <!--ro, opt, bool, whether the device supports synchronizing time with EZ platform, desc:related URI: enabledTiming in the message of
/ISAPI/System/Network/EZVIZ-->true
      </isSupportEZVIZTiming>
      <isSupportTrafficMonitor>
        <!--ro, opt, bool, whether the device supports traffic monitoring, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json-->true
      </isSupportTrafficMonitor>
      <isSupportDMSAuthInfo>
        <!--ro, opt, bool, whether the device supports DMS authentication information management-->true
      </isSupportDMSAuthInfo>
      <isSupportWiredandWirelessTrafficMonitor>
        <!--ro, opt, bool, whether the device supports traffic monitoring (wireless NIC and wired NIC), desc:related URI:
/ISAPI/System/Network/Interfaces/<ID>/trafficMonitor?format=json-->true
      </isSupportWiredandWirelessTrafficMonitor>
      <isSupport4G>
        <!--ro, opt, bool, whether the device supports 4G module-->true
      </isSupport4G>
      <isSupport4GConfig>
        <!--ro, opt, bool, whether the device supports 4G configuration-->true
      </isSupport4GConfig>
      <isSupportGRIDServer>
        <!--ro, opt, bool, N/A-->true
      </isSupportGRIDServer>
      <isSupportGRIDInfo>
        <!--ro, opt, bool, N/A-->true
      </isSupportGRIDInfo>
      <isSupportSipTransmit>
        <!--ro, opt, bool, whether the device supports configuring SIP transmission parameters-->true
      </isSupportSipTransmit>
      <isSupportSip>
        <!--ro, opt, bool, whether the device supports private SIP (VOIP)-->true
      </isSupportSip>
      <isSupportWifiCascade>
        <!--ro, opt, bool, whether the device supports enabling automatic Wi-Fi cascading, desc:related URI: /ISAPI/System/Network/wifiCascade?format=json--
>true
      </isSupportWifiCascade>
      <isSupportRelativeInfo>
        <!--ro, opt, bool, whether the device supports getting the information about the connected devices, desc:related URI:
/ISAPI/System/Network/wifiCascade/relativeInfo?format=json-->true
      </isSupportRelativeInfo>
      <isSupportEmailEncrypt>
        <!--ro, opt, bool, whether the device supports email encryption configuration-->true
      </isSupportEmailEncrypt>
      <isSupportPortFilter>
        <!--ro, opt, bool, whether the device supports port filtering-->true
      </isSupportPortFilter>
      <isSupportPingDeny>
        <!--ro, opt, bool, whether the device supports configuring parameters to prevent the device from being tested using ping command-->true
      </isSupportPingDeny>
      <isSupportDOSDeny>
        <!--ro, opt, bool, whether the device supports configuring parameters to prevent the DoS attack-->true
      </isSupportDOSDeny>
      <isSupportNetworkStatus>
        <!--ro, opt, bool, whether the device supports searching for the network status, desc:related URI: /ISAPI/System/Network/status?format=json-->true
      </isSupportNetworkStatus>
      <isSupportEZVIZQRCode>
        <!--ro, opt, bool, whether the device supports getting the EZ QR code, desc:related URI: /ISAPI/System/Network/EZVIZ/QRCode?format=json-->true
      </isSupportEZVIZQRCode>
      <isSupportEZVIZUnbind>
        <!--ro, opt, bool, whether the device supports unbinding from EZ, desc:/ISAPI/System/Network/EZVIZ/unbind?format=json-->true
      </isSupportEZVIZUnbind>
      <isSupportPingTest>
        <!--ro, opt, bool, whether the device supports ping test, desc:Related URI: /ISAPI/System/Network/pingtest/capabilities-->true
      </isSupportPingTest>
      <isSupportMOTCloudPlateform>
        <!--ro, opt, bool, whether the device supports configuring parameters for accessing the cloud platform service, desc:related URI:
/ISAPI/System/Network/Traffic/MOTCloudPlateform/capabilities?format=json-->true
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/ISAPI/System/Network/Adaptive/MOTCloudPlateform/capabilities?format=json -->true
      </isSupportMOTCloudPlateform>
      <isSupportSpectrumScan>
        <!--ro, opt, bool, whether the device supports frequency spectrum scanning, desc:related URI:
/ISAPI/System/Network/interface/<netCardID>/spectrumScan/capabilities?format=json-->true
      </isSupportSpectrumScan>
      <isSupportFRP>
        <!--ro, opt, bool, whether the device supports FRP (Fast Reverse Proxy), desc:related URI: /ISAPI/Syetem/Network/FRP?format=json-->true
      </isSupportFRP>
      <isSupport5GModule>
        <!--ro, opt, bool, whether the device supports 5G module configuration, desc:related URI: /ISAPI/System/Network/5GModule/capabilities?format=json--
>true
      </isSupport5GModule>
      <isSupportWirelessDialRoute>
        <!--ro, opt, bool, whether the device supports configuring parameters of route forwarding rules of wireless dial-up, desc:related URI:
/ISAPI/System/Network/WirelessDia/route/capabilities?format=json-->true
      </isSupportWirelessDialRoute>
      <isSupportNPlusOne>
        <!--ro, opt, bool, whether the device supports N+1 protocol-->true
      </isSupportNPlusOne>
      <supportEnvironmentalStandards opt="HJ212">
        <!--ro, opt, string, environmental monitoring standard, attr:opt{req, string}, desc:N/A-->true
      </supportEnvironmentalStandards>
      <isSupportIntranetSegment>
        <!--ro, opt, bool, whether the device supports LAN segment configuration, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI:
/ISAPI/System/Network/intranetSegment/capabilities?format=json-->true
      </isSupportIntranetSegment>
      <isSupportDHCPServer>
        <!--ro, opt, bool, whether the device supports DHCP service configuration, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI: /ISAPI/System/Network/DHCPServer/capabilities?
format=json-->true
      </isSupportDHCPServer>
      <isSupportClusterIntercom>
        <!--ro, opt, bool, whether the 7200 intercom protocol is transmitted, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will be returned and its value is true or not be returned.-->true
      </isSupportClusterIntercom>
      <isSupportHydrologicMonitoring>
        <!--ro, opt, bool, whether the device supports hydrologic monitoring data communication, desc:if this function is supported, this node will be
returned and its value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI:
/ISAPI/System/Network/HydrologicMonitoring/capabilities?format=json-->true
      </isSupportHydrologicMonitoring>
      <isWirelessDialMutexWithWlanAndWifi>
        <!--ro, opt, bool, whether the wireless dial-up is mutually exclusive with the WLAN hotspot and Wi-Fi, desc:before a function is enabled, another
one must be disabled first-->true
      </isWirelessDialMutexWithWlanAndWifi>
      <isSupportSpeechTranscription>
        <!--ro, opt, bool, whether the device supports accessing the speech-to-text transcription platform, desc:if this function is supported, this node
will be returned and its value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI:
/ISAPI/System/Network/speechTranscriptionPlatform/capabilities?format=json-->true
      </isSupportSpeechTranscription>
      <isSupportAsynPingTest>
        <!--ro, opt, bool, whether the device supports asynchronous network detection, desc:network detection means testing the network stability and packet
loss rate of the network using the ping command for knowing about the network condition. Related URI: /ISAPI/System/Network/asynPingTest/capabilities?
format=json-->true
      </isSupportAsynPingTest>
      <isSupportBasebandCapture>
        <!--ro, opt, bool, whether the device supports capturing packets of the baseband, desc:you can capture packets transmitted between the 4G module and
the base station using the ping command. Related URI: /ISAPI/System/Network/cellularNetwork/basebandCapture/capabilities?format=json-->true
      </isSupportBasebandCapture>
      <isSupportCellularNetworkSpeedTest>
        <!--ro, opt, bool, whether the device supports testing the connection speed between the device and the cellular network, desc:you can test the
connection speed of the network where the current device is. Related URI: /ISAPI/System/Network/cellularNetwork/speedTest/capabilities?format=json-->true
      </isSupportCellularNetworkSpeedTest>
      <isSupportCellularNetworkCellInfo>
        <!--ro, opt, bool, whether the device supports getting a cell's information of the cellular network, desc:you can get the information of the
cellular network where the device is. Related URI: /ISAPI/System/Network/cellularNetwork/cellInfo?format=json&ecurity=<security>&iv=<iv>-->true
      </isSupportCellularNetworkCellInfo>
      <isSupportMultiBandFrequencyScanning>
        <!--ro, opt, bool, whether the device supports scanning all frequency bands of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/multiBandFrequencyScanning/capabilities?format=json-->true
      </isSupportMultiBandFrequencyScanning>
      <isSupportCellularNetworkLockBand>
        <!--ro, opt, bool, whether the device accessing the locked frequency band of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/lockBand/capabilities?format=json-->true
      </isSupportCellularNetworkLockBand>
      <isSupportRadioFrequencyFluctuation>
        <!--ro, opt, bool, whether the device supports configuring RF parameters of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/radioFrequencyFluctuation/capabilities?format=json-->true
      </isSupportRadioFrequencyFluctuation>
      <isSupportBasebandBlock>
        <!--ro, opt, bool, whether the device supports the unauthorized list of the base station, desc:related URI:
/ISAPI/System/Network/cellularNetwork/basebandBlock/capabilities?format=json-->true
      </isSupportBasebandBlock>
      <isSupportRFIDTagCheckTest>
        <!--ro, opt, bool, whether the device supports the RFID tag disk storage performance test, desc:related URI: GET
/ISAPI/System/Network/RFIDData/GetTagCheckTestInfo?format=json-->true
      </isSupportRFIDTagCheckTest>
      <isSupportQoSDSCPtoLoaclPriorityMapping>
        <!--ro, opt, bool, whether the device supports DSCP and local mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/DSCPtoLoaclPriorityMapping/capabilities?format=json-->true
      </isSupportQoSDSCPtoLoaclPriorityMapping>
      <isSupportQoS802.1pto802.1pPriorityMapping>
        <!--ro, opt, bool, whether the device supports 802.1p mapping priority configuration, desc:related URI:
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/ISAPI/System/Network/qos/802.1pto802.1pPriorityMapping/capabilities?format=json-->true
      </isSupportQoS802.1pto802.1pPriorityMapping>
      <isSupportLinkTestStart>
        <!--ro, opt, bool, whether the network link test is enabled, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Start?format=json-->true
      </isSupportLinkTestStart>
      <isSupportLinkTestResult>
        <!--ro, opt, bool, whether the device supports getting the result of network link test, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Start?format=json-->true
      </isSupportLinkTestResult>
      <isSupportNetworkModuleUpgrade>
        <!--ro, opt, bool, whether it supports upgrading network module, desc:including importing upgrade package
(/ISAPI/System/Network/WirelessDial/ImportUpgradePackage?format=json), getting importing progress
(/ISAPI/System/Network/WirelessDial/GetUpgradePackageImportStatus?format=json), starting upgrade
(/ISAPI/System/Network/WirelessDial/StartUpgradeNetworkModule?format=json)-->true
      </isSupportNetworkModuleUpgrade>
      <isSupportNetworkModuleVersionInfo>
        <!--ro, opt, bool, whether it supports getting the version information of network module, desc:GET
/ISAPI/System/Network/WirelessDial/NetworkModuleVersionInfo?format=json-->true
      </isSupportNetworkModuleVersionInfo>
      <isSupportGreenShieldPlatform>
        <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/GreenShieldPlatform/GreenShieldPlatform?format=json-->true
      </isSupportGreenShieldPlatform>
      <isSupportFusionUniversalApplicationPlatCfg>
        <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/FusionUniversalApplicationPlatform/FusionUniversalApplicationPlatCfg?format=json-->true
      </isSupportFusionUniversalApplicationPlatCfg>
      <isSupportModbusAccess>
        <!--ro, opt, bool, whether the device supports Modbus accessing parameters, desc:GET/PUT /ISAPI/System/Network/ModbusAccess?format=json-->true
      </isSupportModbusAccess>
      <isSupportModbusTestSubordinateStatus>
        <!--ro, opt, bool, whether it supports Modbus testing specified client status, desc:POST /ISAPI/System/Network/ModbusTestSubordinateStatus?
format=json-->true
      </isSupportModbusTestSubordinateStatus>
      <isSupportGetModbusAndRegisterMap>
        <!--ro, opt, bool, whether it supports getting the mapping relationship between Modbus uploaded content and register quantity, desc:GET
/ISAPI/System/Network/GetModbusAndRegisterMap?format=json-->true
      </isSupportGetModbusAndRegisterMap>
      <isSupportTSOverWebSocket>
        <!--ro, opt, bool, whether the device supports transparent transmission via the WebSocket serial port, desc:(TS:TransportSerial)-->true
      </isSupportTSOverWebSocket>
      <isSupportRequestOutlookAuth>
        <!--ro, opt, bool, whether the device supports applying authentication request of Outlook email, desc:POST
/ISAPI/System/Network/Email/RequestOutlookAuth?format=json-->true
      </isSupportRequestOutlookAuth>
      <isSupportOutlookEmailList>
        <!--ro, opt, bool, whether the device supports Outlook email information, desc:GET or PUT /ISAPI/System/Network/Email/OutlookEmailList?format=json--
>true
      </isSupportOutlookEmailList>
      <isSupportDSTPServerCfg>
        <!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET or PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
format=json-->true
      </isSupportDSTPServerCfg>
      <isSupportDSTPProtocolsCap>
        <!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolsCap?format=json-->true
      </isSupportDSTPProtocolsCap>
      <isSupportSearchOutlookAuthResult>
        <!--ro, opt, bool-->true
      </isSupportSearchOutlookAuthResult>
      <RTDROverWebSocket>
        <!--ro, opt, object-->
        <isSupportGasSpectrumRealTimeData>
          <!--ro, opt, bool-->true
        </isSupportGasSpectrumRealTimeData>
        <isSupportDangerousGasSensorRealTimeData>
          <!--ro, opt, bool-->true
        </isSupportDangerousGasSensorRealTimeData>
        <isSupportComposeWindConditionRealTimeData>
          <!--ro, opt, bool-->true
        </isSupportComposeWindConditionRealTimeData>
        <isSupportMeteorologySensorRealTimeData>
          <!--ro, opt, bool-->true
        </isSupportMeteorologySensorRealTimeData>
        <isSupportSPLData>
          <!--ro, opt, bool-->true
        </isSupportSPLData>
      </RTDROverWebSocket>
      <isSupportServerCfg>
        <!--ro, opt, bool-->true
      </isSupportServerCfg>
      <isSupportDelimiterCfg>
        <!--ro, opt, bool-->true
      </isSupportDelimiterCfg>
      <isSupportStartCellNetSelfDiagnosis>
        <!--ro, opt, bool-->true
      </isSupportStartCellNetSelfDiagnosis>
      <isSupportCellNetSelfDiagnosisStatus>
        <!--ro, opt, bool-->true
      </isSupportCellNetSelfDiagnosisStatus>
      <isSupportVoiceTalkOverWebSocket>
        <!--ro, opt, bool-->true
      </isSupportVoiceTalkOverWebSocket>
      <isSupportPlaybackOverWebSocket>
        <!--ro, opt, bool-->true
      </isSupportPlaybackOverWebSocket>
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      </isSupportPlaybackOverWebSocket>
      <WebSocketPlaybackCap>
        <!--ro, opt, object-->
        <cmdType opt="speed,pause,resume,single,stop,seek,startExtracting,stopExtracting,transcoding">
          <!--ro, req, string, attr:opt{opt, string}-->speed
        </cmdType>
        <isSupportPlaybackSeekOverStart>
          <!--ro, opt, bool-->false
        </isSupportPlaybackSeekOverStart>
      </WebSocketPlaybackCap>
      <isSupportHTTPAndWSPortMultiplex>
        <!--ro, opt, bool, dep:and,{$.DeviceCap.SysCap.NetworkCap.isSupportWebSocket,eq,true}-->true
      </isSupportHTTPAndWSPortMultiplex>
      <isSupportHTTPSAndWSSPortMultiplex>
        <!--ro, opt, bool, dep:and,{$.DeviceCap.SysCap.NetworkCap.isSupportWebSocketS,eq,true}-->true
      </isSupportHTTPSAndWSSPortMultiplex>
      <isSupportReverseOverWebSocket>
        <!--ro, opt, bool-->true
      </isSupportReverseOverWebSocket>
    </NetworkCap>
    <IOCap>
      <!--ro, opt, object, I/O capability, desc:related URI: /ISAPI/System/IO/capabilities-->
      <IOInputPortNums>
        <!--ro, opt, int, number of local input ports of the device-->true
      </IOInputPortNums>
      <IOOutputPortNums>
        <!--ro, opt, int, number of local output ports of the device-->true
      </IOOutputPortNums>
      <isSupportStrobeLamp>
        <!--ro, opt, bool, whether the device supports configuring the output of the gate's light, desc:related URI:
/ISAPI/System/IO/outputs/strobeLampConf-->true
      </isSupportStrobeLamp>
      <SoftIOInputPortNums>
        <!--ro, opt, int, number of soft input ports of the device, desc:this node depends on whether the URI /ISAPI/System/IO/softInputs/capabilities?
format=json is supported-->true
      </SoftIOInputPortNums>
      <isSupportIOOutputAdvanceParameter>
        <!--ro, opt, bool, whether the device supports configuring advanced parameters of the output port, desc:this node determines whether the URI
/ISAPI/System/IO/outputs/advanceParameter/capabilities?format=json is supported-->true
      </isSupportIOOutputAdvanceParameter>
      <isSupportSetAllIOOutput>
        <!--ro, opt, bool, whether the device supports configuring all output ports, desc:this node determines whether the PUT method of the URI
/ISAPI/System/IO/outputs is supported-->true
      </isSupportSetAllIOOutput>
      <isSupportCombinationAlarm>
        <!--ro, opt, bool, whether the device supports the combined alarm, desc:CombinationAlarmCap in /ISAPI/System/IO/inputs/capabilities-->true
      </isSupportCombinationAlarm>
      <enabledIOOutputPortNums>
        <!--ro, opt, int, number of enabled alarm outputs-->0
      </enabledIOOutputPortNums>
      <isSupportAlarmKeyParam>
        <!--ro, opt, bool, whether supports configuring the physical keys of one-push alarm, desc:related URI:
/ISAPI/System/IO/inputs/<IOInputID>/AlarmKeyParam/capabilities?format=json-->true
      </isSupportAlarmKeyParam>
      <isSupportWebDisplayAlarmStatus>
        <!--ro, opt, bool, whether the web supports displaying the alarm output status, desc:used for web interface display, true indicates that the web
supports displaying the alarm output status; otherwise, it does not support-->true
      </isSupportWebDisplayAlarmStatus>
    </IOCap>
    <SerialCap>
      <!--ro, opt, object, range of RS-485 serial port numbers supported by the device-->
      <rs485PortNums>
        <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->true
      </rs485PortNums>
      <supportRS232Config>
        <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
      </supportRS232Config>
      <rs422PortNums>
        <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->true
      </rs422PortNums>
      <rs232PortNums>
        <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->true
      </rs232PortNums>
      <rs485WorkMode opt="Led,CaptureTrigger">
        <!--ro, opt, string, RS-485 working modes supported by the device, attr:opt{opt, string}, desc:Led (external LED screen, which is used for LED
display), CaptureTrigger (external device, which is used to trigger the transmission of the captured data)-->test
      </rs485WorkMode>
      <rs232SerialNumber opt="1,2,3">
        <!--ro, opt, string, range of RS-232 serial port numbers supported by the device, attr:opt{opt, string}-->test
      </rs232SerialNumber>
      <rs485SerialNumber opt="4,5,6">
        <!--ro, opt, string, range of RS-485 serial port numbers supported by the device, attr:opt{opt, string}-->test
      </rs485SerialNumber>
      <isSupportAuthenticationService>
        <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
      </isSupportAuthenticationService>
      <isSupportDeviceInfo>
        <!--ro, opt, bool, whether the device supports configuring the information about the serial port device-->true
      </isSupportDeviceInfo>
      <isSupportSearchDeviceInfoRelations>
        <!--ro, opt, bool, whether the device supports searching the linkage information of the serial port device-->true
      </isSupportSearchDeviceInfoRelations>
    </SerialCap>
    <VideoCap>
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        <!--ro, opt, object, video encoding capability, desc:related URI: /ISAPI/System/Video/capabilities-->
        <videoInputPortNums>
          <!--ro, opt, int, number of video input ports-->true
        </videoInputPortNums>
        <videoOutputPortNums>
          <!--ro, opt, int, number of video output ports-->true
        </videoOutputPortNums>
        <menuNums>
          <!--ro, opt, int, number of local menus that can be displayed-->true
        </menuNums>
        <externalChannelNum>
          <!--ro, opt, bool, number of extended analog channels-->true
        </externalChannelNum>
        <isSupportHeatmap>
          <!--ro, opt, bool, whether the device supports heat map-->true
        </isSupportHeatmap>
        <isSupportCounting>
          <!--ro, opt, bool, whether the device supports people counting-->true
        </isSupportCounting>
        <countingType>
          <!--ro, opt, string, statistics type, desc:human (people counting); object (counting)-->test
        </countingType>
        <isSupportPicture>
          <!--ro, opt, bool, whether the device supports OSD picture overlay-->true
        </isSupportPicture>
        <isSupportCapturePicOverlays>
          <!--ro, opt, bool, whether the device supports text overlay on a single picture-->true
        </isSupportCapturePicOverlays>
        <isSupportMergePicOverlays>
          <!--ro, opt, bool, whether the device supports text overlay on composite picture-->true
        </isSupportMergePicOverlays>
        <isSupportPreviewSwitch>
          <!--ro, opt, bool, whether the device supports switching live view-->true
        </isSupportPreviewSwitch>
        <isSupportMultiChannelCounting>
          <!--ro, opt, bool, whether the device supports multichannel statistics-->true
        </isSupportMultiChannelCounting>
        <isSupportCountingCollection>
          <!--ro, opt, bool, whether the device supports people counting ANR-->true
        </isSupportCountingCollection>
        <isSupportHeatmapCollection>
          <!--ro, opt, bool, whether the device supports heat map data ANR-->true
        </isSupportHeatmapCollection>
        <channelFlexible opt="name,online,linknum">
          <!--ro, opt, string, this node is located after the node <isSupportHeatmapCollection>., attr:opt{opt, string}-->true
        </channelFlexible>
        <isSupportOutputsResource>
          <!--ro, opt, bool-->true
        </isSupportOutputsResource>
        <isSupportMixedChannel>
          <!--ro, opt, bool, this node is located after the node <channelFlexible>.-->true
        </isSupportMixedChannel>
        <isSupportMixedChannelStatus>
          <!--ro, opt, bool, this node is located after the node <isSupportMixedChannel>.-->true
        </isSupportMixedChannelStatus>
        <isSupportRecodStatus>
          <!--ro, opt, bool, whether the device supports searching for the recording status-->true
        </isSupportRecodStatus>
        <isSupportPrivacyMask>
          <!--ro, opt, bool, whether the device supports privacy mask-->true
        </isSupportPrivacyMask>
        <isSupportBinocularPreviewSwitch>
          <!--ro, opt, bool, whether the device supports switching live view of the dual-lens camera-->true
        </isSupportBinocularPreviewSwitch>
        <isSupportCalibCheck>
          <!--ro, opt, bool, whether the device supports calibration verification-->true
        </isSupportCalibCheck>
        <isSupportPIP>
          <!--ro, opt, bool, whether the device supports PIP-->true
        </isSupportPIP>
        <isSupportFocusVideoMode>
          <!--ro, opt, bool, whether the device supports the focus mode of the video-->true
        </isSupportFocusVideoMode>
        <isSupportExternalChannel>
          <!--ro, opt, bool, whether the device supports extended analog channels-->true
        </isSupportExternalChannel>
        <isSupportCompressedTranscode>
          <!--ro, opt, bool, whether the device supports configuring the encoding compression of transcoding stream, desc:related URL:
/ISAPI/Streaming/channels/<channelID>/TranscodeCfg/capabilities?format=json-->true
        </isSupportCompressedTranscode>
        <isSupportCompressedStream>
          <!--ro, opt, bool, whether the device supports transcoding compressed stream, desc:it is valid when isSupportCompressedTranscode is true. The device
supports getting real-time stream and real-time compressed stream. The streaming channel stream type can be obtained via id when the serviceType is
transcode (URL: /ISAPI/ContentMgmt/StreamingProxy/channels)-->true
        </isSupportCompressedStream>
        <isSupportLocalStreamCfg>
          <!--ro, opt, bool, whether the device supports configuring the streaming parameters of local device, desc:related URL:
/ISAPI/System/Video/inputs/channels/<channelID>/LocalStreamCfg/capabilities?format=json-->true
        </isSupportLocalStreamCfg>
        <isSupportMenuStatus>
          <!--ro, opt, bool, whether it supports getting the menu status, desc:corresponding URI: /ISAPI/System/Video/Menu/<indexID>/status?format=json, the
range of indexID comes from VideoCap.menuNums-->true
        </isSupportMenuStatus>
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      <isSupportVideoOutputMode>
        <!--ro, opt, bool, whether the device supports video output pattern parameters-->true
      </isSupportVideoOutputMode>
      <pictureChannelList>
        <!--ro, opt, array, subType:object-->
        <channelID>
          <!--ro, opt, int, channel No.-->0
        </channelID>
      </pictureChannelList>
      <transcodeStreamType opt="previewTranscode,recordTranscode">
        <!--ro, opt, string, attr:opt{req, string}-->true
      </transcodeStreamType>
      <isSupportImportPanoramaIOverlayPicture>
        <!--ro, opt, bool, POST /ISAPI/System/Video/channels/<channelID>/ImportPanoramaIOverlayPicture?format=json, desc:POST
/ISAPI/System/Video/channels/<channelID>/ImportPanoramaIOverlayPicture?format=json-->true
      </isSupportImportPanoramaIOverlayPicture>
      <isSupportPanoramaIOverlayPictureParams>
        <!--ro, opt, bool, GET or PUT /ISAPI/System/Video/channels/<channelID>/PanoramaIOverlayPictureParams?format=json, desc:GET or PUT
/ISAPI/System/Video/channels/<channelID>/PanoramaIOverlayPictureParams?format=json-->true
      </isSupportPanoramaIOverlayPictureParams>
    </VideoCap>
    <AudioCap>
      <!--ro, opt, object, audio encoding capability, desc:related URL: /ISAPI/System/Audio/capabilities-->
      <audioInputNums>
        <!--ro, req, int, number of audio inputs-->1
      </audioInputNums>
      <audioOutputNums>
        <!--ro, req, int, number of audio outputs-->1
      </audioOutputNums>
      <mixAudioInSet>
        <!--ro, opt, bool, N/A-->true
      </mixAudioInSet>
      <mixAudioOutSet>
        <!--ro, opt, bool, N/A-->true
      </mixAudioOutSet>
      <isSupportAudioMixing>
        <!--ro, opt, bool, whether the device supports audio mixing-->true
      </isSupportAudioMixing>
      <isSupportRemoveFrequencyPoint>
        <!--ro, opt, bool, whether the device supports clearing the howling frequency point-->true
      </isSupportRemoveFrequencyPoint>
      <isSupportAudioInConfig>
        <!--ro, opt, bool, whether the device supports configuring parameters of all audio inputs, desc:related URI:
/ISAPI/System/Audio/AudioIn/capabilities-->true
      </isSupportAudioInConfig>
      <muteDuringPanTilt opt="MicIn">
        <!--ro, opt, string, whether the device supports muting during motion, attr:opt{opt, string}, desc:muteDuringPanTilt in
/ISAPI/System/TwoWayAudio/channels/ID-->true
      </muteDuringPanTilt>
      <isSupportAudioOutConfig>
        <!--ro, opt, bool, whether the device supports configuring parameters of all audio outputs, desc:related URI:
/ISAPI/System/Audio/AudioOut/capabilities?format=json-->true
      </isSupportAudioOutConfig>
      <isSupportSAIP>
        <!--ro, opt, bool, whether the device supports connecting to the independent audio track channel-->true
      </isSupportSAIP>
      <SAIPChannelsNum>
        <!--ro, opt, int, number of devices that supports connecting to the independent audio track channel, dep:and,{$.AudioCap.isSupportSAIP,eq,true}-->1
      </SAIPChannelsNum>
      <isSupportDataStorage>
        <!--ro, opt, bool, whether the device supports uploading the audio file, desc:related URL: /ISAPI/System/Audio/dataStorage?format=json-->true
      </isSupportDataStorage>
      <isSupportDataRelease>
        <!--ro, opt, bool, whether the device supports releasing the audio file, desc:related URL: /ISAPI/System/Audio/dataRelease?format=json-->true
      </isSupportDataRelease>
      <isSupportAudioPlan>
        <!--ro, opt, bool, whether the device supports the audio scheme plan, desc:related URL: /ISAPI/System/Audio/plan?format=json-->true
      </isSupportAudioPlan>
      <isSupportAudioTask>
        <!--ro, opt, bool, whether the device supports the audio scheme task, desc:related URL: /ISAPI/System/Audio/plan/<PlanID>/task?format=json-->true
      </isSupportAudioTask>
      <isSupportaudioCollection>
        <!--ro, opt, bool, whether the device supports collecting audio files, desc:related URL: /ISAPI/System/Audio/audioCollection/capabilities?
format=json; maintenance function, it is used for collecting audio to analyze audio questions-->true
      </isSupportaudioCollection>
      <isSupportWirelessPickUp>
        <!--ro, opt, bool, whether the device supports connecting wireless pickup, desc:related URL: /ISAPI/System/Audio/wirelessPickUp/capabilities?
format=json-->true
      </isSupportWirelessPickUp>
      <audioOutputBroadcastNums opt="1,2,3,4">
        <!--ro, opt, int, number of audio broadcast outputs, attr:opt{opt, string}, desc:at present, this node is only supported by broadcast audio devices
when opt="1,2,3,4" is returned and , the output depends on the actual node; when not returned, the output starts from 1 by default according to business
type, the audio devices classify the output ports into broadcast output port and audio output port considering the start No. of every output starts from 1,
it should return the specific number of supported broadcast outputs if this node is returned, it indicates that the device supports the audio output volume
configuration; otherwise, the device does not support. Corresponding URI: /ISAPI/System/Audio/AudioOut/broadcast/capabilities?format=json-->0
      </audioOutputBroadcastNums>
      <isSupportAudioOutMixing>
        <!--ro, opt, bool, whether the device supports outputting audio mix, desc:related URL: /ISAPI/System/Audio/AudioOut/mixing/capabilities?format=json-
->true
      </isSupportAudioOutMixing>
      <isSupportRTSPAudio>
        <!--ro, opt, bool, whether the device supports listening to the audio transmitted via RTSP in real time, desc:RTSP URI:
/ISAPI/System/Audio/AudioIn/channels/<audioID>-->true
      </isSupportRTSPAudio>
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      <isSupportRTSPAudioPlaybackByFile>
        <!--ro, opt, bool, whether the device supports playing back audio transmitted via RTSP according to files, desc:the RTSP streaming URL is
playbackURI returned by /ISAPI/ContentMgmt/recordAudio/search?format=json-->true
      </isSupportRTSPAudioPlaybackByFile>
      <isSupportDeviceAudioCapture>
        <!--ro, opt, bool, whether the device supports audio collection, desc:GET/PUT /ISAPI/System/Audio/DeviceAudioCapture?format=json-->true
      </isSupportDeviceAudioCapture>
    </AudioCap>
    <isSupportHolidy>
      <!--ro, opt, bool, this node is located after the node <AudioCap>-->true
    </isSupportHolidy>
    <RebootConfigurationCap>
      <!--ro, opt, object, capability of configuring parameters for rebooting the device, desc:related URI: /ISAPI/System/reboot/capabilities. This node is
located before the node <isSupportSubscribeEvent> and after the node <isSupportHolidy>-->
      <Genetec>
        <!--ro, opt, bool, whether to enable the Genetec protocol-->true
      </Genetec>
      <ONVIF>
        <!--ro, opt, bool, whether to enable the ONVIF protocol-->true
      </ONVIF>
      <RTSP>
        <!--ro, opt, bool, whether to enable the RTSP protocol-->true
      </RTSP>
      <HTTP>
        <!--ro, opt, bool, whether to enable the HTTP protocol-->true
      </HTTP>
      <SADP>
        <!--ro, opt, object, SADP (Search Active Devices Protocol, including the link layer and multicast discovery)-->
        <ISDiscoveryMode>
          <!--ro, opt, bool, mode of the SADP multicast discovery protocol-->true
        </ISDiscoveryMode>
        <PcapMode>
          <!--ro, opt, bool, mode of the SADP link layer discovery protocol-->true
        </PcapMode>
      </SADP>
      <IPCAddStatus>
        <!--ro, opt, bool, IP address status-->true
      </IPCAddStatus>
    </RebootConfigurationCap>
    <isSupportExternalDevice>
      <!--ro, opt, bool, whether the device supports connecting to peripherals: true (support), desc:related URI: /ISAPI/System/externalDevice/capabilities-
->true
    </isSupportExternalDevice>
    <isSupportSoundCfg>
      <!--ro, opt, bool, whether the device supports the sound configuration, desc:related URI: /ISAPI/System/SoundCfg/capabilities-->true
    </isSupportSoundCfg>
    <isSupportSubscribeEvent>
      <!--ro, opt, bool, whether the device supports subscribing to events, desc:related URI: /ISAPI/Event/notification/subscribeEventCap-->true
    </isSupportSubscribeEvent>
    <isSupportSubscribeIOTInfo>
      <!--ro, opt, bool, whether the device supports subscribing to the IoT information-->true
    </isSupportSubscribeIOTInfo>
    <isSupportDiagnosedData>
      <!--ro, opt, bool, whether the device supports exporting the device diagnosis data, desc:true (support), this node is not returned (not support).
Related URI: /ISAPI/System/diagnosedData-->true
    </isSupportDiagnosedData>
    <isSupportChangedUpload>
      <!--ro, opt, bool, whether the device supports uploading changes-->true
    </isSupportChangedUpload>
    <isSupportGettingWorkingStatus>
      <!--ro, opt, bool, whether the device supports getting the device status-->true
    </isSupportGettingWorkingStatus>
    <isSupportGettingChannelInfoByCondition>
      <!--ro, opt, bool, whether the device supports getting the channel information by condition-->true
    </isSupportGettingChannelInfoByCondition>
    <isSupportDiagnosedDataParameter>
      <!--ro, opt, bool, whether the device supports enabling or disabling exporting the diagnosis information-->true
    </isSupportDiagnosedDataParameter>
    <isSupportSimpleDevStatus>
      <!--ro, opt, bool, whether the device supports getting the device's simple status, desc:related URI: /ISAPI/System/workingstatus?format=json-->true
    </isSupportSimpleDevStatus>
    <isSupportFlexible>
      <!--ro, opt, bool, whether the device supports the flexible combination configuration. Related URI: /ISAPI/System/Video/inputs/channels?format=json--
>true
    </isSupportFlexible>
    <isSupportPTZChannels>
      <!--ro, opt, bool, whether the device supports returning the PTZ channel (it is distinguished from the video channel)-->true
    </isSupportPTZChannels>
    <isSupportMetadata>
      <!--ro, opt, bool, whether the device supports metadata, desc:related URI: /ISAPI/Streaming/channels/<ID>/metadata/capabilities-->true
    </isSupportMetadata>
    <isSupportTimeCap>
      <!--ro, opt, bool, whether the device supports the time configuration, desc:related URI: /ISAPI/System/time/capabilities-->true
    </isSupportTimeCap>
    <isSupportThermalStreamData>
      <!--ro, opt, bool, whether the device supports uploading thermal imaging stream data in real time-->true
    </isSupportThermalStreamData>
    <isSupportShutdown>
      <!--ro, opt, bool, whether the device supports the shutdown configuration, desc:related URI: /ISAPI/System/shutdown?format=json-->true
    </isSupportShutdown>
    <isSupportConsumptionMode>
      <!--ro, opt, bool, whether the device supports switching the consumption mode, desc:related URI: /ISAPI/System/consumptionMode/capabilities?
format=json-->true
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    </isSupportConsumptionMode>
    <isSupportManualPowerConsumption>
      <!--ro, opt, bool, whether the device supports controlling the device consumption manually-->true
    </isSupportManualPowerConsumption>
    <isSupportPostUpdateFirmware>
      <!--ro, opt, bool, whether the device supports upgrading firmware by POST method-->true
    </isSupportPostUpdateFirmware>
    <isSupportPostConfigData>
      <!--ro, opt, bool, whether the device supports importing and exporting the configuration file by POST method-->true
    </isSupportPostConfigData>
    <isSupportUserLock>
      <!--ro, opt, bool, whether the device supports locking the user, desc:related URI: /ISAPI/System/userLock/config/capabilities?format=json-->true
    </isSupportUserLock>
    <isSupportModuleLock>
      <!--ro, opt, bool, whether the device supports locking the module, desc:related URI: /ISAPI/System/moduleLock-->true
    </isSupportModuleLock>
    <supportSmartOverlapChannles opt="1">
      <!--ro, opt, bool, whether the device supports configuring the smart event stream, attr:opt{opt, string}, desc:if this function is supported, this
node and the corresponding channel ID will be returned; otherwise, this node will not be returned-->true
    </supportSmartOverlapChannles>
    <isSupportNetworkCapture>
      <!--ro, opt, bool, whether the device supports capturing network packets, desc:this node is used to determine whether the device supports capturing
communication packets transmitted between the device and the external. Related URI: /ISAPI/System/networkCapture/capabilities?format=json-->true
    </isSupportNetworkCapture>
    <isSupportOperationRecord>
      <!--ro, opt, bool, whether the device supports the device operation records, desc:the charging voltage and monthly flow can be recorded in the device
in real-time low-consumption mode and full-consumption mode. Related URI: /ISAPI/System/operationRecord/capabilities?format=json-->true
    </isSupportOperationRecord>
    <isSupportSearchModuleLock>
      <!--ro, opt, bool, whether the device support search module locking parameters, desc:/ISAPI/System/moduleLock/searchconfig?format=json-->true
    </isSupportSearchModuleLock>
    <devSupportUpdateVersion>
      <!--ro, opt, string, supported version of local device upgrade, desc:the device returns the upgrade version to guide tool users (e.g. service
technician) to select an upgrade package for local upgrade-->1
    </devSupportUpdateVersion>
    <notSupportCatalogUpgrade>
      <!--ro, opt, bool-->true
    </notSupportCatalogUpgrade>
  </SysCap>
  <voicetalkNums>
    <!--ro, opt, int, number of two-way audio channels-->2
  </voicetalkNums>
  <isSupportSnapshot>
    <!--ro, opt, bool, whether the device supports capturing pictures-->true
  </isSupportSnapshot>
  <SecurityCap>
    <!--ro, opt, object, encryption capability set-->
    <supportUserNums>
      <!--ro, opt, int, supported maximum number of users-->1
    </supportUserNums>
    <userBondIpNums>
      <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->1
    </userBondIpNums>
    <userBondMacNums>
      <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->1
    </userBondMacNums>
    <isSupCertificate>
      <!--ro, opt, bool, whether the device supports authentication-->true
    </isSupCertificate>
    <issupIllegalLoginLock>
      <!--ro, opt, bool, whether the device supports locking login-->true
    </issupIllegalLoginLock>
    <isSupportOnlineUser>
      <!--ro, opt, bool, whether the device supports the online user configuration-->true
    </isSupportOnlineUser>
    <isSupportAnonymous>
      <!--ro, opt, bool, whether the device supports anonymous login-->true
    </isSupportAnonymous>
    <isSupportStreamEncryption>
      <!--ro, opt, bool, whether the device supports stream encryption-->true
    </isSupportStreamEncryption>
    <securityVersion opt="1,2,3,4,7">
      <!--ro, opt, string, encryption capability, attr:opt{opt, string}, desc:the encryption capability of each version consists of two parts: encryption
algorithm and the range of encrypted nodes. Currently 1 refers to AES128 encryption and 2 refers to AES256 encryption. The range of encrypted nodes is
described in each protocol-->test
    </securityVersion>
    <keyIterateNum>
      <!--ro, opt, int, iteration times, desc:the value is usually between 100 and 1000-->100
    </keyIterateNum>
    <isSupportUserCheck>
      <!--ro, opt, bool, whether the device supports verifying the login password when editing (editing/adding/deleting) user parameters, desc:this node
depends on the node <securityVersion>, which means that it is only valid for the versions that support encrypting the sensitive information-->true
    </isSupportUserCheck>
    <isSupportGUIDFileDataExport>
      <!--ro, opt, bool, whether the device supports exporting the device's GUID file-->true
    </isSupportGUIDFileDataExport>
    <isSupportSecurityQuestionConfig>
      <!--ro, opt, bool, whether the device supports answering security questions-->true
    </isSupportSecurityQuestionConfig>
    <supportSecurityNode opt="wirelessServer,wirelessDial">
      <!--ro, opt, string, additional node for supporting sensitive information encryption, attr:opt{opt, string}, desc:in the earlier version of the
solution for encrypting the sensitive information via ISAPI (i.e., the solution for transmitting plaintext via ISAPI), some URIs containing sensitive
information are left out. Now these URIs need to be encrypted for transmission. This capability node is added to be compatible with old devices-->test
    </supportSecurityNode>
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<isSupportGetOnlineUserListSC>
  <!--ro, opt, bool, whether the device supports searching the online user list-->true
</isSupportGetOnlineUserListSC>
<SecurityLimits>
  <!--ro, opt, object, capability of configuring security limit parameters-->
  <LoginPasswordLenLimit min="1" max="16">
    <!--ro, opt, string, length limit of the user's login password, attr:min{opt, string},max{opt, string}-->test
  </LoginPasswordLenLimit>
  <SecurityAnswerLenLimit min="1" max="128">
    <!--ro, opt, string, length limit of the security question's answer, attr:min{opt, string},max{opt, string}-->test
  </SecurityAnswerLenLimit>
</SecurityLimits>
<RSAKeyLength opt="512,1024,2048,3072" def="3072">
  <!--ro, opt, enum, HTTPS certificate length, subType:string, attr:opt{opt, string},def{opt, string}, desc:512, 1024, 2048-->3072
</RSAKeyLength>
<isSupportONVIFUserManagement>
  <!--ro, opt, bool, whether the device supports ONVIF user management-->true
</isSupportONVIFUserManagement>
<WebCertificateCap>
  <!--ro, opt, object, HTTP authentication capability-->
  <CertificateType opt="basic,digest,digest/basic">
    <!--ro, req, string, certificate type: basic (authentication), attr:opt{opt, string}-->test
  </CertificateType>
  <SecurityAlgorithm>
    <!--ro, opt, object, encryption algorithm, dep:or,{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,degist},
{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
    <algorithmType opt="MD5,SHA256,MD5/SHA256">
      <!--ro, opt, string, algorithm type, attr:opt{req, string}-->test
    </algorithmType>
  </SecurityAlgorithm>
</WebCertificateCap>
<isSupportConfigFileImport>
  <!--ro, opt, bool, whether the device supports importing the configuration file-->true
</isSupportConfigFileImport>
<isSupportConfigFileExport>
  <!--ro, opt, bool, whether the device supports exporting the configuration file-->true
</isSupportConfigFileExport>
<cfgFileSecretKeyLenLimit min="0" max="16">
  <!--ro, opt, string, length limit of the configuration file's verification key, attr:min{opt, string},max{opt, string}-->0
</cfgFileSecretKeyLenLimit>
<supportIPCActivatePassword>
  <!--ro, opt, bool, whether the device supports configuring the password for activating the network camera-->true
</supportIPCActivatePassword>
<isIrreversible>
  <!--ro, opt, bool, whether the device supports irreversible password storage, desc:if this function is not supported, the plaintext password of the
user information will be stored in the device; otherwise, the password will be hashed for storage in the device-->true
</isIrreversible>
<salt>
  <!--ro, opt, string, the specific salt used by the user to log in-->test
</salt>
<isSupportOnvifInfo>
  <!--ro, opt, bool, whether the device supports getting the ONVIF protocol information-->true
</isSupportOnvifInfo>
<isSupportUserNamePasswordCheckUpgrade>
  <!--ro, opt, bool, whether the device supports upgrading the verification of the user name and password-->true
</isSupportUserNamePasswordCheckUpgrade>
<isSupportDeviceCertificatesManagement>
  <!--ro, opt, bool, whether the device supports certificate management-->true
</isSupportDeviceCertificatesManagement>
<isSupportSIPCertificatesManagement>
  <!--ro, opt, bool, whether the device supports SIP certificate management-->true
</isSupportSIPCertificatesManagement>
<isSupportDeviceSelfSignCertExport>
  <!--ro, opt, bool, whether the device supports exporting the self-signed certificate-->true
</isSupportDeviceSelfSignCertExport>
<isSupportSecurityEmail>
  <!--ro, opt, bool, whether the device supports configuring the security email-->true
</isSupportSecurityEmail>
<isSupportRTSPCertificate opt="true,false">
  <!--ro, opt, bool, whether the device supports RTSP authentication, attr:opt{opt, string}-->true
</isSupportRTSPCertificate>
<isSupportPictureURlCertificate opt="true,false">
  <!--ro, opt, bool, whether the device supports configuring parameters for picture URL authentication, attr:opt{opt, string}-->true
</isSupportPictureURlCertificate>
<isSupportGB35114Certificate>
  <!--ro, opt, bool, N/A-->true
</isSupportGB35114Certificate>
<isSupportAuthenticationMode opt="uKey,private">
  <!--ro, opt, enum, verification mode, subType:string, attr:opt{req, string}, desc:"uKey", "private"-->private
</isSupportAuthenticationMode>
<isSupportUnloggedUserPermissionConfig opt="true,false">
  <!--ro, opt, bool, whether the device supports configuring permission for users that have not logged in, attr:opt{opt, string}-->true
</isSupportUnloggedUserPermissionConfig>
<maxIllegalLoginTimes min="3" max="20" def="5">
  <!--ro, opt, string, maximum failed illegal login attempts, attr:min{opt, string},max{opt, string},def{opt, string}, desc:this node is valid when the
value of the node <issupportIllegalLoginLock> is true and will be returned when the maximum failed attempts of illegally logging in to the device reach the
limit-->1
</maxIllegalLoginTimes>
<SecurityAdvanced>
  <!--ro, opt, object, advanced security configuration-->
  <securityEnhanced>
    <!--ro, opt, bool, whether the device supports security reinforcement-->true
  </securityEnhanced>
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      <noOperationEnabled>
        <!--ro, req, bool, whether to enable the control timeout when there is no operation-->true
      </noOperationEnabled>
      <noOperationTime min="1" max="60" def="15">
        <!--ro, req, int, control timeout when there is no operation, unit:分钟, attr:min{opt, string},max{opt, string},def{opt, string}-->1
      </noOperationTime>
      <isSupportDigestStatus>
        <!--ro, opt, bool, digest status-->true
      </isSupportDigestStatus>
    </SecurityAdvanced>
    <LoginLinkNum>
      <!--ro, opt, object, number of users logging in at the same time-->
      <maxLinkNum min="1" max="128" def="50">
        <!--ro, req, int, number of controls logging in at the same time, attr:min{opt, string},max{opt, string},def{opt, string}-->64
      </maxLinkNum>
    </LoginLinkNum>
    <isSupportCCClientCertificate>
      <!--ro, opt, bool, whether the device supports the CC (Client Certificate) function-->true
    </isSupportCCClientCertificate>
    <passwordValidity min="0" max="365">
      <!--ro, opt, string, range of the password validity period, attr:min{opt, string},max{opt, string}, desc:unit: day. If this node is returned, it
indicates that the device supports configuring the password validity period, and only the administrator can edit the configuration-->1
    </passwordValidity>
    <isSupportResetChannelPassword>
      <!--ro, opt, bool, whether the device supports resetting the password of the channel-->true
    </isSupportResetChannelPassword>
    <DoubleVerificationCap>
      <!--ro, opt, object, secondary authentication capability-->
      <isSupportUsersConfig>
        <!--ro, opt, bool, whether the device supports configuring users for secondary authentication-->true
      </isSupportUsersConfig>
      <isSupportUsersPermissionConfig>
        <!--ro, opt, bool, whether the device supports configuring user permission for secondary authentication-->true
      </isSupportUsersPermissionConfig>
      <isSupportFactoryReset>
        <!--ro, opt, bool, whether the device supports restoring secondary authentication to defaults-->true
      </isSupportFactoryReset>
      <factoryResetMode opt="full,basic,other">
        <!--ro, opt, string, attr:opt{req, string}-->full
      </factoryResetMode>
      <isSupportQuestionConfiguration>
        <!--ro, opt, bool, whether it supports configuring a single security question for secondary authentication, desc:PUT
/ISAPI/Security/questionConfiguration/<indexID>-->true
      </isSupportQuestionConfiguration>
    </DoubleVerificationCap>
    <isSupportUserSearch>
      <!--ro, opt, bool, whether the device supports searching users-->true
    </isSupportUserSearch>
    <keypadPassword min="1" max="10">
      <!--ro, opt, int, keypad password length, attr:min{req, int},max{req, int}, desc:when the password length of different user types is different, this
node only indicates the keypad password length of the administrator; otherwise, the keypad password length of all user types is the same and defined by the
value of this node-->1
    </keypadPassword>
    <installerKeypadPassword min="1" max="10">
      <!--ro, opt, int, keypad password length of the installer, attr:min{req, int},max{req, int}-->1
    </installerKeypadPassword>
    <operatorKeypadPassword min="1" max="10">
      <!--ro, opt, int, keypad password length of the operator, attr:min{req, int},max{req, int}-->1
    </operatorKeypadPassword>
    <userOperateType opt="1,2,3">
      <!--ro, opt, enum, user type: 1 (network user), subType:int, attr:opt{req, string}, desc:1 (network user), 2 (keypad user), 3 (network user and keypad
user)-->1
    </userOperateType>
    <isSptUserEnabled>
      <!--ro, opt, bool, whether the device supports configuring parameters for enabling the user-->true
    </isSptUserEnabled>
    <isSptAdminCap>
      <!--ro, opt, bool, whether the device supports getting the admin permission-->true
    </isSptAdminCap>
    <isSupportBackLogCfg>
      <!--ro, opt, bool, whether the device supports configuring basic parameters of the background log-->true
    </isSupportBackLogCfg>
    <isSupportBackLogExport>
      <!--ro, opt, bool, whether the device supports exporting the background log-->true
    </isSupportBackLogExport>
    <isSupportModifyInnerPassword opt="true,false">
      <!--ro, opt, bool, whether the device supports editing the built-in password, attr:opt{req, string}, desc:the device's built-in passwords include the
database password, system account password and so on-->true
    </isSupportModifyInnerPassword>
    <isSupportEncryptCertificate>
      <!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
    </isSupportEncryptCertificate>
    <maxIllegalLoginLockTime min="1" max="120" def="30">
      <!--ro, opt, int, lock time range, range:[1,120], unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
    </maxIllegalLoginLockTime>
    <cloudSensitiveInfoEncryptType>
      <!--ro, opt, string, encryption type of the cloud sensitive information, desc:the current device returns the value "2.1", which means that the method
of generating the key for encrypting the sensitive information is SHA256(calc_sha256(user, salt, SHA256(user+salt2+SHA256(user+salt+password)))+
AaBbCcDd1234!@#$).-->2.1
    </cloudSensitiveInfoEncryptType>
    <isSupportCertificateCustomID>
      <!--ro, opt, bool, whether the device supports using the user's custom ID to configure the certificate-->true
    </isSupportCertificateCustomID>
    <isSupportResetLANPassword>
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<isSupportResetLANPassword>
<!--ro, opt, bool, whether the device supports resetting the device's password in the LAN, desc:related URI: /ISAPI/Security/resetPassword/certification?format=json-->true
</isSupportResetLANPassword>
<isSupportPicHttpsCertificatesManagement>
<!--ro, opt, bool, whether the device supports managing the HTTPS certificate for pictures, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned-->true
</isSupportPicHttpsCertificatesManagement>
<isSupportAKSKImport>
<!--ro, opt, bool, whether the device supports importing AKSK, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned-->true
</isSupportAKSKImport>
<isSupportAKSKExport>
<!--ro, opt, bool, whether the device supports exporting AKSK, desc:If this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned.-->true
</isSupportAKSKExport>
<isSupportAKSKReset>
<!--ro, opt, bool, whether the device supports resetting AKSK, desc:If this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned.-->true
</isSupportAKSKReset>
<ExportCertificateFile>
<!--ro, opt, object, whether the device supports exporting the certificate, desc:related URI: POST /ISAPI/Security/serverCertificate/exportCertificateFile-->
</ExportCertificateFile>
<isSupportGB351114FDWSFCertificate>
<!--ro, opt, bool, whether the device supports FDWSF certificate management, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URIs: /ISAPI/Security/GB351114/FDWSFCertificate/capabilities?format=json, POST /ISAPI/Security/GB351114/FDWSFCertificate?format=json, GET /ISAPI/Security/GB351114/FDWSFCertificate?format=json, and DELETE /ISAPI/Security/GB351114/FDWSFCertificate?format=json.-->true
</isSupportGB351114FDWSFCertificate>
<isSupportMediaObjectEncryption>
<!--ro, opt, bool, whether the device supports setting encryption parameters of the private target media data for the specified video channel, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/MediaObjectEncryption/capabilities?format=json-->true
</isSupportMediaObjectEncryption>
<isSupportDisclaimer>
<!--ro, opt, bool, whether support configuring the product disclaimer, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/disclaimer/capabilities?format=json-->true
</isSupportDisclaimer>
<isSupportSoftwareLicense>
<!--ro, opt, bool, whether support configuring the product open source statement, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/softwareLicense?format=json-->true
</isSupportSoftwareLicense>
<isSupportInstallerCap>
<!--ro, opt, bool, whether the device supports getting the installer permission, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/UserPermission/installer/capabilities?format=json-->true
</isSupportInstallerCap>
<isSupportResetPasswordFile>
<!--ro, opt, bool, whether the device supports exporting and importing the secret key file of device in the LAN, desc:Related URI: /ISAPI/Security/resetPassword/keyFile/capabilities?format=json-->true
</isSupportResetPasswordFile>
<isNotSupportLogSeverCertificate>
<!--ro, opt, bool, whether the device support importing log server certificate, desc:related URI: /ISAPI/Security/serverCertificate/certificate-->true
</isNotSupportLogSeverCertificate>
<isSupportLocalLoginCfg>
<!--ro, opt, bool, whether it supports the capability of device local login parameters, desc:GET/PUT /ISAPI/Security/LocalLoginCfg?format=json-->true
</isSupportLocalLoginCfg>
<isSupportGetProtocolFirewallRulesFileImportStatus>
<!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallRulesFileImportStatus?format=json-->true
</isSupportGetProtocolFirewallRulesFileImportStatus>
<isSupportImportProtocolFirewallRulesFile>
<!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/ImportProtocolFirewallRulesFile?format=json-->true
</isSupportImportProtocolFirewallRulesFile>
<isSupportGetProtocolFirewallEngineUpgradeStatus>
<!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallEngineUpgradeStatus?format=json-->true
</isSupportGetProtocolFirewallEngineUpgradeStatus>
<isSupportUpgradeProtocolFirewallEngine>
<!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/UpgradeProtocolFirewallEngine?format=json-->true
</isSupportUpgradeProtocolFirewallEngine>
<isSupportProtocolFirewallVersionInfo>
<!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/ProtocolFirewallVersionInfo?format=json-->true
</isSupportProtocolFirewallVersionInfo>
<isSupportProtocolFirewallParams>
<!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/Security/ProtocolFirewallParams?format=json-->true
</isSupportProtocolFirewallParams>
<isSupportResetPasswordBySecurityPhoneNo>
<!--ro, opt, bool, whether it supports resetting device password with phone number verification, desc:1. attendance check devices support applying a preset phone number during device activation. The user can reset the device password with the preset phone number, which is similar with password reset via email. The phone number can be changed on the website or local UI of the device
2. if the node is not returned or the returned value is false, it indicates "no"-->true
</isSupportResetPasswordBySecurityPhoneNo>
<isSupportGetIntrusionPreventionEngineUpgradeStatus>
<!--ro, opt, bool, whether the device supports getting the upgrade status of intrusion prevention engine library, desc:GET /ISAPI/Security/GetIntrusionPreventionEngineUpgradeStatus?format=json-->true
</isSupportGetIntrusionPreventionEngineUpgradeStatus>
<isSupportIntrusionPreventionEngineVersionInfo>
<!--ro, opt, bool, whether the device supports version information of intrusion prevention engine, desc:GET /ISAPI/Security/IntrusionPreventionEngineVersionInfo?format=json-->true
</isSupportIntrusionPreventionEngineVersionInfo>
<isSupportIntrusionPreventionParams>
<!--ro, opt, bool, whether the device supports intrusion prevention parameters, desc:GET/PUT /ISAPI/Security/IntrusionPreventionParams?format=json-->true
</isSupportIntrusionPreventionParams>
<isSupportUpgradeIntrusionPreventionEngine>

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      <!--ro, opt, bool, whether the device supports upgrading intrusion prevention engine library, desc:PUT
/ISAPI/Security/UpgradeIntrusionPreventionEngine?format=json-->true
    </isSupportUpgradeIntrusionPreventionEngine>
    <notSupportImportCertificateType opt="CA,CCClentCertificate,GB35114,GB35114Service,https">
      <!--ro, opt, string, file type that cannot be imported, attr:opt{req, string}, desc:if this node does not exist, all supported certificate types can
be imported-->test
    </notSupportImportCertificateType>
    <isSupportPasswordProtection>
      <!--ro, opt, bool, whether it supports parameter configuration of SDK protocol authentication mode, desc:PUT/GET
/ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
    </isSupportPasswordProtection>
  </SecurityCap>
  <EventCap>
    <!--ro, opt, object, event capability-->
    <isSupportHDFull>
      <!--ro, opt, bool, whether the device supports HDD full alarm-->true
    </isSupportHDFull>
    <isSupportHDError>
      <!--ro, opt, bool, whether the device supports HDD error alarm-->true
    </isSupportHDError>
    <isSupportHDBadBlock>
      <!--ro, opt, bool-->true
    </isSupportHDBadBlock>
    <isSupportNicBroken>
      <!--ro, opt, bool, whether the device supports the network disconnected event-->true
    </isSupportNicBroken>
    <isSupportIpConflict>
      <!--ro, opt, bool, whether the device supports the IP address conflict event-->true
    </isSupportIpConflict>
    <isSupportIllAccess>
      <!--ro, opt, bool, whether the device supports the illegal login event-->true
    </isSupportIllAccess>
    <isSupportViException>
      <!--ro, opt, bool, whether the device supports video input exception (only for analog channels)-->true
    </isSupportViException>
    <isSupportViMismatch>
      <!--ro, opt, bool, whether the device supports the event of video standard mismatch-->true
    </isSupportViMismatch>
    <isSupportRecordException>
      <!--ro, opt, bool, whether the device supports the recording exception event-->true
    </isSupportRecordException>
    <isSupportRaidException>
      <!--ro, opt, bool, whether the device supports the array exception event-->true
    </isSupportRaidException>
    <isSupportViResMismatch>
      <!--ro, opt, bool, whether the device supports the event of video resolution mismatch-->true
    </isSupportViResMismatch>
    <isSupportPOCException>
      <!--ro, opt, bool, whether the device supports the event of powering over coaxial exception-->true
    </isSupportPOCException>
    <isSupportSmartDetection>
      <!--ro, opt, bool, whether the device supports the Smart event-->true
    </isSupportSmartDetection>
    <isSupportSpareException>
      <!--ro, opt, bool, whether the device supports hot spare exception-->true
    </isSupportSpareException>
    <isSupportPoePowerException>
      <!--ro, opt, bool, whether the device supports the POE exception event-->true
    </isSupportPoePowerException>
    <isSupportTriggerFocus>
      <!--ro, opt, bool, whether the device supports defocus exception-->true
    </isSupportTriggerFocus>
    <isSupportMotionDetection>
      <!--ro, opt, bool, whether the device supports motion detection-->true
    </isSupportMotionDetection>
    <isSupportVideoLoss>
      <!--ro, opt, bool, whether the device supports video loss events-->true
    </isSupportVideoLoss>
    <isSupportTamperDetection>
      <!--ro, opt, bool, whether the device supports tampering detection-->true
    </isSupportTamperDetection>
    <isSupportFireDetection>
      <!--ro, opt, bool, whether the device supports fire detection events-->true
    </isSupportFireDetection>
    <isSupportRegionEntrance>
      <!--ro, opt, bool, whether the device supports region entrance events-->true
    </isSupportRegionEntrance>
    <isSupportRegionExiting>
      <!--ro, opt, bool, whether the device supports region exiting events-->true
    </isSupportRegionExiting>
    <isSupportLoitering>
      <!--ro, opt, bool, whether the device supports loitering events-->true
    </isSupportLoitering>
    <isSupportGroup>
      <!--ro, opt, bool, whether the device supports people gathering events-->true
    </isSupportGroup>
    <isSupportRapidMove>
      <!--ro, opt, bool, whether the device supports fast moving events-->true
    </isSupportRapidMove>
    <isSupportParking>
      <!--ro, opt, bool, whether the device supports parking detection events-->true
    </isSupportParking>
    <isSupportUnattendedBaggage>
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    <!--ro, opt, bool, whether the device supports unattended baggage events-->true
  </isSupportUnattendedBaggage>
  <isSupportAttendedBaggage>
    <!--ro, opt, bool, whether the device supports object removal events-->true
  </isSupportAttendedBaggage>
  <isSupportHumanAttribute>
    <!--ro, opt, bool, whether the device supports human body features-->true
  </isSupportHumanAttribute>
  <isSupportHumanRecognition>
    <!--ro, opt, bool, whether the device supports target capture events-->true
  </isSupportHumanRecognition>
  <isSupportFaceContrast>
    <!--ro, opt, bool, whether the device supports face picture comparison events-->true
  </isSupportFaceContrast>
  <isSupportFaceLib>
    <!--ro, opt, bool, Front-end devices support linkage by face picture library., desc:related URI: /ISAPI/Event/schedules/faceLib/<ID>/<FDID>-->true
  </isSupportFaceLib>
  <isSupportBlackListFaceContrast>
    <!--ro, opt, bool, whether the device supports events of blocklist face picture comparison-->true
  </isSupportBlackListFaceContrast>
  <isSupportWhiteListFaceContrast>
    <!--ro, opt, bool, whether the device supports events of allowlist face picture comparison-->true
  </isSupportWhiteListFaceContrast>
  <isSupportillegalParking>
    <!--ro, opt, bool, whether the device supports illegal parking detection-->true
  </isSupportillegalParking>
  <isSupportPedestrian>
    <!--ro, opt, bool, whether the device supports pedestrian detection-->true
  </isSupportPedestrian>
  <isSupportTrafficAccident>
    <!--ro, opt, bool, whether the device supports traffic incident detection-->true
  </isSupportTrafficAccident>
  <isSupportConstruction>
    <!--ro, opt, bool, whether the device supports construction detection-->true
  </isSupportConstruction>
  <isSupportRoadBlock>
    <!--ro, opt, bool, whether the device supports roadblock detection-->true
  </isSupportRoadBlock>
  <isSupportAbandonedObject>
    <!--ro, opt, bool, whether the device supports fallen object detection-->true
  </isSupportAbandonedObject>
  <isSupportParallelParking>
    <!--ro, opt, bool, whether the device supports parallel parking detection-->true
  </isSupportParallelParking>
  <isSupportParkingState>
    <!--ro, opt, bool, whether the device supports parking space status detection-->true
  </isSupportParkingState>
  <isSupportTrafficJam>
    <!--ro, opt, bool, whether the device supports traffic jam detection (old version)-->true
  </isSupportTrafficJam>
  <isSupportCongestion>
    <!--ro, opt, bool, whether the device supports traffic jam detection-->true
  </isSupportCongestion>
  <isSupportStudentsStoodUp>
    <!--ro, opt, bool, whether the device supports the student standing up detection event-->true
  </isSupportStudentsStoodUp>
  <isSupportFramesPeopleCounting>
    <!--ro, opt, bool, whether the device supports the regional people counting event-->true
  </isSupportFramesPeopleCounting>
  <isSupportFaceSnap>
    <!--ro, opt, bool, whether the device supports face capture events-->true
  </isSupportFaceSnap>
  <isSupportPersonDensityDetection>
    <!--ro, opt, bool, whether the device supports people density detection-->true
  </isSupportPersonDensityDetection>
  <HFPD>
    <!--ro, opt, bool, whether the device supports frequently appeared person detection-->true
  </HFPD>
  <isSupportPersonQueueDetection>
    <!--ro, opt, bool, whether the device supports queue detection events-->true
  </isSupportPersonQueueDetection>
  <isSupportVehicleStatistics>
    <!--ro, opt, bool, whether the device supports vehicle statistics events-->true
  </isSupportVehicleStatistics>
  <isSupportFaceThermometry>
    <!--ro, opt, bool, whether the device supports face temperature screening-->true
  </isSupportFaceThermometry>
  <isSupportSafetyHelmetDetection>
    <!--ro, opt, bool, whether the device supports hard hat detection-->true
  </isSupportSafetyHelmetDetection>
  <isSupportPowerSupplyCabinet>
    <!--ro, opt, bool, whether the device supports cabinet powering detection-->true
  </isSupportPowerSupplyCabinet>
  <isSupportSensorDetection>
    <!--ro, opt, bool, whether the device supports sensor detection-->true
  </isSupportSensorDetection>
  <isSupportSignAbnormal>
    <!--ro, opt, bool, whether the device supports the physical sign exception detection-->true
  </isSupportSignAbnormal>
  <isSupportTeacherBehaviorDetect>
    <!--ro, opt, bool, whether the device supports teacher's behavior detection-->true
  </isSupportTeacherBehaviorDetect>
  <isSupportCityManagement>
    <!--ro, opt, bool, whether the device supports the city management function-->true
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    </isSupportCityManagement>
    <isSupportMixedTargetDetection>
      <!--ro, opt, bool, whether the device supports multi-target-type detection events-->true
    </isSupportMixedTargetDetection>
    <isSupportFaceSnapModeling>
      <!--ro, opt, bool, whether the device supports face modeling events-->true
    </isSupportFaceSnapModeling>
    <isSupportIntersectionAnalysis>
      <!--ro, opt, bool, whether the device supports the intersection analysis-->true
    </isSupportIntersectionAnalysis>
    <isSupportTriggerCapCheck>
      <!--ro, opt, bool, whether the device supports verifying the capability of triggering linkage actions, desc:this node indicates that the capability
message returned by calling the URI GET /ISAPI/Event/triggersCap has been verified, and the event linkage actions can be displayed exactly according to the
capability-->true
    </isSupportTriggerCapCheck>
    <isSupportLuma>
      <!--ro, opt, bool, VQD brightness exception-->true
    </isSupportLuma>
    <isSupportChroma>
      <!--ro, opt, bool, VQD image color cast-->true
    </isSupportChroma>
    <isSupportSnow>
      <!--ro, opt, bool, VQD snow noise-->true
    </isSupportSnow>
    <isSupportStreak>
      <!--ro, opt, bool, VQD stripe noise-->true
    </isSupportStreak>
    <isSupportFreeze>
      <!--ro, opt, bool, VQD image freeze-->true
    </isSupportFreeze>
    <isSupportSigLose>
      <!--ro, opt, bool, VQD signal loss-->true
    </isSupportSigLose>
    <isSupportClarity>
      <!--ro, opt, bool, VQD resolution exception-->true
    </isSupportClarity>
    <isSupportJitter>
      <!--ro, opt, bool, VQD image flutter-->true
    </isSupportJitter>
    <isSupportBlock>
      <!--ro, opt, bool, VQD video tampering-->true
    </isSupportBlock>
    <isSupportFlowers>
      <!--ro, opt, bool, VQD blurred image alarm-->true
    </isSupportFlowers>
    <isSupportNoise>
      <!--ro, opt, bool, VQD blurred screen-->true
    </isSupportNoise>
    <isSupportGhost>
      <!--ro, opt, bool, VQD image noise-->true
    </isSupportGhost>
    <isSupportPurple>
      <!--ro, opt, bool, VQD purple fringing-->true
    </isSupportPurple>
    <isSupportICR>
      <!--ro, opt, bool, VQD ICP-->true
    </isSupportICR>
    <isSupportProtectiveFilm>
      <!--ro, opt, bool, VQD protective film unremoved-->true
    </isSupportProtectiveFilm>
    <isSupportVoltageInstable>
      <!--ro, opt, bool, whether the device supports the voltage instable exception detection-->true
    </isSupportVoltageInstable>
    <isSupportCertificateRevocation>
      <!--ro, opt, bool, whether the device supports the certificate expiry alarm-->true
    </isSupportCertificateRevocation>
    <isSupportSysStorFull>
      <!--ro, opt, bool, whether the device supports the storage full alarm of the cluster system-->true
    </isSupportSysStorFull>
    <isSupportVersionAbnormal>
      <!--ro, opt, bool, whether the device supports cluster version exception-->true
    </isSupportVersionAbnormal>
    <isSupportNodeOffline>
      <!--ro, opt, bool, whether the device supports device offline detection-->true
    </isSupportNodeOffline>
    <isSupportRecordCycleAbnormal>
      <!--ro, opt, bool, whether the device supports the event of insufficient recording period-->true
    </isSupportRecordCycleAbnormal>
    <isSupportIpcTransferAbnormal>
      <!--ro, opt, bool, whether the device supports the event of network camera migration failure-->true
    </isSupportIpcTransferAbnormal>
    <isSupportDrivingBehaviorDetect>
      <!--ro, opt, bool, whether the device supports driving behavior detection-->true
    </isSupportDrivingBehaviorDetect>
    <isSupportWasteGasDetection>
      <!--ro, opt, bool, whether the device supports waste gas monitoring-->true
    </isSupportWasteGasDetection>
    <isSupportGreyScaleAlarm>
      <!--ro, opt, bool, whether the device supports the gray scale alarm-->true
    </isSupportGreyScaleAlarm>
    <isSupportPTLocking>
      <!--ro, opt, bool, whether the device supports the PTZ locking exception-->true
    </isSupportPTLocking>
```

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      <isSupportNoMaskDetection>
        <!--ro, opt, bool, whether the device supports not wearing mask detection-->true
      </isSupportNoMaskDetection>
      <isSupportTMPA>
        <!--ro, opt, bool, whether the device supports the warning linkage of face temperature screening-->true
      </isSupportTMPA>
      <RuleScheduleCap>
        <!--ro, opt, object, capability of configuring the arming schedule by rule-->
        <isSupportCityManagement>
          <!--ro, opt, bool, whether the device supports configuring the arming schedule by rule for the city management event-->true
        </isSupportCityManagement>
      </RuleScheduleCap>
      <RegionScheduleCap>
        <!--ro, opt, object, capability of configuring the arming schedule by area-->
        <isSupportPersonDensityDetection>
          <!--ro, opt, bool, whether the device supports configuring the arming schedule by area for the people density detection-->true
        </isSupportPersonDensityDetection>
        <isSupportPersonQueueDetection>
          <!--ro, opt, bool, whether the device supports configuring the arming schedule by area for the queue detection (area popularity analytics)-->true
        </isSupportPersonQueueDetection>
      </RegionScheduleCap>
      <isSupportThermalCalibrationFileException>
        <!--ro, opt, bool, whether the device supports the temperature calibration file exception-->true
      </isSupportThermalCalibrationFileException>
      <isSupportTemperatureIntervalMeasurement>
        <!--ro, opt, bool, whether the device supports the temperature range detection-->true
      </isSupportTemperatureIntervalMeasurement>
      <isSupportThermalVehicleDetection>
        <!--ro, opt, bool, whether the device supports thermal imaging vehicle detection-->true
      </isSupportThermalVehicleDetection>
      <isSupportWaterQualityDetection>
        <!--ro, opt, bool, whether the device supports water quality detection-->true
      </isSupportWaterQualityDetection>
      <isSupportWaterLevelDetection>
        <!--ro, opt, bool, whether the device supports water level detection-->true
      </isSupportWaterLevelDetection>
      <isSupportDeviceTemperatureAbnormal>
        <!--ro, opt, bool, whether the device supports temperature exception detection-->true
      </isSupportDeviceTemperatureAbnormal>
      <isSupportObjectAccumulationDetection>
        <!--ro, opt, bool, whether the device supports object accumulation detection-->true
      </isSupportObjectAccumulationDetection>
      <isSupportAppAbnormal>
        <!--ro, opt, bool, whether the device supports the App exception alarm or event-->true
      </isSupportAppAbnormal>
      <isSupportPTEventCfg>
        <!--ro, opt, bool, whether the device supports transparent transmission event configuration event, desc:/ISAPI/Event/PTEventCfg/capabilities?
format=json-->true
      </isSupportPTEventCfg>
      <isSupportLensTemperatureException>
        <!--ro, opt, bool, whether the device supports uploading lens abnormal temperature event-->true
      </isSupportLensTemperatureException>
      <isSupportPTZFXYFinish>
        <!--ro, opt, bool, the event of railway robot and PTZ going to the preset, desc:PTZFXYFinish-->true
      </isSupportPTZFXYFinish>
      <isSupportNormalTemperatureAlarm>
        <!--ro, opt, bool, whether the device supports normal temperature alarm-->true
      </isSupportNormalTemperatureAlarm>
      <isSupportLowTemperatureAlarm>
        <!--ro, opt, bool, whether the device supports low temperature alarm-->true
      </isSupportLowTemperatureAlarm>
      <isSupportLowCoolerLifeEvent>
        <!--ro, opt, bool, whether it supports short life exception event of refrigerating machine-->true
      </isSupportLowCoolerLifeEvent>
      <isSupportObstacleAvoidanceAlarm>
        <!--ro, opt, bool, whether it supports obstacle avoidance alarm-->true
      </isSupportObstacleAvoidanceAlarm>
      <isSupportPTZRotationAbnormal>
        <!--ro, opt, bool, whether it supports PTZ rotation exception event-->true
      </isSupportPTZRotationAbnormal>
      <isSupportBlackListVehContrast>
        <!--ro, opt, bool, whether the vehicle blocklist event is supported-->true
      </isSupportBlackListVehContrast>
      <isSupportWhiteListVehContrast>
        <!--ro, opt, bool, whether the vehicle allowlist event is supported-->true
      </isSupportWhiteListVehContrast>
    </EventCap>
    <ITCCap>
      <!--ro, opt, object, related node and URIs: the node <isSupportVehicleDetection> in /ISAPI/ITC/capability and /ISAPI/ITC/capabilities-->
      <isSupportITC>
        <!--ro, opt, bool, whether the device supports ITC-->true
      </isSupportITC>
      <isSupportIllegalDictionary>
        <!--ro, opt, bool, whether the device supports the violation dictionary-->true
      </isSupportIllegalDictionary>
      <isSupportVehicleDetection>
        <!--ro, opt, bool, whether the device supports vehicle detection-->true
      </isSupportVehicleDetection>
      <isSupportHVTVehicleDetection>
        <!--ro, opt, bool, whether the device supports mixed-traffic vehicle detection-->true
      </isSupportHVTVehicleDetection>
      <isSupportlicencePlateAuditData>
        <!--ro, opt, bool, whether the device supports importing or exporting the license plate blocklist and allowlist data-->true
      </isSupportlicencePlateAuditData>
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    <isSupportSearchLPListAudit>
      <!--ro, opt, bool, whether the device supports searching for the license plate blocklist and allowlist-->true
    </isSupportSearchLPListAudit>
    <isSupportVehicleDetectionManualTrigger opt="false,true">
      <!--ro, opt, bool, whether the device supports manually triggering vehicle detection, attr:opt{opt, string}-->true
    </isSupportVehicleDetectionManualTrigger>
    <isSupportSyncSignalOutput>
      <!--ro, opt, bool, whether the device supports configuring output parameters-->true
    </isSupportSyncSignalOutput>
  </ITCCap>
  <ImageCap>
    <!--ro, opt, object, capability of configuring image parameters-->
    <isSupportRegionalExposure>
      <!--ro, opt, bool, whether the device supports regional exposure-->true
    </isSupportRegionalExposure>
    <isSupportRegionalFocus>
      <!--ro, opt, bool, whether the device supports regional focus-->true
    </isSupportRegionalFocus>
  </ImageCap>
  <RacmCap>
    <!--ro, opt, object, UI before picture search-->
    <isSupportZeroChan>
      <!--ro, opt, bool, whether the device supports channel-zero, desc:related URI: /ISAPI/ContentMgmt/ZeroVideo/channels-->true
    </isSupportZeroChan>
    <inputProxyNums>
      <!--ro, opt, int, supported number of digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->1
    </inputProxyNums>
    <eSATANums>
      <!--ro, opt, int, supported number of eSATA interfaces-->1
    </eSATANums>
    <miniSASNums>
      <!--ro, opt, int, supported number of miniSAS interfaces-->1
    </miniSASNums>
    <nasNums>
      <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->1
    </nasNums>
    <ipSanNums>
      <!--ro, opt, int, supported number of IPSANs-->1
    </ipSanNums>
    <isSupportRaid>
      <!--ro, opt, bool, whether the device supports RAID-->true
    </isSupportRaid>
    <isSupportExtHdCfg>
      <!--ro, opt, bool, whether the device supports advanced HDD management-->true
    </isSupportExtHdCfg>
    <isSupportTransCode>
      <!--ro, opt, bool, whether the device supports live view transcoding-->true
    </isSupportTransCode>
    <isSupportIpcImport>
      <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/ipcConfig-->true
    </isSupportIpcImport>
    <NasMountType>
      <!--ro, opt, object, NAS mounting type-->
      <isNFSSupportAuthentication>
        <!--ro, opt, bool, whether the device supports CIFS authentication-->true
      </isNFSSupportAuthentication>
      <isCIFSSupportAuthentication>
        <!--ro, opt, bool, whether the device supports CIFS authentication-->true
      </isCIFSSupportAuthentication>
    </NasMountType>
    <isSupportIpcStreamType>
      <!--ro, opt, bool, whether the device supports the stream type of the network camera-->true
    </isSupportIpcStreamType>
    <isSupportIOInputProxy>
      <!--ro, opt, bool, whether the device supports dynamic input channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/inputs-->true
    </isSupportIOInputProxy>
    <isSupportIOOutputProxy>
      <!--ro, opt, bool, whether the device supports dynamic output channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/outputs-->true
    </isSupportIOOutputProxy>
    <isSupportPTZRs485Proxy>
      <!--ro, opt, bool, whether the device supports dynamic PTZ RS-485 channels, desc:related URI: /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities-->true
    </isSupportPTZRs485Proxy>
    <isSupportSrcIDSearch>
      <!--ro, opt, bool, whether the device supports stream ID search-->true
    </isSupportSrcIDSearch>
    <isSupportReversePlayback>
      <!--ro, opt, bool, whether the device supports reverse playback-->true
    </isSupportReversePlayback>
    <isSupportSMARTTest>
      <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
    </isSupportSMARTTest>
    <isSupportDownloadByTime>
      <!--ro, opt, bool, whether the device supports downloading by time, desc:related URI: /ISAPI/ContentMgmt/download/capabilities-->true
    </isSupportDownloadByTime>
    <pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
personQueueCounting,personQueueTime,violentMotion,leavePosition,advReachHeight,peopleNumChange,mixedTargetDetection,illegalParking,pedestrian,construction,r
oadblock,
abandonedObject,trafficAccident,fogDetection,smokeDetection,personDensity,failDown,accessController,videoInterCom,GJD,Luminite,OPTEX,securityControlPanel,ge
tUp,toiletTarry,
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playCellphone,vehicleMonitor,vibrationDetection,running,retention,spacingChange,yardTarry,keyPersonGetUp,sitQuietly,standUp,plateRecognition,smoking,onThePhone,fatigueDriving,
notLookingStraightAhead,leaveThePost,seatBelt,sunglasses,identityAbnormal,ForwardCollisionWarning,LaneDepartureWarning,BlindSpotDetection,HeadwayMonitoringWarning,
PedestrianCollisionWarning,SuddenSpeedUp,SuddenSpeedDown,SuddenLeftTurn,SuddenRightTurn,Turnover,NotYieldToPedestrian,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,     illegalHeap,illegalParkingofNon-
MotorVehicle,illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,allPic,ATMPanel,
ATMSurround,ATMFace,ATMSafetyCabin,loitering,smokeDetectAlarm,alarmResult">
      <!--ro, opt, string, picture search type, attr:opt{opt, string}-->test
    </pictureSearchType>
    <recordSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyHelmet,
failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofNon-MotorVehicle,
illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin
">
      <!--ro, opt, string, video search type, attr:opt{opt, string}, desc:"llEvent" (all events), "CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection pictures), "Command", "pir" (pir alarm), "wlsensor" (wireless alarm), "callhelp" (panic alarm), "facedetection" (face detection), "FieldDetection" (intrusion detection), "scenechangedetection" (scene change detection), "LineDetection" (line crossing detection)-->test
    </recordSearchType>
    <isSupportActivateIpc>
      <!--ro, opt, bool, whether the device supports activating the network camera-->true
    </isSupportActivateIpc>
    <isSupportCheckIpcSecurity>
      <!--ro, opt, object, device password detection-->
      <isSupportCheckPassword>
        <!--ro, opt, bool, whether the device supports password detection-->true
      </isSupportCheckPassword>
    </isSupportCheckIpcSecurity>
    <isSupportMainAndSubRecord>
      <!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
    </isSupportMainAndSubRecord>
    <isSupportSyncIPCPassword>
      <!--ro, opt, bool, whether the device supports synchronizing the password of the network camera-->true
    </isSupportSyncIPCPassword>
    <isSupportTransferIPC>
      <!--ro, opt, bool, whether the device supports transparent transmission of the network camera-->true
    </isSupportTransferIPC>
    <isSupportSmartSearch>
      <!--ro, opt, bool, whether the device supports video dual-VCA, desc:related URI: /ISAPI/ContentMgmt/SmartSearch/capabilities-->true
    </isSupportSmartSearch>
    <isSupportPOS>
      <!--ro, opt, bool, whether the device supports POS-->true
    </isSupportPOS>
    <isSupportLinkNum>
      <!--ro, opt, bool, whether the device supports getting the number of links, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/linkNum-->true
    </isSupportLinkNum>
    <isSupportPassBackBasicCfg>
      <!--ro, opt, bool, whether the device supports configuring basic parameters of the CVR copy-back function, desc:related URI: /ISAPI/ContentMgmt/record/passback/basicCfg/capabilities-->true
    </isSupportPassBackBasicCfg>
    <PassBackTaskConfig>
      <!--ro, opt, object, capability of managing tasks of the CVR copy-back function-->
      <isSupportAddTask>
        <!--ro, opt, bool, whether the device supports adding a task, desc:related URI: /ISAPI/ContentMgmt/record/passback/addTask/capabilities-->true
      </isSupportAddTask>
      <isSupportSearchTask>
        <!--ro, opt, bool, whether the device supports searching tasks, desc:related URI: /ISAPI/ContentMgmt/record/passback/searchTask/capabilities-->true
      </isSupportSearchTask>
      <isSupportControlTask>
        <!--ro, opt, bool, whether the device supports controlling the task, desc:related URI: /ISAPI/ContentMgmt/record/passback/controlTask/capabilities-->true
      </isSupportControlTask>
      <isSupportDeleteTask>
        <!--ro, opt, bool, whether the device supports deleting the task, desc:related URI: /ISAPI/ContentMgmt/record/passback/deleteTask/capabilities-->true
      </isSupportDeleteTask>
    </PassBackTaskConfig>
    <PassBackPlanConfig>
      <!--ro, opt, object, capability of configuring the CVR copy-back schedule-->
      <isSupportAddPlan>
        <!--ro, opt, bool, whether the device supports adding a schedule, desc:related URI: /ISAPI/ContentMgmt/record/passback/addPlan/capabilities-->true
      </isSupportAddPlan>
      <isSupportSearchPlan>
        <!--ro, opt, bool, whether the device supports searching schedules, desc:related URI: /ISAPI/ContentMgmt/record/passback/searchPlan/capabilities-->true
      </isSupportSearchPlan>
      <isSupportDeletePlan>
        <!--ro, opt, bool, whether the device supports deleting the schedule, desc:related URI: /ISAPI/ContentMgmt/record/passback/deletePlan/capabilities-->true
      </isSupportDeletePlan>
    </PassBackPlanConfig>
    <IRAIDCap>
      <!--ro, opt, object, network array capability-->
      <isSupportIRAIDRapidConfiguration>
        <!--ro, opt, bool, whether the device supports the rapid IRAID configuration, desc:related URI: /ISAPI/ContentMgmt/Storage/IRAID/RapidConfiguration/capabilities-->true
      </isSupportIRAIDRapidConfiguration>
    </IRAIDCap>
    <isSupportStorageExtraInfo>
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    <isSupportStorageExtraInfo>
      <!--ro, opt, bool, whether the device supports configuring additional storage information, desc:related URI:
/ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities-->true
    </isSupportStorageExtraInfo>
    <isSupportRecordStatus>
      <!--ro, opt, bool, whether the device supports getting the recording status, desc:related URI: /ISAPI/ContentMgmt/channels/<ID>/record/capabilities--
>true
    </isSupportRecordStatus>
    <isSupportRacmChannelsCap opt="true,false">
      <!--ro, opt, bool, whether the device supports the Rcam capability by channel, attr:opt{opt, string}, desc:related URI:
/ISAPI/ContentMgmt/channels/<ID>/capabilities-->true
    </isSupportRacmChannelsCap>
    <LockCap>
      <!--ro, opt, object, video locking and unlocking, desc:related URI: /ISAPI/ContentMgmt/record/control/locks/capabilities-->
    </LockCap>
    <isSupportForamtAll>
      <!--ro, opt, bool, whether the device supports formatting all HDDs, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/format-->true
    </isSupportForamtAll>
    <isSupportExtendCabinetCfg>
      <!--ro, opt, bool, whether the device supports configuring parameters of the extended cabinet. Related URI:
/ISAPI/ContentMgmt/Storage/ExtendCabinet/capabilities-->true
    </isSupportExtendCabinetCfg>
    <diskGroupNums>
      <!--ro, opt, int, supported number of disk groups, desc:related URI: /ISAPI/ContentMgmt/Storage/diskGroup-->1
    </diskGroupNums>
    <ExternalDevice>
      <!--ro, opt, object, external device storage-->
      <USB>
        <!--ro, opt, object, USB information-->
        <debugLogOutput>
          <!--ro, opt, bool, whether to enable the debugging log, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
        </debugLogOutput>
        <isSupportCapturePackage>
          <!--ro, opt, bool, whether the device supports configuring capture and filter function of a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/capturePackage/capabilities?format=json-->true
        </isSupportCapturePackage>
        <isSupportStreamStorage>
          <!--ro, opt, bool, whether the device supports saving stream to a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/streamStorage/capabilities?format=json-->true
        </isSupportStreamStorage>
      </USB>
    </ExternalDevice>
    <customProtocolNums>
      <!--ro, opt, int, number of custom protocols connected via IPC, desc:related URI: /ISAPI/ContentMgmt/InputProxy/customProtocols, supported number of
custom protocols-->1
    </customProtocolNums>
    <isSupportIPCTiming>
      <!--ro, opt, bool, whether the device supports synchronizing the time of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <enableTiming> node supports-->true
    </isSupportIPCTiming>
    <isSupportLogConfig>
      <!--ro, opt, bool, whether the device supports log configuration, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
    </isSupportLogConfig>
    <SecurityLog>
      <!--ro, opt, object, device's capability of security log-->
      <isSupportSecurityLog>
        <!--ro, opt, bool, whether the device supports the security log-->true
      </isSupportSecurityLog>
      <isSupportLogServer>
        <!--ro, opt, bool, whether it supports security log server configuration-->true
      </isSupportLogServer>
      <isSupportLogServerTest>
        <!--ro, opt, bool, whether it supports security log server test-->true
      </isSupportLogServerTest>
      <SecurityLogTypeList>
        <!--ro, req, array, list of security log types, subType:object-->
        <SecurityLogType>
          <!--ro, opt, object, security log type-->
          <primaryType>
            <!--ro, req, string, main log type, desc:"Event", "Operation", "Other", "All"-->test
          </primaryType>
          <secondaryType opt="all">
            <!--ro, req, string, minor log type, attr:opt{opt, string}, desc:for other types, refer to the appendix-->test
          </secondaryType>
        </SecurityLogType>
      </SecurityLogTypeList>
      <isSupportLogServerList>
        <!--ro, opt, bool, whether the device supports batch configuring security log server-->true
      </isSupportLogServerList>
      <logMinItemNum>
        <!--ro, opt, int, range:[1,2147483646]-->100
      </logMinItemNum>
    </SecurityLog>
    <isSupportGetBackupDevice>
      <!--ro, opt, bool, whether the device supports getting the backup device, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/device?format=json--
>true
    </isSupportGetBackupDevice>
    <isSupportAutoBackup>
      <!--ro, opt, bool, whether the device supports automatic backup, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/auto/capabilities?format=json--
>true
    </isSupportAutoBackup>
    <certificateValidationEnabled>
      <!--ro, opt, bool, whether to enable the certification validation, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <certificateValidationEnabled> node supports-->true
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    </certificateValidationEnabled>
    <defaultAdminPortEnabled>
      <!--ro, opt, bool, whether to enable the default communication port, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <defaultAdminPortEnabled> node supports-->true
    </defaultAdminPortEnabled>
    <isSupportLogDataPackage>
      <!--ro, opt, bool, whether the device supports exporting the log files, desc:related URI: /ISAPI/ContentMgmt/logSearch/dataPackage-->true
    </isSupportLogDataPackage>
    <logSearchTimeSpanNums>
      <!--ro, opt, int, number of time periods for log search, desc:related URI: /ISAPI/ContentMgmt/logSearch; corresponding node: timeSpanList in
CMSearchDescription-->1
    </logSearchTimeSpanNums>
    <CloudStorageServerCap>
      <!--ro, opt, object, capability of configuring the cloud storage server-->
      <isSupportCloudStorageParameter>
        <!--ro, opt, bool, whether the device supports configuring the cloud storage parameters, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/capabilities?format=json-->true
      </isSupportCloudStorageParameter>
      <isSupportCloudStoragePool>
        <!--ro, opt, bool, whether the device supports configuring the cloud storage pool, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/pool/capabilities?format=json-->true
      </isSupportCloudStoragePool>
      <supportChannelsNum>
        <!--ro, opt, int, number of channels supported by the cloud storage, desc:if this node is not returned, all channels of the device support; if this
node is returned, several channels of the device supports cloud storage (at present, it will not return the specific number of the channels supporting cloud
storage)-->1
      </supportChannelsNum>
    </CloudStorageServerCap>
    <isSupportWebPrivatePlaybackByUTC>
      <!--ro, opt, bool, whether it supports playback (via private protocol) by UTC-->true
    </isSupportWebPrivatePlaybackByUTC>
    <isSupportFindCommonFileByUTC>
      <!--ro, opt, bool, whether it supports searching for ordinary files (private protocols) by UTC time-->true
    </isSupportFindCommonFileByUTC>
    <isSupportFindEventFileByUTC>
      <!--ro, opt, bool, whether it supports searching for event files (private protocols) by UTC time-->true
    </isSupportFindEventFileByUTC>
    <isSupportSmartSearchRecordByUTC>
      <!--ro, opt, bool, whether it supports smart video search by UTC time-->true
    </isSupportSmartSearchRecordByUTC>
    <isSupportMRDSearchByTimeZone>
      <!--ro, opt, bool, whether it supports searching for files (private protocols) by monthly calendar by UTC time-->true
    </isSupportMRDSearchByTimeZone>
    <isSupportSearchRecordLabelByUTC>
      <!--ro, opt, bool, whether it supports searching for video tags by UTC time-->true
    </isSupportSearchRecordLabelByUTC>
    <isSupportSearchPictureByUTC>
      <!--ro, opt, bool, whether it supports searching for pictures by UTC time-->true
    </isSupportSearchPictureByUTC>
    <isSupportSmartSearchPictureByUTC>
      <!--ro, opt, bool, whether it supports smart pictures search by UTC time-->true
    </isSupportSmartSearchPictureByUTC>
    <isSupportFindLogByUTC>
      <!--ro, opt, bool, whether it supports searching log by UTC time-->true
    </isSupportFindLogByUTC>
    <isSupportUploadRecordByUTC>
      <!--ro, opt, bool, whether it supports uploading files to cloud storage by UTC time-->true
    </isSupportUploadRecordByUTC>
    <isSupportPlaybackByUTC>
      <!--ro, opt, bool, whether it supports playback and locating by UTC time-->true
    </isSupportPlaybackByUTC>
    <isSupportFaceDetecttionAlarmByTimeZone>
      <!--ro, opt, bool, no file is found-->true
    </isSupportFaceDetecttionAlarmByTimeZone>
    <isSupportSourceModify>
      <!--ro, opt, bool, whether it supports configuring the parameters of the LAN IPC, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/sourceModify/capabilities-->true
    </isSupportSourceModify>
    <isSupportCountingSearchByUTC>
      <!--ro, opt, bool, whether it supports searching people counting results by UTC time-->true
    </isSupportCountingSearchByUTC>
    <isSupportCluster>
      <!--ro, opt, bool, whether the device supports the cluster function, desc:related URI: /ISAPI/ContentMgmt/Cluster-->true
    </isSupportCluster>
    <supportAIDTFSType opt="illegalParking,wrongdirection,crosslane,vehicleexist,lanechange,turnround,evidence">
      <!--ro, opt, string, the event picture search type supported by AID and TFS, attr:opt{opt, string}-->test
    </supportAIDTFSType>
    <isSupportMisinfoStateSearch>
      <!--ro, opt, bool, whether it supports searching by false alarm status-->true
    </isSupportMisinfoStateSearch>
    <isSupportOneKeyOperationStorageData>
      <!--ro, opt, bool, whether it supports one-touch configuration for storage data, desc:related URI: /ISAPI/ContentMgmt/Storage/oneKey/capabilities?
format=json-->true
    </isSupportOneKeyOperationStorageData>
    <eventRecordSearch>
      <!--ro, opt, object, whether the device supports searching for videos by event, desc:related URI: /ISAPI/ContentMgmt/eventRecordSearch?format=json-->
      <eventType opt="faceContrast,faceSnap,vehicleMatchResult,hawkResult">
        <!--ro, opt, string, event type, attr:opt{opt, string}-->test
      </eventType>
      <auxEventType opt="faceContrastFailure,faceContrastSuccess">
        <!--ro, opt, string, search sub type of the event, attr:opt{opt, string}-->test
      </auxEventType>
    </eventRecordSearch>
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    <isSupportManualRecord>
      <!--ro, opt, bool, whether the device supports manual recording, desc:related URI: /ISAPI/ContentMgmt/record/control/manual/capabilities?format=json-->true
    </isSupportManualRecord>
    <isSupportRemark>
      <!--ro, opt, bool, whether it supports file remarks-->true
    </isSupportRemark>
    <FileUpload>
      <!--ro, opt, object, automatically upload file-->
      <enabled>
        <!--ro, req, bool, whether it supports uploading the file-->true
      </enabled>
    </FileUpload>
    <PoliceInfoUpload>
      <!--ro, opt, object, automatically upload police officer information-->
      <enabled>
        <!--ro, req, bool, whether the device supports uploading the police officer information-->true
      </enabled>
    </PoliceInfoUpload>
    <PoliceInfo>
      <!--ro, opt, object, police officer information-->
      <policeID>
        <!--ro, opt, bool, whether the device supports police officer ID-->true
      </policeID>
      <policeCode>
        <!--ro, opt, bool, whether the device supports police officer number-->true
      </policeCode>
      <policeName>
        <!--ro, opt, bool, whether the device supports police officer name-->true
      </policeName>
      <password>
        <!--ro, opt, bool, whether the device supports police officer password-->true
      </password>
      <deviceID>
        <!--ro, opt, bool, whether the device supports device ID-->true
      </deviceID>
    </PoliceInfo>
    <isSupportAssignChannelID>
      <!--ro, opt, bool, whether the device supports specifying the access channel No. when adding the network camera, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <defaultAdminPortEnabled> node supports-->true
    </isSupportAssignChannelID>
    <isSupportAssignStreamID>
      <!--ro, opt, bool, whether it supports specifying stream ID of the channel when adding the network camera in channel mode, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <defaultAdminPortEnabled> node supports-->true
    </isSupportAssignStreamID>
    <PictureSearchSubTypeList>
      <!--ro, opt, array, list of picture search sub type, subType:object-->
      <PictureSearchSubType>
        <!--ro, opt, object, the sub type of picture search-->
        <mainType>
          <!--ro, req, string, main event type of the searched picture-->test
        </mainType>
        <subType opt="all,motorVehicle,nonMotorVehicle,pedestrian">
          <!--ro, req, string, sub type of each main event type, attr:opt{opt, string}-->test
        </subType>
      </PictureSearchSubType>
    </PictureSearchSubTypeList>
    <isSupportPlaybackStreamIDByUTC>
      <!--ro, opt, bool, whether it supports playback by stream ID according to UTC time-->true
    </isSupportPlaybackStreamIDByUTC>
    <isSupportTimeLockStreamIDByUTC>
      <!--ro, opt, bool, whether it supports locking and unlocking the video segment by steam ID according to UTC time-->true
    </isSupportTimeLockStreamIDByUTC>
    <isSupportTimeSearch>
      <!--ro, opt, bool, whether it supports searching for recording start and end time by channel, desc:related URI: /ISAPI/ContentMgmt/time/search?format=json-->true
    </isSupportTimeSearch>
    <behaviorEventPicSearch opt="allBehaviorEvent,running,group,violentMotion,failDown,playCellphone,peopleNumChange,leavePosition,retention,sleepOnduty">
      <!--ro, opt, string, abnormal event detection supported by picture search, attr:opt{opt, string}-->test
    </behaviorEventPicSearch>
    <perimeterEventPicSearch opt="allPerimeterEvent,linedetection,fielddetection,regionEntrance,regionExiting,loitering">
      <!--ro, opt, string, perimeter protection events supported by picture search, attr:opt{opt, string}-->test
    </perimeterEventPicSearch>
    <isSupportSourceStatus>
      <!--ro, opt, bool, whether it supports getting the status of access source-->true
    </isSupportSourceStatus>
    <isSupportSSDSMARTTest>
      <!--ro, opt, bool, whether the device supports S.M.A.R.T. detection on SSD, desc:related URI: /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/start-->true
    </isSupportSSDSMARTTest>
    <MobileStorage>
      <!--ro, opt, object, storage capability of the mobile HDD-->
      <isSupportGetFileTime>
        <!--ro, opt, bool, whether it supports getting the file time of the mobile HDD, desc:it supports mobile HDD currently. Related URI: /ISAPI/ContentMgmt/mobileStorage/fileTime/capabilities?format=json-->true
      </isSupportGetFileTime>
    </MobileStorage>
    <isSupportGetChannelList>
      <!--ro, opt, bool, whether it supports getting channels of the HDD, desc:it supports mobile HDD currently. Related URI: /ISAPI/ContentMgmt/Storage/hdd/channelList/capabilities?format=json-->true
    </isSupportGetChannelList>
    <isSupportCascade>
      <!--ro, opt, bool, whether it supports configuring parameters for NVR cascading topology-->true
    </isSupportCascade>
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    <isSupportPictureInfo>
      <!--ro, opt, bool, whether it supports picture search on the SD card of the capture camera-->true
    </isSupportPictureInfo>
    <isSupportCloudStorageModel>
      <!--ro, opt, bool, whether the device supports configuring the cloud storage mode,
desc:/ISAPI/ContentMgmt/Storage/CloudStorageModel/<channelID>/capabilities?format=json-->true
    </isSupportCloudStorageModel>
    <isSupportCloudStorageModelList>
      <!--ro, opt, bool, whether the device supports configuring cloud storage modes in a batch,
desc:/ISAPI/ContentMgmt/Storage/CloudStorageModel/<channelID>?format=json-->true
    </isSupportCloudStorageModelList>
    <isSupportRecordAudio>
      <!--ro, opt, bool, whether the device supports recording the audio, desc:only the audio will be recorded and the video will not be recorded. Related
URI: /ISAPI/ContentMgmt/recordAudio/<audioID>/capabilities?format=json-->true
    </isSupportRecordAudio>
    <isSupportSearchAudio>
      <!--ro, opt, bool, whether the device supports searching for the audio, desc:related URI: /ISAPI/ContentMgmt/recordAudio/search/capabilities?
format=json-->true
    </isSupportSearchAudio>
    <isSupportAudioDailyDistribution>
      <!--ro, opt, bool, whether it supports searching for audio file information by monthly calendar, desc:related URI:
/ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json-->true
    </isSupportAudioDailyDistribution>
    <isSupportDownloadAudio>
      <!--ro, opt, bool, whether the device supports downloading the audio by file name, desc:related URI:
/ISAPI/ContentMgmt/recordAudio/fileNameDownload/capabilities?format=json-->true
    </isSupportDownloadAudio>
    <isSupportDeleteAudio>
      <!--ro, opt, bool, whether the device supports deleting the recorded audio file, desc:related URI: /ISAPI/ContentMgmt/recordAudio/deleteFile?
format=json-->true
    </isSupportDeleteAudio>
    <isSupportSearchRemoteAlbum>
      <!--ro, opt, bool, whether the device supports remotely searching for album information, desc:related URI: /ISAPI/ContentMgmt/searchRemoteAlbum?
format=json-->true
    </isSupportSearchRemoteAlbum>
    <isSupportStreamParam>
      <!--ro, opt, bool, whether the device supports configuring the access stream parameters of all digital channels, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/streamParam/capabilities?format=json-->true
    </isSupportStreamParam>
    <isSupportFusionChan>
      <!--ro, opt, bool, whether the device supports fusion channel, desc:if this node does not exist, it indicates that the device does not support the
function; interaction IP channel: it is a concept from the education sharing system; the education sharing system will allocate IP channels to images of the
connected sub classrooms or meeting rooms. fusion channel: a channel generated when adding the NVR locally or remotely; it supports live view, playback,
storage, and recording schedule/linkage configuration; the channel will take effect by enabling the fusion channel-->true
    </isSupportFusionChan>
    <isSupportRecordMultipleDownload>
      <!--ro, opt, bool, whether the device supports downloading videos in multiples, desc:if this node does not exist, it indicates that the device does
not support the function by default-->true
    </isSupportRecordMultipleDownload>
  </RacmCap>
  <isSupportMI3704>
    <!--ro, opt, bool-->true
  </isSupportMI3704>
  <isSupportLogServer>
    <!--ro, opt, bool, whether it supports security log server configuration-->true
  </isSupportLogServer>
  <isSupportLogServerTest>
    <!--ro, opt, bool, whether it supports security log server test-->true
  </isSupportLogServerTest>
  <PTZCtrlCap>
    <!--ro, opt, object, device's PTZ capability-->
    <isSupportPatrols>
      <!--ro, opt, bool, whether the device supports patrol-->true
    </isSupportPatrols>
    <isSupportCombinedPath>
      <!--ro, opt, bool, whether the device supports the PTZ combined path-->true
    </isSupportCombinedPath>
    <isSupportRailwayRobot>
      <!--ro, opt, bool, whether the device supports railway robot, desc:/ISAPI/PTZCtrl/channels/<channelID>/RailwayRobot/capabilities?format=json-->true
    </isSupportRailwayRobot>
  </PTZCtrlCap>
  <SmartCap>
    <!--ro, opt, object, compact and smart capability, desc:/ISAPI/Smart/capabilities-->
    <isSupportROI>
      <!--ro, opt, bool, whether the device supports ROI encoding configuration-->true
    </isSupportROI>
    <isSupportFaceDetect>
      <!--ro, opt, bool, whether the device supports face detection-->true
    </isSupportFaceDetect>
    <isSupportIntelliTrace>
      <!--ro, opt, bool, whether the device supports smart tracking-->true
    </isSupportIntelliTrace>
    <isSupportFieldDetection>
      <!--ro, opt, bool, whether the device supports intrusion detection-->true
    </isSupportFieldDetection>
    <isSupportDefocusDetection>
      <!--ro, opt, bool, whether the device supports defocus detection-->true
    </isSupportDefocusDetection>
    <isSupportAudioDetection>
      <!--ro, opt, bool, whether the device supports audio detection-->true
    </isSupportAudioDetection>
    <isSupportSceneChangeDetection>
      <!--ro, opt, bool, whether the device supports scene change detection-->true
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    </isSupportSceneChangeDetection>
    <isSupportLineDetection>
      <!--ro, opt, bool, whether the device supports line crossing detection-->true
    </isSupportLineDetection>
    <isSupportRegionEntrance>
      <!--ro, opt, bool, whether the device supports region entrance detection-->true
    </isSupportRegionEntrance>
    <isSupportRegionExiting>
      <!--ro, opt, bool, whether the device supports region exiting detection-->true
    </isSupportRegionExiting>
    <isSupportLoitering>
      <!--ro, opt, bool, whether the device supports loitering detection-->true
    </isSupportLoitering>
    <isSupportGroup>
      <!--ro, opt, bool, whether the device supports people gathering detection-->true
    </isSupportGroup>
    <isSupportRapidMove>
      <!--ro, opt, bool, whether the device supports fast moving detection-->true
    </isSupportRapidMove>
    <isSupportParking>
      <!--ro, opt, bool, whether the device supports parking detection-->true
    </isSupportParking>
    <isSupportUnattendedBaggage>
      <!--ro, opt, bool, whether the device supports unattended baggage detection-->true
    </isSupportUnattendedBaggage>
    <isSupportAttendedBaggage>
      <!--ro, opt, bool, whether the device supports object removal detection-->true
    </isSupportAttendedBaggage>
    <isSupportPeopleDetection>
      <!--ro, opt, bool, whether the device supports human body detection-->true
    </isSupportPeopleDetection>
    <isSupportSmartCalibration>
      <!--ro, opt, bool, whether the device supports calibration filter size-->true
    </isSupportSmartCalibration>
    <isSupportStorageDetection>
      <!--ro, opt, bool, whether the device supports health monitoring storage-->true
    </isSupportStorageDetection>
    <isSupportShipsDetection>
      <!--ro, opt, bool, whether the device supports ship detection-->true
    </isSupportShipsDetection>
    <isSupportShield>
      <!--ro, opt, bool, whether the device supports configuring the compact and smart shielded regions-->true
    </isSupportShield>
    <isSupportAlgVersion>
      <!--ro, opt, bool, whether the device supports algorithm version-->true
    </isSupportAlgVersion>
    <isSupportVideoOverlap>
      <!--ro, opt, bool, whether the device supports configuring information overlay for video stream-->true
    </isSupportVideoOverlap>
    <isSupportParkingState>
      <!--ro, opt, bool, whether the device supports parking status-->true
    </isSupportParkingState>
    <isSupportChannelResource>
      <!--ro, opt, bool, whether the device supports channel resources-->true
    </isSupportChannelResource>
    <isSupportAnalysisUnitSwitch opt="true,false">
      <!--ro, opt, bool, whether the device supports switching analysis unit, attr:opt{opt, string}-->true
    </isSupportAnalysisUnitSwitch>
    <isSupportHFPD>
      <!--ro, opt, bool, whether the device supports frequently appeared person detection, desc:related URI: /ISAPI/SDT/HFPD?format=json-->true
    </isSupportHFPD>
    <isSupportImageROI>
      <!--ro, opt, bool, whether the device supports smartJpeg (image ROI)-->true
    </isSupportImageROI>
    <isSupportLFPD>
      <!--ro, opt, bool, whether the device supports rarely appeared person detection, desc:related URI: /ISAPI/SDT/LFPD?format=json-->true
    </isSupportLFPD>
    <isSupportSmartOverlapParams>
      <!--ro, opt, bool, whether it supports smart overlay parameters, desc:related URI: /ISAPI/Smart/channels/<channelID>/SmartOverlapParams?format=json--
>true
    </isSupportSmartOverlapParams>
  </SmartCap>
  <isSupportEhome>
    <!--ro, opt, bool, whether the device supports ISUP functions-->true
  </isSupportEhome>
  <isSupportStreamingEncrypt>
    <!--ro, opt, bool, whether the device supports stream encryption-->true
  </isSupportStreamingEncrypt>
  <TestCap>
    <!--ro, opt, object, test capability-->
    <isSupportFTPTest>
      <!--ro, opt, bool, whether it supports FTP test-->true
    </isSupportFTPTest>
    <isSupportPingTest>
      <!--ro, opt, bool, whether it supports PING test-->true
    </isSupportPingTest>
    <isSupportNTPTest>
      <!--ro, opt, bool, whether it supports NTP test-->true
    </isSupportNTPTest>
    <isSupportNASTest>
      <!--ro, opt, bool, whether it supports NAS test-->true
    </isSupportNASTest>
    <isSupportEmailTest>
      <!--ro, opt, bool, whether the device supports email test-->true
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      <!--ro, opt, bool, whether the device supports email test-->true
    </isSupportEmailTest>
  </TestCap>
  <ImageMiscCap>
    <!--ro, opt, object, DPC capability-->
    <isSupportDPC>
      <!--ro, opt, bool, whether the device supports DPC-->true
    </isSupportDPC>
  </ImageMiscCap>
  <PanoramaCap>
    <!--ro, opt, object, capability of panoramic channels, desc:the object cannot be extended. You can extend it in /ISAPI/Panorama/capabilities-->
    <isSupportGeneratePanorama>
      <!--ro, opt, bool, whether the device supports generating panoramic images-->true
    </isSupportGeneratePanorama>
    <isSupportPanoramaPosition3D>
      <!--ro, opt, bool, whether the device supports panoramic 3D position-->true
    </isSupportPanoramaPosition3D>
    <isSupportPreset>
      <!--ro, opt, bool, whether the device supports configuring panoramic preset-->true
    </isSupportPreset>
    <isSupportGetPanoramicSplicingStatus>
      <!--ro, opt, bool-->true
    </isSupportGetPanoramicSplicingStatus>
    <isSupportStopPanoramicSplicing>
      <!--ro, opt, bool-->true
    </isSupportStopPanoramicSplicing>
    <isSupportSplicingChannelMotionControl>
      <!--ro, opt, bool-->true
    </isSupportSplicingChannelMotionControl>
    <isSupportGetPanoramaSplicingPictures>
      <!--ro, opt, bool-->true
    </isSupportGetPanoramaSplicingPictures>
    <isSupportImportPanoramaSplicingParamFile>
      <!--ro, opt, bool-->true
    </isSupportImportPanoramaSplicingParamFile>
    <isSupportExportPanoramaSplicingParamFile>
      <!--ro, opt, bool-->true
    </isSupportExportPanoramaSplicingParamFile>
    <isSupportImportPanoramicSplicingPicture>
      <!--ro, opt, bool-->true
    </isSupportImportPanoramicSplicingPicture>
    <isSupportExportPanoramicSplicingPicture>
      <!--ro, opt, bool-->true
    </isSupportExportPanoramicSplicingPicture>
  </PanoramaCap>
  <ThermalCap>
    <!--ro, opt, object, thermal capability, desc:related URI: /ISAPI/Thermal/capabilities-->
    <isSupportFireDetection>
      <!--ro, opt, bool, whether the device supports fire detection, desc:related URI: /ISAPI/Thermal/channels/<ID>/fireDetection/capabilities-->true
    </isSupportFireDetection>
    <isFireFocusZoomSupport>
      <!--ro, opt, bool, whether the device supports visible light lens zooming, desc:related URI: /ISAPI/Thermal/channels/<ID>/fireFocusZoom-->true
    </isFireFocusZoomSupport>
    <isSupportThermometry>
      <!--ro, opt, bool, whether the device supports thermography, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities-->true
    </isSupportThermometry>
    <isSupportRealtimeThermometry>
      <!--ro, opt, bool, whether the device supports uploading real-time temperature measurement data-->true
    </isSupportRealtimeThermometry>
    <isSupportThermIntell>
      <!--ro, opt, bool, whether the device supports configuring smart thermal mutual exclusion, desc:related URI:
/ISAPI/Thermal/channels/<ID>/thermIntell/capabilities-->true
    </isSupportThermIntell>
    <isSupportNVR>
      <!--ro, opt, bool, it is used for client to check-->true
    </isSupportNVR>
    <isSupportThermometrySchedule>
      <!--ro, opt, bool, whether the device supports the arming schedule of temperature measurement, desc:related URI: /ISAPI/Event/schedules/thermometry--
>true
    </isSupportThermometrySchedule>
    <isSupportTemperatureSchedule>
      <!--ro, opt, bool, whether the device supports the arming schedule of temperature difference, desc:related URI: /ISAPI/Event/schedules/temperature--
>true
    </isSupportTemperatureSchedule>
    <isSupportFireDetectionSchedule>
      <!--ro, opt, bool, whether the device supports the arming schedule of fire detection, desc:related URI: /ISAPI/Event/schedules/fireDetections-->true
    </isSupportFireDetectionSchedule>
    <isSupportPower>
      <!--ro, opt, bool, whether the device supports power on/off capability, desc:related URI: /ISAPI/Thermal/Power/capabilities-->true
    </isSupportPower>
    <isSupportRealtimeTempHumi>
      <!--ro, opt, bool, whether the device supports real-time detection of temperature and humidity, desc:related URI:
/ISAPI/Thermal/TempHumi/capabilities-->true
    </isSupportRealtimeTempHumi>
    <ManualThermCap>
      <!--ro, opt, object, the capability of manual temperature measurement-->
      <manualThermRuleNum>
        <!--ro, opt, int, if this node is not returned, it indicates manual temperature measurement is not supported, desc:related URI:
/ISAPI/Thermal/manualThermometry/capabilities-->true
      </manualThermRuleNum>
    </ManualThermCap>
    <isSupportManualThermBasic>
      <!--ro, opt, bool, whether the device supports basic configuration of manual temperature measurement, desc:related URI:
/ISAPI/Thermal/manualThermBasic/capabilities-->true
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    </isSupportManualThermBasic>
    <isSupportFireShieldMask>
      <!--ro, opt, bool, whether the device supports configuring the parameters of fire shielded regions, desc:related URI:
/ISAPI/Thermal/channels/<ID>/fireShieldMask/capabilities-->true
    </isSupportFireShieldMask>
    <isSupportSmokeShieldMask>
      <!--ro, opt, bool, whether the device supports configuring the parameters of smoke shielded regions, desc:related URI:
/ISAPI/Thermal/channels/<ID>/smokeShieldMask/capabilities-->true
    </isSupportSmokeShieldMask>
    <isSupportThermometryMode>
      <!--ro, opt, bool, whether the device supports configuring temperature measurement mode, desc:related URI:
/ISAPI/Thermal/channels/<ID>/thermometryMode/capabilities-->true
    </isSupportThermometryMode>
    <isSupportThermalPip>
      <!--ro, opt, bool, whether the device supports the PIP configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/pip/capabilities-->true
    </isSupportThermalPip>
    <isSupportThermalIntelRuleDisplay>
      <!--ro, opt, bool, whether the device supports the PIP configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/IntelRuleDisplay/capabilities--
>true
    </isSupportThermalIntelRuleDisplay>
    <AlgVersionInfo>
      <!--ro, opt, object, whether the device supports getting the version information of thermal algorithms library, desc:related URI:
/ISAPI/Thermal/AlgVersion-->
      <thermometryAlgName min="1" max="128">
        <!--ro, opt, string, version information of temperature measurement algorithms library, attr:min{opt, string},max{opt, string}-->test
      </thermometryAlgName>
      <shipsAlgName min="1" max="128">
        <!--ro, opt, string, version name of ship detection algorithms library, attr:min{opt, string},max{opt, string}-->test
      </shipsAlgName>
      <fireAlgName min="1" max="128">
        <!--ro, opt, string, algorithm library version name of fire detection, attr:min{opt, int},max{opt, int}-->test
      </fireAlgName>
      <smokeAlgName min="1" max="128">
        <!--ro, opt, string, algorithm library version name of smoke detection, attr:min{opt, int},max{opt, int}-->test
      </smokeAlgName>
      <wasteGasAlgName min="1" max="128">
        <!--ro, opt, string, algorithm library version name of waste gas detection, attr:min{opt, string},max{opt, string}-->test
      </wasteGasAlgName>
      <greyScaleAlarmAlgName min="1" max="128">
        <!--ro, opt, string, algorithm library version name of greyscale alarm, attr:min{opt, string},max{opt, string}-->test
      </greyScaleAlarmAlgName>
    </AlgVersionInfo>
    <isSupportFaceThermometry>
      <!--ro, opt, bool, whether supports temperature screening configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/faceThermometry/capabilities--
>true
    </isSupportFaceThermometry>
    <isSupportThermalBlackBody>
      <!--ro, opt, bool, whether the device supports configuring black body parameters of thermal imaging, desc:related URI:
/ISAPI/Thermal/channels/<ID>/blackBody/capabilities-->true
    </isSupportThermalBlackBody>
    <isSupportThermalStreamParam>
      <!--ro, opt, bool, whether the device supports configuring stream parameters of thermal imaging, desc:related URI:
/ISAPI/Thermal/channels/<ID>/streamParam/capabilities-->true
    </isSupportThermalStreamParam>
    <isSupportBodyTemperatureCompensation>
      <!--ro, opt, bool, whether the device supports temperature compensation configuration, desc:related URI:
/ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation/capabilities-->true
    </isSupportBodyTemperatureCompensation>
    <isSupportClickToThermometry>
      <!--ro, opt, bool, whether the device supports click-to-thermometry, desc:related URI: ISAPI/Thermal/channels/<ID>/clickToThermometry/initialization--
>true
    </isSupportClickToThermometry>
    <isSupportThermometryHistorySearch>
      <!--ro, opt, bool, whether the device supports searching for historical temperature data, desc:related URI:
/ISAPI/Thermal/channels/<ID>/historyTemperature/capabilities-->true
    </isSupportThermometryHistorySearch>
    <isSupportThermometryShieldMask>
      <!--ro, opt, bool, whether the device supports configuring shield parameters of thermography region, desc:related URI:
/ISAPI/Thermal/channels/<ID>/thermometryShieldMask/capabilities-->true
    </isSupportThermometryShieldMask>
    <isSupportTemperatureCorrection>
      <!--ro, opt, bool, whether device supports temperature measurement correction, desc:related URI:
/ISAPI/Thermal/channels/<ID>/CorrectionParam/capabilities-->true
    </isSupportTemperatureCorrection>
    <isSupportBurningPrevention>
      <!--ro, opt, bool, whether device supports burning prevention, desc:related URI: /ISAPI/Thermal/channels/<ID>/burningPrevention/capabilities-->true
    </isSupportBurningPrevention>
    <isSupportTemperatureCollection>
      <!--ro, opt, bool, whether device supports temperature ANR, desc:related URI: ISAPI/Thermal/temperature/collection/capabilities?format=json-->true
    </isSupportTemperatureCollection>
    <isSupportJpegPicWithAppendData>
      <!--ro, opt, bool, whether the device supports capturing pictures, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/jpegPicWithAppendData?
format=json-->true
    </isSupportJpegPicWithAppendData>
    <isSupportRealTimethermometryForHTTP>
      <!--ro, opt, bool, whether device supports real-time temperature measurement, desc:related URI:
/ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules?format=json-->true
    </isSupportRealTimethermometryForHTTP>
    <isSupportthermometryOffLineCapture>
      <!--ro, opt, bool, whether the device supports capturing pictures offline, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/OffLineCapture?
format=json-->true
    </isSupportthermometryOffLineCapture>
    <isSupportShipsDetectionWithScene>
      <!--ro, opt, bool, whether the device supports ship detection by scene, desc:related URI: /ISAPI/Thermal/channels/<ID>/shipsDetection/capabilities?
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<!--ro, opt, bool, whether the device supports ship detection by scene, desc:related URI: /ISAPI/Thermal/channels/<ID>/shipsDetection/capabilities?format=json-->true
<isSupportShipsDetectionWithScene>
<!--ro, opt, bool, whether the device supports waste gas monitoring-->true
<isSupportWasteGasDetection>
<!--ro, opt, bool, whether the device supports the gray scale alarm-->true
<isSupportGreyScaleAlarm>
<!--ro, opt, bool, whether the device supports temperature screening when capturing face pictures-->true
<isSupportFaceSnapThermometry>
<!--ro, opt, bool, whether device supports temperature calibration, desc:related URI: /ISAPI/Thermal/channels/<ID>/temperatureCorrect?format=json-->true
<isSupportThermalTemperatureCorrect>
<!--ro, opt, bool, whether the device supports thermal imaging vehicle detection-->true
<isSupportThermalVehicleDetection>
<!--ro, opt, bool, whether the device supports configuring the sensor temperature measurement parameters, desc:related URI: /ISAPI/Thermal/channels/<channelID>/thermometry/sensorParam/capabilities?format=json-->true
<isSupportSensorParam>
<!--ro, opt, bool, whether the device supports temperature range detection-->true
<isSupportTemperatureIntervalMeasurement>
<!--ro, opt, bool, whether the device supports setting temperature rise compensation in thermal industrial temperature measurement, desc:related URI: /ISAPI/Thermal/channels/<ID>/industryThermometry/temperatureRise?format=json-->true
<isSupportIndustryThermometryTemperatureRise>
<!--ro, opt, bool, whether the device supports getting the temperature data link information, desc:related URI: /ISAPI/Thermal/thermalDataLinkInfo?format=json; /ISAPI/Thermal/thermalDataLinkStatus?format=json-->true
<isSupportThermalDataLinkInfo>
<!--ro, opt, bool, whether the device supports temperature compensation, desc:related URI: /ISAPI/Thermal/channels/<channelID>/temperatureCompensationCalibration?format=json-->true
<isSupportTemperatureCompensationCalib>
<!--ro, opt, bool, whether the web needs to display full-screen temperature parameters, desc:related URI: /ISAPI/Thermal/channels/<channelID>/thermometry/pixelToPixelParam/capabilities-->true
<isSupportWebDisplayPixelToPixelParam>
<!--ro, opt, bool, whether the device supports thermal patrol, desc:related URI: /ISAPI/Thermal/inspection/capabilities?format=json-->true
<isSupportThermalInspection>
<!--ro, opt, bool, whether the device supports searching for black body center point temperature, desc:related URI: /ISAPI/Thermal/channels/<channelID>/blackBodyTemperature?format=json-->true
<isSupportBlackBodyTemperature>
<!--ro, opt, bool, whether the device supports exporting the temperature calibration file, desc:POST /ISAPI/Thermal/ExportThermalCalibrationFile?format=json-->true
<isSupportExportThermalCalibrationFile>
<!--ro, opt, bool, whether the device supports searching for history records of smoke and fire detection, desc:POST /ISAPI/Thermal/fireDetection/SearchSmokeAndFireDetection?format=json-->true
<isSupportSearchSmokeAndFireDetection>
<!--ro, opt, bool, whether the device supports column oven parameters, desc:GET/PUT /ISAPI/OPCA/GC/ColumnOven?format=json-->true
<isSupportColumnOven>
<!--ro, opt, bool, whether the device supports importing the temperature calibration file, desc:POST /ISAPI/Thermal/ImportThermalCalibrationFile?format=json-->true
<isSupportImportThermalCalibrationFile>
<!--ro, opt, bool, whether the device supports the parameters of generating temperature calibration file, desc:GET /ISAPI/Thermal/ThermalCalibrationFileCreatParam?format=json-->true
<isSupportThermalCalibrationFileCreatParam>
<!--ro, opt, bool, whether the device supports the parameters of smoke and fire detection normal rules, desc:GET/PUT /ISAPI/Thermal/fireDetection/channels/<channelID>/NormalRuleParams?format=json-->true
<isSupportNormalRuleParams>
<!--ro, opt, bool, whether the device supports the parameters of environment temperature monitor arming schedule, desc:GET/PUT /ISAPI/Thermal/EnvironmentTemperatureMonitorScheduleParams?format=json-->true
<isSupportEnvironmentTemperatureMonitorScheduleParams>
<!--ro, opt, bool, whether the device supports the parameters of environment temperature monitoring, desc:GET/PUT /ISAPI/Thermal/EnvironmentTemperatureMonitorParam?format=json-->true
<isSupportEnvironmentTemperatureMonitorParam>
<!--ro, opt, bool, whether the device supports the parameters of environment temperature monitoring linkage action, desc:GET/PUT /ISAPI/Thermal/EnvironmentTemperatureMonitorTriggerParams?format=json-->true
<isSupportEnvironmentTemperatureMonitorTriggerParams>
<!--ro, opt, bool, whether the device supports searching for information of specified asynchronous thermal inspection point, desc:POST /ISAPI/Thermal/SearchAsynThermalInspectionTaskTestPoint?format=json-->true
<isSupportSearchAsynThermalInspectionTaskTestPoint>
<!--ro, opt, bool, whether the device supports searching for specified asynchronous thermal inspection task, desc:POST /ISAPI/Thermal/SearchAsynThermalInspectionTask?format=json-->true
<isSupportSearchAsynThermalInspectionTask>
<!--ro, opt, bool, whether the device supports adding information of specified asynchronous thermal inspection point, desc:POST

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/ISAPI/Thermal/AddAsynThermalInspectionTaskTestPoint?format=json-->true
    </isSupportAddAsynThermalInspectionTaskTestPoint>
    <isSupportAddAsynThermalInspectionTask>
      <!--ro, opt, bool, whether the device supports adding asynchronous thermal inspection task, desc:POST /ISAPI/Thermal/AddAsynThermalInspectionTask?
format=json-->true
    </isSupportAddAsynThermalInspectionTask>
    <isSupportImportAsynThermalInspectionTaskBaseMap>
      <!--ro, opt, bool, whether the device supports importing asynchronous thermal inspection task base map, desc:POST
/ISAPI/Thermal/ImportAsynThermalInspectionTaskBaseMap?format=json-->true
    </isSupportImportAsynThermalInspectionTaskBaseMap>
    <isSupportShutterFreeze>
      <!--ro, opt, bool, whether the device supports freezing the shutter, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ShutterFreeze?format=json-->true
    </isSupportShutterFreeze>
  </ThermalCap>
  <WLAlarmCap>
    <!--ro, opt, object, wireless alarm capability-->
    <isSupportTeleControl>
      <!--ro, opt, bool, whether the device supports remote control-->true
    </isSupportTeleControl>
    <isSupportPIR>
      <!--ro, opt, bool, whether the device supports PIR configuration-->true
    </isSupportPIR>
    <isSupportWLSensors>
      <!--ro, opt, bool, N/A-->true
    </isSupportWLSensors>
    <isSupportCallHelp>
      <!--ro, opt, bool, whether the device supports calling for help-->true
    </isSupportCallHelp>
    <WLSensorsNum>
      <!--ro, opt, int, N/A-->1
    </WLSensorsNum>
  </WLAlarmCap>
  <isSupportGIS>
    <!--ro, opt, bool, whether the device supports GIS, desc:related URI: /ISAPI/GIS/channels-->true
  </isSupportGIS>
  <isSupportCompass>
    <!--ro, opt, bool, whether the device supports compass configuration, desc:related URI: /ISAPI/Compass/channels/<ID>/capabilities-->true
  </isSupportCompass>
  <isSupportRoadInfoOverlays>
    <!--ro, opt, bool, whether the device supports overlaying the lane information-->true
  </isSupportRoadInfoOverlays>
  <isSupportFaceCaptureStatistics>
    <!--ro, opt, bool, whether the device supports face capture statistics, desc:related URI: /ISAPI/Intelligent/channels/<ID>/faceCaptureStatistics/search-
->true
  </isSupportFaceCaptureStatistics>
  <isSupportElectronicsEnlarge>
    <!--ro, opt, bool, whether the device supports electronics enlarge-->true
  </isSupportElectronicsEnlarge>
  <isSupportRemoveStorage>
    <!--ro, opt, bool, whether the device supports remote storage-->true
  </isSupportRemoveStorage>
  <isSupportCloud>
    <!--ro, opt, bool, whether the device supports cloud storage-->true
  </isSupportCloud>
  <isSupportRecordHost>
    <!--ro, opt, bool, whether the device supports configuring the education sharing server, desc:related URI: /ISAPI/RecordHost/capabilities-->true
  </isSupportRecordHost>
  <isSupportMetalDetection>
    <!--ro, opt, bool-->true
  </isSupportMetalDetection>
  <isSupportViewshedOverlay>
    <!--ro, opt, bool, whether the device supports overlaying viewshed camera target, desc:related URI:
/ISAPI/System/Video/inputs/channels/<ID>/viewshedOverlay-->true
  </isSupportViewshedOverlay>
  <isSupportGPSLabelTracking>
    <!--ro, opt, bool, whether the device supports GPS tag tracking, desc:related URI:/ISAPI/GIS/channels/<ID>/GPSLabelTracking?format=json-->true
  </isSupportGPSLabelTracking>
  <isSupportEagleEye>
    <!--ro, opt, bool, whether the device supports the PanoVu series camera-->true
  </isSupportEagleEye>
  <isSupportPanorama>
    <!--ro, opt, bool, whether the device supports panoramic mode-->true
  </isSupportPanorama>
  <isSupportFirmwareVersionInfo>
    <!--ro, opt, bool, whether the device supports displaying the firmware version information-->true
  </isSupportFirmwareVersionInfo>
  <isSupportExternalWirelessServer>
    <!--ro, opt, bool, whether the device supports external wireless servers: true (support)-->true
  </isSupportExternalWirelessServer>
  <isSupportSetupCalibration>
    <!--ro, opt, bool, whether the device supports configuring calibration-->true
  </isSupportSetupCalibration>
  <isSupportGetmutexFuncErrMsg>
    <!--ro, opt, bool, whether the device supports the mutual exclusion information-->true
  </isSupportGetmutexFuncErrMsg>
  <isSupportTokenAuthenticate>
    <!--ro, opt, bool, whether the device supports token authentication-->true
  </isSupportTokenAuthenticate>
  <isSupportStreamDualVCA>
    <!--ro, opt, bool, whether the device supports overlaying intelligent private data on the stream, desc:related URI: in the
/ISAPI/Intelligent/channels/<ID>/intelliResource; corresponding node: <isOverlapIntelli>-->true
  </isSupportStreamDualVCA>
  <isSupportLaserSpotManual>
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    <!--ro, opt, bool, whether the device supports configuring laser light spot-->true
  </isSupportLaserSpotManual>
  <isSupportRTMP>
    <!--ro, opt, bool, whether the device supports RTMP configuration, desc:related URI: /ISAPI/Streaming/channels/<ID>/RTMPCfg/capabilities-->true
  </isSupportRTMP>
  <isSupportTraffic>
    <!--ro, opt, bool, traffic service capability-->true
  </isSupportTraffic>
  <isSupportLaserSpotAdjustment>
    <!--ro, opt, bool, adjust the size of laser spot, desc:related URI: /ISAPI/Image/channels/<ID>/laserLight/laserSpotAdjustment/capabilities-->true
  </isSupportLaserSpotAdjustment>
  <VideoIntercomCap>
    <!--ro, opt, object, video intercom capability, desc:related URI: /ISAPI/VideoIntercom/capabilities-->
    <isSupportSceneManage>
      <!--ro, opt, bool, scene management (the configuration port needs to be switched according to various scenes)-->true
    </isSupportSceneManage>
    <isSupportDeviceId>
      <!--ro, opt, bool, whether the device supports device ID-->true
    </isSupportDeviceId>
    <isSupportOperationTime>
      <!--ro, opt, bool, whether the device supports the operation time-->true
    </isSupportOperationTime>
    <isSupportCallerInfo>
      <!--ro, opt, bool, whether the device supports the information management of the calling persons-->true
    </isSupportCallerInfo>
    <isSupportCallSignal>
      <!--ro, opt, bool, whether the device supports the call signaling interaction, desc:related URI: /ISAPI/VideoIntercom/callSignal?format=json-->true
    </isSupportCallSignal>
    <isSupportCallStatus>
      <!--ro, opt, bool, whether the device supports getting call status, desc:related URI: /ISAPI/VideoIntercom/callStatus?format=json-->true
    </isSupportCallStatus>
    <isSupportCallElevator>
      <!--ro, opt, bool, whether the device supports calling for elevator-->true
    </isSupportCallElevator>
    <isSupportRelatedDeviceAdress>
      <!--ro, opt, bool, whether the device supports linking device address-->true
    </isSupportRelatedDeviceAdress>
    <isSupportCardSectorCheck>
      <!--ro, opt, bool, sector encryption verification-->true
    </isSupportCardSectorCheck>
    <isSupportRemoteOpenDoor>
      <!--ro, opt, bool, whether it supports remote unlocking-->true
    </isSupportRemoteOpenDoor>
    <isSupportKeyCfg>
      <!--ro, opt, bool, configuration for pressing button to call for resident-->true
    </isSupportKeyCfg>
    <isSupportWorkModeCfg>
      <!--ro, opt, bool, whether the device supports configuring the working mode, desc:related URI: URL/ISAPI/VideoIntercom/workMode/capabilities-->true
    </isSupportWorkModeCfg>
    <isSupportSystemSwitchCfg>
      <!--ro, opt, bool, configuring the switch of video intercom system-->true
    </isSupportSystemSwitchCfg>
    <isSupportPrivilegePasswordStatus>
      <!--ro, opt, bool, getting the status of the device's permission password-->true
    </isSupportPrivilegePasswordStatus>
    <isSupportSmartHomeRoomsCfg>
      <!--ro, opt, bool, whether the device supports configuring room information-->true
    </isSupportSmartHomeRoomsCfg>
    <isSupportSmartHomeDevices>
      <!--ro, opt, bool, whether the device supports intelligent devices management-->true
    </isSupportSmartHomeDevices>
    <isSupportScenesCfg>
      <!--ro, opt, bool, whether the device supports scene configuration (adding, deleting, editing and searching for)-->true
    </isSupportScenesCfg>
    <isSupportNameModify>
      <!--ro, opt, bool, whether the device supports editing the zone name-->true
    </isSupportNameModify>
    <isSupportElevatorControlCfg>
      <!--ro, opt, bool, whether the device supports elevator control-->true
    </isSupportElevatorControlCfg>
    <isSupportPrivilegePassword>
      <!--ro, opt, bool, whether the device supports configuring passwords for different scenes-->true
    </isSupportPrivilegePassword>
    <isSupportSubModules>
      <!--ro, opt, bool, whether the device supports sub-module management-->true
    </isSupportSubModules>
    <isSupportIssueCards>
      <!--ro, opt, bool, whether the device supports configuring the card issuing function of the master card-->true
    </isSupportIssueCards>
    <isSupportSendCardCfg>
      <!--ro, opt, bool, whether the door station supports configuring card issuing function-->true
    </isSupportSendCardCfg>
    <isSupportRingBackTone>
      <!--ro, opt, bool, whether the device supports importing the ring back tone-->true
    </isSupportRingBackTone>
    <isSupportCallPriority>
      <!--ro, opt, bool, whether the device supports calling permission-->true
    </isSupportCallPriority>
    <isSupportPhoneCfg>
      <!--ro, opt, bool, whether the device supports configuring resident phones-->true
    </isSupportPhoneCfg>
    <isSupportWatchPoint>
      <!--ro, opt, bool, whether the device supports watch point, desc:/ISAPI/VideoIntercom/WatchPoint-->true
    </isSupportWatchPoint>
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    <isSupportVoiceConfiguration>
      <!--ro, opt, bool, whether the device supports the configuration of enabling the voice recognition and prompt-->true
    </isSupportVoiceConfiguration>
    <isSupportSubModulesCfg>
      <!--ro, opt, bool, whether the device supports configuring editing the sub module parameters-->true
    </isSupportSubModulesCfg>
    <isSupportDisplayRegion>
      <!--ro, opt, bool, whether the device supports configuring the area to be displayed-->true
    </isSupportDisplayRegion>
    <isSupportUpdateData>
      <!--ro, opt, bool, whether the device supports importing upgrade files-->true
    </isSupportUpdateData>
    <isSupportUpdateStrategy>
      <!--ro, opt, bool, whether the device supports upgrade configuration-->true
    </isSupportUpdateStrategy>
    <isSupportMetaInfo>
      <!--ro, opt, bool, whether the device supports getting the information of upgrade files-->true
    </isSupportMetaInfo>
    <isSupportStreamAbility>
      <!--ro, opt, bool, whether the device supports IPC management-->true
    </isSupportStreamAbility>
    <platform opt="R0,R1,S29,N10,N8,N4">
      <!--ro, opt, enum, platform type, subType:string, attr:opt{req, string}, desc:"R0", "R1"; it will be returned if the device supports-->R0
    </platform>
    <isSupportSecurityMode>
      <!--ro, opt, bool, whether the device supports switching the version of security protocols-->true
    </isSupportSecurityMode>
    <isSupportDeviceCommunication>
      <!--ro, opt, bool, whether the device supports configuring the parameters of device group-net communication-->true
    </isSupportDeviceCommunication>
    <isSupportAutoSync>
      <!--ro, opt, bool, whether the device supports automatically synchronous parameters configuration-->true
    </isSupportAutoSync>
    <isSupportDifferentKeystrokes>
      <!--ro, opt, bool, whether the device supports distinguishing long press and short press-->true
    </isSupportDifferentKeystrokes>
    <isSupportKeyCfgWeekPlan>
      <!--ro, opt, bool, the week schedule configuration for pressing to call-->true
    </isSupportKeyCfgWeekPlan>
    <isSupportKeyCfgHolidayPlan>
      <!--ro, opt, bool, the holiday schedule configuration for pressing to call-->true
    </isSupportKeyCfgHolidayPlan>
    <isSupportKeyCfgHolidayGroup>
      <!--ro, opt, bool, the holiday group configuration for pressing to call-->true
    </isSupportKeyCfgHolidayGroup>
    <isSupportKeyCfgPlanTemplate>
      <!--ro, opt, bool, the schedule template configuration for pressing to call-->true
    </isSupportKeyCfgPlanTemplate>
    <isSupportCallCfg>
      <!--ro, opt, bool, device call configuration-->true
    </isSupportCallCfg>
    <isSupportAccessParams>
      <!--ro, opt, bool, the application access parameters-->true
    </isSupportAccessParams>
    <isSupportTransmitRequestEventConfirm>
      <!--ro, opt, bool, response to the transparent transmission request-->true
    </isSupportTransmitRequestEventConfirm>
    <isSupportIncomingCallLinkage>
      <!--ro, opt, bool, whether the device supports configuring incoming calls linkage-->true
    </isSupportIncomingCallLinkage>
    <isSupportAdbControl>
      <!--ro, opt, bool, whether the device supports remotely enabling and disabling ADB-->true
    </isSupportAdbControl>
    <isSupportProtocolFileManage>
      <!--ro, opt, bool, whether the device supports the management of the protocol notice files, desc:/ISAPI/VideoIntercom/protocolFileManage/capabilities?
format=json-->true
    </isSupportProtocolFileManage>
    <isSupportStreamSwitch>
      <!--ro, opt, bool, whether the device supports stream switch, desc:/ISAPI/VideoIntercom/streamSwitch/capabilities?format=json-->true
    </isSupportStreamSwitch>
    <isSupportKeywordAudioRecognition>
      <!--ro, opt, bool, whether the device supports keyword voice recognition, desc:/ISAPI/VideoIntercom/keywordAudioRecognition/capabilities?format=json--
>true
    </isSupportKeywordAudioRecognition>
    <isSupportOneWaySpeaking>
      <!--ro, opt, bool, whether the device supports speaking loud, desc:true (support), this node is not returned (not support)-->true
    </isSupportOneWaySpeaking>
    <isSupportPositionInfo>
      <!--ro, opt, bool, whether the device supports configuring the device location information, desc:/ISAPI/VideoIntercom/positionInfo?format=json-->true
    </isSupportPositionInfo>
    <isSupportElevatorControlDisplay>
      <!--ro, opt, bool, whether the device supports configuring displaying elevator control, desc:/ISAPI/VideoIntercom/elevatorControlDisplay?format=json--
>true
    </isSupportElevatorControlDisplay>
    <isSupportRingManage>
      <!--ro, opt, bool, whether the device supports ring tone management, desc:related URI: /ISAPI/VideoIntercom/ringTone?format=json;
/ISAPI/VideoIntercom/ringTone/<ringID>?format=json-->true
    </isSupportRingManage>
    <isSupportPasswordAuthentication>
      <!--ro, opt, bool, whether the device supports verifying the video intercom password, desc:related URI:
/ISAPI/VideoIntercom/passwordAuthentication/capabilitie-->true
    </isSupportPasswordAuthentication>
    <isSupportAlarmControlByPhone>
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      <!--ro, opt, bool, whether the device supports phone arming control, desc:related URI: /ISAPI/SecurityCP/AlarmControlByPhone/capabilities-->true
    </isSupportAlarmControlByPhone>
    <isSupportZoneCfgByScene>
      <!--ro, opt, bool, whether the device supports configuring the zones parameter by scene, desc:related URI:
/ISAPI/SecurityCP/BasicParam/sceneZoneCfg/capabilities-->true
    </isSupportZoneCfgByScene>
    <isSupportAppKeyConfiguration>
      <!--ro, opt, bool, whether the device supports configuring APPKey, desc:related URI: /ISAPI/VideoIntercom/AppKeyConfiguration/capabilities-->true
    </isSupportAppKeyConfiguration>
    <isSupportIntercomProtocolChange>
      <!--ro, opt, bool, whether the device supports configuring the switch of device VoIP protocols, desc:related URI:
/ISAPI/VideoIntercom/IntercomProtocolType/capabilities-->true
    </isSupportIntercomProtocolChange>
    <isSupportGetAnnouncementMessage>
      <!--ro, opt, bool, whether the device supports getting announcement message, desc:related URI: /ISAPI/VideoIntercom/AnnouncementMessage/capabilities--
>true
    </isSupportGetAnnouncementMessage>
    <isSupportGetDeployInfo>
      <!--ro, opt, bool, whether the device supports getting arming information, desc:related URI: /ISAPI/AccessControl/DeployInfo/capabilities-->true
    </isSupportGetDeployInfo>
    <detectorType opt="panicButton,magneticContact,smokeDetector,activeInfraredDetector,passiveInfraredDetector,combustibleGasDetector,doorbell">
      <!--ro, opt, string, detector types supported by the device, attr:opt{req, string}, desc:"panicButton" (panic button), "magneticContact" (magnetic
contact), "smokeDetector" (smoke detector), "activeInfraredDetector" (active infrared detecor), "passiveInfraredDetector" (passice infrared detector),
"combustibleGasDetector" (combustible gas detector)-->test
    </detectorType>
    <isSupportBroadcastRealtime>
      <!--ro, opt, bool, whether the device supports real-time audio broadcast-->true
    </isSupportBroadcastRealtime>
    <isSupportOrganizationNodes>
      <!--ro, opt, bool, whether the device supports organization architecture management-->true
    </isSupportOrganizationNodes>
    <isSupportAudioPlayStatus>
      <!--ro, opt, bool, whether the device supports uploading the audio playing status-->true
    </isSupportAudioPlayStatus>
    <isSupportDeviceIdAnalysisRuleCfg>
      <!--ro, opt, bool, whether the device supports configuring the analysis rule of the device No. (related URI:
/ISAPI/VideoIntercom/AnalysisRuleCfg/capabilities?format=json)-->true
    </isSupportDeviceIdAnalysisRuleCfg>
    <isSupportOperationTimeByChannel>
      <!--ro, opt, bool, whether the device supports configuring the operation time by channel-->true
    </isSupportOperationTimeByChannel>
    <isSupportCallAuthorization>
      <!--ro, opt, bool, whether the device supports configuring permission for incoming calls, desc:related URI:
/ISAPI/VideoIntercom/CallAuthorization/capbilities?format=json-->true
    </isSupportCallAuthorization>
    <isSupportRecordControl>
      <!--ro, opt, bool, whether the device supports audio/video control, desc:related URI: /ISAPI/VideoIntercom/recordControl/capabilities?format=json--
>true
    </isSupportRecordControl>
    <isSupportTrusteeship>
      <!--ro, opt, bool, whether the device supports configuring trusteeship, desc:related URI: /ISAPI/VideoIntercom/trusteeship/capabilities?format=json--
>true
    </isSupportTrusteeship>
    <isSupportAnalogIndoorStation>
      <!--ro, opt, bool, whether the device supports the management of the analog indoor station, desc:related URI:
/ISAPI/VideoIntercom/analogIndoorStation/capabilities?format=json-->true
    </isSupportAnalogIndoorStation>
    <isSupportImportPartitions>
      <!--ro, opt, bool, whether the device supports importing the terminal list, desc:(partition list for the paging microphone devices distributed by the
platform) (Related URI: /ISAPI/VideoIntercom/ImportTerminals/capabilities?format=json)-->true
    </isSupportImportPartitions>
    <isSupportBroadcastPlan>
      <!--ro, opt, bool, whether the device supports full application of schedule broadcast, desc:related URI:
ISAPI/VideoIntercom/broadcast/plan/capabilities?format=json-->true
    </isSupportBroadcastPlan>
    <isSupportBroadcastPlanSearch>
      <!--ro, opt, bool, whether the device supports searching for full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/planSearch?
format=json-->true
    </isSupportBroadcastPlanSearch>
    <isSupportBroadcastPlanPause>
      <!--ro, opt, bool, whether the device supports pausing full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/plan/pause?
format=json-->true
    </isSupportBroadcastPlanPause>
    <isSupportBroadcastPlanResume>
      <!--ro, opt, bool, whether the device supports restoring full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/plan/resume?
format=json-->true
    </isSupportBroadcastPlanResume>
    <isSupportBroadcastPlanTest>
      <!--ro, opt, bool, whether the device supports the test run of full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/plan/test?
format=json-->true
    </isSupportBroadcastPlanTest>
    <isSupportBroadcastPlanEx>
      <!--ro, opt, bool, whether the device supports incremental application of schedule broadcast, desc:related URI:
/ISAPI/VideoIntercom/broadcast/planEx/capabilities?format=json-->true
    </isSupportBroadcastPlanEx>
    <isSupportBroadcastPlanExSearch>
      <!--ro, opt, bool, whether the device supports searching for incremental schedule broadcast, desc:related URI:
/ISAPI/VideoIntercom/broadcast/planExSearch?format=json-->true
    </isSupportBroadcastPlanExSearch>
    <isSupportBroadcastPlanModify>
      <!--ro, opt, bool, whether the device supports editing a single schedule broadcast, desc:it is valid only during incremental application-->true
    </isSupportBroadcastPlanModify>
    <isSupportBroadcastPlanDelete>
      <!--ro, opt, bool, whether the device supports deleting a single schedule broadcast, desc:it is valid only during incremental application-->true
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      <!--req, opt, bool, whether the device supports deleting a single schedule broadcast, desc:it is valid only during incremental application-->true
    </isSupportBroadcastPlanDelete>
    <isSupportAudioInPlayCfg>
      <!--ro, opt, bool, whether the device supports the configuration of playing the audio input sources of the broadcast, desc:URI for getting the
capability: /ISAPI/VideoIntercom/broadcast/AudioInPlayCfg/capabilities?format=json-->true
    </isSupportAudioInPlayCfg>
    <isSupportSearchWardBedInfo>
      <!--ro, opt, bool, whether the device supports searching for ward bed information, desc:URI for getting the capability:
/ISAPI/VideoIntercom/searchWardBedInfo/capabilities?format=json. It is used for the main station to search for whether the bed connected to the sub machine
in the ward is already occupied-->true
    </isSupportSearchWardBedInfo>
    <isSupportPatient>
      <!--ro, opt, bool, whether the device supports applying patient information, desc:URI for getting the capability:
/ISAPI/VideoIntercom/medical/patient/capabilities?format=json-->true
    </isSupportPatient>
    <isSupportIntroduction>
      <!--ro, opt, bool, whether the device supports applying the medical introduction, desc:URI for getting the capability:
/ISAPI/VideoIntercom/medical/introduction/capabilities?format=json-->true
    </isSupportIntroduction>
    <isSupportConstantVoltage>
      <!--ro, opt, bool, whether the device supports constant voltage broadcast, desc:URI for getting the capability:
/ISAPI/VideoIntercom/broadcast/ConstantVoltage/capabilities?format=json-->true
    </isSupportConstantVoltage>
    <isSupportRecordMode>
      <!--ro, opt, bool, whether it supports configuring recording mode, desc:URI for getting the capability: /ISAPI/VideoIntercom/RecordMode/capabilities?
format=json-->true
    </isSupportRecordMode>
    <isSupportCallForwarding>
      <!--ro, opt, bool, whether the device supports active call transfer, desc:URI for getting the capability:
/ISAPI/VideoIntercom/callForwarding/capabilities?format=json-->true
    </isSupportCallForwarding>
    <isSupportMediaLibrary>
      <!--ro, opt, bool, whether the device supports applying media library information, desc:URI for getting the capability:
/ISAPI/VideoIntercom/broadcast/MediaLibrary/capabilities?format=json-->true
    </isSupportMediaLibrary>
    <isSupportBroadcastControl>
      <!--ro, opt, bool, whether the device supports configuring broadcast control parameters, desc:URI for getting the capability:
/ISAPI/VideoIntercom/broadcast/control/capabilities?format=json-->true
    </isSupportBroadcastControl>
    <isSupportGetImportPartitions>
      <!--ro, opt, bool, whether the device supports getting partition list, desc:POST /ISAPI/VideoIntercom/broadcast/GetImportPartitions?format=json-->true
    </isSupportGetImportPartitions>
    <isSupportSearchMediaLibrary>
      <!--ro, opt, bool, whether the device supports searching media library, desc:POST /ISAPI/VideoIntercom/broadcast/SearchMediaLibrary?format=json-->true
    </isSupportSearchMediaLibrary>
    <isSupportImportPartitionsBatch>
      <!--ro, opt, bool, whether the device supports batch importing of partitions, desc:PUT /ISAPI/VideoIntercom/broadcast/ImportPartitionsBatch?
format=json-->true
    </isSupportImportPartitionsBatch>
    <isSupportMediaLibraryBatch>
      <!--ro, opt, bool, whether the device supports batch applying of media library information, desc:PUT /ISAPI/VideoIntercom/broadcast/MediaLibraryBatch?
format=json-->true
    </isSupportMediaLibraryBatch>
    <isSupportPlanScheme>
      <!--ro, opt, bool, whether the device supports broadcast schedules, desc:GET /ISAPI/VideoIntercom/broadcast/PlanScheme/capabilities?format=json-->true
    </isSupportPlanScheme>
    <isSupportRealTimeInsertion>
      <!--ro, opt, bool, whether the device supports real-time cut-in, desc:POST /ISAPI/VideoIntercom/broadcast/RealTimeInsertion?format=json-->true
    </isSupportRealTimeInsertion>
  </VideoIntercomCap>
  <SecurityCPCapabilities>
    <!--ro, opt, object, [TBC], desc:related URI: /ISAPI/SecurityCP/capabilities?format=json-->true
  </SecurityCPCapabilities>
  <isSupportSafetyCabin>
    <!--ro, opt, bool, safety cabinet controller, desc:related URI: /ISAPI/SafetyCabin/capabilities-->true
  </isSupportSafetyCabin>
  <isSupportAcsUpdate>
    <!--ro, opt, bool, whether the device supports peripheral module upgrade (true, support), desc:related URI: /ISAPI/System/AcsUpdate/capabilities-->true
  </isSupportAcsUpdate>
  <isSupportSecurityCP>
    <!--ro, opt, bool, N/A-->true
  </isSupportSecurityCP>
  <isSupportPEA>
    <!--ro, opt, bool, the host capability of one-push alarm-->true
  </isSupportPEA>
  <isSupportCurrentLock>
    <!--ro, opt, bool, whether the device supports the configuration of locking current-->true
  </isSupportCurrentLock>
  <isSupportGuardAgainstTheft>
    <!--ro, opt, bool, whether the device supports configuring device guard against theft, desc:related URI: /ISAPI/System/guardAgainstTheft/capabilities--
>true
  </isSupportGuardAgainstTheft>
  <isSupportPicInfoOverlap>
    <!--ro, opt, bool, whether the device supports overlaying pictures and texts-->true
  </isSupportPicInfoOverlap>
  <isSupportPlay>
    <!--ro, opt, bool, shielded live view-->true
  </isSupportPlay>
  <isSupportPlayback>
    <!--ro, opt, bool, shielded playback-->true
  </isSupportPlayback>
  <UHFRFIDReader>
    <!--ro, opt, object, capability supported by the RFID reader-->
    <isSupportBasicInformation>
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      <!--ro, opt, bool, whether the device supports the basic parameters of the reader-->true
    </isSupportBasicInformation>
    <isSupportHardDiskStorageTest>
      <!--ro, opt, bool, whether the device supports disk storage performance test of the reader-->true
    </isSupportHardDiskStorageTest>
    <isSupportCarReaderConfig>
      <!--ro, opt, bool, whether the device supports configuring the parameters of the electronic license plate reader, desc:related URI:
/ISAPI/Traffic/UHFRFIDReader/carReaderConfig/capabilities?format=json-->true
    </isSupportCarReaderConfig>
  </UHFRFIDReader>
  <isSupportMasterSlaveConfig>
    <!--ro, opt, bool, whether the device supports configuring main and sub device, desc:related URI: /ISAPI/System/masterSlaveConfig?format=json-->true
  </isSupportMasterSlaveConfig>
  <isSupportIntelligentStructureAnalysis>
    <!--ro, opt, bool, whether the device supports intelligent structuralization analysis, desc:related URI:
/ISAPI/Intelligent/structureAnalysis/capabilities-->true
  </isSupportIntelligentStructureAnalysis>
  <isSupportIntelligentAnalysisEngines>
    <!--ro, opt, bool, whether the device supports configuring the intelligent engine, desc:related URI: /ISAPI/Intelligent/analysisEngines/capabilities--
>true
  </isSupportIntelligentAnalysisEngines>
  <PreviewDisplayNum>
    <!--ro, opt, int, number of the browser live view window-->1
  </PreviewDisplayNum>
  <isSupportBoard opt="true,false">
    <!--ro, opt, bool, whether the device supports related sub-board protocols, attr:opt{opt, string}-->true
  </isSupportBoard>
  <ResourceSwitch>
    <!--ro, opt, object, configuration of resource conversion-->
    <workMode opt="4KPreview,educationRecord">
      <!--ro, req, string, the working mode: education record mode, attr:opt{opt, string}-->test
    </workMode>
  </ResourceSwitch>
  <isSupportCustomStream>
    <!--ro, opt, bool, whether the device supports custom stream, desc:related URI: /ISAPI/Streaming/channels/<ID>/customStream/capabilities-->true
  </isSupportCustomStream>
  <isSupportTriggerCapCheck>
    <!--ro, opt, bool, it indicates that the capability of the URL (/ISAPI/Event/triggersCap) is already verified-->true
  </isSupportTriggerCapCheck>
  <isSupportChannelEventCap>
    <!--ro, opt, bool, whether the device supports getting the event capability by channel, desc:related URI: /ISAPI/Event/channels/<ID>/capabilities-->true
  </isSupportChannelEventCap>
  <SensorCap>
    <!--ro, opt, object, capability of configuring sensor parameters, desc:related URI: /ISAPI/System/Sensor/capabilities-->
    <SensorNums>
      <!--ro, opt, int, the maximum number of sensors that can be added-->1
    </SensorNums>
    <isSupportSensorPortConfiguration>
      <!--ro, opt, bool, supports configuring the detection parameters-->true
    </isSupportSensorPortConfiguration>
    <isSupportSensorPortControl>
      <!--ro, opt, bool, supports detection control-->true
    </isSupportSensorPortControl>
    <isSupportAcquireSensorStatus>
      <!--ro, opt, bool, supports getting the real-time status of the sensor-->true
    </isSupportAcquireSensorStatus>
    <isSupportSearchSensorData>
      <!--ro, opt, bool, supports searching the sensor data-->true
    </isSupportSearchSensorData>
  </SensorCap>
  <isSupportEncryption>
    <!--ro, opt, bool, stream encryption capability-->true
  </isSupportEncryption>
  <isSupportTrial>
    <!--ro, opt, bool, whether the device supports configuring inquest device-->true
  </isSupportTrial>
  <isSupportSignInstrument>
    <!--ro, opt, bool, the capability of physical sign monitor-->true
  </isSupportSignInstrument>
  <isSupportLocate>
    <!--ro, opt, bool, whether the device supports location configuration-->true
  </isSupportLocate>
  <isSupportVibration>
    <!--ro, opt, bool, whether the device supports configuring vibration parameters-->true
  </isSupportVibration>
  <isSupportMixedTargetDetection>
    <!--ro, opt, bool, whether the device supports multi-target-type detection events-->true
  </isSupportMixedTargetDetection>
  <SDCardCtrlCap>
    <!--ro, opt, object, the remote control capability of the memory card-->
    <isSupportIOCtrl>
      <!--ro, opt, bool, whether the device supports the capability of control and command upgrade-->true
    </isSupportIOCtrl>
    <isSupportReadCardStatus>
      <!--ro, opt, bool, whether the device supports the capability of card recognition operation-->true
    </isSupportReadCardStatus>
    <isSupportStartUpdate>
      <!--ro, opt, bool, whether the device supports the capability of upgrade start control-->true
    </isSupportStartUpdate>
    <isSupportEndUpdate>
      <!--ro, opt, bool, whether the device supports the capability of upgrade end control-->true
    </isSupportEndUpdate>
    <isSupportPowerCtrl>
      <!--ro, opt, bool, whether the device supports the capability of power on and off control-->true
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    <!--ro, opt, bool, whether the device supports the capability of power on and off control-->true
    </isSupportPowerCtrl>
  </SDCardCtrlCap>
  <isSupportMutiChannelCalibration>
    <!--ro, opt, bool, whether the device supports multi-channel calibration-->true
  </isSupportMutiChannelCalibration>
  <isSupportElevatorFloorCalibration>
    <!--ro, opt, bool, whether the device supports configuring floor display-->true
  </isSupportElevatorFloorCalibration>
  <isSupportPictureCaptureComparision>
    <!--ro, opt, bool, whether the device supports 1V1 comparison of the pictures in the library and from camera-->true
  </isSupportPictureCaptureComparision>
  <isSupportGPSCalibratation>
    <!--ro, opt, bool, whether the device supports the GPS calibration function-->true
  </isSupportGPSCalibratation>
  <isSupportCityManagement>
    <!--ro, opt, bool, whether the device supports city management detection event-->true
  </isSupportCityManagement>
  <isSupportBVCorrect>
    <!--ro, opt, bool, whether the device supports configuring the parameters of camera correction-->true
  </isSupportBVCorrect>
  <isSupportSteamingCodingModeSwitch>
    <!--ro, opt, bool, encoding output mode (true-support), desc:related URI: /ISAPI/System/streamingCoding/mode/capabilities?format=json-->true
  </isSupportSteamingCodingModeSwitch>
  <supportImageChannel opt="1,2,3,4,5,6">
    <!--ro, opt, string, whether the device supports configuring the image channel, attr:opt{req, string},
desc:the configuration of actual image channel supported by the device (URI for picture parameters: /ISAPI/Image/channels/<channelID>), if this node is not
returned, the image channel is the same with the encoding channel (/ISAPI/Streaming/channels) by default
    this node is mainly for the situation when the image channel is inconsistent with the actual stream channel. For example, some device, which altogether
has four sensors and they are all spliced into one encoding channel for output. Every sensor supports configuring image parameters; by default, the image
channel 1 is only valid for the images of sensor1-->test
  </supportImageChannel>
  <isSupportPanoramaCalibration>
    <!--ro, opt, bool, whether the device supports configuring the calibration of the panoramic camera-->true
  </isSupportPanoramaCalibration>
  <isSupportPanoramaMode>
    <!--ro, opt, bool, whether the device supports the mode switch of the panoramic camera-->true
  </isSupportPanoramaMode>
  <supportSnapshotChannel opt="1">
    <!--ro, opt, string, the capability supported by the capture schedule channel, attr:opt{opt, string}-->test
  </supportSnapshotChannel>
  <isSupportPanoramaCorrections>
    <!--ro, opt, bool, whether the device supports configuring the correction algorithm parameters-->true
  </isSupportPanoramaCorrections>
  <isSupportActiveMulticast>
    <!--ro, opt, bool, whether the device supports multicast configuration (/ISAPI/Streaming/channels/<ID>/capabilities)-->true
  </isSupportActiveMulticast>
  <isSupportChannelEventListCap>
    <!--ro, opt, bool, it indicates that whether the device supports getting the event capability of all channels of the device-->true
  </isSupportChannelEventListCap>
  <VCAResourceChannelsCap>
    <!--ro, opt, array, it indicates that the device supports the independent switch of intelligent resources by channel, subType:object-->
    <ChannelsList>
      <!--ro, opt, object, channel list-->
      <channelsID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelsID>
    </ChannelsList>
  </VCAResourceChannelsCap>
  <isSupportTimerVCAResource>
    <!--ro, opt, bool, whether the device supports scheduled switch of intelligent resources-->true
  </isSupportTimerVCAResource>
  <MixedTargetDetection>
    <!--ro, opt, object, the capability of recognizing specified target among mixed targets supported by the device-->
    <isSupportFaceRecognition>
      <!--ro, opt, bool, whether the device supports facial recognition-->true
    </isSupportFaceRecognition>
    <isSupportHumanRecognition>
      <!--ro, opt, bool, whether the device supports human body recognition-->true
    </isSupportHumanRecognition>
    <isSupportVehicleRecognition>
      <!--ro, opt, bool, whether the device supports vehicle recognition-->true
    </isSupportVehicleRecognition>
  </MixedTargetDetection>
  <isSupportDiscoveryMode>
    <!--ro, opt, bool, whether the device supports discovery mode, desc:related URI: /ISAPI/System/discoveryMode/capabilities-->true
  </isSupportDiscoveryMode>
  <isSupportDumpData>
    <!--ro, opt, bool, whether the device supports exporting the Dump file-->true
  </isSupportDumpData>
  <isSupportCameraAccessConfiguration>
    <!--ro, opt, bool, whether the device supports the camera access configuration, desc:related URI: /ISAPI/System/cameraAccess/configuration/capabilities?
format=json-->true
  </isSupportCameraAccessConfiguration>
  <isSupportIntelligentSearch>
    <!--ro, opt, bool, whether the device supports intelligent search, desc:related URI: /ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json--
>true
  </isSupportIntelligentSearch>
  <isSupportAutoMaintenance>
    <!--ro, opt, bool, whether the device supports automatic maintenance, desc:related URI: /ISAPI/System/autoMaintenance/capabilities?format=json-->true
  </isSupportAutoMaintenance>
  <isSupportRtspOverHTTPS>
    <!--ro, opt, bool, supports tspOverHTTPS-->true
  </isSupportRtspOverHTTPS>
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  <isSupportIntersectionAnalysis>
    <!--ro, opt, bool, whether the device supports the intersection analysis, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/intersectionAnalysis/capabilities?format=json-->true
  </isSupportIntersectionAnalysis>
  <isSupportAudioInteract>
    <!--ro, opt, bool, whether the device supports audio interaction, desc:related URI: /ISAPI/Streaming/interact/capabilities?format=json-->true
  </isSupportAudioInteract>
  <isSupportChannelFullEventListCap>
    <!--ro, opt, bool, it indicates that whether the device supports getting the event capability of all channels of the device-->true
  </isSupportChannelFullEventListCap>
  <isSupportAUXInfoCap>
    <!--ro, opt, bool, it indicates that whether the device supports getting the capability of getting the attributes of device's all channels-->true
  </isSupportAUXInfoCap>
  <supportPanoramaChannel opt="2">
    <!--ro, opt, string, the panoramic channels supported by the device, attr:opt{opt, string}-->test
  </supportPanoramaChannel>
  <regionClip>
    <!--ro, opt, object, the channels and streams that supports area cropping-->
    <channels opt="101">
      <!--ro, opt, string, the channels and streams that supports area cropping, attr:opt{opt, string}, desc:101: channel No.*100+stream type-->test
    </channels>
  </regionClip>
  <AIDEventSupport
opt="abandonedObject,pedestrian,congestion,roadBlock,construction,trafficAccident,fogDetection,wrongDirection,illegalParking,SSharpDriving,lowSpeed,dragRaci
ng,obstacle,vehNoYieldPedest,illegalMannedVeh,illegalMannedNonMotorVeh,umbrellaTentInstall,nonMotorVehOnVehLane,wearingNoHelmet,pedestRedLightRunning,pedest
OnNonMotorVehLane,pedestOnVehLane,conflagration,smoke,VehicleSpeedDrop,lowVisibility,animalsOccupy,targetSpeedDetection,emergencyLane,gasser,changeLaneConti
nuously,RegionalVehicleStatistics">
    <!--ro, opt, string, supported AID event, attr:opt{opt, string}, desc:supported AID event-->test
  </AIDEventSupport>
  <TFSEventSupport opt="illegalParking
,wrongDirection,crossLane,laneChange,vehicleExist,turnRound,parallelParking,notKeepDistance,notSlowZebraCrossing,overtakeRightSide,lowSpeed,dragRacing,chang
eLaneContinuously,SSharpDriving,largeVehicleOccupyLine,jamCrossLine,obstacle,blackSmokeVehicle,turnRightStop,occupyDedicatedLane,notDriveInDedicatedLane,TFS
_nonZipperMerge,overSpeed,emergencyLane,gasser">
    <!--ro, opt, string, supported TFS, attr:opt{opt, string}-->test
  </TFSEventSupport>
  <isVehicleStatisticsSupport>
    <!--ro, opt, bool, whether the device supports configuring the parameters for traffic data statistics-->true
  </isVehicleStatisticsSupport>
  <isSupportElevatorFloorName>
    <!--ro, opt, bool, whether the device supports configuring floor custom name-->true
  </isSupportElevatorFloorName>
  <isSupportVehicleFaceRecognition>
    <!--ro, opt, bool, whether the device supports checkpoint vehicle recognition, desc:background: the traffic devices support "facial recognition"
(driver), however, the corresponding node is not added to the device to notify the integration of this capability-->true
  </isSupportVehicleFaceRecognition>
  <VQDEventSupport opt="luma,chroma,snow,streak,freeze,sigLose,clarity,jitter,block,flowers,noise,ghost,purple,ICR,protectiveFilm,blur,contrast">
    <!--ro, opt, string, VQD event list, attr:opt{opt, string}, desc:"Luma" (brightness exception); "Chroma" (image color cast); "Snow" (snow noise);
"Streak" (strip noise); "Freeze" (frozen screen); "SigLose" (signal loss); "Clarity" (resolution exception);  "JItter" (image flutter); "BLock"
(video tampering); "Flowers" (blurred screen); "Noise" (image noise); "Ghost" (ghost image); "Purple" (purple fringing); "ICR" (ICP); "protectiveFilm"
(protective film unremoved)-->test
  </VQDEventSupport>
  <IOTCap>
    <!--ro, opt, object, supports searching for the device status list by page, desc:not returned (not support, get the device status list via GET)-->
    <supportChannelNum opt="5000">
      <!--ro, opt, int, the number of supported IOT device channels, attr:opt{opt, string}-->1
    </supportChannelNum>
    <startChannelNo>
      <!--ro, opt, int, the channel No. of start IOT device-->1
    </startChannelNo>
    <isSupportlinkageChannelsSearch>
      <!--ro, opt, bool, true (support), desc:related URI: (POST /ISAPI/System/IOT/LinkageChannels?format=json)-->true
    </isSupportlinkageChannelsSearch>
    <isSupportIOTChannelCap>
      <!--ro, opt, bool, whether the device supports the channel configuration capability, desc:it returns true for support, related URI: (GET
/ISAPI/System/IOT/channels/<IOTChannelID>/capabilities?format=json)-->true
    </isSupportIOTChannelCap>
    <satusSearchTaskCap>
      <!--ro, opt, object, task search capability-->
      <maxResults>
        <!--ro, opt, int, the maximum number of returned results-->1
      </maxResults>
      <filterType opt="all,online,offline,updateFailed,notPartitioned">
        <!--ro, opt, enum, filtering type, subType:string, attr:opt{req, string}, desc:all, online, offline, updateFailed (upgrading failed)-->all
      </filterType>
      <deviceStatus opt="calling,noCall,medicationChangePrompt,alarming,offline,free,callWaiting">
        <!--ro, opt, string, the value range for the device status search condition, range:[0,32], attr:opt{req, string}, desc:"calling" (calling), "noCall"
(non-calling status), "medicationChangePrompt", "alarming", "offline", "free"; if multiple conditions are specified, their relationship is "and" by default-
->calling
      </deviceStatus>
      <logicRelationship opt="and,or">
        <!--ro, opt, string, the value range for the logic relationship of device status search condition, range:[0,16], attr:opt{req, string}, desc:and-
the relationship between them is "and" (by default), or- the relationship between them is "or"; it is valid when deviceStatus exists-->and
      </logicRelationship>
      <alarmType opt="SOS,toilet">
        <!--ro, opt, string, value range of alarm type, range:[0,16], attr:opt{req, string}, desc:SOS-SOS call, toilet-toilet alarm. It is valid when the
deviceStatus contains Alarming-->test
      </alarmType>
      <callWaitingType opt="normal,cisit">
        <!--ro, opt, string, alarm type, attr:opt{req, string}, desc:normal-normal call waiting, visit-visit call waiting. It is valid when the deviceStatus
contains callWaiting-->test
      </callWaitingType>
    </satusSearchTaskCap>
    <channelConfigList>
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      <!--ro, opt, array, capability list returned according to unitType, subType:object-->
      <channelConfig>
        <!--ro, opt, object, device type capability-->
        <unitType>
          <!--ro, opt, enum, unit type, subType:string, desc:"indoor" (indoor station), "villa" (villa door station), "confirm" (doorphone), "outdoor" (door station), "fence" (outer door station), "doorbell" (doorbell), "manage" (main station), "acs" (access control devices), "wardStation", "bedheadExtension", "bedsideExtension", "terminal", "netAudio" (network sound box), "analogIndoor" (analog indoor station), "decoder" (decoder), "networkFireAlarmMatrix", "amplifier" (network amplifier), "pagingMicrophone" (SIP phone)-->outdoor
        </unitType>
        <floorNumber min="-10" max="999">
          <!--ro, opt, int, floor No., attr:min{req, int},max{req, int}-->0
        </floorNumber>
        <roomNumber min="0" max="999">
          <!--ro, opt, int, room No., attr:min{req, int},max{req, int}-->0
        </roomNumber>
        <isSupportSlaveDevice>
          <!--ro, opt, bool, whether the device supports sub-devices-->true
        </isSupportSlaveDevice>
        <isSupportZeroFloorNumber>
          <!--ro, opt, bool, whether the zero floor is supported, desc:it is used together with node floorNumber-->true
        </isSupportZeroFloorNumber>
        <periodNumber min="1" max="999">
          <!--ro, opt, int, community number, attr:min{req, int},max{req, int}-->0
        </periodNumber>
        <buildingNumber min="1" max="999">
          <!--ro, opt, int, building number, attr:min{req, int},max{req, int}-->0
        </buildingNumber>
        <unitNumber min="0" max="999">
          <!--ro, opt, int, unit number, attr:min{req, int},max{req, int}-->0
        </unitNumber>
        <deviceIndex min="1" max="32">
          <!--ro, opt, string, device No. length range, attr:min{req, int},max{req, int}-->test
        </deviceIndex>
        <industryType opt="buildings,prison,medicalTreatment,broadcasting,general">
          <!--ro, opt, string, industry (scene) type, range:[0,32], attr:opt{req, string}, desc:"buildings" (building), "prison", "medicalTreatment" (medical), "broadcasting", "general"-->test
        </industryType>
        <alarmEnabled opt="true,false">
          <!--ro, opt, bool, whether to enable the arming function of IoT devices, attr:opt{req, string}, desc:this field is used to check if arming is supported when adding devices. (Related field: alarmEnabled; related URI: /ISAPI/System/IOT/channels?format=json)-->true
        </alarmEnabled>
        <priority min="1" max="15">
          <!--ro, opt, int, calling priority, attr:min{req, int},max{req, int}-->0
        </priority>
        <transportType opt="unicast,multicast">
          <!--ro, opt, string, broadcasting method, attr:opt{req, string}, desc:unicast, multicast-->test
        </transportType>
        <audioVolume min="0" max="15">
          <!--ro, opt, int, volume, attr:min{req, int},max{req, int}-->0
        </audioVolume>
        <callForwardingEnabled>
          <!--ro, opt, bool, whether to enable call transfer, desc:true (enable), false (disable). The default value is false-->true
        </callForwardingEnabled>
        <callForwardingType opt="unconditional,busy,noReply,noReachable,calling">
          <!--ro, opt, string, call transfer type, range:[0,32], attr:opt{req, string}, desc:"unconditional" (call transfer under any condition), "busy" (call transfer when busy), "noReply" (call transfer when there is no reply), "noReachable" (call transfer when it is unreachable), "calling" (call transfer when calling)-->test
        </callForwardingType>
        <isSupportCallTerminalNumbe>
          <!--ro, opt, bool, whether the device supports configuring the target master station No. for call forwarding-->true
        </isSupportCallTerminalNumbe>
        <customCommunicationName min="0" max="128">
          <!--ro, opt, string, custom contact name, range:[0,128], attr:min{req, int},max{req, int}, desc:when this node is configured, the custom contact name will be displayed during device calling prior to the terminal number-->test
        </customCommunicationName>
        <isSupportMoreFeatures>
          <!--ro, opt, bool, whether the device supports more functions, desc:it is not supported by default if this node is not returned-->true
        </isSupportMoreFeatures>
        <isNotSupportRegPassword>
          <!--ro, opt, bool, whether the password registration configuration is not supported, desc:it is supported if this node is not returned or false is returned-->true
        </isNotSupportRegPassword>
        <mappingRoomNumber min="0" max="5">
          <!--ro, opt, int, mapping room No., attr:min{req, int},max{req, int}, desc:the mapping room No. is unique and can be called without exposing the actual room No., for privacy requirement for overseas users-->0
        </mappingRoomNumber>
        <deviceID min="0" max="64">
          <!--ro, opt, string, device ID, attr:min{req, int},max{req, int}, desc:supported for ISUP protocol access-->test
        </deviceID>
        <ISUPkey min="1" max="32">
          <!--ro, opt, string, ISUP Key, attr:min{req, int},max{req, int}, desc:supported for ISUP protocol access-->test
        </ISUPkey>
        <dialUpAddress min="1" max="1024">
          <!--ro, opt, string, dial address, attr:min{req, int},max{req, int}, desc:this node is required when <unitType> is "analogIndoor"-->test
        </dialUpAddress>
      </channelConfig>
    </channelConfigList>
    <isSupportConfigTemplate>
      <!--ro, opt, bool, whether the device supports exporting the device list template, desc:related URL: GET /ISAPI/System/IOT/channelConfig/template?format=json-->true
    </isSupportConfigTemplate>
    <isSupportConfigReport>
      <!--ro, opt, bool, whether the device supports getting the importing details, desc:related URL: GET /ISAPI/System/IOT/channelConfig/report?format=json-->true
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    </isSupportConfigReport>
    <unitType
opt="indoor,villa,confirm,outdoor,fence,doorbell,manage,acs,interactive,wardStation,bedheadExtension,bedsideExtension,analogIndoor,decoder,networkFireAlarmM
atrix,amplifier,pagingMicrophone,PDA,netSwitch,entrywayStation,alarmManage,alarmTerminal">
      <!--ro, opt, string, device type, range:[0,32], attr:opt{req, string}, desc:terminal, netAudio (network sound box)-->test
    </unitType>
    <serialNumber min="1" max="32">
      <!--ro, opt, string, the value range of device serial No. length, range:[0,32], attr:min{req, int},max{req, int}-->test
    </serialNumber>
    <isSupportAsyncStatus>
      <!--ro, opt, bool, whether the device supports asynchronously getting the status of non-video channels, desc:URI for device update: PUT
/ISAPI/System/IOT/channels/updateStatusTask?format=json; URI for getting the update progress: GET /ISAPI/System/IOT/channels/updateStatusProgress?
format=json-->true
    </isSupportAsyncStatus>
    <isSupportAddIOTChannelList>
      <!--ro, opt, bool, whether the device supports batch adding channels of IoT devices, desc:URI for batch adding channels of IoT devices: POST
/ISAPI/System/IOT/channels/AddIOTChannelList?format=json-->true
    </isSupportAddIOTChannelList>
    <isSupportSearchMoreIOTSource>
      <!--ro, opt, bool, whether the device supports searching for the IoT devices which can be added (by multiple conditions), desc:related URI: POST
/ISAPI/System/IOT/channels/AddIOTChannelList?format=json-->true
    </isSupportSearchMoreIOTSource>
    <channelConfigReportMode>
      <!--ro, opt, enum, mode of importing the list of IoT devices, subType:string, desc:different devices (main station and door station) process the
imported device list in a different way, which should be returned to the users for prompt. URL for importing the device list:
/ISAPI/System/IOT/channelConfig?format=json; cover; add-->cover
    </channelConfigReportMode>
    <isSupportEncryptImportIOTChannelFile>
      <!--ro, opt, bool, whether it supports encrypting the imported non-video channel files, desc:corresponding URI: PUT /ISAPI/System/IOT/channelConfig?
format=json&security=<security>&iv=<iv>, it is not supported by default if this filed is not returned-->true
    </isSupportEncryptImportIOTChannelFile>
    <isSupportEncryptExportIOTChannelFile>
      <!--ro, opt, bool, whether it supports encrypting exported non-video channel files, desc:corresponding URI: GET /ISAPI/System/IOT/channelConfig?
format=json&security=<security>&iv=<iv>, it is not supported by default if this filed is not returned-->true
    </isSupportEncryptExportIOTChannelFile>
    <isSupportExportAbnormalReport>
      <!--ro, opt, bool, whether it supports exporting error reports for non-video channel devices, desc:corresponding URI: GET
/ISAPI/System/IOT/ChannelConfig/ExportAbnormalReport?format=json&security=<security>&iv=<iv>, it is not supported by default if this filed is not returned--
>true
    </isSupportExportAbnormalReport>
    <isSupportSyncDeviceInfoTask>
      <!--ro, opt, bool-->true
    </isSupportSyncDeviceInfoTask>
  </IOTCap>
  <isSupportLCDScreen>
    <!--ro, opt, bool, whether the device supports LCD display screen-->true
  </isSupportLCDScreen>
  <isSupportBluetooth>
    <!--ro, opt, bool, whether the device supports bluetooth, desc:related URI: /ISAPI/System/Bluetooth/capabilities-->true
  </isSupportBluetooth>
  <isSupportNetScreen>
    <!--ro, opt, bool, whether the device supports network screen (Nova screen), desc:related URI: /ISAPI/System/netScreens/capabilities?format=json-->true
  </isSupportNetScreen>
  <isSupportLocation>
    <!--ro, opt, bool, whether the device supports getting the device location information, desc:related URI: /ISAPI/System/location?format=json-->true
  </isSupportLocation>
  <isSupportDebugInfoExport>
    <!--ro, opt, bool, whether the device supports exporting the debugging information, desc:related URI: /ISAPI/System/debugInfoExport/capabilities?
format=json-->true
  </isSupportDebugInfoExport>
  <isSupportIoTGateway>
    <!--ro, opt, bool, whether the device supports IoT gateway functions, desc:related URI: /ISAPI/System/IoTGateway/capabilities?format=json-->true
  </isSupportIoTGateway>
  <isSupportImageCap>
    <!--ro, opt, bool, whether the device supports configuring image parameters-->true
  </isSupportImageCap>
  <isSupportLogToUSB opt="true,false">
    <!--ro, opt, bool, whether the device supports exporting logs to USB, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/exportLogToUSB/capabilities?format=json-->true
  </isSupportLogToUSB>
  <isSupportWiresharkToUSB opt="true,false">
    <!--ro, opt, bool, whether the device supports exporting packet capture information to USB, attr:opt{opt, string}, desc:related URI:
ISAPI/System/wiresharkToUSB/capabilities?format=json-->true
  </isSupportWiresharkToUSB>
  <CDBurningStreamEncryptionChannels opt="1,2">
    <!--ro, opt, string, return the capability of device recording stream encryption channel, attr:opt{opt, string}, desc:related URI:
/ISAPI/Streaming/channels/<ID>/capabilities-->true
  </CDBurningStreamEncryptionChannels>
  <isSupportOCR>
    <!--ro, opt, bool, whether the device supports OCR detection, desc:related URI: /ISAPI/Intelligent/channels/<ID>/OCR/capabilities?format=json-->true
  </isSupportOCR>
  <isSupportInstallationAngleCalibration>
    <!--ro, opt, bool, whether the device supports the capability of installation angle calibration, desc:related URI:
/ISAPI/System/installationAngleCalibration/channels/<ID>/capabilities-->true
  </isSupportInstallationAngleCalibration>
  <isSupportZeroBiasCalibration>
    <!--ro, opt, bool, whether the device supports the zero angle calibration capability node, desc:related URI:
/ISAPI/System/zeroBiasCalibration/channels/<ID>?format=json-->true
  </isSupportZeroBiasCalibration>
  <isSupportCalibrationFile>
    <!--ro, opt, bool, whether the device supports importing the calibration file, desc:related URI: /ISAPI/System/configurationData-->true
  </isSupportCalibrationFile>
  <isSupportDisplayTrajectory>
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    <!--ro, opt, bool, whether the device supports target track overlay (playing library)-->true
  </isSupportDisplayTrajectory>
  <maximumSuperPositionTime opt="5,10,20,30">
    <!--ro, opt, string, the maximum overlay time, attr:opt{opt, string}, desc:unit: second; it is only returned when the target track overlay is supported-
->true
  </maximumSuperPositionTime>
  <isSupportUnitConfig>
    <!--ro, opt, bool, whether the device supports the unit configuration, desc:related URI: /ISAPI/System/unitConfig/capabilities?format=json-->true
  </isSupportUnitConfig>
  <isSupportShipsDetectionWithScene>
    <!--ro, opt, bool, whether the device supports ship detection by scene-->true
  </isSupportShipsDetectionWithScene>
  <isSupportPictureServer>
    <!--ro, opt, bool, whether the device supports configuring the picture storage server, desc:related URI: /ISAPI/System/PictureServer/capabilities?
format=json-->true
  </isSupportPictureServer>
  <isSupportEmmcInfo>
    <!--ro, opt, bool, the capability of supporting the EMMC configuration, desc:related URI: /ISAPI/System/emmcInfo/capabilities?format=json-->true
  </isSupportEmmcInfo>
  <isSupportVehicleIllegalType opt="1,2...29">
    <!--ro, opt, string, the vehicle violation types supported by the device, attr:opt{opt, string}, desc:the content value is the same as the index value,
refer to NET_ITS_PLATE_RESULT.NET_DVR_VEHICLE_INFO.byIllegalType the capture camera is connected to the new platform, and the device supports "traffic
violation alarm"; however, the corresponding node is not added to the device to notify the integration of this capability; that is, the traffic violation
alarm capability lacks details. If later the platform needs to develop capability with details, the current device will be incompatible with the capability;
and detailed capability is hidden in the configuration of all trigger modes, which will influence overall access-->test
  </isSupportVehicleIllegalType>
  <isSupportRFIDChannels>
    <!--ro, opt, bool, whether the device supports getting the RFID channel, desc:related URI: /ISAPI/System/RFIDChannels?format=json-->true
  </isSupportRFIDChannels>
  <isSupportRfidCollection>
    <!--ro, opt, bool, whether the device supports the event of uploading RFID tag information, desc:related URI:
/ISAPI/System/Network/RFIDData/<channelID>/capabilities, eventType: rfidCollection-->true
  </isSupportRfidCollection>
  <DevInfoSearch opt="tiltAngle">
    <!--ro, opt, string, whether the device supports searching for the device information, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/workingstatus/devInfoSearch?format=json-->test
  </DevInfoSearch>
  <isSupportAIOpenPlatform>
    <!--ro, opt, bool, whether the device supports AI open platform device capability, desc:related URI: /ISAPI/Intelligent/AIOpenPlatform/capabilities?
format=json-->true
  </isSupportAIOpenPlatform>
  <isSupportPictureDownloadError>
    <!--ro, opt, bool, whether the device supports uploading the exception that downloading the picture failed, desc:eventType: pictureDownloadError-->true
  </isSupportPictureDownloadError>
  <isSupportStudentBehavior>
    <!--ro, opt, bool, whether the device supports uploading the data of students in class behavior, desc:eventType: studentBehavior-->true
  </isSupportStudentBehavior>
  <guideEventSupport opt="linkageCapture">
    <!--ro, opt, string, the list of web quick configuration guide, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/guideConfig/<guideEvent>/capabilities?format=json-->test
  </guideEventSupport>
  <supportRemoteCtrl opt="up,down,left,right,enter,menu,num,power,esc,edit,F1,.prev,rec,play,stop,notSupport">
    <!--ro, opt, string, whether the device supports remote control, attr:opt{opt, string}, desc:related URI: /ISAPI/System/remoteCtrl/XXX-->test
  </supportRemoteCtrl>
  <isSupportSSDFileSystemUpgrade>
    <!--ro, opt, bool, whether the device supports updating the SSD file system, desc:related URI: /ISAPI/System/SSDFileSystem/upgrade?format=json-->true
  </isSupportSSDFileSystemUpgrade>
  <isSupportSSDFileSystemFormat>
    <!--ro, opt, bool, whether the device supports formatting SSD, desc:related URI: /ISAPI/System/SSDFileSystem/format?format=json-->true
  </isSupportSSDFileSystemFormat>
  <FaceCounting>
    <!--ro, opt, object, whether the device supports the customer traffic (face)-->
    <ChannelsList>
      <!--ro, opt, array, channel list, subType:object-->
      <channelsID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelsID>
    </ChannelsList>
  </FaceCounting>
  <isSupportCluster>
    <!--ro, opt, bool, related URI:, desc:/ISAPI/ContentMgmt/Cluster/capabilities-->true
  </isSupportCluster>
  <DeviceConfigurationSupport opt="time,network,accessProtocol,recordSchedule,channel,customProtocol">
    <!--ro, opt, string, content supported by device parameters configuration, attr:opt{opt, string}, desc:related URI: /ISAPI/System/deviceConfiguration--
>test
  </DeviceConfigurationSupport>
  <behaviorEventSupport opt="getUp,toiletTarry,playCellphone,advReachHeight">
    <!--ro, opt, string, abnormal event detection types, attr:opt{opt, string}-->test
  </behaviorEventSupport>
  <isSupportBVCorrectPassthrough>
    <!--ro, opt, bool, whether the device supports transparent transmission of camera correction data, desc:related URI:
/ISAPI/System/BVCorrect/Passthrough/channels/<ID>/capabilities?format=json-->true
  </isSupportBVCorrectPassthrough>
  <isSupportChangedCallStatus>
    <!--ro, opt, bool-->true
  </isSupportChangedCallStatus>
  <VehiclePositionControl>
    <!--ro, opt, object, whether the device supports vehicle position arming-->
    <license min="0" max="16">
      <!--ro, req, string, license plate number, attr:min{opt, string},max{opt, string}-->test
    </license>
    <intervalTime min="0" max="65535" def="1">
      <!--ro, req, int, uploading interval, unit:s, attr:min{opt, string},max{opt, string},def{opt, string}-->1
    </intervalTime>
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    </VehiclePositionControl>
    <supportGISEvent opt="AID,TPS,ANPR,mixedTargetDetection">
      <!--ro, opt, string, the capability of supporting getting the GIS information, attr:opt{opt, string}-->test
    </supportGISEvent>
    <isSupportGPSPTZCalibratation>
      <!--ro, opt, bool, whether the device supports GPS PTZ calibration, desc:related URI: /ISAPI/System/GPSPTZCalibratation/channels/<ID>/points?
format=json-->true
    </isSupportGPSPTZCalibratation>
    <isSupportRadar>
      <!--ro, opt, bool, whether the device supports radar functions, desc:related URI: /ISAPI/Radar/capabilities?format=json-->true
    </isSupportRadar>
    <isSupportISUPHttpPassthrough>
      <!--ro, opt, bool, whether the device supports HTTP transparent transmission via ISUP V5.0-->true
    </isSupportISUPHttpPassthrough>
    <isSupportMaterialListQuery>
      <!--ro, opt, bool, whether the device supports uploading the material list, desc:event: materialListQuery-->true
    </isSupportMaterialListQuery>
    <isSupportDebugLogAsync>
      <!--ro, opt, bool, whether the device supports exporting the debugging log asynchronously-->true
    </isSupportDebugLogAsync>
    <isSupportAlgorithmsInfo>
      <!--ro, opt, bool, whether the device supports getting the algorithm version of the device-->true
    </isSupportAlgorithmsInfo>
    <isSupportSecurityChecking>
      <!--ro, opt, bool, whether the device supports configuring security checking, desc:true (support); otherwise, not support-->true
    </isSupportSecurityChecking>
    <isSupportSecurityCheckException>
      <!--ro, opt, bool, whether the device supports the security checking exception event, desc:eventType: securityCheckException-->true
    </isSupportSecurityCheckException>
    <isSupportPictureManage>
      <!--ro, opt, bool, whether the device supports picture management function, desc:related URI: /ISAPI/System/isSupportPictureManage/capabilities?
format=json-->true
    </isSupportPictureManage>
    <isSupportAutoSwitch>
      <!--ro, opt, bool, whether the device supports automatically powering on and powering off, desc:related URI: /ISAPI/System/autoSwitch/capabilities?
format=json-->true
    </isSupportAutoSwitch>
    <SHMCap>
      <!--ro, opt, object, SHM alarm-->
      <isSupportHighHDTemperature>
        <!--ro, opt, bool, whether the device supports HDD high temperature detection, desc:eventType: highHDTemperature-->true
      </isSupportHighHDTemperature>
      <isSupportLowHDTemperature>
        <!--ro, opt, bool, whether the device supports HDD low temperature detection, desc:eventType: lowHDTemperature-->true
      </isSupportLowHDTemperature>
      <isSupportHDImpact>
        <!--ro, opt, bool, whether the device supports HDD impact detection, desc:eventType: hdImpact-->true
      </isSupportHDImpact>
      <isSupportHDBadBlock>
        <!--ro, opt, bool, whether the device supports HDD bad sector detection, desc:eventType: hdBadBlock-->true
      </isSupportHDBadBlock>
      <isSupportSevereHDFailure>
        <!--ro, opt, bool, whether the device supports HDD major fault detection, desc:eventType: severeHDFailure-->true
      </isSupportSevereHDFailure>
    </SHMCap>
    <isSupportLinkStatusEvent>
      <!--ro, opt, bool, whether the device supports uploading the information about the network connection status of the device, desc:eventType: linkEvent--
>true
    </isSupportLinkStatusEvent>
    <isSupportVideoDealStatus>
      <!--ro, opt, bool, whether the device supports uploading the information about the video data processing status, desc:eventType: videoDealStatus-->true
    </isSupportVideoDealStatus>
    <isSupportAudioDealStatus>
      <!--ro, opt, bool, whether the device supports uploading the information about the audio data processing status, desc:eventType: audioDealStatus-->true
    </isSupportAudioDealStatus>
    <isSupportNetPackLossStatus>
      <!--ro, opt, bool, whether the device supports uploading the information about the network lost package status, desc:eventType: netPacketLoss-->true
    </isSupportNetPackLossStatus>
    <isSupportVideoConferenceStatus>
      <!--ro, opt, bool, whether the device supports uploading the information about the video conference status, desc:eventType: videoConference-->true
    </isSupportVideoConferenceStatus>
    <isSupportDataPrealarm>
      <!--ro, opt, bool, whether the device supports network traffic pre-alarm event, desc:eventType: dataPrealarm-->true
    </isSupportDataPrealarm>
    <isSupportClearCache>
      <!--ro, opt, bool, whether the device supports clearing the cache, desc:related URI: /ISAPI/System/clearCache?format=json-->true
    </isSupportClearCache>
    <isSupportIntelligentMode>
      <!--ro, opt, bool, whether the device supports switching the intelligent mode, desc:related URI: /ISAPI/System/IntelligentMode?format=json-->true
    </isSupportIntelligentMode>
    <isSupportFaceTemperatureMeasurementEvent>
      <!--ro, opt, bool, whether the device supports the skin-surface temperature measurement event, desc:eventType: FaceTemperatureMeasurementEvent-->true
    </isSupportFaceTemperatureMeasurementEvent>
    <isSupportQRCodeEvent>
      <!--ro, opt, bool, whether the device supports the QR code event, desc:eventType: QRCodeEvent-->true
    </isSupportQRCodeEvent>
    <isSupportVisitorEvent>
      <!--ro, opt, bool, whether the device supports the visitor event, desc:eventType: VisitorEvent-->true
    </isSupportVisitorEvent>
    <isSupportEmployeeAuthEvent>
      <!--ro, opt, bool, whether the device supports the employee authorization event, desc:eventType: EmployeeAuthEvent-->true
    </isSupportEmployeeAuthEvent>
    <isSupportVisitorInfoSearchEvent>
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    <!--ro, opt, bool, whether the device supports the visitor information search event, desc:eventType: VisitorInfoSearchEvent-->true
  </isSupportVisitorInfoSearchEvent>
  <isSupportConsumptionEvent>
    <!--ro, opt, bool, whether the device supports the transaction record event, desc:eventType: ConsumptionEvent-->true
  </isSupportConsumptionEvent>
  <isSupportTransactionRecordEvent>
    <!--ro, opt, bool, whether the device supports the transaction record event, desc:eventType: TransactionRecordEvent-->true
  </isSupportTransactionRecordEvent>
  <isSupportFaceContrastResultDisplay>
    <!--ro, opt, bool, whether the device supports displaying the face picture comparison result, desc:related URI:
/ISAPI/Intelligent/faceContrast/resultDisplay/capabilities?format=json-->true
  </isSupportFaceContrastResultDisplay>
  <isSupportPatchUpgrade>
    <!--ro, opt, bool, whether it supports incremental upgrade, desc:related URI: /ISAPI/Custom/OpenPlatform/patchUpgrade/capabilities?format=json-->true
  </isSupportPatchUpgrade>
  <isSupportSnapshotAsync>
    <!--ro, opt, bool, whether the device supports capturing pictures asynchronously (the picture URL will be returned), desc:related URI:
/ISAPI/Streaming/channels/<ID>/picture/async?format=json-->true
  </isSupportSnapshotAsync>
  <isSupportLogExportByUSB>
    <!--ro, opt, bool, exporting logs to PC from USB flash, desc:related URI: /ISAPI/System/exportUSBLog?format=json-->true
  </isSupportLogExportByUSB>
  <isSupportShellCmd>
    <!--ro, opt, bool, whether the device supports shell command control, desc:related URI: /ISAPI/System/shellCmd?format=json-->true
  </isSupportShellCmd>
  <faceMonitor opt="dailyFaceMonitor,dailyFaceTracking,VCAFaceMonitor,VCAFaceTracking,manualFaceMonitor,advanceConfiguration,">
    <!--ro, opt, bool, whether the device supports face arming, attr:opt{opt, string}, desc:related URI: /ISAPI/Intelligent/channels/<ID>/dailyFaceMonitor?
format=json-->true
  </faceMonitor>
  <isSupportRelatedVCAResource>
    <!--ro, opt, bool, whether the device supports the system-level intelligent resources configuration protocol, desc:related URI:
/ISAPI/System/RelatedVCAResource/capabilities?format=json; this protocol conficts with thew previous intelligent resources switch protocol-->true
  </isSupportRelatedVCAResource>
  <isSupportVehicleMonitor>
    <!--ro, opt, bool, whether the device supports intelligent arming and tracking of vehicles, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/capabilities?format=json-->true
  </isSupportVehicleMonitor>
  <isSupportManualVehicleMonitor>
    <!--ro, opt, bool, whether the device supports manual arming and tracking of vehicles, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/manual/capabilitis?format=json-->true
  </isSupportManualVehicleMonitor>
  <isSupportDailyVehicleMonitor>
    <!--ro, opt, bool, whether the device supports daily arming and tracking of vehicles, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleDetect/capabilities-->true
  </isSupportDailyVehicleMonitor>
  <isSupportInputProxyDeviceInfo>
    <!--ro, opt, bool, whether the device supports getting the information of the devices added to the proxy device channel, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities-->true
  </isSupportInputProxyDeviceInfo>
  <isSupportAccessControlCap>
    <!--ro, opt, bool, access control capability, desc:related URI: /ISAPI/AccessControl/capabilities-->true
  </isSupportAccessControlCap>
  <isSupportDebugLog>
    <!--ro, opt, bool, whether the device supports exporting the debugging log, desc:related URI: /ISAPI/System/debugLog?format=json-->true
  </isSupportDebugLog>
  <isSupportFaceAppear>
    <!--ro, opt, bool, whether the device supports face appearing detection event, desc:eventType: faceAppear-->true
  </isSupportFaceAppear>
  <isSupportFaceDisappear>
    <!--ro, opt, bool, whether the device supports face disappearing detection event, desc:eventType: faceDisappear-->true
  </isSupportFaceDisappear>
  <isSupportTeacherBehavior>
    <!--ro, opt, bool, whether the device supports the abnormal event detection of teachers, desc:eventType: teacherBehavior-->true
  </isSupportTeacherBehavior>
  <isSupportHeadRiseRate>
    <!--ro, opt, bool, whether the device supports face appearing detection event, desc:eventType: headRiseRate-->true
  </isSupportHeadRiseRate>
  <isSupportIPCUpgrade>
    <!--ro, opt, bool, whether the platform supports upgrading network cameras via NVR and other devices, desc:related URI: /ISAPI/System/IPCUpgrade/<ID>?
format=json-->true
  </isSupportIPCUpgrade>
  <isSupportFaceLibRebuildEvent>
    <!--ro, opt, bool, whether the device supports uploading the modeling rebuilding of the face picture library event, desc:eventType: faceLibRebuild--
>true
  </isSupportFaceLibRebuildEvent>
  <isSupportCalibrationStream>
    <!--ro, opt, bool, supported capability of calibration stream, desc:related URI: /ISAPI/Streaming/calibStream/capabilities?format=json-->true
  </isSupportCalibrationStream>
  <isSupportChannelOccupy>
    <!--ro, opt, bool, whether the device supports outdoor fire engine access occupied by vehicles, desc:eventType: channelOccupy-->true
  </isSupportChannelOccupy>
  <isSupportOffDuty>
    <!--ro, opt, bool, whether to support fire control room person absence detection, desc:eventType: offDuty-->true
  </isSupportOffDuty>
  <isSupportNoCertificate>
    <!--ro, opt, bool, whether to support fire control room person without certificate detection, desc:eventType: noCertificate-->true
  </isSupportNoCertificate>
  <isSupportSmokeAlarm>
    <!--ro, opt, bool, whether the device supports smoke alarm, desc:eventType: smokeAlarm-->true
  </isSupportSmokeAlarm>
  <isSupportBatteryCarDisobey>
    <!--ro, opt, bool, whether the device supports illegal parking detection of electric moped, desc:eventType: batteryCarDisobey-->true
  </isSupportBatteryCarDisobey>
  <isSupportNoFireExtinguisherRecog>
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<isSupportNoFireExtinguisherRecog>
  <!--ro, opt, bool, whether the device supports fire extinguisher loss detection, desc:eventType: noFireExtinguisherRecog-->true
</isSupportNoFireExtinguisherRecog>
<isSupportIndoorPasswayBlock>
  <!--ro, opt, bool, whether the device supports indoor access blocking detection, desc:eventType: indoorPasswayBlock-->true
</isSupportIndoorPasswayBlock>
<isSupportFireSmartFireDetect>
  <!--ro, opt, bool, whether the device supports fire detection, desc:eventType: fireSmartFireDetect-->true
</isSupportFireSmartFireDetect>
<isSupportDetectorRunningStatus>
  <!--ro, opt, bool, whether the device supports getting the detector running status, desc:eventType: detectorRunningStatus-->true
</isSupportDetectorRunningStatus>
<isSupportDetectorOperationStatus>
  <!--ro, opt, bool, whether the device supports getting the detector operation status, desc:eventType: detectorOperationStatus-->true
</isSupportDetectorOperationStatus>
<isSupportDetectorTemperatureAlarm opt="highTemperature,riseTemperature,flame">
  <!--ro, opt, bool, whether the device supports the detector temperature alarm, attr:opt{req, string}-->true
</isSupportDetectorTemperatureAlarm>
<isSupportDetectorShelterAlarm>
  <!--ro, opt, bool, whether the device supports detector sheltering alarm, desc:eventType: detectorShelterAlarm-->true
</isSupportDetectorShelterAlarm>
<isSupportDetectorMotionAlarm>
  <!--ro, opt, bool, whether the device supports the detector moving alarm, desc:eventType: detectorMotionAlarm-->true
</isSupportDetectorMotionAlarm>
<isSupportDetectorTamperAlarm>
  <!--ro, opt, bool, whether the device supports detector tampering alarm, desc:eventType: detectorTamperAlarm-->true
</isSupportDetectorTamperAlarm>
<isSupportDetectorEmergencyAlarm>
  <!--ro, opt, bool, whether the device supports the detector emergency alarm, desc:eventType: detectorEmergencyAlarm-->true
</isSupportDetectorEmergencyAlarm>
<isSupportDetectorSmokeAlarm>
  <!--ro, opt, bool, whether the device supports the detector smoke alarm, desc:eventType: detectorSmokeAlarm-->true
</isSupportDetectorSmokeAlarm>
<isSupportDetectorCombustibleGasAlarm>
  <!--ro, opt, bool, whether the device supports the detector combustible gas alarm, desc:eventType: detectorCombustibleGasAlarm-->true
</isSupportDetectorCombustibleGasAlarm>
<isSupportFireControlData>
  <!--ro, opt, bool, whether the device supports uploading real-time fire control data, desc:eventType:FireControlData-->true
</isSupportFireControlData>
<isSupportFireNoRegulation>
  <!--ro, opt, bool, whether to support people leaving event uploading, desc:eventType: fireNoRegulation-->true
</isSupportFireNoRegulation>
<isSupportSmokeFireRecognize>
  <!--ro, opt, bool, whether the device supports smoke and fire recognition event uploading, desc:eventType: smokeFireRecognize-->true
</isSupportSmokeFireRecognize>
<isSupportClientProxyWEB>
  <!--ro, opt, bool, whether to support WEB page jumping, desc:whether to support WEB page jumping-->true
</isSupportClientProxyWEB>
<WEBLocation>
  <!--ro, opt, string, WEB page placement location, range:[1,32], desc:local-local server, remote-remote server; (if this node is not returned, the WEB page will be placed in the local server by default)-->local
</WEBLocation>
<isSupportWEBLocalPlug>
  <!--ro, opt, bool, whether to support local download of web plug-in, desc:true-yes (by default), false-no; when this node is not returned, it is supported by default-->true
</isSupportWEBLocalPlug>
<deviceId>
  <!--ro, opt, string, device ID, range:[1,32], desc:if WEBLocation is "remote", the unique device ID should be returned to identify the unique corresponding web package. This device ID is managed by the Web configuration package. For a specified device of different versions, the same deviceID will be returned-->R0101webs_4.0_isapi
</deviceId>
<isSupportTimeZone>
  <!--ro, opt, bool, whether the device supports configuring the DST (Daylight Saving Time), desc:related URI: /ISAPI/System/time/timeZone-->true
</isSupportTimeZone>
<isSptHealthDetection>
  <!--ro, opt, bool, whether the device supports health monitoring, desc:related URI: /ISAPI/System/healthDetection-->true
</isSptHealthDetection>
<isSptDiagnosis>
  <!--ro, opt, bool, whether the device supports diagnosis, desc:related URI: /ISAPI/System/diagnosis-->true
</isSptDiagnosis>
<isSptSerialLogCfg>
  <!--ro, opt, bool, whether the device supports configuring parameters of serial port log redirection, desc:related URI: /ISAPI/System/serialLogCfg?format=json-->true
</isSptSerialLogCfg>
<isSptFileExport>
  <!--ro, opt, bool, whether the device supports exporting the device file, desc:related URI: /ISAPI/System/fileExport?format=json-->true
</isSptFileExport>
<isSptCertificationStandard>
  <!--ro, opt, bool, whether the device supports configuring host verification standard, desc:related URI: /ISAPI/SecurityCP/certificationStandard-->true
</isSptCertificationStandard>
<isSptKeypadLock>
  <!--ro, opt, bool, whether the device supports locking the keypad, desc:related URI: /ISAPI/System/keypadLock/config-->true
</isSptKeypadLock>
<isSupportVibrationDetection>
  <!--ro, opt, bool, whether the device supports the vibration detection, desc:related URI: /ISAPI/System/channels/<ID>/vibrationDetection?format=json-->true
</isSupportVibrationDetection>
<isSupportSmokingDetectAlarm>
  <!--ro, opt, bool, whether the device supports the smoking alarm, desc:eventType: smokeDetectAlarm-->true
</isSupportSmokingDetectAlarm>
<isSupportRadarChannels>
  <!--ro, opt, bool, whether the device supports getting the radar channel, desc:related URI: /ISAPI/System/RadarChannels?format=json-->true
</isSupportRadarChannels>
<radarIPDForm>
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    <!--ro, opt, enum, radar-camera status, subType:string, desc:"single" (single radar); "double_diagonal" (double radars arranged diagonally);
"double_vertical" (double radars arranged vertically)-->single
  </radarIPDForm>
  <isSupportRadarFieldDetection>
    <!--ro, opt, bool, whether the device supports radar's area intrusion detection, desc:related URI:
/ISAPI/Smart/RadarChannels/<radarChannelID>/radarFieldDetection/capabilities?format=json-->true
  </isSupportRadarFieldDetection>
  <isSupportRadarLineDetection>
    <!--ro, opt, bool, whether the device supports radar line crossing detection, desc:related URI:
/ISAPI/Smart/RadarChannels/<radarChannelID>/radarLineDetection/capabilities?format=json-->true
  </isSupportRadarLineDetection>
  <mixedTargetDetectionWebNoDisplay>
    <!--ro, opt, bool, multiple targets recognition is not displayed-->true
  </mixedTargetDetectionWebNoDisplay>
  <isSupportRemovePlaybackRecord>
    <!--ro, opt, bool, whether to support removing playback (the device does not support playback), desc:if this function is supported, this node will be
returned and its value is true; otherwise, this node will not be returned. This node will be returned when the value of isSupportRemoveStorage is true;
otherwise, this node will not be returned-->true
  </isSupportRemovePlaybackRecord>
  <isSupportAudioUpload>
    <!--ro, opt, bool, whether the device supports uploading the audio file, desc:related URI: /ISAPI/System/Audio/audioUpload?format=json-->true
  </isSupportAudioUpload>
  <VCAResourceList>
    <!--ro, opt, array, supported smart resources list, subType:object-->
    <VCAResource>
      <!--ro, opt, object, smart resources-->
      <VCAResourceType>
        <!--ro, opt, string, smart resources type-->test
      </VCAResourceType>
      <ChannleVCAResourceList>
        <!--ro, opt, array, supported smart resources list, subType:object-->
        <ChannleVCAResource>
          <!--ro, opt, object, smart channel resources-->
          <channel>
            <!--ro, opt, int, channel No.-->1
          </channel>
          <channleVCAResourceType>
            <!--ro, opt, string, smart channel resources type-->test
          </channleVCAResourceType>
        </ChannleVCAResource>
      </ChannleVCAResourceList>
    </VCAResource>
  </VCAResourceList>
  <isSupportSlotInsertion>
    <!--ro, opt, bool, whether to support slot plugging in and out, desc:eventType: slotInsertion-->true
  </isSupportSlotInsertion>
  <isSupportNetworkLoop>
    <!--ro, opt, bool, whether to support network loop alarm, desc:eventType: networkLoop-->true
  </isSupportNetworkLoop>
  <isSupportApOnOffLine>
    <!--ro, opt, bool, whether to support AP online and offline alarm event uploading, desc:eventType: apOnOffLine-->true
  </isSupportApOnOffLine>
  <isSupportAbnormalWANPort>
    <!--ro, opt, bool, whether to support WAN abnormal alarm event uploading, desc:eventType: abnormalWANPort-->true
  </isSupportAbnormalWANPort>
  <isSupportAPException>
    <!--ro, opt, bool, whether to support AP exception alarm information uploading, desc:eventType: APException-->true
  </isSupportAPException>
  <isSupportACException>
    <!--ro, opt, bool, whether to support AC exception alarm information uploading, desc:eventType: ACException-->true
  </isSupportACException>
  <isSupportCartridgeCapacity>
    <!--ro, opt, bool, whether the device supports the configuration of notifying cartridge capacity, desc:related URL:
/ISAPI/System/printer/cartridgeCapacity/capabilities?format=json-->true
  </isSupportCartridgeCapacity>
  <isSupportCDCakeBoxCapacity>
    <!--ro, opt, bool, whether the device supports notifying CD capacity, desc:related URL: /ISAPI/System/printer/CDCakeBoxCapacity/capabilities?
format=json-->true
  </isSupportCDCakeBoxCapacity>
  <isSupportSleep>
    <!--ro, opt, bool, whether the device supports configuring sleep, desc:related URL: /ISAPI/System/sleep/capabilities?format=json-->true
  </isSupportSleep>
  <isSupportContainerDetection>
    <!--ro, opt, bool, whether the device supports container detection, desc:/ISAPI/Intelligent/containerDetection/capabilities?format=json (for devices
that does not return this node, you can check according to the node returned by the following protocol:
/ISAPI/Traffic/ContentMgmt/InputProxy/channels/<ID>/ocrScene/capabilities)-->true
  </isSupportContainerDetection>
  <isSupportRadarVideoDetection>
    <!--ro, opt, bool, related URI: /ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json-->true
  </isSupportRadarVideoDetection>
  <isSupportRadarVideoDetectionRealtime>
    <!--ro, opt, bool, whether the device supports radar video detection, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json-->true
  </isSupportRadarVideoDetectionRealtime>
  <isSupportBatteryStatusUpload>
    <!--ro, opt, bool, whether the device supports uploading the battery status, desc:eventType: batteryStatus-->true
  </isSupportBatteryStatusUpload>
  <isSupportPowerConsumptionStatusUpload>
    <!--ro, opt, bool, whether the device supports uploading device consumption status-->true
  </isSupportPowerConsumptionStatusUpload>
  <isSupportOnlineUpgradeTask>
    <!--ro, opt, bool, whether device online upgrading is supported-->true
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  </isSupportOnlineUpgradeTask>
  <isSupportPersonArmingTrack>
    <!--ro, opt, bool, whether the device supports person arming configuration, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/personArmingTrack/capabilities?format=json-->true
  </isSupportPersonArmingTrack>
  <isSupportManualPersonArmingTrack>
    <!--ro, opt, bool, whether the device supports manual person arming, desc:related URI: /ISAPI/Intelligent/channels/<ID>/manualPersonArmingTrack?
format=json-->true
  </isSupportManualPersonArmingTrack>
  <isSupportGPSVerification>
    <!--ro, opt, bool, whether the device supports GPS verification points, desc:related URI: /ISAPI/System/GPSVerification/channels/<ID>/points?
format=json-->true
  </isSupportGPSVerification>
  <isSupportHBDLib>
    <!--ro, opt, bool, whether the device supports the human body library, desc:related URI: /ISAPI/Intelligent/HBDLib/capabilities?format=json-->true
  </isSupportHBDLib>
  <isSupportDisinfectionInfo>
    <!--ro, opt, bool, whether the device supports uploading disinfection information, desc:eventType: disinfectionInfo-->true
  </isSupportDisinfectionInfo>
  <isSupportTempQueryDelete>
    <!--ro, opt, bool, whether the device supports deleting temperature query, desc:eventType: tempQueryDelete-->true
  </isSupportTempQueryDelete>
  <isSupportMobilePayment>
    <!--ro, opt, bool, whether the device supports mobile payment, desc:/ISAPI/System/mobilePayment?format=json-->true
  </isSupportMobilePayment>
  <isSupportFileSearch>
    <!--ro, opt, bool, whether the device supports searching files, desc:related URI: /ISAPI/System/fileSearch?format=json-->true
  </isSupportFileSearch>
  <isSupportFileExportAsync>
    <!--ro, opt, bool, whether the device supports exporting files asynchronously, desc:related URI: /ISAPI/System/fileExport/async?format=json-->true
  </isSupportFileExportAsync>
  <isSupportSSDFileSystemCapacity>
    <!--ro, opt, bool, whether the device supports getting the capacity distribution of the SSD system, desc:related URI:
/ISAPI/System/SSDFileSystem/capacity?format=json-->true
  </isSupportSSDFileSystemCapacity>
  <isSupport4GSafetyHelmetInfo>
    <!--ro, opt, bool, whether the device supports the hard hat event, desc:eventType: 4GSafetyHelmetInfo-->true
  </isSupport4GSafetyHelmetInfo>
  <isSupportICCID>
    <!--ro, opt, bool, whether the device supports getting the ICCID information, desc:related URI: /ISAPI/System/deviceInfo/ICCID?format=json-->true
  </isSupportICCID>
  <isSupportFireEscapeDetection>
    <!--ro, opt, bool, whether the device supports fire escape detection, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/fireEscapeDetection/capabilities?format=json-->true
  </isSupportFireEscapeDetection>
  <isSupportTakingElevatorDetection>
    <!--ro, opt, bool, whether the device supports taking elevator detection, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/takingElevatorDetection/capabilities?format=json-->true
  </isSupportTakingElevatorDetection>
  <isSupportVehicleMatchResult>
    <!--ro, opt, bool, whether the device supports license plate comparison, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/licensePlateContrast/capabilities?format=json-->true
  </isSupportVehicleMatchResult>
  <isSupportDataAware>
    <!--ro, opt, bool, the data aware capability, desc:related URI: /ISAPI/System/dataAware/capabilities?format=json-->true
  </isSupportDataAware>
  <isSupportAbnormalReboot>
    <!--ro, opt, bool, whether the device supports abnormal reboot event, desc:eventType: abnormalReboot-->true
  </isSupportAbnormalReboot>
  <isSupportVehicleThermometry>
    <!--ro, opt, bool, whether the device supports vehicle temperature measurement detection, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleDetect/thermometry/capabilities?format=json-->true
  </isSupportVehicleThermometry>
  <AssociationScenesList>
    <!--ro, opt, array, the list of scene adaptive linkage, subType:object, desc:it indicates the device supports the scene adaptive linkage between
channels. The current device has two channels: main channel (box camera) and sub channel (radar camera), however, the device only opens a main channel to
users for scene configuration; the scene configuration of the sub channel is not open for the users; the scene of the sub channel is automatically connected
to the main channel scene; currently, the front-end returns this capability-->
    <AssociationScenes>
      <!--ro, req, object, the channel list of scene adaptive linkage between channels-->
      <channelID>
        <!--ro, req, int, supported scene channel ID, desc:supported scene channel ID-->1
      </channelID>
      <isSupportScene>
        <!--ro, opt, bool, whether the device supports the scene configuration, desc:whether the current channel is the main channel and supports scene
configuration , true (main channel and support), this node is not returned (sub channel and not support)-->true
      </isSupportScene>
    </AssociationScenes>
  </AssociationScenesList>
  <supportRelationVideoEvent opt="PIR,cityManagement,roadMaint">
    <!--ro, opt, string, supports uploading the video of linked triggering event, attr:opt{req, string}, desc:"PIR" (PIR event), "cityManagement" (city
management event), "roadMaint" (road maintenance event)-->test
  </supportRelationVideoEvent>
  <isSupportVCAResourceList>
    <!--ro, opt, bool, whether the device supports configuring intelligent resources in a batch, desc:related URI:
/ISAPI/System/Video/inputs/channels/VCAResourceList?format=json-->true
  </isSupportVCAResourceList>
  <isSupportConvergenceCloud>
    <!--ro, opt, bool, whether the device supports the URI for EZ interconnection channels, desc:URI for capability of all channels:
/ISAPI/Event/convergenceCloud/channels/capabilities?format=json; URI for capability of specified channel:
/ISAPI/Event/convergenceCloud/channels/<ID>/capabilities?format=json-->true
  </isSupportConvergenceCloud>
  <isSupportInterconnection>
    <!--ro, opt, bool, whether the device supports the URI for EZ interconnection devices, desc:related URI: /ISAPI/Event/convergenceCloud/capabilities?
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format=json-->true
  </isSupportInterconnection>
  <isSupportHPVehicleStatus>
    <!--ro, opt, bool, whether the device supports the platform vehicle status event, desc:eventType: HPVehicleStatus-->true
  </isSupportHPVehicleStatus>
  <isSupportHPWorkingStatus>
    <!--ro, opt, bool, whether the device supports the scheduled platform working status event, desc:eventType: HPWorkingStatus-->true
  </isSupportHPWorkingStatus>
  <isSupportHPPeopleDetection>
    <!--ro, opt, bool, whether the device supports the platform people detection event, desc:eventType: HPPeopleDetection-->true
  </isSupportHPPeopleDetection>
  <isSupportHPVehicleDoorDetection>
    <!--ro, opt, bool, whether the device supports the platform door detection event, desc:eventType: HPVehicleDoorDetection-->true
  </isSupportHPVehicleDoorDetection>
  <isSupportRoadMaint>
    <!--ro, opt, bool, whether the device supports road maintenance, desc:related URI: /ISAPI/Intelligent/channels/<ID>/roadMaint?format=json-->true
  </isSupportRoadMaint>
  <isSupportDockStationInfo>
    <!--ro, opt, bool, whether the device supports the event of uploading the dock station information to the media, desc:eventType: dockStationInfo; this
node is not returned if not supported-->true
  </isSupportDockStationInfo>
  <isSupportBodyCameraManualForensics>
    <!--ro, opt, bool, whether the device supports the event of uploading the manual evidence information of the body camera, desc:eventType:
bodyCameraManualForensics; this node is not returned if not supported-->true
  </isSupportBodyCameraManualForensics>
  <isSupportBodyCameraLogin>
    <!--ro, opt, bool, whether the device supports the event of police officer logging in to the body camera, desc:eventType: bodyCameraLogin; this node is
not returned if not supported-->true
  </isSupportBodyCameraLogin>
  <isSupportBodyCameraRunStatus>
    <!--ro, opt, bool, whether the device supports the status of body camera, desc:eventType: bodyCameraRunStatus; this node is not returned if not
supported-->true
  </isSupportBodyCameraRunStatus>
  <isSupportBodyCameraSignIn>
    <!--ro, opt, bool, whether the device supports the event of police officer signing in the body camera, desc:eventType: bodyCameraSignIn; this node is
not returned if not supported-->true
  </isSupportBodyCameraSignIn>
  <isSupportAlertDistribution>
    <!--ro, opt, bool, whether the device supports the alert distribution, desc:eventType: alertDistribution; this node is not returned if not supported--
>true
  </isSupportAlertDistribution>
  <isSupportImageCapture>
    <!--ro, opt, bool, whether the device supports the event of scheduled capture, desc:related URI: /ISAPI/System/Video/inputs/channels/<ID>/imageCapture?
format=json-->true
  </isSupportImageCapture>
  <isSupportScheduledRecordUpload>
    <!--ro, opt, bool, whether the device supports the event of channel scheduled recording, desc:related URI:
/ISAPI/System/Video/inputs/channels/<ID>/scheduledRecordUpload?format=json-->true
  </isSupportScheduledRecordUpload>
  <isSupportTargetPatternCollect>
    <!--ro, opt, bool, whether the device supports the target track collection event, desc:eventType: targetPatternCollect-->true
  </isSupportTargetPatternCollect>
  <isSupportAudioAnalysis>
    <!--ro, opt, bool, whether the device supports the audio analysis event, desc:eventType: audioAnalysis; this node is not returned if not supported--
>true
  </isSupportAudioAnalysis>
  <isSupportDebuggingDataExport>
    <!--ro, opt, bool, whether the device supports importing the parameters (in-out parameters, rule configuration parameters), desc:related URI:
/ISAPI/System/debuggingData/parameters?format=json; this node is not returned if not supported-->true
  </isSupportDebuggingDataExport>
  <isSupportInternalParaAdjust>
    <!--ro, opt, bool, whether the device supports trinocular device; whether the device supports editing in-parameters, desc:related URI:
/ISAPI/System/internalParameters/adjustment?format=json; this node is not returned if not supported-->true
  </isSupportInternalParaAdjust>
  <isSupportManualImageCapture>
    <!--ro, opt, bool, whether the device supports manual capture of the video channel, desc:related URI:
/ISAPI/System/Video/inputs/channels/<ID>/manualImageCapture?format=json; this node is not returned if not supported-->true
  </isSupportManualImageCapture>
  <characteristicCode min="1" max="128">
    <!--ro, opt, string, device attribute code, range:[1,128], attr:min{opt, string},max{opt, string}, desc:related URI:
/ISAPI/System/deviceInfo/characteristicCode?format=json-->test
  </characteristicCode>
  <isSupportSIMCardStatus>
    <!--ro, opt, bool, whether the device supports getting the information of SIM card status, desc:related URI: /ISAPI/System/SIMCardStatus?format=json--
>true
  </isSupportSIMCardStatus>
  <OpenPlatformCap>
    <!--ro, opt, object, detailed HEOP capability supported by the device, desc:related URI: /ISAPI/Custom/OpenPlatform/capabilities-->
    <isSupportOpenPlatform>
      <!--ro, req, bool, whether the device supports HEOP (Embedded Open Platform)-->true
    </isSupportOpenPlatform>
    <maxAPPNum>
      <!--ro, opt, int, the maximum number of third-party application packages allowed to be installed-->0
    </maxAPPNum>
  </OpenPlatformCap>
  <isSupportLogException>
    <!--ro, opt, bool, whether the device supports the log exception alarm, desc:eventType: logException-->true
  </isSupportLogException>
  <isSupportMEF>
    <!--ro, opt, bool, whether the device supports Mobile electronic fence - mobile electronic fence data detection, desc:related URI:
/ISAPI/System/Network/MEF/<ID>?format=json-->true
  </isSupportMEF>
  <manualEvent
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opt="fielddetection,linedetection,ANPR,faceCapture,mixedTargetDetection,studentsStoodUp,headRiseRate,teacherBehavior,ElevatorRunningDetectionEvent,offDuty,batteryCarDisobey,fireExtinguisherRecog,indoorPasswayBlock,fireSmartFireDetect,smokeAlarm,gasTankMonitoring,smokeDetectAlarm,TelephoningBehaviorAlarmEvent,sleepDuty,playCellphone,audioAnalysis,TemperatureEvent,TemperatureEventUp,TemperatureEventDown,TMA">
    <!--ro, opt, string, whether the device supports URI for mannual triggering event, attr:opt{req, string}, desc:related URI: /ISAPI/Intelligent/channels/<ID>/manualEvent/<eventType>?format=json-->test
  </manualEvent>
  <isSupportAppCfg>
    <!--ro, opt, bool, whether the device supports application management, desc:related URI: /ISAPI/System/app/capabilities?format=json-->true
  </isSupportAppCfg>
  <isSupportCleanUp>
    <!--ro, opt, bool, whether the device supports one-click clearing, desc:related URI: /ISAPI/System/cleanup?format=json-->true
  </isSupportCleanUp>
  <isSupportEndUserCheckMsg opt="sleep,reboot,factoryReset,cleanup,parentPickUpQuery,schoolAttendanceQuery,screenShare,screenMirror">
    <!--ro, opt, bool, whether the device supports uploading the confirmation information of end users, attr:opt{req, string}, desc:eventType: endUserCheckMsg-->true
  </isSupportEndUserCheckMsg>
  <isSupportScreenConfig>
    <!--ro, opt, bool, whether the device supports configuring screen parameters, desc:related URI: /ISAPI/System/screenConfig/capabilities?format=json-->true
  </isSupportScreenConfig>
  <isSupportTemperatureMonitor>
    <!--ro, opt, bool, whether the device supports temperature monitoring configuration, desc:related URI: /ISAPI/System/temperatureMonitor/capabilities?format=json-->true
  </isSupportTemperatureMonitor>
  <isSupportObjectServer>
    <!--ro, opt, bool, whether the device supports the object storage service, desc:related URI: /ISAPI/System/objectServer/capabilities?format=json-->true
  </isSupportObjectServer>
  <isSupportCDsStatus>
    <!--ro, opt, bool, whether the device supports notifying disk status, desc:related URI for enabling the function: the bEnableCDsStatus node of /ISAPI/Trial/roompara/capabilities; corresponding event: CDsStatus-->true
  </isSupportCDsStatus>
  <isSupportCoalGangueDetection>
    <!--ro, opt, bool, whether the device supports coal gangue detection, desc:related URI: /ISAPI/Intelligent/channels/<channelID>/coalGangueDetection; this node is not returned if not supported-->true
  </isSupportCoalGangueDetection>
  <isSupportVacantSeatRate>
    <!--ro, opt, bool, whether the device supports seat vacancy detection, desc:related URI: /ISAPI/Intelligent/channels/<channelID>/vacantSeatRate?format=json; corresponding event: VacantSeatRate; this node is not returned if not supported-->true
  </isSupportVacantSeatRate>
  <isSupportObjectsThrownDetection>
    <!--ro, opt, bool, whether the device supports objects thrown detection, desc:related URI: /ISAPI/Intelligent/channels/<channelID>/objectsThrownDetection?format=json; corresponding event: objectsThrownDetection; this node is not returned if not supported-->true
  </isSupportObjectsThrownDetection>
  <isSupportBVCorrectDataSearch>
    <!--ro, opt, bool, whether the device supports searching for the parameters of camera correction, desc:related URI: /ISAPI/System/BVCorrectData/search?format=json; this node is not returned if not supported-->true
  </isSupportBVCorrectDataSearch>
  <isSupportSlagTruckWashingDetection>
    <!--ro, opt, bool, whether the device supports dump truck cleaning detection event, desc:eventType: slagTruckWashingDetection-->true
  </isSupportSlagTruckWashingDetection>
  <isSupportWOL>
    <!--ro, opt, bool, whether the device supports configuring the LAN wakeup parameters, desc:it can be realized via the WOL (Wake On-Lan) function. Related URL: /ISAPI/System/WOL/capabilities?format=json-->true
  </isSupportWOL>
  <isSupportPersonQueueTimingStatistics>
    <!--ro, opt, bool, whether the device supports counting the number of persons in each queue, desc:eventType: isSupportPersonQueueTimingStatistics-->true
  </isSupportPersonQueueTimingStatistics>
  <isSupportLensParamFile>
    <!--ro, opt, bool, whether the device supports importing and exporting camera parameters files-->true
  </isSupportLensParamFile>
  <isSupportVehiclePassingInParkingLot>
    <!--ro, opt, bool, whether the device supports the detection of vehicle passing in parking lot entrance and exit, desc:eventType: vehiclePassingInParkingLot-->true
  </isSupportVehiclePassingInParkingLot>
  <isSupportPowerOutput>
    <!--ro, opt, bool, whether power supply output configuration is supported, desc:related URL: /ISAPI/System/powerOutput/capabilities?format=json-->true
  </isSupportPowerOutput>
  <isSupportIPCUpgradeByFTP>
    <!--ro, opt, bool, it indicates that the device supports upgrading the proxy device channel, such as IPC upgrade via NVR, desc:related URI: /ISAPI/System/IPCUpgrade/<channelID>/FTP?format=json&security=<security>&iv=<iv>-->true
  </isSupportIPCUpgradeByFTP>
  <isSupportMethaneConcentrationException>
    <!--ro, opt, bool, whether the device supports methane concentration exception detection, desc:eventType: methaneConcentrationException-->true
  </isSupportMethaneConcentrationException>
  <isSupportMethaneLightIntensityException>
    <!--ro, opt, bool, whether the device supports methane light intensity exception detection, desc:eventType: methaneLightIntensityException-->true
  </isSupportMethaneLightIntensityException>
  <isSupportLocalButtonPermission>
    <!--ro, opt, bool, whether the device supports controlling local button permission, desc:related URI: /ISAPI/System/localButtonPermission/capabilities?format=json-->true
  </isSupportLocalButtonPermission>
  <isSupportLocalButtonRecord>
    <!--ro, opt, bool, whether the device supports button triggering record, desc:eventType: localButtonRecord-->true
  </isSupportLocalButtonRecord>
  <isSupportLocalButtonSnapshot>
    <!--ro, opt, bool, whether the device supports button triggering snapshot, desc:eventType: localButtonSnapshot-->true
  </isSupportLocalButtonSnapshot>
  <isSupportVoiceTalkEvent>
    <!--ro, opt, bool, whether the device supports voice talk, desc:eventType: voiceTalkEvent-->true
  </isSupportVoiceTalkEvent>
  <isSupportPowerSupply>
    <!--ro, opt, bool, whether the device supports external power supply, desc:related URI: /ISAPI/System/powerSupply/capabilities?format=json-->true
  </isSupportPowerSupply>
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  <isSupportbarrierGateStatus>
    <!--ro, opt, bool, whether the device supports barrier gate status, desc:eventType: barrierGateStatus-->true
  </isSupportbarrierGateStatus>
  <DeployCap>
    <!--ro, opt, object, the capability of arming non-subscription events-->
    <deployID opt="1,2">
      <!--ro, opt, enum, arming type, subType:int, attr:opt{req, string}, desc:0 (client arming); 1 (real-time arming)-->0
    </deployID>
  </DeployCap>
  <isSupportSignalControllerStatus>
    <!--ro, opt, bool, whether the device supports uploading the signal controller status, desc:eventType: signalControllerStatus-->true
  </isSupportSignalControllerStatus>
  <isSupportSignalControllerFault>
    <!--ro, opt, bool, whether the device supports uploading the signal controller fault, desc:eventType: signalControllerFault-->true
  </isSupportSignalControllerFault>
  <isSupportNTPConfig>
    <!--ro, opt, bool, whether the device supports configuring NTP parameters, desc:/ISAPI/System/time/ntp/capabilities?format=json-->true
  </isSupportNTPConfig>
  <isSupportConsumptionQuery>
    <!--ro, opt, bool, whether the device supports uploading the request for searching for consumption, desc:eventType: ConsumptionQuery-->true
  </isSupportConsumptionQuery>
  <StreamingCodingRegionPositionCap>
    <!--ro, opt, object, whether the device supports getting all stream splice area positions information capability,
desc:/ISAPI/System/streamingCoding/channels/<channelID>/regionPosition-->
    <ChannelsList>
      <!--ro, opt, array, channel list, subType:object-->
      <channelsID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelsID>
    </ChannelsList>
  </StreamingCodingRegionPositionCap>
  <isSupportLowPowerPlan>
    <!--ro, opt, bool, whether the device supports configuring low consumption mode schedule, desc:related URI:
/ISAPI/System/consumptionMode/lowPowerPlan/capablities?format=json-->true
  </isSupportLowPowerPlan>
  <isSupportPlateQuaAlarm>
    <!--ro, opt, bool, whether the device supports plate quality detection, desc:eventType: plateQuaAlarm-->true
  </isSupportPlateQuaAlarm>
  <isSupportWiegand>
    <!--ro, opt, bool, whether the device supports the Wiegand protocol, desc:related URI: /ISAPI/System/Wiegand/<wiegandID>/capabilities?format=json-->true
  </isSupportWiegand>
  <zoomFocusWebDisplay
opt="ROI,roadTrafficDetection,SMD,mixedTargetDetection,faceCapture,personCounting,multidimensionalPeopleCounting,AIOP,heelPDC,packingSpaceRecognition,tunnel
Detection,cityManagement">
    <!--ro, opt, string, whether the web supports displaying the zoomfocus page, attr:opt{req, string},
desc:ROI page path: Configuration—Vedio/Audio—ROI
roadTrafficDetection:Configuration—Road Traffic—Detection Configuration; SMD; mixedTargetDetection; faceCapture-->ROI
  </zoomFocusWebDisplay>
  <isSupportInternalParamsAdjustment>
    <!--ro, opt, bool, whether the device supports in-parameters correction by region, desc:true (support), false or this node is not returned (not
support). Related URI: /ISAPI/System/internalParameters/regionAdjustment/capabilities?format=json-->true
  </isSupportInternalParamsAdjustment>
  <isSupportDiagnosticReport>
    <!--ro, opt, bool, whether the device supports diagnostic report function, desc:true (support), false or this node is not returned (not support).
Related URI: /ISAPI/System/diagnosticReport/capabilities?format=json-->true
  </isSupportDiagnosticReport>
  <isSupportWaterQualitySearch>
    <!--ro, opt, bool, whether the device supports the function of searching for water quality, desc:true (support), false or this node is not returned (not
support). Related URI: /ISAPI/Event/waterQualityDetection/search?format=json-->true
  </isSupportWaterQualitySearch>
  <isSupportExhaustDetection>
    <!--ro, opt, bool, whether the device supports configuring gas detection parameters, desc:true (support), false or this node is not returned (not
support). Related URI: /ISAPI/Event/exhaustDetection/channels/<channelID>/capabilities?format=json-->true
  </isSupportExhaustDetection>
  <isSupportTPMSecretKey>
    <!--ro, opt, bool, whether the device supports enabling stream encryption for all TPM keys lists, desc:true (support), false or this node is not
returned (not support). Related URI: /ISAPI/Streaming/encryption/TPMSecretKeyList?format=json&security=<security>&iv=<iv>-->true
  </isSupportTPMSecretKey>
  <isSupportWeatherStation>
    <!--ro, opt, bool, whether the device supports uploading weather station information, desc:eventType: weatherStation; this node is not returned if not
supported-->true
  </isSupportWeatherStation>
  <isSupportRoadVisibilityAnalysis>
    <!--ro, opt, bool, whether the device supports road visibility detection analysis, desc:eventType: weatherStation; this node is not returned if not
supported-->true
  </isSupportRoadVisibilityAnalysis>
  <isSupportRoadWeatherAnalysis>
    <!--ro, opt, bool, whether the device supports road weather detection analysis, desc:eventType: roadWeatherAnalysis; this node is not returned if not
supported-->true
  </isSupportRoadWeatherAnalysis>
  <isSupportRoadVisibilityAbnormal>
    <!--ro, opt, bool, whether the device supports uploading road visibility detection exception, desc:eventType: roadVisibilityAbnormal; this node is not
returned if not supported-->true
  </isSupportRoadVisibilityAbnormal>
  <isSupportRoadWeatherAbnormal>
    <!--ro, opt, bool, whether the device supports uploading road weather detection exception, desc:eventType: roadWeatherAbnormal; this node is not
returned if not supported-->true
  </isSupportRoadWeatherAbnormal>
  <isSupportChannelsTimeSync>
    <!--ro, opt, bool, whether the device supports batch device time synchronization of the IPC/IOT channel, desc:related URL:
/ISAPI/System/time/channelsTimeSync/capabilities?format=json-->true
  </isSupportChannelsTimeSync>
  <isSupport433Sensor>
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    <!--ro, opt, bool, whether the device supports configuring 433 sensor transmission,
desc:related URLs:   /ISAPI/System/Sensor/433sensor/433wirelessTransfer/capabilities?format=json
    /ISAPI/System/Sensor/433sensor/capabilities?format=json
    /ISAPI/System/Sensor/433sensor/sensorDetection/capabilities?format=json
    /ISAPI/System/Sensor/433sensor/sensorPicOverlay/capabilities?format=json
    /ISAPI/System/Sensor/433sensor/rfParamTemplate/capabilities?format=json-->true
  </isSupport433Sensor>
  <isSupportSignalSource>
    <!--ro, opt, bool, whether the device supports configuring signal source parameters, desc:related URL: /ISAPI/System/signalSource/capabilities?
format=json-->true
  </isSupportSignalSource>
  <isSupportUSBConfig>
    <!--ro, opt, bool, whether the device supports configuring USB parameters, desc:related URL: /ISAPI/System/USBConfig/capabilities?format=json-->true
  </isSupportUSBConfig>
  <isSupportStandbyStrategy>
    <!--ro, opt, bool, whether the device supports configuring standby strategy parameters, desc:related URL: /ISAPI/System/standbyStrategy/capabilities?
format=json-->true
  </isSupportStandbyStrategy>
  <isSupportWakeupStrategy>
    <!--ro, opt, bool, whether the device supports configuring wakeup strategy parameters, desc:related URL: /ISAPI/System/wakeupStrategy/capabilities?
format=json-->true
  </isSupportWakeupStrategy>
  <isSupportSeachMutexFunction>
    <!--ro, opt, bool, whether the device supports searching for disabling the mutual exclusion function, desc:related URL: GET
/ISAPI/System/mutexFunction/capabilities?format=json; function description: whether the device supports searching for mutual exclusion function-->true
  </isSupportSeachMutexFunction>
  <isSupportMultiDeviceID>
    <!--ro, opt, bool, whether the device supports multiple long numbers configuration, desc:related URL:
/ISAPI/VideoIntercom/deviceId/batchCfg/capabilities?format=json-->true
  </isSupportMultiDeviceID>
  <isNotSupportMultiDeviceIDCustomName>
    <!--ro, opt, bool, whether the device supports configuring custom name for multiple long numbers, desc:if this function is not supported, the user
cannot configure custom name for the device via the long number configuration API; true (not support), false or this node is not returned (support)-->true
  </isNotSupportMultiDeviceIDCustomName>
  <isSupportMixedSecurityChecking>
    <!--ro, opt, bool, whether the device supports uploading the event of mixed security checking, desc:eventType: mixedSecurityChecking-->true
  </isSupportMixedSecurityChecking>
  <isSupportVisiblePackageEvent>
    <!--ro, opt, bool, whether the device supports visible package event, desc:eventType: visiblePackageEvent-->true
  </isSupportVisiblePackageEvent>
  <isSupportBarCodeEvent>
    <!--ro, opt, bool, whether the device supports the barcode event-->true
  </isSupportBarCodeEvent>
  <isSupportGaugeReading>
    <!--ro, opt, bool, whether the device supports the gauge reading function, desc:related URL: GET
/ISAPI/Event/channels/<channelID>/gaugeReading/capabilities?format=json; this node is not returned if not supported-->true
  </isSupportGaugeReading>
  <isSupportUncertificateCompareEvent>
    <!--ro, opt, bool, whether the device supports the UncertificateCompareEvent event, desc:eventType: UncertificateCompareEvent; this node is not returned
if not supported-->true
  </isSupportUncertificateCompareEvent>
  <isSupportAudioDiacritical>
    <!--ro, opt, bool, whether the device supports audio voice changing function, desc:related URL:
/ISAPI/System/Video/inputs/channels/<channelID>/audioDiacritical/capabilities?format=json-->true
  </isSupportAudioDiacritical>
  <isSupportMEFReportSearch>
    <!--ro, opt, bool, whether the device supports searching for electronic fence data statistics report, desc:URL for getting the capability:
/ISAPI/System/Network/MEF/reportSearch/capabilities?format=json; this node is not returned if not supported-->true
  </isSupportMEFReportSearch>
  <isSupportBroadcastTriggerEvent>
    <!--ro, opt, bool, whether the device supports broadcast triggering event, desc:after the event is uploaded to the platform via the paging microphone,
the platform will broadcast and check according to the uploaded terminal No. list; eventType: broadcastTriggerEvent. URL for checking:
/ISAPI/VideoIntercom/broadcastTriggerConfirm?format=json-->true
  </isSupportBroadcastTriggerEvent>
  <isSupportFishingShipDetection>
    <!--ro, opt, bool, whether the device supports fishing ship detection event, desc:eventType: fishingShipDetection; related URL:
/ISAPI/Thermal/channels/<channelID>/fishingShipDetection/<SID>?format=json-->true
  </isSupportFishingShipDetection>
  <isSupportHPVehicleLoadingRate>
    <!--ro, opt, bool, whether the device supports the platform vehicle loading rate event, desc:eventType: vehicleLoadingRate, this node is not returned if
not supported. Related URL: /ISAPI/Intelligent/channels/<channelID>/HPDetection/capabilities?format=json-->true
  </isSupportHPVehicleLoadingRate>
  <isSupportVideoBarrierGateDetection>
    <!--ro, opt, bool, whether the device supports port and gate video detection, desc:eventType: videoBarrierGateDetection, this node is not returned if
not supported. Related URL: /ISAPI/Traffic/ContentMgmt/videoBarrierGate/capabilities?format=json-->true
  </isSupportVideoBarrierGateDetection>
  <isSupportLogicalResource>
    <!--ro, opt, bool, whether the device supports camera organization and management, desc:related URL: /ISAPI/System/LogicalResource/capabilities?
format=json, this node is not returned if not supported-->true
  </isSupportLogicalResource>
  <isSupportSlagTruckCoverPlate>
    <!--ro, opt, bool, whether the device supports detecting whether back cover the slag truck is covered, desc:eventType: slagTruckCoverPlateDetection,
this node is not returned if not supported. Related URL: /ISAPI/Event/channels/<channelID>/slagTruckCoverPlate/capabilities?format=json-->true
  </isSupportSlagTruckCoverPlate>
  <isSupportTrainDetection>
    <!--ro, opt, bool, whether the device supports train detection, desc:If train detection is not supported, this node will not be returned. Corresponding
eventType: trainDetection-->true
  </isSupportTrainDetection>
  <isNotSupportVideoOverlays>
    <!--ro, opt, bool, whether the device does not support configuring OSD overlay on the device web configuration page, desc:this node is only valid for
checking whether the OSD overlay configuration of the image menu on the person and vehicle verification station is supported or not (this configuration is
supported by default in the protocol and it does not have node control capability). If true is returned, it indicates that the OSD overlay is not supported;
if false or no node is returned, it indicates the OSD overlay function is supported-->true
  </isNotSupportVideoOverlays>
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<isNotSupportVideoOverlays>
<isSupportScheduledCaptureTask>
<!--ro, opt, bool, whether the device supports scheduled capturing tasks, desc:related URI: /ISAPI/Sytem/Video/scheduledCapture/capabilities-->true
</isSupportScheduledCaptureTask>
<isSupportGuideCfg>
<!--ro, opt, bool, whether the device supports the wizard configuration, desc:this node is only valid for the indoor station to check whether to enable wizard configuration, which is provided for the users to perform basic function configuration after activating and logging in to the device (including Internet parameters configuration, dialog configuration and batch adding analog indoor stations). If true is returned, it indicates wizard configuration is supported; if false or no node is returned, it indicates the function is not supported-->true
</isSupportGuideCfg>
<isSupportCameraGroupPeopleCounting>
<!--ro, opt, bool, whether the device supports passenger flow statistics of multiple cameras, desc:/ISAPI/Intelligent/counting/cameraGroup/capabilities?format=json-->true
</isSupportCameraGroupPeopleCounting>
<isSupportAngleDeviationDetection>
<!--ro, opt, bool, whether the device supports tilting detection, desc:related URL: /ISAPI/Event/channels/<channelID>/angleDeviationDetection/capabilities?format=json; this node is not returned if not supported-->true
</isSupportAngleDeviationDetection>
<isSupportCityCode>
<!--ro, opt, bool, whether the device supports getting city code, desc:related URI: /ISAPI/System/cityCode?format=json-->true
</isSupportCityCode>
<isSupportAlcoholDetectionResult>
<!--ro, opt, bool, whether the device supports uploading alcohol test, desc:eventType: alcoholDetectionResult-->true
</isSupportAlcoholDetectionResult>
<isSupportAlcoholDetectionRodBind>
<!--ro, opt, bool, whether the device supports the request of binding alcohol test stick, desc:eventType: alcoholDetectionRodBind-->true
</isSupportAlcoholDetectionRodBind>
<isSupportAlcoholDetectionRodBindQuery>
<!--ro, opt, bool, whether the device supports the query for the request of binding alcohol test stick, desc:eventType: alcoholDetectionRodBindQuery-->true
</isSupportAlcoholDetectionRodBindQuery>
<isSupportAlcoholDetectionRodUnbind>
<!--ro, opt, bool, whether the device supports the event of unbinding alcohol test stick, desc:eventType: alcoholDetectionRodUnbind-->true
</isSupportAlcoholDetectionRodUnbind>
<isSupportAbnormalDriving>
<!--ro, opt, bool, whether the device supports abnormal driving behaviors detection, desc:eventType: abnormalDriving-->true
</isSupportAbnormalDriving>
<isSupportEmanationsEvent>
<!--ro, opt, bool, whether the device supports the event of emanation test, desc:eventType: emanationsEvent; the security inspection system is equipped with an emanation tester, and when the emanation exceeds the threshold, an alarm event will be uploaded. This event is only triggered by the emanation tester, and its configuration is not required-->true
</isSupportEmanationsEvent>
<isSupportClientUpdatePackage>
<!--ro, opt, bool, whether the device supports the client upgrade package management, desc:related URL: /ISAPI/System/clientUpdatePackage/capabilities?format=json-->true
</isSupportClientUpdatePackage>
<isSupportDeviceStatusUploadStrategy>
<!--ro, opt, bool, whether the device supports configuring the parameters of device status uploading strategy, desc:related URL: /ISAPI/System/status/uploadStrategy/capabilities?format=json-->true
</isSupportDeviceStatusUploadStrategy>
<isSupportDialIndicatorReading>
<!--ro, opt, bool, whether the devices supports dial indicator reading, desc:related URL: /ISAPI/Intelligent/channels/<channelID>/dialIndicatorReading/capabilities?format=json; the function refers to reading single pointer dial and uploading the result (currently, it can only recognize dials that do not accumulate the readings such as thermometer and barometer)-->true
</isSupportDialIndicatorReading>
<isNotSupportAudioVideoCompress>
<!--ro, opt, bool, whether the device does not support audio/video encoding, desc:this node is only valid for the person and vehicle verification station; if true is returned, the video/audio page will not be displayed on the device web page. If false or no node is returned, the video/audio page will be displayed on the device web page by default-->true
</isNotSupportAudioVideoCompress>
<isSupportUPLCExceptionEvent>
<!--ro, opt, bool, whether the device supports the event of liquid chromatograph system exception, desc:eventType: UPLCException-->true
</isSupportUPLCExceptionEvent>
<isSupportUrgentTurn>
<!--ro, opt, bool, whether the device supports the sharp turn event, desc:eventType: urgentTurn-->true
</isSupportUrgentTurn>
<isSupportUrgentBrake>
<!--ro, opt, bool, whether the device supports the sudden brake event, desc:eventType: urgentBrake-->true
</isSupportUrgentBrake>
<isSupportUrgentAcceleration>
<!--ro, opt, bool, whether the device supports the rapid acceleration event, desc:eventType: urgentAcceleration-->true
</isSupportUrgentAcceleration>
<isSupportVehiclePanicButton>
<!--ro, opt, bool, whether the device supports the alarm/event of mobile emergency button, desc:eventType: vehiclePanicButton-->true
</isSupportVehiclePanicButton>
<isSupportOverSpeed>
<!--ro, opt, bool, whether the device supports the vehicle speeding alarm/event, desc:eventType: overSpeed-->true
</isSupportOverSpeed>
<isSupportAbnormalAcceleration>
<!--ro, opt, bool, whether the device supports the event of abnormal vehicle custom acceleration alarm, desc:eventType: abnormalAcceleration-->true
</isSupportAbnormalAcceleration>
<isSupportRollover>
<!--ro, opt, bool, whether the device supports the vehicle turnover alarm/event, desc:eventType: rollover-->true
</isSupportRollover>
<isSupportCollision>
<!--ro, opt, bool, whether the device supports the vehicle collision alarm/event, desc:eventType: collision-->true
</isSupportCollision>
<isSupportVehicleOBUInfo>
<!--ro, opt, bool, whether the device supports uploading the vehicle OBU information, desc:eventType: vehicleOBUInfo-->true
</isSupportVehicleOBUInfo>
<isSupportPowerUpPicture>
<!--ro, opt, bool, whether the device supports importing and exporting pictures displayed during startup, desc:/ISAPI/System/powerUpPicture/capabilities?format=json-->true
</isSupportPowerUpPicture>
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  <isSupportPowerUpPictureCfg>
    <!--ro, opt, bool, whether the device supports configuring parameters of pictures displayed during startup,
desc:/ISAPI/System/powerUpPictureCfg/capabilities?format=json-->true
  </isSupportPowerUpPictureCfg>
  <isSupportAlgoPackageScheduling>
    <!--ro, opt, bool, whether the device supports algorithm package scheduling, desc:Related URIs:  /ISAPI/System/AlgoPackageScheduling/capabilities?
format=json and /ISAPI/System/AlgoPackageScheduling/task/capabilities?format=json-->true
  </isSupportAlgoPackageScheduling>
  <isSupportVoiceTalkRequestingAlarm>
    <!--ro, opt, bool, whether the device supports the two-way audio request alarm, desc:eventType: voiceTalkRequestingAlarm-->true
  </isSupportVoiceTalkRequestingAlarm>
  <isSupportSetupStructure>
    <!--ro, opt, bool, whether the device supports configuring construction structure parameters, desc:related URI: /ISAPI/System/setupStructure?
format=json-->true
  </isSupportSetupStructure>
  <isSupportPassNumInfoEvent>
    <!--ro, opt, bool, whether the device supports uploading the number of people passed, desc:eventType: passNumInfoEvent-->true
  </isSupportPassNumInfoEvent>
  <isShowClientRemote>
    <!--ro, opt, bool, whether the device supports displaying remote configuration on the client,
desc:1. If this node is not returned or it is returned and its value is true, it indicates that the device supports this function.
2. If this node is returned and its value is false, it indicates that the device does not support this function.
iVMS-4200 client software supports remote configuration by default. However, there are still some devices that do not support remote configuration via iVMS-
4200 and they can only be configured remotely via the web browser. The capability determined by this node is used to control whether to display the remote
configuration on iVMS-4200 client software-->true
  </isShowClientRemote>
  <isSupportUserManualQRCode>
    <!--ro, opt, bool, whether the device supports getting the QR code of the user manual, desc:related URI: /ISAPI/System/userManualQRCode/capabilities?
format=json-->true
  </isSupportUserManualQRCode>
  <isSupportUnclosedFrontPanel>
    <!--ro, opt, bool, whether the device supports uploading the event of the front panel not closed, desc:related URI for getting the capability:
/ISAPI/Trial/roompara/capabilities; corresponding node: < unclosedFrontPanelEnabled>. Related event: unclosedFrontPanel-->true
  </isSupportUnclosedFrontPanel>
  <isSupportVehicleWeight>
    <!--ro, opt, bool, whether the device supports configuring parameters for vehicle weight and axle detection,
desc:/ISAPI/Traffic/channels/<channelID>/vehicleWeight?format=json-->true
  </isSupportVehicleWeight>
  <isSupportWeightInterface>
    <!--ro, opt, bool, whether it supports integrated overload control parameters, desc:/ISAPI/Traffic/ContentMgmt/WeightInterface?format=json-->true
  </isSupportWeightInterface>
  <isSupportSecurityPanicEvent>
    <!--ro, opt, bool, whether the device supports emergency events of security inspection, desc:eventType: securityPanicEvent-->true
  </isSupportSecurityPanicEvent>
  <isSupportTrafficLightMalfunction>
    <!--ro, opt, bool, whether the device supports the event of detecting faults of traffic lights, desc:eventType: trafficLightMalfunction-->true
  </isSupportTrafficLightMalfunction>
  <isSupportTOFShelterAlarm>
    <!--ro, opt, bool, whether the device supports TOF tampering alarm, desc:/ISAPI/Event/channels/<channelID>/TOFShelterAlarm/capabilities?format=json--
>true
  </isSupportTOFShelterAlarm>
  <isSupportSetMealQuery>
    <!--ro, opt, bool, whether the device supports uploading the event of searching set meals, desc:eventType: SetMealQuery-->true
  </isSupportSetMealQuery>
  <isSupportMultiScreenControlAlarm>
    <!--ro, opt, bool, whether the device supports uploading the multi-screen controller alarm, desc:eventType: multiScreenControlAlarm-->true
  </isSupportMultiScreenControlAlarm>
  <isSupportKeyFrameLoss>
    <!--ro, opt, bool, whether the device supports uploading the event of key frame loss, desc:related URI: /ISAPI/Event/keyFrameLoss/capabilities?
format=json; eventType: keyFrameLoss.-->true
  </isSupportKeyFrameLoss>
  <isSupportWorkStatusCfg>
    <!--ro, opt, bool, whether the device supports working status configuration, desc:/ISAPI/System/WorkStatusCfg?format=json-->true
  </isSupportWorkStatusCfg>
  <isSupportHealthInfoSyncQuery>
    <!--ro, opt, bool, whether the device supports the event of searching for health information, desc:eventType: HealthInfoSyncQuery-->true
  </isSupportHealthInfoSyncQuery>
  <isSupportVisitorPictureSearchEvent>
    <!--ro, opt, bool, whether the device supports the event of searching for visitor profiles, desc:eventType: VisitorPictureSearchEvent-->true
  </isSupportVisitorPictureSearchEvent>
  <isSupportEZVIZParameterTest>
    <!--ro, opt, bool, whether the device supports testing EZ parameters, desc:related URI: /ISAPI/System/Network/EZVIZ/parameterTest?format=json-->true
  </isSupportEZVIZParameterTest>
  <isSupportUnattendedPackageEvent>
    <!--ro, opt, bool, whether the device supports the unattended package event, desc:eventType: unattendedPackageEvent-->true
  </isSupportUnattendedPackageEvent>
  <isSupportPackageFragmentEvent>
    <!--ro, opt, bool, whether the device supports the event of uploading package fragment data, desc:eventType: packageFragmentEvent-->true
  </isSupportPackageFragmentEvent>
  <isSupportPackageBarcodeMismatchEvent>
    <!--ro, opt, bool, whether the device supports the event of package and barcode mismatching, desc:eventType: packageBarcodeMismatchEvent-->true
  </isSupportPackageBarcodeMismatchEvent>
  <isSupportCheckingModeSwitchEvent>
    <!--ro, opt, bool, whether the device supports the event of switching the image checking mode, desc:eventType: checkingModeSwitchEvent-->true
  </isSupportCheckingModeSwitchEvent>
  <FaceSnapPrivacyMask>
    <!--ro, opt, object, whether the device supports privacy mask on faces, desc:based on face picture capture, all faces will be masked to protect the
privacy. Related URI: /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask/capabilities?format=json-->
    <ChannelsList>
      <!--ro, opt, array, channel list, subType:object-->
      <channelsID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelsID>
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    </ChannelsList>
  </FaceSnapPrivacyMask>
  <FaceContrastPrivacyMask>
    <!--ro, opt, object, whether the device supports privacy mask on faces of key persons, desc:based on face picture comparison, the faces of key persons
will be masked to protect the privacy. Related URI: /ISAPI/Intelligent/channels/<channelID>/faceContrast/privacyMask/capabilities?format=json-->
    <ChannelsList>
      <!--ro, opt, array, channel list, subType:object-->
      <channelsID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelsID>
    </ChannelsList>
  </FaceContrastPrivacyMask>
  <isSupportHumanBodyComparison>
    <!--ro, opt, bool, whether the device supports human body comparison events, desc:related URI:
/ISAPI/Event/channels/<channelID>/humanBodyComparison/capabilities?format=json; eventType: humanBodyComparison-->true
  </isSupportHumanBodyComparison>
  <isSupportMedicalInfoSearchEvent>
    <!--ro, opt, bool, whether the device supports the event of searching for medical information, desc:medicalInfoSearchEvent-->true
  </isSupportMedicalInfoSearchEvent>
  <isSupportConsumptionStatusQuery>
    <!--ro, opt, bool, whether the device supports uploading the request for searching for consumption status, desc:eventType: ConsumptionStatusQuery-->true
  </isSupportConsumptionStatusQuery>
  <isSupportShipChannelAbnormal>
    <!--ro, opt, bool, whether the device supports uploading ship channel exception detection event, desc:eventType: shipChannelAbnormal-->true
  </isSupportShipChannelAbnormal>
  <isSupportDepositRetrieveEvent>
    <!--ro, opt, bool, whether the device supports the event of object access in the intelligent cabinet-->true
  </isSupportDepositRetrieveEvent>
  <isSupportDepositTimeoutEvent>
    <!--ro, opt, bool, whether the device supports the event of overtime object deposit in the intelligent cabinet-->true
  </isSupportDepositTimeoutEvent>
  <isSupportCabinetStatusEvent>
    <!--ro, opt, bool, whether the device supports the event of intelligent cabinet's changed status-->true
  </isSupportCabinetStatusEvent>
  <isSupportCabinetAuthorityExpired>
    <!--ro, opt, bool, whether the device supports the event of intelligent cabinet's permission expiration-->true
  </isSupportCabinetAuthorityExpired>
  <isSupportT1Test>
    <!--ro, opt, bool, whether the device supports the configuration of controlling T1 tool-->true
  </isSupportT1Test>
  <isSupportDrivingDirectionAtIntersection>
    <!--ro, opt, bool, whether the device supports intersection driving direction detection, desc:eventType: drivingDirectionAtIntersection; this node is
not returned if not supported-->true
  </isSupportDrivingDirectionAtIntersection>
  <isSupportPlaceSceneExecuteStatus>
    <!--ro, opt, bool, whether the device supports the event of scene execution status, desc:eventType: placeSceneExecuteStatus; this node is not returned
if not supported-->true
  </isSupportPlaceSceneExecuteStatus>
  <isSupportSceneRecordingCtrl>
    <!--ro, opt, bool, whether the device supports the event of controlling scene recording, desc:eventType: sceneRecordingCtrl; this node is not returned
if not supported-->true
  </isSupportSceneRecordingCtrl>
  <isSupportStreamRecordStatus>
    <!--ro, opt, bool, whether the device supports getting the stream ID record status, desc:/ISAPI/System/Video/inputs/streams/recordStatus-->true
  </isSupportStreamRecordStatus>
  <isSupportManualSnapPicture>
    <!--ro, opt, bool, whether the device supports manual picture capturing, desc:related URI:
/ISAPI/Streaming/channels/<trackStreamID>/picture/capabilities?format=json-->true
  </isSupportManualSnapPicture>
  <isSupportVerificationResultEvent>
    <!--ro, opt, bool, whether the device supports the event of human and vehicle verification result, desc:eventType: verificationResultEvent-->true
  </isSupportVerificationResultEvent>
  <OTAP>
    <!--ro, opt, object, the capability of Open Things Access Protocol, desc:OTAP capability-->
    <isSupportOTAPModel>
      <!--ro, opt, bool, whether the device supports getting the virtual No. and device No. of the OTAP, desc:GET /ISAPI/System/Network/OTAP/Model?
format=json-->true
    </isSupportOTAPModel>
  </OTAP>
  <isSupportForeignObjectInRoadDetection>
    <!--ro, opt, bool, whether the device supports the detection of foreign object in road, desc:eventType: foreignObjectInRoadDetection-->true
  </isSupportForeignObjectInRoadDetection>
  <isSupportUPLCMethodOperatingStatus>
    <!--ro, opt, bool, whether the device supports the event of UPLC method operating status, desc:eventType: UPLCMethodOperatingStatus-->true
  </isSupportUPLCMethodOperatingStatus>
  <isSupportMaterialScanTaskStatus>
    <!--ro, opt, bool, whether the device supports material scanning task status, desc:eventType: materialScanTaskStatus-->true
  </isSupportMaterialScanTaskStatus>
  <isSupportSearchInputProxy>
    <!--ro, opt, bool, whether the device supports searching for accessible IPC of device in the LAN, desc:related URI: GET
/ISAPI/ContentMgmt/InputProxy/search?security=<security>&iv=<iv>; true (support), this node is not returned (call the specific API to confirm whether it
supports or not)-->true
  </isSupportSearchInputProxy>
  <isSupportPostureDetection>
    <!--ro, opt, bool, whether the device supports people posture detection, desc:eventType: MeetingLock-->true
  </isSupportPostureDetection>
  <isSupportFans>
    <!--ro, opt, bool, whether the device supports  configuring fan, desc:/ISAPI/System/Fans?format=json-->true
  </isSupportFans>
  <pickUpCalibration>
    <!--ro, opt, object, whether the device supports wireless pickup calibration, desc:related URI:
/ISAPI/System/PickUpCalibration/channels/<channelID>/capabilities?format=json-->
    <ChannelsList>
      <!--ro, opt, array, channel list, subType:object-->
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      <channelsID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelsID>
    </ChannelsList>
  </pickUpCalibration>
  <isSupportSerialLogUpload>
    <!--ro, opt, bool, whether the device supports uploading serial port log, desc:true (support), false (not support)-->true
  </isSupportSerialLogUpload>
  <isSupportLiquidDetectionEvent>
    <!--ro, opt, bool, whether the device supports liquid detection, desc:eventType: liquidDetectionEvent-->true
  </isSupportLiquidDetectionEvent>
  <isSupportETCRSU>
    <!--ro, opt, bool, whether the device supports  ETCRSU configuration, desc:related URI: /ISAPI/Traffic/ETCRSU level protocol-->true
  </isSupportETCRSU>
  <isSupportVideoCodeConvertTool>
    <!--ro, opt, bool, whether the device supports getting the video encoding conversion tool, desc:related URI: /ISAPI/System/videoCodeConvertTool?
format=json protocol-->true
  </isSupportVideoCodeConvertTool>
  <supportQuickConfigType opt="icr,dayNightGate,syncSignalOutput">
    <!--ro, opt, string, the list of supported functions of the quick configuration pop-up on the web, attr:opt{opt, string},
desc:the web quick configuration does not distinguish the configuration order, and it displays all supported function lists on the pop-up on the tab page
    icr refers to ICR switch configuration (related URL: /ISAPI/Image/channels/<channelID>/); icr
   dayNightGate refers to day and night mode switching (related URL: /ISAPI/Image/channels/<channelID>/dayNightGate);
    IOOutputs refers to synchronous signal output (related URL: /ISAPI/System/syncSignalOutput)-->syncSignalOutput
  </supportQuickConfigType>
  <isSupportVisualLocation>
    <!--ro, opt, bool, whether the device supports configuring visual location service, desc:related URI:
/ISAPI/System/LocationMgr/visualLocation/capabilities?format=json-->true
  </isSupportVisualLocation>
  <isSupportVideoAccessAbnormal>
    <!--ro, opt, bool, whether the device supports abnormal video access, desc:eventType: VideoAccessAbnormal; this node is not returned if not supported--
>true
  </isSupportVideoAccessAbnormal>
  <isSupportSubSystemAbnormal>
    <!--ro, opt, bool, whether the device supports abnormal subsystem, desc:eventType: SubSystemAbnormal; this node is not returned if not supported-->true
  </isSupportSubSystemAbnormal>
  <isSupportSearchFileList>
    <!--ro, opt, bool, whether the device supports searching for list of device files, desc:related URI: /ISAPI/System/fileExport/SearchFileList?
format=json-->true
  </isSupportSearchFileList>
  <isSupportDistanceDetectionEvent>
    <!--ro, opt, bool, whether the device supports distance detection, desc:eventType: distanceDetectionEvent; currently it is only used for devices test
tools-->true
  </isSupportDistanceDetectionEvent>
  <isSupportMeetingQueryEvent>
    <!--ro, opt, bool, whether the device supports searching for meetings, desc:eventType: MeetingElementCancel-->true
  </isSupportMeetingQueryEvent>
  <isSupportMeetingLockEvent>
    <!--ro, opt, bool, whether the device supports locking meetings, desc:eventType: MeetingLock-->true
  </isSupportMeetingLockEvent>
  <isSupportMeetingAddEvent>
    <!--ro, opt, bool, whether the device supports reserving meetings, desc:eventType: MeetingAdd-->true
  </isSupportMeetingAddEvent>
  <isSupportMeetingModifyEvent>
    <!--ro, opt, bool, whether the device supports modifying meetings, desc:eventType: MeetingModify-->true
  </isSupportMeetingModifyEvent>
  <isSupportMeetingCancelEvent>
    <!--ro, opt, bool, whether the device supports canceling meetings, desc:eventType: ScreenBindIPC-->true
  </isSupportMeetingCancelEvent>
  <isSupportScreenBindIPCEvent>
    <!--ro, opt, bool, whether the device supports the screen binding IPC, desc:eventType: ScreenBindIPC-->true
  </isSupportScreenBindIPCEvent>
  <isSupportSensorDetection>
    <!--ro, opt, bool, whether the device supports uploading sensor data, desc:eventType: sensorDetection-->true
  </isSupportSensorDetection>
  <isSupportClearLogCache>
    <!--ro, opt, bool, whether the device supports clearing the log cache, desc:related URI: /ISAPI/ContentMgmt/ClearLogCache?format=json-->true
  </isSupportClearLogCache>
  <isSupportRoadOccupyDetection>
    <!--ro, opt, bool, whether the device supports the entrance and exit block detection, desc:related URI:
/ISAPI/ITC/Entrance/channels/<channelID>/roadOccupyDetection/capabilities?format=json-->true
  </isSupportRoadOccupyDetection>
  <isSupportMeetingRoomSignInEvent>
    <!--ro, opt, bool, whether the device supports the MeetingRoomSignIn event, desc:eventType: MeetingRoomSignIn-->true
  </isSupportMeetingRoomSignInEvent>
  <isSupportMeetingRoomSignOutEvent>
    <!--ro, opt, bool, whether the device supports the MeetingRoomSignOut event, desc:eventType: MeetingRoomSignOut-->true
  </isSupportMeetingRoomSignOutEvent>
  <isSupportTimeTypeSTD>
    <!--ro, opt, bool, whether the device supports the time type of ISO8601, desc:if "true" is returned, it indicates that all the devices containing
ISO8601 parameters support adding the timeType=STD, such as /ISAPI/ContentMgmt/search?timeType=STD; the time will be implemented according to the standard--
>true
  </isSupportTimeTypeSTD>
  <isSupportScanner>
    <!--ro, opt, bool, whether the device supports scanning events and configuration, desc:if true is returned, it indicates the device supports the
/ISAPI/System/scanner/capabilities?format=json and scannerInfo event-->true
  </isSupportScanner>
  <isSupportRegionTargetNumberCounting>
    <!--ro, opt, bool, whether the device supports the region target counting, desc:related URI:
/ISAPI/Event/channels/<channelID>/RegionTargetNumberCounting/Capabilities?format=json-->true
  </isSupportRegionTargetNumberCounting>
  <depthMapMaintainMode>
    <!--ro, opt, object, the maintenance mode of depth image, desc:it will influence the smart functions when using depth images for tests, such as vertical
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customer traffic, object accumulation detection and so on enabling the maintenance mode will change the audio and video parameters of the channels related
URI: /ISAPI/Intelligent/channels/<channelID>/DepthMapMaintainModeCfg/capabilities?format=json-->
    <ChannelsList>
      <!--ro, opt, array, channel list, subType:object-->
      <channelID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelID>
    </ChannelsList>
  </depthMapMaintainMode>
  <isSupportExportPipCalibrationFile>
    <!--ro, opt, bool, whether it supports exporting visual and thermal image fusion calibration file, desc:related URI:
/ISAPI/System/ExportPipCalibrationFile?format=json-->true
  </isSupportExportPipCalibrationFile>
  <isSupportCourseUploadInfo>
    <!--ro, opt, bool, whether it supports notifying the course uploading completion, desc:eventType: courseUploadInfo-->true
  </isSupportCourseUploadInfo>
  <isSupportPresetSwitchingInformation>
    <!--ro, opt, bool, whether the device supports uploading the preset switching information, desc:eventType: presetSwitchingInformation-->true
  </isSupportPresetSwitchingInformation>
  <isSupportCabinetChargingQuery>
    <!--ro, opt, bool, whether it supports the request of searching for the charging of intelligent cabinet, desc:eventType: cabinetChargingQuery-->true
  </isSupportCabinetChargingQuery>
  <isSupportSearchVoltageAndPowerConsumptionReport>
    <!--ro, opt, bool, whether the device supports searching for voltage and consumption report, desc:related URI:
/ISAPI/System/SearchVoltageAndPowerConsumptionReport/capabilities?format=json-->true
  </isSupportSearchVoltageAndPowerConsumptionReport>
  <isSupportDeviceDrop>
    <!--ro, opt, bool, whether the device supports the deviceDrop event, desc:corresponding event: deviceDrop-->true
  </isSupportDeviceDrop>
  <isSupportDeviceNotMoved>
    <!--ro, opt, bool, whether the device supports the deviceNotMoved event, desc:corresponding event: deviceNotMoved-->true
  </isSupportDeviceNotMoved>
  <isSupportWalkWrongWayOnEscalator>
    <!--ro, opt, bool, whether the device supports the walkWrongWayOnEscalator event, desc:corresponding event: walkWrongWayOnEscalator-->true
  </isSupportWalkWrongWayOnEscalator>
  <isSupportFallOnEscalator>
    <!--ro, opt, bool, whether the device supports the fallOnEscalator event, desc:corresponding event: fallOnEscalator-->true
  </isSupportFallOnEscalator>
  <isSupportCarryLargeLuggageOnEscalator>
    <!--ro, opt, bool, whether the device supports the carryLargeLuggageOnEscalator event, desc:corresponding event: carryLargeLuggageOnEscalator-->true
  </isSupportCarryLargeLuggageOnEscalator>
  <isSupportPushStrollerOnEscalator>
    <!--ro, opt, bool, whether the device supports the pushStrollerOnEscalator event, desc:corresponding event: pushStrollerOnEscalator-->true
  </isSupportPushStrollerOnEscalator>
  <isSupportTeachingSpeechDetection>
    <!--ro, opt, bool, whether it supports teachingSpeechDetection event, desc:corresponding event: teachingSpeechDetection-->true
  </isSupportTeachingSpeechDetection>
  <updatefirmwareType opt="warningScreenFirmware,warningScreenFont,warningScreenVoice,ledReceiverCard">
    <!--ro, opt, string, RS-485 working modes supported by the device, attr:opt{opt, string}, desc:corresponding to <type> in /ISAPI/System/updateFirmware?
type=<type>; "warningScreenFirmware", "warningScreenFont", "warningScreenVoice"; note that <isSupportUpdatefirmware> is supported when this node is
supported-->warningScreenFirmware
  </updatefirmwareType>
  <isSupportPaymentQRCodeQuery>
    <!--ro, opt, bool, whether the device supports the paymentQRCodeQuery event, desc:corresponding event: paymentQRCodeQuery-->true
  </isSupportPaymentQRCodeQuery>
  <isSupportTopUpEvent>
    <!--ro, opt, bool, whether the device supports the topUpEvent event, desc:corresponding event: topUpEvent-->true
  </isSupportTopUpEvent>
  <isSupportTopUpRequest>
    <!--ro, opt, bool, whether the device supports the topUpRequest event, desc:corresponding event: topUpRequest-->true
  </isSupportTopUpRequest>
  <isSupportTopUpStatusQuery>
    <!--ro, opt, bool, whether the device supports the topUpStatusQuery event, desc:corresponding event: topUpStatusQuery-->true
  </isSupportTopUpStatusQuery>
  <isSupportLogOffRefundRequest>
    <!--ro, opt, bool, whether the device supports the logOffRefundRequest event, desc:corresponding event: logOffRefundRequest-->true
  </isSupportLogOffRefundRequest>
  <isSupportLogOffRefundStatusQuery>
    <!--ro, opt, bool, whether the device supports searching for the refund status of logged-off persons, desc:corresponding event: logOffRefundStatusQuery-
->true
  </isSupportLogOffRefundStatusQuery>
  <isSupportAuthenticationTimeoutEvent>
    <!--ro, opt, bool, whether the device supports the logOffRefundStatusQuery event, desc:corresponding event: logOffRefundStatusQuery-->true
  </isSupportAuthenticationTimeoutEvent>
  <isSupportCertificateCapture>
    <!--ro, opt, bool, whether the device supports the certificateCapture events, desc:corresponding event: certificateCapture-->true
  </isSupportCertificateCapture>
  <isSupportDeviceBatchActivation>
    <!--ro, opt, bool, whether the device supports batch activation, desc:POST /ISAPI/System/activate/DeviceBatchActivation?format=json-->true
  </isSupportDeviceBatchActivation>
  <isSupportPlaceEnvironmentMonitoringRule>
    <!--ro, opt, bool, whether the device supports place environment monitoring rule parameters,
desc:/ISAPI/Event/channels/<channelID>/PlaceEnvironmentMonitoringRule?format=json-->true
  </isSupportPlaceEnvironmentMonitoringRule>
  <isSupportEngineResource>
    <!--ro, opt, bool, whether the device supports engine resources, desc:/ISAPI/System/EngineResource?format=json-->true
  </isSupportEngineResource>
  <isSupportPTModulePowerCfg>
    <!--ro, opt, bool, whether the device supports modifying the power supply parameters of PTZ base, desc:PUT /ISAPI/System/ModifyPTModulePowerCfg?
format=json POST /ISAPI/System/GetPTModulePowerCfg?format=json-->true
  </isSupportPTModulePowerCfg>
  <isSupportGetBoxDoorDetectionCfg>
    <!--ro, opt, bool, whether the device supports getting chassis door status detection configuration, desc:POST /ISAPI/System/GetBoxDoorDetectionCfg?
format=json-->true
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format=json-->true
  </isSupportGetBoxDoorDetectionCfg>
  <isSupportModifyBoxDoorDetectionCfg>
    <!--ro, opt, bool, whether the device supports modifying chassis door status configuration, desc:PUT /ISAPI/System/ModifyBoxDoorDetectionCfg?
format=json-->true
  </isSupportModifyBoxDoorDetectionCfg>
  <isSupportSmartBoxStatus>
    <!--ro, opt, bool, whether the device supports intelligent chassis status, desc:/ISAPI/System/SmartBoxStatus?format=json-->true
  </isSupportSmartBoxStatus>
  <isSupportBatteryStatusCtrl>
    <!--ro, opt, bool, whether the device supports battery status event upload control parameters, desc:/ISAPI/System/BatteryStatusCtrl?format=json-->true
  </isSupportBatteryStatusCtrl>
  <isSupportCooler>
    <!--ro, opt, bool, whether the device supports cooling management, desc:/ISAPI/System/Cooler/-->true
  </isSupportCooler>
  <isSupportBoxDoorDetectionCfg>
    <!--ro, opt, bool, whether the device supports configuring chassis door status detection, desc:/ISAPI/System/BoxDoorDetectionCfg?format=json-->true
  </isSupportBoxDoorDetectionCfg>
  <isSupportBoxDoorStatus>
    <!--ro, opt, bool, whether the device supports chassis door status, desc:/ISAPI/System/BoxDoorStatus?format=json-->true
  </isSupportBoxDoorStatus>
  <isSupportLightningProtectionStatus>
    <!--ro, opt, bool, whether the device supports lightning protection status, desc:/ISAPI/System/LightningProtectionStatus?format=json-->true
  </isSupportLightningProtectionStatus>
  <isSupportDeviceVisualLocalizationDetectionEvent>
    <!--ro, opt, bool, whether the device supports the DeviceVisualLocalizationDetectionEvent event, desc:corresponding event:
DeviceVisualLocalizationDetectionEvent-->true
  </isSupportDeviceVisualLocalizationDetectionEvent>
  <isSupportDeviceVisualLocalizationAnalysisEvent>
    <!--ro, opt, bool, whether the device supports the DeviceVisualLocalizationAnalysisEvent event, desc:corresponding event:
DeviceVisualLocalizationAnalysisEvent-->true
  </isSupportDeviceVisualLocalizationAnalysisEvent>
  <isSupportGetPowerPortWorkParamList>
    <!--ro, opt, bool, whether the device supports getting the working parameter list of power supply interface, desc:POST
/ISAPI/System/GetPowerPortWorkParamList?format=json-->true
  </isSupportGetPowerPortWorkParamList>
  <isSupportUsersLoginNotification>
    <!--ro, opt, bool, whether the device supports parameters of notifications on user login, desc:/ISAPI/System/UsersLoginNotification?format=json-->true
  </isSupportUsersLoginNotification>
  <isSupportLiquidLevelDetectorInfo>
    <!--ro, opt, bool, whether the device supports uploading information of liquid level detector, desc:eventType: LiquidLevelDetectorInfo-->true
  </isSupportLiquidLevelDetectorInfo>
  <isSupportWaterPressureDetectorInfo>
    <!--ro, opt, bool, whether the device supports uploading information of water pressure detector, desc:eventType: WaterPressureDetectorInfo-->true
  </isSupportWaterPressureDetectorInfo>
  <isSupportEncryptionFileUpgradeMgr>
    <!--ro, opt, bool, whether the device supports advanced management of encrypted files, desc:GET /ISAPI/System/EncryptionFileUpgradeMgr/capabilities?
format=json-->true
  </isSupportEncryptionFileUpgradeMgr>
  <isSupportInformedConsentTextParam>
    <!--ro, opt, bool, whether the device supports text parameters of consent confirmation document,
desc:/ISAPI/System/InformedConsentMgr/InformedConsentTextParam?format=json-->true
  </isSupportInformedConsentTextParam>
  <isSupportSendConsolePassword>
    <!--ro, opt, bool, whether the device supports applying control panel password, desc:POST /ISAPI/System/SendConsolePassword?format=json-->true
  </isSupportSendConsolePassword>
  <isSupportAutoLaserRanging>
    <!--ro, opt, bool, whether to support auto laser ranging parameter configuration, desc:/ISAPI/System/AutoLaserRanging?format=json-->true
  </isSupportAutoLaserRanging>
  <isSupportDelayedReboot>
    <!--ro, opt, bool, whether the device supports delayed reboot after power disconnection, desc:POST /ISAPI/System/DelayedReboot?format=json-->true
  </isSupportDelayedReboot>
  <isSupportFalseAlarmMaterialLibMgr>
    <!--ro, opt, bool, whether the device supports false alarm library management, desc:GET /ISAPI/Intelligent/falseAlarmMaterialLibMgr/capabilities?
format=json-->true
  </isSupportFalseAlarmMaterialLibMgr>
  <isSupportGetServerRandom>
    <!--ro, opt, bool, whether the device supports getting random number of Chinese cryptographic server, desc:POST
/ISAPI/Security/ProtocolCertificate/GetServerRandom?format=json-->true
  </isSupportGetServerRandom>
  <isSupportCustomPreviewSplitCfg>
    <!--ro, opt, bool, whether the device supports custom parameters of divided window in live view, desc:GET/PUT /ISAPI/System/Video/CustomPreviewSplitCfg?
format=json-->true
  </isSupportCustomPreviewSplitCfg>
  <isSupportSendTelnetTestCmd>
    <!--ro, opt, bool, whether the device supports applying telnet testing command, desc:POST /ISAPI/System/Network/NetworkAdvancedCfg/SendTelnetTestCmd?
format=json-->true
  </isSupportSendTelnetTestCmd>
  <isSupportTelnetTestCmdStatus>
    <!--ro, opt, bool, whether the device supports the execution status of telnet testing command, desc:GET
/ISAPI/System/Network/NetworkAdvancedCfg/TelnetTestCmdStatus?format=json-->true
  </isSupportTelnetTestCmdStatus>
  <isSupportImportUSBKeyAuthInfo>
    <!--ro, opt, bool, whether the device supports Chinese cryptographic USBKey authentication information, desc:POST
/ISAPI/Security/ProtocolCertificate/ImportUSBKeyAuthInfo?format=json-->true
  </isSupportImportUSBKeyAuthInfo>
  <isSupportSendSupervisionRecordCollection>
    <!--ro, opt, bool, whether the device supports sending completing status of supervision recording replenishment, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SendSupervisionRecordCollection?format=json-->true
  </isSupportSendSupervisionRecordCollection>
  <isSupportSearchPersonnelOperationSupervisionTask>
    <!--ro, opt, bool, whether it supports searching for operation supervision tasks, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SearchPersonnelOperationSupervisionTask?format=json-->true
  </isSupportSearchPersonnelOperationSupervisionTask>
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<isSupportDeletePersonnelOperationSupervisionTask>
<!--ro, opt, bool, whether it supports deleting operation supervision tasks, desc:POST
/ISAPI/Intelligent/channels/<channelID>/DeletePersonnelOperationSupervisionTask?format=json-->true
</isSupportDeletePersonnelOperationSupervisionTask>
<isSupportAddPersonnelOperationSupervisionTask>
<!--ro, opt, bool, whether it supports adding operation supervision tasks, desc:POST
/ISAPI/Intelligent/channels/<channelID>/AddPersonnelOperationSupervisionTask?format=json-->true
</isSupportAddPersonnelOperationSupervisionTask>
<isSupportUserLoginStatusChangeEvent>
<!--ro, opt, bool, whether the device supports the UserLoginStatusChangeEvent event, desc:eventType: UserLoginStatusChangeEvent-->true
</isSupportUserLoginStatusChangeEvent>
<isSupportObstacleAvoidance>
<!--ro, opt, bool, whether it supports obstacle avoidance management, desc:GET/PUT /ISAPI/System/ObstacleAvoidance/-->true
</isSupportObstacleAvoidance>
<isSupportManualSnapFishEyeWallPanoramaView>
<!--ro, opt, bool, whether the device supports manually capturing expanded panorama images by wall-mounted fisheye, desc:GET
/ISAPI/Streaming/channels/<channelID>/ManualSnapFishEyeWallPanoramaView?format=json-->true
</isSupportManualSnapFishEyeWallPanoramaView>
<isSupportPreviewSplitSwitchCfg>
<!--ro, opt, bool, whether the device supports custom parameters of divided window switch, desc:GET/PUT /ISAPI/System/Video/PreviewSplitSwitchCfg?
format=json-->true
</isSupportPreviewSplitSwitchCfg>
<isSupportCountingTargetParam>
<!--ro, opt, bool, whether the device supports target parameters of people counting, desc:GET/PUT
/ISAPI/System/Video/inputs/channels/<channelID>/counting/CountingTargetParam?format=json-->true
</isSupportCountingTargetParam>
<isSupportDebugLogModuleType opt="playService,communicationService,attendanceService,faceService,LCD">
<!--ro, opt, bool, whether the device supports exporting debug log by module, attr:opt{req, string}-->true
</isSupportDebugLogModuleType>
<isSupportLoginPasswordCfg>
<!--ro, opt, bool, whether the device supports login password parameters, desc:GET/PUT /ISAPI/Security/users/LoginPasswordCfg?format=json-->true
</isSupportLoginPasswordCfg>
<isSupportUserDataFile>
<!--ro, opt, bool, whether the device supports importing and exporting the user data file, desc:POST /ISAPI/Security/users/UserDataFile/capabilities?
format=json-->true
</isSupportUserDataFile>
<isSupportNtripAccessCfg>
<!--ro, opt, bool, whether the device supports NTRIP protocol access parameters, desc:GET/PUT /ISAPI/System/Network/Ntrip/NtripAccessCfg?format=json--
>true
</isSupportNtripAccessCfg>
<isSupportNetworkWatchdogParams>
<!--ro, opt, bool, whether the device supports parameters of watchdog for network performance, desc:GET/PUT /ISAPI/System/NetworkWatchdogParams?
format=json-->true
</isSupportNetworkWatchdogParams>
<isSupportAudioRecordPlan>
<!--ro, opt, bool, whether it supports recording schedule, desc:GET/PUT /ISAPI/System/RecordAudio/channels/<audioID>/AudioRecordPlan?format=json-->true
</isSupportAudioRecordPlan>
<isSupportStartVoiceMonitor>
<!--ro, opt, bool, whether it supports enabling audio listening, desc:POST
/ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/StartVoiceMonitor?format=json-->true
</isSupportStartVoiceMonitor>
<isSupportStopVoiceMonitor>
<!--ro, opt, bool, whether it disabling audio listening-->true
</isSupportStopVoiceMonitor>
<voiceMonitorChannelCap>
<!--ro, opt, object-->
<audioIDList>
<!--ro, opt, array, camera list, subType:object-->
<audioID>
<!--ro, req, int, No. of channels supported by the device-->1
</audioID>
</audioIDList>
</voiceMonitorChannelCap>
<isSupportWebPTZCfg>
<!--ro, opt, bool, whether it supports displaying the web page for PTZ configuration, desc:this node is to determine whether to display the web page for
PTZ configuration and is irrelevant to whether PTZ is supported; default process logic: when this node does not exist, whether to display the web page for
PTZ configuration is subject to the device type-->true
</isSupportWebPTZCfg>
<isSupportGaugeReadingEvent>
<!--ro, opt, bool, whether the device supports the GaugeReadingEvent event, desc:eventType: GaugeReadingEvent-->true
</isSupportGaugeReadingEvent>
<isSupportVNCServerCfg>
<!--ro, opt, bool, whether it supports VNC service configuration, desc:GET/PUT /ISAPI/System/VNCServerCfg?format=json-->true
</isSupportVNCServerCfg>
<isSupportResources>
<!--ro, opt, bool, whether it supports resources, desc:GET /ISAPI/System/DynamicResource/Resources?format=json, Resources-->true
</isSupportResources>
<isSupportDeviceGroupMgr>
<!--ro, opt, bool, whether it supports device application resources group management, desc:GET /ISAPI/System/DeviceGroupMgr/capabilities?format=json--
>true
</isSupportDeviceGroupMgr>
<isSupportAsyncResetSubDevicePassword>
<!--ro, opt, bool, whether the device supports resetting the password of sub-device asynchronously,
desc:/ISAPI/Security/resetPassword/AsyncResetSubDevicePassword?format=json-->true
</isSupportAsyncResetSubDevicePassword>
<isSupportAsyncResetSubDevicePasswordStatusList>
<!--ro, opt, bool, whether the device supports the status list of asynchronously resetting sub-devices' passwords,
desc:/ISAPI/Security/resetPassword/AsyncResetSubDevicePasswordStatusList?format=json-->true
</isSupportAsyncResetSubDevicePasswordStatusList>
<isSupportEDRAccessList>
<!--ro, opt, bool, whether the device supports accessing address parameters of EDR service, desc:GET/PUT /ISAPI/System/Network/EDRAccessList?
format=json-->true
</isSupportEDRAccessList>
<isSupportBatteryCanInfoEvent>
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<isSupportBatteryCarInfoEvent>
<!--ro, opt, bool, whether to support the BatteryCarInfoEvent event, desc:eventType: BatteryCarInfoEvent-->true
</isSupportBatteryCarInfoEvent>
<isSupportChildmanage>
<!--ro, opt, bool, whether the device supports gateway sub-device topology management, desc:GET /ISAPI/IoTGateway/Childmanage/capabilities?format=json-->true
</isSupportChildmanage>
<isSupportSearchResources>
<!--ro, opt, bool, whether the device supports searching the resource list, desc:POST /ISAPI/System/DynamicResource/SearchResources?format=json-->true
</isSupportSearchResources>
<isSupportResetPasswordPhoneNo>
<!--ro, opt, bool, whether the device supports the phone No. for resetting password, desc:GET/PUT /ISAPI/Security/resetPassword/ResetPasswordPhoneNo?format=json-->true
</isSupportResetPasswordPhoneNo>
<isSupportGetPhoneNoQRCode>
<!--ro, opt, bool, whether the device supports getting the QR code to be scanned by phone for resetting password, desc:GET /ISAPI/Security/resetPassword/GetPhoneNoQRCode?format=json-->true
</isSupportGetPhoneNoQRCode>
<isSupportResetPasswordByPhoneNoSecurityCode>
<!--ro, opt, bool, whether it supports resetting device password with phone number verification, desc:PUT /ISAPI/Security/resetPassword/ResetPasswordByPhoneNoSecurityCode?format=json-->true
</isSupportResetPasswordByPhoneNoSecurityCode>
<isSupportGaugeReadingOverlapParams>
<!--ro, opt, bool, whether the device supports water level gauge detection information overlay parameters, desc:GET/PUT /ISAPI/Event/gaugeReading/channels/<channelID>/GaugeReadingOverlapParams?format=json-->true
</isSupportGaugeReadingOverlapParams>
<isSupportGaugeReadingRainfallEvent>
<!--ro, opt, bool, whether the device supports the GaugeReadingRainfallEvent event, desc:eventType: GaugeReadingRainfallEvent-->true
</isSupportGaugeReadingRainfallEvent>
<isSupportSearchTrafficLightMalfunctionResult>
<!--ro, opt, bool, whether the device supports searching for traffic light fault detection result, desc:POST /ISAPI/Event/TrafficLightMalfunctionDetection/SearchTrafficLightMalfunctionResult?format=json-->true
</isSupportSearchTrafficLightMalfunctionResult>
<isSupportCheckPhoneNoSecurityCode>
<!--ro, opt, bool, whether the device supports phone security code verification, desc:PUT /ISAPI/Security/resetPassword/CheckPhoneNoSecurityCode?format=json-->true
</isSupportCheckPhoneNoSecurityCode>
<isSupportUserValidExpired>
<!--ro, opt, bool, whether the device supports the UserValidExpired event, desc:eventType: UserValidExpired-->true
</isSupportUserValidExpired>
<isSupportSearchWaterLevelData>
<!--ro, opt, bool, whether the device supports searching for water level detection data, desc:POST /ISAPI/Event/waterQualityDetection/SearchWaterLevelData?format=json-->true
</isSupportSearchWaterLevelData>
<isSupportSearchRecommendPerspective>
<!--ro, opt, bool, whether the device supports searching for recommended FoV, desc:POST /ISAPI/Streaming/channels/channels/<channelID>/SearchRecommendPerspective?format=json-->true
</isSupportSearchRecommendPerspective>
<isSupportGetLegendaryPerspectiveInfo>
<!--ro, opt, bool, whether the device supports getting the information of camera FoV, desc:GET /ISAPI/Streaming/channels/channels/<channelID>/GetLegendaryPerspectiveInfo?format=json-->true
</isSupportGetLegendaryPerspectiveInfo>
<configurationDataType opt="OSD,MODE,IMAGE,IOOUT,OTHERS">
<!--ro, opt, string, "OSD"-import video and image OSD settings, "MODE"-import application mode, "IMAGE"-import image parameters, "IOONT"-import supplement Light Parameters, "OTHERS", attr:opt{req, string}-->test
</configurationDataType>
<isSupportRestoreVideoAdjustmentParam>
<!--ro, opt, bool, whether the device supports restoring video image setting parameters, desc:GET/PUT /ISAPI/System/Video/channels/<channelID>/RestoreVideoAdjustmentParam?format=json-->true
</isSupportRestoreVideoAdjustmentParam>
<isSupportWeatherStationInfo>
<!--ro, opt, bool, whether the device supports weather station information, desc:GET/PUT /ISAPI/Traffic/channels/<channelID>/WeatherStationInfo?format=json-->true
</isSupportWeatherStationInfo>
<isSupportRadarPerimeterAlgCap>
<!--ro, opt, bool, whether the device supports the capability of radar perimeter algorithm parameters, desc:GET /ISAPI/Smart/RadarPerimeterAlg/capabilities?format=json-->true
</isSupportRadarPerimeterAlgCap>
<isSupportBodyCameraOneKeyAlarmEvent>
<!--ro, opt, bool, whether the device supports the BodyCameraOneKeyAlarmEvent event, desc:eventType: BodyCameraOneKeyAlarmEvent; this node is not returned if not supported-->true
</isSupportBodyCameraOneKeyAlarmEvent>
<isSupportGetOnlineTrafficLight>
<!--ro, opt, bool, whether the device supports getting online traffic light information, desc:GET /ISAPI/Event/TrafficLightMalfunctionDetection/GetOnlineTrafficLight?format=json-->true
</isSupportGetOnlineTrafficLight>
<isSupportTrafficLightMalfunctionDefaultParams>
<!--ro, opt, bool, whether the device supports default parameters of traffic light fault detection rules, desc:GET/PUT /ISAPI/Event/TrafficLightMalfunctionDetection/TrafficLightMalfunctionDefaultParams?format=json-->true
</isSupportTrafficLightMalfunctionDefaultParams>
<isSupportFallenObjectsOnTracksDetection>
<!--ro, opt, bool, whether the device supports detection of fallen objects on tracks, desc:GET /ISAPI/Intelligent/channels/<channelID>/FallenObjectsOnTracksDetection/capabilities?format=json-->true
</isSupportFallenObjectsOnTracksDetection>
<isSupportMonitorAuthorization>
<!--ro, opt, bool, whether the device supports device service monitoring authorization information, desc:GET /ISAPI/Security/DeviceServiceMonitor/MonitorAuthorization?format=json-->true
</isSupportMonitorAuthorization>
<isSupportBodyCameraTakeEvent>
<!--ro, opt, bool, whether the device supports the BodyCameraTakeEvent event, desc:eventType: BodyCameraTakeEvent; this node is not returned if not supported-->true
</isSupportBodyCameraTakeEvent>
<isSupportCollaborationModeCfg>
<!--ro, opt, bool, whether the device supports configuring multi-lens coordination mode, desc:GET/PUT /ISAPI/Intelligent/CollaborationModeCfg?

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format=json-->true
  </isSupportCollaborationModeCfg>
  <isSupportAudioChannelVCAResource>
    <!--ro, opt, bool, whether the device supports switching intelligent resources running on audio channels, desc:GET/PUT
/ISAPI/System/VCAResource/channels/<audioID>/AudioChannelVCAResource?format=json-->true
  </isSupportAudioChannelVCAResource>
  <isSupportGasTankMonitoringEvent>
    <!--ro, opt, bool, whether the device supports the GasTankMonitoringEvent event, desc:eventType: GasTankMonitoringEvent-->true
  </isSupportGasTankMonitoringEvent>
  <isSupportChannelClearEvent>
    <!--ro, opt, bool, whether the device supports the ChannelClearEvent event, desc:eventType: ChannelClearEvent-->true
  </isSupportChannelClearEvent>
  <isSupportBackDuty>
    <!--ro, opt, bool, whether the device supports the backDuty event, desc:eventType: backDuty-->true
  </isSupportBackDuty>
  <isSupportSleepDuty>
    <!--ro, opt, bool, whether the device supports the sleepDuty event, desc:eventType: sleepDuty-->true
  </isSupportSleepDuty>
  <isSupportSearchChildDevResourcesStatus>
    <!--ro, opt, bool, whether the device supports searching for sub-device resource status, desc:POST
/ISAPI/System/DynamicResource/SearchChildDevResourcesStatus?format=json-->true
  </isSupportSearchChildDevResourcesStatus>
  <isSupportDeleteResources>
    <!--ro, opt, bool, whether the device supports deleting resources, desc:PUT /ISAPI/System/DynamicResource/DeleteResources?format=json-->true
  </isSupportDeleteResources>
  <isSupportMenuFunctionList>
    <!--ro, opt, bool, whether the device supports the menu list, desc:GET /ISAPI/Thermal/MenuFunctionList?format=json-->true
  </isSupportMenuFunctionList>
  <isSupportKeyFunctionList>
    <!--ro, opt, bool, whether the device supports the button list, desc:GET /ISAPI/Thermal/KeyFunctionList?format=json-->true
  </isSupportKeyFunctionList>
  <isSupportKeyValueList>
    <!--ro, opt, bool, whether the device supports button key list, desc:GET /ISAPI/Thermal/KeyValueList?format=json-->true
  </isSupportKeyValueList>
  <isSupportGetPseudoColorBackgroundPic>
    <!--ro, opt, bool, whether the device supports 3-channel palette background picture, desc:GET
/ISAPI/Streaming/channels/<channelID>/GetPseudoColorBackgroundPic?format=json-->true
  </isSupportGetPseudoColorBackgroundPic>
  <isSupportElevatorRunningDetection>
    <!--ro, opt, bool, whether the device supports elevator running status detection, desc:GET
/ISAPI/Intelligent/channels/<channelID>/ElevatorRunningDetection/capabilities?format=json-->true
  </isSupportElevatorRunningDetection>
  <isSupportSearchPersonQueueRealTimeResult>
    <!--ro, opt, bool, whether the device supports searching for history records of person waiting detection event, desc:POST
/ISAPI/Intelligent/personQueueDetection/SearchPersonQueueRealTimeResult?format=json-->true
  </isSupportSearchPersonQueueRealTimeResult>
  <isSupportSearchLeavePositionEventResult>
    <!--ro, opt, bool, whether the device supports searching for history records of absence detection event, desc:POST
/ISAPI/Intelligent/HehaviorAnalysis/SearchLeavePositionEventResult?format=json-->true
  </isSupportSearchLeavePositionEventResult>
  <isSupportImportElectronicBadgeBackgroundImage>
    <!--ro, opt, bool, whether the device supports importing the background picture of nameplate, desc:POST
/ISAPI/Traffic/ImportElectronicBadgeBackgroundImage?format=json-->true
  </isSupportImportElectronicBadgeBackgroundImage>
  <isSupportVoiceMonitoringEvent>
    <!--ro, opt, bool, whether the device supports uploading the VoiceMonitoringEvent event, desc:eventType: VoiceMonitoringEvent-->true
  </isSupportVoiceMonitoringEvent>
  <isSupportPseudoColorBackgroundParam>
    <!--ro, opt, bool, whether the device supports 3-channel palette background parameters, desc:GET/PUT
/ISAPI/Streaming/channels/<channelID>/PseudoColorBackgroundParam?format=json-->true
  </isSupportPseudoColorBackgroundParam>
  <isSupportGotoPerimeterShieldMaskScene>
    <!--ro, opt, bool, whether the device supports jumping to shielded scenes of specified radar perimeter, desc:POST
/ISAPI/Smart/RadarPerimeterAlg/GotoPerimeterShieldMaskScene?format=json-->true
  </isSupportGotoPerimeterShieldMaskScene>
  <isSupportSearchPersonQueueTimingStatistics>
    <!--ro, opt, bool, whether the device supports the SearchPersonQueueTimingStatistics event, desc:POST
/ISAPI/Intelligent/personQueueDetection/SearchPersonQueueTimingStatistics?format=json-->true
  </isSupportSearchPersonQueueTimingStatistics>
  <isSupportLocalUserChange>
    <!--ro, opt, bool, whether the device supports the LocalUserChange event, desc:eventType: LocalUserChange-->true
  </isSupportLocalUserChange>
  <isSupportSendClientTask>
    <!--ro, opt, bool, whether the device supports applying tasks configured on the client, desc:POST /ISAPI/System/RailwayPatrol/SendClientTask?
format=json-->true
  </isSupportSendClientTask>
  <isSupportGetCurrentClientTaskStatus>
    <!--ro, opt, bool, whether the device supports getting the current execution progress of the client task, desc:GET
/ISAPI/System/RailwayPatrol/GetCurrentClientTaskStatus?format=json-->true
  </isSupportGetCurrentClientTaskStatus>
  <isSupportTestEDRServerConnect>
    <!--ro, opt, bool, whether the device supports testing the connection status of the EDR platform, desc:POST /ISAPI/System/Network/TestEDRServerConnect?
format=json-->true
  </isSupportTestEDRServerConnect>
  <isSupportRealTimeStreamInfo>
    <!--ro, opt, bool, whether the device supports getting the real-time stream information, desc:GET /ISAPI/Streaming/RealTimeStreamInfo?format=json-->true
  </isSupportRealTimeStreamInfo>
  <isSupportIrcutFilterStatusChange>
    <!--ro, opt, bool, whether the device supports the IrcutFilterStatusChange event, desc:eventType: IrcutFilterStatusChange-->true
  </isSupportIrcutFilterStatusChange>
  <isSupportSendRoadMaintPictureAnalysisTask>
    <!--ro, opt, bool, whether the device supports applying tasks of road maintenance picture analysis, desc:POST
/ISAPI/Intelligent/SendRoadMaintPictureAnalysisTask?format=json-->true
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  </isSupportSendRoadMaintPictureAnalysisTask>
  <isSupportFDRemoveDuplicateParams>
    <!--ro, opt, bool, whether the device supports deduplication parameters of custom face picture library, desc:GET/PUT
/ISAPI/System/Video/inputs/counting/FDRemoveDuplicateParams?format=json-->true
  </isSupportFDRemoveDuplicateParams>
  <isSupportRadarMixedTargetSensitivityParams>
    <!--ro, opt, bool, whether the device supports sensitivity parameters of multi-target-type detection via radar, desc:GET/PUT
/ISAPI/Smart/RadarMixedTargetAlg/RadarMixedTargetSensitivityParams?format=json-->true
  </isSupportRadarMixedTargetSensitivityParams>
  <isSupportMixedTargetRadarSensitivityParams>
    <!--ro, opt, bool, whether the device supports sensitivity parameters of multi-target-type detection via video, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/MixedTargetRadarSensitivityParams?format=json-->true
  </isSupportMixedTargetRadarSensitivityParams>
  <isSupportMixedTargetRadarDebugParams>
    <!--ro, opt, bool, whether the device supports algorithm debugging parameters of multi-target-type detection via video, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/MixedTargetRadarDebugParams?format=json-->true
  </isSupportMixedTargetRadarDebugParams>
  <isSupportHttpPreview>
    <!--ro, opt, bool, whether the device supports HTTP live view,
desc:related URI: GET /ISAPI/Streaming/channels/<trackStreamID>/httppreview;
there are 2 steps as follows:
1. check if this node is returned by device and if the returned value is true. If yes, the HTTP live view function is supported by device.
2. if this node is not returned by device, you should go through the stream ID in the API (GET /ISAPI/Streaming/channels/<trackStreamID>/httppreview) of all
channels. If notSupport is returned for all channels, HTTP live view is not supported by device; if at least the success code is returned for at least one
channel, HTTP live view is supported by device.
Note: whether the stream ID (trackStreamID) is supported is determined by the response parameters StreamingChannelList, StreamingChannel, and id from GET
/ISAPI/Streaming/channels-->true
  </isSupportHttpPreview>
  <isSupportGetRecordThumbnail>
    <!--ro, opt, bool, whether the device supports getting the video file thumbnail, desc:POST /ISAPI/ContentMgmt/search/GetRecordThumbnail?format=json--
>true
  </isSupportGetRecordThumbnail>
  <isSupportProxyAIOP>
    <!--ro, opt, bool, whether the device supports proxy access of AIOP (AIOP devices connected with network cameras via NVR transparent transmission),
desc:when this node is supported, full AIOP functions (device AIOP capability, model package management, task rule management, detection data, etc.) can be
accessed via proxy transmission of ISAPI sub-device-->true
  </isSupportProxyAIOP>
  <isSupportStaticObjectPositionOffsetComparison>
    <!--ro, opt, bool, whether the device supports position deviation comparison of static objects, desc:GET
/ISAPI/Intelligent/channels/<channelID>/StaticObjectPositionOffsetComparison/capabilities?format=json-->true
  </isSupportStaticObjectPositionOffsetComparison>
  <isSupportClearFDRemoveDuplicateFDLib>
    <!--ro, opt, bool, whether the device supports manually clearing the custom face picture library on device, desc:POST
/ISAPI/System/Video/inputs/counting/ClearFDRemoveDuplicateFDLib?format=json-->true
  </isSupportClearFDRemoveDuplicateFDLib>
  <isSupportDevPowerStatusReportCtrl>
    <!--ro, opt, bool, whether the device supports control parameters of device battery status upload, desc:GET/PUT /ISAPI/System/DevPowerStatusReportCtrl?
format=json-->true
  </isSupportDevPowerStatusReportCtrl>
  <isSupportIndicatorLight>
    <!--ro, opt, bool, whether the device supports indicator light configuration, desc:GET /ISAPI/System/IndicatorLight/capabilities?format=json-->true
  </isSupportIndicatorLight>
  <isSupportEnvironmentTemperatureMonitorEvent>
    <!--ro, opt, bool, whether the device supports the EnvironmentTemperatureMonitorEvent event, desc:eventType: EnvironmentTemperatureMonitorEvent-->true
  </isSupportEnvironmentTemperatureMonitorEvent>
  <isSupportPhysicalTraining>
    <!--ro, opt, bool, whether the device supports physical training, desc:GET /ISAPI/Intelligent/channels/<channelID>/PhysicalTraining/capabilities?
format=json-->true
  </isSupportPhysicalTraining>
  <isSupportPullUp>
    <!--ro, opt, bool, whether the device supports the PullUp event, desc:eventType: PullUp-->true
  </isSupportPullUp>
  <isSupportSitUpWithHandsBehindHead>
    <!--ro, opt, bool, whether the device supports the SitUpWithHandsBehindHead event, desc:eventType: SitUpWithHandsBehindHead-->true
  </isSupportSitUpWithHandsBehindHead>
  <isSupportSitUpWithArmsOnChest>
    <!--ro, opt, bool, whether the device supports the SitUpWithArmsOnChest event, desc:eventType: SitUpWithArmsOnChest-->true
  </isSupportSitUpWithArmsOnChest>
  <isSupportPushUp>
    <!--ro, opt, bool, whether the device supports the PushUp event, desc:eventType: PushUp-->true
  </isSupportPushUp>
  <isSupportPullUpHold>
    <!--ro, opt, bool, whether the device supports the PullUpHold event, desc:eventType: PullUpHold-->true
  </isSupportPullUpHold>
  <isSupportVerificationPlatformParams>
    <!--ro, opt, bool, whether the device supports accessing parameters for the verification and recording platform, desc:GET/PUT
/ISAPI/System/Network/VerificationPlatformAccess/VerificationPlatformParams?format=json-->true
  </isSupportVerificationPlatformParams>
  <isSupportVehicleMaintenanceDetection>
    <!--ro, opt, bool, whether the device supports vehicle maintenance detection, desc:GET /ISAPI/Intelligent/VehicleMaintenanceDetection/capabilities?
format=json; detects the status of vehicles in the maintenance shop and triggers the alarm when the vehicle is lifted (when detecting chassis) or lowered,
or the hood is opened-->true
  </isSupportVehicleMaintenanceDetection>
  <isSupportPhysicalTrainingStatus>
    <!--ro, opt, bool, whether the device supports the PhysicalTrainingStatus event, desc:eventType: PhysicalTrainingStatus-->true
  </isSupportPhysicalTrainingStatus>
  <isSupportDeviceAbnormalDebug>
    <!--ro, opt, bool, whether the device supports device operation exception debugging, desc:GET /ISAPI/System/DeviceAbnormalDebug/capabilities?
format=json; during device operation exception, the device operation debugging data (debugging log, dump file, etc.) will be collected by technical support-
->true
  </isSupportDeviceAbnormalDebug>
  <isSupportSetWaterLevelResetOfWaterTank>
    <!--ro, opt, bool, whether the device supports manually filling up the water tank and uploading the filled tank information for calculating remaining
water via lens wiper usage, desc:PUT /ISAPI/System/Hardware/SetWaterLevelResetOfWaterTank?format=json-->true
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  </isSupportSetWaterLevelResetOfWaterTank>
  <isSupportGetRemoteAlbumPicture>
    <!--ro, opt, bool, whether the device supports getting pictures from the remote album, desc:POST /ISAPI/ContentMgmt/GetRemoteAlbumPicture?format=json-->true
  </isSupportGetRemoteAlbumPicture>
  <isSupportUltravioletImagingFusionParams>
    <!--ro, opt, bool, whether the device supports UV imaging fusion parameters, desc:GET/PUT /ISAPI/System/Video/channels/<channelID>/UltravioletImagingFusionParams?format=json-->true
  </isSupportUltravioletImagingFusionParams>
  <isSupportAudioAnalysisCfg>
    <!--ro, opt, bool, whether it supports audio analysis parameters, desc:GET/PUT /ISAPI/Intelligent/SensitiveInfoAnalysis/AudioIn/<AudioInID>/AudioAnalysisCfg?format=json-->true
  </isSupportAudioAnalysisCfg>
  <isSupportRuningLogPlan>
    <!--ro, opt, bool, whether the device supports device operation log schedule, desc:GET/PUT /ISAPI/ContentMgmt/RuningLogPlan?format=json-->true
  </isSupportRuningLogPlan>
  <isSupportSearchRuningLogData>
    <!--ro, opt, bool, whether the device supports searching for device operation log data, desc:POST /ISAPI/ContentMgmt/SearchRuningLogData?format=json-->true
  </isSupportSearchRuningLogData>
  <isSupportExportRuningLogData>
    <!--ro, opt, bool, whether the device supports exporting device operation log data, desc:POST /ISAPI/ContentMgmt/ExportRuningLogData?format=json-->true
  </isSupportExportRuningLogData>
  <isSupportCertificateCaptureEvent>
    <!--ro, opt, bool, whether the device supports the CertificateCaptureEvent event, desc:corresponding event: CertificateCaptureEvent-->true
  </isSupportCertificateCaptureEvent>
  <isSupportUpgradeKeywordModel>
    <!--ro, opt, bool, whether the device supports upgrading the sensitive information keyword model, desc:PUT /ISAPI/Intelligent/SensitiveInfoAnalysis/UpgradeKeywordModel?format=json-->true
  </isSupportUpgradeKeywordModel>
  <isSupportGetKeywordModelStatus>
    <!--ro, opt, bool, whether the device supports getting algorithm running status of the sensitive information keyword model, desc:GET /ISAPI/Intelligent/SensitiveInfoAnalysis/GetKeywordModelStatus?format=json-->true
  </isSupportGetKeywordModelStatus>
  <isSupportCloseLensPrivacyCover>
    <!--ro, opt, bool, whether the device supports manually closing the lens privacy cap, desc:PUT /ISAPI/System/channels/<channelID>/CloseLensPrivacyCover?format=json-->true
  </isSupportCloseLensPrivacyCover>
  <isSupportOpenLensPrivacyCover>
    <!--ro, opt, bool, whether the device supports manually opening the lens privacy cap, desc:PUT /ISAPI/System/channels/<channelID>/OpenLensPrivacyCover?format=json-->true
  </isSupportOpenLensPrivacyCover>
  <isSupportLensPrivacyCoverModeParams>
    <!--ro, opt, bool, whether the device supports working mode parameters of the lens privacy cap, desc:GET/PUT /ISAPI/System/channels/<channelID>/LensPrivacyCoverModeParams?format=json-->true
  </isSupportLensPrivacyCoverModeParams>
  <isSupportOTAPService>
    <!--ro, opt, bool, whether the device supports OTAP service configuration, desc:/ISAPI/System/Network/OTAPService/capabilities?format=json-->true
  </isSupportOTAPService>
  <isSupportGetPlaybackStreamInfo>
    <!--ro, opt, bool, whether the device supports getting video playback stream information, desc:GET /ISAPI/Streaming/GetPlaybackStreamInfo?format=json-->true
  </isSupportGetPlaybackStreamInfo>
  <isSupportNBIoTPlateformAccessStatus>
    <!--ro, opt, bool, whether the device supports the NB-IoT platform access status, desc:GET /ISAPI/System/Network/NBIoTPlateformAccessStatus?format=json-->true
  </isSupportNBIoTPlateformAccessStatus>
  <isSupportScannedListEvent>
    <!--ro, opt, bool, whether the device supports Bluetooth scanning list event, desc:eventType: ScannedListEvent-->true
  </isSupportScannedListEvent>
  <isSupportDownloadAudioRecordByTime>
    <!--ro, opt, bool, whether the device supports downloading audio files by time, desc:POST /ISAPI/System/RecordAudio/DownloadAudioRecordByTime?format=json-->true
  </isSupportDownloadAudioRecordByTime>
  <isSupportBackUpConfigurationFile>
    <!--ro, opt, bool, whether the device supports local backup of the configuration file, desc:PUT /ISAPI/System/BackUpConfigurationFile?format=json-->true
  </isSupportBackUpConfigurationFile>
  <isSupportRestoreConfigurationFile>
    <!--ro, opt, bool, whether the device supports restoring the device local configuration file, desc:PUT /ISAPI/System/RestoreConfigurationFile?format=json-->true
  </isSupportRestoreConfigurationFile>
  <isSupportGetCfgFileBackUpInfo>
    <!--ro, opt, bool, whether the device supports getting the local configuration file backup information, desc:GET /ISAPI/System/GetCfgFileBackUpInfo?format=json-->true
  </isSupportGetCfgFileBackUpInfo>
  <isSupportSearchVideoCodeMealParams>
    <!--ro, opt, bool, whether the device supports searching for video encoding plan parameters, desc:POST /ISAPI/Streaming/channels/<channelID>/SearchVideoCodeMealParams?format=json-->true
  </isSupportSearchVideoCodeMealParams>
  <isSupportVideoCodeMealParams>
    <!--ro, opt, bool, whether the device supports video encoding plan parameters, desc:GET/PUT /ISAPI/Streaming/channels/<channelID>/VideoCodeMealParams?format=json-->true
  </isSupportVideoCodeMealParams>
  <isSupportFlameCombustionStatusRuleParams>
    <!--ro, opt, bool, whether the device supports flame burning status detection rule parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusRuleParams?format=json-->true
  </isSupportFlameCombustionStatusRuleParams>
  <isSupportFlameCombustionStatusOverlapParams>
    <!--ro, opt, bool, whether the device supports flame burning status detection information overlay parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusOverlapParams?format=json-->true
  </isSupportFlameCombustionStatusOverlapParams>
  <isSupportAcousticLeakDetection>
    <!--ro, opt, bool, whether the device supports acoustic leak detection capability, desc:related URL: /ISAPI/System/AcousticLeakDetection/capabilities?
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format=json-->true
    </isSupportAcousticLeakDetection>
    <isSupportFlameCombustionStatusSceneList>
      <!--ro, opt, bool, whether the device supports flame burning status detection scene list, desc:GET
/ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusSceneList?format=json-->true
    </isSupportFlameCombustionStatusSceneList>
    <isSupportGasLeakageDetection>
      <!--ro, opt, bool, whether the device supports gas leak event, desc:eventType: GasLeakageDetection-->true
    </isSupportGasLeakageDetection>
    <isSupportDischargeDetection>
      <!--ro, opt, bool, whether the device supports discharging event, desc:eventType: DischargeDetection-->true
    </isSupportDischargeDetection>
    <isSupportSearchBaseMapIDList>
      <!--ro, opt, bool, whether the device supports searching for base map ID list, desc:POST /ISAPI/ContentMgmt/SearchBaseMapIDList?format=json-->true
    </isSupportSearchBaseMapIDList>
    <isSupportRestoreDefaultParam>
      <!--ro, opt, bool, whether the device supports restoring default parameters, desc:PUT /ISAPI/Streaming/RestoreDefaultParam?format=json-->true
    </isSupportRestoreDefaultParam>
    <isSupportSearchEventTargetInfo>
      <!--ro, opt, bool, whether the device supports searching for event detection target information in FoV, desc:GET
/ISAPI/System/Video/inputs/channels/<channelID>/SearchEventTargetInfo?format=json-->true
    </isSupportSearchEventTargetInfo>
    <isSupportElevatorRunningDetectionEvent>
      <!--ro, opt, bool, whether the device supports elevator running event, desc:eventType: ElevatorRunningDetectionEvent-->true
    </isSupportElevatorRunningDetectionEvent>
    <isSupportElevatorOperationStatistics>
      <!--ro, opt, bool, whether the device supports elevator running statistics event, desc:eventType: ElevatorOperationStatistics-->true
    </isSupportElevatorOperationStatistics>
    <isSupportStandingLongJump>
      <!--ro, opt, bool, whether the device supports standing long jump event, desc:eventType: StandingLongJump-->true
    </isSupportStandingLongJump>
    <isSupportJumpRope>
      <!--ro, opt, bool, whether the device supports jumping rope event, desc:eventType: JumpRope-->true
    </isSupportJumpRope>
    <isSupportUserAvailablePwQuery>
      <!--ro, opt, bool, whether the device supports person query through password, desc:eventType:UserAvailablePwQuery-->true
    </isSupportUserAvailablePwQuery>
    <isSupportAddHistoryEventSubscribeCfg>
      <!--ro, opt, bool, whether the device supports adding history event subscription, desc:POST /ISAPI/Event/notification/AddHistoryEventSubscribeCfg?
format=json-->true
    </isSupportAddHistoryEventSubscribeCfg>
    <isSupportHistoryEventSubscribeUploadCtrl>
      <!--ro, opt, bool, whether the device supports history event subscription uploading control, desc:PUT
/ISAPI/Event/notification/HistoryEventSubscribeUploadCtrl?format=json-->true
    </isSupportHistoryEventSubscribeUploadCtrl>
    <isSupportHLSStreamCfg>
      <!--ro, opt, bool, whether the device supports HLS streaming address parameters, desc:GET/PUT /ISAPI/Streaming/channels/<channelID>/HLSStreamCfg?
format=json-->true
    </isSupportHLSStreamCfg>
    <isSupportGetKeywordModelVersionInfo>
      <!--ro, opt, bool, whether the device supports getting sensitive information keyword model version information, desc:GET
/ISAPI/Intelligent/SensitiveInfoAnalysis/GetKeywordModelVersionInfo?format=json-->true
    </isSupportGetKeywordModelVersionInfo>
    <isSupportManualTriggerRemoteAlbumSnapshot>
      <!--ro, opt, bool, whether the device supports manually triggering remote album capture, desc:PUT /ISAPI/ContentMgmt/ManualTriggerRemoteAlbumSnapshot?
format=json-->true
    </isSupportManualTriggerRemoteAlbumSnapshot>
    <isSupportRadarVideoAssistedFocusParams>
      <!--ro, opt, bool, whether the device supports radar-assisted focusing parameters, desc:GET/PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoAssistedFocusParams?format=json-->true
    </isSupportRadarVideoAssistedFocusParams>
    <isSupportCollectionResultUpload>
      <!--ro, opt, bool, whether the device supports the uploading collected result event, desc:eventType: CollectionResultUpload-->true
    </isSupportCollectionResultUpload>
    <isSupportDeveloperMode>
      <!--ro, opt, bool, whether the device supports developer mode, desc:GET/PUT /ISAPI/System/DeveloperMode?format=json-->true
    </isSupportDeveloperMode>
    <isSupportGetDetailsFunction>
      <!--ro, opt, bool, whether the device supports getting detailed functions supported in T1 test, desc:GET /ISAPI/System/T1Test/DetailsFunction?
format=json-->true
    </isSupportGetDetailsFunction>
    <isSupportSearchDeviceIANAList>
      <!--ro, opt, bool, whether the device supports searching for supported IANA list, desc:POST /ISAPI/System/time/SearchDeviceIANAList?format=json-->true
    </isSupportSearchDeviceIANAList>
    <isSupportModifySpecificROIStreamJointParams>
      <!--ro, opt, bool, whether the device supports setting ROI stream jointing parameters, desc:PUT
/ISAPI/Streaming/channels/<channelID>/ModifySpecificROIStreamJointParams?format=json-->true
    </isSupportModifySpecificROIStreamJointParams>
    <isSupportDeviceTamperParams>
      <!--ro, opt, bool, whether the device supports tampering parameters, desc:GET/PUT /ISAPI/System/DeviceTamperParams?format=json-->true
    </isSupportDeviceTamperParams>
    <isSupportDeviceTamperEvent>
      <!--ro, opt, bool, whether the device supports tampering event, desc:eventType: DeviceTamperEvent-->true
    </isSupportDeviceTamperEvent>
    <isSupportRoadVariableLaneCtrlParams>
      <!--ro, opt, bool, whether the device supports dynamic management parameters of changeable lane, desc:GET/PUT
/ISAPI/Traffic/RoadVariableLaneCtrl/channels/<channelID>/RoadVariableLaneCtrlParams?format=json-->true
    </isSupportRoadVariableLaneCtrlParams>
    <isSupportImportModelFile>
      <!--ro, opt, bool, whether the device supports importing the water quality detection model file, desc:POST
/ISAPI/Event/waterQualityDetection/ImportModelFile?format=json-->true
    </isSupportImportModelFile>
    <isSupportSearchImportStatus>
      <!--ro, opt, bool, whether the device supports getting water quality detection model file importing progress, desc:GET
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    <!--ro, opt, bool, whether the device supports getting water quality detection model file importing progress, desc:GET
/ISAPI/Event/waterQualityDetection/SearchImportStatus?format=json-->true
  </isSupportSearchImportStatus>
  <isSupportSafetyWorkwearDetectionOverlapParams>
    <!--ro, opt, bool, whether the device supports parameters of wearing safety work clothes detection information overlay, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/SafetyWorkwearDetectionOverlapParams?format=json-->true
  </isSupportSafetyWorkwearDetectionOverlapParams>
  <isSupportSafetyWorkwearDetectionRuleParams>
    <!--ro, opt, bool, whether the device supports detection rule parameters of wearing safety work clothes, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/SafetyWorkwearDetectionRuleParams?format=json-->true
  </isSupportSafetyWorkwearDetectionRuleParams>
  <isSupportUpdateFirmwareMode>
    <!--ro, opt, bool, whether the device supports the upgrade mode, desc:GET/PUT /ISAPI/System/UpdateFirmwareMode?format=json-->true
  </isSupportUpdateFirmwareMode>
  <isSupportIndoorPasswayRecovery>
    <!--ro, opt, bool, whether the device supports indoor passway block alarm restoring event, desc:eventType: IndoorPasswayRecovery#-->true
  </isSupportIndoorPasswayRecovery>
  <isSupportBatteryCarRecovery>
    <!--ro, opt, bool, whether the device supports electric moped parking violation alarm restoring event, desc:eventType: BatteryCarRecovery-->true
  </isSupportBatteryCarRecovery>
  <isSupportDetectorTemperatureRecovery>
    <!--ro, opt, bool, whether the device supports detector temperature alarm restoring event, desc:eventType: DetectorTemperatureRecovery-->true
  </isSupportDetectorTemperatureRecovery>
  <isSupportGetRFIDReaderWorkStatus>
    <!--ro, opt, bool, whether the device supports getting RFID read/write working status, desc:GET /ISAPI/Traffic/UHFRFIDReader/GetRFIDReaderWorkStatus?
format=json-->true
  </isSupportGetRFIDReaderWorkStatus>
  <isSupportRFIDReaderWorkStatusCtrl>
    <!--ro, opt, bool, whether the device supports RFID read/write working status control, desc:PUT /ISAPI/Traffic/UHFRFIDReader/RFIDReaderWorkStatusCtrl?
format=json-->true
  </isSupportRFIDReaderWorkStatusCtrl>
  <isSupportFlameCombustionStatusAlgorithmsVersion>
    <!--ro, opt, bool, whether the device supports getting flame burning status detection algorithm version, desc:GET
/ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusAlgorithmsVersion?format=json-->true
  </isSupportFlameCombustionStatusAlgorithmsVersion>
  <isSupportTestResultUpload>
    <!--ro, opt, bool, whether the device supports test result uploading event, desc:eventType: TestResultUpload-->true
  </isSupportTestResultUpload>
  <isSupportAutoTestParams>
    <!--ro, opt, bool, whether the device supports automation testing parameters, desc:GET/PUT /ISAPI/System/AutoTest/AutoTestParams?format=json-->true
  </isSupportAutoTestParams>
  <isSupportExportATInstructRecord>
    <!--ro, opt, bool, whether the device supports exporting AT command records, desc:POST /ISAPI/System/ExportATInstructRecord?format=json-->true
  </isSupportExportATInstructRecord>
  <isSupportOfflineVideoFileManagement>
    <!--ro, opt, bool, whether the device supports offline video file management, desc:GET /ISAPI/ContentMgmt/OfflineVideoFileManagement/capabilities?
format=json-->true
  </isSupportOfflineVideoFileManagement>
  <isSupportChannelOccupySceneList>
    <!--ro, opt, bool, whether the device supports getting the scene list of fire escape occupation, desc:GET
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupySceneList?format=json-->true
  </isSupportChannelOccupySceneList>
  <isSupportChannelOccupyAlgorithmsVersion>
    <!--ro, opt, bool, whether the device supports getting the algorithm version of fire escape occupation, desc:GET
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyAlgorithmsVersion?format=json-->true
  </isSupportChannelOccupyAlgorithmsVersion>
  <isSupportChannelOccupyOverlapParams>
    <!--ro, opt, bool, whether the device supports overlay parameters of fire escape occupation, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyOverlapParams?format=json-->true
  </isSupportChannelOccupyOverlapParams>
  <isSupportChannelOccupyRuleParams>
    <!--ro, opt, bool, whether the device supports rule parameters of fire escape occupation, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyRuleParams?format=json-->true
  </isSupportChannelOccupyRuleParams>
  <isSupportGotoChannelOccupyScene>
    <!--ro, opt, bool, whether the device supports jumping to the specified fire escape occupation scene, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/GotoChannelOccupyScene?format=json-->true
  </isSupportGotoChannelOccupyScene>
  <isSupportChannelPassingEvent>
    <!--ro, opt, bool, whether the device supports vehicles passing outdoor fire escape, desc:eventType:channelPassingEvent-->true
  </isSupportChannelPassingEvent>
  <isSupportGunsHandoverRuleParams>
    <!--ro, opt, bool, whether the device supports object handover rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/GunsHandoverRuleParams?format=json-->true
  </isSupportGunsHandoverRuleParams>
  <isSupportGunsHandover>
    <!--ro, opt, bool, whether the device supports object handover event, desc:eventType:GunsHandover-->true
  </isSupportGunsHandover>
  <isSupportBulkUpgradeChildDevice>
    <!--ro, opt, bool, whether the device supports batch upgrading sub-devices, desc:POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json-->true
  </isSupportBulkUpgradeChildDevice>
  <isSupportSearchBulkUpgradeChildDeviceProgress>
    <!--ro, opt, bool, whether the device supports checking the progress of batch upgrading sub-devices, desc:PUT
/ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json-->true
  </isSupportSearchBulkUpgradeChildDeviceProgress>
  <isSupportDash>
    <!--ro, opt, bool, whether the device supports the dash event, desc:eventType:Dash-->true
  </isSupportDash>
  <isSupportStopRadarVideoAssistedFocusCalibrating>
    <!--ro, opt, bool, whether the device supports stopping the calibration of radar-assisted focusing, desc:PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/StopRadarVideoAssistedFocusCalibrating?format=json-->true
  </isSupportStopRadarVideoAssistedFocusCalibrating>
  <isSupportRadarVideoAssistedFocusCalibrating>
    <!--ro, opt, bool, whether the device supports the calibration of radar-assisted focusing, desc:GET/PUT
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/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoAssistedFocusCalibrating?format=json-->true
  </isSupportRadarVideoAssistedFocusCalibrating>
  <isSupportRadarVideoAssistedFocusCalibratingStatus>
    <!--ro, opt, bool, whether the device supports displaying the calibration status of radar-assisted focusing, desc:GET
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoAssistedFocusCalibratingStatus?format=json-->true
  </isSupportRadarVideoAssistedFocusCalibratingStatus>
  <isSupportETCMatchResultUpload>
    <!--ro, opt, bool, whether the device supports uploading ETC vehicle comparison results, desc:eventType:ETCMatchResultUpload-->true
  </isSupportETCMatchResultUpload>
  <isSupportPartsLifeRemind>
    <!--ro, opt, bool, whether the device supports prompting the life span of device component, desc:corresponding event: PartsLifeRemind-->true
  </isSupportPartsLifeRemind>
  <isSupportAuthorizeMicAcquisitionParam>
    <!--ro, opt, bool, whether the device supports audio authorization information, desc:GET/PUT
/ISAPI/System/InformedConsentMgr/AuthorizeMicAcquisitionParam?format=json-->true
  </isSupportAuthorizeMicAcquisitionParam>
  <isSupportTOFDetectEvent>
    <!--ro, opt, bool, whether the device supports TOF detection event, desc:eventType:TOFDetectEvent-->true
  </isSupportTOFDetectEvent>
  <isSupportAudioAnalysisResultOSDOverlay>
    <!--ro, opt, bool, whether the device supports OSD parameters of audio analysis results, desc:GET/PUT
/ISAPI/Intelligent/SensitiveInfoAnalysis/AudioAnalysisResultOSDOverlay?format=json-->true
  </isSupportAudioAnalysisResultOSDOverlay>
  <isSupportChildPasswordStatus>
    <!--ro, opt, bool, whether the device supports getting the activation password status of sub-devices, desc:GET
/ISAPI/System/activate/ChildPasswordStatus?format=json-->true
  </isSupportChildPasswordStatus>
  <isSupportMutexFunctionClose>
    <!--ro, opt, bool, whether the device supports quickly disabling exclusive functions, desc:POST /ISAPI/System/mutexFunctionClose?format=json-->true
  </isSupportMutexFunctionClose>
  <isSupportFirePointAlarmRecovery>
    <!--ro, opt, bool, whether the device supports fire detection alarm restoring event, desc:eventType:FirePointAlarmRecovery-->true
  </isSupportFirePointAlarmRecovery>
  <isSupportSmokeAlarmRecovery>
    <!--ro, opt, bool, whether the device supports smoke alarm restoring event, desc:eventType:FirePointAlarmRecovery-->true
  </isSupportSmokeAlarmRecovery>
  <isSupportTemperatureMeasurementPreviewAlarmRecovery>
    <!--ro, opt, bool, whether the device supports temperature measurement pre-alarm restoring event, desc:eventType:TemperatureMeasurementPreviewRecovery--
>true
  </isSupportTemperatureMeasurementPreviewAlarmRecovery>
  <isSupportTemperatureMeasurementAlarmRecovery>
    <!--ro, opt, bool, whether the device supports temperature measurement alarm restoring event, desc:eventType:TemperatureMeasurementRecovery-->true
  </isSupportTemperatureMeasurementAlarmRecovery>
  <isSupportUVPhotonDetection>
    <!--ro, opt, bool, whether the device supports UV photon detection event, desc:eventType:UVPhotonDetection-->true
  </isSupportUVPhotonDetection>
  <isSupportUVPhotonTimeDetection>
    <!--ro, opt, bool, whether the device supports UV photon scheduled-detection event, desc:eventType:UVPhotonTimeDetection-->true
  </isSupportUVPhotonTimeDetection>
  <isSupportPatientExpensesInfoQuery>
    <!--ro, opt, bool, whether the device supports the patient cost search event, desc:eventType:PatientExpensesInfoQuery-->true
  </isSupportPatientExpensesInfoQuery>
  <isSupportReinforcementCallEvent>
    <!--ro, opt, bool, whether the device supports reinforcement call event, desc:eventType:ReinforcementCallEvent-->true
  </isSupportReinforcementCallEvent>
  <isSupportChargeMode>
    <!--ro, opt, bool, whether the device supports charging mode, desc:GET/PUT /ISAPI/System/ChargeMode?format=json-->true
  </isSupportChargeMode>
  <isSupportCombinationRuleParams>
    <!--ro, opt, bool, whether the device supports combination rule parameters of perimeter detection, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterCombinationDetection/CombinationRuleParams?format=json-->true
  </isSupportCombinationRuleParams>
  <isSupportSwitchHolidayPlanCfg>
    <!--ro, opt, bool, whether the device supports the regular startup/shutdown on holiday, desc:GET/PUT /ISAPI/System/SwitchHolidayPlanCfg?format=json--
>true
  </isSupportSwitchHolidayPlanCfg>
  <isSupportSwitchIrregularHolidayPlanCfg>
    <!--ro, opt, bool, whether the device supports irregular startup/shutdown on holiday, desc:GET/PUT /ISAPI/System/SwitchIrregularHolidayPlanCfg?
format=json-->true
  </isSupportSwitchIrregularHolidayPlanCfg>
  <isSupportSearchPresetConfigInfo>
    <!--ro, opt, bool, whether the device supports searching for the configuration of specified presets, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SearchPresetConfigInfo?format=json-->true
  </isSupportSearchPresetConfigInfo>
  <isSupportSwitchPlanCfg>
    <!--ro, opt, bool, whether the device supports startup/shutdown schedule, desc:GET/PUT /ISAPI/System/SwitchPlanCfg?format=json-->true
  </isSupportSwitchPlanCfg>
  <isSupportVegetationSpectrumDetection>
    <!--ro, opt, bool, whether the device supports vegetation spectrum detection, desc:eventType:VegetationSpectrumDetection-->true
  </isSupportVegetationSpectrumDetection>
  <isSupportCaptureResultAttributeSearch>
    <!--ro, opt, bool, whether the device supports searching for attributes of human body capture, desc:POST
/ISAPI/Intelligent/CaptureResultAttributeSearch?format=json-->true
  </isSupportCaptureResultAttributeSearch>
  <isSupportHealthEducationInfoReadStatus>
    <!--ro, opt, bool, whether the device supports the event of uploading the reading status of health education information,
desc:eventType:HealthEducationInfoReadStatus-->true
  </isSupportHealthEducationInfoReadStatus>
  <isSupportNurseLocationEvent>
    <!--ro, opt, bool, whether the device supports nurse locating event, desc:eventType:NurseLocationEvent-->true
  </isSupportNurseLocationEvent>
  <isSupportAddChildDevResource>
    <!--ro, opt, bool, whether the device supports adding gateway resource information, desc:POST /ISAPI/System/DynamicResource/AddChildDevResource?
format=json-->true
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format=json-->true
    </isSupportAddChildDevResource>
    <isSupportModifyChildDevResource>
      <!--ro, opt, bool, whether the device supports editing gateway resource information, desc:POST /ISAPI/System/DynamicResource/ModifyChildDevResource?
format=json-->true
    </isSupportModifyChildDevResource>
    <isSupportDiagnosisDataCollectionParams>
      <!--ro, opt, bool, whether the device supports diagnosis data statistics parameters, desc:GET/PUT /ISAPI/System/DiagnosisDataCollectionParams?
format=json-->true
    </isSupportDiagnosisDataCollectionParams>
    <isSupportServicePlatformAccountAuth>
      <!--ro, opt, bool, whether the device supports platform account authentication, desc:GET /ISAPI/Security/ServicePlatformAccountAuth/capabilities?
format=json-->true
    </isSupportServicePlatformAccountAuth>
    <isSupportSyncPatientInfoEvent>
      <!--ro, opt, bool, whether the device supports synchronizing patient information, desc:eventType:SyncPatientInfoEvent-->true
    </isSupportSyncPatientInfoEvent>
    <isSupportAuthenticationNetworkCfg>
      <!--ro, opt, bool, whether the device supports the authentication parameters of networking method, desc:GET/PUT
/ISAPI/Security/ProtocolCertificate/AuthenticationNetworkCfg?format=json-->true
    </isSupportAuthenticationNetworkCfg>
    <isSupportPersonnelChannelLogParams>
      <!--ro, opt, bool, whether the device supports turnstile log parameters, desc:GET/PUT /ISAPI/ContentMgmt/PersonnelChannelLogParams?format=json-->true
    </isSupportPersonnelChannelLogParams>
    <isSupportElectricSaveModeParams>
      <!--ro, opt, bool, whether the device supports power-saving mode parameters, desc:GET/PUT /ISAPI/System/ElectricSaveModeParams?format=json-->true
    </isSupportElectricSaveModeParams>
    <isSupportChannelOccupyPOSParams>
      <!--ro, opt, bool, whether the device supports overlaying POS information on the outdoor access blocking detection data, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyPOSParams?format=jsonGET or PUT
/ISAPI/Intelligent/channels/<channelID>/SafetyWorkwearDetectionRuleParams?format=json-->true
    </isSupportChannelOccupyPOSParams>
    <isSupportChildPowerTimePlanParams>
      <!--ro, opt, bool, whether the device supports power-off schedule parameters of sub-devices, desc:GET/PUT /ISAPI/System/ChildPowerTimePlanParams?
format=json-->true
    </isSupportChildPowerTimePlanParams>
    <isSupportPowerIndicatorLightCtrlParams>
      <!--ro, opt, bool, whether the device supports parameters of device battery indicator control, desc:GET/PUT /ISAPI/System/PowerIndicatorLightCtrlParams?
format=json-->true
    </isSupportPowerIndicatorLightCtrlParams>
    <isSupportRadarVideoMapDetectionParams>
      <!--ro, opt, bool, whether the device supports parameters of radar-assisted map detection, desc:GET
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoMapDetectionParams?format=json-->true
    </isSupportRadarVideoMapDetectionParams>
    <isSupportRadarPOSParams>
      <!--ro, opt, bool, whether the device supports overlaying POS information on the radar detection data, desc:GET/PUT
/ISAPI/System/Video/inputs/channels/<channelID>/RadarPOSParams?format=json-->true
    </isSupportRadarPOSParams>
    <isSupportVideoIntercomModeEvent>
      <!--ro, opt, bool, whether the device supports the two-way audio mode event, desc:eventType:VideoIntercomModeEvent-->true
    </isSupportVideoIntercomModeEvent>
    <isSupportSoftwareAndHardwareCtrlParams>
      <!--ro, opt, bool, whether the device supports parameters of hardware and software service control, desc:GET/PUT
/ISAPI/System/FunctionSwitchCtrl/SoftwareAndHardwareCtrlParams?format=json-->true
    </isSupportSoftwareAndHardwareCtrlParams>
    <isSupportUserLoginAbnormal>
      <!--ro, opt, bool, whether the device supports user login exception information, desc:GET/PUT /ISAPI/Security/users/UserLoginAbnormal?format=json-->true
    </isSupportUserLoginAbnormal>
    <isSupportIndustryThermometryAdvancedParam>
      <!--ro, opt, bool, whether the device supports the advanced parameters of industrial temperature measurement, desc:GET/PUT
/ISAPI/Thermal/channels/<channelID>/IndustryThermometryAdvancedParam?format=json-->true
    </isSupportIndustryThermometryAdvancedParam>
    <isSupportGotoVideoRecheckScene>
      <!--ro, opt, bool, whether the device supports jumping to the specified scene of water level recheck detection, desc:POST
/ISAPI/Event/channels/<channelID>/GotoVideoRecheckScene?format=json-->true
    </isSupportGotoVideoRecheckScene>
    <isSupportVideoRecheckRuleParams>
      <!--ro, opt, bool, whether the device supports rule parameters of water level recheck scene, desc:PUT
/ISAPI/Event/channels/<channelID>/VideoRecheckRuleParams?format=json-->true
    </isSupportVideoRecheckRuleParams>
    <isSupportVideoRecheckSceneList>
      <!--ro, opt, bool, whether the device supports getting the list of water level recheck detection scenes, desc:GET
/ISAPI/Event/channels/<channelID>/VideoRecheckSceneList?format=json-->true
    </isSupportVideoRecheckSceneList>
    <isSupportPeopleSedentary>
      <!--ro, opt, bool, whether the device supports sedentary detection, desc:eventType:PeopleSedentary-->true
    </isSupportPeopleSedentary>
    <isSupportPeopleFallingDownDetection>
      <!--ro, opt, bool, whether the device supports person fall detection, desc:eventType: PeopleFallingDownDetection-->true
    </isSupportPeopleFallingDownDetection>
    <isSupportBridgeDeflectionTargetCalibration>
      <!--ro, opt, bool, whether the device supports target calibration of bridge deflection, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionTargetCalibration?format=json-->true
    </isSupportBridgeDeflectionTargetCalibration>
    <isSupportGetBridgeDeflectionTargetCalibrationStatus>
      <!--ro, opt, bool, whether the device supports getting the target calibration status of bridge deflection, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetBridgeDeflectionTargetCalibrationStatus?format=json-->true
    </isSupportGetBridgeDeflectionTargetCalibrationStatus>
    <isSupportBridgeDeflectionMonitoringRuleParams>
      <!--ro, opt, bool, whether the device supports the rule parameters of bridge deflection monitoring, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionMonitoringRuleParams?format=json-->true
    </isSupportBridgeDeflectionMonitoringRuleParams>
    <isSupportHardwareReboot>
      <!--ro, opt, bool, whether the device supports hardware rebooting, desc:PUT /ISAPI/System/HardwareReboot?format=json-->true
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  </isSupportHardwareReboot>
  <isSupportSearchSceneCalibrationStatus>
    <!--ro, opt, bool, whether the device supports searching for calibration status of multiple channels by scene, desc:POST
/ISAPI/System/setupCalibration/channels/<channelID>/SearchSceneCalibrationStatus?format=json-->true
  </isSupportSearchSceneCalibrationStatus>
  <isSupportSearchSceneCalibration>
    <!--ro, opt, bool, whether the device supports searching for calibration status of multiple channels by scene, desc:POST
/ISAPI/System/setupCalibration/channels/<channelID>/SearchSceneCalibration?format=json-->true
  </isSupportSearchSceneCalibration>
  <isSupportSoundSourceLocationRuleParams>
    <!--ro, opt, bool, whether the device supports rule parameters of sound source locating, desc:GET/PUT
/ISAPI/System/SoundSourceLocation/AudioIn/<AudioInID>/SoundSourceLocationRuleParams?format=json-->true
  </isSupportSoundSourceLocationRuleParams>
  <isSupportPersonnelChannelLogDebug>
    <!--ro, opt, bool, whether the device supports debugging commands of turnstile logs, desc:GET /ISAPI/ContentMgmt/PersonnelChannelLogDebug/capabilities?
format=json-->true
  </isSupportPersonnelChannelLogDebug>
  <isSupportCheckDeviceExistNUCFile>
    <!--ro, opt, bool, whether the device supports checking the locally stored the NUC file, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/CheckDeviceExistNUCFile?format=json-->true
  </isSupportCheckDeviceExistNUCFile>
  <isSupportExportSmartDataData>
    <!--ro, opt, bool, whether the device supports exporting smart data, desc:GET /ISAPI/System/VideoQualityMeasurementCollection/ExportSmartDataData?
format=json-->true
  </isSupportExportSmartDataData>
  <isSupportGetCollectImageCalculateResult>
    <!--ro, opt, bool, whether the device supports getting the calculation result of image collection, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/GetCollectImageCalculateResult?format=json-->true
  </isSupportGetCollectImageCalculateResult>
  <isSupportAIOPAlarmTrans>
    <!--ro, opt, bool, whether the device supports uploading AIOP-related events via ISAPI message transmitted by COMM_VCA_ALARM,
desc:AIOP-related events include the following binary alarm types defined in HCNetSDK:
AIOP-accessed video detection alarm information (binary alarm: COMM_UPLOAD_AIOP_VIDEO, ISAPI alarm: AIOP_Video), AIOP-accessed picture detection alarm
information (binary alarm: COMM_UPLOAD_AIOP_PICTURE, ISAPI alarm: AIOP_Picture), AIOP-accessed capture polling detection alarm information (binary alarm:
COMM_UPLOAD_AIOP_POLLING_SNAP, ISAPI alarm: AIOP_Polling_Snap), AIOP-accessed video polling detection alarm information (binary alarm:
COMM_UPLOAD_AIOP_POLLING_VIDEO, ISAPI alarm: AIOP_Polling_Video)) -->true
  </isSupportAIOPAlarmTrans>
  <isSupportLongDistanceRace>
    <!--ro, opt, bool, whether the device supports long-distance race event, desc:eventType:LongDistanceRace-->true
  </isSupportLongDistanceRace>
  <isSupportIndicatorLightAbnormal>
    <!--ro, opt, bool, whether the device supports abnormal alarm indicator event, desc:eventType:IndicatorLightAbnormal-->true
  </isSupportIndicatorLightAbnormal>
  <isSupportVideoLostPictureStatus>
    <!--ro, opt, bool, whether the device supports the status of custom pictures displayed when video loss occurred, desc:GET
/ISAPI/System/Video/videolosses/VideoLostPictureStatus?format=json-->true
  </isSupportVideoLostPictureStatus>
  <DualVCA>
    <!--ro, opt, object, intelligent information overlay-->
    <isSupportdualVCAList>
      <!--ro, opt, bool, whether it supports setting the parameters for overlaying intelligent information of channels in a batch,
desc:/ISAPI/Streaming/channels/dualVCA?format=json-->true
    </isSupportdualVCAList>
    <enabled opt="true,false" def="true">
      <!--ro, opt, bool, whether to enable the function, attr:opt{req, string},def{req, bool}, desc:"ruleRegion" (overlay only rule area information)-->true
    </enabled>
    <channelID min="1" max="256">
      <!--ro, opt, int, channel No., attr:min{req, int},max{req, int}-->1
    </channelID>
    <overlapRangeType opt="ruleRegion,fullScreen" def="ruleRegion">
      <!--ro, opt, string, type of overlay area, attr:opt{req, string},def{req, string}-->ruleRegion
    </overlapRangeType>
  </DualVCA>
  <isSupportGetScopeCalibrationStatus>
    <!--ro, opt, bool, whether the device supports getting the scope calibration status, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetScopeCalibrationStatus?format=json-->true
  </isSupportGetScopeCalibrationStatus>
  <isSupportSearchShootTableUserGroupParams>
    <!--ro, opt, bool, whether the device supports searching for the parameters of specified zeroing profiles, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SearchShootTableUserGroupParams?format=json-->true
  </isSupportSearchShootTableUserGroupParams>
  <isSupportModifyCurrentShootTableParams>
    <!--ro, opt, bool, whether the device supports editing the information of current valid zeroing profiles, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/ModifyCurrentShootTableParams?format=json-->true
  </isSupportModifyCurrentShootTableParams>
  <isSupportCalibrateScopeTargetPointPosition>
    <!--ro, opt, bool, whether the device supports calibrating the absolute position of the sight cross and drop point, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/CalibrateScopeTargetPointPosition?format=json-->true
  </isSupportCalibrateScopeTargetPointPosition>
  <isSupportGetCurrentShootTableParams>
    <!--ro, opt, bool, whether the device supports getting the information of the current valid zeroing profiles, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetCurrentShootTableParams?format=json-->true
  </isSupportGetCurrentShootTableParams>
  <isSupportVideoGunsightRuleParam>
    <!--ro, opt, bool, whether the device supports rule parameters of video scope, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/VideoGunsightRuleParam?format=json-->true
  </isSupportVideoGunsightRuleParam>
  <isSupportFinishCollectImageData>
    <!--ro, opt, bool, whether it supports completing the collection of image data, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/FinishCollectImageData?format=json-->true
  </isSupportFinishCollectImageData>
  <isSupportStartCollectImageData>
    <!--ro, opt, bool, whether it supports starting collecting image data, desc:GET
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/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/StartCollectImageData?format=json-->true
  </isSupportStartCollectImageData>
  <isSupportSearchBridgeDeflectionMonitoringEvent>
    <!--ro, opt, bool, whether it supports searching for the event result of bridge deflection detection, desc:POST
/ISAPI/Intelligent/SearchBridgeDeflectionMonitoringEvent?format=json-->true
  </isSupportSearchBridgeDeflectionMonitoringEvent>
  <isSupportGetAllPIPCofigParams>
    <!--ro, opt, bool, whether it supports getting all picture-in-picture configuration parameters, desc:GET /ISAPI/System/Video/GetAllPIPCofigParams?
format=json-->true
  </isSupportGetAllPIPCofigParams>
  <isSupportManualDetectionCarriageClearStatus>
    <!--ro, opt, bool, whether it supports manual detection of the clearing status of the carriage, desc:POST
/ISAPI/System/channels/<channelID>/ManualDetectionCarriageClearStatus?format=json-->true
  </isSupportManualDetectionCarriageClearStatus>
  <isSupportGetCarriageClearCalibrationStatus>
    <!--ro, opt, bool, whether it supports getting the reference calibration carriage clearing status, desc:GET
/ISAPI/System/channels/<channelID>/GetCarriageClearCalibrationStatus?format=json-->true
  </isSupportGetCarriageClearCalibrationStatus>
  <isSupportModifyCarriageClearCalibration>
    <!--ro, opt, bool, whether it supports setting the reference calibration of carriage clearing, desc:POST
/ISAPI/System/channels/<channelID>/ModifyCarriageClearCalibration?format=json-->true
  </isSupportModifyCarriageClearCalibration>
  <isSupportCarriageClearRuleParams>
    <!--ro, opt, bool, whether it supports setting rule parameters of carriage clearing, desc:GET/PUT
/ISAPI/System/channels/<channelID>/CarriageClearRuleParams?format=json-->true
  </isSupportCarriageClearRuleParams>
  <isSupportRadarFallenObjectsOnTracks>
    <!--ro, opt, bool, whether it supports radar detection of fallen objects on tracks, desc:eventType:RadarFallenObjectsOnTracks-->true
  </isSupportRadarFallenObjectsOnTracks>
  <isSupportTemperatureRiseCalibrationParams>
    <!--ro, opt, bool, whether it supports temperature rise calibration parameters, desc:GET/PUT /ISAPI/Thermal/TemperatureRiseCalibrationParams?
format=json-->true
  </isSupportTemperatureRiseCalibrationParams>
  <isSupportStartTemperatureRiseCalibration>
    <!--ro, opt, bool, whether it supports starting temperature rise calibration, desc:PUT /ISAPI/Thermal/StartTemperatureRiseCalibration?format=json-->true
  </isSupportStartTemperatureRiseCalibration>
  <isSupportGetTemperatureRiseCalibrationStatus>
    <!--ro, opt, bool, whether it supports getting temperature rise calibration status, desc:GET /ISAPI/Thermal/GetTemperatureRiseCalibrationStatus?
format=json-->true
  </isSupportGetTemperatureRiseCalibrationStatus>
  <isSupportExportTemperatureRiseCalibrationFile>
    <!--ro, opt, bool, whether it supports exporting temperature rise calibration file, desc:GET /ISAPI/Thermal/ExportTemperatureRiseCalibrationFile?
format=json-->true
  </isSupportExportTemperatureRiseCalibrationFile>
  <isSupportGetTemperatureRiseCorrectionVersion>
    <!--ro, opt, bool, whether it supports getting temperature rise calibration version information, desc:GET
/ISAPI/Thermal/GetTemperatureRiseCorrectionVersion?format=json-->true
  </isSupportGetTemperatureRiseCorrectionVersion>
  <isSupportSearchImportDeviceDataTaskProgress>
    <!--ro, opt, bool, whether it supports searching for the progress of device data importing task, desc:POST
/ISAPI/System/SearchImportDeviceDataTaskProgress?format=json-->true
  </isSupportSearchImportDeviceDataTaskProgress>
  <isSupportSearchExportDeviceDataTaskProgress>
    <!--ro, opt, bool, whether it supports searching for the progress of device data exporting task, desc:POST
/ISAPI/System/SearchExportDeviceDataTaskProgress?format=json-->true
  </isSupportSearchExportDeviceDataTaskProgress>
  <isSupportAddImportDeviceDataTask>
    <!--ro, opt, bool, whether it supports creating device data importing task, desc:POST /ISAPI/System/AddImportDeviceDataTask?format=json-->true
  </isSupportAddImportDeviceDataTask>
  <isSupportAddExportDeviceDataTask>
    <!--ro, opt, bool, whether it supports creating device data exporting task, desc:POST /ISAPI/System/AddExportDeviceDataTask?format=json-->true
  </isSupportAddExportDeviceDataTask>
  <isSupportSearchRecordStorage>
    <!--ro, opt, bool, whether it supports searching for record storage HDD information, desc:POST /ISAPI/ContentMgmt/search/SearchRecordStorage?
format=json-->true
  </isSupportSearchRecordStorage>
  <isSupportDetectionPathRuleParams>
    <!--ro, opt, bool, whether it supports hazardous gas detection patrol path rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/methaneDetection/DetectionPathRuleParams?format=json-->true
  </isSupportDetectionPathRuleParams>
  <isSupportSearchObjectDistanceFocusInfoList>
    <!--ro, opt, bool, whether it supports searching for the list of object distance focus calibration information, desc:POST
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/SearchObjectDistanceFocusInfoList?format=json-->true
  </isSupportSearchObjectDistanceFocusInfoList>
  <isSupportMoveObjectDistanceFocusLocation>
    <!--ro, opt, bool, whether it supports moving to the point where the focus of the object distance is calibrated, desc:PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/MoveObjectDistanceFocusLocation?format=json-->true
  </isSupportMoveObjectDistanceFocusLocation>
  <isSupportAddObjectDistanceFocusInfo>
    <!--ro, opt, bool, whether it supports adding object distance focus calibration information, desc:POST
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/AddObjectDistanceFocusInfo?format=json-->true
  </isSupportAddObjectDistanceFocusInfo>
  <isSupportModifyObjectDistanceFocusInfo>
    <!--ro, opt, bool, whether it supports editing object distance focus calibration information, desc:PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ModifyObjectDistanceFocusInfo?format=json-->true
  </isSupportModifyObjectDistanceFocusInfo>
  <isSupportObjectDistanceCalibrateAutoFocusParams>
    <!--ro, opt, bool, whether it supports auto-focus parameters of object distance calibration, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceCalibrateAutoFocusParams?format=json-->true
  </isSupportObjectDistanceCalibrateAutoFocusParams>
  <isSupportExpresswayNetworkTollAccessParams>
    <!--ro, opt, bool, whether it supports protocol access parameters of network parking fee charging on highway, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceCalibrateAutoFocusParams?format=json-->true
  </isSupportExpresswayNetworkTollAccessParams>
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  <isSupportControlCameraDetectAction>
    <!--ro, opt, bool, whether it supports controlling the patrol path of camera, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/methaneDetection/ControlCameraDetectAction?format=json-->true
  </isSupportControlCameraDetectAction>
  <isSupportObjectDistanceAccessWay>
    <!--ro, opt, bool, whether it supports the method of getting the object distance, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceAccessWay?format=json-->true
  </isSupportObjectDistanceAccessWay>
  <isSupportCancelTemperatureRiseCalibration>
    <!--ro, opt, bool, whether it supports canceling temperature rise calibration, desc:PUT /ISAPI/Thermal/CancelTemperatureRiseCalibration?format=json--
>true
  </isSupportCancelTemperatureRiseCalibration>
  <isSupportRTCTimeTrigger>
    <!--ro, opt, bool, whether the device supports RTC time exception linkage, desc:GET/PUT /ISAPI/Event/RTCTimeTrigger?format=json-->true
  </isSupportRTCTimeTrigger>
  <isSupportChargingGunAbnormalEvent>
    <!--ro, opt, bool, whether it supports charging station inserted gun exception event, desc:eventType: ChargingGunAbnormalEvent-->true
  </isSupportChargingGunAbnormalEvent>
  <isSupportVideoChannelEnhance>
    <!--ro, opt, bool, whether it supports enhancement configuration of adding video channel, desc:GET/PUT
/ISAPI/ContentMgmt/InputProxy/VideoChannelEnhance?format=json-->true
  </isSupportVideoChannelEnhance>
  <isSupportDeleteObjectDistanceFocusInfo>
    <!--ro, opt, bool, whether it supports deleting the object distance focus calibration information, desc:POST
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/DeleteObjectDistanceFocusInfo?format=json-->true
  </isSupportDeleteObjectDistanceFocusInfo>
  <isSupportGetStartTemperatureRiseCalibrationParams>
    <!--ro, opt, bool, whether the device supports getting parameter of starting temperature rise calibration, desc:GET
/ISAPI/Thermal/GetStartTemperatureRiseCalibrationParams?format=json-->true
  </isSupportGetStartTemperatureRiseCalibrationParams>
  <isSupportSearchAudioRecordPlan>
    <!--ro, opt, bool, whether the device supports searching all recording schedules, desc:POST /ISAPI/System/RecordAudio/SearchAudioRecordPlan?format=json-
->true
  </isSupportSearchAudioRecordPlan>
  <isSupportCleanerDetectionOverlapParams>
    <!--ro, opt, bool, whether the device supports cleaning liquid detection information overlay parameters, desc:GET/PUT
/ISAPI/System/Hardware/channels/<channelID>/CleanerDetectionOverlapParams?format=json-->true
  </isSupportCleanerDetectionOverlapParams>
  <isSupportWaterLevelVerticalAreaDisplayParams>
    <!--ro, opt, bool, whether it supports the display parameters of water level vertical view monitoring area, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/WaterLevelVerticalAreaDisplayParams?format=json-->true
  </isSupportWaterLevelVerticalAreaDisplayParams>
  <isSupportGetRecordStatusList>
    <!--ro, opt, bool, whether the device supports getting the recording status of all channels, desc:POST /ISAPI/ContentMgmt/search/GetRecordStatusList?
format=json-->true
  </isSupportGetRecordStatusList>
  <isSupportWiperClean>
    <!--ro, opt, bool, whether the device supports wiper cleaning event, desc:eventType: WiperClean-->true
  </isSupportWiperClean>
  <isSupportGetNATStatus>
    <!--ro, opt, bool, whether the device supports getting NAT status, desc:GET /ISAPI/System/Network/NetworkAdvancedCfg/GetNATStatus?format=json-->true
  </isSupportGetNATStatus>
  <isSupportRestoreSensitivityDefaultParams>
    <!--ro, opt, bool, whether the device supports restoring radar perimeter sensitivity parameters to the default, desc:PUT
/ISAPI/Smart/RadarPerimeterAlg/RestoreSensitivityDefaultParams?format=json-->true
  </isSupportRestoreSensitivityDefaultParams>
  <isSupportCarryPassengersPricingStatus>
    <!--ro, opt, bool, whether the device supports displaying the status change of meter, desc:eventType: CarryPassengersPricingStatus-->true
  </isSupportCarryPassengersPricingStatus>
  <isSupportCarryPassengersNoPricing>
    <!--ro, opt, bool, Whether to support event of not using the meter when passengers are on board, desc:eventType: CarryPassengersNoPricing-->true
  </isSupportCarryPassengersNoPricing>
  <isSupportNPlusMultiple>
    <!--ro, opt, bool, whether the device supports N+M mode of host and standby device, desc:whether the device supports setting workMode to N+M. Related
API: PUT /ISAPI/System/Network/NPlusOne/capabilities?format=json-->true
  </isSupportNPlusMultiple>
  <ifPopUpNetWorkCfgSetting>
    <!--ro, opt, bool, whether to pop up network configuration interface, desc:PUT/GET /ISAPI/System/Network/interfaces/<interfaceID>. true (if the default
IP address of the device is 192.168.0.100, the device will pop up the network configuration interface), false or node does not exist (the network
configuration interface will not pop up)-->true
  </ifPopUpNetWorkCfgSetting>
  <AudioOutputChannelCfg>
    <!--ro, opt, object, support linking audio output channel with audio file-->
    <AudioOutputChannelIDList>
      <!--ro, opt, array, audio output channel list, subType:object-->
      <AudioOutputChannelID>
        <!--ro, req, int, audio output channel No.-->1
      </AudioOutputChannelID>
    </AudioOutputChannelIDList>
  </AudioOutputChannelCfg>
  <isSupportImportTemperatureRiseCalibrationFile>
    <!--ro, opt, bool, whether the device supports importing the temperature rise calibration file, desc:PUT
/ISAPI/Thermal/ImportTemperatureRiseCalibrationFile?format=json-->true
  </isSupportImportTemperatureRiseCalibrationFile>
  <isSupportGetAPPQRCodePic>
    <!--ro, opt, bool, whether it supports getting QR code picture of the app accessed by the device, desc:GET /ISAPI/System/GetAPPQRCodePic?format=json--
>true
  </isSupportGetAPPQRCodePic>
  <isSupportPersionQueueRealTimeData>
    <!--ro, opt, bool, whether it supports real-time counting of the number of people in the queuing area, desc:GET
/ISAPI/Intelligent/personQueueDetection/channels/<channelID>/PersionQueueRealTimeData?format=json-->true
  </isSupportPersionQueueRealTimeData>
  <isSupportImageVignettingOptimizeParams>
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<!--ro, opt, bool, whether it supports image dark corner optimization parameters, desc:GET/PUT /ISAPI/Streaming/channels/<channelID>/ImageVignettingOptimizeParams?format=json-->true
</isSupportImageVignettingOptimizeParams>
<isSupportStartMultispectralCalibration>
<!--ro, opt, bool, whether it supports enabling multispectral calibration, desc:PUT /ISAPI/System/SpectrumCalibration/channels/<channelID>/StartMultispectralCalibration?format=json-->true
</isSupportStartMultispectralCalibration>
<isSupportGetMultispectralCalibrationStatus>
<!--ro, opt, bool, whether it supports getting the multispectral calibration status, desc:GET /ISAPI/System/SpectrumCalibration/channels/<channelID>/GetMultispectralCalibrationStatus?format=json-->true
</isSupportGetMultispectralCalibrationStatus>
<isSupportStopRadarVideoTargetTracking>
<!--ro, opt, bool, whether it supports stopping target tracking (radar), desc:PUT /ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/StopRadarVideoTargetTracking?format=json-->true
</isSupportStopRadarVideoTargetTracking>
<isSupportGetRadarVideoTargetTrackingStatus>
<!--ro, opt, bool, whether it supports getting the status of target tracking (radar), desc:GET /ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/GetRadarVideoTargetTrackingStatus?format=json-->true
</isSupportGetRadarVideoTargetTrackingStatus>
<isSupportStartRadarVideoTargetTracking>
<!--ro, opt, bool, whether it supports starting target tracking (radar), desc:PUT /ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/StartRadarVideoTargetTracking?format=json-->true
</isSupportStartRadarVideoTargetTracking>
<isSupportBatteryChargeCtrl>
<!--ro, opt, bool, whether it supports battery charging control parameters, desc:GET/PUT /ISAPI/System/BatteryChargeCtrl?format=json-->true
</isSupportBatteryChargeCtrl>
<isUpgrageAuxDevice opt="true,false">
<!--ro, opt, string, whether it supports synchronous upgrade of auxiliary device, attr:opt{opt, string}, desc:it represents support details of isUpgrageAuxDevice in /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>&childDevID=<devIndex>&isUpgrageAuxDevice=<isUpgrageAuxDevice>-->true
</isUpgrageAuxDevice>
<isSupportClearFile>
<!--ro, opt, bool, whether it supports clearing files under the specified path of the device, desc:POST /ISAPI/System/VideoQualityMeasurementCollection/ClearFile?format=json-->true
</isSupportClearFile>
<isSupportRestoreDefaultParamByChannel>
<!--ro, opt, bool, whether it supports restoring default parameters of a single channel, desc:PUT /ISAPI/Streaming/RestoreDefaultParam?format=json&channelID=<channelID>-->true
</isSupportRestoreDefaultParamByChannel>
<isSupportGetMultispectralCalibrationParams>
<!--ro, opt, bool, whether it supports getting multispectral calibration parameters, desc:GET /ISAPI/System/SpectrumCalibration/channels/<channelID>/GetMultispectralCalibrationParams?format=json-->true
</isSupportGetMultispectralCalibrationParams>
<isSupportGetThermometryCalibrationData>
<!--ro, opt, bool, whether it supports getting temperature calibration-related data, desc:GET /ISAPI/Thermal/GetThermometryCalibrationData?format=json-->true
</isSupportGetThermometryCalibrationData>
<isSupportImportEncryptedComponentFile>
<!--ro, opt, bool, whether it supports importing encrypted component files, desc:POST /ISAPI/System/EncryptionFileUpgradeMgr/ImportEncryptedComponentFile?format=json-->true
</isSupportImportEncryptedComponentFile>
<isSupportGetFrameMeanValue>
<!--ro, opt, bool, whether it supports getting frame average range information, desc:GET /ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/GetFrameMeanValue?format=json-->true
</isSupportGetFrameMeanValue>
<isSupportGetBaffleImage>
<!--ro, opt, bool, whether it supports getting image data before and after the image is captured, desc:GET /ISAPI/System/VideoQualityMeasurementCollection/GetBaffleImage?format=json-->true
</isSupportGetBaffleImage>
<isSupportThermometryGainMode>
<!--ro, opt, bool, whether it supports the temperature measurement gain parameters, desc:GET/PUT /ISAPI/Thermal/ThermometryGainMode?format=json-->true
</isSupportThermometryGainMode>
<isSupportGetChannelEnhancePreOperationTips>
<!--ro, opt, bool, whether it supports getting the prompt information before the enhancement configuration of adding the channel, desc:POST /ISAPI/ContentMgmt/InputProxy/GetChannelEnhancePreOperationTips?format=json-->true
</isSupportGetChannelEnhancePreOperationTips>
<isSupportGetDeviceNetworkLinkStatus>
<!--ro, opt, bool, whether it supports getting the network connection status of the device, desc:GET /ISAPI/System/GetDeviceNetworkLinkStatus?format=json-->true
</isSupportGetDeviceNetworkLinkStatus>
<isSupportManualRefreshOnlineUserList>
<!--ro, opt, bool, whether the web needs to display a refresh button, desc:used for web interface display, determine whether the device needs to display the refresh button to refresh the list of online users via this field-->true
</isSupportManualRefreshOnlineUserList>
<isSupportVCAAlarmTrans>
<!--ro, opt, bool, whether it supports uploading events related to abnormal behavior detection through the transmission of ISAPI messages, desc:for all events related to abnormal behavior detection, refer to HCNetSDK enumeration of VCA_RULE_EVENT_TYPE_EX-->true
</isSupportVCAAlarmTrans>
<isSupportHumidityEvent>
<!--ro, opt, bool, whether it supports humidity alarm events, desc:HumidityEvent-->true
</isSupportHumidityEvent>
<isSupportDisplayThermometryFirePrevention>
<!--ro, opt, bool, whether the web needs to display the temperature measurement function as temperature measurement & fire prevention,
desc:if this field is true, the web interface display "Temperature Measurement & Fire Prevention", otherwise, it will be displayed as "Temperature Measurement". If this field is not supported, the web interface will display "Temperature Measurement"-->true
</isSupportDisplayThermometryFirePrevention>
<isSupportPreRemoteCheckEvent>
<!--ro, opt, bool, whether it supports the collection of pre-verification events, desc:eventType:PreRemoteCheckEvent-->true
</isSupportPreRemoteCheckEvent>
<isSupportSearchExportDeviceStatusTaskProgress>
<!--ro, opt, bool, whether it supports getting the progress of the task to export device information and status, desc:POST /ISAPI/System/SearchExportDeviceStatusTaskProgress?format=json-->true

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/ISAPI/System/SearchExportDeviceStatusTaskProgress?format=json"?true
  </isSupportSearchExportDeviceStatusTaskProgress>
  <isSupportAddExportDeviceStatusTask>
    <!--ro, opt, bool, whether it supports creating the task to export device information and status, desc:POST /ISAPI/System/AddExportDeviceStatusTask?
format=json-->true
  </isSupportAddExportDeviceStatusTask>
  <isSupportInsulatorThermometryRuleParams>
    <!--ro, opt, bool, whether it supports the insulator temperature measurement rule parameters, desc:GET/PUT
/ISAPI/Thermal/channels/<channelID>/InsulatorThermometryRuleParams?format=json-->true
  </isSupportInsulatorThermometryRuleParams>
  <isSupportInsulatorCruiseParams>
    <!--ro, opt, bool, whether it supports insulator patrol parameters, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/InsulatorCruiseParams?format=json--
>true
  </isSupportInsulatorCruiseParams>
  <isSupportModifyInsulatorDetectionRegion>
    <!--ro, opt, bool, whether it supports modifying the insulator group detection area, desc:PUT
/ISAPI/Thermal/channels/<channelID>/ModifyInsulatorDetectionRegion?format=json-->true
  </isSupportModifyInsulatorDetectionRegion>
  <isSupportMoveInsulatorPosition>
    <!--ro, opt, bool, whether it supports movement to the specified insulator string position, desc:POST
/ISAPI/Thermal/channels/<channelID>/MoveInsulatorPosition?format=json-->true
  </isSupportMoveInsulatorPosition>
  <isSupportDeleteInsulatorGroup>
    <!--ro, opt, bool, whether it supports deleting a specified insulator group, desc:POST /ISAPI/Thermal/channels/<channelID>/DeleteInsulatorGroup?
format=json-->true
  </isSupportDeleteInsulatorGroup>
  <isSupportGenerateInsulatorRuleRegion>
    <!--ro, opt, bool, whether it supports generating a specified insulator string rule area, desc:POST
/ISAPI/Thermal/channels/<channelID>/GenerateInsulatorRuleRegion?format=json-->true
  </isSupportGenerateInsulatorRuleRegion>
  <isSupportGetInsulatorAllName>
    <!--ro, opt, bool, whether it supports getting all the names of insulator groups, desc:GET /ISAPI/Thermal/channels/<channelID>/GetInsulatorAllName?
format=json-->true
  </isSupportGetInsulatorAllName>
  <isSupportSearchUploadLog>
    <!--ro, opt, bool, whether it supports searching and uploading log files, desc:PUT /ISAPI/ContentMgmt/SearchUploadLog?format=json-->true
  </isSupportSearchUploadLog>
  <isSupportRainfallTelemeter>
    <!--ro, opt, bool, whether it supports remotely monitoring rainfall, desc:eventType:RainfallTelemeter-->true
  </isSupportRainfallTelemeter>
  <isSupportPosOverlayTest>
    <!--ro, opt, bool, whether it supports the test of enabling POS information overlay configuration, desc:POST
/ISAPI/System/Video/inputs/channels/<channelID>/PosOverlayTest?format=json-->true
  </isSupportPosOverlayTest>
  <isSupportFullScreenPosOverlay>
    <!--ro, opt, bool, whether it supports full-screen POS information overlay parameters, desc:GET/PUT
/ISAPI/System/Video/inputs/channels/<channelID>/FullScreenPosOverlay?format=json-->true
  </isSupportFullScreenPosOverlay>
  <isSupportFullScreenPosOverlayCap>
    <!--ro, opt, bool, whether to get the dynamic capability of full-screen POS information overlay parameters, desc:GET
/ISAPI/System/Video/inputs/channels/<channelID>/FullScreenPosOverlayCap?format=json-->true
  </isSupportFullScreenPosOverlayCap>
  <isSupportVCAOptimizeDisplay>
    <!--ro, opt, bool, whether it supports the display of the intelligent resource usability page, desc:if the field is true, the user-friendly interface is
displayed; otherwise, the previous one will be displayed; if the field is not supported, it is determined by the device type-->true
  </isSupportVCAOptimizeDisplay>
  <isSupportSearchInsulatorParams>
    <!--ro, opt, bool, whether it supports searching the parameters of a specified insulator group, desc:POST
/ISAPI/Thermal/channels/<channelID>/SearchInsulatorParams?format=json-->true
  </isSupportSearchInsulatorParams>
  <isSupportSetInsulatorParams>
    <!--ro, opt, bool, whether it supports setting insulator group parameters, desc:PUT /ISAPI/Thermal/channels/<channelID>/SetInsulatorParams?format=json--
>true
  </isSupportSetInsulatorParams>
  <isSupportInsulatorAbnormalAlarm>
    <!--ro, opt, bool, whether it supports insulator exception alarm, desc:eventType:InsulatorAbnormalAlarm-->true
  </isSupportInsulatorAbnormalAlarm>
  <isSupportGetFlashLifespanInfo>
    <!--ro, opt, bool, whether it supports getting the lifespan information of the built-in disk storage medium, desc:GET
/ISAPI/System/GetFlashLifespanInfo?format=json-->true
  </isSupportGetFlashLifespanInfo>
  <isSupportDownloadAudioAnalysisFile>
    <!--ro, opt, bool, whether it supports downloading audio analysis files, desc:POST /ISAPI/Intelligent/SensitiveInfoAnalysis/DownloadAudioAnalysisFile?
format=json-->true
  </isSupportDownloadAudioAnalysisFile>
  <isSupportCrowdDispersing>
    <!--ro, opt, bool, whether it supports crowd dispersing, desc:corresponding event: CrowdDispersing-->true
  </isSupportCrowdDispersing>
  <isSupportPeripheralAbnormalEvent>
    <!--ro, opt, bool, whether it supports peripheral fault event, desc:If it is not supported, this node will not be returned. Corresponding eventType:
PeripheralAbnormalEvent-->true
  </isSupportPeripheralAbnormalEvent>
  <isSupportPeripheralAbnormalTrigger>
    <!--ro, opt, bool, whether it supports peripheral fault linkage, desc:GET/PUT /ISAPI/Event/PeripheralAbnormalTrigger?format=json-->true
  </isSupportPeripheralAbnormalTrigger>
  <isSupportBodyCameraAbnormalEvent>
    <!--ro, opt, bool, whether it supports exception events of body cameras, desc:corresponding event: BodyCameraAbnormalEvent-->true
  </isSupportBodyCameraAbnormalEvent>
  <isSupportDeviceWorkModeAsyncStatusEvent>
    <!--ro, opt, bool, whether it supports the event of enforcement device working mode asynchronous status, desc:corresponding event:
DeviceWorkModeAsyncStatusEvent-->true
  </isSupportDeviceWorkModeAsyncStatusEvent>
  <isSupportDeviceUserAsyncStatusEvent>
    <!--ro, opt, bool, whether it supports asynchronous parameter status event for law enforcement device users, desc:corresponding event:
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DeviceUserAsyncStatusEvent-->true
    </isSupportDeviceUserAsyncStatusEvent>
    <isSupportVideoCodingAsyncStatusEvent>
      <!--ro, opt, bool, whether it supports video encoding asynchronous parameter status event, desc:corresponding event: VideoCodingAsyncStatusEvent-->true
    </isSupportVideoCodingAsyncStatusEvent>
    <isSupportRecordAsyncStatusEvent>
      <!--ro, opt, bool, whether it supports manual recording asynchronous parameter status event, desc:corresponding event: RecordAsyncStatusEvent-->true
    </isSupportRecordAsyncStatusEvent>
    <isSupportISUPAccessAsyncStatusEvent>
      <!--ro, opt, bool, whether it supports asynchronous parameter status event for ISUP protocol access, desc:corresponding event:
ISUPAccessAsyncStatusEvent-->true
    </isSupportISUPAccessAsyncStatusEvent>
    <isSupportWirelessConfigListAsyncStatusEvent>
      <!--ro, opt, bool, whether it supports asynchronous parameters status event for wireless configuration list, desc:corresponding event:
WirelessConfigListAsyncStatusEvent-->true
    </isSupportWirelessConfigListAsyncStatusEvent>
    <isSupportDeviceLocationAsyncStatusEvent>
      <!--ro, opt, bool, whether it supports asynchronous parameter status event for device location, desc:corresponding event:
DeviceLocationAsyncStatusEvent-->true
    </isSupportDeviceLocationAsyncStatusEvent>
    <isSupportVideoCodingAsyncParams>
      <!--ro, opt, bool, whether it supports video encoding async parameters, desc:GET/PUT /ISAPI/Streaming/VideoCodingAsyncParams?format=json-->true
    </isSupportVideoCodingAsyncParams>
    <isSupportMoveShield>
      <!--ro, opt, bool, whether it supports movement to a specified shielded area, desc:POST /ISAPI/Intelligent/channels/<channelID>/MoveShield?format=json--
>true
    </isSupportMoveShield>
    <isSupportStopAutoGenerateShield>
      <!--ro, opt, bool, whether it supports stopping the automatic recognition of smart shielded area, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/StopAutoGenerateShield?format=json-->true
    </isSupportStopAutoGenerateShield>
    <isSupportStartGenerateParams>
      <!--ro, opt, bool, whether it supports starting generating smart shielded area, desc:PUT /ISAPI/Intelligent/channels/<channelID>/StartGenerateParams?
format=json-->true
    </isSupportStartGenerateParams>
    <isSupportIntelligentShieldAlgorithmsVersion>
      <!--ro, opt, bool, whether it supports getting the version information of the intelligent shielding area recognition algorithm library, desc:GET
/ISAPI/Intelligent/channels/<channelID>/IntelligentShieldAlgorithmsVersion?format=json-->true
    </isSupportIntelligentShieldAlgorithmsVersion>
    <isSupportConvertShield>
      <!--ro, opt, bool, whether it supports converting a specified shielded area, desc:PUT /ISAPI/Intelligent/channels/<channelID>/ConvertShield?format=json-
->true
    </isSupportConvertShield>
    <isSupportDeleteShield>
      <!--ro, opt, bool, whether it supports deleting a specified shielded area, desc:PUT /ISAPI/Intelligent/channels/<channelID>/DeleteShield?format=json--
>true
    </isSupportDeleteShield>
    <isSupportGetGenerateParams>
      <!--ro, opt, bool, whether it supports getting the parameters of generating smart shielded areas, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetGenerateParams?format=json-->true
    </isSupportGetGenerateParams>
    <isSupportManualAddShield>
      <!--ro, opt, bool, whether it supports manually adding shielded areas, desc:PUT /ISAPI/Intelligent/channels/<channelID>/ManualAddShield?format=json--
>true
    </isSupportManualAddShield>
    <isSupportIntelligentShieldParams>
      <!--ro, opt, bool, whether it supports smart shielded area parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/IntelligentShieldParams?
format=json-->true
    </isSupportIntelligentShieldParams>
    <isSupportIntelligentShieldListParams>
      <!--ro, opt, bool, whether it supports the smart shielded area list parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/IntelligentShieldListParams?format=json-->true
    </isSupportIntelligentShieldListParams>
    <isSupportDeicingCfg>
      <!--ro, opt, bool, whether it supports lens heating and de-icing control parameters, desc:GET/PUT /ISAPI/System/Hardware/DeicingCfg?format=json-->true
    </isSupportDeicingCfg>
    <isSupportInspectionOverdueEvent>
      <!--ro, opt, bool, whether it supports annual inspection overdue event for alcohol detection enforcement device, desc:corresponding event:
InspectionOverdueEvent-->true
    </isSupportInspectionOverdueEvent>
    <isSupportChannelsPanoramaCalibration>
      <!--ro, opt, bool, whether it supports batch configuring channel calibration distance, desc:GET/PUT /ISAPI/Panorama/channels/calibration/distance?
format=json-->true
    </isSupportChannelsPanoramaCalibration>
    <isSupportRadarPerimeterShieldOverlapParams>
      <!--ro, opt, bool, whether it supports radar perimeter overlay parameters, desc:GET/PUT
/ISAPI/Smart/RadarPerimeterAlg/Radar/<RadarID>/RadarPerimeterShieldOverlapParams?format=json-->true
    </isSupportRadarPerimeterShieldOverlapParams>
    <isSupportAlarmFilteInterval>
      <!--ro, opt, bool, whether it supports configuring alarm filter time interval, desc:corresponding to the "alarmFilteInterval" node in the POST
/ISAPI/IoTGateway/AlarmRules/SetAlarmRules?format=json-->true
    </isSupportAlarmFilteInterval>
    <isSupportImportLicense>
      <!--ro, opt, bool, whether it supports importing the authorization file, desc:POST /ISAPI/System/SoftwareLicense/ImportLicense?format=json-->true
    </isSupportImportLicense>
    <isSupportShipStatisticsAnalysisParams>
      <!--ro, opt, bool, whether it supports ship traffic statistics parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ShipStatisticsAnalysisParams?format=json-->true
    </isSupportShipStatisticsAnalysisParams>
    <isSupportShipStatisticsEvent>
      <!--ro, opt, bool, whether it supports ship traffic statistics event, desc:corresponding event: ShipStatisticsEvent-->true
    </isSupportShipStatisticsEvent>
    <isSupportPeopleVitalSignsDetection>
      <!--ro, opt, bool, whether it supports vital sign detection event, desc:eventType: PeopleVitalSignsDetection-->true
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    <!--ro, opt, bool, whether it supports vital sign detection event, desc:eventType: PeopleVitalSignsDetection-->true
  </isSupportPeopleVitalSignsDetection>
  <isSupportPeopleVitalSignsStatistic>
    <!--ro, opt, bool, whether it supports vital sign statistics event, desc:eventType: PeopleVitalSignsStatistic-->true
  </isSupportPeopleVitalSignsStatistic>
  <isSupportPeopleBodyMoving>
    <!--ro, opt, bool, whether it supports human body movement detection event, desc:eventType: PeopleBodyMoving-->true
  </isSupportPeopleBodyMoving>
  <isSupportPeopleGetUp>
    <!--ro, opt, bool, whether it supports person get-up detection event, desc:eventType: PeopleGetUp (detected a person who got up from the bed during
posture detection)-->true
  </isSupportPeopleGetUp>
  <isSupportPersonLocationDetection>
    <!--ro, opt, bool, whether it supports person location detection event, desc:eventType: PersonLocationDetection-->true
  </isSupportPersonLocationDetection>
  <isSupportPeopleFallingDownRecovery>
    <!--ro, opt, bool, whether it supports person fall restoring event, desc:eventType: PeopleFallingDownRecovery-->true
  </isSupportPeopleFallingDownRecovery>
  <isSupportPersonRetentionEvent>
    <!--ro, opt, bool, whether it supports people overstay event, desc:eventType: PersonRetentionEvent-->true
  </isSupportPersonRetentionEvent>
  <isSupportPersonRetentionRecoveryEvent>
    <!--ro, opt, bool, whether it supports people overstay restoring event, desc:eventType: PersonRetentionRecoveryEvent-->true
  </isSupportPersonRetentionRecoveryEvent>
  <isSupportPeopleLongTimeLeaveEvent>
    <!--ro, opt, bool, whether it supports person long-time off-bed event, desc:eventType: PeopleLongTimeLeaveEvent-->true
  </isSupportPeopleLongTimeLeaveEvent>
  <isSupportPeopleLongTimeLeaveRecoveryEvent>
    <!--ro, opt, bool, whether it supports person long-time off-bed restoring event, desc:eventType: PeopleLongTimeLeaveRecoveryEvent-->true
  </isSupportPeopleLongTimeLeaveRecoveryEvent>
  <isSupportPeopleVitalSignsAbnormalEvent>
    <!--ro, opt, bool, whether it supports vital sign exception event, desc:eventType: PeopleVitalSignsAbnormalEvent-->true
  </isSupportPeopleVitalSignsAbnormalEvent>
  <isSupportPeopleVitalSignsRecoveryEvent>
    <!--ro, opt, bool, whether it supports vital sign exception restoring event, desc:eventType: PeopleVitalSignsRecoveryEvent-->true
  </isSupportPeopleVitalSignsRecoveryEvent>
  <isSupportPeopleVitalSignsExportTaskEvent>
    <!--ro, opt, bool, whether it supports vital sign data exporting task result event, desc:eventType: PeopleVitalSignsExportTaskEvent-->true
  </isSupportPeopleVitalSignsExportTaskEvent>
  <isSupportTOFCalibrate>
    <!--ro, opt, bool, whether it supports configuring TOF detection calibration, desc:PUT /ISAPI/System/Video/inputs/tamperDetection/TOFCalibrate?
format=json-->true
  </isSupportTOFCalibrate>
  <isSupportSearchBulkUpgradeChildDeviceTask>
    <!--ro, opt, bool, whether it supports searching for tasks of batch upgrading sub-devices, desc:/ISAPI/System/SearchBulkUpgradeChildDeviceTask?
format=json-->true
  </isSupportSearchBulkUpgradeChildDeviceTask>
  <isSupportModifyBulkUpgradeChildDeviceTask>
    <!--ro, opt, bool, whether it supports modifying the task of batch upgrading sub-devices, desc:/ISAPI/System/ModifyBulkUpgradeChildDeviceTask?
format=json-->true
  </isSupportModifyBulkUpgradeChildDeviceTask>
  <isSupportDeleteBulkUpgradeChildDeviceTask>
    <!--ro, opt, bool, whether it supports deleting tasks of batch upgrading sub-devices, desc:/ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json--
>true
  </isSupportDeleteBulkUpgradeChildDeviceTask>
  <isSupportAddEventTemplate>
    <!--ro, opt, bool, whether it supports adding event message templates, desc:POST /ISAPI/Event/notification/httpHosts/AddEventTemplate?format=json-->true
  </isSupportAddEventTemplate>
  <isSupportTestEventTemplate>
    <!--ro, opt, bool, whether it supports testing event message templates, desc:POST /ISAPI/Event/notification/httpHosts/TestEventTemplate?format=json--
>true
  </isSupportTestEventTemplate>
  <isSupportEventTemplateParams>
    <!--ro, opt, bool, whether it supports event message template parameters, desc:GET /ISAPI/Event/notification/httpHosts/EventTemplateParams?format=json--
>true
  </isSupportEventTemplateParams>
  <isSupportAddCombineEvents>
    <!--ro, opt, bool, whether it supports adding combined event, desc:POST /ISAPI/Event/CombineEvents/AddCombineEvents?format=json-->true
  </isSupportAddCombineEvents>
  <isSupportModifyCombineEventsParams>
    <!--ro, opt, bool, whether it supports modifying combined event parameters, desc:POST /ISAPI/Event/CombineEvents/ModifyCombineEventsParams?format=json--
>true
  </isSupportModifyCombineEventsParams>
  <isSupportSearchCombineEventsParams>
    <!--ro, opt, bool, whether it supports searching for combined event parameters, desc:POST /ISAPI/Event/CombineEvents/SearchCombineEventsParams?
format=json-->true
  </isSupportSearchCombineEventsParams>
  <isSupportDeleteCombineEventsList>
    <!--ro, opt, bool, whether it supports batch deleting combined event, desc:POST /ISAPI/Event/CombineEvents/DeleteCombineEventsList?format=json-->true
  </isSupportDeleteCombineEventsList>
  <isSupportEventRuleAndTriggerParams>
    <!--ro, opt, bool, whether it supports configuring the rule linkage parameters of a specified event, desc:POST
/ISAPI/Event/EventBatchRulesMgr/EventRuleAndTriggerParams?format=json-->true
  </isSupportEventRuleAndTriggerParams>
  <isSupportSearchEventRuleAndTriggerParams>
    <!--ro, opt, bool, whether it supports searching for the rule linkage parameters of a specified event, desc:POST
/ISAPI/Event/EventBatchRulesMgr/SearchEventRuleAndTriggerParams?format=json-->true
  </isSupportSearchEventRuleAndTriggerParams>
  <isSupportCallMessageHandlingEvent>
    <!--ro, opt, bool, whether it supports the call message handling event, desc:eventType: CallMessageHandlingEvent-->true
  </isSupportCallMessageHandlingEvent>
  <isSupportResetBatchChild>
    <!--ro, opt, bool, whether it supports batch restoring sub-devices, desc:PUT /ISAPI/System/ResetBatchChild?format=json-->true
  </isSupportResetBatchChild>
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  <isSupportSearchResetBatchChildStatus>
    <!--ro, opt, bool, whether it supports searching for the status of batch restoring sub-devices, desc:POST /ISAPI/System/SearchResetBatchChildStatus?format=json-->true
  </isSupportSearchResetBatchChildStatus>
  <isSupportSearchRebootBatchChildStatus>
    <!--ro, opt, bool, whether it supports searching for the status of batch rebooting sub-devices, desc:POST /ISAPI/System/SearchRebootBatchChildStatus?format=json-->true
  </isSupportSearchRebootBatchChildStatus>
  <isSupportRebootBatchChild>
    <!--ro, opt, bool, whether it supports batch restarting sub-devices, desc:PUT /ISAPI/System/RebootBatchChild?format=json-->true
  </isSupportRebootBatchChild>
  <isSupportFlowStatisticsTelemeter>
    <!--ro, opt, bool, whether it supports traffic statistics telemetry event, desc:eventType:FlowStatisticsTelemeter-->true
  </isSupportFlowStatisticsTelemeter>
  <isSupportModifyFlowStatisticsTelemeterUpload>
    <!--ro, opt, bool, whether it supports modifying the parameters of traffic statistics telemetry report strategy, desc:PUT /ISAPI/System/Network/NetworkFlowMonitor/ModifyFlowStatisticsTelemeterUpload?format=json-->true
  </isSupportModifyFlowStatisticsTelemeterUpload>
  <isSupportClearFlowStatisticsTelemeter>
    <!--ro, opt, bool, whether it supports clearing the flow statistics telemetry data, desc:PUT /ISAPI/System/Network/NetworkFlowMonitor/ClearFlowStatisticsTelemeter?format=json-->true
  </isSupportClearFlowStatisticsTelemeter>
  <isSupportFlowStatisticsTelemeterUpload>
    <!--ro, opt, bool, whether it supports the remote telemetry reporting strategy parameters for traffic statistics, desc:GET /ISAPI/System/Network/NetworkFlowMonitor/FlowStatisticsTelemeterUpload?format=json-->true
  </isSupportFlowStatisticsTelemeterUpload>
  <isSupportScreenGraphicsColorControl>
    <!--ro, opt, bool, whether it supports screen image color control, desc:POST /ISAPI/System/ScreenColorQualityDetection/ScreenGraphicsColorControl?format=json-->true
  </isSupportScreenGraphicsColorControl>
  <isSupportSearchScreenColorQualityDetectionResult>
    <!--ro, opt, bool, whether it supports searching for the screen image color quality detection results, desc:POST /ISAPI/System/ScreenColorQualityDetection/SearchScreenColorQualityDetectionResult?format=json-->true
  </isSupportSearchScreenColorQualityDetectionResult>
  <isSupportSearchScreenColorQualityDetectionProcess>
    <!--ro, opt, bool, whether it supports searching for the screen image color quality detection progress, desc:POST /ISAPI/System/ScreenColorQualityDetection/SearchScreenColorQualityDetectionProcess?format=json-->true
  </isSupportSearchScreenColorQualityDetectionProcess>
  <isSupportScreenColorQualityDetectionParams>
    <!--ro, opt, bool, whether it supports the screen image color quality detection parameters, desc:GET/PUT /ISAPI/System/ScreenColorQualityDetection/channels/<channelID>/ScreenColorQualityDetectionParams?format=json-->true
  </isSupportScreenColorQualityDetectionParams>
  <isSupportAddInsulatorGroup>
    <!--ro, opt, bool, whether it supports adding insulator list, desc:GET /ISAPI/Thermal/channels/<channelID>/AddInsulatorGroup?format=json-->true
  </isSupportAddInsulatorGroup>
  <isSupportSuperResolutionReboot>
    <!--ro, opt, bool, whether the device supports rebooting super resolution, desc:whether the device supports restart to complete the conversion: true (yes), false (no). Different devices use different chips. Some devices convert videos/images to ones with super resolution effect when the super resolution function is enabled, while others require a device restart before the conversion is completed-->true
  </isSupportSuperResolutionReboot>
  <isSupportStartAutoTest>
    <!--ro, opt, bool, whether it supports starting automatic test, desc:POST /ISAPI/System/AutoTest/StartAutoTest?format=json-->true
  </isSupportStartAutoTest>
  <isSupportGetUserManualLink>
    <!--ro, opt, bool, whether it supports getting the device user manual link, desc:GET /ISAPI/System/GetUserManualLink?format=json-->true
  </isSupportGetUserManualLink>
  <isSupportExportSignaturePublicKey>
    <!--ro, opt, bool, whether it supports exporting the signed public key, desc:GET /ISAPI/Security/ProtocolCertificate/ExportSignaturePublicKey?format=json-->true
  </isSupportExportSignaturePublicKey>
  <isSupportCipherCardRunningStatus>
    <!--ro, opt, bool, whether it supports the password card running status, desc:GET /ISAPI/System/CipherCardRunningStatus?format=json-->true
  </isSupportCipherCardRunningStatus>
  <isSupportBridgeDeflectionMonitoringCapture>
    <!--ro, opt, bool, whether it supports manual triggering of bridge deflection monitoring and capturing, desc:POST /ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionMonitoringCapture?format=json-->true
  </isSupportBridgeDeflectionMonitoringCapture>
  <isSupportAdaptiveVideoStreamParams>
    <!--ro, opt, bool, whether it supports adaptive video stream control parameters, desc:GET/PUT /ISAPI/ContentMgmt/InputProxy/AdaptiveVideoStreamParams?format=json-->true
  </isSupportAdaptiveVideoStreamParams>
  <isSupportSearchAudioDetectionTaskProcess>
    <!--ro, opt, bool, whether it supports searching for the progress of audio detection tasks, desc:POST /ISAPI/System/DeviceAudioDetection/SearchAudioDetectionTaskProcess?format=json-->true
  </isSupportSearchAudioDetectionTaskProcess>
  <isSupportSendAudioDetectionTask>
    <!--ro, opt, bool, whether it supports applying audio detection tasks, desc:POST /ISAPI/System/DeviceAudioDetection/SendAudioDetectionTask?format=json-->true
  </isSupportSendAudioDetectionTask>
  <isSupportMedicalImageURLExpiredEvent>
    <!--ro, opt, bool, whether it supports medical image URL expiration event, desc:eventType:MedicalImageURLExpiredEvent-->true
  </isSupportMedicalImageURLExpiredEvent>
  <isSupportBridgeDeflectionMonitoringEvent>
    <!--ro, opt, bool, whether it supports bridge deflection monitoring event, desc:eventType:BridgeDeflectionMonitoringEvent-->true
  </isSupportBridgeDeflectionMonitoringEvent>
  <isSupportBridgeDeflectionTargetLossEvent>
    <!--ro, opt, bool, whether it supports bridge deflection target loss event, desc:eventType:BridgeDeflectionTargetLossEvent-->true
  </isSupportBridgeDeflectionTargetLossEvent>
  <isSupportNobodyDetectionRuleParams>
    <!--ro, opt, bool, whether it supports unmanned detection rule parameters, desc:GET/PUT /ISAPI/Event/channels/<channelID>/NobodyDetectionRuleParams?format=json-->true
  </isSupportNobodyDetectionRuleParams>
  <isSupportNobodyDetectionRegionShieldParams>
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<!--ro, opt, bool, whether it supports the shielding parameters for unmanned detection areas, desc:GET/PUT /ISAPI/Event/channels/<channelID>/NobodyDetectionRegionShieldParams?format=json-->true</isSupportNobodyDetectionRegionShieldParams>
<isSupportReduceTemperatureFilterParams>
<!--ro, opt, bool, whether it supports the cooling filter parameter, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ReduceTemperatureFilterParams?format=json-->true</isSupportReduceTemperatureFilterParams>
<isSupportGetStreamOverlayLogDebugStatus>
<!--ro, opt, bool, whether it supports getting the status of overlaying log debugging status on stream, desc:GET /ISAPI/ContentMgmt/channels/<channelID>/GetStreamOverlayLogDebugStatus?format=json-->true</isSupportGetStreamOverlayLogDebugStatus>
<isSupportStreamOverlayLogDebugParams>
<!--ro, opt, bool, whether it supports the parameters of overlaying log debugging information on stream, desc:GET/PUT /ISAPI/ContentMgmt/channels/<channelID>/StreamOverlayLogDebugParams?format=json-->true</isSupportStreamOverlayLogDebugParams>
<isSupportBridgeDeflectionAbnormalRuleParams>
<!--ro, opt, bool, whether it supports the bridge deflection exception detection rule parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionAbnormalRuleParams?format=json-->true</isSupportBridgeDeflectionAbnormalRuleParams>
<isSupportBridgeDeflectionAbnormalEvent>
<!--ro, opt, bool, whether it supports bridge deflection exception event, desc:eventType:BridgeDeflectionAbnormalEvent-->true</isSupportBridgeDeflectionAbnormalEvent>
<isSupportRestoreBackup>
<!--ro, opt, bool, whether it supports data restoring and backup, desc:POST /ISAPI/System/SystemSnapshot/RestoreBackup?format=json-->true</isSupportRestoreBackup>
<isSupportDeleteBackupFile>
<!--ro, opt, bool, whether it supports deleting backup files, desc:POST /ISAPI/System/SystemSnapshot/DeleteBackupFile?format=json-->true</isSupportDeleteBackupFile>
<isSupportSearchRestoreBackupProgress>
<!--ro, opt, bool, whether it supports searching for the progress of restoring backup files, desc:POST /ISAPI/System/SystemSnapshot/SearchRestoreBackupProgress?format=json-->true</isSupportSearchRestoreBackupProgress>
<isSupportGetBackupsFileList>
<!--ro, opt, bool, whether it supports getting the list of backup files, desc:POST /ISAPI/System/SystemSnapshot/GetBackupsFileList?format=json-->true</isSupportGetBackupsFileList>
<isSupportAutoBackupParams>
<!--ro, opt, bool, whether it supports automatic backup parameters, desc:GET/PUT /ISAPI/System/SystemSnapshot/AutoBackupParams?format=json-->true</isSupportAutoBackupParams>
<isSupportDeviceSoftwareRunningProgramRestart>
<!--ro, opt, bool, whether it supports restarting the device software program, desc:POST /ISAPI/System/DeviceSoftwareRunningProgramRestart?format=json-->true</isSupportDeviceSoftwareRunningProgramRestart>
<isSupportSendScreenColorQualityDetectionRequest>
<!--ro, opt, bool, whether it supports sending a screen color quality detection request, desc:POST /ISAPI/System/ScreenColorQualityDetection/SendScreenColorQualityDetectionRequest?format=json-->true</isSupportSendScreenColorQualityDetectionRequest>
<isSupportTelephoningDetectionRuleParams>
<!--ro, opt, bool, whether it supports phone call detection rule parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/telephoningDetect/TelephoningDetectionRuleParams?format=json-->true</isSupportTelephoningDetectionRuleParams>
<isSupportTelephoningBehaviorAlarmEvent>
<!--ro, opt, bool, whether it supports phone call alarm event, desc:eventType:TelephoningBehaviorAlarmEvent-->true</isSupportTelephoningBehaviorAlarmEvent>
<isSupportTelephoningBehaviorRecoveryEvent>
<!--ro, opt, bool, whether it supports phone call restoring event, desc:eventType:TelephoningBehaviorRecoveryEvent-->true</isSupportTelephoningBehaviorRecoveryEvent>
<isSupportSleepDutyRecoveryEvent>
<!--ro, opt, bool, whether the device supports absence or sleeping on duty restoring event, desc:eventType:sleepDutyRecoveryEvent-->true</isSupportSleepDutyRecoveryEvent>
<isSupportGetBridgeDeflectionTargetOffsetResetStatus>
<!--ro, opt, bool, whether it supports getting the zeroing result of bridge deflection target displacement, desc:GET /ISAPI/Intelligent/channels/<channelID>/GetBridgeDeflectionTargetOffsetResetStatus?format=json-->true</isSupportGetBridgeDeflectionTargetOffsetResetStatus>
<isSupportBridgeDeflectionTargetOffsetReset>
<!--ro, opt, bool, whether it supports bridge deflection target displacement reset, desc:PUT /ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionTargetOffsetReset?format=json-->true</isSupportBridgeDeflectionTargetOffsetReset>
<isSupportBridgeDeflectionTargetLossRuleParams>
<!--ro, opt, bool, whether it supports bridge deflection target loss rule parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionTargetLossRuleParams?format=json-->true</isSupportBridgeDeflectionTargetLossRuleParams>
<isSupportGetAnalogOutputParams>
<!--ro, opt, bool, whether it supports getting all analog output parameters, desc:GET /ISAPI/System/AnalogOutput/GetAnalogOutputParams?format=json-->true</isSupportGetAnalogOutputParams>
<isSupportSystemReset>
<!--ro, opt, bool, whether the device supports restoring system to default settings, desc:POST /ISAPI/System/SystemReset?format=json-->true</isSupportSystemReset>
<isSupportFakePlateDetection>
<!--ro, opt, bool, whether it supports fake license plate detection event, desc:corresponding event: FakePlateDetection-->true</isSupportFakePlateDetection>
<isSupportSafetyWorkwearDetectionEvent>
<!--ro, opt, bool, whether it supports the event of wearing safety work clothes detection, desc:corresponding event: SafetyWorkwearDetectionEvent-->true</isSupportSafetyWorkwearDetectionEvent>
<isSupportVehicleMaintenanceStatusEvent>
<!--ro, opt, bool, whether it supports vehicle maintenance status events, desc:corresponding event: VehicleMaintenanceStatusEvent-->true</isSupportVehicleMaintenanceStatusEvent>
<isSupportEventRetransmissionInfo>
<!--ro, opt, bool, whether it supports reloading event information, desc:GET /ISAPI/System/EventRetransmissionInfo?format=json-->true</isSupportEventRetransmissionInfo>
<isSupportGetEnvironmentCaptureStatus>
<!--ro, opt, bool, whether it supports getting the environmental data collection status, desc:POST /ISAPI/Event/channels/<channelID>/objectAccumulationDetection/GetEnvironmentCaptureStatus?format=json-->true</isSupportGetEnvironmentCaptureStatus>

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  <isSupportEnvironmentCapture>
    <!--ro, opt, bool, whether it supports environmental data collection, desc:POST
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/EnvironmentCapture?format=json-->true
  </isSupportEnvironmentCapture>
  <isSupportSearchCPUDetailInfo>
    <!--ro, opt, bool, whether it supports searching for detailed CPU performance information, desc:POST /ISAPI/System/SearchCPUDetailInfo?format=json--
>true
  </isSupportSearchCPUDetailInfo>
  <isSupportPanelControlParams>
    <!--ro, opt, bool, whether it supports device hardware panel configuration parameters, desc:GET/PUT /ISAPI/System/PanelControlParams?format=json-->true
  </isSupportPanelControlParams>
  <isSupportSearchGoodsIncAndDecStatistics>
    <!--ro, opt, bool, whether it supports searching for cargo increase/decrease statistics, desc:POST
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/SearchGoodsIncAndDecStatistics?format=json-->true
  </isSupportSearchGoodsIncAndDecStatistics>
  <isSupportCustomAlarmHTTPParams>
    <!--ro, opt, bool, whether it supports custom alarm HTTP parameters, desc:GET/PUT /ISAPI/Event/notification/httpHosts/CustomAlarmHTTPParams?format=json-
->true
  </isSupportCustomAlarmHTTPParams>
  <isSupportTestCustomAlarmHTTP>
    <!--ro, opt, bool, whether it supports custom alarm HTTP request test, desc:PUT /ISAPI/Event/notification/httpHosts/TestCustomAlarmHTTP?format=json--
>true
  </isSupportTestCustomAlarmHTTP>
  <isSupportCustomAlarmHTTPLinkageParams>
    <!--ro, opt, bool, whether it supports customizing the alarm HTTP linkage parameters, desc:GET/PUT /ISAPI/Thermal/CustomAlarmHTTPLinkageParams?
format=json-->true
  </isSupportCustomAlarmHTTPLinkageParams>
  <isSupportAccumulationDetectionPeriodicReport>
    <!--ro, opt, bool, whether it supports scheduled reporting of pile detection events, desc:corresponding event: AccumulationDetectionPeriodicReport--
>true
  </isSupportAccumulationDetectionPeriodicReport>
  <isSupportImportDiskCompatibilityCertificate>
    <!--ro, opt, bool, whether it supports importing disk compatibility certificate, desc:PUT /ISAPI/Security/serverCertificate/certificate?type=
<type>&platformNo=<platformNo>, and type is DiskCompatibility-->true
  </isSupportImportDiskCompatibilityCertificate>
  <isSupportGetAcumulationDetectionAlgorithmicVersion>
    <!--ro, opt, bool, whether it supports getting the version of the stacking detection algorithm, desc:GET
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/GetAcumulationDetectionAlgorithmicVersion?format=json-->true
  </isSupportGetAcumulationDetectionAlgorithmicVersion>
  <isSupportSnapshotUpgradeDifferenceFile>
    <!--ro, opt, bool, whether it supports getting snapshot upgrade difference files, desc:GET /ISAPI/System/SystemSnapshot/SnapshotUpgradeDifferenceFile?
format=json-->true
  </isSupportSnapshotUpgradeDifferenceFile>
  <isSupportGroupRecoveryEvent>
    <!--ro, opt, bool, whether it supports people gathering restoring event-->true
  </isSupportGroupRecoveryEvent>
  <isSupportElectronicEyepieceParams>
    <!--ro, opt, bool, whether it supports electron microscope channel parameters, desc:GET/PUT /ISAPI/System/Video/ElectronicEyepieceParams?format=json--
>true
  </isSupportElectronicEyepieceParams>
  <isSupportSmokingDetectionRecoverEvent>
    <!--ro, opt, bool, whether it supports smoking detection restoring event, desc:fire alarm upload in progress, eventType: SmokingDetectionRecoverEvent--
>true
  </isSupportSmokingDetectionRecoverEvent>
  <isSupportDeviceRegionParams>
    <!--ro, opt, bool, whether it supports device area parameters, desc:GET/PUT /ISAPI/System/DeviceRegionParams?format=json-->true
  </isSupportDeviceRegionParams>
  <isSupportTemperatureEvent>
    <!--ro, opt, bool, whether it supports temperature alarm event, desc:TemperatureEvent-->true
  </isSupportTemperatureEvent>
  <isSupportTemperatureRecoverEvent>
    <!--ro, opt, bool, whether it supports temperature restoring event, desc:TemperatureRecoverEvent-->true
  </isSupportTemperatureRecoverEvent>
  <isSupportWebSSH>
    <!--ro, opt, bool, whether it supports WebSSH, desc:if the node is not returned, it indicates no-->true
  </isSupportWebSSH>
  <isSupportCreateDebugChannel>
    <!--ro, opt, bool, whether it supports creating WebSSH debugging channel, desc:if the node is not returned, it indicates no-->true
  </isSupportCreateDebugChannel>
  <isSupportExportDebugLog>
    <!--ro, opt, bool, whether it supports exporting the debugging log, desc:if the node is not returned, it indicates no-->true
  </isSupportExportDebugLog>
  <isSupportAcumulationDetectionEffectiveAreaCfgParams>
    <!--ro, opt, bool, whether it supports the configuration parameters of pile detection effective area, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/AcumulationDetectionEffectiveAreaCfgParams?format=json-->true
  </isSupportAcumulationDetectionEffectiveAreaCfgParams>
  <isSupportGetGoalDryWetBallTempCurve>
    <!--ro, opt, bool, whether it supports getting the target dry-wet bulb temperature curve, desc:POST
/ISAPI/System/TobaccoBakeMgr/GetGoalDryWetBallTempCurve?format=json-->true
  </isSupportGetGoalDryWetBallTempCurve>
  <isSupportBakeTobaccoRuleParams>
    <!--ro, opt, bool, whether it supports tobacco baking rule parameters, desc:GET/PUT /ISAPI/System/TobaccoBakeMgr/BakeTobaccoRuleParams?format=json--
>true
  </isSupportBakeTobaccoRuleParams>
  <isSupportStartBakeTobacco>
    <!--ro, opt, bool, whether it supports starting baking tobacco, desc:POST /ISAPI/System/TobaccoBakeMgr/StartBakeTobacco?format=json-->true
  </isSupportStartBakeTobacco>
  <isSupportEndBakeTobacco>
    <!--ro, opt, bool, whether it supports stopping baking tobacco, desc:POST /ISAPI/System/TobaccoBakeMgr/EndBakeTobacco?format=json-->true
  </isSupportEndBakeTobacco>
  <isSupportDryWetBallTempEventRuleParams>
    <!--ro, opt, bool, whether it supports the dry-wet bulb temperature reporting rule parameters, desc:GET/PUT
/ISAPI/System/TobaccoBakeMgr/DryWetBallTempEventRuleParams?format=json-->true
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    </isSupportDryWetBallTempEventRuleParams>
    <isSupportDevAccessServerNetworkStatus>
      <!--ro, opt, bool, whether it supports device connection status with multi-access service network, desc:GET /ISAPI/System/DevAccessServerNetworkStatus?
format=json-->true
    </isSupportDevAccessServerNetworkStatus>
    <isSupportMaintenanceServiceAuthorization>
      <!--ro, opt, bool, whether it supports device maintenance service authorization information, desc:GET/PUT /ISAPI/System/MaintenanceServiceAuthorization?
format=json-->true
    </isSupportMaintenanceServiceAuthorization>
    <isSupportUploadIndividualCallSignalOperation>
      <!--ro, opt, bool, whether it supports uploading single-mode call signaling events, desc:UploadIndividualCallSignalOperation-->true
    </isSupportUploadIndividualCallSignalOperation>
    <isSupportUploadGroupCallSignalOperation>
      <!--ro, opt, bool, whether it supports uploading group call signaling event, desc:UploadGroupCallSignalOperation-->true
    </isSupportUploadGroupCallSignalOperation>
    <isSupportDrivingBehaviorVoiceReminder>
      <!--ro, opt, bool, whether it supports configuring intelligent voice reminders for driving behavior, desc:GET/PUT
/ISAPI/Event/DrivingBehaviorVoiceReminder?format=json-->true
    </isSupportDrivingBehaviorVoiceReminder>
    <isSupportImportMultiCameraCalibratationFile>
      <!--ro, opt, bool, whether it supports importing image calibration files of multiple cameras, desc:POST
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/ImportMultiCameraCalibratationFile?format=json-->true
    </isSupportImportMultiCameraCalibratationFile>
    <isSupportDeletePlayerFileList>
      <!--ro, opt, bool, whether it supports batch deleting player files, desc:POST /ISAPI/System/Video/DeletePlayerFileList?format=json-->true
    </isSupportDeletePlayerFileList>
    <isSupportPlayerInfoParams>
      <!--ro, opt, bool, whether it supports player information parameters, desc:GET /ISAPI/System/Video/PlayerInfoParams?format=json-->true
    </isSupportPlayerInfoParams>
    <isSupportBurnPlayerParams>
      <!--ro, opt, bool, whether it supports burning player parameters, desc:GET/PUT /ISAPI/System/Video/BurnPlayerParams?format=json-->true
    </isSupportBurnPlayerParams>
    <isSupportOneKeyVoiceAbnormalDetection>
      <!--ro, opt, bool, whether it supports quick voice exception detection, desc:POST
/ISAPI/System/VoiceAbnormalDetection/channels/<channelID>/OneKeyVoiceAbnormalDetection?format=json-->true
    </isSupportOneKeyVoiceAbnormalDetection>
    <isSupportVoiceAbnormalDetectionRuleParams>
      <!--ro, opt, bool, whether it supports quick voice exception detection parameters, desc:GET/PUT
/ISAPI/System/VoiceAbnormalDetection/channels/<channelID>/VoiceAbnormalDetectionRuleParams?format=json-->true
    </isSupportVoiceAbnormalDetectionRuleParams>
    <isSupportSearchOneKeyVoiceAbnormalDetectionProcess>
      <!--ro, opt, bool, whether it supports searching for the progress of quick voice exception detection, desc:POST
/ISAPI/System/VoiceAbnormalDetection/SearchOneKeyVoiceAbnormalDetectionProcess?format=json-->true
    </isSupportSearchOneKeyVoiceAbnormalDetectionProcess>
    <isSupportSearchOneKeyAudioDetection>
      <!--ro, opt, bool, whether it supports searching for the progress of quick audio detection, desc:POST
/ISAPI/Smart/AudioExceptionDetection/SearchOneKeyAudioDetection?format=json-->true
    </isSupportSearchOneKeyAudioDetection>
    <isSupportOneKeyAudioDetection>
      <!--ro, opt, bool, whether it supports quick audio detection, desc:POST /ISAPI/Smart/AudioExceptionDetection/channels/<channelID>/OneKeyAudioDetection?
format=json-->true
    </isSupportOneKeyAudioDetection>
    <isNotSupportScheduledRecordUpload>
      <!--ro, opt, bool, whether it supports the event of scheduled recording of channel, desc:if this node does not exist or returns false (default value),
it indicates for support; if it returns true, it indicates no-->false
    </isNotSupportScheduledRecordUpload>
    <isSupportDevEnvironmentalCtrlMaterielInfoDetect>
      <!--ro, opt, bool, whether it supports the detection of equipment environmental control material information?, desc:POST
/ISAPI/System/DeviceModuleDiagnosis/DevEnvironmentalCtrlMaterielInfoDetect?format=json-->true
    </isSupportDevEnvironmentalCtrlMaterielInfoDetect>
    <isSupportWaterRescueDetectionRuleParams>
      <!--ro, opt, bool, whether it supports water rescue detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/WaterRescueDetectionRuleParams?format=json-->true
    </isSupportWaterRescueDetectionRuleParams>
    <isSupportWaterRescueDetectionEvent>
      <!--ro, opt, bool, whether it supports water rescue detection event, desc:corresponding event: WaterRescueDetectionEvent-->true
    </isSupportWaterRescueDetectionEvent>
    <isSupportGetActualDryWetBallTempCurve>
      <!--ro, opt, bool, whether it supports getting the actual wet-bulb temperature curve, desc:POST
/ISAPI/System/TobaccoBakeMgr/GetActualDryWetBallTempCurve?format=json-->true
    </isSupportGetActualDryWetBallTempCurve>
    <isSupportPerimeterAlgoTargetFilterParams>
      <!--ro, opt, bool, whether it supports target filtering parameters of perimeter algorithm, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterAlgoMgr/PerimeterAlgoTargetFilterParams?format=json-->true
    </isSupportPerimeterAlgoTargetFilterParams>
    <isSupportSendDeviceMaintenanceStatus>
      <!--ro, opt, bool, whether it supports applying device maintenance status, desc:PUT /ISAPI/System/SendDeviceMaintenanceStatus?format=json-->true
    </isSupportSendDeviceMaintenanceStatus>
    <videoEncodeChannel opt="1,2,3,4,5,6">
      <!--ro, opt, string, channel supporting video parameter display on the web, attr:opt{req, string}, desc:N/A-->1,2
    </videoEncodeChannel>
    <isSupportLocalButtonRecordControl>
      <!--ro, opt, bool, whether it supports controlling local recording button on the device, desc:/ISAPI/System/localButtonRecord?format=json-->true
    </isSupportLocalButtonRecordControl>
    <isSupportLocalButtonSnapshotControl>
      <!--ro, opt, bool, whether it supports controlling local capture button on the device, desc:/ISAPI/System/localButtonSnapshot?format=json-->true
    </isSupportLocalButtonSnapshotControl>
    <isSupportSecurityDisclaimerParams>
      <!--ro, opt, bool, whether it supports the security disclaimer parameters, desc:GET/PUT /ISAPI/System/InformedConsentMgr/SecurityDisclaimerParams?
format=json-->true
    </isSupportSecurityDisclaimerParams>
    <isSupportThermometryGatewayParams>
      <!--ro, opt, bool, whether it supports configuring parameters of the temperature measurement gateway, desc:GET/PUT
/ISAPI/Thermal/ThermometryGatewayParams?format=json-->true
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/ISAPI/Thermal/ThermometryGatewayParams?format=json">true
  </isSupportThermometryGatewayParams>
  <isSupportSearchRegionDefaultInfo>
    <!--ro, opt, bool, whether it supports searching for the default parameters for the region, desc:POST /ISAPI/System/SearchRegionDefaultInfo?format=json-
->true
  </isSupportSearchRegionDefaultInfo>
  <isSupportAntiCollisionWarningEvent>
    <!--ro, opt, bool, whether it supports anti-collision pre-alarm event, desc:AntiCollisionWarningEvent-->true
  </isSupportAntiCollisionWarningEvent>
  <isSupportMovementWorkStatus>
    <!--ro, opt, bool, whether it supports gas cloud module working status, desc:GET /ISAPI/System/MovementMgr/MovementWorkStatus?format=json-->true
  </isSupportMovementWorkStatus>
  <isSupportMovementTimingPowerOffParams>
    <!--ro, opt, bool, whether it supports scheduled shutdown parameters of the gas cloud imaging module, desc:GET/PUT
/ISAPI/System/MovementMgr/MovementTimingPowerOffParams?format=json-->true
  </isSupportMovementTimingPowerOffParams>
  <isSupportFireExtinguisherDetectionRuleParams>
    <!--ro, opt, bool, whether it supports fire extinguisher detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/FireExtinguisherDetectionRuleParams?format=json-->true
  </isSupportFireExtinguisherDetectionRuleParams>
  <isSupportFireExtinguisherRecoveryEvent>
    <!--ro, opt, bool, whether it supports fire extinguisher detection restoring event, desc:eventType:FireExtinguisherRecoveryEvent-->true
  </isSupportFireExtinguisherRecoveryEvent>
  <isSupportGasTankMonitoringRecoveryEvent>
    <!--ro, opt, bool, whether it supports gas tank detection restoring event, desc:eventType:GasTankMonitoringRecoveryEvent-->true
  </isSupportGasTankMonitoringRecoveryEvent>
  <isSupportSearchImportPlayerFileProcess>
    <!--ro, opt, bool, whether it supports searching for the progress of importing player files, desc:POST
/ISAPI/System/Video/SearchImportPlayerFileProcess?format=json-->true
  </isSupportSearchImportPlayerFileProcess>
  <isSupportStatisticalDataFlashStorageParams>
    <!--ro, opt, bool, whether it supports data statistics for Flash storage parameters, desc:GET/PUT
/ISAPI/System/dataAware/StatisticalDataFlashStorageParams?format=json-->true
  </isSupportStatisticalDataFlashStorageParams>
  <eventRecheckCap>
    <!--ro, opt, object, event review capability, desc:low-end cameras depend on powerful NVRs, due to the limitation of computing power for secondary
review of low-end camera alarms to reduce false alarms-->
    <algorithmInfoList>
      <!--ro, opt, array, list of event detection algorithm information, subType:object, range:[1,5]-->
      <algorithmInfoitem>
        <!--ro, req, object, event detection algorithm element-->
        <algorithmType>
          <!--ro, req, enum, event detection algorithm type, subType:string, desc:"VMD", "regionEntrance", "regionExiting", "linedetection" (line crossing),
"fielddetection" (intrusion detection)-->VMD
        </algorithmType>
        <algorithmVersion>
          <!--ro, req, string, event detection algorithm version, desc:N/A-->1B
        </algorithmVersion>
      </algorithmInfoitem>
    </algorithmInfoList>
  </eventRecheckCap>
  <isSupportModifyWindowGroup>
    <!--ro, opt, bool, whether it supports modifying the specified window group of the scope, desc:PUT
/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/ModifyWindowGroup?format=json-->true
  </isSupportModifyWindowGroup>
  <isSupportCalibrateWindowGroup>
    <!--ro, opt, bool, whether it supports calibrating specified window group of the scope, desc:PUT
/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/CalibrateWindowGroup?format=json-->true
  </isSupportCalibrateWindowGroup>
  <isSupportAllWindowParams>
    <!--ro, opt, bool, whether it supports all scope window parameters, desc:GET /ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/AllWindowParams?
format=json-->true
  </isSupportAllWindowParams>
  <isSupportWindowCalibrationStatus>
    <!--ro, opt, bool, whether it supports getting the scope window calibration status, desc:GET
/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/WindowCalibrationStatus?format=json-->true
  </isSupportWindowCalibrationStatus>
  <isSupportOBUModuleInfo>
    <!--ro, opt, bool, whether it supports OBU module information, desc:GET /ISAPI/Traffic/ETC/OBUModuleInfo?format=json-->true
  </isSupportOBUModuleInfo>
  <isSupportExportOBUDetectionLogFile>
    <!--ro, opt, bool, whether it supports exporting the OBU label detection module logs, desc:POST /ISAPI/Traffic/ETC/ExportOBUDetectionLogFile?
format=json-->true
  </isSupportExportOBUDetectionLogFile>
  <isSupportOBUDetectionReboot>
    <!--ro, opt, bool, whether it supports restarting OBU label detection module, desc:PUT /ISAPI/Traffic/ETC/OBUDetectionReboot?format=json-->true
  </isSupportOBUDetectionReboot>
  <isSupportGetOBURebootStatus>
    <!--ro, opt, bool, whether it supports getting the restart status of the OBU module, desc:GET /ISAPI/Traffic/ETC/GetOBURebootStatus?format=json-->true
  </isSupportGetOBURebootStatus>
  <isSupportSearchOBUStatisticsReport>
    <!--ro, opt, bool, whether it supports searching for the statistical data of OBU tag detection, desc:POST /ISAPI/Traffic/ETC/SearchOBUStatisticsReport?
format=json-->true
  </isSupportSearchOBUStatisticsReport>
  <isSupportSearchTargetEnhancementInfo>
    <!--ro, opt, bool, whether it supports searching for target prominence parameters, desc:POST
/ISAPI/System/Video/TargetEnhancement/channels/<channelID>/SearchTargetEnhancementInfo?format=json-->true
  </isSupportSearchTargetEnhancementInfo>
  <isSupportTargetEnhancementParams>
    <!--ro, opt, bool, whether it supports target prominence parameters, desc:GET/PUT
/ISAPI/System/Video/TargetEnhancement/channels/<channelID>/TargetEnhancementParams?format=json-->true
  </isSupportTargetEnhancementParams>
  <isSupportPicSearchByCoordinate>
    <!--ro, opt, bool, whether the device supports configuring AcuSearch parameters, desc:GET/PUT /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json--
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>true
  </isSupportPicSearchByCoordinate>
  <AcuSearchCap>
    <!--ro, opt, object, AcuSearch capability-->
    <targetType opt="face,human,vehicle">
      <!--ro, opt, enum, supported target types of AcuSearch, subType:string, attr:opt{req, string}, desc:"face", "human", "vehicle"-->face
    </targetType>
    <dataType opt="url,binary,modelData">
      <!--ro, opt, enum, supported image transmission format types, subType:string, attr:opt{req, string}, desc:"url", "binary", "modelData"-->url
    </dataType>
    <maxPicSize>
      <!--ro, opt, int, maximum size of image can be searched, range:[0,5], desc:unit: MB-->1
    </maxPicSize>
    <maxPicWidthSize>
      <!--ro, opt, int, maximum width of image can be searched, range:[0,4096]-->4096
    </maxPicWidthSize>
    <maxPicHeightSize>
      <!--ro, opt, int, maximum height of image can be searched, range:[0,4096]-->4096
    </maxPicHeightSize>
  </AcuSearchCap>
  <isSupportSendDevRemoteGUICtrl>
    <!--ro, opt, bool, whether it supports applying parameter settings of remote GUI control of devices, desc:POST
/ISAPI/System/RemoteCtrlDevGUI/SendDevRemoteGUICtrl?format=json-->true
  </isSupportSendDevRemoteGUICtrl>
  <isSupportStreamTemperatureParams>
    <!--ro, opt, bool, whether the device supports configuring stream temperature parameters of thermal imaging, desc:GET/PUT
/ISAPI/Thermal/channels/<channelID>/StreamTemperatureParams?format=json-->true
  </isSupportStreamTemperatureParams>
  <moduleTypeSupportCap>
    <!--ro, opt, object, capability supported by device module-->
    <isSupportMaintenanceModule>
      <!--ro, opt, bool, whether it supports maintenance module, desc:the wire products are embedded with the business module (businessModule) and
maintenance module (maintenanceModule), which supports independent access to EZVIZ cloud, etc.; at the same time, they support the cloud platform to send
maintenance instructions to the maintenance module-->true
    </isSupportMaintenanceModule>
  </moduleTypeSupportCap>
  <isSupportSearchEventResources>
    <!--ro, opt, bool, whether it supports searching for event resource information, desc:POST /ISAPI/Event/EventBatchRulesMgr/SearchEventResources?
format=json-->true
  </isSupportSearchEventResources>
  <isSupportRestorePerimeterAlgoDefaultParam>
    <!--ro, opt, bool, whether it supports restoring parameters of perimeter algorithm, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterAlgoMgr/RestorePerimeterAlgoDefaultParam?format=json-->true
  </isSupportRestorePerimeterAlgoDefaultParam>
  <isSupportGetDeviceMaintenanceStatus>
    <!--ro, opt, bool, whether it supports getting device maintenance status, desc:GET /ISAPI/System/GetDeviceMaintenanceStatus?format=json-->true
  </isSupportGetDeviceMaintenanceStatus>
  <isSupportImportConfigurationFileByModel>
    <!--ro, opt, bool, whether it supports importing configuration files by device model, desc:POST /ISAPI/System/ImportConfigurationFileByModel?
format=json-->true
  </isSupportImportConfigurationFileByModel>
  <isSupportRadarVideoDetectionStorageParams>
    <!--ro, opt, bool, whether it supports the parameters of saving radar-assisted target detection result, desc:GET/PUT
/ISAPI/Intelligent/radarVideoDetection/RadarVideoDetectionStorageParams?format=json-->true
  </isSupportRadarVideoDetectionStorageParams>
  <isSupportBodyCameraMediaInfo>
    <!--ro, opt, bool, whether it supports body camera reporting media information, desc:eventType: BodyCameraMediaInfo-->true
  </isSupportBodyCameraMediaInfo>
  <isSupportRadarVideoPOSOverlayParams>
    <!--ro, opt, bool, whether it supports smart radar-assisted POS information overlay parameters, desc:GET/PUT
/ISAPI/Intelligent/radarVideoDetection/RadarVideoPOSOverlayParams?format=json-->true
  </isSupportRadarVideoPOSOverlayParams>
  <isSupportDeleteNetworkCaptureFile>
    <!--ro, opt, bool, whether it supports deleting captured packet files, desc:POST /ISAPI/System/DeleteNetworkCaptureFile?format=json-->true
  </isSupportDeleteNetworkCaptureFile>
  <isSupportRecoverCustomEventTemplate>
    <!--ro, opt, bool, whether it supports restoring a custom event message template, desc:POST
/ISAPI/Event/notification/httpHosts/RecoverCustomEventTemplate?format=json-->true
  </isSupportRecoverCustomEventTemplate>
  <isSupportModCustomEventTemplateParams>
    <!--ro, opt, bool, whether it supports modifying a specified template of custom event message, desc:PUT
/ISAPI/Event/notification/httpHosts/ModCustomEventTemplateParams?format=json-->true
  </isSupportModCustomEventTemplateParams>
  <isSupportSearchAbnormalStreamInfo>
    <!--ro, opt, bool, whether it supports searching for streaming exception information, desc:POST /ISAPI/Streaming/SearchAbnormalStreamInfo?format=json--
>true
  </isSupportSearchAbnormalStreamInfo>
  <isSupportAbnormalStreamEvent>
    <!--ro, opt, bool, whether it supports real-time streaming exception report-->true
  </isSupportAbnormalStreamEvent>
  <isSupportBatchDeleteNetworkCaptureFile>
    <!--ro, opt, bool, whether it supports batch deleting captureed packet files, desc:PUT /ISAPI/System/BatchDeleteNetworkCaptureFile?format=json-->true
  </isSupportBatchDeleteNetworkCaptureFile>
  <isSupportWindowRuleParams>
    <!--ro, opt, bool, whether it supports the scope window rule parameters, desc:GET/PUT
/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/WindowRuleParams?format=json-->true
  </isSupportWindowRuleParams>
  <isSupportGetMultiRemoteAlbumPicture>
    <!--ro, opt, bool, whether the device supports getting pictures from the remote album under specified path, desc:POST
/ISAPI/Thermal/GetMultiRemoteAlbumPicture?format=json-->true
  </isSupportGetMultiRemoteAlbumPicture>
  <isSupportGetDeviceTokenInfo>
    <!--ro, opt, bool, whether the device supports getting the device token information, desc:/ISAPI/System/ImportConfigurationFileByURL?format=json
/ISAPI/System/DeviceAuth/GetDeviceTokenInfo?format=json-->true
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/ISAPI/System/DeviceAuth/GetDeviceTokenInfo?format=json-->true
    </isSupportGetDeviceTokenInfo>
    <isSupportDisplayEnableCloudServicePrompt>
      <!--ro, opt, bool, whether it supports displaying the prompt for enabling cloud services, desc:false by default. When the device returns true, it indicates that displaying a prompt of enabling cloud services is supported. Scenario example: After your device is connected to the Internet and the cloud service is enabled, we will provide you with better cloud device management and maintenance upgrade services. Therefore, we will collect relevant device information such as the device's IP address, device name, and device serial No. If you have any questions, you can contact customer service at any time-->true
    </isSupportDisplayEnableCloudServicePrompt>
    <isSupportFiredoorDetectionEvent>
      <!--ro, opt, bool, whether it supports fireproof door detection event, desc:eventType:FiredoorDetectionEvent-->true
    </isSupportFiredoorDetectionEvent>
    <isSupportImportConfigurationFileByURL>
      <!--ro, opt, bool, whether it supports importing configuration file via URL, desc:POST /ISAPI/System/ImportConfigurationFileByURL?format=json-->true
    </isSupportImportConfigurationFileByURL>
    <isSupportPlayCellphoneRecoveryEvent>
      <!--ro, opt, bool, whether it supports playing mobile restoring event, desc:eventType:PlayCellphoneRecoveryEvent-->true
    </isSupportPlayCellphoneRecoveryEvent>
    <isSupportDetectorShelterAlarmRecovery>
      <!--ro, opt, bool, whether it supports detector occlusion detection restoring event, desc:eventType:DetectorShelterAlarmRecovery-->true
    </isSupportDetectorShelterAlarmRecovery>
    <isSupportExportThermalMovementFile>
      <!--ro, opt, bool, whether it supports exporting the device temperature measuring module template, desc:POST /ISAPI/Thermal/ExportThermalMovementFile?format=json-->true
    </isSupportExportThermalMovementFile>
    <isSupportUPSAbnormalAutoShutDownParams>
      <!--ro, opt, bool, whether it supports parameters of UPS power supply exception and auto power off, desc:GET/PUT /ISAPI/System/UPSAbnormalAutoShutDownParams?format=json-->true
    </isSupportUPSAbnormalAutoShutDownParams>
    <isSupportThermalLensDriveParams>
      <!--ro, opt, bool, whether it supports configuring thermal lens driver parameters, desc:GET/PUT /ISAPI/Thermal/ThermalLensDriveParams?format=json-->true
    </isSupportThermalLensDriveParams>
    <isSupportBatchDeleteUserInfoCfg>
      <!--ro, opt, bool, whether it supports batch deleting user information configurations, desc:PUT /ISAPI/Security/users/BatchDeleteUserInfoCfg?format=json-->true
    </isSupportBatchDeleteUserInfoCfg>
    <isSupportAddUserGroup>
      <!--ro, opt, bool, whether it supports adding user groups, desc:POST /ISAPI/System/UserGroupingMgr/AddUserGroup?format=json-->true
    </isSupportAddUserGroup>
    <isSupportModifyUserGroup>
      <!--ro, opt, bool, whether it supports editing user groups, desc:PUT /ISAPI/System/UserGroupingMgr/ModifyUserGroup?format=json-->true
    </isSupportModifyUserGroup>
    <isSupportSearchUserGroup>
      <!--ro, opt, bool, whether it supports searching user groups, desc:POST /ISAPI/System/UserGroupingMgr/SearchUserGroup?format=json-->true
    </isSupportSearchUserGroup>
    <isSupportDeleteUserGroup>
      <!--ro, opt, bool, whether it supports deleting user groups, desc:PUT /ISAPI/System/UserGroupingMgr/DeleteUserGroup?format=json-->true
    </isSupportDeleteUserGroup>
    <isSupportReadUpgradePackageData>
      <!--ro, opt, bool, whether it supports notifying the device to read upgrade package data, desc:PUT /ISAPI/System/SystemUpgrade/ReadUpgradePackageData?format=json-->true
    </isSupportReadUpgradePackageData>
    <isSupportWebDisplayThermometryDistanceTips>
      <!--ro, opt, bool, whether it supports displaying the prompt of temperature measurement distance on the web page, desc:the temperature measurement distance not only affects the temperature measurement results, but also affects the accuracy of temperature measurement location-->true
    </isSupportWebDisplayThermometryDistanceTips>
    <isSupportFlashLifespanProtectParams>
      <!--ro, opt, bool, whether it supports flash life protection parameter, desc:GET/PUT /ISAPI/System/FLASHMgr/FlashLifespanProtectParams?format=json-->true
    </isSupportFlashLifespanProtectParams>
    <isSupportIlluminanceTelemeter>
      <!--ro, opt, bool, whether it supports light intensity telemetry, desc:eventType:IlluminanceTelemeter-->true
    </isSupportIlluminanceTelemeter>
    <isSupportWindDirectionInDegreesTelemeter>
      <!--ro, opt, bool, whether it supports wind direction telemetry, desc:eventType:WindDirectionInDegreesTelemeter-->true
    </isSupportWindDirectionInDegreesTelemeter>
    <isSupportWindSpeedTelemeter>
      <!--ro, opt, bool, whether it supports wind telemetry, desc:eventType:WindSpeedTelemeter-->true
    </isSupportWindSpeedTelemeter>
    <isSupportAirPressureTelemeter>
      <!--ro, opt, bool, whether it supports air pressure telemetry, desc:eventType:AirPressureTelemeter-->true
    </isSupportAirPressureTelemeter>
    <isSupportBatteryPercentageTelemeter>
      <!--ro, opt, bool, whether it supports battery telemetry, desc:eventType:BatteryPercentageTelemeter-->true
    </isSupportBatteryPercentageTelemeter>
    <isSupportTemperaturesTelemeter>
      <!--ro, opt, bool, whether it supports temperature telemetry, desc:eventType:TemperaturesTelemeter-->true
    </isSupportTemperaturesTelemeter>
    <isSupportHumiditysTelemeter>
      <!--ro, opt, bool, whether it supports humidity telemetry, desc:eventType:HumiditysTelemeter-->true
    </isSupportHumiditysTelemeter>
    <isSupportDipAngleTelemeter>
      <!--ro, opt, bool, whether it supports angle telemetry, desc:eventType:DipAngleTelemeter-->true
    </isSupportDipAngleTelemeter>
    <isSupportLocationEvent>
      <!--ro, opt, bool, whether it supports the device location information uploading event, desc:eventType:Location-->true
    </isSupportLocationEvent>
    <isSupportSoilMoistureTelemeter>
      <!--ro, opt, bool, whether it supports soil moisture telemetry, desc:eventType:SoilMoistureTelemeter-->true
    </isSupportSoilMoistureTelemeter>
    <isSupportLitterTelemeter>
      <!--ro, opt, bool, whether it supports litter telemetry, desc:eventType: LitterTelemeter-->true
    </isSupportLitterTelemeter>
    <isSupportSearchRealStreamStatus>
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    <!--ro, opt, bool, whether it supports searching for the real-time streaming status, desc:POST /ISAPI/Streaming/SearchRealStreamStatus?format=json-->true
  </isSupportSearchRealStreamStatus>
  <isSupportSearchStreamBandwidthInfo>
    <!--ro, opt, bool, whether it supports searching for stream bandwidth status, desc:POST /ISAPI/Streaming/SearchStreamBandwidthInfo?format=json-->true
  </isSupportSearchStreamBandwidthInfo>
  <isSupportExportThermometryData>
    <!--ro, opt, bool, whether it supports exporting visible light temperature measurement data, desc:POST /ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ExportThermometryData?format=json-->true
  </isSupportExportThermometryData>
  <isSupportModifyClickToThermometry>
    <!--ro, opt, bool, whether it supports configuring parameters of visible light click-to-thermometry rules, desc:PUT /ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ModifyClickToThermometry?format=json-->true
  </isSupportModifyClickToThermometry>
  <isSupportExpertThermometryAlarmParams>
    <!--ro, opt, bool, whether it supports alarm parameters of temperature measurement in expert mode, desc:GET/PUT /ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ExpertThermometryAlarmParams?format=json-->true
  </isSupportExpertThermometryAlarmParams>
  <isSupportExpertThermometryRuleParams>
    <!--ro, opt, bool, whether it supports the rule parameters of temperature measurement in expert mode, desc:GET/PUT /ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ExpertThermometryRuleParams?format=json-->true
  </isSupportExpertThermometryRuleParams>
  <isSupportNormalThermometryRuleParams>
    <!--ro, opt, bool, whether it supports the rule parameters of temperature measurement in normal mode, desc:GET/PUT /ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/NormalThermometryRuleParams?format=json-->true
  </isSupportNormalThermometryRuleParams>
  <isSupportModeParams>
    <!--ro, opt, bool, whether it supports mode parameters of visible light radiation temperature measurement, desc:GET/PUT /ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ModeParams?format=json-->true
  </isSupportModeParams>
  <isSupportBasicParams>
    <!--ro, opt, bool, whether it supports basic parameters of visible light radiation temperature measurement, desc:GET/PUT /ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/BasicParams?format=json-->true
  </isSupportBasicParams>
  <isSupportImportPlayerFile>
    <!--ro, opt, bool, whether it supports importing player files, desc:PUT /ISAPI/System/Video/ImportPlayerFile?format=json-->true
  </isSupportImportPlayerFile>
  <isSupportSearchQuickQRCode>
    <!--ro, opt, bool, whether it supports searching for quick configuration codes, desc:POST /ISAPI/System/QuickConfigQRCodeMgr/SearchQuickQRCode?format=json-->true
  </isSupportSearchQuickQRCode>
  <isSupportQuickQRCodeEncryptionParams>
    <!--ro, opt, bool, whether it supports the encryption parameters of quick configuration codes, desc:PUT /ISAPI/System/QuickConfigQRCodeMgr/QuickQRCodeEncryptionParams?format=json-->true
  </isSupportQuickQRCodeEncryptionParams>
  <isUnsynchronizedUpgradeKVMIntput>
    <!--ro, opt, bool, whether it supports upgrading KVM input node synchronously, desc:1. For distributed main control, if this field is not returned or returned as false, the KVM input node added to the main control will be upgraded. 2. If this field is returned as true, the KVM input node will not be upgraded.-->true
  </isUnsynchronizedUpgradeKVMIntput>
  <isSupportWaterRescueDetectionOverlapParams>
    <!--ro, opt, bool, whether it supports intelligent information overlay parameters of water rescue, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/WaterRescueDetectionOverlapParams?format=json-->true
  </isSupportWaterRescueDetectionOverlapParams>
  <isSupportSearchRecordStatusList>
    <!--ro, opt, bool, whether it supports searching for the channel recording status, desc:POST /ISAPI/ContentMgmt/search/SearchRecordStatusList?format=json-->true
  </isSupportSearchRecordStatusList>
  <isSupportWiperSelfCheckParams>
    <!--ro, opt, bool, whether it supports the wiper self-checking parameters, desc:GET/PUT /ISAPI/System/Hardware/channels/<channelID>/WiperSelfCheckParams?format=json-->true
  </isSupportWiperSelfCheckParams>
  <isSupportBlackoutNotification>
    <!--ro, opt, bool, whether it supports power-off notification, desc:PUT /ISAPI/System/channels/<channelID>/BlackoutNotification?format=json-->true
  </isSupportBlackoutNotification>
  <isSupportGotoWaterFlowScene>
    <!--ro, opt, bool, whether it supports jumping to the specified water flow detection scene, desc:POST /ISAPI/Event/channels/<channelID>/GotoWaterFlowScene?format=json-->true
  </isSupportGotoWaterFlowScene>
  <isSupportHVTVehicleDetectUploadChannel>
    <!--ro, opt, bool, whether it supports configuring the channel of uploading mixed-traffic detection information to the FTP server, desc:GET/PUT /ISAPI/System/Network/ftp/HVTVehicleDetectUpload?format=json&channelID=<channelID>-->true
  </isSupportHVTVehicleDetectUploadChannel>
  <isSupportWaterFlowAdvancedParams>
    <!--ro, opt, bool, whether it supports advanced parameters of water flow detection, desc:GET/PUT /ISAPI/Event/channels/<channelID>/WaterFlowAdvancedParams?format=json-->true
  </isSupportWaterFlowAdvancedParams>
  <isSupportWaterFlowSceneList>
    <!--ro, opt, bool, whether it supports water flow detection scene list, desc:GET /ISAPI/Event/channels/<channelID>/WaterFlowSceneList?format=json-->true
  </isSupportWaterFlowSceneList>
  <isSupportWaterFlowSceneRuleParams>
    <!--ro, opt, bool, whether it supports water flow detection scene rule parameters, desc:PUT /ISAPI/Event/channels/<channelID>/WaterFlowSceneRuleParams?format=json-->true
  </isSupportWaterFlowSceneRuleParams>
  <isSupportRainfallScheduledUpload>
    <!--ro, opt, bool, whether it supports scheduled rainfall information uploading event, desc:RainfallScheduledUpload-->true
  </isSupportRainfallScheduledUpload>
  <isSupportFlowVelocityScheduledUpload>
    <!--ro, opt, bool, whether it supports scheduled water flow speed uploading event, desc:FlowVelocityScheduledUpload-->true
  </isSupportFlowVelocityScheduledUpload>
  <isSupportFlowVelocityAlarm>
    <!--ro, opt, bool, whether it supports water flow speed alarm event, desc:FlowVelocityAlarm-->true
  </isSupportFlowVelocityAlarm>
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<isSupportVolumeFlowRateScheduledUpload>
<!--ro, opt, bool, whether it supports scheduled water flow rate uploading event, desc:VolumeFlowRateScheduledUpload-->true
</isSupportVolumeFlowRateScheduledUpload>
<isSupportVolumeFlowRateAlarm>
<!--ro, opt, bool, whether it supports water flow rate alarm event, desc:VolumeFlowRateAlarm-->true
</isSupportVolumeFlowRateAlarm>
<isSupportSecurityDisclaimerResultEvent>
<!--ro, opt, bool, whether it supports security disclaimer result uploading, desc:SecurityDisclaimerResultEvent-->true
</isSupportSecurityDisclaimerResultEvent>
<isSupportInfusionAlarmEvent>
<!--ro, opt, bool, whether it supports infusion alarm event, desc:InfusionAlarmEvent-->true
</isSupportInfusionAlarmEvent>
<isSupportBodyCameraBatteryTamperEvent>
<!--ro, opt, bool, whether it supports body camera battery tampering event, desc:eventType:BodyCameraBatteryTamperEvent-->true
</isSupportBodyCameraBatteryTamperEvent>
<isSupportShipSafeOperationDetectionEvent>
<!--ro, opt, bool, whether it supports ship operation safety event detection, desc:eventType:ShipSafeOperationDetectionEvent-->true
</isSupportShipSafeOperationDetectionEvent>
<isSupportTrafficLightStatusDetectionEvent>
<!--ro, opt, bool, whether it supports traffic light status detection event, desc:eventType:TrafficLightStatusDetectionEvent-->true
</isSupportTrafficLightStatusDetectionEvent>
<isSupportSoundPressureDetectionEvent>
<!--ro, opt, bool, whether it supports triggering overload sound pressure alarm, desc:eventType:SoundPressureDetectionEvent-->true
</isSupportSoundPressureDetectionEvent>
<isSupportInvigilatorBehaviorRuleParams>
<!--ro, opt, bool, whether it supports driving license exam proctoring behavior detection rule parameters, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/InvigilatorBehaviorRuleParams?format=json-->true
</isSupportInvigilatorBehaviorRuleParams>
<isSupportInvigilatorBehaviorEvent>
<!--ro, opt, bool, whether it supports driving license exam proctoring behavior detection event, desc:eventType: InvigilatorBehaviorEvent-->true
</isSupportInvigilatorBehaviorEvent>
<isSupportPeopleSleepingDetectionReport>
<!--ro, opt, bool, whether it supports sleep detection report event, desc:eventType:PeopleSleepingDetectionReport-->true
</isSupportPeopleSleepingDetectionReport>
<isSupportPhysicalMatchMgr>
<!--ro, opt, bool, whether it supports sports competitions management, desc:GET /ISAPI/Intelligent/channels/<channelID>/PhysicalMatchMgr/capabilities?
format=json-->true
</isSupportPhysicalMatchMgr>
<isSupportSearchNetworkResources>
<!--ro, opt, bool, whether it supports searching for network resource information, desc:POST
/ISAPI/System/Network/NetworkAdvancedCfg/SearchNetworkResources?format=json-->true
</isSupportSearchNetworkResources>
<isSupportTriggerPeripheralAbnormalDetection>
<!--ro, opt, bool, whether it supports triggering the device to check component fault, desc:PUT /ISAPI/Event/TriggerPeripheralAbnormalDetection?
format=json-->true
</isSupportTriggerPeripheralAbnormalDetection>
<isSupportDeviceCabinetDoorOpenAlarm>
<!--ro, opt, bool, whether it support device chassis open alarm event, desc:eventType:DeviceCabinetDoorOpenAlarm-->true
</isSupportDeviceCabinetDoorOpenAlarm>
<isSupportSurplusPower>
<!--ro, opt, bool, whether it supports remaining battery information, desc:GET/PUT /ISAPI/System/SurplusPower?format=json-->true
</isSupportSurplusPower>
<isSupportPeripheralAbnormalScheduledUpload>
<!--ro, opt, bool, whether it supports scheduled device component fault uploading event, desc:eventType:PeripheralAbnormalScheduledUpload-->true
</isSupportPeripheralAbnormalScheduledUpload>
<isSupportBandwidthLimitWarningParams>
<!--ro, opt, bool, whether it supports the bandwidth limit alarm parameters, desc:GET/PUT
/ISAPI/System/Network/NetworkFlowMonitor/BandwidthLimitWarningParams?format=json-->true
</isSupportBandwidthLimitWarningParams>
<isSupportRoadMaintSceneList>
<!--ro, opt, bool, whether it supports road maintenance scene list, desc:GET /ISAPI/Intelligent/channels/<channelID>/RoadMaintSceneList?format=json--
>true
</isSupportRoadMaintSceneList>
<isSupportSearchIntelligencePerformance>
<!--ro, opt, bool, whether it supports searching post-intelligence performance parameters, desc:POST
/ISAPI/System/VCAResource/SearchIntelligencePerformance?format=json-->true
</isSupportSearchIntelligencePerformance>
<isSupportRainfallRuleParams>
<!--ro, opt, bool, whether it supports rainfall detection rule parameters, desc:GET/PUT /ISAPI/Event/channels/<channelID>/RainfallRuleParams?
format=json-->true
</isSupportRainfallRuleParams>
<isSupportRadarFlowmeterOverlapParams>
<!--ro, opt, bool, whether it supports radar detection information overlay parameters, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/RadarFlowmeterOverlapParams/?format=json-->true
</isSupportRadarFlowmeterOverlapParams>
<isSupportCurrentComStatus>
<!--ro, opt, bool, whether it supports getting the device communication status, desc:GET /ISAPI/System/CurrentComStatus?format=json-->true
</isSupportCurrentComStatus>
<isSupportRoadMaintAlgorithmVersion>
<!--ro, opt, bool, whether it supports road maintenance algorithm version, desc:GET /ISAPI/Intelligent/channels/<channelID>/RoadMaintAlgorithmVersion?
format=json-->true
</isSupportRoadMaintAlgorithmVersion>
<isSupportRoadMaintOverlapParams>
<!--ro, opt, bool, whether it supports road maintenance overlay parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/RoadMaintOverlapParams?
format=json-->true
</isSupportRoadMaintOverlapParams>
<isSupportActionExecuteStatus>
<!--ro, opt, bool, whether it supports action execution status uploading, desc:eventType: ActionExecuteStatus-->true
</isSupportActionExecuteStatus>
<isSupportExportConfigurationFile>
<!--ro, opt, bool, whether it supports exporting the configuration file, desc:GET /ISAPI/System/ExportConfigurationFile?format=json-->true
</isSupportExportConfigurationFile>
<isSupportMultiChannelStreamMergeParams>
<!--ro, opt, bool, whether it supports the parameters of merging the  encoding of multiple panoramic channels, desc:GET/PUT
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/ISAPI/System/Video/inputs/MultiChannelStreamMergeParams?format=json-->true
  </isSupportMultiChannelStreamMergeParams>
  <isSupportAnalysisEngineCfgTest>
    <!--ro, opt, bool, whether it supports engine analysis parameters configuration restarting detection, desc:POST
/ISAPI/System/VCAResource/AnalysisEngineCfgTest?format=json-->true
  </isSupportAnalysisEngineCfgTest>
  <isSupportModifyPlaybackSpeed>
    <!--ro, opt, bool, whether it supports video speed playback of the streaming device, desc:POST
/ISAPI/Streaming/channels/<channelID>/ModifyPlaybackSpeed?format=json-->true
  </isSupportModifyPlaybackSpeed>
  <isSupportSeekToSegsPlayback>
    <!--ro, opt, bool, whether it supports dragging to the time period during streaming video playback, desc:POST
/ISAPI/Streaming/channels/<channelID>/SeekToSegsPlayback?format=json-->true
  </isSupportSeekToSegsPlayback>
  <isSupportIntersectionTrafficFlowStatisticsEvent>
    <!--ro, opt, bool, whether it supports event for counting motor vehicle traffic at an intersection, desc:eventType:
IntersectionTrafficFlowStatisticsEvent-->true
  </isSupportIntersectionTrafficFlowStatisticsEvent>
  <isSupportLensCoverAbnormalEvent>
    <!--ro, opt, bool, whether it supports lens cover abnormal event, desc:event type: LensCoverAbnormalEvent.-->true
  </isSupportLensCoverAbnormalEvent>
  <isSupportCrackMonitoringEvent>
    <!--ro, opt, bool, whether it supports geological cracks detection event, desc:event type: CrackMonitoringEvent-->true
  </isSupportCrackMonitoringEvent>
  <isSupportExportNetworkCaptureFile>
    <!--ro, opt, bool, whether it supports exporting captured packets, desc:POST /ISAPI/System/ExportNetworkCaptureFile?format=json-->true
  </isSupportExportNetworkCaptureFile>
  <isSupportGetNetworkCaptureStatus>
    <!--ro, opt, bool, whether it supports getting captured packet status, desc:POST /ISAPI/System/GetNetworkCaptureStatus?format=json-->true
  </isSupportGetNetworkCaptureStatus>
  <isSupportStartNetworkCapture>
    <!--ro, opt, bool, whether it supports starting capturing network packets, desc:POST /ISAPI/System/StartNetworkCapture?format=json-->true
  </isSupportStartNetworkCapture>
  <isSupportStopNetworkCapture>
    <!--ro, opt, bool, whether it supports stopping capturing network packets, desc:POST /ISAPI/System/StopNetworkCapture?format=json-->true
  </isSupportStopNetworkCapture>
  <isSupportObjectAccumulationAlgorithmParams>
    <!--ro, opt, bool, whether it supports stacking detection algorithm parameters, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/ObjectAccumulationAlgorithmParams?format=json-->true
  </isSupportObjectAccumulationAlgorithmParams>
  <isSupportGetTobaccoStageStatus>
    <!--ro, opt, bool, whether it supports getting tobacco baking status, desc:POST /ISAPI/System/TobaccoBakeMgr/GetTobaccoStageStatus?format=json-->true
  </isSupportGetTobaccoStageStatus>
  <isSupportMultipleIntelligentParams>
    <!--ro, opt, bool, whether it supports multi-intelligent mode parameters, desc:GET/PUT /ISAPI/System/VCAResource/MultipleIntelligentParams?format=json--
>true
  </isSupportMultipleIntelligentParams>
  <isSupportStartSystemEnvDetection>
    <!--ro, opt, bool, whether it supports starting system environment check, desc:POST /ISAPI/System/StartSystemEnvDetection?format=json-->true
  </isSupportStartSystemEnvDetection>
  <isSupportSearchSystemEnvDetectionProcess>
    <!--ro, opt, bool, whether it supports searching for system environment checking progress, desc:POST /ISAPI/System/SearchSystemEnvDetectionProcess?
format=json-->true
  </isSupportSearchSystemEnvDetectionProcess>
  <isSupportEVACInputParams>
    <!--ro, opt, bool, whether it supports EVAC input parameters, desc:GET/PUT /ISAPI/System/EVACInput/EVACInputParams?format=json-->true
  </isSupportEVACInputParams>
  <isSupportModifyT1TestParams>
    <!--ro, opt, bool, whether it editing T1 test parameters, desc:POST /ISAPI/System/AutoTest/ModifyT1TestParams?format=json-->true
  </isSupportModifyT1TestParams>
  <isSupportDetectionRegionRuleParams>
    <!--ro, opt, bool, whether it supports rule parameters of hazardous gas area scanning detection, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/methaneDetection/DetectionRegionRuleParams?format=json-->true
  </isSupportDetectionRegionRuleParams>
  <isSupportFaceApproach>
    <!--ro, opt, bool, whether it supports face approaching event, desc:eventType:FaceApproach-->true
  </isSupportFaceApproach>
  <isSupportHardwareEnergySaving>
    <!--ro, opt, bool, whether it supports device hardware energy-saving parameters, desc:GET/PUT /ISAPI/System/consumptionMode/HardwareEnergySaving?
format=json-->true
  </isSupportHardwareEnergySaving>
  <isSupportDeviceHighTemperature>
    <!--ro, opt, bool, whether it supports device overheat event, desc:eventType:DeviceHighTemperature-->true
  </isSupportDeviceHighTemperature>
  <isSupportExportAudioAndVideoFile>
    <!--ro, opt, bool, whether it supports exporting audio and video files, desc:POST /ISAPI/System/ExportAudioAndVideoFile?format=json-->true
  </isSupportExportAudioAndVideoFile>
  <isSupportGetExportAudioAndVideoFileStatus>
    <!--ro, opt, bool, whether it supports getting the export status of video and audio files, desc:GET /ISAPI/System/GetExportAudioAndVideoFileStatus?
format=json-->true
  </isSupportGetExportAudioAndVideoFileStatus>
  <isSupportStartExportAudioAndVideoFile>
    <!--ro, opt, bool, whether it supports starting exporting audio and video files, desc:PUT /ISAPI/System/StartExportAudioAndVideoFile?format=json-->true
  </isSupportStartExportAudioAndVideoFile>
  <isSupportStopExportAudioAndVideoFile>
    <!--ro, opt, bool, whether it supports stopping exporting audio and video files, desc:PUT /ISAPI/System/StopExportAudioAndVideoFile?format=json-->true
  </isSupportStopExportAudioAndVideoFile>
  <isSupportAsyncExportSearchEventResult>
    <!--ro, opt, bool, whether it supports exporting the results of searching for event-related videos asynchronously, desc:POST
/ISAPI/ContentMgmt/search/AsyncExportSearchEventResult?format=json-->true
  </isSupportAsyncExportSearchEventResult>
  <isRebootAfterUpgradeFailed>
    <!--ro, opt, bool, whether restarting device is required after upgrading failed, desc:if this field does not exist, the device will not be restarted
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after upgrading failed-->true
  </isRebootAfterUpgradeFailed>
  <isSupportGetAsyncExportSearchEventResultProgress>
    <!--ro, opt, bool, whether it supports getting the async exporting progress of event videos search result, desc:POST
/ISAPI/ContentMgmt/search/GetAsyncExportSearchEventResultProgress?format=json-->true
  </isSupportGetAsyncExportSearchEventResultProgress>
  <isSupportManualTriggerWipe>
    <!--ro, opt, bool, whether it supports manually starting the wiper, desc:POST /ISAPI/System/Hardware/ManualTriggerWipe?format=json-->true
  </isSupportManualTriggerWipe>
  <isSupportWiperCleanMode>
    <!--ro, opt, bool, whether it supports wiper cleaning mode, desc:GET/PUT /ISAPI/System/Hardware/WiperCleanMode?format=json-->true
  </isSupportWiperCleanMode>
  <isSupportRadarWaterFlowRuleParams>
    <!--ro, opt, bool, whether it supports radar-assisted water flow detection rule parameters, desc:GET/PUT
/ISAPI/Event/Radar/<RadarID>/RadarWaterFlowRuleParams?format=json-->true
  </isSupportRadarWaterFlowRuleParams>
  <isSupportRadarWaterFlowAdvancedParams>
    <!--ro, opt, bool, whether it supports advanced parameters of radar-assisted water flow detection, desc:GET/PUT
/ISAPI/Event/Radar/<RadarID>/RadarWaterFlowAdvancedParams?format=json-->true
  </isSupportRadarWaterFlowAdvancedParams>
  <isSupportLaserSourceSwitch>
    <!--ro, opt, bool, whether it supports laser light source switching, desc:GET/PUT /ISAPI/System/channels/<channelID>/LaserSourceSwitch?format=json--
>true
  </isSupportLaserSourceSwitch>
  <isSupportLensStainConfigParams>
    <!--ro, opt, bool, whether it supports lens dirt detection configuration parameters, desc:GET/PUT /ISAPI/System/Hardware/LensStainConfigParams?
format=json-->true
  </isSupportLensStainConfigParams>
  <setupParamType opt="PTZLaser">
    <!--ro, opt, object, whether it supports construction parameters of device sub modules, attr:opt{req, string}, desc:whether /ISAPI/System/SetupParam?
type=<type> supports the optional parameter "type"-->
  </setupParamType>
  <isSupportLensStain>
    <!--ro, opt, bool, whether it supports lens stain event, desc:eventType:LensStain-->true
  </isSupportLensStain>
  <isSupportLensStainRecovered>
    <!--ro, opt, bool, whether it supports lens stain restoring event, desc:eventType:LensStainRecovered-->true
  </isSupportLensStainRecovered>
  <isSupportDeviceCircuitFault>
    <!--ro, opt, bool, whether it supports device circuit fault event, desc:eventType:DeviceCircuitFault-->true
  </isSupportDeviceCircuitFault>
  <isSupportDeviceCircuitRestored>
    <!--ro, opt, bool, whether it supports device circuit fault restoring event, desc:eventType:DeviceCircuitRestored-->true
  </isSupportDeviceCircuitRestored>
  <isSupportTurnOffBuiltInBattery>
    <!--ro, opt, bool, whether it supports disabling the built-in battery, desc:PUT /ISAPI/System/TurnOffBuiltInBattery?format=json-->true
  </isSupportTurnOffBuiltInBattery>
  <isSupportElevatorStartMove>
    <!--ro, opt, bool, whether it supports elevator starting running event, desc:eventType:ElevatorStartMove-->true
  </isSupportElevatorStartMove>
  <isSupportElevatorStopMove>
    <!--ro, opt, bool, whether it supports elevator stopping running event, desc:eventType:ElevatorStopMove-->true
  </isSupportElevatorStopMove>
  <isSupportOpticalFiberRasterAlarm>
    <!--ro, opt, bool, whether it supports fiber and raster alarm event, desc:corresponding event: OpticalFiberRasterAlarm-->true
  </isSupportOpticalFiberRasterAlarm>
  <isSupportSensorRealTimeData>
    <!--ro, opt, bool, whether it supports real-time sensor data uploading event, desc:corresponding event: SensorRealTimeData-->true
  </isSupportSensorRealTimeData>
  <isSupportGNSSWorkStatus>
    <!--ro, opt, bool, whether it supports GNSS working status uploading event, desc:corresponding event: GNSSWorkStatus.-->true
  </isSupportGNSSWorkStatus>
  <isSupportDevOnPlatformInfoEvent>
    <!--ro, opt, bool, whether it supports on-platform device information request event, desc:corresponding event: DevOnPlatformInfoEvent-->true
  </isSupportDevOnPlatformInfoEvent>
  <isSupportUpdateScheduleEvent>
    <!--ro, opt, bool, whether it supports request event of updating the schedule, desc:corresponding event: UpdateScheduleEvent-->true
  </isSupportUpdateScheduleEvent>
  <isSupportIntelligentApply>
    <!--ro, opt, bool, whether it support intelligent mode application parameters, desc:GET/PUT /ISAPI/System/VCAResource/IntelligentApply?format=json--
>true
  </isSupportIntelligentApply>
  <isSupportMCUDevSerialWrite>
    <!--ro, opt, bool, whether it supports writing MCU device serial number, desc:POST /ISAPI/System/MCUDevSerialWrite?format=json-->true
  </isSupportMCUDevSerialWrite>
  <isSupportSambaServerParams>
    <!--ro, opt, bool, whether it supports for Samba server parameters, desc:GET/PUT /ISAPI/System/SambaServerManagement/SambaServerParams?format=json--
>true
  </isSupportSambaServerParams>
  <isSupportToolingBoardConfigParams>
    <!--ro, opt, bool, whether it supports tooling plate configuration parameters, desc:PUT /ISAPI/System/AutoTest/ToolingBoardConfigParams?format=json--
>true
  </isSupportToolingBoardConfigParams>
  <isSupportAlumFlocObservationParams>
    <!--ro, opt, bool, whether it supports alum floc detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/AlumFlocObservation/channels/<channelID>/AlumFlocObservationParams?format=json-->true
  </isSupportAlumFlocObservationParams>
  <isDisplayRebootTips>
    <!--ro, opt, bool, whether to prompt users to restart the device when switching non-uniformity correction, desc:For devices with low performance,
switching non-uniformity correction will take effect only after restarting the device. The web should inform users of this. For devices with high
performance, restarting is not required. You can determine whether the web needs to prompt users to restart the device via this field. If this field does
not exist, the web needs to prompt by default.-->true
  </isDisplayRebootTips>
  <isSupportATOPMPAndResChanged>
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<isSupportAIOPMPAndResChanged>
<!--ro, opt, bool, whether the meaning of AIOP model package and resource interface is changed, desc:If the returned value is true, it indicates that the meaning of the AIOP API has changed, that is, the AIOP API for model package and engine management have replaced that of algorithm package scheduling and of the algorithm and engine management in algorithm application scheduling. For example, /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engine/<engineID>?format=json and /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engines?format=json will return all the engines for scheduling the algorithm packages and applications, and /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/management?format=json will return the information of the AIOP model package, algorithm application scheduling, and algorithm package scheduling.-->true
</isSupportAIOPMPAndResChanged>
<isSupportRoadVisibilityAnalysisCfg>
<!--ro, opt, bool, whether it supports road visibility detection analysis parameters, desc:GET/PUT /ISAPI/Traffic/channels/<channelID>/RoadVisibilityAnalysisCfg?format=json-->true
</isSupportRoadVisibilityAnalysisCfg>
<isSupportHotStandbyParams>
<!--ro, opt, bool, whether it supports hot spare parameters, desc:GET/PUT /ISAPI/System/MainSpareDeviceSettings/HotStandbyParams?format=json-->true
</isSupportHotStandbyParams>
<isSupportHotStandbySwitchParams>
<!--ro, opt, bool, whether it supports hot spare switching, desc:GET/PUT /ISAPI/System/MainSpareDeviceSettings/HotStandbySwitchParams?format=json-->true
</isSupportHotStandbySwitchParams>
<isSupportManualControlHotStandby>
<!--ro, opt, bool, whether it supports manually taking over hot spare, desc:POST /ISAPI/System/MainSpareDeviceSettings/ManualControlHotStandby?format=json-->true
</isSupportManualControlHotStandby>
<isSupportGetHotStandbyStatus>
<!--ro, opt, bool, whether it supports getting hot spare status, desc:GET /ISAPI/System/MainSpareDeviceSettings/GetHotStandbyStatus?format=json-->true
</isSupportGetHotStandbyStatus>
<isSupportRestoreStreamCollectionInfo>
<!--ro, opt, bool, whether it supports confirmation for collecting the raw stream data without mosaic mask, desc:GET/PUT /ISAPI/System/InformedConsentMgr/RestoreStreamCollectionInfo?format=json-->true
</isSupportRestoreStreamCollectionInfo>
<isSupportFileNameFormat>
<!--ro, opt, bool, whether it supports file name format, desc:GET/PUT /ISAPI/System/FileNameFormat?format=json-->true
</isSupportFileNameFormat>
<isSupportThermalInspectionParams>
<!--ro, opt, bool, whether it supports thermal inspection parameters, desc:GET/PUT /ISAPI/Thermal/ThermalInspectionParams?format=json-->true
</isSupportThermalInspectionParams>
<isSupportThermalRecordParams>
<!--ro, opt, bool, whether it supports configuring the recording parameters of specified thermal imaging channel, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ThermalRecordParams?format=json-->true
</isSupportThermalRecordParams>
<isSupportThermalLensParams>
<!--ro, opt, bool, whether it supports configuring thermal imaging lens parameters, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ThermalLensParams?format=json-->true
</isSupportThermalLensParams>
<isSupportThermalInfoDisplayParam>
<!--ro, opt, bool, whether it supports the display parameters of temperature measurement information, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ThermalInfoDisplayParam?format=json-->true
</isSupportThermalInfoDisplayParam>
<isSupportHotMapParam>
<!--ro, opt, bool, whether it supports thermal image capturing parameters, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/HotMapParam?format=json-->true
</isSupportHotMapParam>
<isSupportGetLaserRangingResult>
<!--ro, opt, bool, whether it supports getting the laser ranging results, desc:GET /ISAPI/System/GetLaserRangingResult?format=json-->true
</isSupportGetLaserRangingResult>
<isSupportLaserRangingOverlay>
<!--ro, opt, bool, whether it supports parameters of overlaying laser ranging information, desc:GET/PUT /ISAPI/System/LaserRangingOverlay?format=json-->true
</isSupportLaserRangingOverlay>
<isSupportPersonQueueRegionRealTime>
<!--ro, opt, bool, whether it supports real-time uploading of number of people in queue detection, desc:eventType:PersonQueueRegionRealTime-->true
</isSupportPersonQueueRegionRealTime>
<isSupportFullSpectrumAnalysis>
<!--ro, opt, bool, whether it supports full spectrum analysis parameters, desc:GET /ISAPI/FullSpectrumAnalysis/capabilities?format=json-->true
</isSupportFullSpectrumAnalysis>
<isSupportColorDetectionActionDelayTimeParams>
<!--ro, opt, bool, whether it supports the parameters of screen color quality detection action delay, desc:GET/PUT /ISAPI/System/ScreenColorQualityDetection/ColorDetectionActionDelayTimeParams?format=json-->true
</isSupportColorDetectionActionDelayTimeParams>
<isSupportAudioDetectionActionDelayTimeParams>
<!--ro, opt, bool, whether it supports the parameters of audio detection action delay, desc:GET/PUT /ISAPI/System/DeviceAudioDetection/AudioDetectionActionDelayTimeParams?format=json-->true
</isSupportAudioDetectionActionDelayTimeParams>
<isSupportCustomVideoChannelStatusIconParams>
<!--ro, opt, bool, whether it supports configuring the parameters of customizing the video channel status icon, desc:GET/PUT /ISAPI/ContentMgmt/InputProxy/CustomVideoChannelStatusIconParams?format=json-->true
</isSupportCustomVideoChannelStatusIconParams>
<isSupportImportVideoChannelStatusIcon>
<!--ro, opt, bool, whether it supports importing custom video channel status icons, desc:POST /ISAPI/ContentMgmt/InputProxy/ImportVideoChannelStatusIcon?format=json-->true
</isSupportImportVideoChannelStatusIcon>
<isSupportDeleteVideoChannelStatusIcon>
<!--ro, opt, bool, whether it supports deleting custom video channel status icons, desc:POST /ISAPI/ContentMgmt/InputProxy/DeleteVideoChannelStatusIcon?format=json-->true
</isSupportDeleteVideoChannelStatusIcon>
<isSupportSmokeScopeTelemeter>
<!--ro, opt, bool, whether it supports smoke density telemetry, desc:eventType:SmokeScopeTelemeter-->true
</isSupportSmokeScopeTelemeter>
<isSupportTempDataScheduledUpload>
<!--ro, opt, bool, whether it supports scheduled temperature measurement data uploading, desc:eventType:TempDataScheduledUpload-->true
</isSupportTempDataScheduledUpload>
<isSupportSafetyRopeDetectionRuleParams>
<!--ro, opt, bool, whether it supports safety rope detection rule parameters, desc:GET/PUT /ISAPI/Event/channels/<channelID>/SafetyRopeDetectionRuleParams?format=json-->true
</isSupportSafetyRopeDetectionRuleParams>

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<isSupportSafetyRopeDetectionEvent>
<!--ro, opt, bool, whether it supports safety rope detection event, desc:eventType: SafetyRopeDetectionEvent-->true
</isSupportSafetyRopeDetectionEvent>
<isSupportAddPositionShieldMask>
<!--ro, opt, bool, whether it supports adding the shielded area of a specified PTZ absolute position, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/AddPositionShieldMask?format=json-->true
</isSupportAddPositionShieldMask>
<isSupportPRPDUpload>
<!--ro, opt, bool, whether it supports uploading PRPD data, desc:eventType:PRPDUpload-->true
</isSupportPRPDUpload>
<isSupportAsynchronousUploadGNSSInitDataResult>
<!--ro, opt, bool, whether it supports the result of uploading GNSS data asynchronously to the storage server, desc:eventType:
AsynchronousUploadGNSSInitDataResult-->true
</isSupportAsynchronousUploadGNSSInitDataResult>
<isSupportVirtualChannelExtendedParams>
<!--ro, opt, bool, whether it supports virtual channel extension parameters, desc:GET/PUT /ISAPI/ContentMgmt/InputProxy/VirtualChannelExtendedParams?
format=json-->true
</isSupportVirtualChannelExtendedParams>
<isSupportSystemIdentify>
<!--ro, opt, bool, whether it supports verification request, desc:POST /ISAPI/System/identify?format=json-->true
</isSupportSystemIdentify>
<isSupportPerimeterAnalysisParams>
<!--ro, opt, bool, whether it supports perimeter algorithm analysis mode parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterAlgoMgr/PerimeterAnalysisParams?format=json-->true
</isSupportPerimeterAnalysisParams>
<isSupportCallCaptureImageEvent>
<!--ro, opt, bool, whether it supports the event of capturing in call, desc:eventType: CallCaptureImageEvent-->true
</isSupportCallCaptureImageEvent>
<isSupportTDLASModuleInfoMgr>
<!--ro, opt, bool, whether it supports managing hazardous gas TDLAS module information, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/methaneDetection/TDLASModuleInfoMgr?format=json-->true
</isSupportTDLASModuleInfoMgr>
<isSupportPhysicalMatchWonderfulMoments>
<!--ro, opt, bool-->true
</isSupportPhysicalMatchWonderfulMoments>
<isSupportHighResolutionLowPerformanceTip>
<!--ro, opt, bool-->true
</isSupportHighResolutionLowPerformanceTip>
<isSupportPeopleNumChangeOverlapParams>
<!--ro, opt, bool, GET or PUT /ISAPI/Smart/PeopleNumChange/channels/<channelID>/PeopleNumChangeOverlapParams?format=json, desc:GET or PUT
/ISAPI/Smart/PeopleNumChange/channels/<channelID>/PeopleNumChangeOverlapParams?format=json-->true
</isSupportPeopleNumChangeOverlapParams>
<isSupportAudioAlarmParam>
<!--ro, opt, bool, GET or PUT /ISAPI/Smart/AudioExceptionDetection/AudioIn/<AudioInID>/AudioAlarmParam?format=json, desc:GET or PUT
/ISAPI/Smart/AudioExceptionDetection/AudioIn/<AudioInID>/AudioAlarmParam?format=json-->true
</isSupportAudioAlarmParam>
<isSupportCrackMonitoringCalibration>
<!--ro, opt, bool, POST /ISAPI/Event/CrackMonitoringCalibration?format=json, desc:POST /ISAPI/Event/CrackMonitoringCalibration?format=json-->true
</isSupportCrackMonitoringCalibration>
<isSupportFileUploadServerList>
<!--ro, opt, bool, GET or PUT /ISAPI/System/FileUploadServer/FileUploadServerList?format=json, desc:GET or PUT
/ISAPI/System/FileUploadServer/FileUploadServerList?format=json-->true
</isSupportFileUploadServerList>
<isSupportFileUploadServerTest>
<!--ro, opt, bool, POST /ISAPI/System/FileUploadServer/FileUploadServerTest?format=json, desc:POST /ISAPI/System/FileUploadServer/FileUploadServerTest?
format=json-->true
</isSupportFileUploadServerTest>
<isSupportBestDetectionArea>
<!--ro, opt, bool, GET /ISAPI/Event/channels/<channelID>/objectAccumulationDetection/BestDetectionArea?format=json, desc:GET
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/BestDetectionArea?format=json-->true
</isSupportBestDetectionArea>
<isSupportAnalysisEngineEnabledTest>
<!--ro, opt, bool, POST /ISAPI/System/VCAResource/AnalysisEngineEnabledTest?format=json, desc:POST /ISAPI/System/VCAResource/AnalysisEngineEnabledTest?
format=json-->true
</isSupportAnalysisEngineEnabledTest>
<isSupportAlarmTriggeredByPanicButton>
<!--ro, opt, bool-->true
</isSupportAlarmTriggeredByPanicButton>
<isSupportSecretKeyFileParam>
<!--ro, opt, bool, GET or PUT /ISAPI/Security/resetPassword/SecretKeyFileParam?format=json, desc:GET or PUT
/ISAPI/Security/resetPassword/SecretKeyFileParam?format=json-->true
</isSupportSecretKeyFileParam>
<isSupportSearchEventRuleAndScheduleParams>
<!--ro, opt, bool, POST /ISAPI/Event/EventBatchRulesMgr/SearchEventRuleAndScheduleParams?format=json, desc:POST
/ISAPI/Event/EventBatchRulesMgr/SearchEventRuleAndScheduleParams?format=json-->true
</isSupportSearchEventRuleAndScheduleParams>
<isSupportEventRuleAndScheduleParams>
<!--ro, opt, bool, POST /ISAPI/Event/EventBatchRulesMgr/EventRuleAndScheduleParams?format=json, desc:POST
/ISAPI/Event/EventBatchRulesMgr/EventRuleAndScheduleParams?format=json-->true
</isSupportEventRuleAndScheduleParams>
<deviceReboot>
<!--ro, opt, object, restart devices-->
<module opt="algorithmProgram,radar">
<!--ro, opt, string, attr:opt{opt, string}-->radar
</module>
</deviceReboot>
<factoryReset>
<!--ro, opt, object-->
<module opt="radar">
<!--ro, opt, string, attr:opt{opt, string}-->radar
</module>
</factoryReset>
<isSupportCancelAsyncExportSearchEventResult>
<!--ro, opt, bool, POST /ISAPI/ContentMgmt/search/CancelAsyncExportSearchEventResult?format=json, desc:POST
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    <!--ro, opt, bool, POST /ISAPI/ContentMgmt/search/cancelAsyncExportSearchEventResult?format=json, desc:POST
/ISAPI/ContentMgmt/search/CancelAsyncExportSearchEventResult?format=json-->true
    </isSupportCancelAsyncExportSearchEventResult>
    <isSupportGetIntellAndTemperatureRangeMap>
      <!--ro, opt, bool, GET /ISAPI/System/VCAResource/channels/<channelID>/GetIntellAndTemperatureRangeMap?format=json, desc:GET
/ISAPI/System/VCAResource/channels/<channelID>/GetIntellAndTemperatureRangeMap?format=json-->true
    </isSupportGetIntellAndTemperatureRangeMap>
    <isSupportSendScreenColorEnvErrorCalibrationTask>
      <!--ro, opt, bool, POST /ISAPI/System/ScreenColorQualityDetection/SendScreenColorEnvErrorCalibrationTask?format=json, desc:POST
/ISAPI/System/ScreenColorQualityDetection/SendScreenColorEnvErrorCalibrationTask?format=json-->true
    </isSupportSendScreenColorEnvErrorCalibrationTask>
    <isSupportSearchScreenColorEnvErrorCalibrationTask>
      <!--ro, opt, bool, POST /ISAPI/System/ScreenColorQualityDetection/SearchScreenColorEnvErrorCalibrationTask?format=json, desc:POST
/ISAPI/System/ScreenColorQualityDetection/SearchScreenColorEnvErrorCalibrationTask?format=json-->true
    </isSupportSearchScreenColorEnvErrorCalibrationTask>
    <isSupportCurrentScreenColorEnvErrorCalibrationParams>
      <!--ro, opt, bool, GET /ISAPI/System/ScreenColorQualityDetection/CurrentScreenColorEnvErrorCalibrationParams?format=json, desc:GET
/ISAPI/System/ScreenColorQualityDetection/CurrentScreenColorEnvErrorCalibrationParams?format=json-->true
    </isSupportCurrentScreenColorEnvErrorCalibrationParams>
    <isSupportSynchronizationRulesBetweenChannels>
      <!--ro, opt, bool, POST /ISAPI/System/IntelligentRulesMgr/channels/<channelID>/SynchronizationRulesBetweenChannels?format=json, desc:POST
/ISAPI/System/IntelligentRulesMgr/channels/<channelID>/SynchronizationRulesBetweenChannels?format=json-->true
    </isSupportSynchronizationRulesBetweenChannels>
    <isSupportGestureRecognitionRuleParams>
      <!--ro, opt, bool, GET or PUT /ISAPI/Event/channels/<channelID>/GestureRecognitionRuleParams?format=json, desc:GET or PUT
/ISAPI/Event/channels/<channelID>/GestureRecognitionRuleParams?format=json-->true
    </isSupportGestureRecognitionRuleParams>
    <isSupportSolidBallSport>
      <!--ro, opt, bool-->true
    </isSupportSolidBallSport>
    <isSupportVolleyballSport>
      <!--ro, opt, bool-->true
    </isSupportVolleyballSport>
    <isSupportBasketballSport>
      <!--ro, opt, bool-->true
    </isSupportBasketballSport>
    <isSupportSoccerSport>
      <!--ro, opt, bool-->true
    </isSupportSoccerSport>
    <isSupportPersonnelGestureDetection>
      <!--ro, opt, bool-->true
    </isSupportPersonnelGestureDetection>
    <isSupportExportLensThermalCalibrationFile>
      <!--ro, opt, bool-->true
    </isSupportExportLensThermalCalibrationFile>
    <isSupportModifyPresetRatioCombination>
      <!--ro, opt, bool-->true
    </isSupportModifyPresetRatioCombination>
    <isSupportExportChannelConfigAbnormalReport>
      <!--ro, opt, bool-->true
    </isSupportExportChannelConfigAbnormalReport>
    <isSupportExportChannelConfigTemplate>
      <!--ro, opt, bool-->true
    </isSupportExportChannelConfigTemplate>
    <isSupportSoftStorageInterfaceParams>
      <!--ro, opt, bool-->true
    </isSupportSoftStorageInterfaceParams>
    <isSupportImportSceneFile>
      <!--ro, opt, bool-->true
    </isSupportImportSceneFile>
    <isSupportExportSceneFile>
      <!--ro, opt, bool-->true
    </isSupportExportSceneFile>
    <isSupportRadarWaterLevelAdvancedParams>
      <!--ro, opt, bool-->true
    </isSupportRadarWaterLevelAdvancedParams>
    <isSupportOpenAndCloseDoorLearningEvent>
      <!--ro, opt, bool-->true
    </isSupportOpenAndCloseDoorLearningEvent>
    <isSupportResetScreenColorEnvErrorCalibration>
      <!--ro, opt, bool-->true
    </isSupportResetScreenColorEnvErrorCalibration>
    <isSupportConsumptionPermissionSearchEvent>
      <!--ro, opt, bool-->true
    </isSupportConsumptionPermissionSearchEvent>
    <isSupportDeviceCap>
      <!--ro, opt, bool-->true
    </isSupportDeviceCap>
    <isSupportVideoPipList>
      <!--ro, opt, bool-->true
    </isSupportVideoPipList>
    <maxProtocolConcurrencyNum>
      <!--ro, opt, int-->10
    </maxProtocolConcurrencyNum>
    <isSupportReplayShotImage>
      <!--ro, opt, bool-->true
    </isSupportReplayShotImage>
    <isSupportColumnSpeakers>
      <!--ro, opt, bool-->true
    </isSupportColumnSpeakers>
    <isSupportPublish>
      <!--ro, opt, bool-->true
    </isSupportPublish>
    <isSupportGetDevicePartsLife>
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    <!--ro, opt, bool-->true
  </isSupportGetDevicePartsLife>
  <isSupportVisibilityTelemeter>
    <!--ro, opt, bool-->true
  </isSupportVisibilityTelemeter>
  <isSupportVisibilityEvent>
    <!--ro, opt, bool-->true
  </isSupportVisibilityEvent>
  <isSupportRestoreRadarWaterLevelDefaultParams>
    <!--ro, opt, bool-->true
  </isSupportRestoreRadarWaterLevelDefaultParams>
  <isSupportDeviceCreateUserNum>
    <!--ro, opt, bool-->true
  </isSupportDeviceCreateUserNum>
  <isSupportSetStreamingSecretKey>
    <!--ro, opt, bool-->true
  </isSupportSetStreamingSecretKey>
  <isSupportSearchClickToThermometry>
    <!--ro, opt, bool-->true
  </isSupportSearchClickToThermometry>
  <isSupportResourceChildParams>
    <!--ro, opt, bool-->true
  </isSupportResourceChildParams>
  <isSupportSearchThermometryInfo>
    <!--ro, opt, bool-->true
  </isSupportSearchThermometryInfo>
  <isSupportSendScanInfoToCamera>
    <!--ro, opt, bool-->true
  </isSupportSendScanInfoToCamera>
  <isSupportRtspPreRecord>
    <!--ro, opt, bool-->true
  </isSupportRtspPreRecord>
  <isSupportGetClientAccessCapabilities>
    <!--ro, opt, bool-->true
  </isSupportGetClientAccessCapabilities>
  <isSupportISUPChileDevHttp>
    <!--ro, opt, bool-->true
  </isSupportISUPChileDevHttp>
  <isSupportRemoteDesktopParams>
    <!--ro, opt, bool-->true
  </isSupportRemoteDesktopParams>
  <isSupportPassbackRecordUpload>
    <!--ro, opt, bool-->true
  </isSupportPassbackRecordUpload>
  <alertDistributionAddMode opt="cover,append" def="append">
    <!--ro, opt, bool, attr:opt{req, string},def{req, string}-->append
  </alertDistributionAddMode>
  <isSupportSkeletonKeypointRuleParams>
    <!--ro, opt, bool-->true
  </isSupportSkeletonKeypointRuleParams>
  <isSupportHumanSkeletonKeypointDetection>
    <!--ro, opt, bool-->true
  </isSupportHumanSkeletonKeypointDetection>
  <isSupportWaterQualityCountingRule>
    <!--ro, opt, bool-->true
  </isSupportWaterQualityCountingRule>
  <isSupportWaterQualityReport>
    <!--ro, opt, bool-->true
  </isSupportWaterQualityReport>
  <isOnlySupportAnalogChannel>
    <!--ro, opt, bool-->true
  </isOnlySupportAnalogChannel>
  <isSupportClarityLevel>
    <!--ro, opt, bool-->true
  </isSupportClarityLevel>
  <isSupportDeviceNetworkDiscoverySearchParams>
    <!--ro, opt, bool-->true
  </isSupportDeviceNetworkDiscoverySearchParams>
  <isSupportConfigurationFileAsyncUpload>
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  <isSupportDoorStatusSceneList>
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```

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```
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    <timeOut>
      <!--ro, req, int-->60
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Portuguese, Spanish, Romanian, Turkish, Japanese, Danish, Swedish, Norwegian, Finnish, Korean, Thai, Estonia, Vietnamese, Hebrew, Latvian, Arabic,
Slovenian, Croatian, Lithuanian, Serbian, BrazilianPortuguese, Indonesian, Ukrainian, EURSpanish, Sovenian, Uzbek, Kazak, Kirghiz, Farsi, Azerbaidzhan,
Burmese, Mongolian, Anglicism, HongKongTraChinese, Albanian, LatinAmericanSpanish">
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  </supportHomepositionChannel>
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  <isSupportShipDetectionPOSParams>
    <!--ro, opt, bool-->true
```

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<isSupportAnalyzerSwitchControl>
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</isSupportAnalyzerSwitchControl>
```

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```

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<isSupportStopOrchExportTask>

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```
      <!--ro, opt, bool-->true
    </isSupportStopOrchExportTask>
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    </isSupportPIPDataCfg>
    <isSupportReocrdTranscodeParams>
      <!--ro, opt, bool-->true
    </isSupportReocrdTranscodeParams>
  </DeviceCap>
```

15.1.6.3 Export device configuration file

Request URL

GET /ISAPI/System/configurationData?secretkey=<secretkey>&mode=<mode>

Query Parameter

Parameter Name	Parameter Type	Description
secretkey	string	The verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing.
mode	enum	--

Request Message

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None

Response Message

Binary Data

15.1.6.4 Get the device attribute code

Request URL

GET /ISAPI/System/deviceInfo/characteristicCode?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "Characteristic": {
    /*ro, req, object, attribute code, desc:attribute code*/
        "code":  "+/"
        /*ro, opt, string, device attribute code, range:[1,128], desc:device attribute code*/
    }
}
```

15.1.6.5 Get the mutual exclusion information when device function exception occurred

Request URL

GET /ISAPI/System/mutexFunctionErrorMsg

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MutexFunctionErrorMsg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, mutual exclusion information, attr:version{req, string, protocolVersion}-->
  <MutexFunctionList size="10">
    <!--ro, opt, array, mutual exclusion information list, subType:object, attr:size{req, int}, desc:get the functions that are mutually exclusive to the
enabled functions-->
    <MutexFunction>
      <!--ro, opt, object, mutual exclusion message-->
      <id>
        <!--ro, req, int, No.-->1
      </id>
      <describe>
        <!--ro, req, enum, description, subType:string, desc:"SMD", "vehicleDetection" (vehicle detection), "counting" (people counting), "heatMap" (heat
map statistics), "fieldDetection" (intrusion detection), "lineDetection" (line crossing detection), "sceneChangeDetection" (scene change detection),
"VMDHumanVehicle" (motion detection), "regionEntrance" (region entrance), "regionExiting" (region exiting), "faceCapture" (face capture), "faceRecognition"
(face recognition), "IPEnhance" (IP enhancement), "SVCEnhance" (SVC enhancement), "4KOutput" (4K output), "ZCR" (zero channel), "misinfoFilter" (alarm
filtering), "transcodeCfg" (stream transcoding), "AIOP_Video" (AI video task analysis)-->SMD
      </describe>
      <channelNo>
        <!--ro, req, string, channel No. corresponding to the mutually exclusive function, desc:multiple channel numbers are supported and should be
separated by commas-->1
      </channelNo>
    </MutexFunction>
  </MutexFunctionList>
</MutexFunctionErrorMsg>
```

15.1.6.6 Get the network service capability

Request URL

GET /ISAPI/System/Network/capabilities

Query Parameter

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None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<NetworkCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, get the network service capability, attr:version{opt, string, protocolVersion}-->
  <isSupportWireless>
    <!--ro, req, bool, whether the device supports wireless network-->true
  </isSupportWireless>
  <isSupportWAN>
    <!--ro, opt, bool, whether the device supports WAN (Wide Area Network)-->true
  </isSupportWAN>
  <isSupportPPPoE>
    <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
  </isSupportPPPoE>
  <isSupportBond>
    <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
  </isSupportBond>
  <isSupport802_1x>
    <!--ro, req, bool, whether the device supports 802.1x protocol-->true
  </isSupport802_1x>
  <isSupportNtp>
    <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
  </isSupportNtp>
  <isSupportFtp>
    <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
  </isSupportFtp>
  <isSupportUpnp>
    <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play) protocol-->true
  </isSupportUpnp>
  <isSupportPNP>
    <!--ro, opt, bool, whether the device supports plug-and-play protocol-->true
  </isSupportPNP>
  <isSupportDdns>
    <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
  </isSupportDdns>
  <isSupportHttps>
    <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
  </isSupportHttps>
  <SnmpCap>
    <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
    <isSupport>
      <!--ro, req, bool, whether the device supports SNMP-->true
    </isSupport>
  </SnmpCap>
  <isSupportExtNetCfg>
    <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
  </isSupportExtNetCfg>
  <isSupportIPFilter>
    <!--ro, opt, bool, whether the device supports IP filtering-->true
  </isSupportIPFilter>
  <isSupportNetPreviewStrategy>
    <!--ro, opt, bool, whether the device supports network live view strategy-->true
  </isSupportNetPreviewStrategy>
  <isSupportEZVIZ>
    <!--ro, opt, bool, whether the device supports EZ protocol-->true
  </isSupportEZVIZ>
  <isSupportMACFilter>
    <!--ro, opt, bool, whether the device supports physical address filtering-->true
  </isSupportMACFilter>
  <GB28181Cap>
    <!--ro, opt, object, N/A-->
    <isSupportGB28181Service>
      <!--ro, opt, bool, N/A-->true
    </isSupportGB28181Service>
  </GB28181Cap>
  <VerificationCodeModification>
    <!--ro, opt, object, verification code editing-->
    <verificationCodeType opt="normal,empty">
      <!--ro, opt, string, verification code type, attr:opt{req, string}-->test
    </verificationCodeType>
    <isSupportDeclarationURL>
      <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
    </isSupportDeclarationURL>
    <isSupportPrivacyPolicyURL>
      <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
    </isSupportPrivacyPolicyURL>
    <verificationCodeModify opt="true,false">
      <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{req, string}-->true
    </verificationCodeModify>
    <Hyperlinks>
      <!--ro, opt, object, hyperlink-->
      <declarationURL>
        <!--ro, opt, string, hyperlink to the Terms of Service-->test
      </declarationURL>
      <privacyPolicyURL>
```

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```
        <!--ro, opt, string, hyperlink to the Privacy Policy-->test
      </privacyPolicyURL>
    </Hyperlinks>
    <isSupportVerificationCodeCheck>
      <!--ro, opt, bool, whether the device supports verifying the verification code-->true
    </isSupportVerificationCodeCheck>
  </VerificationCodeModification>
  <isSupportIntegrate>
    <!--ro, opt, bool, whether the device supports access protocol configuration-->true
  </isSupportIntegrate>
  <isSupportPlatformAccess>
    <!--ro, opt, bool, whether the device supports platform access, desc:GET /ISAPI/System/Network/platformAccessFilter/capabilities (platform access IP
address filtering configuration (Ehome allowlist configuration))-->true
  </isSupportPlatformAccess>
  <isSupportResourceStatistics>
    <!--ro, opt, bool, whether the device supports getting the network resource information-->true
  </isSupportResourceStatistics>
  <isSupportBandwidthLimit>
    <!--ro, opt, bool, whether the device supports bandwidth limit-->true
  </isSupportBandwidthLimit>
  <isSupportGetLinkSocketIP>
    <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
  </isSupportGetLinkSocketIP>
  <isSupportWebSocket>
    <!--ro, opt, bool, whether the device supports WebSocket-->true
  </isSupportWebSocket>
  <isSupportVideoImgDB>
    <!--ro, opt, bool, whether the device supports image and video library configuration-->true
  </isSupportVideoImgDB>
  <isSupportEventDataOverWebSocket opt="true,false">
    <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{req, string}-->true
  </isSupportEventDataOverWebSocket>
  <isSupportEZVIZTiming>
    <!--ro, opt, bool, whether the device supports EZ platform synchronization (enabledTiming of message returned by/ISAPI/System/Network/EZVIZ)-->true
  </isSupportEZVIZTiming>
  <isSupportPingTest>
    <!--ro, opt, bool, whether the device supports ping test, desc:related URI: /ISAPI/System/Network/pingtest/capabilities-->true
  </isSupportPingTest>
</NetworkCap>
```

15.1.6.7 Get the thermal capability

Request URL

GET /ISAPI/Thermal/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ThermalCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, thermal capability, attr:version{req, string, protocolVersion}-->
  <isSupportFireDetection>
    <!--ro, opt, bool, whether the device supports fire detection, desc:Related URI: /ISAPI/Thermal/channels/<ID>/fireDetection/capabilities-->true
  </isSupportFireDetection>
  <isSupportThermometry>
    <!--ro, opt, bool, whether the device supports thermography, desc:Related URI: /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities-->true
  </isSupportThermometry>
  <isSupportRealtimeThermometry>
    <!--ro, opt, bool, whether supports uploading real-time temperature measurement data-->true
  </isSupportRealtimeThermometry>
  <isSupportNVR>
    <!--ro, opt, bool-->true
  </isSupportNVR>
  <isSupportThermometrySchedule>
    <!--ro, opt, bool-->true
  </isSupportThermometrySchedule>
  <isSupportTemperatureSchedule>
    <!--ro, opt, bool-->true
  </isSupportTemperatureSchedule>
  <isSupportFireDetectionSchedule>
    <!--ro, opt, bool-->true
  </isSupportFireDetectionSchedule>
  <isSupportTemperatureCollection>
    <!--ro, opt, bool, whether device supports temperature ANR, desc:whether device supports temperature ANR-->true
  </isSupportTemperatureCollection>
</ThermalCap>
```

15.1.6.8 Upload device status change

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EventType:changedStatus

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, req, int, times that the same alarm has been uploaded, step:1, desc:the event triggering frequency. The pulse event is an event which is triggered continuously according to the detection frequency of the device, e.g., the motion detection. The instant event is the event triggered when object recognition occurs, and the recognition of each object will trigger the event once, e.g., the face capture. When the pulse events are triggered, this node is used to tell whether the events are triggered by the same source. For example, if it is motion detection, the alarms will be uploaded continuously according to the device detection frequency; if it changes to another source, then the counting will start again. It can be regarded as the event triggering frequency*/
    "eventType":  "changedStatus",
    /*ro, req, string, event type*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid event or event starts), "inactive" (invalid event or the event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "change status",
    /*ro, req, string, event description*/
    "channelName":  "ABC",
    /*ro, opt, string, channel name (camera name), desc:it should be the same as the channelName in the message of URI (/ISAPI/Streaming/channels/<ID>)*/
    "deviceID":  "test0123",
    /*ro, opt, string, PUID, range:[1,64]*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers*/
    "ChangedStatus": {
    /*ro, req, object, status upload message structure*/
        "seq":  0,
        /*ro, req, int, alarm No., desc:a 32-character integer*/
        "devStatus":  0,
        /*ro, opt, enum, value of device status, subType:int, desc:0-normal, 1-CPU usage, higher than 85%, 2-hardware error (e.g., serial port exception)*/
        "devCapHash":  "fdsafdsfdsafdsfji",
        /*ro, opt, string, device capability full hash value, desc:generated by md5, and encoded by base 64. For NVR, the hash value of the managed network camera should be the calculation source*/
        "zeroChanEnable":  true,
        /*ro, opt, bool, whether to enable zero-channel, desc:true-enabled, false-disabled*/
        "cycleRecordEnable":  true,
        /*ro, opt, bool, whether to enable overwritten recording, desc:true-enabled, false-disabled*/
        "chanChanged": [1, 2, 3],
        /*ro, opt, array, channel status change, subType:int, desc:if channels of the device are deleted, edited, or added, the hash value will change*/
        "IoTChanChanged": [1, 2, 3],
        /*ro, opt, array, IoT channel status change, subType:int, desc:if channels of the device are deleted, edited, or added, the hash value will change*/
        "hdChanged": [1, 2, 3],
        /*ro, opt, array, device HDD status, subType:int, desc:device HDD is deleted or added*/
        "ChanStatus": [
        /*ro, opt, array, channel change, subType:object*/
            {
                "chanNo":  1,
                /*ro, opt, int, channel No., step:1*/
                "enable":  1,
                /*ro, opt, enum, whether to enable the analog channel, subType:int, desc:0 (disable), 1 (enable). It is valid for analog channel only*/
                "online":  1,
                /*ro, opt, enum, online status, subType:int, desc:0 (offline), 1 (online), 2 (sleeping), 3 (waking up)*/
                "record":  1,
                /*ro, opt, enum, whether the device is recording, subType:int, desc:0 (not recording), 1 (recording)*/
                "recordStatus":  0,
                /*ro, opt, enum, recording status, subType:int, desc:0 (normal), 1 (abnormal, the device is not recording according to the recording schedule), 2 (abnormal, manual recording failed), 3 (abnormal, the disk error), 4 (abnormal, the network camera is offline), 5 (abnormal, other cause)*/
                "signal":  0,
                /*ro, opt, enum, signal status, subType:int, desc:0 (normal), 1 (signal loss)*/
                "arming":  0
                /*ro, opt, enum, network camera arming status by NVR, subType:int, desc:0 (succeeded), 1 (failed)*/
            }
        ],
        "HDStatus": [
        /*ro, opt, array, HDD status list, subType:object*/
            {
                "hdNo":  1,
                /*ro, opt, int, HDD No.*/
                "status":  0
                /*ro, opt, enum, HDD status, subType:int, desc:0 (active), 1 (sleeping), 2 (abnormal), 3 (sleepy HDD error), 4 (unformatted), 5 (disconnected, for network HDD), 6 (formatting), 7 (the HDD is full), 8 (other error)*/
            }
        ],
        "CapChanged": [
        /*ro, opt, array, capability set change, subType:object*/
```

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```
            {
                "name":  "FireDetection",
                /*ro, req, string, root node that shows the capability set change*/
                "chan":  1,
                /*ro, opt, int, channel No., desc:it is required for getting channel capability, and it can be empty for getting device capability*/
                "url":  "/ISAPI/Thermal/channels/1/fireDetection/capabilities",
                /*ro, opt, string, URL of the capability set, desc:required for ISAPI capability set*/
                "abilityType":  "0x700"
                /*ro, opt, string, capability type, desc:required for capability outside ISAPI*/
            }
        ],
        "IoTChanStatus": [
        /*ro, opt, array, IoT channel status list, subType:object*/
            {
                "IoTChannel":  1,
                /*ro, req, int, IoT channel No.*/
                "online":  1,
                /*ro, req, enum, online status, subType:int, desc:0 (offline), 1 (online)*/
                "channelName":  "test"
                /*ro, req, string, channel name, range:[1,32]*/
            }
        ],
        "intranetIPAddressStatus": {
        /*ro, opt, object, intranet IP address, desc:upload intranet IP address change. It is returned only in EZ link. Application scenario: the technical
support can operate devices by the IP address after remotely connecting to the LAN of the device. It can be used to reduce unnecessary communication and
improve efficiency. Currently, it is uploaded every five minutes*/
            "intranetIPAddress": [
            /*ro, opt, array, intranet IP address list, subType:object*/
                {
                    "id":  1,
                    /*ro, req, int, network interface ID, range:[1,4]*/
                    "ipVersion":  "ipv4",
                    /*ro, opt, enum, IP address type, subType:string, desc:"ipv4", "ipv6"*/
                    "ipv4Address":  "test",
                    /*ro, req, string, IPv4 address*/
                    "ipv6Address":  "test"
                    /*ro, req, string, IPv6 address*/
                }
            ]
        }
    }
}
```

15.1.7 时间管理

15.1.7.1 Get device time synchronization management parameters

Request URL

GET /ISAPI/System/time

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--ro, req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--ro, opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}-->2019-02-28T10:50:44+08:30
  </localTime>
  <timeZone>
    <!--ro, opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:daylight saving time configuration in time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the device time returned currently is in DST (Daylight Saving Time) system, desc:whether the device time returned currently is in DST (Daylight Saving Time) system-->true
  </isSummerTime>
  <platformType>
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform-->EZVIZ
  </platformType>
  <platformNo>
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--ro, opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--ro, opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--ro, opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

15.1.7.2 Set device time synchronization management parameters

Request URL

PUT /ISAPI/System/time

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}, desc:the time difference between local time and coordinated universal time (UTC) is configured via the timeZone field-->2019-02-28T10:50:44
  </localTime>
  <timeZone>
    <!--opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the time returned by the current device is that in the DST (daylight saving time), desc:whether the time returned by the current device is that in the DST (daylight saving time)-->true
  </isSummerTime>
  <platformType>
    <!--opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform, related URI: /ISAPI/System/Network/EZVIZ-->EZVIZ
  </platformType>
  <platformNo>
    <!--opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <FailedNodeInfoList>
    <!--ro, opt, object, information list of failed nodes, desc:for the manual time synchronization of central analysis cluster, this field is returned if time synchronization failed-->
    <FailedNodeInfo>
      <!--ro, opt, object, information of failed nodes-->
      <nodeID>
        <!--ro, req, string, node ID, range:[0,64]-->test
      </nodeID>
      <nodeIP>
        <!--ro, req, string, node IP, range:[0,20]-->test
      </nodeIP>
      <reason>
        <!--ro, opt, string, reason why the node failed to synchronize time, range:[0,128]-->test
      </reason>
    </FailedNodeInfo>
  </FailedNodeInfoList>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>
```

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15.1.7.3 Get the capability of device time synchronization management

Request URL

GET /ISAPI/System/time/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management capability set, attr:version{opt, string, protocolVersion}-->
  <timeMode opt="NTP,manual,satellite,SDK,28181,ONVIF,ALL(任何支持的校时方式都允许校时),NONE(不允校时或无校时源),platform,PTP">
    <!--ro, req, enum, time synchronization mode, subType:string, attr:opt{opt, string}, desc:“NTP” (NTP time synchronization), “manual” (manual time
synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or
no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime min="0" max="256">
    <!--ro, opt, string, local time, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTime>
  <timeZone min="0" max="256">
    <!--ro, opt, string, time zone, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </timeZone>
  <satelliteInterval min="0" max="3600">
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, attr:min{opt, string},max{opt, string}, desc:unit: minute-->60
  </satelliteInterval>
  <timeType opt="local,UTC">
    <!--ro, opt, enum, time type, subType:string, attr:opt{opt, string}, desc:“local” (local time), “UTC” (UTC time)-->local
  </timeType>
  <platformType opt="EZVIZ">
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, attr:opt{opt, string}, desc:platform type-->EZVIZ
  </platformType>
  <platformNo min="1" max="2">
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, attr:min{req, int},max{req, int}, desc:it is the only ID, which is
configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <isSupportHistoryTime>
    <!--ro, opt, bool, supported capability of the historical time synchronization list, desc:related URI: /ISAPI/System/time/historyInfo?format=json-->true
  </isSupportHistoryTime>
  <isSupportTimeFilter>
    <!--ro, opt, bool, supported capability of filtering time synchronization, desc:related URI: /ISAPI/System/time/filter/capabilities?format=json-->true
  </isSupportTimeFilter>
  <displayFormat opt="MM/dd/yyyy hh:mm,mm,dd-MM-yyyy hh,MM-dd-yyyy hh:mm,yyyy-MM-dd hh:mm,MM/dd/yyyy hh:mm:ss,dd-MM-yyyy hh:mm:ss,MM-dd-yyyy hh:mm:ss,yyyy-
MM-dd hh:mm:ss,yyyy-MM-dd,dd-MM-yyyy,MM-dd-yyyy">
    <!--ro, opt, enum, time display format, subType:string, attr:opt{req, string}, desc:if this node is returned, it indicates that the device supports
configuring time display format, related URI: /ISAPI/System/time/timeType?format=json-->MM/dd/yyyy hh:mm
  </displayFormat>
  <isSupportSyncDeviceNTPInfoToCamera>
    <!--ro, opt, bool, the capability of synchronizing device’s NTP service information with the camera, desc:related URI:
/ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json-->true
  </isSupportSyncDeviceNTPInfoToCamera>
  <isSupportNTPService>
    <!--ro, opt, bool-->true
  </isSupportNTPService>
  <isSupportTimeZoneListInfo>
    <!--ro, opt, bool-->true
  </isSupportTimeZoneListInfo>
  <isSupportAutoEnterSummerTime>
    <!--ro, opt, bool-->true
  </isSupportAutoEnterSummerTime>
  <ethernetPort min="0" max="255">
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}, attr:min{req, int},max{req, int}-->1
  </ethernetPort>
  <IANA
opt="Africa/Abidjan,Africa/Accra,Africa/Addis_Ababa,Africa/Algiers,Africa/Blantyre,Africa/Brazzaville,Africa/Cairo,Africa/Casablanca,Africa/Conakry,Africa/D
akar,Africa/Dar_es_Salaam,Africa/Djibouti,Africa/Freetown,Africa/Gaborone,Africa/Harare,Africa/Johannesburg,Africa/Kampala,Africa/Khartoum,Africa/Kigali,Afr
ica/Kinshasa,Africa/Lagos,Africa/Maseru,Africa/Mogadishu,Africa/Nairobi,Africa/Sao_Tome,Africa/Timbuktu,Africa/Tripoli,Africa/Tunis,America/Anchorage,Americ
a/Aruba,America/Asuncion,America/Barbados,America/Belize,America/Bogota,America/Buenos_Aires,America/Cancun,America/Caracas,America/Cayman,America/Chicago,A
merica/Curacao,America/Dawson_Creek,America/Denver,America/Detroit,America/Dominica,America/Edmonton,America/El_Salvador,America/Fortaleza,America/Grand_Tur
k,America/Grenada,America/Guatemala,America/Guyana,America/Halifax,America/Havana,America/Indiana/Indianapolis,America/Indiana/Knox,America/Indiana/Marengo,
America/Indiana/Petersburg,America/Indiana/Tell_City,America/Indiana/Vevay,America/Indiana/Vincennes,America/Indiana/Winamac,America/Jamaica,America/La_Paz,
America/Lima,America/Los_Angeles,America/Louisville,America/Managua,America/Martinique,America/Mendoza,America/Metlakatla,America/Mexico_City,America/Monter
rey,America/Montevideo,America/Montreal,America/Nassau,America/New_York,America/North_Dakota/Beulah,America/North_Dakota/Center,America/North_Dakota/New_Sal
em,America/Panama,America/Phoenix,America/Port_of_Spain,America/Port-au-
Prince,America/Puerto_Rico,America/Santo_Domingo,America/Sao_Paulo,America/St_Johns,America/St_Kitts,America/St_Lucia,America/St_Thomas,America/Tijuana,Amer
ica/Toronto,America/Vancouver,America/Winnipeg,Antarctica/South_Pole,Arctic/Longyearbyen,Asia/Almaty,Asia/Amman,Asia/Anadyr,Asia/Aqtau,Asia/Baghdad,Asia/Bah
rain,Asia/Baku,Asia/Bangkok,Asia/Beirut,Asia/Calcutta,Asia/Damascus,Asia/Dhaka,Asia/Dubai,Asia/Gaza,Asia/Hebron,Asia/Ho_Chi_Minh,Asia/Hong_Kong,Asia/Jakarta
,Asia/Jerusalem,Asia/Kabul,Asia/Karachi,Asia/Kathmandu,Asia/Kuala_Lumpur,Asia/Kuwait,Asia/Macau(China),Asia/Manila,Asia/Muscat,Asia/Phnom_Penh,Asia/Pyongyan
g,Asia/Rangoon,Asia/Riyadh,Asia/Seoul,Asia/Shanghai,Asia/Singapore,Asia/Taipei,Asia/Tehran,Asia/Tel_Aviv,Asia/Tokyo,Asia/Ulaanbaatar,Atlantic/Bermuda,Atlant
ic/Canary,Atlantic/Cape_Verde,Atlantic/Reykjavik,Atlantic/Stanley,Australia/Adelaide,Australia/Brisbane,Australia/Canberra,Australia/Darwin,Australia/Melbou
rne,Australia/NSW,Australia/Perth,Australia/Queensland,Australia/Sydney,Australia/Victoria,Canada/Newfoundland,Canada/Saskatchewan,Chile/EasterIsland,Europe
/Amsterdam,Europe/Andorra,Europe/Athens,Europe/Belfast,Europe/Belgrade,Europe/Berlin,Europe/Bratislava,Europe/Brussels,Europe/Bucharest,Europe/Budapest,Euro
pe/Chisinau,Europe/Copenhagen,Europe/Dublin,Europe/Gibraltar,Europe/Helsinki,Europe/Isle_of_Man,Europe/Istanbul,Europe/Kiev,Europe/Lisbon,Europe/London,Euro
```

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```
pe/Luxembourg,Europe/Madrid,Europe/Malta,Europe/Minsk,Europe/Monaco,Europe/Moscow,Europe/Oslo,Europe/Paris,Europe/Prague,Europe/Riga,Europe/Rome,Europe/San_
Marino,Europe/Sarajevo,Europe/Simferopol,Europe/Skopje,Europe/Sofia,Europe/Stockholm,Europe/Tallinn,Europe/Vatican,Europe/Vienna,Europe/Vilnius,Europe/Warsa
w,Europe/Zagreb,Europe/Zurich,Indian/Antananarivo,Indian/Maldives,Indian/Mauritius,Pacific/Auckland,Pacific/Fiji,Pacific/Guam,Pacific/Honolulu,Pacific/Kirit
imati,Pacific/Noumea,America/Antigua,MST,Asia/Kolkata,America/Santiago,Africa/Monrovia,Asia/Colombo,America/Chihuahua,America/Tegucigalpa,America/Paramaribo
,Europe/Ljubljana,Asia/Ashgabat,Africa/Asmara,Asia/Brunei,Africa/Bangui,Africa/Banjul,Asia/Bishkek,Africa/Bissau,Africa/Bujumbura,Asia/Dili,Asia/Dushanbe,Pa
cific/Funafuti,Pacific/Guadalcanal,Africa/Juba,America/St_Vincent,Africa/Libreville,Africa/Lome,Africa/Luanda,Africa/Lusaka,Pacific/Majuro,Africa/Malabo,Afr
ica/Maputo,Africa/Mbabane,Indian/Comoro,Africa/Ndjamena,Pacific/Palau,Africa/Niamey,Europe/Nicosia,Africa/Nouakchott,Pacific/Tongatapu,Africa/Ouagadougou,Am
erica/Lower_Princes,Europe/Podgorica,Pacific/Efate,Africa/Porto-
Novo,Asia/Tashkent,Asia/Tbilisi,Asia/Thimphu,Europe/Tirane,Europe/Vaduz,Asia/Vientiane,Africa/Windhoek,Africa/Douala,Pacific/Nauru,Asia/Yerevan,Etc/GMT+12,E
tc/GMT+11,America/Adak,Pacific/Marquesas,Etc/GMT+9,Etc/GMT+8,America/Whitehorse,Pacific/Easter,America/Regina,America/Indianapolis,America/Cuiaba,America/Ar
aguaina,America/Cayenne,America/Godthab,America/Punta_Arenas,America/Miquelon,America/Bahia,Etc/GMT+2,Atlantic/Azores,Etc/UTC,Europe/Kaliningrad,Europe/Astr
akhan,Europe/Samara,Europe/Saratov,Europe/Volgograd,Asia/Yekaterinburg,Asia/Qyzylorda,Asia/Katmandu,Asia/Omsk,Asia/Barnaul,Asia/Hovd,Asia/Krasnoyarsk,Asia/N
ovosibirsk,Asia/Tomsk,Asia/Irkutsk,Australia/Eucla,Asia/Chita,Asia/Yakutsk,Pacific/Port_Moresby,Australia/Hobart,Asia/Vladivostok,Australia/Lord_Howe,Pacifi
c/Bougainville,Asia/Srednekolymsk,Asia/Magadan,Pacific/Norfolk,Asia/Sakhalin,Asia/Kamchatka,Etc/GMT-12,Pacific/Chatham,Etc/GMT-13,Pacific/Apia">
    <!--ro, opt, enum, IANA synchronization, subType:string, attr:opt{opt, string}-->Africa/Abidjan
  </IANA>
  <isSupportSearchDeviceIANAList>
    <!--ro, opt, bool, whether the device supports searching for supported IANA list, desc:POST /ISAPI/System/time/SearchDeviceIANAList?format=json-->true
  </isSupportSearchDeviceIANAList>
  <windowsZone opt="Dateline Standard Time,UTC-11,Aleutian Standard Time,Hawaiian Standard Time,Marquesas Standard Time,Alaskan Standard Time,UTC-09,Pacific
Standard Time (Mexico),UTC-08,Pacific Standard Time,US Mountain Standard Time,Mountain Standard Time (Mexico),Mountain Standard Time,Yukon Standard
Time,Central America Standard Time,Central Standard Time,Easter Island Standard Time,Central Standard Time (Mexico),Canada Central Standard Time,SA Pacific
Standard Time,Eastern Standard Time (Mexico),Eastern Standard Time,Haiti Standard Time,Cuba Standard Time,US Eastern Standard Time,Turks And Caicos Standard
Time,Paraguay Standard Time,Atlantic Standard Time,Venezuela Standard Time,Central Brazilian Standard Time,SA Western Standard Time,Pacific SA Standard
Time,Newfoundland Standard Time,Tocantins Standard Time,E. South America Standard Time,SA Eastern Standard Time,Argentina Standard Time,Greenland Standard
Time,Montevideo Standard Time,Magallanes Standard Time,Saint Pierre Standard Time,Bahia Standard Time,UTC-02,Azores Standard Time,Cape Verde Standard
Time,UTC,GMT Standard Time,Greenwich Standard Time,Sao Tome Standard Time,Morocco Standard Time,W. Europe Standard Time,Central Europe Standard Time,Romance
Standard Time,Central European Standard Time,W. Central Africa Standard Time,Jordan Standard Time,GTB Standard Time,Middle East Standard Time,Egypt Standard
Time,E. Europe Standard Time,Syria Standard Time,West Bank Standard Time,South Africa Standard Time,FLE Standard Time,Israel Standard Time,South Sudan
Standard Time,Kaliningrad Standard Time,Sudan Standard Time,Libya Standard Time,Namibia Standard Time,Arabic Standard Time,Turkey Standard Time,Arab
Standard Time,Belarus Standard Time,Russian Standard Time,E. Africa Standard Time,Iran Standard Time,Arabian Standard Time,Astrakhan Standard
Time,Azerbaijan Standard Time,Russia Time Zone 3,Mauritius Standard Time,Saratov Standard Time,Georgian Standard Time,Volgograd Standard Time,Caucasus
Standard Time,Afghanistan Standard Time,West Asia Standard Time,Ekaterinburg Standard Time,Pakistan Standard Time,Qyzylorda Standard Time,India Standard
Time,Sri Lanka Standard Time,Nepal Standard Time,Central Asia Standard Time,Bangladesh Standard Time,Omsk Standard Time,Myanmar Standard Time,SE Asia
Standard Time,Altai Standard Time,W. Mongolia Standard Time,North Asia Standard Time,N. Central Asia Standard Time,Tomsk Standard Time,China Standard
Time,North Asia East Standard Time,Singapore Standard Time,W. Australia Standard Time,Taipei Standard Time,Ulaanbaatar Standard Time,Aus Central W. Standard
Time,Transbaikal Standard Time,Tokyo Standard Time,North Korea Standard Time,Korea Standard Time,Yakutsk Standard Time,Cen. Australia Standard Time,AUS
Central Standard Time,E. Australia Standard Time,AUS Eastern Standard Time,West Pacific Standard Time,Tasmania Standard Time,Vladivostok Standard Time,Lord
Howe Standard Time,Bougainville Standard Time,Russia Time Zone 10,Magadan Standard Time,Norfolk Standard Time,Sakhalin Standard Time,Central Pacific
Standard Time,Russia Time Zone 11,New Zealand Standard Time,UTC+12,Fiji Standard Time,Chatham Islands Standard Time,UTC+13,Tonga Standard Time,Samoa
Standard Time,Line Islands Standard Time">
    <!--ro, opt, enum, subType:string, attr:opt{opt, string}-->Dateline Standard Time
  </windowsZone>
  <isSupportTimeStateParam>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeStateParam?format=json-->true
  </isSupportTimeStateParam>
  <localTimeDST min="0" max="256">
    <!--ro, opt, string, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTimeDST>
  <isSupportStandard>
    <!--ro, opt, bool-->true
  </isSupportStandard>
  <isSupportPTPParams>
    <!--ro, opt, bool, desc:GET/PUT /ISAPI/System/time/PTPParams?format=json-->true
  </isSupportPTPParams>
  <isSupportTimeZoneLock>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeZoneLock?format=json-->true
  </isSupportTimeZoneLock>
</Time>
```


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15.1.7.4 Get device local time

Request URL

GET /ISAPI/System/time/localTime

Query Parameter

None

Request Message

None

Response Message

Binary Data

15.1.7.5 Set device local time

Request URL

PUT /ISAPI/System/time/localTime

Query Parameter

None

Request Message

Binary Data

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

15.1.7.6 Set parameters of all NTP servers

Request URL

PUT /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--opt, object, NTP server information-->
    <id>
      <!--req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

15.1.7.7 Get the parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{req, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--ro, req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--ro, opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--ro, opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

15.1.7.8 Set the parameters of a NTP server

Request URL

PUT /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

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Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, NTP server domain name地址去掉, range:[1,64]-->12345
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,32], desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--opt, bool-->false
  </enabled>
</NTPServer>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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15.1.7.9 Get the parameters of a NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	NTP server No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--ro, req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, NTP server domain name, range:[1,64], dep:and,{$.NTPServer.addressingFormatType,eq,hostname}-->xxx12345
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,32], dep:and,{$.NTPServer.addressingFormatType,eq,ipAddress}-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--ro, opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--ro, opt, bool-->false
  </enabled>
  <devicePortNoValid>
    <!--ro, opt, enum, subType:string-->yes
  </devicePortNoValid>
  <portType>
    <!--ro, opt, enum, subType:string-->auto
  </portType>
  <customPortNo>
    <!--ro, opt, int, range:[1,65535], step:1, dep:and,{$.NTPServer.portType,eq,custom}-->1
  </customPortNo>
  <hostNameExampleList>
    <!--ro, opt, array, subType:object-->
    <hostNameExample>
      <!--ro, opt, object-->
      <hostName>
        <!--ro, opt, string-->xxx12345
      </hostName>
    </hostNameExample>
  </hostNameExampleList>
</NTPServer>
```

15.1.7.10 Get the configuration capability of a specific NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id min="1" max="1">
    <!--ro, req, string, ID, range:[0,1], attr:min{req, int},max{req, int}-->1
  </id>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, IP address type of NTP server, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)--
>hostname
  </addressingFormatType>
  <hostName min="0" max="68">
    <!--ro, opt, string, NTP server domain name, range:[0,68], attr:min{req, int},max{req, int}-->time.windows.com
  </hostName>
  <ipAddress min="0" max="32">
    <!--ro, opt, string, IPv4 address, range:[0,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address min="0" max="128">
    <!--ro, opt, string, IPv6 address, range:[0,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo min="0" max="65535">
    <!--ro, opt, int, port No., range:[0,65535], attr:min{req, int},max{req, int}, desc:port No.-->123
  </portNo>
  <synchronizeInterval min="0" max="10800">
    <!--ro, opt, int, time synchronization interval, range:[0,10800], unit:min, attr:min{req, int},max{req, int}, desc:NTP time synchronization interval,
unit: minute-->1440
  </synchronizeInterval>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <portType opt="auto,custom" def="auto">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->auto
  </portType>
  <customPortNo min="1" max="65535">
    <!--ro, opt, int, range:[0,65535], dep:and,{$.NTPServer.portType,eq,custom}, attr:min{req, int},max{req, int}-->123
  </customPortNo>
</NTPServer>
```

15.1.7.11 Get the configuration capability of parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers/capabilities

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, NTP server address type, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName min="1" max="64">
      <!--ro, opt, string, NTP server domain name, range:[1,64], attr:min{req, int},max{req, int}-->12345
    </hostName>
    <ipAddress min="1" max="32">
      <!--ro, opt, string, IPv4 address, range:[1,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
    </ipAddress>
    <ipv6Address min="1" max="128">
      <!--ro, opt, string, IPv6 address, range:[1,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo min="1" max="65535">
      <!--ro, opt, int, port No., range:[1,65535], attr:min{req, int},max{req, int}, desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval min="1" max="10800">
      <!--ro, opt, int, time synchronization interval, unit:min, attr:min{req, int},max{req, int}-->1440
    </synchronizeInterval>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable, attr:opt{req, string}, desc:true (enable), false (disable)-->true
    </enabled>
  </NTPServer>
</NTPServerList>
```

15.1.7.12 Set the device time type

Request URL

PUT /ISAPI/System/time/timeType?format=json

Query Parameter

None

Request Message

```
{
    "TimeType": {
    /*req, object, time type information, desc:time type information*/
        "type":  "UTC",
        /*req, enum, time type, subType:string, desc:"UTC, local "*/
        "displayFormat":  "MM/dd/yyyy hh:mm"
        /*opt, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

15.1.7.13 Get the device time type

Request URL

GET /ISAPI/System/time/timeType?format=json

Query Parameter

None

Request Message

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None

Response Message

```
{
    "TimeType": {
    /*ro, req, object*/
        "type":  "UTC",
        /*ro, req, enum, value: local,UTC, subType:string, desc:"UTC,local"*/
        "displayFormat":  "MM/dd/yyyy hh:mm"
        /*ro, opt, enum, subType:string*/
    }
}
```

15.1.7.14 Get time zone

Request URL

GET /ISAPI/System/time/timeZone

Query Parameter

None

Request Message

None

Response Message

None

15.1.7.15 Set time zone

Request URL

PUT /ISAPI/System/time/timeZone

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

15.1.8 设备本地升级

15.1.8.1 Get the upgrade package uploading speed

Request URL

GET /ISAPI/System/updateFirmware?format=json

Query Parameter

None

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Request Message

None

Response Message

```
{
    "requestURL":  "/ISAPI/Streaming/channels/1",
    /*ro, opt, string, request URL*/
    "statusCode":  "test",
    /*ro, req, string, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, this field is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, detailed error description, this field is required when the value of statusCode is not 1*/
    "percent":  22
    /*ro, opt, int, uploading process*/
}
```

15.1.8.2 Upgrade devices

Request URL

POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>&childDevID=<devIndex>&isUpdateAuxDevice=<isUpdateAuxDevice>&upgradeToken=<upgradeToken>&upgradeSubOption=<upgradeSubOption>&doubleControlType=<doubleControlType>

Query Parameter

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Parameter Name	Parameter Type	Description
type	enum	device type 1.alarm device: "keypad" (keypad), "wirelessRecv" (wireless receiver module), "wiredZone" (wired zone module), "sirenIndoor" (indoor sounder), "sirenOutdoor" (outdoor sounder), "sirenAudio" (sounder two-way audio), "repeater" (repeater), "bluetoothModule" (bluetooth module), "single" (single output module), "wallSwitch" (wall mounted switch), "smartPlug" (smart plug), "detector" (detector), "transmitter" (transmitter peripheral), "remoteCtrl" (keyfob), "subModule" (sub module), "dispModule" (display module), "netReader" (network card reader), "faceModule" (face module), "touchScreenModule" (touch screen module), "temperatureModule" (temperature measurement module), "lockControlBoard" (lock control board), "networkZoneModule" (network zone module), "userInterfaceBoard" (interface board), "subPermissionController" (sub permission controller), "electricLock" (electric lock), "heatingModule" (heating module), "RS485Module" (RS-485 module), "QRCodeModule" (QR code module), "electricGenie" (electric Genie), "zigbeeModule" (zigbee module), "PMRModule" (PMR module), "networkInputAndOutputModule" (network input and output module via network TAP) 2. access control device: "cardReader" (485 card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm program of card reader), "bluetoothModule" (bluetooth module), "MCU"-MCU upgrade, "temperatureModule" (temperature measurement module), "faceModule" (face module), "touchScreenModule" (touch screen module), "netReader" (network card reader), "userInterfaceBoard" (interface board), "subPermissionController" (sub permission controller), "QRCodeModule" (QR code module) 3. intelligent cabinet: "lockControlBoard" (lock control board) 4. "ledReceiverCard"-receiving card; 5. "subsystem"-sub system; 6. radar-assisted device: "warningScreenFirmware"-firmware upgrade of pre-alarm screen, "warningScreenFont"-font library upgrade of pre-alarm screen, "warningScreenVoice"-audio upgrade of pre-alarm screen; 7. "electricGenie"-electric Genie
moduleAddress	string	module address is an exclusive node for Ax Hybrid example: keypad: GET /ISAPI/SecurityCP/Configuration/outputs?format=json; relay: GET /ISAPI/SecurityCP/Configuration/outputs?format=json; sounder: GET /ISAPI/SecurityCP/Configuration/wirelessSiren?format=json; output module: GET /ISAPI/SecurityCP/Configuration/outputModules?format=json; extension module: GET /ISAPI/SecurityCP/Configuration/extensionModule?format=json; zone: GET /ISAPI/SecurityCP/Configuration/zones?format=json
indexID	string	device No. 1. AX Hybrid Pro and wireless security control upgrade by ID; 2. access control devices upgrade by ID; 3. intelligent cabinet upgrade by IS
devIndex	string	--
isUpdateAuxDevice	string	--
upgradeToken	string	--
upgradeSubOption	string	--
doubleControlType	enum	--

Request Message

Binary Data

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:it describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information, range:[0,1024], desc:this node is used for debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

15.1.8.3 Start upgrading the specified analysis unit (device of the cluster)

Request URL

POST /ISAPI/System/upgradeStatus/startUpgrade?format=json&type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
type	string	--

Request Message

```
{
    "idList": [
    /*opt, array, ID list, subType:object*/
        {
            "id":  "test"
            /*req, string, analysis unit ID*/
        }
]
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:this field is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

15.1.8.4 Get the device upgrade progress

Request URL

GET /ISAPI/System/upgradeStatus?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "requestURL":  "/ISAPI/Streaming/channels/1",
    /*ro, opt, string, request URL*/
    "statusCode":  "test",
    /*ro, req, string, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, This field is required when the value of statusCode is not 1, and it corresponds to subStatusCode.*/
    "errorMsg":  "ok",
    /*ro, opt, string, This field is required when the value of statusCode is not 1. Detailed error description of a certain parameter can be provided*/
    "upgrading":  "TRUE",
    /*ro, opt, string, whether the device is upgrading: “TRUE" (upgrading), "FALSE" (not in upgrading)*/
    "percent":  22,
    /*ro, opt, int, upgrade progress (% complete)*/
    "idList": [
    /*ro, opt, array, ID list, subType:object*/
        {
            "id":  "test",
            /*ro, req, string, analysis unit ID*/
            "percent":  22,
            /*ro, opt, int, upgrade progress (% complete)*/
            "status":  "test"
            /*ro, opt, string, "backingUp" (backing up upgrade)*/
        }
    ]
}
```

15.1.8.5 Get the device upgrading status and progress

Request URL

GET /ISAPI/System/upgradeStatus?type=<type>&childDevID=<childDevID>&id=<id>

Query Parameter

Parameter Name	Parameter Type	Description
type	string	"usbmic" (array microphone module)
childDevID	string	--
id	string	The id indicates the peripheral No. to be upgraded.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<upgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, upgrade status and result, attr:version{req, string, protocolVersion}-->
  <upgrading>
    <!--ro, req, bool, upgrade status-->true
  </upgrading>
  <percent>
    <!--ro, req, int, upgrade progress (% complete), range:[0,100]-->1
  </percent>
  <reboot>
    <!--ro, opt, bool, dep:and,{$.upgradeStatus.percent,eq,100}-->false
  </reboot>
  <upgradeStage>
    <!--ro, opt, enum, subType:string-->software
  </upgradeStage>
</upgradeStatus>
```

15.2 视频通用

15.2.1 实时取流

15.2.1.1 Set the multicast address

Request URL

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PUT /ISAPI/System/Network/extension

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<networkExtension xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, req, object, advanced network parameters, attr:version{req, string, protocolVersion}-->
  <multicastAddress>
    <!--wo, opt, object, multicast address-->
    <ipVersion>
      <!--wo, req, enum, IP address version, subType:string, desc:"v4" (IPv4), "v6" (IPv6), "dual" (IPv4 and IPv6)-->v4
    </ipVersion>
    <ipAddress>
      <!--wo, opt, string, IPv4 address-->172.6.64.7
    </ipAddress>
    <ipv6Address>
      <!--wo, opt, string, IPv6 address-->1080:0:0:0:8:800:200C:417A
    </ipv6Address>
  </multicastAddress>
  <enVirtualHost>
    <!--wo, opt, bool, whether to enable the virtual host-->true
  </enVirtualHost>
</networkExtension>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

15.2.2 手动抓拍

15.2.2.1 Asynchronously get the picture captured manually

Request URL

GET /ISAPI/Streaming/channels/<channelID>/picture/async?format=json&imageType=<imageType>&URLType=<URLType>&channelFormat=<channelFormat>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
imageType	enum	--
URLType	enum	--
channelFormat	string	--

Request Message

None

Response Message

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```
{
    "PictureData": {
    /*ro, opt, object, data of the captured picture*/
        "url":  "test"
        /*ro, req, string, URL of the pictured stored in the picture storage server, range:[1,256]*/
    }
}
```

15.2.3 场景变更

15.2.3.1 Sudden scene change detection

EventType:scenechangedetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"scenechangedetection" (sudden scene change detection)-->scenechangedetection
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
</EventNotificationAlert>
```

15.2.4 视频通道资源管理

15.2.4.1 Set the device parameters of a digital channel

Request URL

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PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/deviceInfo

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, device information, attr:version{opt, string, protocolVersion}-->
  <deviceName>
    <!--req, string, device name-->test
  </deviceName>
  <deviceID>
    <!--req, string, UUID device No.-->fb8d4000-8158-11b5-8450-b4a38245654c
  </deviceID>
  <deviceDescription>
    <!--opt, string, device description-->test
  </deviceDescription>
  <deviceLocation>
    <!--opt, string, device location-->hangzhou
  </deviceLocation>
  <deviceStatus>
    <!--opt, enum, device status, subType:string, desc:"normal", "abnormal"-->normal
  </deviceStatus>
  <DetailAbnormalStatus>
    <!--opt, object, error status details-->
    <hardDiskFull>
      <!--opt, bool, the disk is full-->true
    </hardDiskFull>
    <hardDiskError>
      <!--opt, bool, the disk error-->true
    </hardDiskError>
    <ethernetBroken>
      <!--opt, bool, network disconnected-->true
    </ethernetBroken>
    <ipaddrConflict>
      <!--opt, bool, IP address is conflicted-->true
    </ipaddrConflict>
    <illegalAccess>
      <!--opt, bool, illegal access-->true
    </illegalAccess>
    <recordError>
      <!--opt, bool, recording exception-->true
    </recordError>
    <raidLogicDiskError>
      <!--opt, bool, virtual disk exception in the array-->true
    </raidLogicDiskError>
    <spareWorkDeviceError>
      <!--opt, bool, backup device exception-->true
    </spareWorkDeviceError>
  </DetailAbnormalStatus>
  <systemContact>
    <!--opt, string, manufacturer-->STD-CGI
  </systemContact>
  <model>
    <!--req, string, device model-->iDS-9632NX-I8/X
  </model>
  <serialNumber>
    <!--req, string, serial No.-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
  </serialNumber>
  <macAddress>
    <!--req, string, MAC address-->44:47:cc:c8:d9:e4
  </macAddress>
  <firmwareVersion>
    <!--req, string, firmware version No.-->V4.1.40
  </firmwareVersion>
  <firmwareReleasedDate>
    <!--opt, string, firmware release date-->build 181129
  </firmwareReleasedDate>
  <encoderVersion>
    <!--opt, string, device encoder version No.-->V7.3
  </encoderVersion>
  <encoderReleasedDate>
    <!--opt, string, compile date of the device encoder version-->build 20161223
  </encoderReleasedDate>
  <bootVersion>
    <!--opt, string, boot version-->V1.3.4
  </bootVersion>
  <bootReleasedDate>
    <!--opt, string, release date of boot-->100316
  </bootReleasedDate>
  <panelVersion>
    <!--opt, string, panel version-->V1.0
  </panelVersion>
  <hardwareVersion>
    <!--opt, string, hardware version-->0x0
```

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```
    <!--opt, string, hardware version-->0x0
  </hardwareVersion>
  <decoderVersion>
    <!--opt, string, device decoder version-->V1.0
  </decoderVersion>
  <decoderReleasedDate>
    <!--opt, string, compile date of the device decoder version-->build 20161223
  </decoderReleasedDate>
  <softwareVersion>
    <!--opt, string, software version-->build 181129
  </softwareVersion>
  <capacity>
    <!--opt, int, device capacity, unit: MB-->10240
  </capacity>
  <usedCapacity>
    <!--opt, int, capacity usage, unit: MB-->5120
  </usedCapacity>
  <telecontrolID>
    <!--opt, int, remote control control ID, range:[1,255]-->1
  </telecontrolID>
  <supportBeep>
    <!--opt, bool, whether to support buzzer-->true
  </supportBeep>
  <supportVideoLoss>
    <!--opt, bool, whether to support video loss-->true
  </supportVideoLoss>
  <firmwareVersionInfo>
    <!--opt, string, firmware version information-->1
  </firmwareVersionInfo>
  <actualFloorNum>
    <!--opt, int, actual number of floors, range:[1,255]-->1
  </actualFloorNum>
  <localZoneNum>
    <!--opt, int, number of local zones-->1
  </localZoneNum>
  <alarmOutNum>
    <!--opt, int, number of alarm outputs-->1
  </alarmOutNum>
  <distanceResolution>
    <!--opt, float, distance resolution-->0.000
  </distanceResolution>
  <angleResolution>
    <!--opt, float, angle resolution-->0.000
  </angleResolution>
  <speedResolution>
    <!--opt, float, speed resolution-->0.000
  </speedResolution>
  <detectDistance>
    <!--opt, float, detection distance-->0.000
  </detectDistance>
  <relayNum>
    <!--opt, int, number of local relays-->1
  </relayNum>
  <electroLockNum>
    <!--opt, int, number of local electronic locks-->1
  </electroLockNum>
  <RS485Num>
    <!--opt, int, number of local RS-485-->1
  </RS485Num>
  <radarVersion>
    <!--opt, string, radar version-->test
  </radarVersion>
  <cameraModuleVersion>
    <!--opt, string, camera module version-->test
  </cameraModuleVersion>
  <mainversion>
    <!--opt, int, main version No., range:[1,255]-->1
  </mainversion>
  <subversion>
    <!--opt, int, sub version No., range:[1,255]-->1
  </subversion>
  <upgradeversion>
    <!--opt, int, updated version No., range:[1,255]-->1
  </upgradeversion>
  <customizeversion>
    <!--opt, int, customized version No., range:[1,255]-->1
  </customizeversion>
  <companyName>
    <!--opt, string, manufacturer name-->test
  </companyName>
  <copyright>
    <!--opt, string, copyright Information-->test
  </copyright>
  <systemName>
    <!--opt, enum, storage system name, subType:string, desc:"storageManagement" (storage management system), "distributedStorageManagement" (distributed
storage management system)-->storageManagement
  </systemName>
  <systemStatus>
    <!--opt, enum, system status, subType:string, desc:"configured", "unConfigured" (not configured)-->configured
  </systemStatus>
  <isLeaderDevice>
    <!--opt, bool, whether it is the corresponding device of the resource IP address-->true
  </isLeaderDevice>
  <clusterVersion>
```

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```
    <!--opt, string, system cluster version, desc:this node is valid when the value of isLeaderDevice is true-->test
  </clusterVersion>
  <centralStorageVersion>
    <!--opt, string, center storage version, range:[1,16]-->test
  </centralStorageVersion>
  <powerOnMode>
    <!--opt, enum, startup mode, subType:string, desc:"button" (press button to power on), "adapter" (connect adapter to power on)-->button
  </powerOnMode>
  <customizedInfo>
    <!--opt, string, customized project No., desc:order No. of the customization project. For baseline devices,this node is empty; for custom devices,the
order No. of the customization project will be returned by this node-->test
  </customizedInfo>
  <charEncodeFormat>
    <!--opt, enum, character encoding, subType:string, desc:GB2312:GB2312;GBK:GBK-->GB2312
  </charEncodeFormat>
  <languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech">
    <!--opt, enum, language type, subType:string, attr:opt{req, string}, desc:"Chinese", "English", "Spanish", "Portuguese", "Italian", "French", "Russian",
"Turkish", "Greek", "Czech"-->chinese
  </languageType>
  <DockStation>
    <!--opt, object, dock station information configuration-->
    <Platform>
      <!--opt, object, platform dock station information configuration-->
      <ip>
        <!--opt, string, IP address-->test
      </ip>
      <port>
        <!--opt, int, communication port-->1
      </port>
      <userName>
        <!--req, string, user name-->test
      </userName>
      <password>
        <!--req, string, password-->test
      </password>
    </Platform>
    <centralStorageBackupEnabled>
      <!--opt, bool, whether to enable center storage backup-->true
    </centralStorageBackupEnabled>
  </DockStation>
  <webVersion>
    <!--opt, string, Web version No.-->test
  </webVersion>
  <deviceRFProgramVersion>
    <!--opt, string, version No. of the device's RF (Radio Frequency) program-->test
  </deviceRFProgramVersion>
  <securityModuleSerialNo>
    <!--opt, string, security module serial No.-->test
  </securityModuleSerialNo>
  <securityModuleVersion>
    <!--opt, string, security module version No.-->test
  </securityModuleVersion>
  <securityChipVersion>
    <!--opt, string, security chip version No.-->test
  </securityChipVersion>
  <securityModuleKeyVersion>
    <!--opt, string, security module key version No.-->test
  </securityModuleKeyVersion>
  <UIDLampRecognition>
    <!--opt, object, information of the UID lamp recognition device-->
    <enabled>
      <!--opt, bool, whether to enable the function-->true
    </enabled>
  </UIDLampRecognition>
  <confDeviceIdPrefix>
    <!--opt, bool, whether the meeting uses the device name as its prefix-->true
  </confDeviceIdPrefix>
  <OEMCode>
    <!--opt, enum, manufacturer OME code, subType:int, desc:1 (Hikvision device), 0 (neutral device)-->1
  </OEMCode>
  <bootTime>
    <!--opt, datetime, system boot time-->1970-01-01T00:00:00+08:00
  </bootTime>
  <intelligentAnalysisEngineModel>
    <!--opt, string, engine model-->test
  </intelligentAnalysisEngineModel>
</DeviceInfo>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

15.2.4.2 Get the capability of activating digital channels

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/activate/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoSourceActivationCapability xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <ActivateByChannel>
    <!--ro, opt, object-->
  </ActivateByChannel>
  <ActivateByAddress>
    <!--ro, opt, object-->
  </ActivateByAddress>
</VideoSourceActivationCapability>
```

15.2.4.3 Activate the network devices

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/activate

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<VideoSourceActivation xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <userName>
    <!--req, string, user name, range:[1,32]-->test
  </userName>
  <passwordType>
    <!--opt, enum, password types, subType:string, desc:"default" (NVR password), "selfDefine", "IPCActivatePasswd" (activation password)-->default
  </passwordType>
  <password>
    <!--req, string, password, range:[1,16]-->test
  </password>
  <ActivationDevice>
    <!--opt, object-->
    <mode>
      <!--req, enum, "channel,descriptor", subType:string, desc:"channel, descriptor"-->channel
    </mode>
    <channel>
      <!--opt, object, channel information, desc:channel information-->
      <dynVideoInputChannelID>
        <!--req, int-->1
      </dynVideoInputChannelID>
    </channel>
    <descriptor>
      <!--opt, object-->
      <macAddress>
        <!--req, string, MAC address, range:[1,48]-->test
      </macAddress>
      <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"hostname" (host name), "ipaddress" (ip address)-->hostname
      </addressingFormatType>
      <hostName>
        <!--opt, string, host name, range:[1,64]-->test
      </hostName>
      <ipAddress>
        <!--opt, string, IPv4 address, range:[1,32]-->test
      </ipAddress>
      <ipv6Address>
        <!--opt, string, IPv6 address, range:[1,128]-->test
      </ipv6Address>
      <portNo>
        <!--req, int, port No., range:[1,65535]-->1
      </portNo>
    </descriptor>
  </ActivationDevice>
</VideoSourceActivation>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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15.2.4.4 Get status of all digital channels in pages

Request URL

POST /ISAPI/ContentMgmt/InputProxy/channels/statusPage

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CMSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, search description, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of search results this time-->1
  </maxResults>
  <searchResultType>
    <!--opt, enum, search type, subType:string, desc:"all" (search all, default value), "boundAudio" (only search the binding relations between videos and audio channels, and only boundAudioChannelList will be returned in the search result)-->all
  </searchResultType>
</CMSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search result, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <responseStatus>
    <!--ro, req, bool, search result-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"MORE" (searching for more data), "NO MATCHES" (no matched data), "OK" (searching completed)-->MORE
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <InputProxyChannelStatusList>
    <!--ro, opt, array, channel status list, subType:object-->
    <InputProxyChannelStatus>
      <!--ro, opt, object, channel status-->
      <id>
        <!--ro, req, string, index-->test
      </id>
      <sourceInputPortDescriptor>
        <!--ro, req, object, input channel description-->
        <addressingFormatType>
          <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress". It is invalid when the protocol type is GB28181-->hostname
        </addressingFormatType>
        <hostName>
          <!--ro, opt, string, domain name, range:[1,64], desc:it is invalid when the protocol type is GB28181-->test
        </hostName>
        <ipAddress>
          <!--ro, opt, string, IPv4 address, range:[1,32], desc:it is invalid when the protocol type is GB28181-->test
        </ipAddress>
        <ipv6Address>
          <!--ro, opt, string, IPv6 address, range:[1,128], desc:it is invalid when the protocol type is GB28181-->test
        </ipv6Address>
        <managePortNo>
          <!--ro, req, int, port No. under management, range:[1,65535]-->1
        </managePortNo>
        <srcInputPort>
          <!--ro, req, string, device channel No., range:[1,16]-->test
        </srcInputPort>
        <userName>
          <!--ro, req, string, user name, range:[1,32]-->test
        </userName>
        <password>
          <!--ro, req, string, password, range:[1,16], desc:this field is empty-->test
        </password>
        <connMode>
          <!--ro, opt, enum, connection mode, subType:string, desc:"plugplay" (plug and play), "manual"-->plugplay
        </connMode>
        <streamType>
          <!--ro, opt, enum, streaming protocol type, subType:string, desc:"auto", "TCP", "UDP"-->auto
        </streamType>
        <deviceID>
          <!--ro, opt, string, device ID, range:[0,20], desc:the node exists only when the protocol type is GB28181-->test
        </deviceID>
        <getStreamMode>
          <!--ro, opt, enum, streaming mode, subType:string, desc:"active", "passive". The node exists only when the protocol type is GB28181-->active
        </getStreamMode>
        <cameraType>
          <!--ro, opt, enum, camera type, subType:string, desc:"", "", "teacherFull", "studentFull", "media", "blackboard", "teacherLocation",
```

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```
"studentLocation", "blackboardLocation", "none"-->none
        </cameraType>
        <isDoubleAdd>
          <!--ro, opt, bool, whether to add two streams of a same camera, desc:it is used when DeepinView is added to education sharing server-->true
        </isDoubleAdd>
        <streamUrl>
          <!--ro, opt, string, RTSP streaming URL, range:[1,256], desc:if this node exists, use this URL for streaming; if not, follow the original rule--
>test
        </streamUrl>
        <macAddress>
          <!--ro, opt, string, MAC address, range:[1,48]-->test
        </macAddress>
        <deviceName>
          <!--ro, opt, string, device name-->DS-2XM6736FWD-IM/ND
        </deviceName>
        <firmwareVersion>
          <!--ro, opt, string, software version-->V5.5.62 build 180822
        </firmwareVersion>
        <model>
          <!--ro, opt, string, device model-->iDS-9632NX-I8/X
        </model>
        <serialNumber>
          <!--ro, opt, string, serial No.-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
        </serialNumber>
        <subnetMask>
          <!--ro, opt, string, IPv4 subnet mask, desc:subnet mask for IPv4 address-->test
        </subnetMask>
        <bitMask>
          <!--ro, opt, int, IPv6 subnet mask, desc:bitmask IPv6 address-->1
        </bitMask>
        <DefaultGateway>
          <!--ro, opt, object, gateway-->
          <ipAddress>
            <!--ro, opt, string, IPv4 address-->test
          </ipAddress>
          <ipv6Address>
            <!--ro, opt, string, IPv6 address-->test
          </ipv6Address>
        </DefaultGateway>
        <SDKMapPort>
          <!--ro, opt, int-->8000
        </SDKMapPort>
        <SSHMapPort>
          <!--ro, opt, int-->22
        </SSHMapPort>
        <cloudStrageCfg>
          <!--ro, opt, object, dep:or,{$.InputProxyChannel.sourceInputPortDescriptor.proxyProtocol,eq,cloudStrage}-->
          <accessKey>
            <!--ro, opt, string, range:[1,64]-->test
          </accessKey>
          <secretKey>
            <!--ro, opt, string, range:[1,64]-->test
          </secretKey>
          <encoderID>
            <!--ro, opt, string, range:[1,64]-->test
          </encoderID>
          <protocolType>
            <!--ro, opt, string, range:[1,64]-->V3.1
          </protocolType>
        </cloudStrageCfg>
      </sourceInputPortDescriptor>
      <online>
        <!--ro, req, bool, whether the channel is online-->true
      </online>
      <supportCreateStream>
        <!--ro, opt, bool, whether creating stream is supported-->true
      </supportCreateStream>
      <streamingProxyChannelIdList>
        <!--ro, req, array, streaming channel list, subType:object-->
        <streamingProxyChannelId>
          <!--ro, opt, string, streaming channel-->test
        </streamingProxyChannelId>
      </streamingProxyChannelIdList>
      <relatedIOProxy>
        <!--ro, opt, object, linked IO port-->
        <inputProxyPortIdList>
          <!--ro, opt, array, input port No. list, subType:object-->
          <inputProxyPortId>
            <!--ro, opt, string, input port No.-->test
          </inputProxyPortId>
        </inputProxyPortIdList>
        <outputProxyPortIdList>
          <!--ro, opt, array, output port No. list, subType:object-->
          <outputProxyPortId>
            <!--ro, opt, string, output port No.-->test
          </outputProxyPortId>
        </outputProxyPortIdList>
      </relatedIOProxy>
      <chanDetectResult>
        <!--ro, opt, enum, channel detection status, subType:string, desc:"connect" (connected), "overSysBandwidth" (exceeded the system bandwidth),
"domainError" (domain error), "ipcStreamFail" (network camera streaming failed), "connecting" (connecting), "chanNoError" (incorrect channel number),
"ipAddrConflictWithDev" (IP address is conflicted with current device's), "ipAddrConflicWithIpc" (IP address is conflicted with other device's.),
"errorUserNameOrPasswd" (incorrect user name or password), "netUnreachable" (network is unreachable), "unknownError" (unknown error), "notExist" (not
exist), "ipcStreamTypeNotSupport" (the network camera stream type is not supported), "ipcResolutionNotSupport" (the network camera resolution is not
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```
exist)", "ipcStreamTypeNotSupport" (the network camera stream type is not supported), "ipcResolutionNotSupport" (the network camera resolution is not
supported), "userLocked" (the user is locked), "userNotExist" (the user does not exist), "ipcUnregistered" (network camera is not registered),
"ipcNotActivated" (network camera is inactivated), "poePortDetecting" (detecting POE port), "uploadingCloudFile" (uploading files to cloud storage),
"certificateValidationError" (certification validation failed), "domainResolutionFail" (domain resolution failed), "sleep", "wakeUping" (waking up)--
>connect
      </chanDetectResult>
      <SecurityStatus>
        <!--ro, opt, object, security status-->
        <PasswordStatus>
          <!--ro, opt, enum, password status, subType:string, desc:"ipcNotActivated" (IPC is not activated), "risk", "weak", "medium", "strong", "invalid"--
>ipcNotActivated
        </PasswordStatus>
      </SecurityStatus>
      <chanStatus>
        <!--ro, opt, enum, channel status, subType:string, desc:"locate", "getStream"-->locate
      </chanStatus>
      <channelName>
        <!--ro, opt, string, channel name-->--
      </channelName>
      <IP>
        <!--ro, opt, string, IP address, range:[1,128]-->test
      </IP>
      <deviceName>
        <!--ro, opt, string, device name-->test
      </deviceName>
      <recordInformation>
        <!--ro, opt, object, recording parameters-->
        <status>
          <!--ro, req, enum, recording status, subType:string, desc:"recording", "noVideo"" (no video recording), "recordingFailed" (recording failed)--
>recording
        </status>
        <recordType>
          <!--ro, opt, enum, recording type, subType:string, desc:"CMR" (scheduled recording), "MOTION" (motion detection recording), "AllEvent" (event
recording), "MANUAL" (manual recording)-->CMR
        </recordType>
        <failedReason>
          <!--ro, opt, enum, recording failed reason, subType:string, desc:"diskFull" (disk full), "diskError" (disk error), "noAssociatedDiskGroup" (not
linked to disk group), "other"-->diskFull
        </failedReason>
      </recordInformation>
      <isSettingRecordPlan>
        <!--ro, opt, bool, whether to configure recording schedule-->true
      </isSettingRecordPlan>
      <videoPlanStatus>
        <!--ro, opt, enum, recording schedule status, subType:string, desc:"normal", "busy", "abnormal", "planless" (unplanned)-->normal
      </videoPlanStatus>
      <videoPassBackStatus>
        <!--ro, opt, enum, recording copy-back status, subType:string, desc:"normal", "busy", "abnormal", "planless" (unplanned)-->normal
      </videoPassBackStatus>
      <videoPassBackDetail>
        <!--ro, opt, string, recording copy-back details-->test
      </videoPassBackDetail>
      <videoFirstStartTime>
        <!--ro, opt, datetime, earliest recording time-->1970-01-01T00:00:00+08:00
      </videoFirstStartTime>
      <nodeId>
        <!--ro, opt, string, node ID-->test
      </nodeId>
      <bitStreamType>
        <!--ro, req, enum, stream type, subType:string, desc:"mainStream", "subStream"-->mainStream
      </bitStreamType>
      <rateType>
        <!--ro, req, enum, bit rate type, subType:string, desc:"CBR" (constant bit rate), "VBR" (variable bit rate)-->CBR
      </rateType>
      <codeParam>
        <!--ro, req, enum, encoding parameters, subType:string, desc:"H.264", "H.265"-->H.264
      </codeParam>
      <enabled>
        <!--ro, opt, bool, enable the channel, desc:the default value is true-->true
      </enabled>
      <chanExceptionDetails>
        <!--ro, opt, string, channel detection status detailed information, dep:and,
{$.InputProxyChannelStatusList[*].InputProxyChannelStatus.chanDetectResult,ue,connect}, desc:can be used to locate channel access exceptions-->incorrect
username or password
      </chanExceptionDetails>
      <batteryPercentage>
        <!--ro, opt, int, channel camera's remaining battery percentage, range:[0,100], desc:it is valid when a solar camera is accessed-->0
      </batteryPercentage>
    </InputProxyChannelStatus>
  </InputProxyChannelStatusList>
  <boundAudioChannelList>
    <!--ro, opt, array, linked audio channel status list, subType:object-->
    <boundAudioChannel>
      <!--ro, opt, object, linked audio channel information-->
      <channelID>
        <!--ro, req, int, video channel No.-->test
      </channelID>
      <channelName>
        <!--ro, opt, string, channel name.-->test
      </channelName>
      <boundAudioChannelID>
        <!--ro, opt, int, linked audio channel No., desc:if this node does not exist, no audio channel is linked to-->1
      </boundAudioChannelID>
    </boundAudioChannel>
```

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```
  </boundAudioChannelList>
</SearchResult>
```

15.2.4.5 Get the status capability of all digital channels in pages

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/statusPage/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CMSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, search conditions, attr:version{req, string, protocolVersion}-->
  <searchID min="1" max="10">
    <!--ro, req, string, search ID, attr:min{req, int},max{req, int}, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition min="1" max="10">
    <!--ro, req, int, the start position of the search result in the result list, attr:min{req, int},max{req, int}, desc:the initial value is 0-->1
  </searchResultPosition>
  <maxResults min="1" max="10">
    <!--ro, req, int, number of results in this search, attr:min{req, int},max{req, int}, desc:number of channels in a single search in this protocol-->1
  </maxResults>
  <searchResultType opt="all,boundAudio" def="all">
    <!--ro, opt, enum, search type, subType:string, attr:opt{opt, string},def{req, bool}, desc:"all" (search all), "boundAudio" (only search the binding relations between videos and audio channels)-->all
  </searchResultType>
</CMSearchDescription>
```

15.2.4.6 Search for device information of all digital channels

Request URL

POST /ISAPI/ContentMgmt/InputProxy/deviceInfo

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ProxyDevicesInfoCondition xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of search results this time-->1
  </maxResults>
</ProxyDevicesInfoCondition>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ProxyDevicesInfoResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <responseStatus>
    <!--ro, req, bool, search status-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, search status description, subType:string, desc:"MORE" (searching for more data), "NO MATCHES" (no matched data),"OK" (searching completed)-->MORE
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
```

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```
</totalMatches>
<DeviceInfoList>
  <!--ro, opt, array, refer to XML_DeviceInfo, subType:object-->
  <DeviceInfo>
    <!--ro, opt, object-->
    <deviceName>
      <!--ro, req, string, device name-->test
    </deviceName>
    <deviceID>
      <!--ro, req, string, uuid-->fb8d4000-8158-11b5-8450-b4a38245654c
    </deviceID>
    <deviceDescription>
      <!--ro, opt, string, device description-->test
    </deviceDescription>
    <deviceLocation>
      <!--ro, opt, string, device location-->hangzhou
    </deviceLocation>
    <deviceStatus>
      <!--ro, opt, enum, device status, desc:"normal", "abnormal"-->normal
    </deviceStatus>
    <DetailAbnormalStatus>
      <!--ro, opt, object, abnormal status-->
      <hardDiskFull>
        <!--ro, opt, bool, whether disk is full-->true
      </hardDiskFull>
      <hardDiskError>
        <!--ro, opt, bool, whether disk has error-->true
      </hardDiskError>
      <ethernetBroken>
        <!--ro, opt, bool, whether it is disconnected-->true
      </ethernetBroken>
      <ipaddrConflict>
        <!--ro, opt, bool, whether IP address conflicts-->true
      </ipaddrConflict>
      <illegalAccess>
        <!--ro, opt, bool, whether it is illegal access-->true
      </illegalAccess>
      <recordError>
        <!--ro, opt, bool, whether video exception occurred-->true
      </recordError>
      <raidLogicDiskError>
        <!--ro, opt, bool, whether raid logic disk exception occurred-->true
      </raidLogicDiskError>
      <spareWorkDeviceError>
        <!--ro, opt, bool, whether spare device exception occurred-->true
      </spareWorkDeviceError>
    </DetailAbnormalStatus>
    <systemContact>
      <!--ro, opt, string, manufacturer-->STD-CGI
    </systemContact>
    <model>
      <!--ro, req, string, device model-->iDS-9632NX-I8/X
    </model>
    <serialNumber>
      <!--ro, req, string, serial No.-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
    </serialNumber>
    <macAddress>
      <!--ro, req, string, physical address-->44:47:cc:c8:d9:e4
    </macAddress>
    <firmwareVersion>
      <!--ro, req, string, firmware version-->V4.1.40
    </firmwareVersion>
    <firmwareReleasedDate>
      <!--ro, opt, string-->build 181129
    </firmwareReleasedDate>
    <encoderVersion>
      <!--ro, opt, string, device encoder version No.-->V7.3
    </encoderVersion>
    <encoderReleasedDate>
      <!--ro, opt, string-->build 20161223
    </encoderReleasedDate>
    <bootVersion>
      <!--ro, opt, string, boot version-->V1.3.4
    </bootVersion>
    <bootReleasedDate>
      <!--ro, opt, string, release date of boot-->100316
    </bootReleasedDate>
    <panelVersion>
      <!--ro, opt, string, panel version-->V1.0
    </panelVersion>
    <hardwareVersion>
      <!--ro, opt, string, hardware version-->0x0
    </hardwareVersion>
    <decoderVersion>
      <!--ro, opt, string, device decoder version-->V1.0
    </decoderVersion>
    <decoderReleasedDate>
      <!--ro, opt, string, compile date of the device decoder version-->build 20161223
    </decoderReleasedDate>
    <softwareVersion>
      <!--ro, opt, string, software version-->build 181129
    </softwareVersion>
    <capacity>
      <!--ro, opt, int, device capacity (MB)-->10240
```

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```
        </capacity>
        <usedCapacity>
          <!--ro, opt, int, used capacity (MB)-->5120
        </usedCapacity>
        <deviceType>
          <!--ro, req, enum, device type, subType:string, desc:read-only,"IPCamera,IPDome,DVR,HybirdNVR,NVR,DVS,IPZoom,CVR"-->Blade
        </deviceType>
        <telecontrolID>
          <!--ro, opt, int, "1-255", range:[1,255]-->1
        </telecontrolID>
        <supportBeep>
          <!--ro, opt, bool, whether to support buzzer-->true
        </supportBeep>
        <supportVideoLoss>
          <!--ro, opt, bool, whether it supports video loss-->true
        </supportVideoLoss>
        <firmwareVersionInfo>
          <!--ro, opt, string, firmware version information-->1
        </firmwareVersionInfo>
        <actualFloorNum>
          <!--ro, opt, int, "1-128", range:[1,255]-->1
        </actualFloorNum>
        <localZoneNum>
          <!--ro, opt, int, number of local zones-->1
        </localZoneNum>
        <alarmOutNum>
          <!--ro, opt, int, number of alarm outputs-->1
        </alarmOutNum>
        <distanceResolution>
          <!--ro, opt, float, distance resolution-->0.000
        </distanceResolution>
        <angleResolution>
          <!--ro, opt, float, angle resolution-->0.000
        </angleResolution>
        <speedResolution>
          <!--ro, opt, float, speed resolution-->0.000
        </speedResolution>
        <detectDistance>
          <!--ro, opt, float, detection distance-->0.000
        </detectDistance>
        <relayNum>
          <!--ro, opt, int, number of local relays-->1
        </relayNum>
        <electroLockNum>
          <!--ro, opt, int, number of local electronic locks-->1
        </electroLockNum>
        <RS485Num>
          <!--ro, opt, int, number of local RS-485-->1
        </RS485Num>
        <radarVersion>
          <!--ro, opt, string, radar version-->test
        </radarVersion>
        <cameraModuleVersion>
          <!--ro, opt, string, camera module version-->test
        </cameraModuleVersion>
        <mainversion>
          <!--ro, opt, int, main version No., range:[1,255]-->1
        </mainversion>
        <subversion>
          <!--ro, opt, int, sub version No., range:[1,255]-->1
        </subversion>
        <upgradeversion>
          <!--ro, opt, int, update version No., range:[1,255]-->1
        </upgradeversion>
        <customizeversion>
          <!--ro, opt, int, custom version No., range:[1,255]-->1
        </customizeversion>
        <companyName>
          <!--ro, opt, string, manufacturer name-->test
        </companyName>
        <copyright>
          <!--ro, opt, string, copyright Information-->test
        </copyright>
        <systemName>
          <!--ro, opt, enum, storage system name, subType:string, desc:"storageManagement (storage management system), "distributedStorageManagement"
(distributed storage management system)-->storageManagement
        </systemName>
        <systemStatus>
          <!--ro, opt, enum, system status, subType:string, desc:"configured" (system has been configured), "unConfigured" (system has not been configured)--
>configured
        </systemStatus>
        <isLeaderDevice>
          <!--ro, opt, bool, whether it is the corresponding device of the resource IP-->true
        </isLeaderDevice>
        <clusterVersion>
          <!--ro, opt, string, system cluster version, desc:this field is valid only when isLeaderDevice is "true"-->test
        </clusterVersion>
        <manufacturer>
          <!--ro, opt, enum, manufacturer information, subType:string, desc:it is used to differentiate OEM devices-->test
        </manufacturer>
        <centralStorageVersion>
          <!--ro, opt, string, center storage version, range:[1,16]-->test
        </centralStorageVersion>
```

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```
        <powerOnMode>
          <!--ro, opt, enum, startup mode, subType:string, desc:"button" (press button to power on), "adapter" (connect adapter to power on)-->button
        </powerOnMode>
        <customizedInfo>
          <!--ro, opt, string, custom project No., desc:custom project information-->test
        </customizedInfo>
        <charEncodeFormat>
          <!--ro, opt, enum, subType:string-->GB2312
        </charEncodeFormat>
        <languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->chinese
        </languageType>
        <DockStation>
          <!--ro, opt, object, dock station information configuration-->
          <Platform>
            <!--ro, opt, object, platform dock station information configuration-->
            <type>
              <!--ro, req, enum, platform which is connected to the dock station, subType:string, desc:"none", "9533", "8618", "ISAPI"-->none
            </type>
            <ip>
              <!--ro, opt, string, IP address-->test
            </ip>
            <port>
              <!--ro, opt, int, communication port-->1
            </port>
            <userName>
              <!--ro, req, string, user name-->test
            </userName>
            <password>
              <!--ro, req, string, password-->test
            </password>
          </Platform>
          <centralStorageBackupEnabled>
            <!--ro, opt, bool-->true
          </centralStorageBackupEnabled>
        </DockStation>
        <webVersion>
          <!--ro, opt, string, Web version No.-->test
        </webVersion>
        <deviceRFProgramVersion>
          <!--ro, opt, string-->test
        </deviceRFProgramVersion>
        <securityModuleSerialNo>
          <!--ro, opt, string, security module serial No.-->test
        </securityModuleSerialNo>
        <securityModuleVersion>
          <!--ro, opt, string-->test
        </securityModuleVersion>
        <securityChipVersion>
          <!--ro, opt, string-->test
        </securityChipVersion>
        <securityModuleKeyVersion>
          <!--ro, opt, string-->test
        </securityModuleKeyVersion>
        <UIDLampRecognition>
          <!--ro, opt, object, recognize the device information by UID-->
          <enabled>
            <!--ro, opt, bool, whether to enable the function-->true
          </enabled>
        </UIDLampRecognition>
        <confDeviceIdPrefix>
          <!--ro, opt, bool, whether the meeting uses the device name as its prefix-->true
        </confDeviceIdPrefix>
        <OEMCode>
          <!--ro, opt, enum, manufacturer OME code, subType:int, desc:1 (standard device), 0 (neutral device)-->1
        </OEMCode>
        <bootTime>
          <!--ro, opt, datetime, system boot time-->1970-01-01T00:00:00+08:00
        </bootTime>
        <intelligentAnalysisEngineModel>
          <!--ro, opt, string, engine model-->test
        </intelligentAnalysisEngineModel>
        <channel>
          <!--ro, opt, int, channel No. of the NVR with the network camera-->1
        </channel>
      </DeviceInfo>
    </DeviceInfoList>
  </ProxyDevicesInfoResult>
```

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15.2.4.7 Add a patrol of a digital channel

Request URL

POST /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--req, string-->test
  </id>
  <patrolName>
    <!--req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--opt, enum, type, subType:string, desc:“relative”, “absolute”-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--req, array, subType:object-->
    <PatrolSequence>
      <!--req, object-->
      <presetID>
        <!--req, string, preset ID-->test
      </presetID>
      <seqSpeed>
        <!--req, string, patrol speed-->test
      </seqSpeed>
      <delay>
        <!--req, int, delay time, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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15.2.4.8 Add a preset to a digital PTZ channel

Request URL

POST /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPreset xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, PTZ preset, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--req, string, listening host ID-->test
  </id>
  <presetName>
    <!--req, string, preset name-->test
  </presetName>
</PTZPreset>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

15.3 客流统计

15.3.1 客流实时统计

15.3.1.1 Get the people counting parameters of one video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/counting

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Counting xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <MountingConfiguration>
    <!--ro, opt, object, installation parameters-->
    <viewingAngle>
      <!--ro, req, enum, "vertical,tilt", subType:string, desc:"tilt", "vertical"-->vertical
    </viewingAngle>
    <mountHeight>
      <!--ro, opt, int, unit: cm, range:[1,1000], unit:cm-->300
```

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```
    </mountHeight>
    <horizontalDistance>
      <!--ro, opt, int, unit: cm, range:[1,1000], unit:cm-->30
    </horizontalDistance>
    <focalLength>
      <!--ro, opt, int, range:[1,1000], unit:mm-->1
    </focalLength>
  </MountingConfiguration>
  <OverlayConfiguration>
    <!--ro, opt, object, counting OSD parameters-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
    <OSDType>
      <!--ro, opt, enum, OSD type, subType:string, desc:it is valid when enabled is "true","enter,leave,entreLeave,peoplePassing"-->none
    </OSDType>
    <child>
      <!--ro, opt, bool, whether to enable children counting for OSD-->true
    </child>
    <OSDList>
      <!--ro, opt, array, subType:object, range:[0,9]-->
      <OSDType>
        <!--ro, opt, enum, subType:string-->enter
      </OSDType>
    </OSDList>
  </OverlayConfiguration>
  <Demarcation>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <DemarcationRegionList>
      <!--ro, req, array, subType:object-->
      <DemarcationRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <RegionCoordinatesList>
          <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </DemarcationRegion>
    </DemarcationRegionList>
    <DemarcationLine>
      <!--ro, req, object-->
      <StartPoint>
        <!--ro, req, object, start point, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </StartPoint>
      <EndPoint>
        <!--ro, req, object, end point, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </EndPoint>
    </DemarcationLine>
  </Demarcation>
  <CountingRegionType>
    <!--ro, req, enum, detection type, subType:string, desc:read-only,detection type: "region,line"-->region
  </CountingRegionType>
  <detectionPartType>
    <!--ro, opt, enum, subType:string-->headAndShoulders
  </detectionPartType>
  <CountingRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <CountingRegion>
      <!--ro, opt, object, detection area-->
      <id>
        <!--ro, req, int, detection area ID-->1
      </id>
```

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```
        <direction>
          <!--ro, opt, object-->
          <startPoint>
            <!--ro, req, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->1
            </positionY>
          </startPoint>
          <endPoint>
            <!--ro, req, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->1
            </positionY>
          </endPoint>
        </direction>
        <sensitivityLevel>
          <!--ro, opt, int, target detection sensitivity, range:[1,100]-->50
        </sensitivityLevel>
        <spaceGenerationSpeed>
          <!--ro, opt, int, range:[1,100]-->50
        </spaceGenerationSpeed>
        <timeGenerationSpeed>
          <!--ro, opt, int, range:[1,100]-->50
        </timeGenerationSpeed>
        <countingSpeed>
          <!--ro, opt, int, range:[1,100]-->50
        </countingSpeed>
        <detectionType>
          <!--ro, opt, enum, detection type, subType:string, desc:"auto", "head", "shoulder"-->auto
        </detectionType>
        <objectSizeCorrection>
          <!--ro, opt, int, range:[1,100]-->50
        </objectSizeCorrection>
        <RegionCoordinatesList>
          <!--ro, opt, array, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <RuleRegionCoordinatesList>
          <!--ro, opt, array, rule frame, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RuleRegionCoordinatesList>
        <RuleRegionLimit>
          <!--ro, opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
          <height>
            <!--ro, req, int, height, range:[0,1000]-->1
          </height>
          <width>
            <!--ro, req, int, width, range:[0,1000]-->1
          </width>
        </RuleRegionLimit>
        <enabled>
          <!--ro, opt, bool, whether to enable the function, desc:whether to enable the function-->true
        </enabled>
        <regionColor>
          <!--ro, opt, enum, subType:string-->blue
        </regionColor>
        <TrajectoryCountFilter>
          <!--ro, opt, object-->
          <enabled>
            <!--ro, req, bool-->true
          </enabled>
          <residenceTime>
            <!--ro, opt, int, range:[1,60], unit:s-->10
          </residenceTime>
        </TrajectoryCountFilter>
      </CountingRegion>
```

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```
    </CountingRegionList>
    <CountingLineItemList>
      <!--ro, opt, array, list of detection lines, subType:object-->
      <CountingLineItem>
        <!--ro, opt, object, detection line-->
        <id>
          <!--ro, req, int, detection line ID-->1
        </id>
        <direction>
          <!--ro, req, object, detection line direction-->
          <startPoint>
            <!--ro, req, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->1
            </positionY>
          </startPoint>
          <endPoint>
            <!--ro, req, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->1
            </positionY>
          </endPoint>
        </direction>
        <sensitivityLevel>
          <!--ro, opt, int, target detection sensitivity, range:[1,100], desc:the default value is 50-->50
        </sensitivityLevel>
        <spaceGenerationSpeed>
          <!--ro, opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
        </spaceGenerationSpeed>
        <timeGenerationSpeed>
          <!--ro, opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
        </timeGenerationSpeed>
        <countingSpeed>
          <!--ro, opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
        </countingSpeed>
        <detectionType>
          <!--ro, opt, enum, detection type, subType:string, desc:"auto", "head", "shoulder"-->auto
        </detectionType>
        <objectSizeCorrection>
          <!--ro, opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
        </objectSizeCorrection>
        <LineCoordinatesList>
          <!--ro, opt, array, subType:object, range:[0,4]-->
          <Coordinates>
            <!--ro, req, object, coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </Coordinates>
        </LineCoordinatesList>
        <polyline>
          <!--ro, opt, bool, this node should be set to "true" when displaying polyline; otherwise,this node does not exist, desc:this node should be set to
"true" when displaying polyline; otherwise, this node does not exist-->true
        </polyline>
      </CountingLineItem>
    </CountingLineItemList>
    <dataUploadCycle>
      <!--ro, opt, int, interval for uploading data, unit:min-->1
    </dataUploadCycle>
    <SECUploadEnabled>
      <!--ro, opt, bool, uploading mechanism for each second-->true
    </SECUploadEnabled>
    <InterferenceSuppression>
      <!--ro, opt, object-->
      <shadow>
        <!--ro, opt, bool-->true
      </shadow>
      <loitering>
        <!--ro, opt, bool-->true
      </loitering>
      <cart>
        <!--ro, opt, bool-->true
      </cart>
    </InterferenceSuppression>
    <EmailReport>
      <!--ro, opt, object-->
      <DayReport>
        <!--ro, opt, bool, daily report-->true
      </DayReport>
      <WeekReport>
        <!--ro, opt, bool, weekly report-->true
      </WeekReport>
      <MonthReport>
        <!--ro, opt, bool, monthly report-->true
```

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```
    </MonthReport>
    <YearReport>
      <!--ro, opt, bool, yearly report-->true
    </YearReport>
    <reportFormat>
      <!--ro, opt, string, email report format, desc:"excel" (default), "csv", "txt", "xml"-->excel
    </reportFormat>
  </EmailReport>
  <CountingCalibrate>
    <!--ro, opt, object-->
    <calibrateType>
      <!--ro, opt, enum, calibration type, subType:string, desc:"automatic,manual,no",calibration types,including Auto,Manual,and None-->automatic
    </calibrateType>
    <SetupParam>
      <!--ro, req, object, installation parameters-->
      <height>
        <!--ro, opt, float, height, range:[200.0,500.0], unit:cm, desc:the default value is 300-->300.0
      </height>
      <tiltAngle>
        <!--ro, opt, float, angle of pitch, range:[-90.0,90.0], unit:°, desc:the default value is 0-->0.0
      </tiltAngle>
      <heelAngle>
        <!--ro, opt, float, angle of inclination, range:[0.0,180.0], unit:°, desc:the default value is 0-->0.0
      </heelAngle>
    </SetupParam>
    <CountingArea>
      <!--ro, req, object, counting area (in red frame)-->
      <RegionCoordinatesList>
        <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </CountingArea>
    <AutomaticCalib>
      <!--ro, opt, object, it is valid when calibrateType is "automaticCalib"-->
      <CalibRegion>
        <!--ro, opt, object, calibration area (in green frame)-->
        <RegionCoordinatesList>
          <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </CalibRegion>
    </AutomaticCalib>
  </CountingCalibrate>
  <HeightFilterOverlay>
    <!--ro, opt, object, height filter-->
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <heightFilter>
      <!--ro, opt, int, height filter, range:[0,200], unit:cm-->0
    </heightFilter>
  </HeightFilterOverlay>
  <countingType>
    <!--ro, opt, enum, method to trigger counting, subType:string, desc:“none”, “alarmInputTrigger”, “videoTrigger"-->none
  </countingType>
  <signalType>
    <!--ro, opt, enum, signal type, subType:string, desc:"level”, “pulses"-->level
  </signalType>
  <RS485TransmissionEnabled>
    <!--ro, opt, bool, "false,true"-->true
  </RS485TransmissionEnabled>
  <dailyResetTime>
    <!--ro, opt, time, e.g.,"22:00:00+08:00", desc:e.g.,"22:00:00+08:00"-->22:00:00+08:00
  </dailyResetTime>
  <streamOverlayRuleInfos>
    <!--ro, opt, bool, whether to enable overlaying rule information in stream-->true
  </streamOverlayRuleInfos>
  <detectThreshold>
    <!--ro, opt, int, global detection threshold, range:[0,100], desc:the default value is 90-->90
  </detectThreshold>
  <aidedTrackEnabled>
    <!--ro, opt, bool, whether to enable auxiliary tracking, desc:it is enabled by default-->true
  </aidedTrackEnabled>
  <TargetRegionCoordinatesList>
    <!--ro, opt, array, target frame, subType:object, range:[0,4], desc:only supports square-->
    <RegionCoordinates>
      <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
```

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```
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </TargetRegionCoordinatesList>
  <ChildFilter>
    <!--ro, opt, object, counting children-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <heightThreshold>
      <!--ro, opt, int, height threshold of children,the counting will start when the height of children is higher than configured height threshold,unit: cm,value range: [100,200], range:[100,200], unit:cm, desc:height threshold of children,the counting will start when the height of children is higher than configured height threshold,unit: cm,value range: [100,200]-->150
    </heightThreshold>
  </ChildFilter>
  <MisinfoFilter>
    <!--ro, opt, object, false alarm filtering-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <threshold>
      <!--ro, opt, int, threshold of false alarm filtering, range:[0,100], desc:the value ranges from 0 to 100 and the default value is 15. Larger threshold value corresponds to higher precision-->15
    </threshold>
    <timesLimit>
      <!--ro, opt, int, judge times of false alarm, range:[1,5], desc:value range: [1,5], default value: 3, more attempts corresponds to higher precision-->3
    </timesLimit>
  </MisinfoFilter>
  <detectionMode>
    <!--ro, opt, enum, target detection mode, subType:string, desc:detection mode: single-detect based on tracking algorithm,double-detect based on depth map,double_single-detect based on depth map mainly and tracking algorithm secondarily,single_double-detect based on tracking algorithm mainly and depth map secondarily-->single
  </detectionMode>
  <TrajectoryCountFilter>
    <!--ro, opt, object, pattern counting filtering-->
    <enabled>
      <!--ro, req, bool, whether to enable the function, desc:whether to enable the function-->true
    </enabled>
    <movementDisplacement>
      <!--ro, opt, int, displacement range, range:[0,200], unit:cm-->40
    </movementDisplacement>
    <residenceTime>
      <!--ro, opt, float, dwell time duration, range:[0.0,10.0], unit:s-->0.1
    </residenceTime>
  </TrajectoryCountFilter>
  <RegionsDirectionList>
    <!--ro, opt, array, counting direction, subType:object-->
    <RegionsDirection>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->1
      </id>
      <StartRegionCoordinatesList>
        <!--ro, req, array, start region, subType:object, range:[3,10], desc:start region-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </StartRegionCoordinatesList>
      <EndRegionCoordinatesList>
        <!--ro, req, array, end region, subType:object, range:[3,10], desc:end region-->
        <RegionCoordinates>
          <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,10]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,10]-->1
          </positionY>
        </RegionCoordinates>
      </EndRegionCoordinatesList>
    </RegionsDirection>
  </RegionsDirectionList>
  <maintenanceModeEnabled>
    <!--ro, opt, bool, whether to enable maintenance mode-->true
  </maintenanceModeEnabled>
  <OverlayConfigurationList>
    <!--ro, opt, array, subType:object-->
    <OverlayConfiguration>
      <!--ro, opt, object, counting OSD parameters-->
      <id>
        <!--ro, req, int-->1
      </id>
      <name>
```

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```
        <!--ro, opt, string-->test
      </name>
      <enabled>
        <!--ro, req, bool, whether to enable OSD overlay, desc:whether to enable OSD overlay-->true
      </enabled>
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
      </positionY>
      <OSDType>
        <!--ro, opt, enum, OSD type, subType:string, desc:it is valid when enabled is "true","enter,leave,entreLeave,peoplePassing"-->enter
      </OSDType>
      <OSDList>
        <!--ro, opt, array, subType:object, range:[0,9]-->
        <OSDType>
          <!--ro, opt, enum, subType:string-->enter
        </OSDType>
      </OSDList>
    </OverlayConfiguration>
  </OverlayConfigurationList>
  <analysisMode>
    <!--ro, opt, enum, analysis mode, subType:string, desc:"realTimeMode"-real time, "autoSwitchMode"-auto switch-->realTimeMode
  </analysisMode>
  <countingSceneMode>
    <!--ro, opt, enum, subType:string-->entrance
  </countingSceneMode>
  <flowDirectionAnalysis>
    <!--ro, opt, bool-->true
  </flowDirectionAnalysis>
  <dataUploadType>
    <!--ro, opt, object-->
    <enterEnabled>
      <!--ro, opt, bool-->true
    </enterEnabled>
    <leaveEnabled>
      <!--ro, opt, bool-->true
    </leaveEnabled>
  </dataUploadType>
  <InterPupileRegion>
    <!--ro, opt, object-->
    <positionX>
      <!--ro, req, int, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--ro, req, int, range:[0,1000]-->50
    </positionY>
    <width>
      <!--ro, req, int, range:[0,1000]-->50
    </width>
    <height>
      <!--ro, req, int, range:[0,1000]-->50
    </height>
  </InterPupileRegion>
  <MaxInterPupilRegion>
    <!--ro, opt, object-->
    <positionX>
      <!--ro, req, int, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--ro, req, int, range:[0,1000]-->50
    </positionY>
    <width>
      <!--ro, req, int, range:[0,1000]-->50
    </width>
    <height>
      <!--ro, req, int, range:[0,1000]-->50
    </height>
  </MaxInterPupilRegion>
  <SizeFilter>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <filterMultiple>
      <!--ro, opt, int, range:[4,6]-->0
    </filterMultiple>
  </SizeFilter>
  <expertModeEnabled>
    <!--ro, opt, bool-->false
  </expertModeEnabled>
  <debugEnabled>
    <!--ro, opt, bool-->false
  </debugEnabled>
  <congestion>
    <!--ro, opt, object-->
    <minPeopleNum>
      <!--ro, opt, int, range:[0,999]-->100
    </minPeopleNum>
    <maxFlowRate>
      <!--ro, opt, int, range:[0,9999]-->1000
    </maxFlowRate>
```

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```
    </congestion>
  </Counting>
```

15.3.1.2 Set the people counting parameters for a specific channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Counting xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <MountingConfiguration>
    <!--opt, object, installation parameters-->
    <viewingAngle>
      <!--req, enum, "vertical,tilt", subType:string, desc:"tilt", "vertical"-->vertical
    </viewingAngle>
    <mountHeight>
      <!--opt, int, unit: cm, range:[1,1000], unit:cm-->300
    </mountHeight>
    <horizontalDistance>
      <!--opt, int, unit: cm, range:[1,1000], unit:cm-->30
    </horizontalDistance>
    <focalLength>
      <!--opt, int, range:[1,1000], unit:mm-->1
    </focalLength>
  </MountingConfiguration>
  <OverlayConfiguration>
    <!--opt, object, counting OSD parameters-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
    <OSDType>
      <!--opt, enum, OSD type, subType:string, desc:it is valid when enabled is "true","enter,leave,entreLeave,peoplePassing"-->none
    </OSDType>
    <child>
      <!--opt, bool, whether to enable children counting for OSD-->true
    </child>
  </OverlayConfiguration>
  <Demarcation>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <DemarcationRegionList>
      <!--req, array, subType:object-->
      <DemarcationRegion>
        <!--opt, object-->
        <id>
          <!--req, int-->1
        </id>
        <RegionCoordinatesList>
          <!--opt, array, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
```

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```
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </DemarcationRegion>
    </DemarcationRegionList>
    <DemarcationLine>
      <!--req, object-->
      <StartPoint>
        <!--req, object, start point, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </StartPoint>
      <EndPoint>
        <!--req, object, end point, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </EndPoint>
    </DemarcationLine>
  </Demarcation>
  <CountingRegionType>
    <!--req, enum, detection type, subType:string, desc:"region", "line"-->region
  </CountingRegionType>
  <CountingRegionList>
    <!--opt, array, detection area list, subType:object-->
    <CountingRegion>
      <!--opt, object, detection area-->
      <id>
        <!--req, int, detection area ID-->1
      </id>
      <direction>
        <!--req, object-->
        <startPoint>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </startPoint>
        <endPoint>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </endPoint>
      </direction>
      <sensitivityLevel>
        <!--opt, int, target detection sensitivity, range:[1,100]-->50
      </sensitivityLevel>
      <spaceGenerationSpeed>
        <!--opt, int, range:[1,100]-->50
      </spaceGenerationSpeed>
      <timeGenerationSpeed>
        <!--opt, int, range:[1,100]-->50
      </timeGenerationSpeed>
      <countingSpeed>
        <!--opt, int, range:[1,100]-->50
      </countingSpeed>
      <detectionType>
        <!--opt, enum, detection type, subType:string, desc:"auto", "head", "shoulder"-->auto
      </detectionType>
      <objectSizeCorrection>
        <!--opt, int, range:[1,100]-->50
      </objectSizeCorrection>
      <RegionCoordinatesList>
        <!--opt, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <RuleRegionCoordinatesList>
        <!--opt, array, rule frame, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
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```
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RuleRegionCoordinatesList>
      <RuleRegionLimit>
        <!--opt, object, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
        <height>
          <!--req, int, height, range:[0,1000]-->1
        </height>
        <width>
          <!--req, int, width, range:[0,1000]-->1
        </width>
      </RuleRegionLimit>
      <enabled>
        <!--opt, bool-->true
      </enabled>
    </CountingRegion>
  </CountingRegionList>
  <CountingLineItemList>
    <!--opt, array, list of detection lines, subType:object-->
    <CountingLineItem>
      <!--opt, object, detection line-->
      <id>
        <!--req, int, detection line ID-->1
      </id>
      <direction>
        <!--req, object, detection line direction-->
        <startPoint>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </startPoint>
        <endPoint>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </endPoint>
      </direction>
      <sensitivityLevel>
        <!--opt, int, target detection sensitivity, range:[1,100], desc:the default value is 50-->50
      </sensitivityLevel>
      <spaceGenerationSpeed>
        <!--opt, int, range:[1,100], desc:the default value is 50-->50
      </spaceGenerationSpeed>
      <timeGenerationSpeed>
        <!--opt, int, range:[1,100], desc:the default value is 50-->50
      </timeGenerationSpeed>
      <countingSpeed>
        <!--opt, int, range:[1,100], desc:the default value is 50-->50
      </countingSpeed>
      <detectionType>
        <!--opt, enum, detection type, subType:string, desc:"auto", "head", "shoulder"-->auto
      </detectionType>
      <objectSizeCorrection>
        <!--opt, int, range:[1,100], desc:the default value is 50-->50
      </objectSizeCorrection>
      <LineCoordinatesList>
        <!--opt, array, subType:object, range:[0,2]-->
        <Coordinates>
          <!--req, object, coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </Coordinates>
      </LineCoordinatesList>
      <polyline>
        <!--opt, bool, this node should be set to "true" when displaying polyline; otherwise,this node does not exist, desc:this node should be set to
"true" when displaying polyline; otherwise, this node does not exist-->true
      </polyline>
    </CountingLineItem>
  </CountingLineItemList>
  <dataUploadCycle>
    <!--opt, int, interval for uploading data, unit:min-->1
  </dataUploadCycle>
```

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```
<SECUploadEnabled>
  <!--opt, bool, uploading data for each second-->true
</SECUploadEnabled>
<InterferenceSuppression>
  <!--opt, object-->
  <shadow>
    <!--opt, bool-->true
  </shadow>
  <loitering>
    <!--opt, bool-->true
  </loitering>
  <cart>
    <!--opt, bool-->true
  </cart>
</InterferenceSuppression>
<EmailReport>
  <!--opt, object-->
  <DayReport>
    <!--opt, bool, daily report-->true
  </DayReport>
  <WeekReport>
    <!--opt, bool, weekly report-->true
  </WeekReport>
  <MonthReport>
    <!--opt, bool, monthly report-->true
  </MonthReport>
  <YearReport>
    <!--opt, bool, yearly report-->true
  </YearReport>
  <reportFormat>
    <!--opt, string, email report format, desc:"excel" (default), "csv", "txt", "xml"-->excel
  </reportFormat>
</EmailReport>
<CountingCalibrate>
  <!--opt, object-->
  <calibrateType>
    <!--opt, enum, calibration type, subType:string, desc:"automatic,manual,no",calibration types,including Auto,Manual,and None-->automatic
  </calibrateType>
  <SetupParam>
    <!--req, object, installation parameters-->
    <height>
      <!--opt, float, height, range:[200.0,500.0], unit:cm, desc:the default value is 300-->300.0
    </height>
    <tiltAngle>
      <!--opt, float, angle of pitch, range:[-90.0,90.0], unit:°, desc:the default value is 0-->0.0
    </tiltAngle>
    <heelAngle>
      <!--opt, float, angle of inclination, range:[0.0,180.0], unit:°, desc:the default value is 0-->0.0
    </heelAngle>
  </SetupParam>
  <CountingArea>
    <!--req, object, counting area (in red frame)-->
    <RegionCoordinatesList>
      <!--opt, array, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </CountingArea>
  <AutomaticCalib>
    <!--opt, object, auto calibration-->
    <CalibRegion>
      <!--opt, object, calibration area (in green frame)-->
      <RegionCoordinatesList>
        <!--opt, array, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </CalibRegion>
  </AutomaticCalib>
</CountingCalibrate>
<HeightFilterOverlay>
  <!--opt, object-->
  <enabled>
    <!--opt, bool-->true
  </enabled>
  <heightFilter>
    <!--opt, int, filter height, range:[0,200], unit:cm-->0
  </heightFilter>
</HeightFilterOverlay>
<countingType>
```

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```
    <!--opt, enum, method to trigger counting, subType:string, desc:method to trigger counting: "none,alarmInputTrigger,videoTrigger"-->none
  </countingType>
  <signalType>
    <!--opt, enum, type of the signal, subType:string, desc:signal type: "level,pulses",this node is valid when <countingType> is set to "videoTrigger" or
"alarmInputTrigger"-->level
  </signalType>
  <RS485TransmissionEnabled>
    <!--opt, bool, "false,true"-->true
  </RS485TransmissionEnabled>
  <dailyResetTime>
    <!--opt, time, e.g.,"22:00:00+08:00", desc:e.g.,"22:00:00+08:00"-->22:00:00+08:00
  </dailyResetTime>
  <streamOverlayRuleInfos>
    <!--opt, bool, whether to enable overlaying rule information in stream-->true
  </streamOverlayRuleInfos>
  <detectThreshold>
    <!--opt, int, global detection threshold, range:[0,100], desc:the default value is 90-->90
  </detectThreshold>
  <aidedTrackEnabled>
    <!--opt, bool, whether to enable auxiliary tracking, desc:by default, it is enabled-->true
  </aidedTrackEnabled>
  <TargetRegionCoordinatesList>
    <!--opt, array, target frame, subType:object, range:[0,4], desc:only supports square-->
    <RegionCoordinates>
      <!--req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </TargetRegionCoordinatesList>
  <ChildFilter>
    <!--opt, object, counting children-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <heightThreshold>
      <!--opt, int, threshold of children's height, range:[100,200], unit:cm, desc:the counting will start when the height of children is higher than the
configured height threshold-->150
    </heightThreshold>
  </ChildFilter>
  <MisinfoFilter>
    <!--opt, object, whether to filter false alarms-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <threshold>
      <!--opt, int, threshold of false alarm filtering, range:[0,100], desc:the value ranges from 0 to 100 and the default value is 15. Larger threshold
value corresponds to higher precision-->15
    </threshold>
    <timesLimit>
      <!--opt, int, limit of times for checking false alarms, range:[1,5], desc:the value ranges from 1 to 5 and the default value is 3. Larger value
corresponds to higher precision-->3
    </timesLimit>
  </MisinfoFilter>
  <detectionMode>
    <!--opt, enum, target detection mode, subType:string, desc:"single" (detect based on tracking algorithm), "double" (detect based on depth map),
"double_single" (detect based on depth map mainly and tracking algorithm secondarily), "single_double" (detect based on tracking algorithm mainly and depth
map secondarily)-->single
  </detectionMode>
  <TrajectoryCountFilter>
    <!--opt, object, track counting filtering-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <movementDisplacement>
      <!--opt, int, displacement range, range:[0,200], unit:cm-->40
    </movementDisplacement>
    <residenceTime>
      <!--opt, float, dwell time duration, range:[0.0,10.0], unit:s-->0.1
    </residenceTime>
  </TrajectoryCountFilter>
  <RegionsDirectionList>
    <!--opt, array, counting direction, subType:object-->
    <RegionsDirection>
      <!--opt, object, area direction-->
      <id>
        <!--req, int, detection area ID-->1
      </id>
      <StartRegionCoordinatesList>
        <!--req, array, start area, subType:object, range:[3,10], desc:start region-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </StartRegionCoordinatesList>
```

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```
        <EndRegionCoordinatesList>
          <!--req, array, end area, subType:object, range:[3,10], desc:end region-->
          <RegionCoordinates>
            <!--req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,10]-->1
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,10]-->1
            </positionY>
          </RegionCoordinates>
        </EndRegionCoordinatesList>
      </RegionsDirection>
    </RegionsDirectionList>
    <maintenanceModeEnabled>
      <!--opt, bool, whether to enable maintenance mode-->true
    </maintenanceModeEnabled>
    <OverlayConfigurationList>
      <!--opt, array, subType:object-->
      <OverlayConfiguration>
        <!--opt, object, counting OSD parameters-->
        <id>
          <!--req, int-->1
        </id>
        <name>
          <!--opt, string-->test
        </name>
        <enabled>
          <!--req, bool, whether to enable OSD overlay-->true
        </enabled>
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
        </positionY>
        <OSDType>
          <!--opt, enum, OSD type, subType:string, desc:it is valid when enabled is "true","enter,leave,entreLeave,peoplePassing"-->enter
        </OSDType>
      </OverlayConfiguration>
    </OverlayConfigurationList>
    <analysisMode>
      <!--opt, enum, analysis mode, subType:string, desc:"realTimeMode"-real time,"autoSwitchMode"-auto switch-->realTimeMode
    </analysisMode>
    <countingSceneMode>
      <!--opt, enum, subType:string-->entrance
    </countingSceneMode>
    <flowDirectionAnalysis>
      <!--opt, bool-->true
    </flowDirectionAnalysis>
    <dataUploadType>
      <!--opt, object-->
      <enterEnabled>
        <!--opt, bool-->true
      </enterEnabled>
      <leaveEnabled>
        <!--opt, bool-->true
      </leaveEnabled>
    </dataUploadType>
  </Counting>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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15.3.1.3 Get capability of getting people counting statistics

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CountingCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <CountingRegionType opt="line,region,lineAndRegion">
    <!--ro, opt, string, area type, attr:opt{opt, string}, desc:"line"-->line
  </CountingRegionType>
  <dataUploadCycle opt="1,5,10,15,20,30,60">
    <!--ro, opt, enum, subType:int, attr:opt{req, string}-->1
  </dataUploadCycle>
  <isSupportSECUpload>
    <!--ro, opt, bool-->true
  </isSupportSECUpload>
  <isSupportRecommendValue>
    <!--ro, opt, bool, "true,false"-->true
  </isSupportRecommendValue>
  <isSupportFlashRemoveCouting>
    <!--ro, opt, bool, "true"-->true
  </isSupportFlashRemoveCouting>
  <OSDType opt="enter,leave,enterLeave,
none,enterPeoplePassing,pepolePassing,enterDuplicatePeople,enterLeaveDuplicatePeople,enterLeaveDuplicatePassByShop,enterLeaveDuplicatePassing" def="none">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{opt, string}-->enter
  </OSDType>
  <OverlayConfiguration>
    <!--ro, opt, object-->
    <OSDList size="9">
      <!--ro, opt, object, attr:size{req, int}-->
      <OSDType opt="enter,leave,none">
        <!--ro, opt, string, attr:opt{req, string}-->enter
      </OSDType>
    </OSDList>
  </OverlayConfiguration>
  <InterferenceSuppression>
    <!--ro, opt, object-->
    <shadow opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </shadow>
    <loitering opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </loitering>
    <cart opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </cart>
  </InterferenceSuppression>
  <EmailReport>
    <!--ro, opt, object-->
    <DayReport opt="true,false">
      <!--ro, opt, bool, daily report, attr:opt{req, bool}-->true
    </DayReport>
    <WeekReport opt="true,false">
      <!--ro, opt, bool, weekly report, attr:opt{req, bool}-->true
    </WeekReport>
    <MonthReport opt="true,false">
      <!--ro, opt, bool, monthly report, attr:opt{req, bool}-->true
    </MonthReport>
    <YearReport opt="true,false">
      <!--ro, opt, bool, yearly report, attr:opt{req, bool}-->true
    </YearReport>
    <reportFormat opt="excel,csv,txt,xml">
      <!--ro, opt, string, email report format, attr:opt{req, string}-->excel
    </reportFormat>
  </EmailReport>
  <CountingCalibrate>
    <!--ro, opt, object-->
    <calibrateType opt="automatic,manual,no,smart">
      <!--ro, opt, enum, calibration type, subType:string, attr:opt{req, string}, desc:calibration type-->automatic
    </calibrateType>
    <SetupParam>
      <!--ro, req, object-->
      <height min="200.0" max="500.0">
        <!--ro, opt, float, height, range:[200.0,500.0], unit:cm, attr:min{req, float},max{req, float}-->200.0
      </height>
      <tiltAngle min="-90.0" max="90.0">
        <!--ro, opt, float, range:[-90,90], unit:°, attr:min{req, float},max{req, float}-->-90
      </tiltAngle>
```

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```
      <heelAngle min="0.0" max="180.0">
        <!--ro, opt, float, range:[0,180], unit:°, attr:min{req, float},max{req, float}-->0
      </heelAngle>
    </SetupParam>
    <CountingArea>
      <!--ro, req, object, counting area (red frame)-->
      <RegionCoordinatesList size="4">
        <!--ro, opt, array, rectangle, subType:object, range:[0,4], attr:size{req, string}, desc:rectangle-->
        <RegionCoordinates>
          <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </CountingArea>
    <AutomaticCalib>
      <!--ro, opt, object, automatic calibration-->
      <CalibRegion>
        <!--ro, opt, object, calibration area (green frame)-->
        <RegionCoordinatesList size="4">
          <!--ro, opt, array, rectangle, subType:object, range:[0,4], attr:size{req, string}, desc:rectangle-->
          <RegionCoordinates>
            <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </CalibRegion>
    </AutomaticCalib>
  </CountingCalibrate>
  <HeightFilterOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false" default="false">
      <!--ro, opt, bool, attr:opt{req, bool},default{req, bool}-->true
    </enabled>
    <heightFilter min="0" max="250" default="120">
      <!--ro, opt, int, range:[0,250], unit:cm, attr:min{req, int},max{req, int},default{req, int}-->0
    </heightFilter>
  </HeightFilterOverlay>
  <isSupportCalibrate opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isSupportCalibrate>
  <isSupportPosInfoOverlay opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isSupportPosInfoOverlay>
  <countingType opt="none,alarmInputTrigger,videoTrigger">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->none
  </countingType>
  <alarmInputTrigger_signalType opt="level,pulses">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->level
  </alarmInputTrigger_signalType>
  <videoTrigger_RS485Transmission opt="true,false">
    <!--ro, opt, bool, attr:opt{req, bool}-->true
  </videoTrigger_RS485Transmission>
  <isSupportCountingStatus>
    <!--ro, opt, bool-->true
  </isSupportCountingStatus>
  <isSupportTriggerPeopleCountingDataSearch>
    <!--ro, opt, bool-->true
  </isSupportTriggerPeopleCountingDataSearch>
  <isSupportDailyResetTime>
    <!--ro, opt, bool, whether it supports resetting and clearing by schedule-->true
  </isSupportDailyResetTime>
  <Polyline>
    <!--ro, opt, object, whether it supports polyline-->
    <maxPointNumber>
      <!--ro, req, int-->1
    </maxPointNumber>
  </Polyline>
  <isSupportBusDetectionArea>
    <!--ro, opt, bool-->true
  </isSupportBusDetectionArea>
  <isSupportStreamOverlayRuleInfos>
    <!--ro, opt, bool, whether it supports overlaying intelligent information in stream-->true
  </isSupportStreamOverlayRuleInfos>
  <isSupportMobileDescribeChange>
    <!--ro, opt, bool, this node is for marking the editing of the displayed parameter name of people counting camera-->true
  </isSupportMobileDescribeChange>
  <isSupportAdvanceConfiguration>
    <!--ro, opt, bool, this node is for marking whether the camera supports advanced configuration-->true
  </isSupportAdvanceConfiguration>
  <isSupportSearchDoorStates>
    <!--ro, opt, bool, this node if for supporting the detection of door status triggered by signal of vehicle-mounted device-->true
  </isSupportSearchDoorStates>
  <isSupportRuleRegion>
    <!--ro, opt, bool, whether it supports setting counting area-->true
```

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```
    <!--ro, opt, bool, whether it supports setting counting area-->true
  </isSupportRuleRegion>
  <notSupportStatisticType>
    <!--ro, opt, enum, this node is valid when <isSupportTriggerPeopleCountingDataSearch> is set to "true", subType:string, desc:this node is valid when
<isSupportTriggerPeopleCountingDataSearch> is set to "true"-->enternum
  </notSupportStatisticType>
  <detectThreshold min="0" max="100" def="90">
    <!--ro, opt, int, detection threshold, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->90
  </detectThreshold>
  <aidedTrackEnabled opt="true,false" def="true">
    <!--ro, opt, bool, whether to enable auxiliary tracking, attr:opt{req, bool},def{req, string}-->true
  </aidedTrackEnabled>
  <SceneLimitCapList size="2">
    <!--ro, opt, array, capability list of scene limit, subType:object, attr:size{req, string}-->
    <SceneLimitCap>
      <!--ro, opt, object, capability of a scene limit-->
      <minTargetWidth>
        <!--ro, opt, int, normalized minimum width of target-->1
      </minTargetWidth>
      <maxTargetWidth>
        <!--ro, opt, int, normalized maximum width of target-->1
      </maxTargetWidth>
      <maxRuleRegionArea>
        <!--ro, opt, int, normalized maximum area of rule frame-->1
      </maxRuleRegionArea>
    </SceneLimitCap>
  </SceneLimitCapList>
  <targetRegionEdgesNum min="4" max="4">
    <!--ro, opt, int, capability of edge quantity of target region frame, attr:min{req, string},max{req, int}, desc:capability of frame edge quantity of
targetregion-->4
  </targetRegionEdgesNum>
  <ruleRegionEdgesNum min="4" max="10">
    <!--ro, opt, int, capability of edge quantity of rule frame, range:[4,10], attr:min{req, int},max{req, int}-->4
  </ruleRegionEdgesNum>
  <ChildFilter>
    <!--ro, opt, object, children counting-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <heightThreshold min="100" max="200" def="150">
      <!--ro, opt, int, threshold of the height of children, range:[100,200], unit:cm, attr:min{req, int},max{req, int},def{req, int}-->150
    </heightThreshold>
  </ChildFilter>
  <MisinfoFilter>
    <!--ro, opt, object, false alarm filtering-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <threshold min="0" max="100" def="15">
      <!--ro, opt, int, filtering threshold, range:[0,100], attr:min{req, int},max{req, int},def{req, int}, desc:the larger the threshold value, the higher
the precision-->15
    </threshold>
    <timesLimit min="1" max="5" def="3">
      <!--ro, opt, int, prevention attempts of false alarm, range:[1,5], attr:min{req, int},max{req, int},def{req, int}, desc:the larger the number of
attempts, the higher the precision-->3
    </timesLimit>
  </MisinfoFilter>
  <isSupportReverseAlarm>
    <!--ro, opt, bool, whether it supports setting reverse counting alarm-->true
  </isSupportReverseAlarm>
  <detectionMode opt="single,double,double_single,single_double" def="double_single">
    <!--ro, opt, enum, detection mode, subType:string, attr:opt{req, string},def{req, string}, desc:"single" (detect based on tracking algorithm), "double"
(detect based on depth map), "double_single" (detect based on depth map mainly and tracking algorithm secondarily), "single_double" (detect based on
tracking algorithm mainly and depth map secondarily)-->double_single
  </detectionMode>
  <TrajectoryCountFilter>
    <!--ro, opt, object, pattern counting filtering-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <movementDisplacement min="0" max="200" def="40">
      <!--ro, opt, int, motion displacement range, range:[0,200], attr:min{req, int},max{req, int},def{req, int}-->40
    </movementDisplacement>
    <residenceTime min="0.0" max="10.0" def="0.1">
      <!--ro, opt, float, dwell time duration, range:[0.0,10.0], unit:s, attr:min{req, float, unit:s},max{req, float, unit:s},def{req, float, unit:s}-->0.1
    </residenceTime>
  </TrajectoryCountFilter>
  <isSupportParameterExport>
    <!--ro, opt, bool, whether it supports one-touch exporting people counting parameters-->true
  </isSupportParameterExport>
  <MultiRegions min="1" max="8">
    <!--ro, opt, int, number of counting regions, attr:min{req, int},max{req, int}, desc:this node will not be returned if people counting in multiple
regions is not supported-->1
  </MultiRegions>
  <isSupportRegionsDirection>
    <!--ro, opt, bool, whether it supports configuring counting direction-->true
  </isSupportRegionsDirection>
  <maintenanceModeEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable maintenance mode, attr:opt{req, bool}-->true
  </maintenanceModeEnabled>
  <MultiOverlayConfiguration>
    <!--ro, opt, object-->
    <id min="1" max="8">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
```

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```
      </id>
      <name min="1" max="20">
        <!--ro, opt, string, range:[1,20], attr:min{req, string},max{opt, string}-->1
      </name>
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->false
      </enabled>
      <positionX min="0" max="1000">
        <!--ro, req, int, X-coordinate, range:[0,1000], attr:min{req, int},max{req, int}, desc:the origin is the lower-left corner of the screen-->1
      </positionX>
      <positionY min="0" max="1000">
        <!--ro, req, int, Y-coordinate, range:[0,1000], attr:min{opt, string},max{opt, string}, desc:the origin is the lower-left corner of the screen-->1
      </positionY>
      <OSDType opt="enter,leave,enterLeave,enterLeavePassing">
        <!--ro, opt, enum, OSD type, subType:string, attr:opt{req, string}, desc:OSD type-->enter
      </OSDType>
      <OSDList size="9">
        <!--ro, opt, object, attr:size{req, int}-->
        <OSDType opt="enter,leave,none">
          <!--ro, opt, string, attr:opt{req, string}-->enter
        </OSDType>
      </OSDList>
    </MultiOverlayConfiguration>
    <RuleRegionLimit>
      <!--ro, opt, object-->
      <width>
        <!--ro, req, int, width-->1
      </width>
      <height>
        <!--ro, req, int, height-->1
      </height>
    </RuleRegionLimit>
    <isSupportMultiRegionsEnabled>
      <!--ro, opt, bool-->true
    </isSupportMultiRegionsEnabled>
    <analysisMode opt="realTimeMode,autoSwitchMode">
      <!--ro, opt, string, analysis mode, attr:opt{req, string}, desc:"realTimeMode" (real time), "autoSwitchMode" (auto switch). The engine also has its
analysis mode (refer to the <analysisMode> node in /ISAPI/Intelligent/analysisEngines/<ID>). The device will be added to the engine with the same analysis
mode via the load balancing method. If the engine is not configured correspondingly, this node will be failed to be returned, and the error code of
taskNotMatchEngine will be returned-->test
    </analysisMode>
    <countingSceneMode opt="entrance,exit">
      <!--ro, opt, string, attr:opt{req, string}-->entrance
    </countingSceneMode>
    <flowDirectionAnalysis opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->false
    </flowDirectionAnalysis>
    <isSupportRemoveDuplicate>
      <!--ro, opt, bool-->true
    </isSupportRemoveDuplicate>
    <isSupportCaptureOverlap>
      <!--ro, opt, bool-->true
    </isSupportCaptureOverlap>
    <dataUploadType>
      <!--ro, opt, object-->
      <enterEnabled opt="true,false" def="true">
        <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
      </enterEnabled>
      <leaveEnabled opt="true,false" def="true">
        <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
      </leaveEnabled>
    </dataUploadType>
    <InterPupileRegion>
      <!--ro, opt, object-->
      <positionX>
        <!--ro, req, int, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,1000]-->50
      </positionY>
      <width min="0" max="1000">
        <!--ro, req, int, range:[0,1000], attr:min{opt, string},max{opt, string}-->50
      </width>
      <height>
        <!--ro, req, int, range:[0,1000]-->50
      </height>
    </InterPupileRegion>
    <MaxInterPupilRegion>
      <!--ro, opt, object-->
      <positionX>
        <!--ro, req, int, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,1000]-->50
      </positionY>
      <width min="50" max="50">
        <!--ro, req, int, range:[0,1000], attr:min{opt, string},max{opt, string}-->50
      </width>
      <height>
        <!--ro, req, int, range:[0,1000]-->50
      </height>
    </MaxInterPupilRegion>
    <SizeFilter>
      <!--ro, opt, object-->
```

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```
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <filterMultiple min="4" max="6">
      <!--ro, opt, int, range:[4,6], attr:min{req, int},max{req, int}-->0
    </filterMultiple>
  </SizeFilter>
  <expertModeEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, bool}-->false
  </expertModeEnabled>
  <CountingRegionList>
    <!--ro, opt, object-->
    <CountingRegion>
      <!--ro, opt, object-->
      <regionColor opt="blue,red" def="yellow">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->blue
      </regionColor>
      <TrajectoryCountFilter>
        <!--ro, opt, object-->
        <enabled opt="true,false" def="false">
          <!--ro, req, bool, attr:opt{req, string},def{req, bool}-->true
        </enabled>
        <residenceTime min="1" max="60" def="10">
          <!--ro, opt, int, range:[1,60], unit:s, attr:min{req, int},max{req, int},def{req, int}-->10
        </residenceTime>
      </TrajectoryCountFilter>
    </CountingRegion>
  </CountingRegionList>
  <advanceParams>
    <!--ro, opt, object-->
    <algorithmBrightnessThreshold>
      <!--ro, opt, int-->1
    </algorithmBrightnessThreshold>
    <faceExposurEnable opt="true,false" def="false">
      <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
    </faceExposurEnable>
    <referenceBright min="0" max="100" def="50">
      <!--ro, opt, int, range:[0,100], dep:and,{$.CountingCap.advanceParams.faceExposurEnable,eq,true}, attr:min{opt, string},max{opt, string},def{req,
int}-->50
    </referenceBright>
    <minContinueTime min="1" max="60" def="10">
      <!--ro, opt, int, range:[1,60], attr:min{opt, string},max{opt, string},def{req, int}-->10
    </minContinueTime>
    <faceFilterTime min="0" max="100" def="0">
      <!--ro, opt, int, range:[0,100], attr:min{opt, string},max{opt, string},def{req, int}-->0
    </faceFilterTime>
    <algorithmSceneType opt="indoor,outdoor" def="indoor">
      <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->indoor
    </algorithmSceneType>
  </advanceParams>
  <debugEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->false
  </debugEnabled>
  <congestion>
    <!--ro, opt, object-->
    <minPeopleNum min="0" max="999" def="100">
      <!--ro, opt, int, range:[0,999], attr:min{req, int},max{req, int},def{req, int}-->100
    </minPeopleNum>
    <maxFlowRate min="0" max="9999" def="1000">
      <!--ro, opt, int, range:[0,9999], attr:min{req, int},max{req, int},def{req, int}-->1000
    </maxFlowRate>
  </congestion>
  <detectionPartType opt="headAndShoulders,feet">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->headAndShoulders
  </detectionPartType>
  <enabled opt="true,false" def="true">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
  </enabled>
</CountingCap>
```

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15.3.1.4 People counting event

EventType:PeopleCounting

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, people counting event details, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPv4 address-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 Address-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, device's communication port No.-->1
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol, subType:string, desc:"HTTP, HTTPS, EHome". The value should be "EHome" when ISAPI protocol is
```

```
transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->test
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No.-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, event occurred time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, req, int, event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"PeopleCounting" (people counting)-->PeopleCounting
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:"active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, PUID, range:[1,64], desc:required in Ehome alarm, corresponding to deviceID in /ISAPI/System/Network/Ehome-->test
  </deviceID>
  <peopleCounting>
    <!--ro, opt, object, people counting-->
    <statisticalMethods>
      <!--ro, opt, enum, people counting method, subType:string, desc:"realTime" (real-time people counting), "timeRange" (scheduled people counting),
"signalTrigger" (alarm input triggered people counting)-->realTime
    </statisticalMethods>
    <RealTime>
      <!--ro, opt, object, real-time, desc:it is required when statisticalMethods is "realTime"-->
      <time>
        <!--ro, req, datetime, people counting time-->1970-01-01T00:00:00+08:00
      </time>
    </RealTime>
    <TimeRange>
      <!--ro, opt, object, time period for real-time people counting, desc:it is required when statisticalMethods is "timeRange" or "signalTrigger".-->
      <startTime>
        <!--ro, req, datetime, start time of people counting-->1970-01-01T00:00:00+08:00
      </startTime>
      <endTime>
        <!--ro, req, datetime, end time of people counting-->1970-01-01T00:00:00+08:00
      </endTime>
    </TimeRange>
    <enter>
      <!--ro, opt, int, number of entered people-->1
    </enter>
    <exit>
      <!--ro, opt, int, number of exited people-->1
    </exit>
    <pass>
      <!--ro, opt, int, passers-by number, desc:passers-by number-->1
    </pass>
    <duplicatePeople>
      <!--ro, opt, int, number of duplicated people, desc:number of duplicated people-->1
    </duplicatePeople>
    <FlowStatisticsList>
      <!--ro, opt, array, flow direction counting list, subType:object-->
      <FlowStatistics>
        <!--ro, opt, object, flow direction counting, desc:for deduplicated people counting statistics, the flow direction analysis is only valid in the
following circumstances: the rule area is a quadrilateral, the counting broken line is formed with four points, and the counting line has only two points
that overlap with the rule area line-->
        <countingLineID>
          <!--ro, opt, int, counting line ID-->1
        </countingLineID>
        <pass>
          <!--ro, opt, int, passers-by number, desc:people who enter the counting area but do not cross the counting line-->1
        </pass>
        <movingDirectionList>
          <!--ro, opt, array, people moving direction list, subType:object, desc:count the moving direction of people who enter the counting area and cross
the counting line-->
          <MovingDirection>
            <!--ro, opt, object, people moving direction-->
            <countingLineSegmentID>
              <!--ro, opt, int, area ID, range:[1,3], desc:the four points of the broken line are marked as 1, 2, 3, and 4 respectively in counterclockwise
direction from the counting direction. Draw two lines parallel to the counting direction respectively starting from point 2 and point 3 to intersect the
counting rule area. The formed three areas are marked as 1, 2, and 3 in counterclockwise direction from the counting direction-->1
            </countingLineSegmentID>
            <count>
              <!--ro, opt, int, the number of people-->0
            </count>
```

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```
            </count>
          </MovingDirection>
        </movingDirectionList>
      </FlowStatistics>
    </FlowStatisticsList>
    <countingSceneMode>
      <!--ro, opt, enum, people counting scene mode, subType:string, desc:"entrance", "exit". The default mode is entrance-->entrance
    </countingSceneMode>
    <deDuplicateEnter>
      <!--ro, opt, int, deduplicated number of entered people-->0
    </deDuplicateEnter>
    <deDuplicateExit>
      <!--ro, opt, int, deduplicated number of exited people-->0
    </deDuplicateExit>
    <faceLibDuplicatePeopleList>
      <!--ro, opt, array, list of duplicated people in the face picture library, subType:object, desc:count the number of duplicated people in every face
picture library-->
      <faceLibDuplicatePeople>
        <!--ro, opt, object, number of duplicated people in the face picture library-->
        <FDID>
          <!--ro, req, string, face picture library ID, desc:check details of the library with /ISAPI/Intelligent/FDLib/search?format=json-->1
        </FDID>
        <name>
          <!--ro, req, string, face picture library name-->11
        </name>
        <duplicate>
          <!--ro, opt, int, number of duplicated people in the library-->1
        </duplicate>
      </faceLibDuplicatePeople>
    </faceLibDuplicatePeopleList>
  </peopleCounting>
  <childCounting>
    <!--ro, opt, object, children counting-->
    <enter>
      <!--ro, opt, int, number of entered people-->1
    </enter>
    <exit>
      <!--ro, opt, int, number of exited people-->1
    </exit>
  </childCounting>
  <FaceExpressionList>
    <!--ro, opt, array, facial expression statistics, subType:object-->
    <FaceExpression>
      <!--ro, opt, object, facial expression-->
      <value>
        <!--ro, req, enum, facial attributes, subType:string, desc:"unknown", "poker-faced" (no facial expression), "happy", "surprised", "disgusted",
"sad", "angry", "contemptuous", "panic", "all"-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people corresponding to the face expression-->1
      </count>
    </FaceExpression>
  </FaceExpressionList>
  <GenderList>
    <!--ro, opt, array, gender statistics, subType:object-->
    <Gender>
      <!--ro, opt, object, gender-->
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "female", "male", "all"-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people corresponding to the gender-->1
      </count>
    </Gender>
  </GenderList>
  <MaskList>
    <!--ro, opt, array, mask statistics, subType:object-->
    <Mask>
      <!--ro, opt, object, mask wearing status-->
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "yes" (wearing a mask), "no" (not wearing a mask)-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people corresponding to the masks wearing status-->1
      </count>
    </Mask>
  </MaskList>
  <AgeGroupList>
    <!--ro, opt, array, age group statistics, subType:object-->
    <AgeGroup>
      <!--ro, opt, object, age group-->
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "child", "young", "middle", "old", "infant", "kid", "teenager", "prime", "middleAged"
(the middle age), "all"-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people corresponding to the age group-->1
      </count>
    </AgeGroup>
  </AgeGroupList>
  <GlassList>
    <!--ro, opt, array, glasses statistics, subType:object-->
    <Glass>
      <!--ro, opt, object, glasses wearing status-->
```

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```
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "yes" (wearing glasses), "no" (not wearing glasses), "sunglasses" (wearing sunglasses),
"all"-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people corresponding to the glass wearing attribute-->1
      </count>
    </Glass>
  </GlassList>
  <RegionList>
    <!--ro, opt, array, people counting information in the area, subType:object-->
    <Region>
      <!--ro, opt, object, area-->
      <id>
        <!--ro, req, int, area ID-->1
      </id>
      <name>
        <!--ro, opt, string, area name-->test
      </name>
      <enter>
        <!--ro, opt, int, number of entered people-->1
      </enter>
      <exit>
        <!--ro, opt, int, number of exited people-->1
      </exit>
      <totalEnter>
        <!--ro, opt, int, total number of entered people, desc:before resetting on the current day, the number of people will accumulate the number will be
initialized only after manual resetting on the current day or auto resetting on the next day (if the device supports duplicated people counting, the number
will be the deduplicated number of people of the current day)-->1
      </totalEnter>
      <totalExit>
        <!--ro, opt, int, total number of exited people, desc:before resetting on the current day, the number of people will accumulate the number will be
initialized only after manual resetting on the current day or auto resetting on the next day (if the device supports duplicated people counting, the number
will be the deduplicated number of people of the current day)-->1
      </totalExit>
      <totalPass>
        <!--ro, opt, int-->1
      </totalPass>
    </Region>
  </RegionList>
  <targetAttrs>
    <!--ro, opt, string, target attributes, range:[1,1024],
desc:transparent transmission field, which corresponds to targetAttrs in applying task, format:
    <![CDATA[{
    /*opt, transparent transmission field, the following three fileds are recommended*/
    "deviceId": "1",
    /*opt, device ID, string, maximum length 64*/
    "deviceChannel": 1,
    /*opt, device channel, integer32*/
    "deviceName": "shebei"
    /*opt, device name, string, maximum length 128*/
    }]]>, example: <![CDATA[{ "deviceId": "1", "deviceChannel": 1, "deviceName": "shebei"}]]>-->test
  </targetAttrs>
  <targetInfo>
    <!--ro, opt, object, target information, desc:corresponding to targetInfo in heat analysis task /ISAPI/SDT/Management/HeatAnalysis/video/AddTask?
format=json. Either targetInfo or targetAttrs is required-->
    <cameraInfo>
      <!--ro, opt, string, camera information-->test
      <cameraIndexCode>
        <!--ro, req, string, camera ID-->test
      </cameraIndexCode>
      <cameraName>
        <!--ro, opt, string, camera name-->test
      </cameraName>
    </cameraInfo>
    <mapFileID>
      <!--ro, opt, string, map file ID-->test
    </mapFileID>
    <storeID>
      <!--ro, opt, string, store ID-->test
    </storeID>
  </targetInfo>
  <customInfo>
    <!--ro, opt, string, custom information-->test
  </customInfo>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle), 2-"hightPerformanceVehicle" (high-performance
vehicle), 3-"personStructModel" (person structural modeling), 4-"vehicleAndPersonStructModel" (person and vehicle structural modeling), 5-"videoFaceDetect"
(video face detection), 6-"videoFaceDetectModel" (video face detection and modeling), 7-"smart" (face detection + human body detection + behavior
detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
```

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```
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime", "historyvideo", "Localvideo"-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to
128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm, desc:ISO 8601 format, e.g., 2019-12-30T15:00:00.23+08:00-->test
  </timeStamp>
  <DetectionRegionList>
    <!--ro, opt, array, detection area, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity level, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <HatList>
    <!--ro, opt, array, wearing hat statistics, subType:object-->
    <Hat>
      <!--ro, opt, object, whether the person wears a hat-->
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "yes" (wearing a hat), "no" (not wearing a hat), "helmet" (wearing a helmet)-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people wearing hats-->1
      </count>
    </Hat>
  </HatList>
  <ThingsList>
    <!--ro, opt, array, carrying objects statistics, subType:object-->
    <Things>
      <!--ro, opt, object, carrying objects-->
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "yes", "no"-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people carrying objects-->1
      </count>
    </Things>
  </ThingsList>
  <isDataRetransmission>
    <!--ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be
uploaded again after the device recovers-->true
  </isDataRetransmission>
</EventNotificationAlert>
```

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15.3.1.5 People reverse entering alarm

EventType:reverseEntrance

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "test",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, string, protocol type, desc:"HTTP", "HTTPS"*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime":  "2009-11-14T15:27Z",
    /*ro, req, string, alarm trigger time*/
    "activePostCount":  1,
    /*ro, req, int, times that the same alarm has been uploaded*/
    "eventType":  "reverseEntrance",
    /*ro, req, string, event type*/
    "eventState":  "active",
    /*ro, req, string, event status, desc:"active" (triggered), "inactive" (not triggered, the heartbeat data)*/
    "eventDescription":  "reverseEntrance alarm",
    /*ro, req, string, event description*/
    "url":  "test"
    /*ro, opt, string, URL of reverse entering alarm picture, desc:this field is not returned when the data is transmitted in binary format*/
}
```

15.4 异常行为检测

15.4.1 快速移动

15.4.1.1 Fast moving

EventType:rapidMove

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, transmitted parameters, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome".
The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP--
>HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"rapidMove" (fast moving detection)-->rapidMove
  </eventType>
```

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```
<eventState>
  <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
</eventState>
<eventDescription>
  <!--ro, req, string, event description-->test
</eventDescription>
<deviceID>
  <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
</deviceID>
<DetectionRegionList>
  <!--ro, opt, array, detection area list, subType:object-->
  <DetectionRegionEntry>
    <!--ro, opt, object, detection area-->
    <regionID>
      <!--ro, req, string, area ID-->test
    </regionID>
    <sensitivityLevel>
      <!--ro, opt, int, sensitivity, range:[0,100]-->0
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </DetectionRegionEntry>
</DetectionRegionList>
<channelName>
  <!--ro, opt, string, channel name-->test
</channelName>
<Extensions>
  <!--ro, opt, object, extended information-->
  <serialNumber>
    <!--ro, opt, string, serial No.-->test
  </serialNumber>
  <eventPush>
    <!--ro, opt, string, event notification-->test
  </eventPush>
</Extensions>
<detectionPictureTransType>
  <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
</detectionPictureTransType>
<detectionPicturesNumber>
  <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
</detectionPicturesNumber>
<bkgUrl>
  <!--ro, opt, string, background picture URL, desc:background picture URL-->test
</bkgUrl>
<URLCertificationType>
  <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
</URLCertificationType>
<pId>
  <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
</pId>
<targetAttrs>
  <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
</targetAttrs>
<TaskInfo>
  <!--ro, opt, object, task information submitted by data center products-->
  <taskID>
    <!--ro, req, string, task ID, range:[0,64]-->test
  </taskID>
  <algorithmType>
    <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
  </algorithmType>
  <taskName>
    <!--ro, opt, string, task name-->test
  </taskName>
  <algorithmID>
    <!--ro, opt, string, algorithm package ID-->test
  </algorithmID>
  <monitorPointID>
    <!--ro, opt, string, camera No.-->test
  </monitorPointID>
  <monitorPointName>
    <!--ro, opt, string, camera name-->test
  </monitorPointName>
  <streamType>
    <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
  </streamType>
```

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```
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, datetime, timestamp of the frame, which triggers the alarm-->1970-01-01T00:00:00+08:00
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

15.4.2 徘徊侦测

15.4.2.1 Loitering detection

EventType:loitering

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"loitering" (loitering detection)-->loitering
  </eventType>
  <eventState>
```

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```
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, detection area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human"-->human
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
```

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```
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

15.4.3 物品拿取

15.4.3.1 Object removal

EventType:attendedBaggage

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, object removal alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, event triggering frequency, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"attendedBaggage" (object removal)-->attendedBaggage
```

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```
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"others"-->others
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
```

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```
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm, desc:ISO 8601 format, e.g., 2019-12-30T15:00:00.23+08:00-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

15.4.4 人员聚集

15.4.4.1 People gathering

EventType:group

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
```

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```
<activePostCount>
<eventType>
<!--ro, req, enum, event type, subType:string, desc:"group" (people gathering)-->group
</eventType>
<eventState>
<!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
</eventState>
<eventDescription>
<!--ro, req, string, event description-->test
</eventDescription>
<deviceID>
<!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
</deviceID>
<DetectionRegionList>
<!--ro, opt, array, detection area list, subType:object-->
<DetectionRegionEntry>
<!--ro, opt, object, detection area-->
<regionID>
<!--ro, req, string, area ID-->test
</regionID>
<sensitivityLevel>
<!--ro, opt, int, sensitivity, range:[0,100]-->1
</sensitivityLevel>
<RegionCoordinatesList>
<!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
<RegionCoordinates>
<!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
<positionX>
<!--ro, req, int, X-coordinate, range:[0,1000]-->50
</positionX>
<positionY>
<!--ro, req, int, Y-coordinate, range:[0,1000]-->50
</positionY>
</RegionCoordinates>
</RegionCoordinatesList>
<objectOccupation>
<!--ro, opt, int, percentage, range:[1,100]-->1
</objectOccupation>
</DetectionRegionEntry>
</DetectionRegionList>
<channelName>
<!--ro, opt, string, channel name, range:[1,64]-->test
</channelName>
<Extensions>
<!--ro, opt, object, extended information-->
<serialNumber>
<!--ro, opt, string, serial No.-->test
</serialNumber>
<eventPush>
<!--ro, opt, string, event notification-->test
</eventPush>
</Extensions>
<detectionPictureTransType>
<!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
</detectionPictureTransType>
<detectionPicturesNumber>
<!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
</detectionPicturesNumber>
<bkgUrl>
<!--ro, opt, string, background picture URL, desc:background picture URL-->test
</bkgUrl>
<URLCertificationType>
<!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
</URLCertificationType>
<pId>
<!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
</pId>
<targetAttrs>
<!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
</targetAttrs>
<TaskInfo>
<!--ro, opt, object, task information submitted by data center products-->
<taskID>
<!--ro, req, string, task ID, range:[0,64]-->test
</taskID>
<algorithmType>
<!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
</algorithmType>
<taskName>
<!--ro, opt, string, task name-->test
</taskName>
<algorithmID>
<!--ro, opt, string, algorithm package ID-->test
</algorithmID>
<monitorPointID>
<!--ro, opt, string, camera No.-->test
</monitorPointID>
<monitorPointName>
```

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```
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
group	[Message content]	application/xml	--	--	--
.value	[Binary picture data]	image/jpeg	随机值	pic.jpeg	--
video	[Video data]	video/mp4	随机值	video.mp4	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.4.5 物品遗留

15.4.5.1 Unattended baggage detection

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EventType:unattendedBaggage

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, unattended baggage alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, event triggering frequency, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"unattendedBaggage" (unattended baggage)-->unattendedBaggage
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"others"-->others
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
```

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```
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm, desc:ISO 8601 format, e.g., 2019-12-30T15:00:00.23+08:00-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

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15.5 人脸抓拍

15.5.1 人脸1V1比对

15.5.1.1 Set 1V1 face picture comparison parameters and start comparing

Request URL

POST /ISAPI/Intelligent/imagesComparision/face

Query Parameter

None

Request Message

```
{
    "dataType": "URL",
    /*opt, enum, picture data type, subType:string, desc:"URL" (image URL, default type), "modelData" (model data), "binary" (binary image)*/
    "algorithmID": "test",
    /*opt, string, algorithm package ID*/
    "TargetImage": {
    /*req, object, target picture, desc:size of the target picture should be smaller than 8MB. JPEG, JPG, PNG, TIFF, and BMP formats are supported*/
        "URL": "test",
        /*opt, string, target picture URL, desc:the maximum length is 256*/
        "targetModelData": "test",
        /*opt, string, target model data, desc:target model data*/
        "picData": "base64转义的图片二进制数据",
        /*opt, string, picture binary data, desc:this node is required when the value of datatype is "base64". The picture data is transmitted in string
format*/
        "faceRect": {
        /*opt, object, face frame in picture, desc:the origin is the upper-left corner of the screen*/
            "height": 0.810,
            /*req, float, height, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
            "width": 0.430,
            /*req, float, width, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
            "x": 0.130,
            /*req, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
            "y": 0.870
            /*req, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
        }
    },
    "ContrastImage": {
    /*req, object, comparison picture, desc:size of the comparison picture should be smaller than 8MB. JPEG, JPG, PNG, TIFF, and BMP formats are supported*/
        "URL": "test",
        /*opt, string, comparison picture URL, desc:the maximum length is 256*/
        "targetModelData": "test",
        /*opt, string, target model data, desc:target model data*/
        "picData": "base64转义的图片二进制数据",
        /*opt, string, picture data, desc:picture data*/
        "faceRect": {
        /*opt, object, face frame in picture, desc:the origin is the upper-left corner of the screen*/
            "height": 0.810,
            /*req, float, height, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
            "width": 0.430,
            /*req, float, width, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
            "x": 0.130,
            /*req, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
            "y": 0.870
            /*req, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
        }
    }
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, req, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "similarity": 1.0
    /*ro, opt, float, similarity, range:[0,1]*/
}
```

15.5.1.2 Get the 1V1 face picture comparison capability

Request URL

GET /ISAPI/Intelligent/imagesComparision/face/capabilities?format=json

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Query Parameter

None

Request Message

None

Response Message

```
{
    "imagesComparision": {
    /*ro, opt, object, 1V1 face picture comparison*/
        "dataType":  "URL,modelData,binary",
        /*ro, opt, string, picture data type*/
        "algorithmID": {
        /*ro, opt, object, algorithm package ID*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  64
            /*ro, opt, int*/
        },
        "faceRect": {
        /*ro, opt, object, face frame in picture, desc:the origin is the upper-left corner of the screen*/
            "height": {
            /*ro, opt, object, height*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            },
            "width": {
            /*ro, opt, object, width*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            },
            "x": {
            /*ro, opt, object, x-coordinate*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            },
            "y": {
            /*ro, opt, object, y-coordinate*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            }
        }
    }
}
```

15.5.2 人脸抓拍

15.5.2.1 Set the arming schedule of face modeling for all channels

Request URL

PUT /ISAPI/Event/schedules/faceSnapModeling

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapModelingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--req, object-->
    <id>
      <!--req, string-->faceSnapModeling-1
    </id>
    <eventType>
      <!--opt, enum, subType:string-->faceSnapModeling
    </eventType>
    <inputIOPortID>
      <!--opt, string-->test
    </inputIOPortID>
    <outputIOPortID>
      <!--opt, string-->test
    </outputIOPortID>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--opt, object-->
          <vehicleDetectSceneID>
            <!--opt, int-->0
          </vehicleDetectSceneID>
          <sceneID>
            <!--opt, int-->0
          </sceneID>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceSnapModelingScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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15.5.2.2 Get the arming schedule of face modeling for all channels

Request URL

GET /ISAPI/Event/schedules/faceSnapModeling

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapModelingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, req, object-->
    <id>
      <!--ro, req, string-->faceSnapModeling-1
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->faceSnapModeling
    </eventType>
    <inputIOPortID>
      <!--ro, opt, string-->test
    </inputIOPortID>
    <outputIOPortID>
      <!--ro, opt, string-->test
    </outputIOPortID>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--ro, opt, object-->
          <vehicleDetectSceneID>
            <!--ro, opt, int-->0
          </vehicleDetectSceneID>
          <sceneID>
            <!--ro, opt, int-->0
          </sceneID>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceSnapModelingScheduleList>
```

15.5.2.3 Set the arming schedule of face modeling for a specified channel

Request URL

PUT /ISAPI/Event/schedules/faceSnapModeling/faceSnapModeling-<channelID>

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Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->faceSnapModeling-1
  </id>
  <eventType>
    <!--opt, enum, subType:string-->faceSnapModeling
  </eventType>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <outputIOPortID>
    <!--ro, opt, string-->test
  </outputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
      <ScheduleProperity>
        <!--opt, object-->
        <vehicleDetectSceneID>
          <!--opt, int-->0
        </vehicleDetectSceneID>
        <sceneID>
          <!--opt, int-->0
        </sceneID>
      </ScheduleProperity>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

15.5.2.4 Get the arming schedule of face modeling for a specified channel

Request URL

GET /ISAPI/Event/schedules/faceSnapModeling/faceSnapModeling-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceSnapModeling-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->faceSnapModeling
  </eventType>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <outputIOPortID>
    <!--ro, opt, string-->test
  </outputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
      <ScheduleProperity>
        <!--ro, opt, object-->
        <vehicleDetectSceneID>
          <!--ro, opt, int-->0
        </vehicleDetectSceneID>
        <sceneID>
          <!--ro, opt, int-->0
        </sceneID>
      </ScheduleProperity>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

15.5.2.5 Get the linkage parameters of face capture

Request URL

GET /ISAPI/Event/triggers/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Device channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, linkage configuration, attr:version{req, string, protocolVersion}-->
```

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```
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"ADAS", "ADASAlarm", "AID", "ANPR", "AccessControllerEvent", "CDsStatus", "DBD", "GPSUpload", "HFP",
"IO", "IOTD", "LES", 'LFPD", "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup", "TMA", "TMPAM", "VMD", "abnormalAcceleration",
"abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult",
"blindSpotDetection", "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus",
"diskUnformat", "diskerror", "diskfull", "driverConditionMonitor", "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown",
"faultAlarm", "fielddetection", "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp", "group", "hdBadBlock", "hdImpact",
"heatmap", "highHDTemperature", "highTempAlarm", "hotSpare", "illaccess", "ipcTransferAbnorma", "ipconflict", "keyPersonGetUp", "LeavePosition",
"linedetection", "ListSyncException", "loitering", "LowHDTemperature", "mixedTargetDetection", "modelError", "nicbroken", "nodeOffline",
"nonPoliceIntrusio", "overSpeed", "overtimeTarry", "parking", "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm", "personDensityDetection",
"personQueueCounting", "personQueueDetection", "personQueueRealTime", "personQueueTime", "playCellphone", "pocException", "poe", "policeAbsent",
"radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule", "radarTargetDetection", "radarVideoDetection", "raidException",
"rapidMove", "reachHeight", "recordCycleAbnormal", "recordException", "regionEntrance", "regionExiting", "retention", "rollover", "running",
"safetyHelmetDetection", "scenechangedetection", "sensorAlarm", "severeHDFailure", "shelteralarm", "shipsDetection", "sitQuietly", "smokeAndFireDetection",
"smokeDetection", "softIO", "spacingChange", "sysStorFull", "takingElevatorDetection", "targetCapture", "temperature", "thermometry", "thirdPartyException",
"toiletTarry", "tollCodeInfo", "tossing", "unattendedBaggage", "vehicleMatchResult", "vehicleRcogResult", "versionAbnormal", "videoException", "videoloss",
"violationAlarm", "violentMotion", "yardTarry", "faceSnap"-->faceSnap
  </eventType>
  <eventDescription>
    <!--ro, opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel ID, desc:only valid when event types are VMD, videoloss, shelteralarm, regionEntrance, regionExiting,
loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage, thermal, zoneTriggeredAlarm, criticalAlerts, businessCounselingLinkage-->test
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, event capture interval, step:1, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string, wireless ID-->test
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, step:1, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->test
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->test
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, req, int, range:[1,60], step:1, unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <childDevID>
        <!--ro, opt, string-->1
      </childDevID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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15.5.2.6 Set the face capture rule parameters of a device

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceRule

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, face capture rule capability, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--req, enum, whether to enable the function, subType:string, desc:“true”, “false”-->true
  </enabled>
  <FaceDetectionRegion>
    <!--opt, object, detection area-->
    <RegionCoordinatesList>
      <!--opt, array, list of region coordinates, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--req, object, area point coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </FaceDetectionRegion>
  <InterPupileRegion>
    <!--opt, object, rectangle frame of minimum pupil distance, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->50
    </positionY>
    <width>
      <!--req, int, pupil distance, range:[0,1000]-->50
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->50
    </height>
  </InterPupileRegion>
  <MaxInterPupilRegion>
    <!--opt, object, the maximum pupil distance, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->50
    </positionY>
    <width>
      <!--req, int, pupil distance, range:[0,1000]-->50
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->50
    </height>
  </MaxInterPupilRegion>
  <SizeFilter>
    <!--opt, object, size filter-->
    <enabled>
      <!--req, bool, whether to enable size filter-->true
    </enabled>
    <ratio>
      <!--opt, float, ratio of head-shoulder size filter to minimum pupil distance, range:[4.0,6.0]-->5.0
    </ratio>
    <MaxTargetSize>
      <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000], desc:X-coordinate-->0
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
      <width>
        <!--req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->0
      </height>
    </MaxTargetSize>
    <MinTargetSize>
      <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
```

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```
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
      <width>
        <!--req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->0
      </height>
    </MinTargetSize>
  </SizeFilter>
  <detectionType>
    <!--opt, string, face type for detection, desc:"abnormalFace" (abnormal face),"normalFace" (normal face),"multiFace" (multiple faces),"sunglassesFace"
(face,on which wearing sunglasses),"callFace" (face of a person who is making a phone call). Multiple values are separated by commas--
>abnormalFace,normalFace
  </detectionType>
  <autoROIEnable>
    <!--opt, bool, whether to enable face automatic ROI, desc:the value of this node is false by default-->false
  </autoROIEnable>
</FaceRule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

15.5.2.7 Get the face capture rule parameters of a device

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceRule?defualt=<defualt>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
defualt	string	Get default parameters via FaceRule.defualt in /ISAPI/Intelligent/channels//faceRule/capabilities.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, face capture rule capability, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--ro, req, enum, whether to enable, subType:string, desc:“true”, “false”-->true
  </enabled>
  <FaceDetectionRegion>
    <!--ro, opt, object, detection area-->
    <RegionCoordinatesList>
      <!--ro, opt, array, list of region coordinates, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, req, object, area point coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
```

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```
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </FaceDetectionRegion>
    <InterPupileRegion>
      <!--ro, opt, object, rectangle frame of minimum pupil distance, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
      </positionY>
      <width>
        <!--ro, req, int, pupillary distance, range:[0,1000]-->50
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->50
      </height>
    </InterPupileRegion>
    <MaxInterPupilRegion>
      <!--ro, opt, object, the maximum pupillary distance, desc:The origin is the lower-left corner of the screen.-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
      </positionY>
      <width>
        <!--ro, req, int, pupillary distance, range:[0,1000]-->50
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->50
      </height>
    </MaxInterPupilRegion>
    <SizeFilter>
      <!--ro, opt, object, size filter-->
      <enabled>
        <!--ro, req, bool, whether to enable size filter-->true
      </enabled>
      <ratio>
        <!--ro, opt, float, ratio of head-shoulder size filter to minimum pupil distance, range:[4.0,6.0]-->5.0
      </ratio>
      <MaxTargetSize>
        <!--ro, opt, object, target frame on the largest face, desc:The origin is the lower-left corner of the screen.-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000], desc:Value range: [0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
        <width>
          <!--ro, req, int, width, range:[0,1000]-->0
        </width>
        <height>
          <!--ro, req, int, height, range:[0,1000]-->0
        </height>
      </MaxTargetSize>
      <MinTargetSize>
        <!--ro, opt, object, target frame on the smallest face, desc:The origin is the lower-left corner of the screen.-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
        <width>
          <!--ro, req, int, width, range:[0,1000]-->0
        </width>
        <height>
          <!--ro, req, int, height, range:[0,1000]-->0
        </height>
      </MinTargetSize>
    </SizeFilter>
    <detectionType>
      <!--ro, opt, string, face type for detection, desc:abnormalFace (abnormal face), normalFace (normal face), multiFace (multiple faces), sunglassesFace
(face with sunglasses), callFace (face of a person who is making a phone call, which is used to detect faces in ATM mode). This node supports to be set to
multiple values and they should be separated by commas.-->abnormalFace,normalFace
    </detectionType>
    <autoROIEnable>
      <!--ro, opt, bool, whether to enable automatic face ROI, desc:By default it is false.-->false
    </autoROIEnable>
    <minInterPupileDistance>
      <!--ro, opt, int, range:[13,666], step:1, unit:px-->58
    </minInterPupileDistance>
  </FaceRule>
```

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15.5.2.8 Get the capability of configuring advanced parameters of face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/advancedParam/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
}
```

15.5.2.9 Get the capability of configuring parameters for capturing face pictures and overlaying information on the captured face pictures.

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/overlapPic/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
}
```

15.5.2.10 Get the parameters of capturing face pictures and overlaying information on the captured face pictures.

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/overlapPic?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "OverlapPic": {
    /*ro, opt, object, parameters of overlaying information on the picture*/
        "AddIntelInfo": {
        /*ro, opt, object, overlay the VCA information*/
            "streamWithIntelInfo":  true,
            /*ro, opt, bool, overlay the VCA information on the stream*/
            "alarmWithTargetInfo":  false
            /*ro, opt, bool, overlay the target frame on the alarm picture*/
        },
        "TargetPicParam": {
        /*ro, opt, object, target picture parameters*/
            "targetPicMode":  "custom",
            /*ro, opt, enum, target picture mode, subType:string, desc:"custom", "head" (head shot), "headShoulders" (head and shoulders picture), "human"
(full-body shot)*/
```

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```
            "targetPicWidth": 3.0,
            /*ro, opt, float, width*/
            "headHeight": 2.0,
            /*ro, opt, float, head height*/
            "bodyHeight": 7.0,
            /*ro, opt, float, body height*/
            "TargetPicHeight": {
            /*ro, opt, object, target picture with fixed pixels*/
                "enable": true,
                /*ro, req, bool, whether to enable*/
                "height": 100
                /*ro, req, int, height*/
            },
            "FaceBeautification": {
            /*ro, opt, object, face beautification*/
                "enable": true,
                /*ro, req, bool, whether to enable*/
                "level": 100
                /*ro, req, int, beautification level*/
            },
            "faceSharpeningLevel": 50,
            /*ro, opt, int, range:[0,100]*/
            "faceEnhancement": true,
            /*ro, opt, bool, whether to enable face enhancement*/
            "faceEnhancementLevel": 50,
            /*ro, opt, int, range:[0,100]*/
            "faceCloseupRatio": {
            /*ro, opt, object*/
                "upRatio": 20,
                /*ro, opt, int*/
                "downRatio": 20,
                /*ro, opt, int*/
                "leftRatio": 20,
                /*ro, opt, int*/
                "rightRatio": 20
                /*ro, opt, int*/
            },
            "humanEyeLighteningEnabled": false,
            /*ro, opt, bool*/
            "colorReproduction": true
            /*ro, opt, bool*/
        },
        "AlarmPicParam": {
        /*ro, opt, object, alarm picture parameters*/
            "backgroundPicUpload": true,
            /*ro, opt, bool, whether to upload the background picture*/
            "picQuality": "general",
            /*ro, opt, enum, picture quality, subType:string, desc:"best", "better", "general"*/
            "PicSize": {
            /*ro, req, object, picture size*/
                "width": 1920,
                /*ro, req, int, width*/
                "height": 1080
                /*ro, req, int, height*/
            },
            "facePictureUpload": true
            /*ro, opt, bool, whether to upload the close-up face picture (head and shoulders picture), desc:If this field does not exist, the close-up face
picture will be uploaded by default.*/
        },
        "AlarmOsdParam": [
        /*ro, opt, array, parameters of overlaying OSD on the alarm picture, subType:object*/
            {
                "osdIndex": "cameraNo",
                /*ro, opt, enum, overlaid items, subType:string, desc:"cameraNo" (channel No.), "positionInfo" (position information), "snapTimeOsd"
(capture time), "facingPupil" (pupillary distance)*/
                "osdValue": "test",
                /*ro, opt, string, overlaid content, desc:If the value of this field is empty, it indicates that the content needs to be overlaid by DSP.*/
                "enabled": true
                /*ro, opt, bool, whether to enable overlaying the item*/
            }
        ],
        "faceNoiseReduction": true,
        /*ro, opt, bool*/
        "faceMosaic": false,
        /*ro, opt, bool*/
        "backgroundCompress": {
        /*ro, opt, object*/
            "enabled": true,
            /*ro, req, bool*/
            "backgroundCompressLevel": "middle"
            /*ro, opt, enum, subType:string*/
        }
    }
}
```

15.5.2.11 Get the face picture privacy mask configuration capability of a specified channel

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask/capabilities?format=json

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Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "enabled": {
    /*ro, req, object, whether to enable*/
        "@opt": [true, false],
        /*ro, req, array, range, subType:bool*/
        "@def":  false
        /*ro, opt, bool, default value*/
    },
    "privacyMaskLevel": {
    /*ro, opt, object*/
        "@opt": ["high", "mid", "low", "auto"]
        /*ro, opt, array, subType:string*/
    },
    "modeType": {
    /*ro, opt, object*/
        "@opt": [1, 2, 3]
        /*ro, opt, array, subType:int*/
    },
    "nonMosaicFaceProperty": {
    /*ro, opt, object*/
        "@size":  7,
        /*ro, opt, int*/
        "@opt": ["glass", "mask", "hat", "glassAndMask", "glassAndHat", "maskAndHat", "glassAndMaskAdnHat"]
        /*ro, opt, array, subType:string*/
    },
    "nonMosaicFaceLib": {
    /*ro, req, object*/
        "@opt": [true, false],
        /*ro, req, array, subType:bool*/
        "@def":  false
        /*ro, opt, bool*/
    }
}
```

15.5.2.12 Set the privacy mask parameters of a specified channel

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "enabled":  true
    /*req, bool, whether to enable*/
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

15.5.2.13 Restore the face capture algorithm to defaults

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/restoreAlgParam?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

15.5.2.14 Get the capability of configuring shielded area of face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/shieldRegion/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
{
    "ShieldRegionCap": {
    /*ro, opt, object*/
        "@size":  8,
        /*ro, req, int*/
        "enable": {
        /*ro, req, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "PointF": {
        /*ro, opt, object*/
            "@minSize":  0,
            /*ro, req, int*/
            "@maxSize":  4,
            /*ro, req, int*/
            "x": {
            /*ro, req, object*/
                "@min":  0.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@def":  0.001
                /*ro, req, float, range:[0.000,1.000]*/
            },
            "y": {
            /*ro, req, object*/
                "@min":  0.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@def":  0.001
                /*ro, req, float, range:[0.000,1.000]*/
            }
        }
    }
}
```

15.5.2.15 Set the intelligent resource parameters of face capture channel

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/intelliResource

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliResource xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <BehaviorInfo>
    <!--opt, object-->
    <IntelliImage>
      <!--opt, object-->
      <isUpload>
        <!--req, enum, subType:string-->true
      </isUpload>
      <imgQuality>
        <!--opt, enum, subType:string-->best
      </imgQuality>
      <imgResolutionWidth>
        <!--opt, int-->1
      </imgResolutionWidth>
      <imgResolutionHeight>
        <!--opt, int-->1
      </imgResolutionHeight>
      <isImgTargetOverlap>
        <!--req, bool-->true
      </isImgTargetOverlap>
      <isImgRuleOverlap>
        <!--req, bool-->true
      </isImgRuleOverlap>
      <isImgSizeOverlap>
        <!--opt, bool-->bool
      </isImgSizeOverlap>
    </IntelliImage>
    <VideoOverlapInfo>
      <!--opt, object-->
      <isOverlapIntelli>
        <!--req, bool-->true
```

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```
      </isOverlapIntelli>
      <isOverlapTarget>
        <!--req, bool-->true
      </isOverlapTarget>
      <isOverlapRule>
        <!--req, bool-->true
      </isOverlapRule>
    </VideoOverlapInfo>
    <IntelliAnalysisEnable>
      <!--req, bool-->true
    </IntelliAnalysisEnable>
    <DisplayTrajectory>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <durationLimit>
        <!--opt, enum, subType:string-->5s
      </durationLimit>
    </DisplayTrajectory>
    <targetMarkingColorEnabled>
      <!--req, bool-->true
    </targetMarkingColorEnabled>
    <picRecordEnable>
      <!--opt, bool-->true
    </picRecordEnable>
  </BehaviorInfo>
  <FaceCaptureInfo>
    <!--opt, object-->
    <imgQuality>
      <!--opt, enum, subType:string-->best
    </imgQuality>
    <isImgTargetOverlap>
      <!--req, bool-->true
    </isImgTargetOverlap>
    <VideoOverlapInfo>
      <!--opt, object-->
      <isOverlapIntelli>
        <!--req, bool-->true
      </isOverlapIntelli>
      <isOverlapTarget>
        <!--opt, bool-->true
      </isOverlapTarget>
      <isOverlapRule>
        <!--opt, bool-->true
      </isOverlapRule>
    </VideoOverlapInfo>
    <backgroundUpload>
      <!--opt, bool-->true
    </backgroundUpload>
    <imgResolutionWidth>
      <!--req, int-->1
    </imgResolutionWidth>
    <imgResolutionHeight>
      <!--req, int-->1
    </imgResolutionHeight>
    <TargetImgInfo>
      <!--opt, object-->
      <targetImgType>
        <!--opt, enum, subType:string-->custom
      </targetImgType>
      <TargetImgSize>
        <!--opt, object-->
        <widthRatio>
          <!--req, float-->0.000
        </widthRatio>
        <upperPartHeightRatio>
          <!--req, float-->0.000
        </upperPartHeightRatio>
        <underPartHeightRatio>
          <!--req, float-->0.000
        </underPartHeightRatio>
      </TargetImgSize>
      <fixedPixelEnable>
        <!--opt, bool-->true
      </fixedPixelEnable>
      <heightPixel>
        <!--opt, int-->1
      </heightPixel>
      <FaceBeautification>
        <!--opt, object-->
        <enabled>
          <!--opt, bool-->true
        </enabled>
        <level>
          <!--opt, int, range:[0,100]-->1
        </level>
      </FaceBeautification>
      <faceEnhancement>
        <!--opt, bool-->true
      </faceEnhancement>
    </TargetImgInfo>
    <backgroundIntervalUpload>
```

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```
        <!--opt, int, range:[0,65534], unit:s-->1
      </backgroundIntervalUpload>
      <picRecordEnable>
        <!--opt, bool-->true
      </picRecordEnable>
    </FaceCaptureInfo>
    <AlgVersionInfo>
      <!--opt, object-->
      <AlgItem>
        <!--opt, object-->
        <id>
          <!--opt, string, index-->test
        </id>
        <algName>
          <!--opt, string-->test
        </algName>
      </AlgItem>
    </AlgVersionInfo>
    <JudiciaryInfo>
      <!--opt, object-->
      <IntelliImage>
        <!--req, object-->
        <isUpload>
          <!--req, enum, subType:string-->true
        </isUpload>
        <imgQuality>
          <!--opt, enum, subType:string-->best
        </imgQuality>
        <imgResolutionWidth>
          <!--req, int-->1
        </imgResolutionWidth>
        <imgResolutionHeight>
          <!--req, int-->1
        </imgResolutionHeight>
      </IntelliImage>
      <VideoOverlapInfo>
        <!--opt, object-->
        <isOverlapIntelli>
          <!--req, bool-->true
        </isOverlapIntelli>
      </VideoOverlapInfo>
    </JudiciaryInfo>
    <QRCodeUrl>
      <!--opt, object-->
      <vcaVideoQRCodeUrl>
        <!--opt, string, range:[0,256]-->http://127.0.0.1
      </vcaVideoQRCodeUrl>
    </QRCodeUrl>
  </IntelliResource>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

15.5.2.16 Get the configuration capability of intelligent

resources by channel **Request URL**

GET /ISAPI/Intelligent/channels/<channelID>/intelliResource/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

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None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IntelliResourceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
</IntelliResourceCap>
```

15.5.2.17 Get the configuration capability of manual face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/manualFaceSnap/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "ManualFaceSnapCap": {
    /*ro, req, object*/
        "enabled": {
        /*ro, req, object*/
            "@opt":  "true,false",
            /*ro, req, string*/
            "#text":  true
            /*ro, opt, bool*/
        }
    }
}
```

15.5.2.18 Get the parameters of manual face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/manualFaceSnap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "ManualFaceSnap": {
    /*ro, req, object*/
        "enabled":  true
        /*ro, req, bool*/
    }
}
```

15.5.2.19 Get the face modeling configuration capability

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/modeling/capabilities?format=json

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "ModelingCap": {
    /*ro, opt, object*/
        "FaceSnapModeling": {
        /*ro, req, object*/
            "enabled":  "true,false",
            /*ro, req, string*/
            "QuickModeling": {
            /*ro, opt, object*/
                "enabled":  "true,false",
                /*ro, opt, string*/
                "threshold": {
                /*ro, opt, object*/
                    "@min":  20,
                    /*ro, req, int*/
                    "@max":  80,
                    /*ro, req, int*/
                    "@def":  60
                    /*ro, req, int*/
                },
                "snapTime": {
                /*ro, opt, object*/
                    "@min":  0.0,
                    /*ro, req, float*/
                    "@max":  300.0,
                    /*ro, req, float*/
                    "@def":  2.0
                    /*ro, req, float*/
                }
            },
            "LivenessDetection": {
            /*ro, req, object*/
                "enabled":  "true,false"
                /*ro, req, string*/
            },
            "alarmStorageEnable":  "true,false",
            /*ro, opt, string*/
            "mixedTargetDetectionWithFaceSnapModeling":  "true,false"
            /*ro, opt, string*/
        },
        "uploadPictureEnabled":  "true,false"
        /*ro, opt, string*/
    }
}
```

15.5.2.20 Get the face modeling parameters

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/modeling?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
{
    "Modeling": {
    /*ro, opt, object*/
        "FaceSnapModeling": {
        /*ro, req, object*/
            "enabled":  true,
            /*ro, opt, bool*/
            "QuickModeling": {
            /*ro, req, object*/
                "enabled":  true,
                /*ro, opt, bool*/
                "threshold":  60,
                /*ro, opt, int*/
                "snapTime":  2.0
                /*ro, opt, float, unit:s*/
            },
            "LivenessDetection": {
            /*ro, req, object*/
                "enabled":  true
                /*ro, opt, bool*/
            },
            "alarmStorageEnable":  true,
            /*ro, opt, bool*/
            "mixedTargetDetectionWithFaceSnapModeling":  true
            /*ro, opt, bool*/
        },
        "uploadPictureEnabled":  false
        /*ro, opt, bool*/
    }
}
```

15.5.2.21 Set the face modeling parameters

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/modeling?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "Modeling": {
    /*opt, object*/
        "FaceSnapModeling": {
        /*req, object*/
            "enabled":  true,
            /*opt, bool*/
            "QuickModeling": {
            /*req, object*/
                "enabled":  true,
                /*opt, bool*/
                "threshold":  60,
                /*opt, int*/
                "snapTime":  2.0
                /*opt, float, unit:s*/
            },
            "LivenessDetection": {
            /*req, object*/
                "enabled":  true
                /*opt, bool*/
            },
            "alarmStorageEnable":  true,
            /*opt, bool*/
            "mixedTargetDetectionWithFaceSnapModeling":  true
            /*opt, bool*/
        },
        "uploadPictureEnabled":  false
        /*opt, bool*/
    }
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

15.5.2.22 Face capture and recognition event

EventType:faceCapture

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, req, int, times that the same alarm has been uploaded*/
    "eventType":  "faceCapture",
    /*ro, req, string, event type*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)*/
    "eventDescription":  "faceCapture",
    /*ro, req, string, event description, desc:face capture data upload*/
    "channelName":  "ABC",
    /*ro, req, string, channel name (camera name)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID (PUID)*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished*/
    "faceCapture": [
    /*ro, opt, array, face capture and recognition results, one result can contain multiple pictures and one picture can contains multiple faces,
subType:object*/
        {
            "errorCode":  1,
            /*ro, opt, int, error code*/
            "errorMsg":  "ok",
            /*ro, opt, string, error code description*/
            "taskid":  "temp_dfasdffsadfdsadada",
            /*ro, opt, string, task ID*/
            "faceTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, datetime, capture time*/
            "traceUuid":  "sddddddddff",
            /*ro, opt, string, target ID, the same person in multiple capture results corresponds to same target ID*/
            "traceIdx":  1,
            /*ro, opt, int, No. of picture with highest comparison similarity*/
            "targetAttrs": {
            /*ro, opt, object, face attributes*/
                "deviceId":  "ee3c4000-a22d-11b4-8334-5803fb77894c",
                /*ro, opt, string, device UUID*/
                "deviceChannel":  1,
                /*ro, opt, int, device channel No.*/
                "deviceName":  "shebei",
                /*ro, opt, string, device name*/
                "faceTime":  "2004-05-03T17:30:08+08:00",
                /*ro, opt, datetime, capture time*/
                "bkgUrl":  "test",
                /*ro, opt, string, background picture URL*/
                "thermalURL":  "test",
                /*ro, opt, string, thermal picture URL*/
                "contentID":  "backgroundImage",
                /*ro, opt, string, background picture (binary)*/
                "pId":  "F4F665D6A18E41308CE9934DCDDD1111"
                /*ro, opt, string*/
            },
            "faces": [
            /*ro, opt, array, face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on,
subType:object*/
                {
```

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```
                    "faceId":  1,
                    /*ro, opt, int, face ID*/
                    "faceRect": {
                    /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "age": {
                    /*ro, opt, object, age*/
                        "range":  5,
                        /*ro, opt, int, age deviation*/
                        "value":  23,
                        /*ro, opt, int, age*/
                        "ageGroup":  "old"
                        /*ro, opt, string, age group*/
                    },
                    "gender": {
                    /*ro, opt, object, gender*/
                        "confidence":  99.9999,
                        /*ro, opt, float, confidence*/
                        "value":  "male"
                        /*ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
                    },
                    "glass": {
                    /*ro, opt, object, whether the person wears glasses*/
                        "confidence":  99.945,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no", "yes"*/
                    },
                    "mask": {
                    /*ro, opt, object, whether the person wears a mask*/
                        "confidence":  99.945,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"no", "yes"*/
                    },
                    "smile": {
                    /*ro, opt, object, whether the person is smiling*/
                        "confidence":  99.945,
                        /*ro, opt, float, confidence*/
                        "value":  "no"
                        /*ro, opt, enum, whether the person is smiling, subType:string, desc:"no", "yes"*/
                    },
                    "faceExpression": {
                    /*ro, opt, object, facial expression*/
                        "value":  "unknown"
                        /*ro, opt, enum, facial expression, subType:string, desc:"unknown", "surprised", "panic", "disgusted", "happy", "sad", "angry",
"poker-faced"*/
                    },
                    "hat": {
                    /*ro, opt, object, whether the person wears a hat*/
                        "value":  "no"
                        /*ro, opt, enum, whether the person wears a hat, subType:string, desc:"no", "yes", "unknown"*/
                    },
                    "URL":  "xxxxxxx",
                    /*ro, opt, string, face picture thumbnail URL*/
                    "contentID":  "faceImage",
                    /*ro, opt, string, face picture thumbnail (binary)*/
                    "stayDuration":  20,
                    /*ro, opt, int, dwell time duration in the image, unit:ms*/
                    "faceScore":  30,
                    /*ro, req, int, face score (face picture quality), range:[0,100]*/
                    "captureEndMark":  true,
                    /*ro, req, bool, end flag of capture, desc:true (ended), false (not ended)*/
                    "targetID":  "xxxxxxx",
                    /*ro, opt, string, face and human body linkage ID*/
                    "FacePictureRect": {
                    /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
                        "height":  1.000,
                        /*ro, req, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, req, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "blockingState":  "noBlocking",
                    /*ro, opt, enum, target face shielding status, subType:string, desc:"noBlocking" (unshielded), "instantaneousInsignificantBlocking"
(lightly shielded), "continuousInsignificantBlocking" (keep shielding), "significantBlocking" (seriously shielded), "OSD", "unknown"*/
                    "swingAngle":  -90,
                    /*ro, opt, int, panning angle, range:[-90,90]*/
                    "tiltAngle":  -90,
                    /*ro, opt, int, tilting angle, range:[-90,90]*/
                    "pupilDistance":  10,
                    /*ro, opt, int, pupil distance of target face, desc:the maximum value is 10 bytes, and the maximum value is (resolution width/1.6),
unit: pixel*/
```


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```
unit: pixel*/
                    "livenessDetectionStatus":  "liveFace",
                    /*ro, opt, enum, face anti-spoofing detection status, subType:string, desc:"liveFace" (live face), "notLiveFace" (the face is not real),
"detectionFailed" (detection failed), "notEnabled" (disabled).*/
                    "pId":  "F4F665D6A18E41308CE9934DCDDD2222",
                    /*ro, opt, string*/
                    "enterTime":  "2004-05-03T17:30:08+08:00",
                    /*ro, opt, datetime*/
                    "exitTime":  "2004-05-03T17:30:08+08:00",
                    /*ro, opt, datetime*/
                    "perClassify":  0,
                    /*ro, opt, int, face data presorting*/
                    "faceSnapThermometryEnabled":  true,
                    /*ro, opt, bool, whether to enable uploading captured face picture with temperature information*/
                    "currTemperature":  36.8,
                    /*ro, opt, float, face temperature*/
                    "isAbnomalTemperature":  true,
                    /*ro, opt, bool, whether the temperature is abnormal or not: true, false*/
                    "thermometryUnit":  "celsius",
                    /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
                    "alarmTemperature":  36.7,
                    /*ro, opt, float, temperature threshold to trigger alarm*/
                    "interiorPosition":  "driver"
                    /*ro, opt, enum, subType:string*/
                }
            ],
            "uid":  "20190520103922026O0Kb4s3iQPzLignc1sRUbFT0hR039Y0YzuLt0VLfl99yZi",
            /*ro, opt, string, the unique ID of event uploading*/
            "OperatorInfo": {
            /*ro, opt, object, operator information, desc:operator information*/
                "personUUID":  "test",
                /*ro, opt, string, range:[1,32]*/
                "personName":  "test"
                /*ro, opt, string, range:[1,64]*/
            },
            "MEFData": [
            /*ro, opt, array, subType:object*/
                {
                    "time":  "2004-05-03T17:30:08.000+08:00",
                    /*ro, req, datetime, it should be accurate to the millisecond, desc:it should be accurate to the millisecond*/
                    "IMSI":  "460001357924680",
                    /*ro, req, string, range:[1,15]*/
                    "bandIdx":  11
                    /*ro, opt, enum, subType:int*/
                }
            ],
            "isNeedVerification":  true
            /*ro, opt, bool*/
        }
    ],
    "GPS": {
    /*ro, opt, object*/
        "divisionEW":  "E",
        /*ro, req, enum, "E" (Eastern Hemisphere), W" (West Hemisphere), subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)*/
        "longitude":  100,
        /*ro, req, int, longitude, desc:longitude*/
        "divisionNS":  "N",
        /*ro, req, enum, "S" (Southern Hemisphere), "N" (North Hemisphere), subType:string, desc:"S" (Southern Hemisphere), "N" (North Hemisphere)*/
        "latitude":  100
        /*ro, req, int, latitude, desc:latitude*/
    },
    "URLCertificationType":  "digest"
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (no authentication, it is used for the cloud protocol), "digest" (digest
authentication, it is used for local picture URL returned by NVR or DVR)*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceCapture	[Message content]	application/json	--	--	--
..	[Binary picture data]	image/jpeg	backgroundImage	backgroundImage.jpeg	--
..	[Binary picture data]	image/jpeg	faceImage	faceImage.jpeg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.5.3 人脸侦测

15.5.3.1 Face detection

EventType:facedetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"facedetection" (face detection)-->facedetection
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:face detection-->facedetection
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, additional information-->
    <serialNumber>
      <!--ro, req, string, device serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, req, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <facePictureRectList>
```

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```
    <!--ro, opt, array, subType:object, range:[1,30]-->
    <facePictureRect>
      <!--ro, opt, object, the origin is the upper-left corner of the screen, desc:the origin is the upper-left corner of the screen-->
      <height>
        <!--ro, req, float, width, range:[0.000,1.000]-->0.000
      </height>
      <width>
        <!--ro, req, float, range:[0.000,1.000]-->0.000
      </width>
      <positionX>
        <!--ro, req, float, X-coordinate of the upper-left corner of the frame, range:[0.000,1.000]-->0.000
      </positionX>
      <positionY>
        <!--ro, req, float, Y-coordinate of the upper-left corner of the frame, range:[0.000,1.000]-->0.000
      </positionY>
    </facePictureRect>
  </facePictureRectList>
  <backgroundImage>
    <!--ro, opt, object, background picture-->
    <resourcesContentType>
      <!--ro, opt, string-->binary
    </resourcesContentType>
    <resourcesContent>
      <!--ro, opt, string, when the returned alarm message contains additional resources (e.g., pictures), this field should be returned. When the value of resourcesContentType is "binary", it must be the same as Content-ID of the resource. When the value of resourcesContentType is "url", its value is the resource URL, range:[1,128], desc:when the returned alarm message contains additional resources (e.g., pictures), this field should be returned. When the value of resourcesContentType is "binary", it must be the same as Content-ID of the resource. When the value of resourcesContentType is "url", its value is the resource URL-->backgroundImage
    </resourcesContent>
  </backgroundImage>
</EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
facedetection	[Message content]	application/xml	--	--	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.5.4 人脸建模比对

15.5.4.1 Face picture analysis (modeling)

Request URL

POST /ISAPI/SDT/Face/pictureAnalysis

Query Parameter

None

Request Message

```
{
    "imagesType":  "URL",
    /*req, enum, picture type, subType:string, desc:picture type: "URL"-picture URL*/
    "imagesData":  "test",
    /*opt, string, data type, desc:picture URL,this node is valid only when imagesType is "URL"*/
    "httpAuthenticationMethod":  "MD5digest",
    /*opt, enum, authentication method, subType:string, desc:"MD5digest" (MD5 digest authentication)*/
    "userName":  "admin",
    /*opt, string, digest authentication user name, range:[0,32]*/
    "password":  "12345",
    /*opt, string, digest authentication password, range:[0,16]*/
```

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```
    "algorithmType":  "faceModel",
    /*req, enum, algorithm type, subType:string, desc:"faceModel" (face modeling), "faceStruct" (face structurization), "faceDetect" (ace detection), if there are multiple types, you can separate each one by comma*/
    "algorithm": [
    /*opt, array, algorithm type, subType:object, desc:currently only one algorithm type is allowed for one task. Either algorithm or algorithmType is required. algorithmID is used to specify a unique algorithm while classificationID is used for fuzzy search*/
        {
            "targetType":  "1",
            /*req, enum, algorithm target type, subType:string, desc:1 (face), 2 (vehicle), 3 (human body and vehicle), 4 (human body)*/
            "analysisType":  "1",
            /*opt, enum, algorithm analysis type, subType:string, desc:1 (detection), 2 (attribute), 3 (modeling), 4 (comparison), 5 (behavior), see details in Data Dictionary, separate multiple items with commas*/
            "analysisSourceType":  "video,picture",
            /*req, string, VCA source type*/
            "algorithmID":  "test",
            /*opt, string, algorithm package ID, desc:either algorithmID or classificationID is required*/
            "classificationID":  "test",
            /*opt, string, algorithm category ID, desc:either algorithmID or classificationID is required*/
            "version":  "test",
            /*opt, string, algorithm version, desc:this node is valid when classificationID exists*/
            "vendor":  "test"
            /*opt, string, algorithm manufacturer, desc:this node is valid when classificationID exists*/
        }
    ],
    "mode":  "mutiface",
    /*opt, enum, face detection mode, subType:string, desc:face detection mode: "singleface"-detect one face(default),"mutiface"-detect multiple faces. By default,if no value is assigned to this node,the face detection mode is "singleface".*/
    "roi": {
    /*opt, object, region of interest (ROI), desc:if roi,faceRect,and faceMark are all configured,faceRect and faceMark are preferred*/
        "height":  0.850,
        /*opt, float, height, range:[0.000,1.000], desc:it is accurate to three decimal places*/
        "width":  0.850,
        /*opt, float, width, range:[0.000,1.000], desc:it is accurate to three decimal places*/
        "x":  0.120,
        /*opt, float, X-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
        "y":  0.120
        /*opt, float, Y-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
    },
    "faceRect": {
    /*opt, object, face frame in picture, desc:the origin is the upper-left corner of the screen*/
        "height":  0.81,
        /*opt, float, height, range:[0.000,1.000], desc:it is accurate to three decimal places*/
        "width":  0.43,
        /*opt, float, width, range:[0.000,1.000], desc:it is accurate to three decimal places*/
        "x":  0.13,
        /*opt, float, X-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
        "y":  0.87
        /*opt, float, Y-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
    },
    "faceMark": {
    /*opt, object, face features in the picture*/
        "leftEye": {
        /*opt, object, left eye's coordinates, desc:the origin is the upper-left corner of the screen*/
            "x":  0.13,
            /*opt, float, X-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
            "y":  0.87
            /*opt, float, Y-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
        },
        "rightEye": {
        /*opt, object, The location of right eye., desc:the origin is the upper-left corner of the screen*/
            "x":  0.78,
            /*opt, float, X-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
            "y":  0.50
            /*opt, float, Y-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
        },
        "noseTip": {
        /*opt, object, nose tip's coordinates, desc:the origin is the upper-left corner of the screen*/
            "x":  0.78,
            /*opt, float, X-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
            "y":  0.50
            /*opt, float, Y-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
        },
        "leftMouth": {
        /*opt, object, left mouth corner's coordinates, desc:the origin is the upper-left corner of the screen*/
            "x":  0.78,
            /*opt, float, X-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
            "y":  0.50
            /*opt, float, Y-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
        },
        "rightMouth": {
        /*opt, object, right mouth corner's coordinates, desc:the origin is the upper-left corner of the screen*/
            "x":  0.78,
            /*opt, float, X-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
            "y":  0.50
            /*opt, float, Y-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
        }
    }
}
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
pictureAnalysis	[Message content]	application/json	--	--	--
facePicture	[Binary picture data]	image/jpeg		facePicture.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "requestURL": "test",
    /*ro, opt, string, request URL*/
    "statusCode": 1,
    /*ro, req, int, status code*/
    "statusString": "test",
    /*ro, req, string, status description*/
    "subStatusCode": "test",
    /*ro, req, string, sub status code*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:this node is required when statusCode is not 1 it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error description, desc:this field is required when statusCode is not 1, and can be the error information of a specific parameter*/
    "targets": [
    /*ro, opt, array, picture analysis results, subType:object, desc:face picture information of the target*/
        {
            "id": 1,
            /*ro, req, int, No.*/
            "age": 23,
            /*ro, opt, int, age*/
            "ageRange": 5,
            /*ro, opt, int, age range, desc:it is valid only when the age node exists*/
            "ageGroup": "child",
            /*ro, opt, string, age group, desc:for details, refer to the Dictionaries*/
            "gender": "male",
            /*ro, opt, string, gender, desc:for details, refer to the Dictionaries*/
            "glasses": "yes",
            /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no", "yes", "sunglasses"*/
            "smile": "yes",
            /*ro, opt, string, whether the person wears a smile*/
            "facePose": {
            /*ro, opt, object, face angle*/
                "pitch": 0.998,
                /*ro, opt, float, face pitch angle, range:[-90,90], desc:face pitching up is regarded as positive, range: [-90, 90], it is accurate to three decimal places. This node does not exist if not supported*/
                "yaw": 0.876,
                /*ro, opt, float, face yaw angle, face yawing left is regarded as positive, range:[-90,90], desc:face pitching up is regarded as positive, range: [-90, 90], it is accurate to three decimal places. This node does not exist if not supported*/
                "roll": 0.623
                /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive, range:[-90,90], desc:face pitching up is regarded as positive, range: [-90, 90], it is accurate to three decimal places. This node does not exist if not supported*/
            },
            "targetModelData": "test",
            /*ro, opt, string, target modeling data, desc:target modeled data,it is valid when the algorithmType in the JSON_FacePicAnalysisInput message is "faceModel"*/
            "faceRect": {
            /*ro, opt, object, face frame in picture*/
                "height": 0.81,
                /*ro, opt, float, height, desc:it is accurate to three decimal places*/
                "width": 0.43,
                /*ro, opt, float, width, desc:it is accurate to three decimal places*/
                "x": 0.13,
```

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```
                /*ro, opt, float, X-coordinate, desc:it is accurate to three decimal places*/
                "y":  0.87
                /*ro, opt, float, Y-coordinate, desc:it is accurate to three decimal places*/
            },
            "recommendFaceRect": {
            /*ro, opt, object, recommended face frame coordinates, desc:it is used for image matting by expanding the face thumbnail frame*/
                "height":  0.81,
                /*ro, opt, float, height, desc:it is accurate to three decimal places*/
                "width":  0.43,
                /*ro, opt, float, width, desc:it is accurate to three decimal places*/
                "x":  0.13,
                /*ro, opt, float, X-coordinate, desc:it is accurate to three decimal places*/
                "y":  0.87
                /*ro, opt, float, Y-coordinate, desc:it is accurate to three decimal places*/
            },
            "faceMark": {
            /*ro, opt, object, face features in the picture, desc:face features in the picture*/
                "leftEye": {
                /*ro, opt, object, left eye's coordinates*/
                    "x":  0.13,
                    /*ro, opt, float, X-coordinate, desc:it is accurate to three decimal places*/
                    "y":  0.87
                    /*ro, opt, float, Y-coordinate, desc:it is accurate to three decimal places*/
                },
                "rightEye": {
                /*ro, opt, object, The location of right eye.*/
                    "x":  0.78,
                    /*ro, opt, float, X-coordinate, desc:it is accurate to three decimal places*/
                    "y":  0.50
                    /*ro, opt, float, Y-coordinate, desc:it is accurate to three decimal places*/
                },
                "noseTip": {
                /*ro, opt, object, nose tip's coordinates*/
                    "x":  0.78,
                    /*ro, opt, float, X-coordinate, desc:it is accurate to three decimal places*/
                    "y":  0.50
                    /*ro, opt, float, Y-coordinate, desc:it is accurate to three decimal places*/
                },
                "leftMouth": {
                /*ro, opt, object, float,left mouth corner's position*/
                    "x":  0.78,
                    /*ro, opt, float, X-coordinate, desc:it is accurate to three decimal places*/
                    "y":  0.50
                    /*ro, opt, float, Y-coordinate, desc:it is accurate to three decimal places*/
                },
                "rightMouth": {
                /*ro, opt, object, float,left mouth corner's position*/
                    "x":  0.78,
                    /*ro, opt, float, X-coordinate, desc:it is accurate to three decimal places*/
                    "y":  0.50
                    /*ro, opt, float, Y-coordinate, desc:it is accurate to three decimal places*/
                }
            },
            "mask":  "yes",
            /*ro, opt, string, whether the person wears a mask, desc:see details in the Data Dictionary*/
            "faceScore":  0.000,
            /*ro, opt, float, face score (face picture quality), range:[0.000,1.000], desc:float,face score,value range: [0.000,1.000]*/
            "faceExpression":  "test"
            /*ro, opt, string, facial expression, desc:see details in the Data Dictionary*/
        }
    ]
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
targets	[Message content]	application/json	--	--	--
facePic	[Binary picture data]	image/jpeg	facePic	facePic.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.

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- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.5.4.2 Get the capability of detecting and modeling face pictures in a batch

Request URL

GET /ISAPI/SDT/Face/pictureAnalysis/batch/capabilities

Query Parameter

None

Request Message

None

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:This field is required when the value of statusCode is not 1, and it corresponds to subStatusCode.*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:This field is required when the value of statusCode is not 1, and it can be the error of a certain parameter.*/
    "taskInfo": {
    /*ro, req, object, task information*/
        "algorithmType": [
        /*ro, opt, array, algorithm type, subType:object*/
            {
                "value":  "faceDetect"
                /*ro, req, enum, algorithm type value, subType:string, desc:"faceModel" (face modeling), "faceStruct" (face structuralization), "faceDetect"
(face detection)*/
            }
        ],
        "model": [
        /*ro, opt, array, face detection mode, subType:object*/
            {
                "value":  "singleface"
                /*ro, opt, enum, value of the face detection mode, subType:string, desc:"singleface" (single-face detection), "mutiface" (multi-face
detection). If this field is not configured, the default value is "singleface".*/
            }
        ],
        "destination": [
        /*ro, req, array, result recipient information, subType:object*/
            {
                "value":  "faceListLib"
                /*ro, opt, string, list library of face picture big data*/
            }
        ],
        "maxPictureNum":  0,
        /*ro, opt, int, the maximum number of pictures*/
        "image": {
        /*ro, opt, object, picture streaming information*/
            "imageInfos": [
            /*ro, req, array, picture stream information, subType:object*/
                {
                    "dataType": [
                    /*ro, req, array, picture data type, subType:object*/
                        {
                            "value":  "URL"
                            /*ro, opt, enum, picture URL, subType:string, desc:"URL" (picture URL), "binaryData" (binary picture data)*/
                        }
                    ],
                    "httpAuthenticationMethod": {
                    /*ro, opt, object, authentication method*/
                        "@opt": ["MD5digest", "none"]
                        /*ro, opt, array, options, subType:string*/
                    },
                    "userName": {
                    /*ro, opt, object, user name for digest authentication*/
                        "@min":  0,
                        /*ro, opt, int, the minimum value*/
                        "@max":  0
                        /*ro, opt, int, the maximum value*/
                    },
                    "password": {
                    /*ro, opt, object, password for digest authentication*/
                        "@min":  0,
                        /*ro, opt, int, the minimum value*/
                        "@max":  0
                        /*ro, opt, int, the maximum value*/
                    },
                    "total":  100,
                    /*ro, req, int, the maximum number of pictures of the same person*/
                    "targetAttrs":  2048
                    /*ro, opt, int, the maximum size of the string that can be transmitted*/
                }
            ]
        }
    }
}
```

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15.5.4.3 Batch search for the inbound status

Request URL

POST /ISAPI/SDT/Face/pictureAnalysis/batch/status

Query Parameter

None

Request Message

```
{
    "BatchCond": {
    /*opt, object, batch importing into library*/
        "searchResultPosition": 0,
        /*req, int, the start position of the search result in the result list, desc:when there are multiple records and you cannot get all search results at a time, you can search for the rest records after the specified position next time*/
        "maxResults": 100,
        /*req, int, the number of results returned this time*/
        "FDID": "-1",
        /*opt, string, list library ID, desc:list library ID, separated by the port No.*/
        "lastRowkey": "test",
        /*opt, string, the last record returned in last search, desc:the last record returned in last search*/
        "status": "success"
        /*opt, enum, inbound status, subType:string, desc:inbound status, string "all" (searching for all); "success" (putting status code into the library succeeded); "failure" (putting status code into the library failed);*/
    }
}
```

Response Message

```
{
    "BatchStatus": {
    /*ro, opt, object, batch importing into library*/
        "responseStatusStrg": "OK",
        /*ro, opt, enum, search status, subType:string, desc:search status "OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data), the maximum length is 32*/
        "numOfMatches": 100,
        /*ro, opt, int, number of records returned this time*/
        "totalMatches": 200,
        /*ro, opt, int, total number of matched records*/
        "lastRowkey": "test",
        /*ro, opt, string, the last returned record of this search, desc:the last returned record of this search*/
        "MatchList": [
        /*ro, opt, array, list of matched results, subType:object*/
            {
                "FPID": "test",
                /*ro, req, string, face record ID*/
                "status": "success",
                /*ro, req, enum, inbound status, subType:string, desc:"success" (importing status code into the library succeeded); "failure" (importing status code into the library failed);*/
                "errorCode": 1,
                /*ro, opt, int, error code*/
                "errorMsg": "ok"
                /*ro, opt, string, error description, desc:it is required when the status is "failure"*/
            }
        ]
    }
}
```

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15.5.4.4 Submit face detection and modeling tasks in a batch

Request URL

POST /ISAPI/SDT/Face/pictureAnalysis/batch

Query Parameter

None

Request Message

```
{
    "taskInfo": {
    /*req, object, task information*/
        "algorithmType": "faceModel",
        /*req, enum, algorithm type, subType:string, desc:"faceModel" (face modeling), "faceStruct" (face structurization), "faceDetect" (face detection), separate multiple algorithm types with commas*/
        "algorithm": [
        /*opt, array, algorithm type information, subType:object, desc:currently only one algorithm type is allowed for one task. Either algorithm or algorithmType is required. algorithmID is used to specify a unique algorithm while classificationID is used for fuzzy search*/
            {
                "targetType": "1",
                /*req, enum, algorithm target type, subType:string, desc:"1" (face), "2" (vehicle), "3" (active target (human body and vehicle)), "4" (human body)*/
                "analysisType": "1",
                /*opt, enum, algorithm analysis type, subType:string, desc:"1" (detection), "2" (attribute), "3" (modeling), "4" (comparison), "5" (behavior), separate multiple analysis types with commas*/
                "analysisSourceType": "picture",
                /*req, string, VCA source type*/
                "algorithmID": "test",
```

```
                /*opt, string, algorithm package ID*/
                "classificationID":  "test",
                /*opt, string, algorithm category ID, desc:either algorithmID or classificationID is required*/
                "version":  "test",
                /*opt, string, algorithm version, desc:this node is valid only when classificationID exists*/
                "vendor":  "test"
                /*opt, string, algorithm manufacturer, desc:this node is valid only when classificationID exists*/
            }
        ],
        "imageInfo": {
        /*opt, object*/
            "imageSize":  200,
            /*opt, int, range:[10,512], unit:KB*/
            "imageRes":  3
            /*opt, enum, subType:int*/
        },
        "mode":  "singleface",
        /*opt, enum, face detection mode, subType:string, desc:"singleface" (detect one face), "mutiface" (detect multiple faces), if no value is assigned
to this node, the face detection mode is "singleface" by default*/
        "destination": [
        /*req, array, result recipient information, subType:object*/
            {
                "destinationType":  "faceListLib",
                /*opt, enum, result recipient type, subType:string, desc:"faceListLib" (face big data list library), "KafkaFaceAnalysis" (face data imported
to library and merged with big data)*/
                "destinationUrl":  "test",
                /*opt, string, recipient URL description, desc:recipient description,which is in URL format: "http://IP:PORT/face-
web/ws/rest/BlackListInfoApply/batchUpladBlacklistRecords"-face data list library,"{"topic":"HUMAN_INFO_TOPIC","url":"172.10.201.156:2181"}"-face data
static library. The maximum string length is 1023 bytes.*/
                "userName":  "test",
                /*opt, string, user name of the recipient*/
                "password":  "test"
                /*opt, string, password of the recipient, desc:password of the recipient*/
            }
        ],
        "image": {
        /*req, object, picture information*/
            "imageInfos": [
            /*req, array, picture streaming information, subType:object*/
                {
                    "dataType":  "URL",
                    /*req, string, picture data type*/
                    "URL":  "test",
                    /*req, string, picture URL, desc:picture URL*/
                    "picID":  "test",
                    /*opt, string*/
                    "httpAuthenticationMethod":  "MD5digest",
                    /*opt, enum, authentication method, subType:string, desc:"MD5digest" (Digest-MD5 authentication)*/
                    "userName":  "admin",
                    /*opt, string, user name of the recipient, range:[0,32]*/
                    "password":  "xxx12345",
                    /*opt, string, password of the recipient, range:[0,16], desc:password of the recipient*/
                    "roi": {
                    /*opt, object, region of interest (ROI), desc:if roi, faceRect, and faceMark are all configured, faceRect and faceMark are preferred*/
                        "height":  1.0,
                        /*opt, float, height*/
                        "width":  1.0,
                        /*opt, float, width*/
                        "x":  0.0,
                        /*opt, float, X-coordinate*/
                        "y":  0.0
                        /*opt, float, Y-coordinate*/
                    },
                    "faceRect": {
                    /*opt, object, face thumbnail frame*/
                        "height":  1.0,
                        /*opt, float, height*/
                        "width":  1.0,
                        /*opt, float, width*/
                        "x":  0.0,
                        /*opt, float, X-coordinate*/
                        "y":  0.0
                        /*opt, float, Y-coordinate*/
                    },
                    "faceMark": {
                    /*opt, object, face features*/
                        "leftEye": {
                        /*opt, object, position of the left eye*/
                            "x":  0.13,
                            /*opt, float, X-coordinate, desc:normalized coordinate, which is accurate to three decimal places*/
                            "y":  0.87
                            /*opt, float, Y-coordinate, desc:normalized coordinate, which is accurate to three decimal places*/
                        },
                        "rightEye": {
                        /*opt, object, position of the right eye*/
                            "x":  0.78,
                            /*opt, float, X-coordinate, desc:normalized coordinate, which is accurate to three decimal places*/
                            "y":  0.50
                            /*opt, float, Y-coordinate, desc:normalized coordinate, which is accurate to three decimal places*/
                        },
                        "noseTip": {
                        /*opt, object, position of the nose tip*/
                            "x":  0.78,
                            /*opt, float, X-coordinate, desc:normalized coordinate, which is accurate to three decimal places*/
```

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```
                        "y":  0.50
                        /*opt, float, Y-coordinate, desc:normalized coordinate, which is accurate to three decimal places*/
                    },
                    "leftMouth": {
                    /*opt, object, position of the left mouth corner*/
                        "x":  0.78,
                        /*opt, float, X-coordinate, desc:normalized coordinate, which is accurate to three decimal places*/
                        "y":  0.50
                        /*opt, float, Y-coordinate, desc:normalized coordinate, which is accurate to three decimal places*/
                    },
                    "rightMouth": {
                    /*opt, object, position of the right mouth corner*/
                        "x":  0.78,
                        /*opt, float, X-coordinate, desc:normalized coordinate, which is accurate to three decimal places*/
                        "y":  0.50
                        /*opt, float, Y-coordinate, desc:normalized coordinate, which is accurate to three decimal places*/
                    }
                },
                "targetAttrs": {
                /*opt, object, transmitted parameters, desc:the transmitted parameters are different for different values of "destination"*/
                    "blackListId":  "test",
                    /*req, string, face picture library ID, desc:separate multiple face picture library IDs with commas*/
                    "humanId":  "test",
                    /*req, string, face record ID in the list library*/
                    "reserveField": {
                    /*opt, object, reserved area*/
                        "name":  "张三",
                        /*opt, string, person name*/
                        "gender":  "male",
                        /*opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
                        "bornTime":  "2004-05-03",
                        /*opt, string, birth date of the person in the face picture*/
                        "createDate":  "2004-05-03",
                        /*opt, string, record added date*/
                        "city":  "130100",
                        /*opt, string, city code of birth, desc:city code of birth,the maximum string length is 32 bytes*/
                        "certificateType":  "officerID",
                        /*opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "other" (other certificate);
the maximum length is 10*/
                        "certificateNumber":  "test",
                        /*opt, string, certificate number*/
                        "address":  "test",
                        /*opt, string, person address*/
                        "customInfo":  "test"
                        /*opt, string, custom information*/
                    }
                }
            }
        ]
    }
}
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "taskUID":  "test"
    /*ro, opt, string, task ID*/
}
```

15.5.4.5 Get capability of analyzing and modeling the face picture

Request URL

GET /ISAPI/SDT/Face/pictureAnalysis/capabilities

Query Parameter

None

Request Message

None

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:error code*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this field is required when statusCode is not 1, and can be the error information of a specific parameter*/
    "imagesType": [
    /*ro, req, array, picture type, subType:object*/
        {
            "value":  "URL"
            /*ro, opt, enum, URL: picture URL; binaryData: picture binary data;, subType:string, desc:URL: picture URL; binaryData: picture binary data;*/
        }
    ],
    "httpAuthenticationMethod": {
    /*ro, opt, object, authentication method*/
        "@opt": ["MD5digest", "none"]
        /*ro, opt, array, options, subType:string*/
    },
    "userName": {
    /*ro, opt, object, digest authentication user name*/
        "@min":  0,
        /*ro, opt, int, the minimum value*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "password": {
    /*ro, opt, object, digest authentication password*/
        "@min":  0,
        /*ro, opt, int, the minimum value*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "modelRectSupport":  true,
    /*ro, req, bool, whether supports configuring frame for analyzing and modeling face target, desc:true (support), false (not support)*/
    "algorithmType": [
    /*ro, req, array, algorithm type, subType:object*/
        {
            "value":  "faceModel"
            /*ro, opt, enum, faceModel: face modeling; faceStruct: face structurization; faceDetect: face detection;, subType:string, desc:faceModel: face
modeling; faceStruct: face structurization; faceDetect: face detection;*/
        }
    ],
    "mode": [
    /*ro, opt, array, face detection mode, subType:object*/
        {
            "value":  "singleface"
            /*ro, opt, enum, singleface: detect one face (default); mutiface: detect multiple faces;, subType:string, desc:singleface: detect one face
(default); mutiface: detect multiple faces;*/
        }
    ]
}
```

15.5.4.6 Get the capability of configuring the modeling mode

Request URL

GET /ISAPI/SDT/Management/modellingMode/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "ModellingMode": {
    /*ro, opt, object, modeling mode*/
        "syncFaceAutoChip": {
        /*ro, opt, object, whether to automatically specify a chip for synchronous face picture analysis*/
            "@opt": [true, false]
            /*ro, opt, array, whether to automatically specify a chip for synchronous face picture analysis, subType:bool*/
        },
        "chip": {
        /*ro, opt, object, chip type*/
            "@min":  0,
            /*ro, opt, int, the minimum length*/
            "@max":  64
            /*ro, opt, int, the maximum length*/
        },
        "algorithmID": {
        /*ro, opt, object, algorithm package ID*/
            "@min":  0,
            /*ro, opt, int, the minimum length*/
            "@max":  64
            /*ro, opt, int, the maximum length*/
        }
    }
}
```

15.5.4.7 Edit the parameters of modeling mode

Request URL

PUT /ISAPI/SDT/Management/modellingMode?format=json

Query Parameter

None

Request Message

```
{
    "ModellingMode": {
    /*opt, object, modeling mode configuration*/
        "syncFaceAutoChip":  true
        /*opt, bool, whether to automatically specify a chip for synchronous face picture analysis*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

15.5.4.8 Get modeling mode

Request URL

GET /ISAPI/SDT/Management/modellingMode?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "ModellingMode": {
    /*ro, opt, object, configuration of modeling mode*/
        "syncFaceAutoChip":  true,
        /*ro, opt, bool, whether to automatically specify a chip for synchronous face picture analysis*/
        "chip":  "test",
        /*ro, opt, string, chip type*/
        "algorithmID":  "test"
        /*ro, opt, string, algorithm package ID*/
    }
}
```

15.5.4.9 Get the capability for the task of face picture comparison in a batch

Request URL

GET /ISAPI/SDT/Management/Task/facePicture/batch/capabilities

Query Parameter

None

Request Message

None

Response Message

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```
{
    "taskInfoCap": {
    /*ro, req, object, task information*/
        "algorithmType":  "faceModel",
        /*ro, req, enum, algorithm type, subType:string, desc:algorithm type: "faceModel"-face modeling,"faceStruct"-face structurization,"faceDetect"-face
detection,"faceModelContrast"-face model comparison,multiple algorithm types should be separated by commas*/
        "mode":  "singleface",
        /*ro, opt, enum, face detection mode, subType:string, desc:"singleface"-detect one face,"mutiface"-detect multiple faces; by default,if no value is
assigned to this node,the face detection mode is "singleface"”*/
        "destination": {
        /*ro, req, object, result recipient list*/
            "destinationType":  "faceCaptrueLib",
            /*ro, req, enum, result recipient type, subType:string, desc:"faceCaptureLib"-captured face picture library,"faceAlarmLib"-list alarm library
for face picture big data,"kafkaWhiteAlarmLib"-Kafka allowlist alarm library,"faceCaptureUpload"-captured picture uploaded by ISAPI,"faceAlarmUpload"-list
alarm uploaded by ISAPI,"faceAlarmSdkUpload"-list alarm uploaded by SDK*/
            "destinationUrl": {
            /*ro, opt, object, recipient URL description*/
                "@min":  1,
                /*ro, opt, int, the minimum length*/
                "@max":  60
                /*ro, req, int, the maximum length*/
            },
            "userName": {
            /*ro, opt, object, user name of the recipient*/
                "@min":  1,
                /*ro, opt, int, the minimum length*/
                "@max":  60
                /*ro, opt, int, the maximum length*/
            },
            "password": {
            /*ro, opt, object, recipient password*/
                "@min":  1,
                /*ro, opt, int, the minimum length*/
                "@max":  60
                /*ro, opt, int, the maximum length*/
            }
        },
        "relateGroup": {
        /*ro, opt, object, specified face picture list relationship*/
            "ctrlID": {
            /*ro, opt, object, arming ID*/
                "@min":  0,
                /*ro, opt, int, the minimum length*/
                "@max":  32
                /*ro, opt, int, the maximum length*/
            },
            "min":  0,
            /*ro, req, int, the minimum threshold*/
            "max":  1000,
            /*ro, req, int, the maximum threshold*/
            "relate": {
            /*ro, req, object, face picture library ID*/
                "@min":  63,
                /*ro, opt, int, the minimum length*/
                "@max":  63
                /*ro, opt, int, the maximum length*/
            },
            "top":  5,
            /*ro, req, int, number of alarm pictures*/
            "type":  [1, 2],
            /*ro, opt, enumarray, alarm category, subType:int, desc:1-blocklist (default), 2-allowlist*/
        }
    }
}
```

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15.5.4.10 Submit the task of comparing and recognizing face pictures in a batch

Request URL

POST /ISAPI/SDT/Management/Task/facePicture/batch

Query Parameter

None

Request Message

```
{
    "taskIDList": [
    /*req, array, list of batch face picture comparison task, subType:object*/
        {
            "taskInfo": {
            /*req, object, task information*/
                "algorithm": [
                /*opt, array, algorithm type, subType:object, desc:currently only one algorithm type is allowed for one task. Either algorithm or
algorithmType is required. algorithmID is used to specify a unique algorithm while classificationID is used for fuzzy search*/
                    {
                        "targetType":  "1"
```

```
                        "targetType":  1,
                        /*req, string, algorithm target type*/
                        "analysisType":  1,
                        /*opt, enum, algorithm analysis type, subType:int, desc:1 (detection), 2 (attribute), 3 (modeling), separate multiple types with
commas*/
                        "analysisSourceType":  "video,picture",
                        /*req, string, VCA source type*/
                        "algorithmID":  "test",
                        /*opt, string, algorithm package ID,either algorithm or algorithmType is required,the maximum size is 64 bytes*/
                        "classificationID":  "test",
                        /*opt, string, algorithm category ID*/
                        "version":  "test",
                        /*opt, string, algorithm version*/
                        "vendor":  "test"
                        /*opt, string, algorithm manufacturer*/
                    }
                ],
                "algorithmType":  "faceModel",
                /*req, enum, algorithm type, subType:string, desc:"faceModel" (face modeling), "faceStruct" (face structurization), "faceDetect" (face
detection), "faceModelContrast" (face model comparison, multiple algorithm types should be separated by commas), "trajectory" (person track)*/
                "mode":  "singleface",
                /*opt, enum, face detection mode, subType:string, desc:"singleface" (detect one face), "mutiface" (detect multiple faces). By default,if no
value is assigned to this node, the face detection mode is "singleface"*/
                "faceSelectStrategy":  "best",
                /*opt, enum, "best"-the best face,"middest"-the face in the middle,"biggest"-the biggest face, subType:string, desc:"best" (the best face),
"middest" (the face in the middle), "biggest" (the biggest face)*/
                "destination": [
                /*req, array, result recipient list, subType:object*/
                    {
                        "destinationType":  "faceCaptureLib",
                        /*opt, enum, result recipient type, subType:string, desc:"faceCaptureLib" (captured face picture library), "faceAlarmLib" (list
alarm library for face picture big data), "kafkaWhiteAlarmLib" (Kafka allowlist alarm library), "faceCaptureUpload" (captured picture uploaded by ISAPI),
"faceAlarmUpload" (list alarm uploaded by ISAPI), "faceAlarmSdkUpload" (list alarm uploaded by SDK), "KafkaFaceSnap" (captured face picture merged with big
data), "kafkaRedAlarm" (red list alarm)*/
                        "destinationUrl":  "test",
                        /*opt, string, recipient URL description, desc:recipient URL description,the maximum length is 1023 bytes*/
                        "userName":  "test",
                        /*opt, string, user name of the recipient, desc:recipient user name,the maximum length is 32 bytes. This node can be set to NULL if
recipient user name does not exist*/
                        "password":  "test"
                        /*opt, string, recipient password, desc:recipient password,the maximum length is 32 bytes. This node can be set to NULL if recipient
password does not exist*/
                    }
                ],
                "compareSearchType":  "planQuery",
                /*opt, enum, subType:string*/
                "relateGroup": [
                /*opt, array, specified face picture list relationship, subType:object*/
                    {
                        "ctrlID":  "test",
                        /*req, string, arming ID, range:[1,63], desc:this parameter is UUID. Due to historical reasons, the old version of  device only has
arming id. In the multi-analysis task scenes, task ID and id parameters need to work together to identify a unique relation. Later, considering the new
needs of the operation and management platform, id is replaced by the ctrlID (uuid) the current device supports ctrlID and id (check whether it is ctrlID or
id according to the node returned by the capability set). ctrlID is recommended, and users can configure ctrlID or id. Users are not allowed to configure
both ctrlID and id; otherwise, the device will return error message*/
                        "id":  1,
                        /*req, int, serial No., range:[1,2147483647], step:1, desc:URL for getting the capability set:
/ISAPI/SDT/Management/Task/facePicture/batch/capabilities, if ctrlID is not returned, it indicates that only id is supported*/
                        "relate":  "xxxxxx-xxx",
                        /*opt, string, face picture library ID, range:[1,10240], desc:for relate node, "-1" represents all faces, face picture library ID,
separate multiple items with comma, compare the captured picture with the pictures in the capture library, restrict the threshold according to min. value
and max. value; when the value of "type" is 1 or 2, relate is required; when it is 3 (frequently appeared person), relate is optional and it works as a
frequently appeared person alarm. When relate is configured, the captured picture will be compared with the pictures in the corresponding library; if the
captured picture matches with the picture in the library, the person will be regarded as normal person, and the frequently appeared person alarm will not be
triggered. If relate is not configured, this step will be skipped*/
                        "min":  0,
                        /*opt, int, the minimum threshold, range:[0,1000], desc:when relate exists, the min is required and the max must be larger than the
min*/
                        "max":  1000,
                        /*opt, int, the maximum threshold, range:[0,1000], desc:when relate exists, the max is required; when the value of "type" is 1 or 3,
the fixed value is 1,000*/
                        "top":  1,
                        /*opt, int, number of alarm pictures, range:[1,10], desc:when relate exists, top is required; the top value is 1 by default for the
allowlist alarm and the configured value is invalid*/
                        "type":  1,
                        /*opt, enum, alarm type, subType:int, desc:alarm type: 1-blocklist (default),2-allowlist,3-frequently appeared person*/
                        "similarity":  80,
                        /*opt, int, similarity, range:[0,100], desc:it is required when the value of "type" is 3, the default value is 85, and the
recommended value is between 80 to 100; it is used for the face match between the captured picture and the library of captured pictures; if it matches, its
frequency will be added by 1*/
                        "occurrences":  10,
                        /*opt, int, occurrences, range:[1,100], desc:it is required when the value of "type" is 2, integer31, the default value is 10*/
                        "captureTimeInterval":  10,
                        /*opt, int, capture interval, range:[0,60], desc:if it is set to 0, the target will be counted every time when it appears*/
                        "analysisDays":  1,
                        /*opt, int, number of days of a specified channel, range:[1,30], desc:it is required when the value of "type" is 3, integer32, the
number of days of a specified channel (it is seven days by default)*/
                        "timeRange": [
                        /*opt, array, arming time period, subType:object, desc:up to 5 time periods are supported, by default, the whole day is armed*/
                            {
                                "startTime":  "17:30:08",
                                /*req, time, arming start time*/
                                "endTime":  "17:30:08"
                                /*req, time, arming end time*/
```

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```
                        }
                    ],
                    "channels": [
                    /*opt, array, camera list, subType:object, desc:this node is required for frequently appeared person alarm*/
                        {
                            "id":  "d9bcd1a09f0d44a18425429a0fe0f6c5"
                            /*opt, string, camera ID*/
                        }
                    ]
                }
            ],
            "image": {
            /*req, object, picture streaming information*/
                "imageInfos": [
                /*req, array, picture stream information, subType:object*/
                    {
                        "dataType":  "URL",
                        /*req, enum, picture data type, subType:string, desc:"URL" (picture URL), "modelData" (picture modeled data (currently only
supported by the comparison server)*/
                        "URL":  "test",
                        /*opt, string, picture URL, range:[1,1024], desc:picture URL,this node is required when dataType is "URL",the maximum length is
256 bytes*/
                        "captureTime":  "2017-04-22T15:39:01.000+08:00",
                        /*opt, datetime*/
                        "FacePostureList": [
                        /*opt, array, list of frame for human face, subType:object, desc:person track task; when there are multiple pictures of a single
target person (of a same person), it represents other face gesture pictures of a same person if uploading FacePostureList*/
                            {
                                "URL":  "test",
                                /*opt, string, picture URL, range:[1,1024]*/
                                "faceTime":  "2017-04-22T15:39:01.000+08:00",
                                /*opt, datetime, capture time*/
                                "facePosture":  "sideFace"
                                /*opt, enum, face gesture, subType:string, desc:"sideFace", "largeDepressionAngle", "mask"*/
                            }
                        ],
                        "httpAuthenticationMethod":  "MD5digest",
                        /*opt, enum, authentication method, subType:string, desc:"MD5digest" (MD5 digest authentication)*/
                        "userName":  "admin",
                        /*opt, string, digest authentication user name, range:[0,32]*/
                        "password":  "xxx12345",
                        /*opt, string, digest authentication password, range:[0,16], desc:digest authentication password*/
                        "modelData":  "test",
                        /*opt, string, model data*/
                        "traceUuid":  "xxxxxxxx",
                        /*req, string, target ID, which is same when the target person is in multiple pictures, desc:target tracking ID,it is used to
mark the same person in multiple pictures,the maximum length is 63 bytes*/
                        "traceIdx":  1,
                        /*req, int, No. of picture that contains person target*/
                        "traceTotal":  8,
                        /*opt, int, total picture number of a person*/
                        "roi": {
                        /*opt, object, region of interest (ROI)*/
                            "height":  1.0,
                            /*opt, float, height*/
                            "width":  1.0,
                            /*opt, float, width*/
                            "x":  0.0,
                            /*opt, float, x-coordinate*/
                            "y":  0.0
                            /*opt, float, y-coordinate*/
                        },
                        "faceRect": {
                        /*opt, object, face thumbnail frame*/
                            "height":  1.0,
                            /*opt, float, height*/
                            "width":  1.0,
                            /*opt, float, width*/
                            "x":  0.0,
                            /*opt, float, x-coordinate*/
                            "y":  0.0
                            /*opt, float, y-coordinate*/
                        },
                        "faceMark": {
                        /*opt, object, face information*/
                            "leftEye": {
                            /*opt, object, left eye coordinates*/
                                "x":  0.13,
                                /*opt, float, x-coordinate*/
                                "y":  0.87
                                /*opt, float, y-coordinate*/
                            },
                            "rightEye": {
                            /*opt, object, right eye coordinates*/
                                "x":  0.78,
                                /*opt, float, x-coordinate*/
                                "y":  0.50
                                /*opt, float, y-coordinate*/
                            },
                            "noseTip": {
                            /*opt, object, nose tip coordinates*/
                                "x":  0.78,
                                /*opt, float, x-coordinate*/
```

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```
                    "y":  0.50
                    /*opt, float, y-coordinate*/
                },
                "leftMouth": {
                /*opt, object, left mouth corner's position*/
                    "x":  0.78,
                    /*opt, float, x-coordinate*/
                    "y":  0.50
                    /*opt, float, y-coordinate*/
                },
                "rightMouth": {
                /*opt, object, left mouth corner's position*/
                    "x":  0.78,
                    /*opt, float, x-coordinate*/
                    "y":  0.50
                    /*opt, float, y--coordinate*/
                }
            },
            "humanImages": [
            /*opt, array, human body picture linked with the face picture, subType:object*/
                {
                    "dataType":  "URL",
                    /*req, string, picture data type*/
                    "URL":  "test",
                    /*req, string, picture URL*/
                    "humanRect": {
                    /*opt, object, face thumbnail rectangle frame*/
                        "height":  1.0,
                        /*opt, float, height*/
                        "width":  1.0,
                        /*opt, float, width*/
                        "x":  0.0,
                        /*opt, float, x-coordinate*/
                        "y":  0.0
                        /*opt, float, y-coordinate*/
                    },
                    "reservedFields": {
                    /*opt, object, human body information, desc:transmitted parameters of human body information. Related examples are as follows; specified report format is defined by the platform big data and the track partition (area)*/
                        "timeStamp":  "test",
                        /*opt, string, timestamp, desc:the number of seconds between current time and the Unix Epoch. It is accurate to millisecond*/
                        "posture":  0,
                        /*opt, enum, human body posture, subType:int, desc:human body posture: "0" (back); "1" (left); "2" (right); "3" (forward)*/
                        "poseScore":  65,
                        /*opt, int, human body posture rating*/
                        "postureMaskURL":  "test",
                        /*opt, string, the picture URL of the human body posture mask*/
                        "confidence":  1,
                        /*opt, int, human body direction confidence, range:[0,100]*/
                        "HumanBodyClarity": {
                        /*opt, object, human body definition*/
                            "score":  20,
                            /*opt, int, rating, range:[0,1000]*/
                            "confidence":  1
                            /*opt, int, confidence, range:[0,100]*/
                        },
                        "UpperBodyIntegrity": {
                        /*opt, object, the completeness of the upper body*/
                            "score":  20,
                            /*opt, int, rating, range:[0,1000]*/
                            "confidence":  1
                            /*opt, int, confidence, range:[0,100]*/
                        },
                        "LowerBodyIntegrity": {
                        /*opt, object, the completeness of the lower body*/
                            "score":  20,
                            /*opt, int, rating, range:[0,1000]*/
                            "confidence":  1
                            /*opt, int, confidence, range:[0,100]*/
                        },
                        "HeanAndShoulderIntegrity": {
                        /*opt, object, the completeness of head and shoulder*/
                            "score":  20,
                            /*opt, int, rating, range:[0,1000]*/
                            "confidence":  1
                            /*opt, int, confidence, range:[0,100]*/
                        }
                    }
                }
            ],
            "targetAttrs": {
            /*opt, object, target features (transparent transmission field),different recipient addresses assigned in destination correspond to different transparent transmission fields,and the transparent transmission field can be customized if uploading to the third-party platform, desc:target features (transparent transmission field),different recipient addresses assigned in destination correspond to different transparent transmission fields,and the transparent transmission field can be customized if uploading to the third-party platform*/
                "deviceId":  "1",
                /*opt, string, device ID*/
                "deviceSerialNumber":  "iDS-9632NX-I8/X1620181209CCRRC77605411WCVU",
                /*ro, opt, string, device serial No., range:[1,48], desc:when the cameras are managed by the mobile server, users can confirm the device by unique device serial No.; if the device is NVR or multicast camera, users need to confirm via unique device serial No. and device channel No.*/
                "deviceChannel":  1,
```


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```
                            /*opt, int, device channel No.*/
                            "deviceName":  "shebei",
                            /*opt, string, device name*/
                            "faceTime":  "2016-09-10T01:01:00.000Z",
                            /*opt, string, capture time*/
                            "rect": {
                            /*opt, object, rectangle frame on the face picture*/
                                "height":  1.0,
                                /*opt, float, height*/
                                "width":  1.0,
                                /*opt, float, width*/
                                "x":  0.0,
                                /*opt, float, x-coordinate*/
                                "y":  0.0
                                /*opt, float, y-coordinate*/
                            },
                            "bkgUrl":  "http://absddsdfsdfsdk.jpg"
                            /*opt, string, background image*/
                        },
                        "customInfo":  "test",
                        /*opt, string, user-defined information*/
                        "attrFilter": {
                        /*opt, object, attribute filter*/
                            "gender":  "male"
                            /*opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
                        }
                    }
                ]
            }
        }
    ],
    "faceLibType":  "staticFD"
    /*opt, enum, face picture library type, subType:string, desc:"staticFD" (static library; used for clustering, alarm and picture searching),
"blocklistFD" (list library; used for alarm), "faceConfirmLib" (face picture library of identity verification; used for picture searching); for the server
before version 1.3.0 and other devices, the device has only one container to store the list library of synchronizing big data for the server of version
1.3.0, the device has two containers respectively for the static library and the newly added list library; for the server of 1.4.0 version, the device
supports three containers, respectively for the static library, list library and face picture library of identity verification;*/
}
```

Response Message

```
{
    "taskIDList": [
    /*ro, req, array, task ID, subType:object*/
        {
            "taskID":  "test"
            /*ro, opt, string, task ID*/
        }
    ]
}
```

15.5.4.11 Get the capability of face picture comparison task

Request URL

GET /ISAPI/SDT/Management/Task/facePicture/capabilities

Query Parameter

None

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  0,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description*/
    "taskInfo": {
    /*ro, req, object, task information*/
        "algorithmType": [
        /*ro, req, array, algorithm type, subType:object, desc:algorithm type:*/
            {
```

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```
            {
                "value":  "faceDetect"
                /*ro, opt, string, address of capture library*/
            }
        ],
        "faceLibType": {
        /*ro, opt, object, face picture library type*/
            "@opt":  ["staticFD", "blocklistFD", "faceConfirmLib"]
            /*ro, opt, enumarray, face picture library type, subType:string, desc:"staticFD" (static library); "blocklistFD" (list library); "aceConfirmLib"
(face picture library of identity verification)*/
        },
        "model": [
        /*ro, req, array, face detection mode, subType:object, desc:"singleface"-detect a face (default),"mutiface"-detect multiple faces,by default,if no
value is assigned to this node,the mode is "singleface"*/
            {
                "value":  "singleface"
                /*ro, opt, string, address of capture library*/
            }
        ],
        "faceSelectStrategy": {
        /*ro, opt, object, face selection strategy, desc:"best"-the best face,"middest"-the face in the middle,"biggest"-the biggest face*/
            "@opt": ["best", "middest", "biggest"]
            /*ro, opt, array, face selection strategy, subType:string*/
        },
        "destination": [
        /*ro, req, array, result recipient list, subType:object, desc:result receiver information*/
            {
                "value":  "faceCaptrueLib"
                /*ro, opt, string, address of capture library*/
            }
        ],
        "relateGroup": [
        /*ro, req, array, specified face picture list relationship, subType:object*/
            {
                "ctrlID": {
                /*ro, opt, object, arming ID*/
                    "@min":  0,
                    /*ro, opt, int, the minimum threshold*/
                    "@max":  32
                    /*ro, opt, int, the maximum threshold*/
                },
                "min":  0,
                /*ro, req, int, the minimum threshold*/
                "max":  1000,
                /*ro, req, int, the maximum threshold*/
                "top":  5,
                /*ro, req, int, number of alarm pictures*/
                "type": [1, 2, 3],
                /*ro, opt, array, alarm category, subType:int, desc:1: blocklist (default); 2: allowlist; 3-frequently appeared person; 4-one person one
archive*/
                "similarity": {
                /*ro, opt, object, similarity, desc:this node is valid when the value of "type" is 3, integer32, the value ranges from 80 to 100 and the
default value is 85*/
                    "@min":  80,
                    /*ro, opt, int, the minimum value*/
                    "@max":  100,
                    /*ro, opt, int, the maximum value*/
                    "@def":  85
                    /*ro, opt, int, the default value*/
                },
                "occurrences": {
                /*ro, opt, object, number of appearances, desc:this node is valid when the value of "type" is 3, integer32, the value ranges from 5 to 100
and the default value is 10*/
                    "@min":  5,
                    /*ro, opt, int, the minimum value*/
                    "@max":  100,
                    /*ro, opt, int, the maximum value*/
                    "@def":  10
                    /*ro, opt, int, the default value*/
                },
                "captureTimeInterval": {
                /*ro, opt, object, capture interval, desc:this node is valid when the value of "type" is 3, integer32, the value ranges from 0 to 60 and the
default value is 10. When the value is set to 0, every appearance will be counted*/
                    "@min":  0,
                    /*ro, opt, int, the minimum value*/
                    "@max":  60,
                    /*ro, opt, int, the maximum value*/
                    "@def":  10
                    /*ro, opt, int, the default value*/
                },
                "analysisDays": {
                /*ro, opt, object, number of days of a specified channel, desc:this node is valid when the value of "type" is 3. The value ranges from 1 to
30 and the default value is 7*/
                    "@min":  1,
                    /*ro, opt, int, the minimum value*/
                    "@max":  30,
                    /*ro, opt, int, the maximum value*/
                    "@def":  7
                    /*ro, opt, int, the default value*/
                },
                "timeRange": {
                /*ro, opt, object, time period*/
                    "@size":  1,
                    /*ro, opt, int, maximum number of time periods, desc:up to 5 time periods are supported,by default,the whole day is armed*/
```

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```
                    "startTime":  "HH:mm:ss",
                    /*ro, req, string, arming start time*/
                    "endTime":  "HH:mm:ss"
                    /*ro, req, string, arming end time*/
                },
                "channels": {
                /*ro, opt, object, camera list, desc:this node is required for frequently appeared person alarm*/
                    "@size":  64,
                    /*ro, opt, int, the maximum supported number of channels*/
                    "id": {
                    /*ro, opt, object, the length of the camera ID*/
                        "@min":  0,
                        /*ro, opt, int, the minimum length of the camera ID*/
                        "@max":  0
                        /*ro, opt, int, the maximum length of the camera ID*/
                    }
                }
            }
        ],
        "maxPictureNum":  0,
        /*ro, opt, int, the maximum number of pictures*/
        "image": {
        /*ro, opt, object, information about streaming pictures*/
            "imageInfos": [
            /*ro, req, array, picture stream information, subType:object*/
                {
                    "dataType": [
                    /*ro, req, array, picture data type, subType:object, desc:picture data type:*/
                        {
                            "value":  "URL"
                            /*ro, opt, string, address of capture library*/
                        }
                    ],
                    "httpAuthenticationMethod": {
                    /*ro, opt, object, authentication method*/
                        "@opt": ["MD5digest", "none"]
                        /*ro, opt, array, option, subType:string*/
                    },
                    "userName": {
                    /*ro, opt, object, digest authentication user name*/
                        "@min":  0,
                        /*ro, opt, int, the minimum value*/
                        "@max":  0
                        /*ro, opt, int, the maximum value*/
                    },
                    "password": {
                    /*ro, opt, object, digest authentication password*/
                        "@min":  0,
                        /*ro, opt, int, the minimum value*/
                        "@max":  0
                        /*ro, opt, int, the maximum value*/
                    },
                    "total":  100,
                    /*ro, req, int, total face pictures' number of a person*/
                    "targetAttrs":  2048
                    /*ro, opt, int, the maximum length of the transmitted parameters*/
                }
            ]
        }
    }
}
```

15.5.4.12 Get the capability set of synchronous face picture comparison task

Request URL

GET /ISAPI/SDT/Management/Task/facePicture/sync/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "taskInfoCap": {
    /*ro, req, object, picture information*/
        "dataType":  "URL",
        /*ro, req, enum, picture data type, subType:string, desc:"URL", (picture URL), "modelData" (model data), "binary" (picture binary form), "base64"
(picture data encoded via base64)*/
        "URL": {
        /*ro, opt, object, picture URL*/
            "@min":  1,
            /*ro, opt, int, the minimum length*/
            "@max":  1
```

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```
        /*ro, opt, int, the maximum length*/
    },
    "httpAuthenticationMethod": {
    /*ro, opt, object, authentication method*/
        "@opt": ["MD5digest", "none"]
        /*ro, opt, array, options, subType:string*/
    },
    "userName": {
    /*ro, opt, object, digest authentication user name*/
        "@min":  0,
        /*ro, opt, int, the minimum length*/
        "@max":  0
        /*ro, opt, int, the maximum length*/
    },
    "password": {
    /*ro, opt, object, digest authentication password*/
        "@min":  0,
        /*ro, opt, int, the minimum length*/
        "@max":  0
        /*ro, opt, int, the maximum length*/
    },
    "modelData": {
    /*ro, opt, object, model data*/
        "@min":  1,
        /*ro, opt, int, the minimum length*/
        "@max":  1
        /*ro, opt, int, the maximum length*/
    },
    "faceRect": {
    /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
        "height": {
        /*ro, opt, object, height*/
            "@min":  0.000,
            /*ro, opt, float, the minimum length, range:[0.000,1.000]*/
            "@max":  1.000
            /*ro, opt, float, the maximum length, range:[0.000,1.000]*/
        },
        "width": {
        /*ro, opt, object, width*/
            "@min":  0.000,
            /*ro, opt, float, the minimum length, range:[0.000,1.000]*/
            "@max":  1.000
            /*ro, opt, float, the maximum length, range:[0.000,1.000]*/
        },
        "x": {
        /*ro, opt, object, X-coordinate*/
            "@min":  0.000,
            /*ro, opt, float, the minimum length, range:[0.000,1.000]*/
            "@max":  1.000
            /*ro, opt, float, the maximum length, range:[0.000,1.000]*/
        },
        "y": {
        /*ro, opt, object, Y-coordinate*/
            "@min":  0.000,
            /*ro, opt, float, the minimum length, range:[0.000,1.000]*/
            "@max":  1.000
            /*ro, opt, float, the maximum length, range:[0.000,1.000]*/
        }
    },
    "relateGroup": {
    /*ro, opt, object, relation among specified lists*/
        "id": {
        /*ro, opt, object, No., desc:either "id" or "ctrlID" is required*/
            "@min":  0,
            /*ro, opt, int, the minimum length*/
            "@max":  0
            /*ro, req, int, the maximum length*/
        },
        "ctrlID": {
        /*ro, opt, object, arming ID, desc:either "id" or "ctrlID" is required*/
            "@min":  0,
            /*ro, opt, int, the minimum length*/
            "@max":  0
            /*ro, req, int, the maximum length*/
        },
        "min": {
        /*ro, req, object, the minimum threshold*/
            "min":  1,
            /*ro, opt, int, the minimum length*/
            "max":  1
            /*ro, opt, int, the maximum length*/
        },
        "max": {
        /*ro, opt, object, the maximum threshold*/
            "min":  1,
            /*ro, opt, int, the minimum length*/
            "max":  1
            /*ro, opt, int, the maximum length*/
        },
        "relateLen": {
        /*ro, req, object, the length of the face picture library ID*/
            "min":  1,
            /*ro, opt, int, the minimum length*/
```

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```
                "max":  1
                /*ro, opt, int, the maximum length*/
            },
            "top": {
            /*ro, req, object, number of alarm pictures*/
                "min":  1,
                /*ro, opt, int, the minimum length*/
                "max":  1
                /*ro, opt, int, the maximum length*/
            },
            "type":  "1,2"
            /*ro, opt, string, alarm category: 1-blocklist (default), 2-allowlist*/
        },
        "faceLibType": {
        /*ro, opt, object, face picture library type*/
            "@opt":  ["staticFD", "blocklistFD", "faceConfirmLib"]
            /*ro, opt, enumarray, face picture library type, subType:string, desc:"staticFD" (static library); "blocklistFD" (list library); "aceConfirmLib"
(face picture library of identity verification)*/
        }
    }
}
```

15.5.4.13 Submit the face picture comparison task in asynchronous mode

Request URL

POST /ISAPI/SDT/Management/Task/facePicture/sync?format=json

Query Parameter

None

Request Message

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```
{
    "taskInfo": {
    /*req, object, task information*/
        "image": {
        /*req, object, picture information*/
            "dataType":  "URL",
            /*req, enum, picture data type, subType:string, desc:"URL"-picture URL, "modelData"-model data, "binary"-picture in binary form format,
"base64"-picture data converted by Base64*/
            "URL":  "test",
            /*opt, string, picture URL*/
            "httpAuthenticationMethod":  "MD5digest",
            /*opt, enum, authentication method, subType:string, desc:"MD5digest" (MD5 digest authentication)*/
            "userName":  "admin",
            /*opt, string, digest authentication user name, range:[0,32]*/
            "password":  "12345",
            /*opt, string, digest authentication password, range:[0,16]*/
            "modelData":  "test",
            /*opt, string, picture model data in string format converted by Base64, dep:and,{$.taskInfo.image.dataType,eq,modelData}, desc:modeldata is
required when the dataType is modelData*/
            "picData":  "base64转义的图片二进制数据",
            /*opt, string, binary picture data converted by Base64, desc:this node is required when the value of datatype is "base64". The picture data is
transmitted in string format*/
            "rect": {
            /*opt, object, face thumbnail frame, desc:when there are multiple face thumbnails in the submitted picture, the face in the frame will be
analyzed; if this node is not transmitted, by default, the face thumbnail with optimal quality will be outputted. The origin is the upper-left corner of the
screen*/
                "height":  1.000,
                /*opt, float, height, range:[0.000,1.000]*/
                "width":  1.000,
                /*opt, float, width, range:[0.000,1.000]*/
                "x":  0.000,
                /*opt, float, X-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*opt, float, Y-coordinate, range:[0.000,1.000]*/
            },
            "attrFilter": {
            /*opt, object, attribute filter*/
                "gender":  "male"
                /*opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
            }
        },
        "relateGroup": {
        /*opt, object, specified face picture list relationship*/
            "id":  1,
            /*req, int, No., range:[1,2147483647], step:1, desc:either this node or ctrlID is required*/
            "ctrlID":  "test",
            /*req, string, arming ID, range:[1,63], desc:either this node or id is required*/
            "min":  1,
            /*req, int, the minimum threshold, range:[0,1000], step:1*/
            "max":  1000,
            /*req, int, the maximum threshold, range:[0,1000], step:1, desc:this parameter is only valid in allowlist scenes (invalid in blocklist scenes),
and the fixed value is 1,000*/
            "relate":  "xxxxxx-xxx",
            /*req, string, face picture library ID, desc:separate multiple items with comma*/
            "top":  1,
            /*req, int, number of alarm pictures, range:[1,100], step:1, desc:for allowlist, the default value of this node is 1 and the settings will not
take effect*/
            "type":  1
            /*opt, enum, alarm type, subType:int, desc:1-blocklist (default), 2-allowlist*/
        },
        "faceLibType":  "staticFD"
        /*opt, enum, face picture library type, subType:string, desc:"staticFD" (static library; used for clustering, alarm and picture searching),
"blocklistFD" (list library; used for alarm), "faceConfirmLib" (face picture library of identity verification; used for picture searching); for the server
before version 1.3.0 and other devices, the device has only one container to store the list library of synchronizing big data for the server of version
1.3.0, the device has two containers respectively for the static library and the newly added name library; for the server of 1.4.0 version, the device
supports three containers, respectively for the static library, list library and face picture library of identity verification;*/
    }
}
```

Response Message

```
{
    "taskInfo": {
    /*ro, req, object, task information*/
        "targetModelData":  "test",
        /*ro, opt, string, model data*/
        "ageGroup":  "child",
        /*ro, opt, string, age group*/
        "gender":  "male",
        /*ro, opt, string, gender*/
        "glass":  "yes",
        /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no", "yes", "sunglasses"*/
        "smile":  "yes",
        /*ro, opt, string, whether the person wears a smile*/
        "mask":  "yes",
        /*ro, opt, string, whether the person wears a mask*/
        "faceScore":  0.000,
        /*ro, opt, float, face score (face picture quality)*/
        "faceExpression":  "test",
```

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```
        /*ro, opt, string, facial expression*/
        "facePose": {
        /*ro, opt, object, face angle*/
            "pitch":  0.998,
            /*ro, opt, float, face pitch angle, face pitching up is regarded as positive, range:[-90,90], desc:it is accurate to three decimal places. This
node does not exist if not supported*/
            "yaw":  0.876,
            /*ro, opt, float, face yaw angle, face yawing left is regarded as positive, range:[-90,90], desc:it is accurate to three decimal places. This
node does not exist if not supported*/
            "roll":  0.623
            /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive, range:[-90,90], desc:it is accurate to three decimal places.
This node does not exist if not supported*/
        },
        "faceRect": {
        /*ro, opt, object, face frame in picture, desc:the origin is the upper-left corner of the screen*/
            "height":  1.000,
            /*ro, opt, float, height, range:[0.000,1.000]*/
            "width":  1.000,
            /*ro, opt, float, width, range:[0.000,1.000]*/
            "x":  0.000,
            /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
            "y":  0.000
            /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
        },
        "recommendFaceRect": {
        /*ro, opt, object, recommended face frame coordinates, desc:it is used for image matting by expanding the face thumbnail frame*/
            "height":  1.000,
            /*ro, opt, float, height, range:[0.000,1.000]*/
            "width":  1.000,
            /*ro, opt, float, width, range:[0.000,1.000]*/
            "x":  0.000,
            /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
            "y":  0.000
            /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
        },
        "faceMark": {
        /*ro, opt, object, face features in the picture, desc:faceRect and faceMark must be configured together*/
            "leftEye": {
            /*ro, opt, object, left eye's coordinates, desc:the origin is the upper-left corner of the screen*/
                "x":  0.130,
                /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
                "y":  0.870
                /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
            },
            "rightEye": {
            /*ro, opt, object, right eye's coordinates, desc:the origin is the upper-left corner of the screen*/
                "x":  0.130,
                /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
                "y":  0.870
                /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
            },
            "noseTip": {
            /*ro, opt, object, nose tip's coordinates, desc:the origin is the upper-left corner of the screen*/
                "x":  0.130,
                /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
                "y":  0.870
                /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
            },
            "leftMouth": {
            /*ro, opt, object, left mouth corner's coordinates, desc:the origin is the upper-left corner of the screen*/
                "x":  0.130,
                /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
                "y":  0.870
                /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
            },
            "rightMouth": {
            /*ro, opt, object, right mouth corner's coordinates, desc:the origin is the upper-left corner of the screen*/
                "x":  0.130,
                /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
                "y":  0.870
                /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:it is accurate to three decimal places*/
            }
        },
        "totalMatches":  1,
        /*ro, opt, int, total number of matched results*/
        "MatchList": [
        /*ro, opt, array, the matched data information, subType:object*/
            {
                "similarity":  0.2,
                /*ro, req, float, similarity, range:[0,1], desc:it is accurate to two decimal places*/
                "similarityRange":  "1",
                /*ro, opt, enum, similarity range, subType:string, desc:0 (smaller than the minimum value), 1 (between the minimum value and maximum value),
2 (larger than the maximum value). If the similarity is smaller than the minimum value, the person will be marked*/
                "listType":  "blackList",
                /*ro, opt, enum, list type, subType:string, desc:"blackList" (blocklist), "whiteList" (allowlist), the maximum value length is 20*/
                "FPID":  "test",
                /*ro, opt, string, face record ID*/
                "FDID":  "test",
                /*ro, opt, string, face picture library ID*/
                "FDName":  "test",
                /*ro, opt, string, face picture library name*/
                "faceURL":  "test",
                /*ro, opt, string, face picture URL*/
                "name":  "test"
```

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```
                "name":  "test",
                /*ro, opt, string, name*/
                "gender":  "male",
                /*ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
                "bornTime":  "2004-05-03",
                /*ro, opt, string, birth date of the person in the face picture*/
                "city":  "130100",
                /*ro, opt, string, code of the city of birth, desc:city code of birth for the person in the face picture,the maximum size is 32 bytes*/
                "certificateType":  "officerID",
                /*ro, opt, enum, certificate type, subType:string, desc:"officerID"-officer ID, "ID"-identify card, "other"-other certificate*/
                "certificateNumber":  "test",
                /*ro, opt, string, certificate No.*/
                "caseInfo":  "test",
                /*ro, opt, string, remarks*/
                "tag":  "aa,bb,cc,dd",
                /*ro, opt, string, custom tag, desc:up to 4 tags can be added and tags should be separated with comma, the maximum length for 4 tags is 195
(48 for each)*/
                "address":  "test",
                /*ro, opt, string, person address, desc:the maximum length is 192*/
                "customInfo":  "test"
                /*ro, opt, string, custom information, desc:the maximum length is 192*/
            }
        ]
    }
}
```

15.5.4.14 Submit a face picture comparison task to return taskUID

Request URL

POST /ISAPI/SDT/Management/Task/facePicture

Query Parameter

None

Request Message

```
{
    "taskInfo": {
    /*req, object, task information*/
        "algorithmType":  "faceModel",
        /*req, enum, algorithm type, subType:string, desc:"faceModel" (face modeling), "faceStruct" (face structurization), "faceDetect" (face detection),
multiple algorithm types should be separated by commas*/
        "faceLibType":  "staticFD",
        /*opt, enum, face picture library type, subType:string, desc:"staticFD" (static library; used for clustering, alarm and picture searching),
"blocklistFD" (list library; used for alarm), "faceConfirmLib" (face picture library of identity verification; used for picture searching); for the server
before version 1.3.0 and other devices, the device has only one container to store the list library of synchronizing big data for the server of version
1.3.0, the device has two containers respectively for the static library and the newly added name library; for the server of 1.4.0 version, the device
supports three containers, respectively for the static library, list library and face picture library of identity verification;*/
        "algorithm": [
        /*opt, array, algorithm type, subType:object, desc:currently only one algorithm type is allowed for one task. Either algorithm or algorithmType is
required. AlgorithmID is used to specify a unique algorithm while classificationID is used for fuzzy search*/
            {
                "targetType":  "1",
                /*req, string, algorithm target type, desc:see details in Data Dictionary*/
                "analysisType":  1,
                /*req, enum, analysis type, subType:int, desc:1 (detection), 2 (attribute), 3 (modeling), separate multiple types with commas*/
                "analysisSourceType":  "video,picture",
                /*req, string, VCA source type*/
                "algorithmID":  "test",
                /*opt, string, algorithm package ID, desc:either algorithmID or classificationID is required*/
                "classificationID":  "test",
                /*opt, string, algorithm category ID, desc:either algorithmID or classificationID is required*/
                "version":  "test",
                /*opt, string, algorithm version, desc:this node is valid when classificationID exists*/
                "vendor":  "test"
                /*opt, string, algorithm manufacturer, desc:this node is valid only when classificationID exists*/
            }
        ],
        "mode":  "mutiface",
        /*opt, enum, face detection mode, subType:string, desc:"singleface" (detect one face, by default), "mutiface" (detect multiple faces). By default,
if no value is assigned to this node, the face detection mode is "singleface"*/
        "faceSelectStrategy":  "best",
        /*opt, enum, face selection strategy, subType:string, desc:"best" (the best face), "middest" (the face in the middle), "biggest" (the biggest
face)*/
        "destination": [
        /*req, array, result recipient list, subType:object*/
            {
                "destinationType":  "faceCaptureLib",
                /*opt, enum, result recipient type, subType:string, desc:"faceCaptureLib" (captured picture library), "faceAlarmLib" (list library for
saving face alarms), "faceCaptureUpload" (service for uploading captured face picture to), "faceAlarmUpload" (service for uploading face comparison results
to based on ISAPI protocol), "faceAlarmSdkUpload" (service for uploading face comparison results to based on private protocol), "kafkaRedAlarm" (red list
alarm), "KafkaFaceLabel"*/
                "destinationUrl":  "test",
                /*opt, string, recipient URL description, desc:recipient description,which is in URL format:
"{"topic":"SNAP_IMAGE_INFO_TOPIC","url":"172.10.201.156:2181"}","{"topic":"ALARM_INFO_TOPIC","url":"172.10.201.156:2181"}","http://IP:PORT/{customDesc}","IP
:PORT". The maximum string length is 1023 bytes.*/
                "userName":  "test",
                /*opt, string, user name of the recipient, desc:recipient user name,the maximum string length is 32 bytes. If there is no account,keep this
```

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```
                /* opt, string, user name of the recipient, desc:recipient user name,the maximum string length is 32 bytes. If there is no account,keep this
node as none.*/
                "password":  "test"
                /*opt, string, recipient password, desc:recipient password,the maximum string length is 32 bytes. If there is no account,keep this node as
none.*/
            }
        ],
        "compareSearchType":  "planQuery",
        /*opt, enum, subType:string*/
        "relateGroup": [
        /*opt, array, the specified relation between cameras and face picture libraries, subType:object*/
            {
                "id":  1,
                /*req, int, ID, range:[1,2147483647], step:1, desc:either this node or ctrlID is required*/
                "ctrlID":  "test",
                /*req, string, arming ID, range:[1,63], desc:either this node or id is required*/
                "min":  0,
                /*req, int, the minimum threshold*/
                "max":  1000,
                /*req, int, the maximum threshold*/
                "relate":  "xxxxxx-xxx",
                /*req, string, face picture library ID, range:[0,10240], desc:separate multiple items with comma*/
                "top":  1,
                /*req, int, number of alarm pictures, range:[1,100], desc:for allowlist, the default value of this node is 1 and the settings will not take
effect*/
                "type":  "1",
                /*opt, enum, alarm type, subType:string, desc:"1" (blocklist, by default), "2" (allowlist), "3" (high frequency), "4" (one archive for one
person), 3 and 4 are only supported by Intelligent Fusion Server*/
                "similarity":  80,
                /*opt, int, similarity, range:[80,100], desc:this node is valid only when the value of "type" is 3*/
                "occurrences":  10,
                /*opt, int, occurrence*/
                "captureTimeInterval":  10,
                /*opt, int, capture interval, range:[0,60], unit:min, desc:when the value is set to 0, every appearance will be counted*/
                "analysisDays":  7,
                /*opt, int, number of days of a specified channel*/
                "timeRange": [
                /*opt, array, arming time period, up to 5 time periods are supported, by default, the whole day is armed, subType:object*/
                    {
                        "startTime":  "17:30:08",
                        /*req, string, arming start time*/
                        "endTime":  "17:30:08"
                        /*req, string, arming end time*/
                    }
                ],
                "channels": [
                /*opt, array, camera list, subType:object, desc:this node is required for frequently appeared person alarm*/
                    {
                        "id":  "d9bcd1a09f0d44a18425429a0fe0f6c5"
                        /*opt, string, camera ID*/
                    }
                ]
            }
        ],
        "image": {
        /*req, object, picture streaming information*/
            "imageInfos": [
            /*req, array, picture stream information, subType:object*/
                {
                    "dataType":  "URL",
                    /*req, enum, picture data type, subType:string, desc:"URL" (picture URL), "modelData" (picture modeled data (currently only supported by
the comparison server)*/
                    "URL":  "test",
                    /*opt, string, picture URL*/
                    "httpAuthenticationMethod":  "MD5digest",
                    /*opt, enum, authentication method, subType:string, desc:"MD5digest" (MD5 digest authentication)*/
                    "userName":  "admin",
                    /*opt, string, digest authentication user name, range:[0,32]*/
                    "password":  "xxx12345",
                    /*opt, string, digest authentication password, range:[0,16], desc:digest authentication password*/
                    "modelData":  "test",
                    /*opt, string, model data, desc:here the model data string has been transferred by Base64 and extra processing is unnecessary*/
                    "traceUuid":  "xxxxxxxx",
                    /*req, string, target ID, desc:target tracking ID,which is used to mark all pictures of a person,and the maximum string length is 63
bytes*/
                    "traceIdx":  1,
                    /*req, int, No. of picture that contains the same target, desc:the picture No. in all pictures of a person*/
                    "traceTotal":  8,
                    /*opt, int, total number of  pictures of the same target, desc:total picture number of a person*/
                    "roi": {
                    /*opt, object, region of interest (ROI), desc:If roi, faceRect, and faceMark are all configured, faceRect and faceMark are preferred*/
                        "height":  1.0,
                        /*opt, float, region height*/
                        "width":  1.0,
                        /*opt, float, region width*/
                        "x":  0.0,
                        /*opt, float, X-coordinate*/
                        "y":  0.0
                        /*opt, float, Y-coordinate*/
                    },
                    "faceRect": {
                    /*opt, object, face thumbnail frame*/
                        "height":  1.0,
                        /*opt, float, region height*/
```


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```
                    "width":  1.0,
                    /*opt, float, region width*/
                    "x":  0.0,
                    /*opt, float, X-coordinate*/
                    "y":  0.0
                    /*opt, float, Y-coordinate*/
                },
                "faceMark": {
                /*opt, object, face features in the picture*/
                    "leftEye": {
                    /*opt, object, left eye's coordinates*/
                        "x":  0.13,
                        /*opt, float, X-coordinate, desc:normalized coordinates, it is accurate to three decimal places*/
                        "y":  0.87
                        /*opt, float, Y-coordinate, desc:normalized coordinates, it is accurate to three decimal places*/
                    },
                    "rightEye": {
                    /*opt, object, right eye's coordinates*/
                        "x":  0.78,
                        /*opt, float, X-coordinate, desc:normalized coordinates, it is accurate to three decimal places*/
                        "y":  0.50
                        /*opt, float, Y-coordinate, desc:normalized coordinates, it is accurate to three decimal places*/
                    },
                    "noseTip": {
                    /*opt, object, nose tip's coordinates*/
                        "x":  0.78,
                        /*opt, float, X-coordinate, desc:normalized coordinates, it is accurate to three decimal places*/
                        "y":  0.50
                        /*opt, float, Y-coordinate, desc:normalized coordinates, it is accurate to three decimal places*/
                    },
                    "leftMouth": {
                    /*opt, object, left mouth corner's coordinates*/
                        "x":  0.78,
                        /*opt, float, X-coordinate, desc:normalized coordinates, it is accurate to three decimal places*/
                        "y":  0.50
                        /*opt, float, Y-coordinate, desc:normalized coordinates, it is accurate to three decimal places*/
                    },
                    "rightMouth": {
                    /*opt, object, right mouth corner's position*/
                        "x":  0.78,
                        /*opt, float, X-coordinate, desc:normalized coordinates, it is accurate to three decimal places*/
                        "y":  0.50
                        /*opt, float, Y-coordinate, desc:normalized coordinates, it is accurate to three decimal places*/
                    }
                },
                "targetAttrs": {
                /*opt, object, target attributes (to be passed through), desc:the transmitted parameters are different for different values of
"destination"*/
                    "deviceId":  "1",
                    /*opt, string, device ID*/
                    "deviceSerialNumber":  "iDS-9632NX-I8/X1620181209CCRRC77605411WCVU",
                    /*ro, opt, string, device serial No., range:[1,48], desc:when the cameras are managed by the mobile server, users can confirm the
device by unique device serial No.; if the device is NVR or multicast camera, users need to confirm via unique device serial No. and device channel No.*/
                    "deviceChannel":  1,
                    /*opt, int, device channel No.*/
                    "deviceName":  "shebei",
                    /*opt, string, device name*/
                    "faceTime":  "2016-09-10T01:01:00.000Z",
                    /*opt, string, capture time*/
                    "rect": {
                    /*opt, object, background picture frame information of face*/
                        "height":  1.0,
                        /*opt, float, region height*/
                        "width":  1.0,
                        /*opt, float, region width*/
                        "x":  0.0,
                        /*opt, float, X-coordinate*/
                        "y":  0.0
                        /*opt, float, Y-coordinate*/
                    },
                    "bkgUrl":  "http://absddsdfsdfsdk.jpg",
                    /*opt, string, background picture*/
                    "name":  "张三",
                    /*opt, string, person name*/
                    "registerGender":  "male",
                    /*opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
                    "nation":  1,
                    /*opt, int, nation*/
                    "bornTime":  "2016-01-30",
                    /*opt, string, date of birth*/
                    "address":  "中国浙江省",
                    /*opt, string, address*/
                    "certificateNumber":  "111111111111111111",
                    /*opt, string, certificate number (person ID)*/
                    "startDate":  "2009-11-14T15:27:00.000Z",
                    /*opt, string, start time of certificate effective period*/
                    "endDate":  "2009-11-14T15:27:00.000Z",
                    /*opt, string, end time of certificate effective period*/
                    "credImgUrl":  "http://XXX",
                    /*opt, string, URL of person profile on certificate*/
                    "matchResult":  0,
                    /*opt, int, matched results*/
                    "alarmState":  true,
                    /*opt, bool, whether the person is armed*/
```

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```
                    /*opt, bool, whether the person is armed*/
                    "caseInfo":  "XXX"
                    /*opt, string, remarks*/
                },
                "customInfo":  "test",
                /*opt, string, user custom information, desc:it will be transmitted with the result*/
                "attrFilter": {
                /*opt, object, attributes filter*/
                    "gender":  "male"
                    /*opt, enum, gender: "male", "female", "unknown", subType:string, desc:"male", "female", "unknown"*/
                }
            }
        ]
    },
    "enableAlgorithmConversion":  true
    /*opt, enum, whether to enable algorithm transformation, subType:bool, desc:true: yes, to make sure the model data in task is same as that of list library; false: no*/
  }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskUID":  "test"
    /*ro, opt, string, task ID*/
}
```

15.5.4.15 Search for the task status information

Request URL

POST /ISAPI/SDT/Management/Task/status

Query Parameter

None

Request Message

```
{
    "taskInfo": [
    /*opt, array, task information, subType:object*/
        {
            "taskID":  "fdb4465d7aa5495884845127a8bbbe3c"
            /*req, string, task ID*/
        }
    ]
}
```

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description*/
    "status": [
    /*ro, opt, array, task status, it is valid when errorCode is 1 and errorMsg is "ok", subType:object*/
        {
            "errorCode":  1,
            /*ro, req, int, 1 indicates success*/
            "errorMsg":  "ok",
            /*ro, req, string, "ok" indicates success*/
            "taskID":  "test",
            /*ro, req, string, task ID*/
            "taskStatus":  1,
            /*ro, req, enum, task status, subType:int, desc:1-"NotDispatched" (not scheduled), 2-"Waiting" (waiting for the task), 3-"Executing" (executing
the task), 4-"Completed" (the task is completed), 5-"Deleted" (the task is deleted), 6-"Offline" (the node is offline), 7-"ServerDown" (service downtime),
8-"Stopping" (stopping the task), 9-"Stopped" (the task is stopped), 10-"Rebooting", 11-"Paused", 12-"Pausing", 13-"Failed", 14 (the task ID does not
exist)*/
            "process":  22,
            /*ro, opt, int, task progress*/
            "demonstrateStatus":  "demonstrating",
            /*ro, req, enum, presentation status, subType:string, desc:"noDemonstrate" (no presentation), "demonstrateWait" (waiting for the presentation),
"demonstrating" (in presentation), "demonstrateFail" (presentation failed), "demonstrateStop" (presentation stopped)*/
            "estimatedTotalTime":  10
            /*ro, opt, int, estimated total time consumption, range:[0,2147483647], step:1, unit:min*/
        }
    ]
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
taskStatus	[Message content]	application/json	--	--	--
picID-algorithmID	[Binary picture data]	image/jpeg	FaceMattingImage	FaceMattingImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.5.5 人脸比对

15.5.5.1 Set the arming schedule of multiple stranger detection alarms

Request URL

PUT /ISAPI/Event/schedules/whiteListFaceContrast

Query Parameter

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None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteListFaceContrastScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--req, object-->
    <id>
      <!--req, string-->whiteListFaceContrast-1
    </id>
    <eventType>
      <!--opt, enum, subType:string-->whiteListFaceContrast
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</WhiteListFaceContrastScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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15.5.5.2 Get the arming schedule of allowlist face picture comparison for all channels

Request URL

GET /ISAPI/Event/schedules/whiteListFaceContrast

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteListFaceContrastScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, req, object-->
    <id>
      <!--ro, req, string-->whiteListFaceContrast-1
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->whiteListFaceContrast
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</WhiteListFaceContrastScheduleList>
```

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15.5.5.3 Set the arming schedule of allowlist face picture comparison for a specified channel

Request URL

PUT /ISAPI/Event/schedules/whiteListFaceContrast/whiteListFaceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->whiteListFaceContrast-1
  </id>
  <eventType>
    <!--opt, string-->whiteListFaceContrast
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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15.5.5.4 Get the arming schedule of allowlist face picture comparison for a specified channel

Request URL

GET /ISAPI/Event/schedules/whiteListFaceContrast/whiteListFaceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->whiteListFaceContrast-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->whiteListFaceContrast
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

15.5.5.5 Set the linkage parameters of allowlist face picture comparison

Request URL

PUT /ISAPI/Event/triggers/whiteListFaceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->whiteListFaceContrast-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->whiteListFaceContrast
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

15.5.5.6 Search face picture analysis

Request URL

POST /ISAPI/Intelligent/analysisImage/face?dataType=<dataType>&modeling=<modeling>

Query Parameter

Parameter Name	Parameter Type	Description
dataType	string	--
modeling	enum	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceContrastData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <URL>
    <!--req, string, picture URL, desc:image/jpeg-->test
  </URL>
</FaceContrastData>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FaceContrastTargetsList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FaceContrastTarget>
    <!--ro, opt, object-->
    <Rect>
      <!--ro, opt, object, the origin is the upper-left corner of the screen, desc:the origin is the upper-left corner of the screen-->
      <height>
        <!--ro, req, float, height, range:[0.000,1.000]-->0.000
      </height>
      <width>
        <!--ro, req, float, width, range:[0.000,1.000]-->0.000
      </width>
      <x>
        <!--ro, req, float, x-coordinate, range:[0.000,1.000]-->0.000
      </x>
      <y>
        <!--ro, req, float, y-coordinate, range:[0.000,1.000]-->0.000
      </y>
    </Rect>
    <modeData>
      <!--ro, opt, string, target modeling data, desc:during transmission process, the binary non-modeling data will be encrypted by Base64 method. This
node is invalid when the URI contains the parameter modeling=false-->test
    </modeData>
    <recommendFaceRect>
      <!--ro, opt, object, recommended face frame coordinates, desc:recommended face frame coordinates-->
      <height>
        <!--ro, req, float, height, range:[0.000,1.000], desc:height-->0.000
      </height>
      <width>
        <!--ro, req, float, width, range:[0.000,1.000], desc:width-->0.000
      </width>
      <x>
        <!--ro, req, float, x-coordinate, range:[0.000,1.000], desc:x-coordinate-->0.000
      </x>
      <y>
        <!--ro, req, float, Y-coordinate, range:[0.000,1.000], desc:Y-coordinate-->0.000
      </y>
    </recommendFaceRect>
  </FaceContrastTarget>
</FaceContrastTargetsList>
```

15.5.5.7 Set the face picture comparison parameters

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceContrast

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceContrastList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FaceContrast>
    <!--opt, object-->
    <id>
      <!--req, int-->1
    </id>
    <enable>
      <!--opt, bool-->true
    </enable>
    <faceContrastType>
      <!--req, enum, face picture comparison type, subType:string-->faceContrast
    </faceContrastType>
    <thresholdValue>
      <!--opt, int, detection threshold, range:[0,100]-->1
    </thresholdValue>
    <FDLibList>
      <!--opt, array, subType:object-->
      <FDLib>
        <!--opt, object, face picture library-->
        <id>
          <!--req, int-->1
        </id>
        <FDID>
          <!--req, string, a unique ID of face picture library-->test
        </FDID>
        <thresholdValue>
```

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```
      <!--opt, int, range:[0,100]-->1
    </thresholdValue>
    <customFaceLibID>
      <!--opt, string, custom face picture library ID-->test
    </customFaceLibID>
  </FDLib>
</FDLibList>
<faceSnapDataUpload>
  <!--opt, bool, upload captured face image, desc:true by default-->true
</faceSnapDataUpload>
<AttendanceSaveEnable>
  <!--opt, bool, desc:true by default-->true
</AttendanceSaveEnable>
<contrastFailureAlarmUpload>
  <!--opt, bool, desc:true by default-->true
</contrastFailureAlarmUpload>
<alarmOutType>
  <!--opt, enum, subType:string-->off
</alarmOutType>
<plusesTime>
  <!--opt, int, unit:ms-->1
</plusesTime>
<contrastSuccessAlarmUpload>
  <!--opt, bool-->true
</contrastSuccessAlarmUpload>
<LivenessDetection>
  <!--opt, object-->
  <enable>
    <!--req, bool-->true
  </enable>
</LivenessDetection>
<voiceEnabled>
  <!--opt, bool-->true
</voiceEnabled>
<faceSnapDataSave>
  <!--opt, bool-->true
</faceSnapDataSave>
<QuickContrast>
  <!--opt, object-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <snapTime>
    <!--req, float, unit:s-->0.000
  </snapTime>
  <threshold>
    <!--opt, int-->1
  </threshold>
  <quickConfigMode>
    <!--opt, enum, subType:string-->custom
  </quickConfigMode>
  <Custom>
    <!--opt, object-->
    <timeOutMode>
      <!--opt, enum, subType:string-->infinite
    </timeOutMode>
    <timeOut>
      <!--opt, int, range:[1,99], unit:s-->1
    </timeOut>
    <duplicateContrastMode>
      <!--opt, enum, subType:string-->success
    </duplicateContrastMode>
    <duplicateContrastTimes>
      <!--opt, int, range:[1,99]-->1
    </duplicateContrastTimes>
  </Custom>
</QuickContrast>
<mixedTargetDetectionWithFaceContrast>
  <!--opt, bool-->true
</mixedTargetDetectionWithFaceContrast>
<brokenNetHttp>
  <!--opt, bool-->true
</brokenNetHttp>
<relationId>
  <!--opt, string, arming ID-->test
</relationId>
<alarmStorageEnable>
  <!--opt, bool-->true
</alarmStorageEnable>
<realTimeContrast>
  <!--opt, bool-->true
</realTimeContrast>
<analysisTime>
  <!--opt, datetime-->1970-01-01T00:00:00+08:00
</analysisTime>
<contrastSuccessAlarmout>
  <!--opt, bool, desc:true by default-->true
</contrastSuccessAlarmout>
<contrastFailureAlarmout>
  <!--opt, bool, desc:true by default-->true
</contrastFailureAlarmout>
<removeDuplicateEnabled>
  <!--opt, bool-->true
```

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```
    </removeDuplicateEnabled>
    <removeDuplicateTime>
      <!--opt, float, unit:s-->0.000
    </removeDuplicateTime>
    <repetitionTime>
      <!--opt, int-->1
    </repetitionTime>
    <topOverlayNameEnabled>
      <!--opt, bool-->true
    </topOverlayNameEnabled>
    <personListShowEnabled>
      <!--opt, bool-->true
    </personListShowEnabled>
    <refreshListTime>
      <!--opt, int, unit:s-->1
    </refreshListTime>
    <personCountShowEnabled>
      <!--opt, bool-->true
    </personCountShowEnabled>
    <FaceROI>
      <!--opt, object-->
      <enabled>
        <!--opt, bool-->true
      </enabled>
      <level>
        <!--opt, int-->1
      </level>
    </FaceROI>
    <uploadResultCount>
      <!--opt, int-->1
    </uploadResultCount>
    <normalTemperatureAlarmout>
      <!--opt, bool-->true
    </normalTemperatureAlarmout>
    <abnormalTemperatureAlarmout>
      <!--opt, bool-->true
    </abnormalTemperatureAlarmout>
    <matchType>
      <!--opt, enum, subType:int-->0
    </matchType>
    <alarmSourceType>
      <!--opt, enum, subType:string-->timing
    </alarmSourceType>
    <timingInterval>
      <!--opt, enum, subType:int, dep:and,{$.FaceContrastList[*].FaceContrast.alarmSourceType,eq,timing}-->5
    </timingInterval>
  </FaceContrast>
</FaceContrastList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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15.5.5.8 Start packaging and exporting the searched face picture information

Request URL

POST /ISAPI/Intelligent/FDLib/FCSearch/dataPackage

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two
searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of search results this time-->1
  </maxResults>
  <snapStartTime>
    <!--opt, datetime, capture start time with ISO8601 time format-->1970-01-01T00:00:00+08:00
  </snapStartTime>
  <snapEndTime>
    <!--opt, datetime, capture end time with ISO8601 time format-->1970-01-01T00:00:00+08:00
  </snapEndTime>
  <faceMatchInfoEnable>
    <!--opt, bool, whether to only upload data with matched information-->true
  </faceMatchInfoEnable>
  <monitorName>
    <!--opt, string, camera name-->test
  </monitorName>
  <similarity>
    <!--opt, float, similarity, range:[0.0,100.0]-->0.0
  </similarity>
  <eventType>
    <!--opt, enum, event type, subType:string, desc:event type-->unlimit
  </eventType>
  <FaceModeList>
    <!--opt, array, face picture information list, subType:object-->
    <FaceMode>
      <!--opt, object, face picture information-->
      <ModeInfo>
        <!--opt, object, model information-->
        <similarity>
          <!--opt, float, similarity, range:[0.0,100.0]-->0.0
        </similarity>
        <modeData>
          <!--opt, string, target model data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method--
>test
        </modeData>
      </ModeInfo>
    </FaceMode>
  </FaceModeList>
  <PictureInfoList>
    <!--opt, array, list of the picture information, subType:object, desc:list of the picture information-->
    <PictureInfo>
      <!--opt, object, picture information-->
      <FDID>
        <!--req, string, a unique ID of face picture library-->test
      </FDID>
      <PID>
        <!--req, string, picture data ID-->test
      </PID>
    </PictureInfo>
  </PictureInfoList>
  <ChannelList>
    <!--opt, array, searched channel list, subType:object-->
    <Channel>
      <!--opt, object, searched channels-->
      <channelID>
        <!--req, int, channel No.-->1
      </channelID>
    </Channel>
  </ChannelList>
  <totalSearchResults>
    <!--opt, int, the total number of results in a single search, desc:the total number of results in a single search-->1
  </totalSearchResults>
  <exportPathType>
    <!--opt, enum, subType:string-->hyperlink
  </exportPathType>
</FCSearchDescription>
```

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Response Message

Binary Data

15.5.5.9 Search for the face picture comparison alarm information

Request URL

POST /ISAPI/Intelligent/FDLib/FCSearch

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same
one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of search results this time, desc:the maximum number of search results this time-->1
  </maxResults>
  <snapStartTime>
    <!--opt, datetime, capture start time with ISO8601 time format-->1970-01-01T00:00:00+08:00
  </snapStartTime>
  <snapEndTime>
    <!--opt, datetime, capture end time with ISO8601 time format-->1970-01-01T00:00:00+08:00
  </snapEndTime>
  <faceMatchInfoEnable>
    <!--opt, bool, whether to only upload data with matched information-->true
  </faceMatchInfoEnable>
  <monitorName>
    <!--opt, string, camera name-->test
  </monitorName>
  <similarity>
    <!--opt, float, similarity, range:[0.0,100.0]-->0.0
  </similarity>
  <eventType>
    <!--opt, enum, event type, subType:string, desc:event type-->unlimit
  </eventType>
  <FaceModeList>
    <!--opt, array, face picture information list, subType:object-->
    <FaceMode>
      <!--opt, object, face picture information-->
      <ModeInfo>
        <!--opt, object, model information-->
        <similarity>
          <!--opt, float, similarity, range:[0.0,100.0]-->0.0
        </similarity>
        <modeData>
          <!--opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method-
->test
        </modeData>
      </ModeInfo>
    </FaceMode>
  </FaceModeList>
  <PictureInfoList>
    <!--opt, array, list of the picture information, subType:object, desc:list of the picture information-->
    <PictureInfo>
      <!--opt, object, picture information-->
      <FDID>
        <!--req, string-->test
      </FDID>
      <PID>
        <!--req, string-->test
      </PID>
    </PictureInfo>
  </PictureInfoList>
  <ChannelList>
    <!--opt, array, search channel list, subType:object-->
    <Channel>
      <!--opt, object-->
      <channelID>
        <!--req, int, channel No.-->1
      </channelID>
    </Channel>
  </ChannelList>
  <totalSearchResults>
    <!--opt, int, the total number of results in a single search, desc:the total number of results in a single search-->1
  </totalSearchResults>
  <FDIDList>
    <!--opt, array, face picture list, subType:object-->
    <FDID>
      <!--opt, string, face picture library ID-->test
    </FDID>
  </FDIDList>
  <sortord>
    <!--opt, enum, subType:string-->similarity
  </sortord>
  <sex>
    <!--opt, enum, gender, subType:string, desc:gender-->male
  </sex>
  <glass>
```

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```
    <glass>
      <!--opt, enum, whether the person wears glasses, subType:string, desc:whether the person wears glasses-->yes
    </glass>
    <smile>
      <!--opt, enum, whether the person wears a smile, subType:string, desc:whether the person wears a smile-->yes
    </smile>
    <high_risk_group>
      <!--opt, enum, subType:string-->yes
    </high_risk_group>
    <importLib>
      <!--opt, enum, subType:string-->yes
    </importLib>
    <CustomFaceLibIDList>
      <!--opt, array, custom list of face picture library ID, subType:object-->
      <customFaceLibID>
        <!--opt, string, custom face picture library ID-->test
      </customFaceLibID>
    </CustomFaceLibIDList>
    <thermometryUnit>
      <!--opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
    </thermometryUnit>
    <currTemperature>
      <!--opt, float, face temperature, range:[-20.0,105.0], unit:℃, desc:face temperature-->36.7
    </currTemperature>
    <temperatureCondition>
      <!--opt, enum, face temperature status, subType:string, desc:face temperature status-->normal
    </temperatureCondition>
    <temperatureValue>
      <!--opt, float, temperature value, range:[-20,105], unit:℃, dep:and,{$.FCSearchDescription.temperatureCondition,eq,above}-->37.2
    </temperatureValue>
    <humanIDList>
      <!--opt, array, subType:object-->
      <humanID>
        <!--req, string-->test
      </humanID>
    </humanIDList>
    <collectEnabled>
      <!--opt, enum, subType:string-->yes
    </collectEnabled>
  </FCSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search results, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <responseStatus>
    <!--ro, req, enum, search result, subType:string, desc:search result-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"MORE" (searching for more data), "NO MATCHES" (no matched data),"OK" (searching completed)-->MORE
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <progress>
    <!--ro, opt, int, range:[0,100]-->1
  </progress>
  <MatchList>
    <!--ro, opt, array, matched results list, subType:object-->
    <MatchElement>
      <!--ro, opt, object, matched results-->
      <snapPicURL>
        <!--ro, opt, string, URL of captured background picture-->test
      </snapPicURL>
      <snapTime>
        <!--ro, opt, datetime, capture time-->1970-01-01T00:00:00+08:00
      </snapTime>
      <facePicURL>
        <!--ro, opt, string, face sub-picture URL-->test
      </facePicURL>
      <FaceMatchInfoList>
        <!--ro, opt, array, subType:object-->
        <FaceMatchInfo>
          <!--ro, opt, object-->
          <FDID>
            <!--ro, req, string-->test
          </FDID>
          <FDname>
            <!--ro, opt, string, face picture library name-->test
          </FDname>
          <thresholdValue>
```

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```
            <!--ro, opt, int, detection threshold, range:[0,100], desc:detection threshold-->1
          </thresholdValue>
          <bornTime>
            <!--ro, opt, date, date of birth-->1970-01-01+08:00
          </bornTime>
          <name>
            <!--ro, opt, string, name-->test
          </name>
          <sex>
            <!--ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
          </sex>
          <province>
            <!--ro, req, string-->0
          </province>
          <city>
            <!--ro, opt, string, city, desc:see the data dictionary for details-->11
          </city>
          <certificateType>
            <!--ro, req, enum, certificate type, subType:string, desc:certificate type-->ID
          </certificateType>
          <certificateNumber>
            <!--ro, opt, string, certificate No., range:[8,32], desc:certificate No.-->test
          </certificateNumber>
          <picURL>
            <!--ro, req, string-->test
          </picURL>
          <PID>
            <!--ro, req, string, picture ID-->test
          </PID>
          <PersonInfoExtendList>
            <!--ro, opt, array, person extended information list, subType:object-->
            <PersonInfoExtend>
              <!--ro, opt, object, person extended information-->
              <id>
                <!--ro, req, int, serial No. of person custom extension information-->1
              </id>
              <enable>
                <!--ro, req, bool, enable person extended information or not-->true
              </enable>
              <name>
                <!--ro, opt, string, person tag extended information name-->test
              </name>
              <value>
                <!--ro, opt, string, person tag extended information-->test
              </value>
            </PersonInfoExtend>
          </PersonInfoExtendList>
          <phoneNumber>
            <!--ro, opt, string, phone No.-->test
          </phoneNumber>
          <caseInfo>
            <!--ro, opt, string, case description-->test
          </caseInfo>
          <similarity>
            <!--ro, opt, float, similarity, range:[0.0,100.0]-->0.0
          </similarity>
          <modelStatus>
            <!--ro, opt, enum, subType:string-->modeling
          </modelStatus>
          <ModelingStatus>
            <!--ro, opt, object-->
            <FDID>
              <!--ro, opt, string, face picture library ID-->test
            </FDID>
            <PID>
              <!--ro, opt, string, picture ID-->test
            </PID>
            <name>
              <!--ro, req, string, picture name-->test
            </name>
            <status>
              <!--ro, req, enum, subType:string-->abnormal
            </status>
            <reason>
              <!--ro, req, enum, subType:string-->analysisFailed
            </reason>
            <customHumanID>
              <!--ro, opt, string-->123456789012345678901234567890l2
            </customHumanID>
            <customFaceLibID>
              <!--ro, opt, string, custom face picture library ID, desc:custom face picture library ID-->123456789012345678901234567890l2
            </customFaceLibID>
            <modeData>
              <!--ro, opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64
method-->test
            </modeData>
          </ModelingStatus>
        </FaceMatchInfo>
      </FaceMatchInfoList>
      <monitorName>
        <!--ro, opt, string, camera information-->test
      </monitorName>
      <glass>
        <!--ro, req, enum, whether the person wears glasses, subType:string, desc:"any" (unlimited condition which is used in search), "no" (not wear
```

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```
      <!--ro, req, enum, whether the person wears glasses, subType:string, desc: any" (unlimited condition which is used in search), "no" (not wear glasses), "sunglasses" (wears sunglasses), "unknown"-->any
    </glass>
    <smile>
      <!--ro, req, enum, whether the person wears a smile, subType:string, desc:whether the person wears a smile-->no
    </smile>
    <high_risk_group>
      <!--ro, opt, enum, subType:string-->yes
    </high_risk_group>
    <similarity>
      <!--ro, opt, float, similarity, range:[0.0,100.0], desc:similarity-->0.0
    </similarity>
    <Rect>
      <!--ro, opt, object-->
      <height>
        <!--ro, req, float, width, range:[0.000,1.000]-->0.000
      </height>
      <width>
        <!--ro, req, float, height, range:[0.000,1.000]-->0.000
      </width>
      <x>
        <!--ro, req, float, X-coordinate, range:[0.000,1.000], desc:the reference origin is the upper left corner of image-->0.000
      </x>
      <y>
        <!--ro, req, float, Y-coordinate, range:[0.000,1.000], desc:the reference origin is the upper left corner of image-->0.000
      </y>
    </Rect>
    <channelID>
      <!--ro, opt, int, channel No., desc:channel No.-->1
    </channelID>
    <name>
      <!--ro, opt, string-->test
    </name>
    <sex>
      <!--ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
    </sex>
    <ageGroup>
      <!--ro, req, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime", "teenager", "unknown", "young"-->any
    </ageGroup>
    <mask>
      <!--ro, req, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"-->no
    </mask>
    <imageQuality>
      <!--ro, opt, int, picture quality-->1
    </imageQuality>
    <faceExpression>
      <!--ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-faced", "sad", "surprised", "unknown"-->angry
    </faceExpression>
    <contrastStatus>
      <!--ro, opt, bool, comparison result-->true
    </contrastStatus>
    <age>
      <!--ro, opt, int, age, range:[1,255]-->1
    </age>
    <targetID>
      <!--ro, opt, string, target ID, range:[1,48]-->test
    </targetID>
    <thermometryUnit>
      <!--ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
    </thermometryUnit>
    <currTemperature>
      <!--ro, opt, float, face temperature, range:[-20.0,105.0], unit:℃, desc:face temperature-->36.7
    </currTemperature>
    <temperatureCondition>
      <!--ro, opt, enum, face temperature status, subType:string, desc:face temperature status-->normal
    </temperatureCondition>
    <temperatureValue>
      <!--ro, opt, float, temperature value, range:[-20,105], unit:℃, dep:and,{$.FCSearchResult.MatchList[*].MatchElement.temperatureCondition,eq,above}-->37.2
    </temperatureValue>
    <MEFData>
      <!--ro, opt, array, subType:object-->
      <MEF>
        <!--ro, opt, object, electronic fence data-->
        <time>
          <!--ro, req, datetime-->2004-05-03T17:30:08+08:00.000
        </time>
        <IMSI>
          <!--ro, req, string, range:[1,15]-->test
        </IMSI>
      </MEF>
    </MEFData>
    <alarmTemperature>
      <!--ro, opt, float, alarm temperature threshold, range:[-20.0,60.0], unit:℃-->37.6
    </alarmTemperature>
    <collectEnabled>
      <!--ro, opt, bool-->false
    </collectEnabled>
  </MatchElement>
 </MatchList>
</FCSearchResult>
```

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15.5.5.10 Get the prompt when the stranger face picture comparison succeeded

Request URL

GET /ISAPI/Intelligent/FDLib/whiteListFaceContrastPrompt/channels/<channelID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "prompt":  "test"
    /*ro, req, string, range:[0,16]*/
}
```

15.5.5.11 Set the prompt when the stranger face picture comparison succeeded

Request URL

PUT /ISAPI/Intelligent/FDLib/whiteListFaceContrastPrompt/channels/<channelID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "prompt":  "test"
    /*req, string, range:[0,16]*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, req, int*/
    "errorMsg":  "ok"
    /*ro, req, string*/
}
```

15.5.5.12 Face picture comparison event

EventType:alarmResult

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm, range:[1,32]*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm, range:[1,128]*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm, range:[1,65535]*/
    "protocol":  "HTTP",
    /*ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address, range:[1,32]*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm*/
```

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```
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or playback on the platform*/
    "dateTime": "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount": 1,
    /*ro, req, int, times that the same alarm has been uploaded*/
    "eventType": "alarmResult",
    /*ro, req, enum, event type, subType:string, desc:"alarmResult" (face picture comparison)*/
    "eventState": "active",
    /*ro, req, enum, continuous event status, subType:string, desc:active (valid event); inactive (invalid event)*/
    "eventDescription": "alarmResult",
    /*ro, req, string, event description, range:[1,128], desc:event description*/
    "channelName": "ABC",
    /*ro, req, string, channel name (camera name), range:[1,128]*/
    "deviceID": "test0123",
    /*ro, opt, string, device ID (PUID), range:[1,64], desc:this field must be returned when ISAPI event information is transmitted via ISUP*/
    "isDataRetransmission": true,
    /*ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers*/
    "alarmResult": [
    /*ro, opt, array, face picture comparison results, subType:object, desc:face picture comparison results*/
        {
            "errorCode": 1,
            /*ro, opt, int, when analyzing face picture failed, the corresponding error code will be returned*/
            "errorMsg": "ok",
            /*ro, opt, string, when analyzing face picture failed, the corresponding error information will be returned, desc:when analyzing face picture failed, the corresponding error information will be returned*/
            "image": " http://10.7.52.174:8080/kms/services/rest/dataInfoService/downloadFile?id=00000001/temp001/011_163957_23006&token=7a57a5a7ffffffffc1a0316369671314",
            /*ro, opt, string, picture URL*/
            "modelData": "test",
            /*ro, opt, string, modeling data of background picture, which is encoded by Base64, range:[1,2048]*/
            "traceUuid": "sddddddddff",
            /*ro, opt, string, target ID, desc:one person corresponds to one target ID in multiple alarms (related URI: /ISAPI/SDT/Management/Task/facePicture); the maximum size*/
            "traceIdx": 1,
            /*ro, opt, int, picture index, desc:one person can correspond to multiple pictures*/
            "targetAttrs": {
            /*ro, opt, object, target attributes*/
                "deviceId": "ee3c4000-a22d-11b4-8334-5803fb77894c",
                /*ro, opt, string, device UUID, range:[1,64]*/
                "deviceChannel": 1,
                /*ro, opt, int, device channel No.*/
                "deviceName": "shebei",
                /*ro, opt, string, device name*/
                "faceTime": "2016-09-10 00:00:00.000",
                /*ro, opt, string, capture time, range:[1,64]*/
                "rect": {
                /*ro, opt, object, background picture frame information of face, desc:the origin is the upper-left corner of the screen*/
                    "height": 1.000,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width": 1.000,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "x": 0.000,
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                    "y": 0.000
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "contentID": "backgroundImage",
                /*ro, opt, string, contentID of the picture data binary data, range:[1,128], desc:the bkgURL cannot be returned when this field is returned*/
                "bkgUrl": "http://absddsdfsdfsdk.jpg",
                /*ro, opt, string, background picture URL*/
                "pId": "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, captured background picture ID, range:[1,32], desc:captured background picture ID*/
                "ThermalImage":{
                /*ro, opt, object, thermal imaging picture information*/
                    "resourcesContentType": "binary",
                    /*ro, opt, string, resource transmission type, desc:"url" (transmitted via URL), "binary" (transmitted in binary format). When the returned alarm message contains additional resource (e.g., pictures), this field should be returned*/
                    "resourcesContent": "test"
                    /*ro, opt, string, resource ID, desc:when the returned alarm message contains additional resources (e.g., pictures), this field should be returned. When the value of resourcesContentType is "binary", it must be the same as Content-ID of the resource. When the value of resourcesContentType is "url", its value is the resource URL*/
                }
            },
            "faces": [
            /*ro, opt, array, face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on, subType:object, desc:face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on*/
                {
                    "faceId": 1,
                    /*ro, opt, int, face ID, desc:the face ID cannot be duplicated in the same picture*/
                    "faceRect": {
                    /*ro, opt, object, face thumbnail frame, desc:it is used for modeling. The origin is the upper-left corner of the screen*/
                        "height": 1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width": 1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x": 0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y": 0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
```

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```
                    },
                    "faceMark": {
                    /*ro, opt, object, face features in the picture*/
                        "leftEye": {
                        /*ro, opt, object, left eye coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "rightEye": {
                        /*ro, opt, object, right eye's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "noseTip": {
                        /*ro, opt, object, nose tip's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "leftMouth": {
                        /*ro, opt, object, left mouth corner's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "rightMouth": {
                        /*ro, opt, object, right mouth corner's position, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        }
                    },
                    "recommendFaceRect": {
                    /*ro, opt, object, recommended face frame coordinates, desc:recommended face frame coordinates*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "facePose": {
                    /*ro, opt, object, face angle*/
                        "pitch":  0.998,
                        /*ro, opt, float, face pitch angle, face pitching up is regarded as positive, range:[-90.000,90.000], desc:this field does not
exists if it is invalid. It is accurate to three decimal places*/
                        "yaw":  0.876,
                        /*ro, opt, float, face yaw angle, face yawing left is regarded as positive, range:[-90.000,90.000], desc:this field does not exists
if it is invalid. It is accurate to three decimal places*/
                        "roll":  0.623
                        /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive, range:[-90.000,90.000], desc:this field does not
exists if it is invalid. It is accurate to three decimal places*/
                    },
                    "age": {
                    /*ro, opt, object, age*/
                        "range":  5,
                        /*ro, opt, int, age deviation; if the age deviation already exists, the age is required*/
                        "value":  23,
                        /*ro, opt, int, age*/
                        "ageGroup":  "any"
                        /*ro, opt, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old",
"prime", "teenager", "unknown", "young"*/
                    },
                    "gender": {
                    /*ro, opt, object, gender*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "male"
                        /*ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
                    },
                    "glass": {
                    /*ro, opt, object, whether the person wears glasses*/
                        "confidence":  99.94499999999999,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no", "yes", "sunglasses"*/
                    },
                    "smile": {
                    /*ro, opt, object, whether the person is smiling*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person is smiling, subType:string, desc:"no", "yes"*/
                    },
                    "mask": {
```


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```
                    "mask": {
                    /*ro, opt, object, whether the person wears a mask*/
                        "confidence": 99.94499999999999,
                        /*ro, opt, float, confidence*/
                        "value": "yes"
                        /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"no", "yes"*/
                    },
                    "faceExpression": {
                    /*ro, opt, object, facial expression*/
                        "confidence": 99.99990,
                        /*ro, opt, float, confidence*/
                        "value": "angry"
                        /*ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-
faced", "sad", "surprised", "unknown"*/
                    },
                    "hat": {
                    /*ro, opt, object, whether the person wears a hat*/
                        "confidence": 99.99990,
                        /*ro, opt, float, confidence*/
                        "value": "unknown"
                        /*ro, opt, enum, whether the person wears a hat, subType:string, desc:"no", "yes", "unknown"*/
                    },
                    "score": {
                    /*ro, opt, object, face score (face picture quality)*/
                        "confidence": 99.99990,
                        /*ro, opt, float, confidence*/
                        "value": 65
                        /*ro, opt, int, score value, range:[1,100]*/
                    },
                    "modeldata": "test",
                    /*ro, opt, string, face model data (encoded by base64)*/
                    "AbsoluteHigh": {
                    /*ro, opt, object, PTZ position*/
                        "elevation": -900,
                        /*ro, opt, int, tilting angle, range:[-900,2700]*/
                        "azimuth": 22,
                        /*ro, opt, int, azimuth, range:[0,3600]*/
                        "absoluteZoom": 1
                        /*ro, opt, int, zooming ratio, range:[1,1000]*/
                    },
                    "identify": [
                    /*ro, opt, array, arming linkage information, subType:object*/
                        {
                            "relationId": "9887",
                            /*ro, opt, string, arming linkage ID, range:[0,64], desc:arming linkage ID*/
                            "maxsimilarity": 0.780,
                            /*ro, opt, float, the maximum similarity of face comparison results, range:[0.000,1.000], step:0.001*/
                            "similarityRange": 1,
                            /*ro, opt, enum, similarity range, subType:int, desc:0 (smaller than the minimum value), 1 (between the minimum value and
maximum value), 2 (larger than the maximum value). If the similarity is smaller than the minimum value, the person will be marked*/
                            "candidate": [
                            /*ro, opt, array, matched face picture information, subType:object*/
                                {
                                    "alarmId": 1,
                                    /*ro, opt, int, alarm ID, desc:it cannot be duplicated of one arming linkage*/
                                    "blacklist_id": "2323",
                                    /*ro, opt, string, face picture library ID, range:[1,64], desc:it corresponds to the FDID of face picture library. For
cameras, only the default FDID will be returned, the custom face picture library ID will be returned in customFaceLibID*/
                                    "human_data": [
                                    /*ro, opt, array, person information of matched face pictures, subType:object*/
                                        {
                                            "face_id": "123445",
                                            /*ro, opt, string, face ID, range:[1,32]*/
                                            "face_picurl": "http://127.0.0.1:6120/face_picurl",
                                            /*ro, opt, string, face picture URL*/
                                            "pId": "F4F665D6A18E41308CE9934DCDDD766F",
                                            /*ro, opt, string, face thumbnail ID, range:[1,64], desc:face thumbnail ID*/
                                            "face_rect": {
                                            /*ro, opt, object, face thumbnail coordinates, desc:the origin is the upper-left corner of the screen*/
                                                "height": 1.000,
                                                /*ro, opt, float, height, range:[0.000,1.000]*/
                                                "width": 1.000,
                                                /*ro, opt, float, width, range:[0.000,1.000]*/
                                                "x": 0.000,
                                                /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                                                "y": 0.000
                                                /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                                            },
                                            "similarity": 0.997000,
                                            /*ro, opt, float, face picture similarity*/
                                            "isNoSaveFDPicture": true,
                                            /*ro, opt, bool, whether to not save the picture of face picture library, desc:true (do not save); if this node
is not returned, it indicates that saving the picture*/
                                            "contentID": "faceLibImage",
                                            /*ro, opt, string, picture in the face picture library, desc:picture in the face picture library*/
                                            "bkg_picurl": "http://10639103:80/FDLib?
FDID=529FB9EAC2A04041A603E18F300CD3A5&pId=F4F665D6A18E41308CE9934DCDDD766F&size=22332"
                                            /*ro, opt, string, background picture URL*/
                                        }
                                    ],
                                    "human_id": "123",
                                    /*ro, opt, string, person ID, desc:person ID in the list, which corresponds to FPID in the face picture library*/
                                    "reserve_field": {
                                    /*ro, opt, object, reserved (transmitted data)*/
```


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```
                                    "name":  "张三",
                                    /*ro, req, string, name, range:[1,256]*/
                                    "gender":  "any",
                                    /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition which is used for search), "male",
"female", "unknown"*/
                                    "bornTime":  "2004-05-03",
                                    /*ro, req, string, date of birth*/
                                    "city":  "130100",
                                    /*ro, opt, string, city code of person birth, range:[1,32], desc:see the data dictionary for details*/
                                    "certificateType":  "officerID",
                                    /*ro, opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "passport"
(passport), "other" (the other certificate)*/
                                    "certificateNumber":  "test",
                                    /*ro, opt, string, certificate No., range:[0,32]*/
                                    "ageGroup":  "any",
                                    /*ro, opt, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle
age), "old", "prime", "teenager", "unknown", "young"*/
                                    "phoneNumber":  "test",
                                    /*ro, opt, string, phone number, range:[1,64]*/
                                    "cityName":  "石家庄市",
                                    /*ro, opt, string, city (corresponding to the field "city"), range:[0,32], desc:city (corresponding to the field
"city")*/
                                    "provinceName":  "河北省"
                                    /*ro, opt, string, province, range:[0,32], desc:province*/
                                },
                                "similarity":  0.780,
                                /*ro, opt, float, range:[0.000,1.000]*/
                                "listType":  "blackList",
                                /*ro, opt, enum, list type, subType:string, desc:"blacklist" (blocklist), "whitelist" (allowlist)*/
                                "extendData": [
                                /*ro, opt, array, person extended information, subType:object*/
                                    {
                                        "extendID":  1,
                                        /*ro, opt, int, person extended information ID, desc:it starts from 1*/
                                        "enable":  true,
                                        /*ro, opt, bool, enable person extended information or not*/
                                        "name":  "test",
                                        /*ro, opt, string, name of person tag extended information, range:[1,96]*/
                                        "value":  "test"
                                        /*ro, opt, string, person tag extended information, range:[0,64]*/
                                    }
                                ],
                                "FDLibName":  "faceLib1",
                                /*ro, opt, string, face picture library name, desc:face picture library name*/
                                "FDLibThreshold":  50,
                                /*ro, opt, int, detection threshold of face picture library, range:[0,100]*/
                                "customFaceLibID":  "test",
                                /*ro, opt, string, custom face picture library ID*/
                                "customHumanID":  "test"
                                /*ro, opt, string, custom person ID*/
                            }
                        ],
                        "alarmRecordID":  "test",
                        /*ro, opt, string, alarm record ID*/
                        "libAttribute":  "general"
                        /*ro, opt, enum, library type, subType:string, desc:"general" (normal library), "blackList" (blocklist library), "VIP" (VIP
library), "passerby" (allowlist library)*/
                    }
                ],
                "contentID":  "faceImage",
                /*ro, opt, string, background picture ID, desc:background picture ID*/
                "URL":  "test",
                /*ro, opt, string, face picture thumbnail URL, range:[1,256], desc:face picture thumbnail URL*/
                "pId":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, picture ID in face picture library, range:[1,32]*/
                "stayDuration":  20,
                /*ro, opt, int, dwell time duration in the image, unit:ms*/
                "prompt":  "test",
                /*ro, opt, string, prompt contents of face picture comparison, range:[1,16]*/
                "contrastStatus":  true,
                /*ro, opt, bool, comparison result*/
                "realTimeContrast":  true,
                /*ro, opt, bool, whether to enable real-time face picture comparison*/
                "remainReportNum":  0,
                /*ro, opt, int, the remaining times of uploading the comparison alarm of the current target, desc:the remaining times of uploading the
comparison alarm of the current target*/
                "UID":  "test",
                /*ro, opt, string*/
                "UUPID":  "test",
                /*ro, opt, string, background picture ID of face picture library, range:[1,64], desc:it consists of device serial No., imported time,
and random digits*/
                "frontMatchType":  1,
                /*ro, opt, enum, face picture library type, subType:int, desc:1 (cluster library), 0 (normal library)*/
                "isImportFD":  true,
                /*ro, opt, bool, whether to import the captured face data to the library, desc:true (yes), false (no). When face picture comparison
failed, the face picture quality meets the requirements, and is imported to the library, true will be returned. When face picture comparison failed, and the
face picture quality does not meet the requirements, false will be returned*/
                "faceSnapThermometryEnabled":  true,
                /*ro, opt, bool, whether to enable temperature screening when capturing face pictures*/
                "currTemperature":  36.7,
                /*ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃, desc:it should be used with thermometryUnit together*/
                "isAbnomalTemperature":  true,
                /*ro, opt, bool, whether the temperature is abnormal, desc:true (yes), false (no)*/
                "thermometryUnit":  "celsius"
```


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```
                    "thermometryUnit": "celsius",
                    /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
                    "alarmTemperature": 36.7,
                    /*ro, opt, float, range:[-20.0,150.0], unit:℃*/
                    "facePicQuality": "low"
                    /*ro, opt, enum, subType:string*/
                }
            ],
            "alarmRecordID": "test",
            /*ro, opt, string, alarm record ID*/
            "uid": "test",
            /*ro, opt, string, the unique ID of event uploading, range:[1,64], desc:it can be composed of time (accurate to the millisecond) and random
digits*/
            "remainReportNum": 0,
            /*ro, opt, int, the remaining times of uploading the comparison alarm of the current target*/
            "algorithmInfo": {
            /*ro, opt, object, algorithm version information, desc:this field is required for secondary analysis*/
                "algorithmId": "test",
                /*ro, opt, string, algorithm package ID*/
                "algoVendorId": "test",
                /*ro, opt, string, manufacturer ID*/
                "algoVersionId": "test",
                /*ro, opt, string, algorithm version No.*/
                "algoChipType": "test",
                /*ro, opt, string, chip type*/
                "algoModelVersion": "test"
                /*ro, opt, string, model version*/
            }
        }
    ],
    "URLCertificationType": "digest",
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no"-no authentication, "digest"-digest authentication (it is used for local
picture URL returned by NVR or DVR)*/
    "GPS": {
    /*ro, opt, object, device GPS information*/
        "divisionEW": "E",
        /*ro, req, enum, "E" (Eastern Hemisphere), W" (West Hemisphere), subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)*/
        "longitude": 100,
        /*ro, req, int, longitude, desc:longitude*/
        "divisionNS": "N",
        /*ro, req, enum, subType:string*/
        "latitude": 100
        /*ro, req, int, latitude, desc:latitude*/
    }
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
alarmResult	[Message content]	application/json	--	--	--
..value	[Binary picture data]	image/jpeg	backgroundImage	backgroundImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
faceLibImage	[Binary picture data]	image/jpeg	faceLibImage	faceLibImage.jpg	--
...value	[Binary picture data]	image/jpeg			--
..value	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit

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name.

- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.5.6 人脸库管理

15.5.6.1 Get the parameters of person custom information

Request URL

GET /ISAPI/Intelligent/faceContrast/personInfoExtend

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PersonInfoExtendList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PersonInfoExtend>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->1
    </id>
    <enable>
      <!--ro, req, bool-->true
    </enable>
    <name>
      <!--ro, opt, string-->test
    </name>
    <value>
      <!--ro, opt, string-->test
    </value>
  </PersonInfoExtend>
</PersonInfoExtendList>
```

15.5.6.2 Set the parameters of person custom information

Request URL

PUT /ISAPI/Intelligent/faceContrast/personInfoExtend

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PersonInfoExtendList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PersonInfoExtend>
    <!--opt, object-->
    <id>
      <!--req, int-->1
    </id>
    <enable>
      <!--req, bool-->true
    </enable>
    <name>
      <!--opt, string-->test
    </name>
    <value>
      <!--opt, string-->test
    </value>
  </PersonInfoExtend>
</PersonInfoExtendList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

15.5.6.3 Get the capability of person custom information

Request URL

GET /ISAPI/Intelligent/faceContrast/personInfoExtend/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PersonInfoExtendList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PersonInfoExtend>
    <!--ro, opt, object-->
    <id min="1" max="10">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </id>
    <enable>
      <!--ro, req, bool-->true
    </enable>
    <name min="1" max="10">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </name>
    <value min="1" max="10">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </value>
  </PersonInfoExtend>
</PersonInfoExtendList>
```

15.5.6.4 Delete all face picture libraries

Request URL

DELETE /ISAPI/Intelligent/FDLib

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

15.5.6.5 Get the progress of packaging alarm information

Request URL

GET /ISAPI/Intelligent/FDLib/FCSearch/dataPackage/progress?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	Task ID, and the length range is 1 to 64 characters.

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL, range:[1,256]*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "progress":  22,
    /*ro, opt, int, packaging progress, range:[0,100]*/
    "URL":  "test"
    /*ro, opt, string, storage URL generated by device after packaging, range:[1,128], desc:you can download the data files via the URL*/
}
```

15.5.6.6 Start packaging and exporting the searched information about face picture comparison alarm

Request URL

POST /ISAPI/Intelligent/FDLib/FCSearch/dataPackage?format=json

Query Parameter

None

Request Message

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```
{
    "searchID":  "test",
    /*opt, string, search ID, desc:it is unchanged when the search condition remains the same*/
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, range:[0,4294967295], desc:in a single search, if you cannot get all the records
in the result list, you can mark the end position and get the following records after the marked position in the next search*/
    "maxResultNumber":  100,
    /*req, int, the maximum records to be found, range:[1,4294967295]*/
    "maxResults":  1,
    /*req, int, the maximum results per search, range:[1,100]*/
    "FDID":  "aa,bb",
    /*req, string, face picture library ID, range:[0,63], desc:"-1" (all libraries). If there are multiple libraries, you can separate them by commas*/
    "cameraID":  "aa,bb",
    /*opt, string, camera ID (channel ID), range:[0,4095], desc:"-1" (all cameras). If there are multiple cameras, you can separate each one by commas*/
    "snapStartTime":  "1970-01-01T00:00:00Z",
    /*opt, datetime, capture start time*/
    "snapEndTime":  "1970-01-01T01:00:00Z",
    /*opt, datetime, capture end time*/
    "maxSimilarity":  0.99,
    /*opt, float, the maximum similarity, range:[0.00,1.00]*/
    "minSimilarity":  0.99,
    /*opt, float, the minimum similarity, range:[0.00,1.00]*/
    "sortType":  "similarity",
    /*opt, enum, sorting type, subType:string, desc:"similarity" (by similarity), "time" (by time), "attendanceNumber" (by check-in counts)*/
    "Attendance": {
    /*opt, object, attendance information*/
        "Schedules": [
        /*opt, array, attendance time, subType:object*/
            {
                "startTime":  "00:00:00+08:00",
                /*opt, time, start time*/
                "endTime":  "00:00:00+08:00",
                /*opt, time, end time*/
                "dayOfWeek": [1, 2]
                /*opt, array, days of the week, subType:int, dep:and,{$.Attendance.type,eq,attendance}, desc:from Monday to Sunday, multiple selections are
supported*/
            }
        ],
        "MatchFace": [
        /*opt, array, matched face information, subType:object*/
            {
                "FDID":  "test",
                /*opt, string, face picture library ID, range:[0,63]*/
                "PID":  "test"
                /*opt, string, picture ID, range:[0,64]*/
            }
        ],
        "type":  "check-in"
        /*opt, enum, mode switch, subType:string, desc:"attendance", "check-in"*/
    },
    "listType":  "blackList"
    /*opt, enum, list type, subType:string, desc:"blacklist" (blocklist), "whitelist" (allowlist)*/
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL, range:[1,256]*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "test"
    /*ro, opt, string, packaging task ID, range:[0,32], desc:it can be used for packaging progress search*/
}
```

15.5.6.7 Search for the face picture comparison alarm information

Request URL

POST /ISAPI/Intelligent/FDLib/FCSearch?format=json

Query Parameter

None

Request Message

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```
{
    "searchID":  "test",
    /*req, string, search ID*/
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list*/
    "maxResultNumber":  100,
    /*req, int, the maximum records to be found*/
    "maxResults":  100,
    /*req, int, the maximum search results,and up to 100 results are allowed*/
    "FDID":  "aa,bb",
    /*req, string, face picture library ID, desc:if there are multiple libraries,you can separate each one by comma*/
    "cameraID":  "aa,bb",
    /*opt, string, camera ID, range:[1,4096], desc:up to 4095 bytes are allowed. -1 (all libraries). If there are multiple cameras,you can separate each one
by comma, e.g.,"aa,bb"*/
    "snapStartTime":  "2004-05-03T17:30:08+08:00",
    /*opt, string, capture start time, range:[1,32]*/
    "snapEndTime":  "2004-05-03T17:30:08+08:00",
    /*opt, string, capture end time, range:[1,32]*/
    "maxSimilarity":  0.99,
    /*opt, float, the maximum similarity, range:[0,1], desc:its value ranges from 0 to 1 and it is accurate to two decimal places*/
    "minSimilarity":  0.99,
    /*opt, float, the minimum similarity, range:[0,1], desc:its value ranges from 0 to 1 and it is accurate to two decimal places*/
    "sortType":  "similarity",
    /*opt, enum, sorting type, subType:string, desc:"similarity"-by similarity, "time"-by time*/
    "Attendance": {
    /*opt, object*/
        "Schedules": [
        /*opt, array, subType:object*/
            {
                "startTime":  "17:00:00",
                /*opt, time, start time*/
                "endTime":  "00:00:00+08:00",
                /*opt, time*/
                "dayOfWeek": [1, 2]
                /*opt, array, subType:int*/
            }
        ],
        "MatchFace": [
        /*opt, array, subType:object*/
            {
                "FDID":  "test",
                /*opt, string, face picture library ID, range:[1,64], desc:face picture library ID,and up to 63 bytes are allowed,"-1"-all libraries. If
there are multiple libraries,you can separate each one by comma. E.g.,"aa,bb"*/
                "PID":  "test"
                /*opt, string, picture ID, range:[1,64], desc:the maximum length is 64*/
            }
        ],
        "type":  "check-in"
        /*opt, enum, subType:string*/
    },
    "listType":  "blackList",
    /*opt, enum, list type, subType:string*/
    "alarmAckState":  "acknowledged",
    /*opt, enum, alarm acknowledgment status, subType:string*/
    "facialAttendanceStatus":  "success",
    /*opt, enum, subType:string*/
    "temperatureCondition":  "normal"
    /*opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "activeNumMax",
    /*ro, req, enum, sub status code, subType:string*/
    "errorCode":  1,
    /*ro, opt, int, error code*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this field is required when statusCode is not 1, and can be the error information of a specific parameter*/
    "responseStatusStrg":  "OK",
    /*ro, opt, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched
data), the maximum length is 32*/
    "numOfMatches":  100,
    /*ro, opt, int, number of results returned this time, desc:number of total returned results after searching,this node is valid only when errorCode is 1
and errorMsg is "ok"*/
    "totalMatches":  200,
    /*ro, opt, int, total number of matched results, desc:number of total matched results,this node is valid only when errorCode is 1 and errorMsg is "ok"*/
    "progress":  100,
    /*ro, opt, int*/
    "MatchList": [
    /*ro, opt, array, matched results details, subType:object*/
        {
            "bkg_picurl":  "http://10.7.52.174:8080/kms/services/rest/dataInfoService/downloadFile?
```

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```
id=00000001/temp001/011_163957_23006&token=7a57a5a7ffffffffc1a0316369671314",
            /*ro, opt, string, captured background picture URL, range:[1,128]*/
            "face_picurl":  "http://10.7.52.174:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/011_163957_23006&token=7a57a5a7ffffffffc1a0316369671314",
            /*ro, opt, string, captured face thumbnail URL, range:[1,128]*/
            "videoUrl":  "xxx",
            /*ro, opt, string*/
            "device_name":  "test123456789",
            /*ro, opt, string, camera name, range:[1,64]*/
            "monitorId":  "1",
            /*ro, opt, string, camera ID, range:[1,64]*/
            "alarm_date":  "1479529411000",
            /*ro, opt, string, alarm date, range:[1,64]*/
            "ageGroup":  "old",
            /*ro, opt, enum, age group, subType:string, desc:age groups: "unknown","child","young","middle","old",the maximum string length is 32 bytes*/
            "gender":  "unknown",
            /*ro, opt, enum, gender, subType:string, desc:"male"; "female"; "unknown";*/
            "glass":  "yes",
            /*ro, opt, enum, whether wears glasses, subType:string, desc:"no"; "yes"; "unknown"*/
            "alarm_max_similarity":  0.5,
            /*ro, req, float, maximum threshold of similarity, desc:the default value is 1.0,but the value cannot be smaller than 0*/
            "smile":  "yes",
            /*ro, opt, enum, whether the person wears a smile, subType:string*/
            "mask":  "yes",
            /*ro, opt, enum, whether the person wears a mask, subType:string*/
            "expression":  "unknown",
            /*ro, opt, enum, facial expression, subType:string*/
            "modeldata":  "test",
            /*ro, opt, string*/
            "videourl":  "xxx",
            /*ro, opt, string*/
            "AbsoluteHigh": {
            /*ro, opt, object, PTZ position*/
                "elevation":  11,
                /*ro, opt, int, range:[-900,2700]*/
                "azimuth":  22,
                /*ro, opt, int, panning angle, range:[0,3600]*/
                "absoluteZoom":  33
                /*ro, opt, int, zooming ratio, range:[1,1000]*/
            },
            "human_data": [
            /*ro, opt, array, person information, subType:object, desc:up to 5 persons are supported*/
                {
                    "human_name":  "123",
                    /*ro, opt, string, person name, range:[1,128]*/
                    "human_id":  "20170303D1488853914D14871",
                    /*ro, opt, string, person ID, range:[1,64], desc:person ID,the maximum ID length is 64 bytes,e.g.,20170303D1488853914D14871*/
                    "FDID":  "test",
                    /*ro, opt, string, face picture library ID*/
                    "gender":  "male",
                    /*ro, opt, enum, gender, subType:string, desc:the maximum length is 10; "male","female","unknown"*/
                    "bornTime":  "2004-05-03",
                    /*ro, opt, date, desc:the maximum length is 32*/
                    "city":  "130100",
                    /*ro, opt, string, range:[1,32], desc:see the data dictionary for details*/
                    "certificateType":  "officerID",
                    /*ro, opt, enum, certificate type, subType:string*/
                    "certificateNumber":  "test",
                    /*ro, opt, string, desc:the maximum length is 32*/
                    "ageGroup":  "old",
                    /*ro, opt, enum, age group, subType:string, desc:age groups: "unknown","child","young","middle","old",the maximum string length is 32
bytes*/
                    "phoneNumber":  "test",
                    /*ro, opt, string*/
                    "face_data": [
                    /*ro, opt, array, face pictures, subType:object*/
                        {
                            "face_similarity":  0.763567,
                            /*ro, req, float, similarity*/
                            "bl_picurl":  "http://10.7.52.31:8080/km"
                            /*ro, opt, string, face picture URL*/
                        }
                    ],
                    "Attendance": {
                    /*ro, opt, object*/
                        "totalNumbers":  0,
                        /*ro, opt, int*/
                        "DaliyList": [
                        /*ro, opt, array, subType:object*/
                            {
                                "id":  1,
                                /*ro, opt, int*/
                                "date":  "2004-05-03",
                                /*ro, opt, string*/
                                "numbers":  0
                                /*ro, opt, int*/
                            }
                        ],
                        "CheckIn": [
                        /*ro, opt, array, subType:object*/
                            {
                                "checkInDay":  7,
                                /*ro, opt, int*/
                                "nocheckInDay":  0
```

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```
                        "noCheckInDay": 0
                        /*ro, opt, int*/
                    }
                ],
                "AttendStatus": [
                /*ro, opt, array, subType:object*/
                    {
                        "normal": 7,
                        /*ro, opt, int*/
                        "late": 0,
                        /*ro, opt, int*/
                        "leaveEarly": 0,
                        /*ro, opt, int*/
                        "absenteeism": 0,
                        /*ro, opt, int*/
                        "withOutAttend": 0,
                        /*ro, opt, int*/
                        "abnormalRate": 1.00
                        /*ro, opt, float*/
                    }
                ]
            }
        }
    ],
    "alarmRecordID": "test",
    /*ro, opt, string, alarm record ID*/
    "alarmAckState": "acknowledged",
    /*ro, opt, enum, alarm acknowledgment status, subType:string, desc:"acknowledged","pending" (to be acknowledged. This value can be used when the user cannot acknowledge the alarm when it is triggered but wants to prioritize acknowledging it later),"notAcknowledged" (not acknowledgment);*/
    "alarmCustomInfo": "test",
    /*ro, opt, string, custom information, range:[1,64]*/
    "thermometryUnit": "celsius",
    /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
    "currTemperature": "38.0",
    /*ro, opt, string, face temperature*/
    "temperatureCondition": "normal",
    /*ro, opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
    "facialAttendanceStatus": "success",
    /*ro, opt, enum, subType:string*/
    "SimilarFaceList": [
    /*ro, opt, array, subType:object*/
        {
            "SimilarFace": [
            /*ro, opt, array, subType:object*/
                {
                    "human_name": "123",
                    /*ro, opt, string, person name, desc:the maximum length is 128*/
                    "facelibPictureURL": "test"
                    /*ro, opt, string*/
                }
            ]
        }
    ]
  }
 ]
}
```


15.5.6.8 Search face picture data

Request URL

POST /ISAPI/Intelligent/FDLib/FDSearch?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.

Request Message


```
<?xml version="1.0" encoding="UTF-8"?>

<FDSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <searchID>
```


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```
  <searchID>
    <!--req, string, search ID, desc:it is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device to speed up the next search-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, initial position of search result list, desc:if it cannot get all records in one time searching,you can search the records followed specified position for next search. For video intercom devices,this field can only be set to 0 as the picture will be returned along with the message-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, maximum number of records for single searching-->1
  </maxResults>
  <FDID>
    <!--opt, string, face picture library ID-->test
  </FDID>
  <startTime>
    <!--opt, date, start time-->1970-01-01+08:00
  </startTime>
  <endTime>
    <!--opt, date, end time-->1970-01-01+08:00
  </endTime>
  <name>
    <!--opt, string, name, range:[1,32], desc:name-->test
  </name>
  <sex>
    <!--opt, enum, gender, subType:string, desc:gender-->male
  </sex>
  <province>
    <!--opt, string-->11
  </province>
  <city>
    <!--opt, string, city, desc:city code of birth for the person in the face picture,the maximum size is 32 bytes-->11
  </city>
  <certificateType>
    <!--opt, enum, certificate type, subType:string, desc:the maximum size is 10 bytes,certificate type: "officerID"-officer ID,"ID"-identify card,passport,other-->officerID
  </certificateType>
  <certificateNumber>
    <!--opt, string, certificate No., range:[8,32], desc:certificate No.,the maximum size is 32 bytes-->test
  </certificateNumber>
  <phoneNumber>
    <!--opt, string, range:[1,64]-->test
  </phoneNumber>
  <FaceModeList>
    <!--opt, array, subType:object-->
    <FaceMode>
      <!--opt, object-->
      <ModeInfo>
        <!--opt, object-->
        <similarity>
          <!--opt, float, similarity, range:[0.0,100.0]-->0.0
        </similarity>
        <modeData>
          <!--opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method-->test
        </modeData>
      </ModeInfo>
    </FaceMode>
  </FaceModeList>
  <modelStatus>
    <!--opt, enum, subType:string-->modeling
  </modelStatus>
  <modelingStatus>
    <!--opt, enum, subType:string-->success
  </modelingStatus>
  <customFaceLibID>
    <!--opt, string, custom face picture library ID, desc:custom face picture library ID-->test
  </customFaceLibID>
  <sortord>
    <!--opt, enum, subType:string-->time
  </sortord>
  <order>
    <!--opt, enum, subType:string-->DESC
  </order>
  <StrangerLibFaceAppendData>
    <!--opt, object-->
    <PersonInfoExtendList size="4">
      <!--opt, array, subType:object, attr:size{req, int}-->
      <PersonInfoExtend>
        <!--opt, object, person extended information-->
        <id>
          <!--req, int, serial No. of person custom extension information, desc:serial No. of person custom extension information-->1
        </id>
        <enable>
          <!--req, bool, enable person extended information or not-->true
        </enable>
        <name>
          <!--opt, string, person tag extended information name, desc:person tag extended information name-->test
        </name>
        <value>
          <!--opt, string, person tag extended information, desc:person tag extended information-->test
        </value>
      </PersonInfoExtend>
    </PersonInfoExtendList>
  </StrangerLibFaceAppendData>
```

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```
  <OccurrencesInfo>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <occurrences>
      <!--opt, int-->1
    </occurrences>
    <occurrencesSearchType>
      <!--opt, enum, subType:string-->greaterThanOrEqual
    </occurrencesSearchType>
  </OccurrencesInfo>
  <faceScore>
    <!--opt, int, face score (face picture quality), range:[-1,100], desc:face score (face picture quality)-->1
  </faceScore>
  <FDIDList>
    <!--opt, object-->
    <FDID>
      <!--req, string, range:[1,64]-->test
    </FDID>
  </FDIDList>
  <faceScoreMax>
    <!--opt, int, range:[-1,100]-->0
  </faceScoreMax>
  <pictureStatus>
    <!--opt, enum, subType:string-->normal
  </pictureStatus>
  <importStartTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </importStartTime>
  <importEndTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </importEndTime>
  <faceContrastSuccessStartTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </faceContrastSuccessStartTime>
  <faceContrastSuccessEndTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </faceContrastSuccessEndTime>
  <userType>
    <!--opt, enum, subType:string-->normal
  </userType>
  <associatedPlateType>
    <!--opt, enum, subType:string-->greenChannel
  </associatedPlateType>
</FDSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, search result, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, search ID, desc:it is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device to speed up the next search-->test
  </searchID>
  <responseStatus>
    <!--ro, req, enum, search result, subType:string, desc:search result-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data).-->OK
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <progress>
    <!--ro, opt, int, range:[0,100]-->1
  </progress>
  <MatchList>
    <!--ro, opt, array, searched matched data list, subType:object-->
    <MatchElement>
      <!--ro, opt, object-->
      <FDID>
        <!--ro, opt, string, face picture library ID-->test
      </FDID>
      <thresholdValue>
        <!--ro, opt, int, range:[0,100]-->1
      </thresholdValue>
      <bornTime>
        <!--ro, opt, date, birthday of person in the face picture-->1970-01-01+08:00
      </bornTime>
      <name>
        <!--ro, req, string, name, range:[1,32], desc:name-->test
      </name>
      <sex>
        <!--ro, opt, enum, gender, subType:string, desc:male; female:-->male
```

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```
        <!--ro, opt, enum, gender, subType:string, desc:male, female-->male
      </sex>
      <province>
        <!--ro, opt, string-->11
      </province>
      <city>
        <!--ro, opt, string, city, desc:city code of birth for the person in the face picture,the maximum size is 32 bytes-->11
      </city>
      <certificateType>
        <!--ro, opt, enum, certificate type, subType:string, desc:the max. size is 10 bytes,certificate type: "officerID"-officer ID,"ID"-identify
card,passport,other-->officerID
      </certificateType>
      <certificateNumber>
        <!--ro, opt, string, certificate No., range:[8,32], desc:certificate No.,the max. size is 32 bytes-->test
      </certificateNumber>
      <picURL>
        <!--ro, req, string-->test
      </picURL>
      <PID>
        <!--ro, req, string, picture ID-->test
      </PID>
      <PersonInfoExtendList>
        <!--ro, opt, array, subType:object-->
        <PersonInfoExtend>
          <!--ro, opt, object, person extended information-->
          <id>
            <!--ro, req, int, serial No. of person custom extension information-->1
          </id>
          <enable>
            <!--ro, req, bool, enable person extended information or not-->true
          </enable>
          <name>
            <!--ro, opt, string, person tag extended information name, desc:person tag extended information name-->test
          </name>
          <value>
            <!--ro, opt, string, person tag extended information, desc:person tag extended information-->test
          </value>
        </PersonInfoExtend>
      </PersonInfoExtendList>
      <similarity>
        <!--ro, opt, float, similarity, range:[0.0,100.0]-->0.0
      </similarity>
      <ModelingStatus>
        <!--ro, opt, object-->
        <FDID>
          <!--ro, opt, string, face picture library ID-->test
        </FDID>
        <PID>
          <!--ro, opt, string, picture ID-->test
        </PID>
        <name>
          <!--ro, req, string, picture name-->test
        </name>
        <status>
          <!--ro, req, enum, subType:string-->success
        </status>
        <reason>
          <!--ro, opt, enum, subType:string-->noface
        </reason>
        <customHumanID>
          <!--ro, opt, string-->123456789012345678901234567890 12
        </customHumanID>
        <customFaceLibID>
          <!--ro, opt, string-->123456789012345678901234567890 12
        </customFaceLibID>
        <modeData>
          <!--ro, opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64
method-->test
        </modeData>
      </ModelingStatus>
      <isNoSaveFDPicture>
        <!--ro, opt, bool-->true
      </isNoSaveFDPicture>
      <humanId>
        <!--ro, opt, string-->test
      </humanId>
      <modelStatus>
        <!--ro, opt, enum, subType:string-->modeling
      </modelStatus>
      <FMCount>
        <!--ro, opt, int-->1
      </FMCount>
      <importTime>
        <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
      </importTime>
      <UUPID>
        <!--ro, opt, string-->test
      </UUPID>
      <occurrences>
        <!--ro, opt, int-->1
      </occurrences>
      <IMSIList>
        <!--ro, opt, array, subType:object-->
        <IMSIItem>
```

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```
            <!--ro, opt, object-->
            <IMSI>
              <!--ro, opt, string-->123456789012345
            </IMSI>
            <count>
              <!--ro, opt, int-->10
            </count>
            <lastFMTime>
              <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
            </lastFMTime>
          </IMSIItem>
        </IMSIList>
        <lastFMTime>
          <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
        </lastFMTime>
        <faceScore>
          <!--ro, opt, int, range:[-1,100]-->1
        </faceScore>
        <pictureStatus>
          <!--ro, opt, enum, subType:string-->normal
        </pictureStatus>
        <userType>
          <!--ro, opt, enum, subType:string-->normal
        </userType>
        <associatedPlateType>
          <!--ro, opt, enum, subType:string-->greenChannel
        </associatedPlateType>
        <contrastStatus>
          <!--ro, opt, enum, subType:string-->contrast
        </contrastStatus>
      </MatchElement>
    </MatchList>
  </FDSearchResult>
```

15.5.6.9 Get the capability of configuring face picture pre-classification parameters.

Request URL

GET /ISAPI/Intelligent/FDLib/presort/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "presortInfo": {
    /*ro, req, object, face picture pre-classification parameters*/
        "presortSpeedupTimes":  "1,20,50"
        /*ro, req, string, value range of times of speeding up pre-classification*/
    }
}
```

15.5.6.10 Set presorting parameters of human face pictures

Request URL

PUT /ISAPI/Intelligent/FDLib/presort?format=json

Query Parameter

None

Request Message

```
{
    "presortInfo": {
    /*req, object*/
        "presortSpeedupTimes":  1
        /*opt, enum, 1 (1 days), 15 (15 days), 50 (50 days), subType:int, desc:1 (1 days), 15 (15 days), 50 (50 days)*/
    }
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

15.5.6.11 Get presorting parameters of face pictures

Request URL

GET /ISAPI/Intelligent/FDLib/presort?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "presortInfo": {
    /*ro, req, object*/
        "presortSpeedupTimes":  1
        /*ro, opt, enum, subType:int*/
    }
}
```

15.5.6.12 Set the face picture library parameters

Request URL

PUT /ISAPI/Intelligent/FDLib?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibBaseCfgList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, face picture library information list, subType:object, attr:version{req, string, protocolVersion}-->
  <FDLibBaseCfg>
    <!--opt, object, face picture library information-->
    <id>
      <!--opt, int, range:[1,15]-->1
    </id>
    <FDID>
      <!--opt, string, unique ID of face picture library-->test
    </FDID>
    <name>
      <!--req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <thresholdValue>
      <!--opt, int, detection threshold, range:[0,100]-->50
    </thresholdValue>
    <customInfo>
      <!--opt, string, additional information of face picture library, range:[1,128]-->test
    </customInfo>
    <customFaceLibID>
      <!--opt, string, custom face picture library ID, range:[1,64]-->test
    </customFaceLibID>
    <faceLibType>
      <!--ro, opt, enum, face picture library type, subType:string, desc:"ordinary" (normal library), "private" (private library), "stranger" (stranger
library), "allowable" (allowlist library), "privacyMask" (privacy mask library)-->allowable
    </faceLibType>
    <autoUpdata>
      <!--opt, bool-->true
    </autoUpdata>
    <qualityThreshold>
      <!--opt, int, range:[0,100]-->50
    </qualityThreshold>
    <importFDThreshold>
      <!--opt, int, range:[0,100]-->40
    </importFDThreshold>
    <Age>
      <!--opt, object-->
      <judgementCycleType>
        <!--req, enum, subType:string-->day
      </judgementCycleType>
      <statisticalCycleType>
        <!--ro, req, enum, subType:string-->month
      </statisticalCycleType>
      <statisticalCycle>
        <!--req, int, range:[0,99999]-->1
      </statisticalCycle>
      <minFMCount>
        <!--req, int, range:[0,99999]-->1
      </minFMCount>
      <FMCountEnabled>
        <!--opt, bool-->true
      </FMCountEnabled>
      <FMCount>
        <!--opt, int, range:[0,99999]-->1
      </FMCount>
      <backgoundPicThreshold>
        <!--opt, int, range:[1,100]-->90
      </backgoundPicThreshold>
      <judgementCycle>
        <!--opt, int, range:[1,365], unit:d, dep:and,{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->30
      </judgementCycle>
    </Age>
    <FaceCounting>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <uploadType>
        <!--ro, req, enum, subType:string-->noUpload
      </uploadType>
    </FaceCounting>
    <faceMonitor>
      <!--ro, opt, bool-->true
    </faceMonitor>
    <monitorFaceNum>
      <!--ro, opt, int-->1
    </monitorFaceNum>
    <customNoConfig>
      <!--ro, opt, bool-->true
    </customNoConfig>
    <uploadID>
      <!--opt, string, range:[1,128], dep:and,{$.FDLibBaseCfglist.FDLibBaseCfg.autoUpdata,eq,true}-->test
    </uploadID>
  </FDLibBaseCfg>
</FDLibBaseCfgList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

15.5.6.13 Create a face picture library

Request URL

POST /ISAPI/Intelligent/FDLib?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64 characters. This parameter stands for the identity key of the operation platform. The device can know the operator's identity by checking this parameter in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<CreateFDLibList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, create the list of face picture library, subType:object, attr:version{req, string, protocolVersion}-->
  <CreateFDLib>
    <!--opt, object, create a face picture library-->
    <id>
      <!--opt, int, sub item No. in the list; it increases from 1, range:[1,15]-->1
    </id>
    <name>
      <!--req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <thresholdValue>
      <!--opt, int, detection threshold, range:[0,100]-->50
    </thresholdValue>
    <customInfo>
      <!--opt, string, custom information, range:[1,128]-->test
    </customInfo>
    <customFaceLibID>
      <!--opt, string, custom face picture library ID, range:[1,64]-->test
    </customFaceLibID>
    <autoUpdata>
      <!--opt, bool, whether to enable auto update of background picture in face picture library-->true
    </autoUpdata>
    <qualityThreshold>
      <!--opt, int, picture quality threshold, range:[0,100]-->50
    </qualityThreshold>
    <importFDThreshold>
      <!--opt, int, inbound similarity threshold, range:[0,100]-->40
    </importFDThreshold>
    <Age>
      <!--opt, object, the aggregate aging parameters configuration of face picture library; it is valid when autoUpdata is true-->
      <judgementCycleType>
        <!--req, enum, the time type for aging judgement cycle of the aggregate data, subType:string, desc:"day", "week", "month", "duration" (judge
according to actual duration)-->day
      </judgementCycleType>
      <statisticalCycleType>
        <!--req, enum, statistics cycle type of clustering matches in face picture library, subType:string, desc:"day", "week", "month"-->month
      </statisticalCycleType>
      <statisticalCycle>
        <!--req, int, statistics cycle of face picture matches, range:[0,99999]-->1
      </statisticalCycle>
      <minFMCount>
        <!--req, int, min. number of face picture matches within statistics cycle, range:[0,99999]-->1
      </minFMCount>
      <FMCountEnabled>
        <!--opt, bool, whether to enable total number of face picture matches-->true
      </FMCountEnabled>
      <FMCount>
        <!--opt, int, total number of face picture matches, range:[0,99999]-->1
      </FMCount>
      <backgroundPicThreshold>
        <!--opt, int, range:[1,100]-->90
      </backgroundPicThreshold>
      <judgementCycle>
        <!--opt, int, evaluation cycle range of clustering data aging in face picture library, range:[1,365], unit:d, dep:and,
{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->60
      </judgementCycle>
      <agingExecDataTime>
        <!--opt, time-->00:00:00+08:00
      </agingExecDataTime>
      <automaticClearEnabled>
        <!--opt, bool-->true
      </automaticClearEnabled>
    </Age>
    <FaceCounting>
      <!--opt, object, the configuration for customer traffic (face) deduplication of face picture comparison-->
      <enabled>
        <!--req, bool, whether to enable customer traffic (face) deduplication-->true
      </enabled>
      <uploadType>
        <!--req, enum, uploading mode after the customer traffic (face) deduplication is enabled, subType:string, desc:"out_repetition" (uploading outside
the deduplication period); "noUpload" (not uploading outside the deduplication period)-->noUpload
      </uploadType>
    </FaceCounting>
    <faceLibType>
      <!--opt, enum, face picture library type, subType:string, desc:if this node is not returned, a normal library is created by default; "ordinary"
(normal library); "privacyMask" (privacy mask library)-->ordinary
    </faceLibType>
  </CreateFDLib>
</CreateFDLibList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, create the result list of face picture library, subType:object, attr:version{req, string, protocolVersion}-->
  <FDLibInfo>
    <!--ro, opt, object, face picture library information-->
    <id>
      <!--ro, req, int, sub item No. in the list; it increases from 1 and corresponds to the id of CreateFDLibList-->1
    </id>
    <name>
      <!--ro, req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <FDID>
      <!--ro, req, string, returned face picture library ID when it created, the library ID of the same type is unique, range:[1,64]-->test
    </FDID>
    <statusCode>
      <!--ro, opt, enum, status code; it is returned when creating a face picture library failed, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3
(Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->1
    </statusCode>
    <subStatusCode>
      <!--ro, opt, string, sub status code; it is returned when creating a face picture library failed-->existSameNameFacelib
    </subStatusCode>
  </FDLibInfo>
</FDLibInfoList>
```

15.5.6.14 Pause the task according to task ID

Request URL

GET /ISAPI/SDT/Management/Task/pause/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	Task ID

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field must be returned when the value of statusCode is not 1*/
}
```

15.5.6.15 Obtain the task progress

Request URL

GET /ISAPI/SDT/Management/Task/progress/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

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```
{
    "progress": {
    /*ro, opt, object, task progress*/
        "value":  22
        /*ro, req, int, value of task searching progress, range:[0,100]*/
    }
}
```

15.5.6.16 Resume the task according to task ID

Request URL

GET /ISAPI/SDT/Management/Task/resume/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	Task ID

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:status code*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:status description*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:which corresponds to subStatusCode, this field is required when statusCode is not 1. The returned value is the
transformed decimal number*/
    "errorMsg":  "ok"
    /*ro, opt, string, error details, desc:this field is required when statusCode is not 1*/
}
```

15.5.7 人脸高频检测

15.5.7.1 Set the arming schedule of frequently appeared person detection

Request URL

PUT /ISAPI/Event/schedules/HFPD

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<HFPDScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, undefined, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string, ID, desc:ID-->HFPD
    </id>
    <eventType>
      <!--opt, string, event type-->HFPD
    </eventType>
    <inputIOPortID>
      <!--opt, string-->1
    </inputIOPortID>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object, arming the task by day each week-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object, arming the task by day each week-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</HFPDScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
</ResponseStatus>
```

15.5.7.2 Get the arming schedule of the frequently appeared person detection

Request URL

GET /ISAPI/Event/schedules/HFPD

Query Parameter

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None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HFPDScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, arming schedule of the frequently appeared person detection, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->HFPD
    </id>
    <eventType>
      <!--ro, opt, string, event type-->HFPD
    </eventType>
    <inputIOPortID>
      <!--ro, opt, string-->1
    </inputIOPortID>
    <outputIOPortID>
      <!--ro, opt, string-->1
    </outputIOPortID>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</HFPDScheduleList>
```

15.5.7.3 Get the alarm information of the frequently appeared person detection in real time

Request URL

GET /ISAPI/SDT/HFPD/alarmInfo?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "faceCapture": [
    /*ro, opt, array, face capture result information, subType:object*/
        {
            "errorCode":  1,
```

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```
            /*ro, opt, int, error code, desc:when analyzing face picture failed, the corresponding error code will be returned*/
            "errorMsg":  "ok",
            /*ro, opt, string, error description, desc:when analyzing face picture failed, the corresponding error information will be returned*/
            "targetAttrs": {
            /*ro, opt, object, target attributes*/
                "deviceId":  "1",
                /*ro, opt, string, device ID*/
                "deviceChannel":  1,
                /*ro, opt, int, device channel No.*/
                "deviceName":  "shebei",
                /*ro, opt, string, device name*/
                "faceTime":  "1970-01-01T00:00:00+08:00",
                /*ro, opt, datetime, capture time*/
                "isSummerTime":  false,
                /*ro, opt, bool, whether the time returned currently is in DST (Daylight Saving Time) system*/
                "rect": {
                /*ro, opt, object, rectangle frame of the big face picture, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "bkgUrl":  "http://absddsdfsdfsdk.jpg"
                /*ro, opt, string, background picture*/
            },
            "faces": [
            /*ro, opt, array, face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on,
subType:object, desc:array of object,face information including detection results,attributes,modeling,and matched lists. There may be multiple faces in one
picture*/
                {
                    "faceId":  1,
                    /*ro, opt, int, face picture ID*/
                    "faceRect": {
                    /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "faceMark": {
                    /*ro, opt, object*/
                        "leftEye": {
                        /*ro, opt, object, left eye coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.130,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                            "y":  0.870
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                        },
                        "rightEye": {
                        /*ro, opt, object, right eye coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.130,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                            "y":  0.870
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                        },
                        "noseTip": {
                        /*ro, opt, object, nose tip coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.130,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                            "y":  0.870
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                        },
                        "leftMouth": {
                        /*ro, opt, object, left mouth corner coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.130,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                            "y":  0.870
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                        },
                        "rightMouth": {
                        /*ro, opt, object, right mouth corner coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.130,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                            "y":  0.870
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                        }
                    },
                    "recommendFaceRect": {
                    /*ro, opt, object, recommended face frame coordinates, desc:the origin is the upper-left corner of the screen*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
```

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```
                        "y": 0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "facePose": {
                    /*ro, opt, object, face angle*/
                        "pitch": 0.998,
                        /*ro, opt, float, face pitch angle, face pitching up is regarded as positive, range:[-90,90], desc:face pitching up is regarded as positive, range:*/
                        "yaw": 0.876,
                        /*ro, opt, float, face yaw angle, face yawing left is regarded as positive, range:[-90,90], desc:float,yawing angle outside the plane,value range: [-90,90],the value is accurate to three decimal places. Facing left is positive. This node does not exist if it is invalid*/
                        "roll": 0.623
                        /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive, range:[-90,90], desc:float,rolling angle inside the plane,value range: [-90,90],the value is accurate to three decimal places. The clockwise rotation is positive. This node does not exist if it is invalid*/
                    },
                    "age": {
                    /*ro, opt, object, age*/
                        "range": 5,
                        /*ro, opt, int, age deviation, desc:age deviation. If this node exists,the node value is required*/
                        "value": 23,
                        /*ro, opt, int, age*/
                        "ageGroup": "any"
                        /*ro, req, enum, age group, subType:string, desc:“any”, “child”, “infant”, “kid”, “middle”, “middleAged” (the middle age), “old”, “prime”, “teenager”, “unknown”, “young”*/
                    },
                    "gender": {
                    /*ro, opt, object, gender*/
                        "confidence": 99.99990,
                        /*ro, opt, float, confidence*/
                        "value": "male"
                        /*ro, opt, enum, gender, subType:string, desc:"male", "female",*/
                    },
                    "glass": {
                    /*ro, opt, object, whether the person wears glasses*/
                        "confidence": 99.94499999999999,
                        /*ro, opt, float, confidence*/
                        "value": "yes"
                        /*ro, opt, enum, value, subType:string, desc:"yes", "no"*/
                    },
                    "smile": {
                    /*ro, opt, object, whether the person wears a smile*/
                        "confidence": 99.99990,
                        /*ro, opt, float, confidence*/
                        "value": "yes"
                        /*ro, opt, enum, value, subType:string, desc:"yes", "no"*/
                    },
                    "modeldata": "xxxx",
                    /*ro, opt, string, target model data. This node is valid for captured results only, desc:the maximum size is 5*1024 bytes*/
                    "URL": "test",
                    /*ro, opt, string, face thumbnail URL, the maximum size is 256*/
                    "stayDuration": 20,
                    /*ro, opt, int, Dwell Time, unit:ms*/
                    "occurrences": 7
                    /*ro, opt, int, number of appearances for the target*/
                }
            ],
            "recordDataIndex": "test",
            /*ro, req, string, index for searching for history alarms of the frequently appeared person detection, desc:related URI: GET /ISAPI/SDT/HFPD/searchTask/result/recordData/<ID>?format=json*/
            "recordData": [
            /*ro, opt, array, history data records, subType:object*/
                {
                    "captureTime": "1970-01-01T00:00:00+08:00",
                    /*ro, opt, datetime, capture time, desc:ISO 8601 format*/
                    "captureSite": "3#",
                    /*ro, opt, string, capture site (camera name)*/
                    "similarity": 1.000,
                    /*ro, opt, float, similarity, range:[0.000,1.000]*/
                    "picUrl": "xxx",
                    /*ro, opt, string, background picture URL*/
                    "subpicUrl": "xxx"
                    /*ro, opt, string, face picture thumbnail URL*/
                }
            ],
            "alarmAckState": "acknowledged",
            /*ro, opt, enum, alarm acknowledgment status, subType:string, desc:"acknowledged", "pending" (to be acknowledged. This value can be used when the user cannot acknowledge the alarm when it is triggered but wants to prioritize acknowledging it later), "notAcknowledged" (not acknowledgment)*/
            "alarmCustomInfo": "test"
            /*ro, opt, string, custom information, range:[1,64]*/
        }
    ],
    "URLCertificationType": "digest"
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (for the cloud storage protocol), "digest" (digest authentication, for the device local storage)*/
}
```

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15.5.7.4 Frequently Appeared Person Detection Capability Set

Request URL

GET /ISAPI/SDT/HFPD/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "HFPDCap": {
    /*ro, opt, object, Capability of frequently appeared person detection.*/
        "analysisDays": {
        /*ro, opt, object, analysis duration*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  30,
            /*ro, opt, int, the maximum value*/
            "@def":  7
            /*ro, opt, int, the default value*/
        },
        "timeRange": {
        /*ro, opt, object, time period*/
            "size":  8
            /*ro, opt, int, analysis duration returns to the time period of arming collision detected each day*/
        },
        "occurrences": {
        /*ro, opt, object, frequently appeared person detection times*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  100,
            /*ro, opt, int, the maximum value*/
            "@def":  10
            /*ro, opt, int, the default value*/
        },
        "similarity": {
        /*ro, opt, object, similarity of face picture library comparison for frequently appeared person detection*/
            "@min":  80,
            /*ro, opt, int, the minimum value*/
            "@max":  100,
            /*ro, opt, int, the maximum value*/
            "@def":  85
            /*ro, opt, int, the default value*/
        },
        "captureTimeInterval": {
        /*ro, opt, object, capture time interval for checking duplicate in capture library*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  60,
            /*ro, opt, int, the maximum value*/
            "@def":  10
            /*ro, opt, int, the default value*/
        },
        "monitorIds": {
        /*ro, opt, object, camera*/
            "size":  4,
            /*ro, req, int, the maximum supported cameras*/
            "@opt":  "1,2,3"
            /*ro, opt, string, return to the camera which supports frequently appeared person detection*/
        },
        "taskID": {
        /*ro, opt, object, task ID*/
            "size":  4
            /*ro, opt, int, task ID number*/
        },
        "FaceLibrary": {
        /*ro, opt, object, face picture library*/
            "size":  4,
            /*ro, opt, int, array size*/
            "similarity": {
            /*ro, opt, object, similarity of face picture library comparison for frequently appeared person detection*/
                "@min":  80,
                /*ro, opt, int, the minimum value*/
                "@max":  100,
                /*ro, opt, int, the maximum value*/
                "@def":  85
                /*ro, opt, int, the default value*/
            },
            "FDID": {
            /*ro, opt, object, face picture library*/
                "@min":  80,
                /*ro, opt, int, minimum string length*/
                "@max":  100
                /*ro, opt, int, maximum string length*/
            },
            "isSupportStrangerFaceLib":  false
            /*ro, opt, bool, supports stranger library configuration*/
        },
```

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```
        "eventRecordDataSearch": {
        /*ro, opt, object, search the event records of frequently appeared people detection, desc:indicates that supporting POST
/ISAPI/SDT/HFPD/searchTask/eventRecordData?format=json*/
            "isSupportSearchTime":  true,
            /*ro, opt, bool, whether to support searching cross time periods*/
            "isSupportRecordDataSearchSync":  true
            /*ro, opt, bool, search the event records of frequently appeared people detection, desc:indicates supporting POST
/ISAPI/SDT/HFPD/searchTask/eventRecordData?format=json*/
        },
        "captureLibCollisionSearch": {
        /*ro, opt, object, returning indicates that supporting search task of frequently appeared person capture library comparison(duplicate check in the
library), desc:indicates that supporting POST /ISAPI/SDT/HFPD/searchTask/captureLibCollision?format=json*/
            "isSupportSearchTime":  true,
            /*ro, opt, bool, whether to support searching cross time periods*/
            "captureTimeInterval": {
            /*ro, opt, object, capture time interval for checking duplicate in capture library*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  60,
                /*ro, opt, int, the maximum value*/
                "@def":  10
                /*ro, opt, int, the default value*/
            },
            "similarity": {
            /*ro, opt, object, similarity of face picture library comparison for frequently appeared person detection*/
                "@min":  80,
                /*ro, opt, int, the minimum value*/
                "@max":  100,
                /*ro, opt, int, the maximum value*/
                "@def":  85
                /*ro, opt, int, the default value*/
            },
            "occurrences": {
            /*ro, opt, object, frequently appeared person detection times*/
                "@min":  5,
                /*ro, opt, int, the minimum value*/
                "@max":  100,
                /*ro, opt, int, the maximum value*/
                "@def":  7
                /*ro, opt, int, the default value*/
            }
        },
        "multiLibCollisionSearch": {
        /*ro, opt, object, search the multiple libraries comparison, desc:related URL: URL/ISAPI/SDT/HFPD/searchTask/multiLibCollision?format=json*/
            "FD": {
            /*ro, opt, object, face picture library*/
                "size":  4
                /*ro, opt, int, the maximum number of supported face picture library comparison*/
            },
            "priority": {
            /*ro, opt, object, priority*/
                "@min":  0,
                /*ro, opt, int, the minimum value*/
                "@max":  100,
                /*ro, opt, int, the maximum value*/
                "@def":  80
                /*ro, opt, int, the default value*/
            },
            "similarity": {
            /*ro, opt, object, similarity*/
                "@min":  80,
                /*ro, opt, int, the minimum value*/
                "@max":  100,
                /*ro, opt, int, the maximum value*/
                "@def":  85
                /*ro, opt, int, the default value*/
            }
        },
        "isSupportDataPackage":  true,
        /*ro, opt, bool, support exporting data of frequently appeared person detection data*/
        "BlackListFaceLibrary": {
        /*ro, opt, object, blocklist face picture library configuration for frequently appeared person detection*/
            "size":  4,
            /*ro, req, int, the maximum number of supported face picture library*/
            "similarity": {
            /*ro, opt, object, similarity of comparison library  for frequently appeared person detection*/
                "@min":  70,
                /*ro, opt, int, the minimum value*/
                "@max":  100,
                /*ro, opt, int, the maximum value*/
                "@def":  80
                /*ro, opt, int, the default value*/
            }
        },
        "isSupportCorrectionStrategy":  true
        /*ro, opt, bool, whether to support face picture correction strategy search, desc:related URL: URL/ISAPI/SDT/HFPD/searchTask/multiLibCollision?
format=json*/
    }
}
```

15.5.7.5 Get Exported Results of Frequently Appeared Person Event Alarm

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Request URL

GET /ISAPI/SDT/HFPD/search/dataPackage/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	The value is the same with TaskInfo->id value in the response message of POST /ISAPI/SDT/HFPD/search/dataPackage?format=json.

Request Message

None

Response Message

```
{
    "progress":  22,
    /*ro, req, int, packaging progress, range:[0,100]*/
    "URL":  "test"
    /*ro, opt, string, storage URL generated by device after packaging, desc:you can download the data files via the URL*/
}
```

15.5.7.6 Set Condition of Exporting Frequently Appeared Person Alarm

Request URL

POST /ISAPI/SDT/HFPD/search/dataPackage?format=json

Query Parameter

None

Request Message

```
{
    "DataPackageCond": {
    /*opt, object, condition*/
        "searchResultPosition":  0,
        /*opt, int, the start position of search result in result list, desc:In a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search.*/
        "maxResults":  1,
        /*opt, int, the maximum number of search results this time, range:[1,1000], desc:by default, it equals to maximum value which does not exceed 1000 iterms*/
        "taskID":  "test",
        /*opt, string, search task returned ID, desc:corresponding node TaskInfo->id in the returned message POST /ISAPI/SDT/HFPD/searchTask/eventRecordData?format=json*/
        "Description": {
        /*opt, object, Packaging search condition, desc:"Description" and "taskID" overlap in meaning. When they coexist, "taskid" shall prevail*/
            "startTime":  "1970-01-01T00:00:00Z",
            /*opt, datetime, start time of search*/
            "endTime":  "1970-01-01T01:00:00Z",
            /*opt, datetime, end time of search*/
            "channels": [1, 2, 3],
            /*opt, array, array of the linked channel No., subType:int, desc:all channels are searched if channel information is not specified*/
            "monitorIds": ["11111", "22222"],
            /*opt, array, array information of camera ID, subType:string, desc:all channels are searched if channel information is not specified*/
            "occurrences":  7,
            /*opt, int, appearing times of frequently appeared person*/
            "lowestOccurrences":  1
            /*opt, int, the minimum appeared times, range:[1,30], desc:default value: 1*/
        }
    }
}
```

Response Message

```
{
    "TaskInfo": {
    /*ro, opt, object, task information*/
        "id":  "test"
        /*ro, opt, string, It is required for success, desc:control task operation via this ID*/
    }
}
```

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15.5.7.7 Search Frequently Appeared Person Detection Events Synchronously

Request URL

POST /ISAPI/SDT/HFPD/searchTask/eventRecordData/sync?format=json

Query Parameter

None

Request Message

```
{
    "EventRecordDataDescription": {
    /*req, object, search condition*/
        "startTime":  "1970-01-01T00:00:00Z",
        /*opt, datetime, start time*/
        "endTime":  "1970-01-01T01:00:00Z",
        /*opt, datetime, end time*/
        "monitorIds": ["11111", "22222"],
        /*opt, array, array information of camera ID, subType:string*/
        "occurrences":  7,
        /*opt, int, appearing times of frequently appeared person*/
        "searchResultPosition":  1,
        /*req, int, the start position of the search result in the result list, desc:In a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search.*/
        "maxResults":  100,
        /*req, int, the maximum number of search results this time*/
        "alarmAckState":  "acknowledged",
        /*opt, enum, alarm acknowledgment status, subType:string, desc:"acknowledged","pending" (to be acknowledged. This value can be used when the user cannot acknowledge the alarm when it is triggered but wants to prioritize acknowledging it later),"notAcknowledged" (not acknowledgment),"all".*/
        "lowestOccurrences":  1
        /*opt, int, the minimum appeared times, range:[1,30]*/
    }
}
```

Response Message

```
{
    "HFPDSearchResult": {
    /*ro, req, object, search result*/
        "responseStatusStrg":  "OK",
        /*ro, opt, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data).*/
        "numOfMatches":  1,
        /*ro, opt, int, number of results returned this time*/
        "totalMatches":  1,
        /*ro, opt, int, total number of matched results*/
        "targetInfo": [
        /*ro, req, array, target information, subType:object*/
            {
                "recordDataIndex":  "test",
                /*ro, req, string, ID of searched record detailed information, desc:corresponds to the node <indexID> in the URI(GET /ISAPI/SDT/HFPD/searchTask/result/recordData/<indexID>?format=json). Retrun the detailed information of the frequently search target when returning the URL at the same time*/
                "occurrences":  7,
                /*ro, req, int, appearing times of frequently appeared person*/
                "analysisTime":  "1970-01-01T00:00:00Z",
                /*ro, req, datetime, alarm analysis time*/
                "timeDifference":  "+08:00",
                /*ro, opt, string, time difference information, desc:starttime and endtime are in TZ format, indicating UTC time. This field indicates the time difference from UTC (equivalent to the last time difference information in T+ format). Added in the Intelligent Application Server V1.2.0.*/
                "URL":  "test",
                /*ro, req, string, face sub-picture URL, desc:IP and port are local. If the devices cross domains, the upper layer needs to be reassembled according to the device mapping*/
                "alarmAckState":  "acknowledged",
                /*ro, opt, enum, alarm acknowledgment status, subType:string, desc:"acknowledged","pending" (to be acknowledged. This value can be used when the user cannot acknowledge the alarm when it is triggered but wants to prioritize acknowledging it later),"notAcknowledged" (not acknowledgment),"all".*/
                "alarmCustomInfo":  "test"
                /*ro, opt, string, custom information, range:[1,64]*/
            }
        ]
    }
}
```

15.5.7.8 Search Frequently Appeared Person Detection Events Asynchronously

Request URL

POST /ISAPI/SDT/HFPD/searchTask/eventRecordData?format=json

Query Parameter

None

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Request Message

```
{
    "EventRecordDataDescription": {
    /*opt, object, search condition*/
        "startTime":  "1970-01-01T00:00:00Z",
        /*opt, datetime, start time*/
        "endTime":  "1970-01-01T01:00:00Z",
        /*opt, datetime, end time*/
        "monitorIds": ["11111", "22222"],
        /*opt, array, array information of camera ID, subType:string*/
        "occurrences":  7,
        /*opt, int, appearing times of frequently appeared person*/
        "alarmAckState":  "acknowledged",
        /*opt, enum, alarm acknowledgment status, subType:string, desc:"acknowledged","pending" (to be acknowledged. This value can be used when the user
cannot acknowledge the alarm when it is triggered but wants to prioritize acknowledging it later),"notAcknowledged" (not acknowledgment),"all".*/
        "lowestOccurrences":  1
        /*opt, int, the minimum appeared times, range:[1,30]*/
    }
}
```

Response Message

```
{
    "TaskInfo": {
    /*ro, opt, object, task information*/
        "id":  "test"
        /*ro, opt, string, task ID, desc:Call the following URL to control the task: suspend the task: GET /ISAPI/SDT/Management/Task/pause/<taskID>?
format=json; restore task: GET /ISAPI/SDT/Management/Task/resume/<taskID>?format=json; delete task: DELETE /ISAPI/SDT/Management/Task/<taskID>?format=json;
search task progress: GET /ISAPI/SDT/Management/Task/progress/<taskID>?format=json; cancel task: GET /ISAPI/SDT/Management/Task/cancel/<taskID>?
format=json,*/
    }
}
```

15.5.7.9 Get the search results of frequently appeared person alarms

Request URL

GET /ISAPI/SDT/HFPD/searchTask/result/recordData/<indexID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	Data index

Request Message

None

Response Message

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```
{
    "targets": [
    /*ro, req, array, list of matched records, subType:object*/
        {
            "id": 1,
            /*ro, req, int, No.*/
            "captureTime": "1970-01-01T00:00:00+08:00",
            /*ro, opt, datetime, capture time*/
            "isSummerTime": false,
            /*ro, opt, bool*/
            "organizationID": "test123456789",
            /*ro, opt, string, range:[1,32]*/
            "captureSite": "3#大门处",
            /*ro, opt, string, capture site (camera name)*/
            "monitorId": "1",
            /*ro, opt, string, camera ID (CVR encoder stream ID)*/
            "ageGroup": "any",
            /*ro, req, enum, age group, subType:string, desc:“any”, “child”, “infant”, “kid”, “middle”, “middleAged” (the middle age), “old”, “prime”,
“teenager”, “unknown”, “young”*/
            "gender": "any",
            /*ro, req, enum, gender, subType:string, desc:"any” (unlimited condition which is used in search), "male”, “female”, “unknown"*/
            "glasses": "no",
            /*ro, req, enum, whether the person wears glasses, subType:string, desc:“no” (not wear glasses), “sunglasses” (wears sunglasses), “unknown"*/
            "smile": "no",
            /*ro, req, enum, whether the person wears a smile, subType:string, desc:"no", "unknown", "yes”*/
            "mask": "no",
            /*ro, req, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes”*/
            "picUrl": "http://127.0.0.1",
            /*ro, opt, string, background picture URL*/
            "subpicUrl": "http://127.0.0.1",
            /*ro, opt, string, face thumbnail URL*/
            "rect": {
            /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
                "height": 0.514,
                /*ro, opt, float, height, range:[0.000,1.000]*/
                "width": 0.2909,
                /*ro, opt, float, width, range:[0.000,1.000]*/
                "y": 0.3669,
                /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                "x": 0.183
                /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
            },
            "face_pose": {
            /*ro, opt, object, face angle*/
                "pitch": 0.514,
                /*ro, opt, float, face pitch angle, face pitching up is regarded as positive*/
                "yaw": 0.2909,
                /*ro, opt, float, face yaw angle, face yawing left is regarded as positive*/
                "roll": 0.3669
                /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive*/
            },
            "face_rect": {
            /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                "height": 0.514,
                /*ro, opt, float, height, range:[0.000,1.000]*/
                "width": 0.2909,
                /*ro, opt, float, width, range:[0.000,1.000]*/
                "y": 0.3669,
                /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                "x": 0.183
                /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
            },
            "similarity": 1.00,
            /*ro, opt, float, similarity*/
            "range": 5,
            /*ro, opt, int, age deviation*/
            "value": 23,
            /*ro, opt, int, age*/
            "faceExpression": {
            /*ro, opt, object, facial expression*/
                "confidence": 99.99990,
                /*ro, opt, float, confidence*/
                "value": "angry"
                /*ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-faced",
"sad", "surprised", "unknown"*/
            },
            "thermometryUnit": "celsius",
            /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
            "currTemperature": 38.0,
            /*ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃*/
            "temperatureCondition": "normal"
            /*ro, opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
        }
    ]
}
```

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15.5.7.10 Search Results of Frequently Appeared Person Events

Request URL

POST /ISAPI/SDT/HFPD/searchTask/result?format=json

Query Parameter

None

Request Message

```
{
    "PagingCond": {
    /*opt, object, paging condition*/
        "taskID":  "test",
        /*req, string, task ID*/
        "searchResultPosition":  0,
        /*req, int, the start position of the search result in the result list, desc:When there are multiple records and you cannot get all search results
at a time, you can search for the records after the specified position next time.*/
        "maxResults":  100
        /*req, int, the maximum number of search results this time*/
    }
}
```

Response Message

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```
{
    "HFPDSearchResult": {
    /*ro, opt, object, searching result*/
        "responseStatusStrg":  "MORE",
        /*ro, req, enum, searching status description, subType:string, desc:"MORE" (searching for more data), "NO MATCHES" (no matched data),"OK" (searching
completed),*/
        "numOfMatches":  1,
        /*ro, opt, int, number of results returned this time*/
        "totalMatches":  1,
        /*ro, opt, int, total number of matched results*/
        "targetInfo": [
        /*ro, opt, array, target information, subType:object*/
            {
                "recordDataIndex":  "test",
                /*ro, req, string, ID of searched record detailed information, desc:corresponds to the node <indexID> in the URI(GET
/ISAPI/SDT/HFPD/searchTask/result/recordData/<indexID>?format=json). Retrun the detailed information of the frequently search target when returning the URL
at the same time*/
                "occurrences":  7,
                /*ro, req, int, appearing times of frequently appeared person*/
                "analysisTime":  "2004-01-01T00:00:00Z",
                /*ro, req, datetime, alarm analysis time*/
                "timeDifference":  "+08:00",
                /*ro, opt, string*/
                "isSummerTime":  false,
                /*ro, opt, bool*/
                "URL":  "test",
                /*ro, req, string, face sub-picture URL, desc:IP and port are local. If the devices cross domains, the upper layer needs to be reassembled
according to the device mapping*/
                "targetId":  "test",
                /*ro, opt, string, target ID, desc:The person target ID in the frequently appeared person detection time is different for different target
ID in each detection*/
                "FaceInfo": [
                /*ro, opt, array, the matched face picture list, subType:object, desc:the same arming linkage corresponds to multiple matched person*/
                    {
                        "FDID":  "2323",
                        /*ro, req, string, face picture library ID, range:[1,64]*/
                        "PID":  "123",
                        /*ro, opt, string, person ID of face picture library, range:[0,64]*/
                        "face_picurl":  "http://127.0.0.1",
                        /*ro, opt, picurl, face picture URL*/
                        "FaceAppendData": {
                        /*ro, opt, object, face picture extended information*/
                            "name":  "张三",
                            /*ro, req, string, name, range:[1,96]*/
                            "gender":  "any",
                            /*ro, req, enum, gender, subType:string, desc:"any"-unlimited condition (used in search),"male", "female", "unknown",*/
                            "bornTime":  "1970-01-01",
                            /*ro, req, date, date of birth*/
                            "city":  "130100",
                            /*ro, opt, string, city NO. of person birth, range:[1,32], desc:see the list of China's cities in all provinces for details*/
                            "certificateType":  "officerID",
                            /*ro, opt, enum, certificate type, subType:string, desc:"officerID"-officer ID,"ID"-identify card,"passport"-passport,"other"-
other certificate*/
                            "certificateNumber":  "test",
                            /*ro, opt, string, certificate No., range:[1,32]*/
                            "ageGroup":  "any",
                            /*ro, req, enum, age group, subType:string, desc:any, child, infant, kid, middle,  middleAged (the middle age), old, prime,
teenager, unknown, young,*/
                            "phoneNumber":  "test"
                            /*ro, opt, string, phone number, range:[1,64]*/
                        },
                        "similarity":  0.780,
                        /*ro, opt, float, the maximum similarity of all matched pictures, range:[0.000,1.000]*/
                        "extendData": [
                        /*ro, opt, array, person extended information, subType:object*/
                            {
                                "extendID":  "1",
                                /*ro, opt, string, person extended information ID*/
                                "enable":  "true",
                                /*ro, opt, string, enable person extended information or not*/
                                "name":  "test",
                                /*ro, opt, string, person tag extended information name, range:[1,96]*/
                                "value":  "test"
                                /*ro, opt, string, person tag extended information, range:[1,64]*/
                            }
                        ]
                    }
                ]
            }
        ]
    }
}
```

15.5.7.11 Get default or configured parameters of frequently appeared person detection

Request URL

GET /ISAPI/SDT/HFPD?format=json

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dc.io.jin@gmail.com

Query Parameter

None

Request Message

None

Response Message

```
{
    "HFPD": {
    /*ro, opt, object*/
        "enabled":  true,
        /*ro, req, bool, whether to enable the function*/
        "analysisDays":  1,
        /*ro, req, int, days of unit time for specified channel, range:[1,30], unit:d, desc:days of unit time for specified channel,range: [1,30],it is 7 days by default*/
        "occurrences":  5,
        /*ro, req, int, face picture frequency, range:[1,100], desc:the default value is 10*/
        "similarity":  80,
        /*ro, req, int, similarity of comparing captured picture with stranger library, range:[0,100], desc:similarity of comparing captured picture with stranger library,range: [80,100],it is 85 by default*/
        "captureTimeInterval":  10,
        /*ro, req, int, capture time interval, range:[0,60], unit:min, desc:it is 10 by default; When the value is set to 0, every appearance will be counted*/
        "monitorIds": ["11111", "22222"],
        /*ro, opt, array, the ID array of camera, subType:string*/
        "taskID": ["1111", "2222"],
        /*ro, opt, array, task ID, subType:string, desc:either task ID or monitorIds is required. Task ID refers to task linkage mode*/
        "FaceLibrary": [
        /*ro, opt, array, face picture library in allowlist for frequently appeared person detection, subType:object, desc:e.g., when comparing the captured face picture with the face picture library, if the similarity reaches configured threshold, the alarm will not be triggered*/
            {
                "FDID":  "test",
                /*ro, req, string, face picture library ID, desc:multiple libraries are separated by commas*/
                "similarity":  80
                /*ro, req, int, similarity of comparing captured picture with stranger library, range:[80,100], desc:it is 85 by default*/
            }
        ],
        "BlackListFaceLibrary": [
        /*ro, opt, array, face picture library in blocklist for frequently appeared person detection, subType:object*/
            {
                "FDID":  "test",
                /*ro, req, string, face picture library ID, desc:multiple libraries are separated by commas*/
                "similarity":  80
                /*ro, req, int, similarity of comparing captured picture with stranger library, range:[80,100], desc:the default value is 85*/
            }
        ]
    }
}
```

15.5.7.12 Set frequently appeared person detection

Request URL

PUT /ISAPI/SDT/HFPD?format=json

Query Parameter

None

Request Message

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```
{
    "HFPD": {
    /*opt, object*/
        "enabled": true,
        /*req, bool, whether to enable the function*/
        "analysisDays": 1,
        /*req, int, number of days of a specified channel, range:[1,30], unit:d, desc:number of days of a specified channel*/
        "occurrences": 5,
        /*req, int, number of appearances, range:[1,100], desc:the default value is 10*/
        "similarity": 80,
        /*req, int, similarity of comparing captured picture with stranger library, range:[0,100], desc:it is 85 by default*/
        "captureTimeInterval": 10,
        /*req, int, capture interval, range:[0,60], unit:min, desc:capture interval*/
        "monitorIds": ["11111", "22222"],
        /*opt, array, array information of camera ID, subType:string*/
        "taskID": ["1111", "2222"],
        /*opt, array, task ID, subType:string, desc:either taskID or monitorIds is required. TaskID refers to task linkage mode*/
        "FaceLibrary": [
        /*opt, array, face picture library in allowlist for frequently appeared person detection, subType:object, desc:e.g., when comparing the captured face picture with the face picture library, if the similarity reaches configured threshold, the alarm will not be triggered*/
            {
                "FDID": "test",
                /*req, string, face picture library ID, desc:face picture library ID*/
                "similarity": 80
                /*req, int, similarity, range:[80,100], desc:similarity*/
            }
        ],
        "BlackListFaceLibrary": [
        /*opt, array, face picture library in blocklist for frequently appeared person detection, subType:object*/
            {
                "FDID": "test",
                /*req, string, face picture library ID, desc:face picture library ID*/
                "similarity": 80
                /*req, int, similarity, range:[80,100], desc:the default value is 85*/
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

15.5.7.13 Frequently appeared person detection

EventType:HFPD

```
{
    "ipAddress": "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address": "test",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo": 80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol": "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress": "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID": 1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or playback on the platform*/
    "dateTime": "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time, desc:alarm trigger time*/
    "activePostCount": 1,
    /*ro, req, int, times that the same alarm has been uploaded, desc:alarm/event frequency*/
    "eventType": "HFPD",
    /*ro, req, enum, event type, subType:string, desc:alarm/event type,"HFPD"-frequently appeared person event,the maximum size is 128 bytes*/
    "eventState": "active",
    /*ro, req, enum, continuous event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the
```

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```
event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "High frequency personnel detection",
    /*ro, req, string, event description, desc:frequently appeared person detection*/
    "channelName":  "12345",
    /*ro, opt, string, channel name*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID*/
    "faceCapture": [
    /*ro, opt, array, subType:object*/
        {
            "errorCode":  1,
            /*ro, opt, int, error code, desc:when analyzing face picture failed, the corresponding error code will be returned*/
            "errorMsg":  "ok",
            /*ro, opt, string, error description, desc:when analyzing face picture failed, the corresponding error information will be returned*/
            "image":  "DSS",
            /*ro, req, enum, picture, subType:string, desc:picture URL for face comparison and recognization; the maximum size is 256 bytes,and it is valid
only when errcode is 1 and errMsg is "ok"; e.g.,http://10.7.52.174:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/011_163957_23006&token=7a57a5a7ffffffffc1a0316369671314*/
            "traceUuid":  "sddddddddff",
            /*ro, opt, string, target ID, desc:one person corresponds to one target ID in multiple alarms (related URI:
/ISAPI/SDT/Management/Task/facePicture); the maximum size is 32*/
            "traceIdx":  1,
            /*ro, opt, int, picture index, desc:index No. of picture that has the highest similarity in the matched pictures of a person (related URI:
/ISAPI/SDT/Management/Task/facePicture)*/
            "targetAttrs": {
            /*ro, opt, object, target attributes*/
                "deviceId":  "1",
                /*ro, opt, string, device ID*/
                "deviceChannel":  1,
                /*ro, opt, int, device channel No.*/
                "deviceName":  "shebei",
                /*ro, opt, string, device name*/
                "faceTime":  "1970-01-01T00:00:00+08:00",
                /*ro, opt, datetime, capture time*/
                "rect": {
                /*ro, opt, object, face picture frame, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "bkgUrl":  "http://absddsdfsdfsdk.jpg"
                /*ro, opt, string, background picture URL*/
            },
            "faces": [
            /*ro, opt, array, face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on,
subType:object, desc:face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on*/
                {
                    "faceId":  1,
                    /*ro, opt, int, face picture ID*/
                    "relateID":  "255",
                    /*ro, opt, string, arming relation ID*/
                    "faceRect": {
                    /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "faceMark": {
                    /*ro, opt, object*/
                        "leftEye": {
                        /*ro, opt, object, left eye's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.130,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                            "y":  0.870
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                        },
                        "rightEye": {
                        /*ro, opt, object, right eye's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.130,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                            "y":  0.870
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                        },
                        "noseTip": {
                        /*ro, opt, object, nose tip's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.130,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                            "y":  0.870
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                        },
                        "leftMouth": {
                        /*ro, opt, object, left mouth corner's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.130,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
```

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```
                    "y":  0.870
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                },
                "rightMouth": {
                /*ro, opt, object, right mouth corner's coordinates, desc:the origin is the upper-left corner of the screen*/
                    "x":  0.130,
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                    "y":  0.870
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates*/
                }
            },
            "recommendFaceRect": {
            /*ro, opt, object, recommended face frame coordinates, desc:the origin is the upper-left corner of the screen*/
                "height":  1.000,
                /*ro, opt, float, height, range:[0.000,1.000]*/
                "width":  1.000,
                /*ro, opt, float, width, range:[0.000,1.000]*/
                "x":  0.000,
                /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
            },
            "facePose": {
            /*ro, opt, object, face angle*/
                "pitch":  0.998,
                /*ro, opt, float, face pitch angle, range:[-90,90], desc:face pitching up is regarded as positive range: [-90,90]*/
                "yaw":  0.876,
                /*ro, opt, float, face yaw angle, face yawing left is regarded as positive, range:[-90,90], desc:float,face siding left or right
angle,range: [-90,90],face siding left is regarded as positive,correct to three decimals. This node does not exists if it is invalid*/
                "roll":  0.623
                /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive, range:[-90,90], desc:range: [-90,90]*/
            },
            "age": {
            /*ro, opt, object, age*/
                "range":  5,
                /*ro, opt, int, age deviation, desc:age deviation*/
                "value":  23,
                /*ro, opt, int, age*/
                "ageGroup":  "any"
                /*ro, req, enum, age group, subType:string, desc:“any”, “child”, “infant”, “kid”, “middle”,  “middleAged” (the middle age), “old”,
“prime”, “teenager”, “unknown”, “young”*/
            },
            "gender": {
            /*ro, opt, object, gender*/
                "confidence":  99.99990,
                /*ro, opt, float, confidence*/
                "value":  "male"
                /*ro, opt, enum, gender, subType:string, desc:"male", "female"*/
            },
            "glass": {
            /*ro, opt, object, whether the person wears glasses*/
                "confidence":  99.94499999999999,
                /*ro, opt, float, confidence*/
                "value":  "yes"
                /*ro, opt, enum, value, subType:string, desc:"yes", "no"*/
            },
            "smile": {
            /*ro, opt, object, whether the person wears a smile*/
                "confidence":  99.99990,
                /*ro, opt, float, confidence*/
                "value":  "yes"
                /*ro, opt, enum, value, subType:string, desc:"yes", "no"*/
            },
            "modeldata":  "xxxx",
            /*ro, opt, string, target model data, desc:the maximum size is 5 × 1024 bytes*/
            "URL":  "test",
            /*ro, opt, string, face thumbnail URL, the maximum size is 256*/
            "stayDuration":  20,
            /*ro, opt, int, dwell time duration in the image, unit:ms*/
            "occurrences":  7,
            /*ro, opt, int, number of appearances for the target*/
            "faceExpression": {
            /*ro, opt, object, facial expression*/
                "confidence":  99.99990,
                /*ro, opt, float, confidence*/
                "value":  "angry"
                /*ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-
faced", "sad", "surprised", "unknown"*/
            },
            "algorithmInfo": {
            /*ro, opt, object, algorithm version information, this field is required for secondary analysis*/
                "algorithmId":  "test",
                /*ro, opt, string, algorithm package ID*/
                "algoVendorId":  "test",
                /*ro, opt, string, manufacturer ID*/
                "algoVersionId":  "test",
                /*ro, opt, string, algorithm version No.*/
                "algoChipType":  "test",
                /*ro, opt, string, chip type*/
                "algoModelVersion":  "test"
                /*ro, opt, string, model version*/
            }
        }
```


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```
            ],
            "recordDataIndex":  "test",
            /*ro, req, string, search index No. of frequently appeared person alarm, desc:search index No. of frequently appeared person alarm*/
            "recordData": [
            /*ro, opt, array, history data records, subType:object*/
                {
                    "captureTime":  "1970-01-01T00:00:00+08:00",
                    /*ro, opt, datetime, capture time, desc:in ISO 8601 format*/
                    "captureSite":  "3#",
                    /*ro, opt, string, capture site (camera name)*/
                    "similarity":  1.000,
                    /*ro, opt, float, similarity, range:[0.000,1.000]*/
                    "picUrl":  "xxx",
                    /*ro, opt, string, background picture URL*/
                    "subpicUrl":  "xxx",
                    /*ro, opt, string, face picture thumbnail URL*/
                    "rowKey":  "test"
                    /*ro, opt, string, main key of face picture library, desc:searching by rowKey can be more efficient*/
                }
            ],
            "targetId":  "test",
            /*ro, opt, string, target ID, desc:the person target ID during the frequently appeared person detection, the ID varies according to different
targets*/
            "correctionTargetIds": ["XXX1", "XXX2"],
            /*ro, opt, array, device internal cluster correction of targetIds, subType:string*/
            "FaceInfo": [
            /*ro, opt, array, person information of matched face pictures, subType:object, desc:array of object,matched face picture information. Multiple
matched face pictures can be linked*/
                {
                    "FDID":  "2323",
                    /*ro, req, string, face picture library ID*/
                    "faceLibType":  "ordinary",
                    /*ro, opt, enum, face picture library type, subType:string, desc:"ordinary" (normal library), "stranger" (stranger library). The default
value is "ordinary" ,and no alarm will be uploaded if the captured face picture matches that in the normal face picture library*/
                    "PID":  "123",
                    /*ro, req, string, person ID of face picture library, desc:the maximum size is 64*/
                    "face_picurl":  "http://127.0.0.1:6120/face_picurl",
                    /*ro, opt, string, face picture URL*/
                    "FaceAppendData": {
                    /*ro, opt, object, extended information about the face*/
                        "name":  "张三",
                        /*ro, req, string, name, desc:the maximum length is 96*/
                        "gender":  "any",
                        /*ro, req, enum, gender, subType:string, desc:"any"-unlimited condition, "male", “female”, “unknown"*/
                        "bornTime":  "2004-05-03",
                        /*ro, req, string, date of birth*/
                        "city":  "130100",
                        /*ro, opt, string, city code of person birth, desc:city No. of person birth*/
                        "certificateType":  "officerID",
                        /*ro, opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "passport" (passport), "other" (the other
certificate)*/
                        "certificateNumber":  "test",
                        /*ro, opt, string, certificate No., desc:the maximum length is 32*/
                        "ageGroup":  "any",
                        /*ro, req, enum, age group, subType:string, desc:“any”, “child”, “infant”, “kid”, “middle”,  “middleAged” (the middle age), “old”,
“prime”, “teenager”, “unknown”, “young”*/
                        "phoneNumber":  "test",
                        /*ro, opt, string, phone number, desc:the maximum length is 64*/
                        "cityName":  "石家庄市",
                        /*ro, opt, string, city (corresponding to the field "city"), range:[0,32], desc:city (corresponding to the field "city)*/
                        "provinceName":  "河北省"
                        /*ro, opt, string, province, range:[0,32], desc:province*/
                    },
                    "similarity":  0.780,
                    /*ro, opt, float, the maximum similarity of all matched pictures for the person, range:[0.000,1.000]*/
                    "extendData": [
                    /*ro, opt, array, person extended information, subType:object*/
                        {
                            "extendID":  "1",
                            /*ro, opt, string, person extended information ID, desc:it starts from 1*/
                            "enable":  true,
                            /*ro, opt, bool, enable person extended information or not*/
                            "name":  "test",
                            /*ro, opt, string, name of person tag extended information, desc:the maximum length is 96*/
                            "value":  "test"
                            /*ro, opt, string, person tag extended information, desc:the maximum length is 64*/
                        }
                    ]
                }
            ]
        }
    ],
    "URLCertificationType":  "digest"
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (it is for the cloud storage protocol), "digest" (digest authentication,
it is for the device local storage)*/
}
```

15.5.8 人员实时聚档

15.5.8.1 Edit the parameters of clearing personnel archives automatically

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Request URL

PUT /ISAPI/SDT/personnelFile/autoClear?format=json

Query Parameter

None

Request Message

```
{
    "enabled":  true,
    /*req, bool, whether to enable clearing automatically*/
    "timeOfDay":  "00:00:00",
    /*opt, string, time to execute automatic clearing, desc:If this node is not configured, the default time is 01:00:00*/
    "interval":  10,
    /*opt, int, execution interval, range:[1,180], unit:s, desc:the default value is 10*/
    "frequency":  10
    /*opt, int, frequency, range:[1,10], desc:the default value is 1*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

15.5.8.2 Get the parameters of clearing personnel archives automatically

Request URL

GET /ISAPI/SDT/personnelFile/autoClear?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "enabled":  true,
    /*ro, req, bool, whether to enable clearing automatically*/
    "timeOfDay":  "00:00:00",
    /*ro, opt, string, time to execute automatic clearing., desc:If this node is not configured, the default time is 01:00:00*/
    "interval":  10,
    /*ro, opt, int, execution interval, the value is between 1 and 180 and the default value is 10,unit: day, range:[1,180], unit:s, desc:the default value
is 10*/
    "frequency":  10
    /*ro, opt, int, frequency, range:[1,10], desc:the default value is 1*/
}
```

15.5.8.3 Get personnel archive capability

Request URL

GET /ISAPI/SDT/personnelFile/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "personnelFileCap": {
    /*ro, req, object, personnel archive capability*/
        "enabled":  "true,false",
        /*ro, req, string, whether to enable the function*/
        "similarity": {
        /*ro, req, object, similarity, desc:default value: 85*/
            "@min":  0,
            /*ro, opt, int, minimum value, range:[0,100]*/
            "@max":  100
            /*ro, opt, int, the maximum value, range:[0,100]*/
        },
        "monitorIds": {
        /*ro, req, object, array information of camera ID*/
            "@min":  128,
            /*ro, opt, int, the minimum value*/
            "@max":  1024
            /*ro, opt, int, the maximum value*/
        },
        "FDID": {
        /*ro, req, object, face picture library ID*/
            "@min":  64,
            /*ro, opt, int, the minimum value*/
            "@max":  1024
            /*ro, opt, int, the maximum value*/
        },
        "taskID": {
        /*ro, opt, object, task ID, desc:either taskID or monitorIds should exist, taskID implies task linkage mode*/
            "size":  4
            /*ro, opt, int, array size*/
        },
        "strangerEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false],
            /*ro, opt, array, optional value, subType:bool*/
            "def":  true
            /*ro, opt, bool, the default value*/
        },
        "autoClear": {
        /*ro, opt, object, whether it supports clearing personnel archives automatically, desc:related URI: /ISAPI/SDT/personnelFile/autoClear?format=json*/
            "enabled": {
            /*ro, req, object, whether to enable personnel archive*/
                "@opt": [true, false]
                /*ro, opt, array, optional value, subType:bool*/
            },
            "timeOfDay": {
            /*ro, opt, object, execution time in a day. If this node is not configured,the default value is 00:00:00, desc:execution time in a day. If this
node is not configured,the default value is 00:00:00*/
                "@min":  0,
                /*ro, opt, int, the minimum value*/
                "@max":  0
                /*ro, opt, int, the maximum value*/
            },
            "interval": {
            /*ro, opt, object, execution time interval, unit: day, desc:the default value is 10*/
                "@min":  1,
                /*ro, opt, int, the minimum value, range:[1,180]*/
                "@max":  180
                /*ro, opt, int, the maximum value, range:[1,180]*/
            },
            "frequency": {
            /*ro, opt, object, frequency, desc:the default value is 10*/
                "@min":  0,
                /*ro, opt, int, the minimum value, range:[0,10]*/
                "@max":  10
                /*ro, opt, int, the maximum value, range:[0,10]*/
            }
        },
        "recordSearch": {
        /*ro, opt, object*/
            "searchResultPosition": {
            /*ro, req, object, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the
result list, you can mark the end position and get the following records after the marked position in the next search*/
                "@min":  0,
                /*ro, opt, int, the minimum value*/
                "@max":  0
                /*ro, opt, int, the maximum value*/
            },
            "maxResults": {
            /*ro, req, object, the maximum number of search results this time*/
                "@min":  0,
                /*ro, opt, int, the minimum value*/
                "@max":  0
                /*ro, opt, int, the maximum value*/
            },
            "startCreateTime ": {
            /*ro, opt, object, start time in ISO 8601 time format*/
                "@min":  0,
                /*ro, opt, int, the minimum value*/
                "@max":  0
                /*ro, opt, int, the maximum value*/
            },
```

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```
            "endCreateTime" : {
            /*ro, opt, object, end time, desc:ISO8601 time format*/
                "@min":  0,
                /*ro, opt, int, the minimum value*/
                "@max":  0
                /*ro, opt, int, the maximum value*/
            }
        },
        "isSupportPersonStrategy":  true,
        /*ro, opt, bool, whether it supports configuring the strategy for archiving personnel, desc:related URI:
/ISAPI/SDT/personnelFile/personStrategy/capabilities?format=json*/
        "isSupportTrajectoryAnalysisCfg":  true
        /*ro, opt, bool*/
    }
}
```

15.5.8.4 Get people counting statistics reports

Request URL

GET /ISAPI/SDT/personnelFile/dataStatistics/task/<taskID>/result?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "StatisticsTaskResult": {
    /*ro, opt, object*/
        "taskID":  "test",
        /*ro, req, string, task ID*/
        "reportType":  "AnnualReport",
        /*ro, req, enum, report type, subType:string, desc:"AnnualReport", "MonthlyReport", "WeeklyReport", "DailyReport"*/
        "Report": [
        /*ro, opt, array, subType:object*/
            {
                "dateTime":  "1970-01-01T00:00:00+08:00",
                /*ro, req, datetime, statistics time, desc:statistics time*/
                "Gender": {
                /*ro, opt, object, statistics by gender*/
                    "female":  0,
                    /*ro, req, int, female*/
                    "male":  0,
                    /*ro, req, int, male*/
                    "unknown":  0
                    /*ro, req, int, unknown*/
                },
                "AgeGroup": {
                /*ro, opt, object, statistics by age group*/
                    "child":  0,
                    /*ro, req, int, children*/
                    "young":  0,
                    /*ro, req, int, young age*/
                    "middle":  0,
                    /*ro, req, int, middle age*/
                    "old":  0,
                    /*ro, req, int, old age*/
                    "unknown":  0
                    /*ro, req, int, unknown*/
                },
                "PeopleCounting": {
                /*ro, opt, object, people counting*/
                    "peopleNum":  0
                    /*ro, req, int, the number of people*/
                }
            }
        ]
    }
}
```

15.5.8.5 Generate people counting statistics report

Request URL

GET /ISAPI/SDT/personnelFile/dataStatistics/task?format=json

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Query Parameter

None

Request Message

None

Response Message

```
{
    "StatisticsTaskId": [
    /*ro, opt, array, subType:object*/
        {
            "taskID":  "test",
            /*ro, req, string, task ID*/
            "progress":  100
            /*ro, req, int, task progress of generating statistics report*/
        }
    ]
}
```

15.5.8.6 Create people counting statistics reports

Request URL

POST /ISAPI/SDT/personnelFile/dataStatistics/task?format=json

Query Parameter

None

Request Message

```
{
    "StatisticsTaskCond": {
    /*req, object*/
        "monitorPointIDList": [""],
        /*req, array, camera No., subType:string*/
        "reportType":  "AnnualReport",
        /*req, enum, report type, subType:string, desc:"AnnualReport", "MonthlyReport", "WeeklyReport", "DailyReport"*/
        "statisticsTime":  "1970-01-01T00:00:00+08:00",
        /*req, datetime, statistics time*/
        "statisticsSource":  "realname",
        /*opt, enum, statistics resource, subType:string, desc:"realname", "stranger"*/
        "statisticsAttr":  "gender",
        /*req, enum, statistics mode, subType:string, desc:"gender", "ageGroup", "peopleCounting"*/
        "FDIDList": ["aaa", "bbb"]
        /*opt, array, subType:string*/
    }
}
```

Response Message

```
{
    "StatisticsTaskId": {
    /*ro, req, object*/
        "taskID":  "test"
        /*ro, req, string, task ID*/
    }
}
```

15.5.8.7 Delete people counting statistics

report **Request URL**

DELETE /ISAPI/SDT/personnelFile/dataStatistics/task?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

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Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when statusCode is not 1*/
}
```

15.5.8.8 Perform personnel archive statistics

Request URL

POST /ISAPI/SDT/personnelFile/dataStatistics?format=json

Query Parameter

None

Request Message

```
{
    "StatisticsCond": {
    /*opt, object*/
        "startDate":  "1970-01-01",
        /*req, date, start date*/
        "endDate":  "1970-01-01"
        /*req, date, end date*/
    }
}
```

Response Message

```
{
    "StatisticsInfo": {
    /*ro, opt, object, statistics information*/
        "totalNum":  123,
        /*ro, req, int, total number of archived persons*/
        "identifiedNum":  123,
        /*ro, req, int, number of persons with identities confirmed*/
        "passerbyNum":  123,
        /*ro, req, int, number of pedestrians*/
        "personnelFileNum": [
        /*ro, opt, array, the number of items of person information, subType:object*/
            {
                "date":  "1970-01-01",
                /*ro, req, date, date*/
                "identifiedNum":  123,
                /*ro, req, int, number of persons with identities confirmed*/
                "passerbyNum":  123
                /*ro, req, int, number of pedestrians*/
            }
        ]
    }
}
```

15.5.8.9 Get the information of all algorithm packages

Request URL

GET /ISAPI/SDT/personnelFile/personStrategy/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "PersonStrategy": {
```

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```
    /*ro, opt, object*/
        "enabled": {
        /*ro, opt, object, whether to enable the error correction strategy of face pictures*/
            "@opt": [true, false],
            /*ro, opt, array, options, subType:bool*/
            "@def":  true
            /*ro, opt, bool, the default value*/
        },
        "Similarity": {
        /*ro, opt, object, similarity*/
            "compareCommonSimilarity": {
            /*ro, opt, object, comparison similarity threshold*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,100]*/
                "@max":  100,
                /*ro, req, int, the maximum value, range:[0,100]*/
                "@def":  79
                /*ro, req, int, the default value, range:[0,100]*/
            },
            "compareCommonChildSimilarity": {
            /*ro, opt, object, comparison similarity threshold*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,100]*/
                "@max":  100,
                /*ro, req, int, maximum value, range:[0,100]*/
                "@def":  79
                /*ro, req, int, range:[0,100]*/
            },
            "addFusedSimilarity": {
            /*ro, opt, object, model fusion threshold*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,100]*/
                "@max":  100,
                /*ro, req, int, the maximum value, range:[0,100]*/
                "@def":  86
                /*ro, req, int, the default value, range:[0,100]*/
            },
            "addFusedChildSimilarity": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,100]*/
                "@max":  100,
                /*ro, req, int, the maximum value, range:[0,100]*/
                "@def":  95
                /*ro, req, int, range:[0,100]*/
            },
            "mergeSimilarity": {
            /*ro, opt, object, similarity threshold for merging personnel archives*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,100]*/
                "@max":  100,
                /*ro, req, int, the maximum value, range:[0,100]*/
                "@def":  86
                /*ro, req, int, the default value, range:[0,100]*/
            },
            "mergeChildSimilarity": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,100]*/
                "@max":  100,
                /*ro, req, int, the maximum value, range:[0,100]*/
                "@def":  100
                /*ro, req, int, range:[0,100]*/
            },
            "mergeMinSimilarity": {
            /*ro, opt, object*/
                "@min": 0,
                /*ro, req, int, the minimum value, range:[0,100]*/
                "@max": 100,
                /*ro, req, int, the maximum value, range:[0,100]*/
                "@def":  86
                /*ro, req, int, range:[0,100]*/
            },
            "addCommonSimilarity": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,100]*/
                "@max":  100,
                /*ro, req, int, the maximum value, range:[0,100]*/
                "@def":  88
                /*ro, req, int, range:[0,100]*/
            },
            "addCoverSimilarity": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,100]*/
                "@max":  100,
                /*ro, req, int, the maximum value, range:[0,100]*/
                "@def":  93
                /*ro, req, int, range:[0,100]*/
            },
            "addCoverChildSimilarity": {
            /*ro, opt, object*/
```

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```
            "@min":  0,
            /*ro, req, int, the minimum value, range:[0,100]*/
            "@max":  100,
            /*ro, req, int, the maximum value, range:[0,100]*/
            "@def":  93
            /*ro, req, int, range:[0,100]*/
        },
        "compareCoverSimilarity": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, the minimum value, range:[0,100]*/
            "@max":  100,
            /*ro, req, int, the maximum value, range:[0,100]*/
            "@def":  79
            /*ro, req, int, range:[0,100]*/
        }
    },
    "Score": {
    /*ro, opt, object, score parameters*/
        "SnapScore": {
        /*ro, opt, object, captured picture score*/
            "visibleScore": {
            /*ro, opt, object, visibility score*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                "@def":  0.200
                /*ro, req, float, the default value, range:[0.000,1.000]*/
            },
            "totalScore": {
            /*ro, opt, object, total score*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                "@def":  0.100
                /*ro, req, float, the default value, range:[0.000,1.000]*/
            },
            "clearityScore": {
            /*ro, opt, object, clarity score*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                "@def":  0.100
                /*ro, req, float, the default value, range:[0.000,1.000]*/
            }
        },
        "StrangerScore": {
        /*ro, opt, object, stranger score*/
            "pupillaryDistance": {
            /*ro, opt, object, pupillary distance*/
                "@min":  18,
                /*ro, req, int, the minimum value, range:[18,719]*/
                "@max":  719,
                /*ro, req, int, the maximum value, range:[18,719]*/
                "@def":  18
                /*ro, req, int, the default value, range:[18,719]*/
            },
            "pitchAngle": {
            /*ro, opt, object, pitch angle*/
                "@min":  1.000,
                /*ro, req, float, the minimum value, range:[0.000,90.000]*/
                "@max":  90.000,
                /*ro, req, float, the maximum value, range:[0.000,90.000]*/
                "@def":  25.000
                /*ro, req, float, the default value, range:[0.000,90.000]*/
            },
            "yawAngle": {
            /*ro, opt, object, yawAngle*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,90.000]*/
                "@max":  90.0000,
                /*ro, req, float, the maximum value, range:[0.000,90.000]*/
                "@def":  30.000
                /*ro, req, float, the default value, range:[0.000,90.000]*/
            },
            "clearityScore": {
            /*ro, opt, object, clarity score*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                "@def":  0.220
                /*ro, req, float, the default value, range:[0.000,1.000]*/
            },
            "visibleScore": {
            /*ro, opt, object, visibility score*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                "@def":  0.510
```

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```
            /*ro, req, float, the default valu, range:[0.000,1.000]*/
        },
        "totalScore": {
        /*ro, opt, object, total score*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "@def":  0.100
            /*ro, req, float, the default value, range:[0.000,1.000]*/
        }
    },
    "FusedScore": {
    /*ro, opt, object, model fusion score*/
        "pupillaryDistance": {
        /*ro, opt, object, pupillary distance*/
            "@min":  18,
            /*ro, req, int, the minimum value, range:[18,719]*/
            "@max":  719,
            /*ro, req, int, the maximum value, range:[18,719]*/
            "@def":  18
            /*ro, req, int, the default value, range:[18,719]*/
        },
        "pitchAngle": {
        /*ro, opt, object, pitch angle*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,90.000]*/
            "@max":  90.000,
            /*ro, req, float, the maximum value, range:[0.000,90.000]*/
            "@def":  25.000
            /*ro, req, float, the default value, range:[0.000,90.000]*/
        },
        "yawAngle": {
        /*ro, opt, object, yaw angle*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,90.000]*/
            "@max":  90.000,
            /*ro, req, float, the maximum value, range:[0.000,90.000]*/
            "@def":  30.000
            /*ro, req, float, the default value, range:[0.000,90.000]*/
        },
        "clearityScore": {
        /*ro, opt, object, clarity score*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "@def":  0.200
            /*ro, req, float, the default value, range:[0.000,1.000]*/
        },
        "visibleScore": {
        /*ro, opt, object, visibility score*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "@def":  0.500
            /*ro, req, float, the default value, range:[0.000,1.000]*/
        },
        "totalScore": {
        /*ro, opt, object, total score*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "@def":  0.100
            /*ro, req, float, the default value, range:[0.000,1.000]*/
        }
    },
    "CoverScore": {
    /*ro, opt, object*/
        "pupillaryDistance": {
        /*ro, opt, object*/
            "@min":  18,
            /*ro, req, int, the minimum value*/
            "@max":  719,
            /*ro, req, int, the maximum value*/
            "@def":  18
            /*ro, req, int*/
        },
        "pitchAngle": {
        /*ro, opt, object*/
            "@min":  0.0,
            /*ro, req, float, the minimum value*/
            "@max":  90.0,
            /*ro, req, float, the maximum value*/
            "@def":  25.0
            /*ro, req, float*/
        },
        "yawAngle": {
        /*ro, opt, object*/
            "@min":  0.0,
            /*ro, req, float, the minimum value*/
```

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```
                    "@max":  90.0,
                    /*ro, req, float, the maximum value*/
                    "@def":  30.0
                    /*ro, req, float*/
                },
                "clearityScore": {
                /*ro, opt, object*/
                    "@min":  0.0,
                    /*ro, req, float, the minimum value*/
                    "@max":  1.0,
                    /*ro, req, float, the maximum value*/
                    "@def":  0.2
                    /*ro, req, float*/
                },
                "visibleScore": {
                /*ro, opt, object*/
                    "@min":  0.0,
                    /*ro, req, float, the minimum value*/
                    "@max":  1.0,
                    /*ro, req, float, the maximum value*/
                    "@def":  0.5
                    /*ro, req, float*/
                },
                "totalScore": {
                /*ro, opt, object*/
                    "@min":  0.0,
                    /*ro, req, float, the minimum value*/
                    "@max":  1.0,
                    /*ro, req, float, the maximum value*/
                    "@def":  0.1
                    /*ro, req, float*/
                }
            }
        },
        "Fused": {
        /*ro, opt, object, model fusion parameters, desc:object,model fusion parameters*/
            "constantFusedModelNum": {
            /*ro, opt, object, number of models to be fused, desc:it is between 1 and 10 and the default value is 5*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  10,
                /*ro, req, int, the maximum value*/
                "@def":  5
                /*ro, req, int, the default value*/
            },
            "constantFusedModelTime": {
            /*ro, opt, object, update time of model fusion, desc:the value is between 0 and 12 and the default value is 0 (0 means model fusion will not be
updated), unit: month. when this node is configured, the latest constantFusedModelNum faces should be fused again after the constantFusedModelTime*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  12,
                /*ro, req, int, the maximum value*/
                "@def":  0
                /*ro, req, int, the default value*/
            },
            "constantChildAge": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  18,
                /*ro, req, int, the maximum value*/
                "@def":  8
                /*ro, req, int, the default value*/
            },
            "switchIqaEmptyFilter": {
            /*ro, opt, object*/
                "@opt": [true, false],
                /*ro, req, array, options, subType:bool*/
                "@def": true
                /*ro, req, bool, the default value*/
            },
            "switchCoverCheckGender": {
            /*ro, opt, object*/
                "@opt": [true, false],
                /*ro, req, array, options, subType:bool*/
                "@def":  true
                /*ro, req, bool, the default value*/
            },
            "switchMergeDealChild": {
            /*ro, opt, object*/
                "@opt": [true, false],
                /*ro, req, array, options, subType:bool*/
                "@def":  false
                /*ro, req, bool, the default value*/
            },
            "switchStrangerAddChild": {
            /*ro, opt, object*/
                "@opt": [true, false],
                /*ro, req, array, options, subType:bool*/
                "@def":  false
                /*ro, req, bool, the default value*/
            },
            "handleType": {
            /*ro, opt, object*/
```

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```
            /*ro, opt, object*/
                "@opt": ["cover", "fuse"],
                /*ro, req, array, options, subType:string*/
                "@def":  "cover"
                /*ro, req, string, the default value*/
            }
        }
    },
    "cacheCompareSimilarity": {
    /*ro, opt, object*/
        "@min":  0,
        /*ro, opt, int, the minimum value, range:[0,100]*/
        "@max":  100,
        /*ro, opt, int, the maximum value, range:[0,100]*/
        "@def":  50
        /*ro, req, int, the default value, range:[0,100]*/
    },
    "strangeSwitch": {
    /*ro, opt, object*/
        "@opt": [true, false],
        /*ro, req, array, options, subType:bool*/
        "@def":  true
        /*ro, req, bool, the default value*/
    },
    "coverNum": {
    /*ro, opt, object*/
        "@min":  1,
        /*ro, opt, int, the minimum value, range:[1,10]*/
        "@max":  10,
        /*ro, opt, int, the maximum value, range:[1,10]*/
        "@def":  1
        /*ro, req, int, the default value, range:[1,10]*/
    },
    "humanCompareCommonSimilarity": {
    /*ro, opt, object*/
        "@min":  0,
        /*ro, req, int, the minimum value, range:[0,100]*/
        "@max":  100,
        /*ro, req, int, the maximum value, range:[0,100]*/
        "@def":  80
        /*ro, req, int, the default value, range:[0,100]*/
    }
}
```

15.5.8.10 Get the strategy parameters for archiving personnel archives

Request URL

GET /ISAPI/SDT/personnelFile/personStrategy?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "PersonStrategy": {
    /*ro, opt, object*/
        "enabled":  true,
        /*ro, opt, bool, whether to enable the error correction strategy of face pictures*/
        "Similarity": {
        /*ro, opt, object, similarity*/
            "compareCommonSimilarity":  79,
            /*ro, opt, int, comparison similarity threshold, range:[0,100]*/
            "compareCommonChildSimilarity":  79,
            /*ro, opt, int, range:[0,100]*/
            "addFusedSimilarity":  86,
            /*ro, opt, int, model fusion threshold, range:[0,100]*/
            "addFusedChildSimilarity":  95,
            /*ro, opt, int, range:[0,100]*/
            "mergeSimilarity":  86,
            /*ro, opt, int, similarity threshold for merging personnel archives, range:[0,100]*/
            "mergeChildSimilarity":  100,
            /*ro, opt, int, range:[0,100]*/
            "mergeMinSimilarity":  83,
            /*ro, opt, int, range:[0,100]*/
            "addCommonSimilarity":  88,
            /*ro, opt, int, range:[0,100]*/
            "addCoverSimilarity":  93,
            /*ro, opt, int, range:[0,100]*/
            "addCoverChildSimilarity":  93,
            /*ro, opt, int, range:[0,100]*/
            "compareCoverSimilarity":  79
            /*ro, opt, int, range:[0,100]*/
```

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```
        },
        "Score": {
        /*ro, opt, object, score parameters*/
            "SnapScore": {
            /*ro, opt, object, captured picture score*/
                "visibleScore":  0.1,
                /*ro, opt, float, visibility score, range:[0.000,1.000]*/
                "totalScore":  0.1,
                /*ro, opt, float, total score, range:[0.000,1.000]*/
                "clearityScore":  0.1
                /*ro, opt, float, sharpness score, range:[0.00,1.000]*/
            },
            "StrangerScore": {
            /*ro, opt, object, stranger score*/
                "pupillaryDistance":  18,
                /*ro, opt, int, pupil distance, range:[18,719]*/
                "pitchAngle":  25.0,
                /*ro, opt, float, angle of pitch, range:[0.000,90.000]*/
                "yawAngle":  30.000,
                /*ro, opt, float, yaw angle, range:[0.000,90.000]*/
                "clearityScore":  0.22,
                /*ro, opt, float, Sharpness Score, range:[0.000,1.000]*/
                "visibleScore":  0.51,
                /*ro, opt, float, visibility score, range:[0.000,1.000]*/
                "totalScore":  0.1
                /*ro, opt, float, total score, range:[0.000,1.000]*/
            },
            "FusedScore": {
            /*ro, opt, object, model fusion score*/
                "pupillaryDistance":  18,
                /*ro, opt, int, pupil distance, range:[18,719]*/
                "pitchAngle":  25.0,
                /*ro, opt, float, angle of pitch, range:[0.000,90.000]*/
                "yawAngle":  30.000,
                /*ro, opt, float, yaw angle, range:[0.000,90.000]*/
                "clearityScore":  0.2,
                /*ro, opt, float, sharpness score, range:[0.000,1.000]*/
                "visibleScore":  0.5,
                /*ro, opt, float, visibility score, range:[0.000,1.000]*/
                "totalScore":  0.1
                /*ro, opt, float, total score, range:[0.000,1.000]*/
            },
            "CoverScore": {
            /*ro, opt, object*/
                "pupillaryDistance":  18,
                /*ro, opt, int, pupil distance, range:[18,719]*/
                "pitchAngle":  25.0,
                /*ro, opt, float, angle of pitch, range:[0.000,90.000]*/
                "yawAngle":  30.000,
                /*ro, opt, float, yaw angle, range:[0.000,90.000]*/
                "clearityScore":  0.2,
                /*ro, opt, float, sharpness score, range:[0.000,1.000]*/
                "visibleScore":  0.5,
                /*ro, opt, float, visibility score, range:[0.000,1.000]*/
                "totalScore":  0.1
                /*ro, opt, float, total score, range:[0.000,1.000]*/
            }
        },
        "Fused": {
        /*ro, opt, object, model fusion parameters, desc:object,model fusion parameters*/
            "constantFusedModelNum":  5,
            /*ro, opt, int, number of models to be fused*/
            "constantFusedModelTime":  0,
            /*ro, opt, int, update time of model fusion, desc:update time of model fusion. When this node is configured,the latest constantFusedModelNum
faces should be fused again after the constantFusedModelTime. The value is between 0 and 12 and the default value is 0 (0 means model fusion will not be
updated),unit: month*/
            "constantChildAge":  8,
            /*ro, opt, int, range:[1,18]*/
            "switchIqaEmptyFilter":  true,
            /*ro, opt, bool*/
            "switchCoverCheckGender":  true,
            /*ro, opt, bool*/
            "switchMergeDealChild":  false,
            /*ro, opt, bool*/
            "switchStrangerAddChild":  false,
            /*ro, opt, bool*/
            "handleType":  "cover"
            /*ro, opt, enum, subType:string*/
        }
    },
    "cacheCompareSimilarity":  50,
    /*ro, opt, int, range:[0,100]*/
    "strangeSwitch":  true,
    /*ro, opt, bool*/
    "coverNum":  1,
    /*ro, opt, int, range:[1,10]*/
    "humanCompareCommonSimilarity":  80
    /*ro, opt, int, range:[0,100]*/
}
```

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15.5.8.11 Set the strategy parameters for archiving personnel archives

Request URL

PUT /ISAPI/SDT/personnelFile/personStrategy?format=json

Query Parameter

None

Request Message

```
{
    "PersonStrategy": {
    /*opt, object*/
        "enabled":  true,
        /*opt, bool, whether to enable the error correction strategy of face pictures*/
        "Similarity": {
        /*opt, object, similarity*/
            "compareCommonSimilarity":  79,
            /*opt, int, comparison similarity threshold, range:[0,100]*/
            "compareCommonChildSimilarity":  79,
            /*opt, int, range:[0,100]*/
            "addFusedSimilarity":  86,
            /*opt, int, model fusion threshold, range:[0,100]*/
            "addFusedChildSimilarity":  95,
            /*opt, int, range:[0,100]*/
            "mergeSimilarity":  86,
            /*opt, int, similarity threshold for merging personnel archives, range:[0,100]*/
            "mergeChildSimilarity":  100,
            /*opt, int, range:[0,100]*/
            "mergeMinSimilarity":  83,
            /*opt, int, range:[0,100]*/
            "addCommonSimilarity":  88,
            /*opt, int, range:[0,100]*/
            "addCoverSimilarity":  93,
            /*opt, int, range:[0,100]*/
            "addCoverChildSimilarity":  93,
            /*opt, int, range:[0,100]*/
            "compareCoverSimilarity":  79
            /*opt, int, range:[0,100]*/
        },
        "Score": {
        /*opt, object, score parameters*/
            "SnapScore": {
            /*opt, object, captured picture score*/
                "visibleScore":  0.1,
                /*opt, float, visibility score, range:[0.000,1.000]*/
                "totalScore":  0.1,
                /*opt, float, total score, range:[0.000,1.000]*/
                "clearityScore":  0.1
                /*opt, float, Sharpness Score, range:[0.00,1.000]*/
            },
            "StrangerScore": {
            /*opt, object, stranger score*/
                "pupillaryDistance":  18,
                /*opt, int, pupil distance, range:[18,719]*/
                "pitchAngle":  25.0,
                /*opt, float, pitch angle, range:[0.000,90.000]*/
                "yawAngle":  30.000,
                /*opt, float, yaw angle, range:[0.000,90.000]*/
                "clearityScore":  0.22,
                /*opt, float, Sharpness Score, range:[0.000,1.000]*/
                "visibleScore":  0.51,
                /*opt, float, visibility score, range:[0.000,1.000]*/
                "totalScore":  0.1
                /*opt, float, total score, range:[0.000,1.000]*/
            },
            "FusedScore": {
            /*opt, object, model fusion score*/
                "pupillaryDistance":  18,
                /*opt, int, pupil distance, range:[18,719]*/
                "pitchAngle":  25.0,
                /*opt, float, pitch angle, range:[0.000,90.000]*/
                "yawAngle":  30.000,
                /*opt, float, yaw angle, range:[0.000,90.000]*/
                "clearityScore":  0.2,
                /*opt, float, Sharpness Score, range:[0.000,1.000]*/
                "visibleScore":  0.5,
                /*opt, float, visibility score, range:[0.000,1.000]*/
                "totalScore":  0.1
                /*opt, float, total score, range:[0.000,1.000]*/
            },
            "CoverScore": {
            /*opt, object*/
                "pupillaryDistance":  18,
                /*opt, int, pupil distance, range:[18,719]*/
                "pitchAngle":  25.0,
                /*opt, float, pitch angle, range:[0.000,90.000]*/
                "yawAngle":  30.000,
                /*opt, float, range:[0.000,90.000]*/
                "clearityScore":  0.2,
                /*opt, float, Sharpness Score, range:[0.000,1.000]*/
```

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```
                "visibleScore":  0.5,
                /*opt, float, range:[0.000,1.000]*/
                "totalScore":  0.1
                /*opt, float, total score, range:[0.000,1.000]*/
            }
        },
        "Fused": {
        /*opt, object, model fusion parameters, desc:object,model fusion parameters*/
            "constantFusedModelNum":  5,
            /*opt, int, number of models to be fused*/
            "constantFusedModelTime":  0,
            /*opt, int, update time of model fusion, desc:update time of model fusion. When this node is configured,the latest constantFusedModelNum faces
should be fused again after the constantFusedModelTime. The value is between 0 and 12 and the default value is 0 (0 means model fusion will not be
updated),unit: month*/
            "constantChildAge":  8,
            /*opt, int, range:[1,18]*/
            "switchIqaEmptyFilter":  true,
            /*opt, bool*/
            "switchCoverCheckGender":  true,
            /*opt, bool*/
            "switchMergeDealChild":  false,
            /*opt, bool*/
            "switchStrangerAddChild":  false,
            /*opt, bool*/
            "handleType":  "cover"
            /*opt, enum, subType:string*/
        }
    },
    "cacheCompareSimilarity":  50,
    /*opt, int, range:[0,100]*/
    "strangeSwitch":  true,
    /*opt, bool*/
    "coverNum":  1,
    /*opt, int, range:[1,10]*/
    "humanCompareCommonSimilarity":  80
    /*opt, int, range:[0,100]*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
}
```

15.5.8.12 Search for the personnel merging record

Request URL

POST /ISAPI/SDT/personnelFile/recordSearch?format=json

Query Parameter

None

Request Message

```
{
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list*/
    "maxResults":  100,
    /*req, int, the maximum number of search results this time*/
    "startCreateTime ":  "2017-09-18T02:00:00+08:00",
    /*req, string, start time in ISO 8601 time format*/
    "endCreateTime":  "2017-09-18T02:00:00+08:00"
    /*req, string, end time*/
}
```

Response Message

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```
{
    "numOfMatches":  100,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches":  200,
    /*ro, opt, int, total number of matched results*/
    "recordList": [
    /*ro, opt, array, search results, subType:object*/
        {
            "combinationTime":  "2017-09-18T02:00:00+08:00 ",
            /*ro, req, string, merging time*/
            "humanList": [
            /*ro, req, array, records that have been merged each time, subType:object*/
                {
                    "srcFDID":  "test",
                    /*ro, req, string, source face picture library ID*/
                    "srcPID":  "test",
                    /*ro, req, string, source face picture ID*/
                    "dstFDID":  "test",
                    /*ro, req, string, destination face picture library ID*/
                    "dstPID":  "test"
                    /*ro, req, string, destination face picture ID*/
                }
            ]
        }
    ]
}
```

15.5.8.13 Get personnel archive configuration parameters

Request URL

GET /ISAPI/SDT/personnelFile?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "personnelFile": {
    /*ro, req, object*/
        "enabled":  true,
        /*ro, req, bool, whether to enable personnel archive*/
        "similarity":  85,
        /*ro, req, int, similarity, range:[0,100], desc:default value: 85*/
        "monitorIds": ["11111", "22222"],
        /*ro, opt, array, array of camera ID, subType:string*/
        "FDID":  "test",
        /*ro, req, string, face picture library ID, desc:multiple IDs should be separated by commas*/
        "taskID": ["1111", "2222"],
        /*ro, opt, array, task ID, subType:string, desc:either taskID or monitorIds should exist, taskID implies task linkage mode*/
        "strangerEnabled":  true
        /*ro, opt, bool*/
    }
}
```

15.5.8.14 Set personnel archive parameters

Request URL

PUT /ISAPI/SDT/personnelFile?format=json

Query Parameter

None

Request Message

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```
{
    "personnelFile": {
    /*req, object*/
        "enabled":  true,
        /*req, bool, whether to enable the function*/
        "similarity":  85,
        /*req, int, similarity, range:[0,100], desc:default value: 85*/
        "monitorIds": ["11111", "22222"],
        /*opt, array, array information of camera ID, subType:string*/
        "FDID":  "test",
        /*req, string, face picture library ID, desc:face picture library ID*/
        "taskID": ["1111", "2222"],
        /*opt, array, task ID, subType:string, desc:task ID*/
        "strangerEnabled":  true
        /*opt, bool*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

15.5.9 人脸特征搜图

15.5.9.1 Get the capability set of searching by picture in captured picture library

Request URL

GET /ISAPI/SDT/Face/searchByPic/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, string, search start time*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, string, search end time*/
    "channelNum":  100,
    /*ro, req, int, number of cameras can be selected*/
    "resultMaxNum":  500,
    /*ro, req, int, maximum results can be searched*/
    "modelThreshold": [
    /*ro, opt, array, maximum and minimum threshold for comparison, subType:object*/
        {
            "min":  0,
            /*ro, opt, int, the minimum value*/
            "max":  100
            /*ro, req, int, the maximum value*/
        }
    ],
    "ageGroup": [
    /*ro, opt, array, age group, subType:object*/
        {
            "value":  "child"
```

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```
            /*ro, opt, string, options*/
        }
    ],
    "gender": [
    /*ro, req, array, gender, subType:object*/
        {
            "value":  "male"
            /*ro, opt, string, options*/
        }
    ],
    "glasses": [
    /*ro, req, array, wear glasses or not, subType:object*/
        {
            "value":  "yes"
            /*ro, opt, string, options*/
        }
    ],
    "smile": [
    /*ro, req, array, smile or not, subType:object*/
        {
            "value":  "yes"
            /*ro, opt, string, options*/
        }
    ],
    "isSupportSyncSearch":  true,
    /*ro, opt, bool, whether supports searching by picture in captured picture library*/
    "isSupportPictureMerge":  true,
    /*ro, opt, bool, whether supports displaying picture in merge*/
    "generateMaxNum":  1234,
    /*ro, opt, int, maximum number of records can be exported*/
    "modelMaxNum":  100,
    /*ro, opt, int, maximum number of modeled results*/
    "faceURL": {
    /*ro, opt, object, length of picture URL*/
        "min":  0,
        /*ro, opt, int, the minimum value*/
        "max":  100
        /*ro, opt, int, the maximum value*/
    },
    "targetModelData": {
    /*ro, opt, object, length of target model data*/
        "min":  0,
        /*ro, opt, int, the minimum value*/
        "max":  100
        /*ro, opt, int, the maximum value*/
    },
    "sortType":  "similarity,time",
    /*ro, opt, string, sorting type, desc:“similarity” (by similarity), “time” (by time)*/
    "faceExpression": {
    /*ro, opt, object, whether supports expression*/
        "@opt": ["unknown", "poker-faced", "happy", "surprised", "panic", "sad", "angry", "contemptuous", "disgusted"]
        /*ro, opt, array, options, subType:string*/
    },
    "mask": {
    /*ro, opt, object, whether with gauze mask*/
        "@opt": ["unknown", "yes", "no"]
        /*ro, opt, array, options, subType:string*/
    },
}
```

15.5.9.2 Get task progress of using picture to search picture

Request URL

GET /ISAPI/SDT/Face/searchByPic/progress?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "ffc98d57-c156-40a8-ae64-84cb98dcee10",
    /*ro, req, string, task ID of using picture to search picture*/
    "progress":  100,
    /*ro, req, int, task progress of using picture to search picture*/
    "analysisModelNum":  100
    /*ro, req, int, total number of models to be compared*/
}
```

15.5.9.3 Search for the results of asynchronously using picture to search picture

Request URL

POST /ISAPI/SDT/Face/searchByPic/result?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

```
{
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults":  100
    /*req, int, the maximum number of search results this time*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "activeNumMax",
    /*ro, req, enum, sub status code, subType:string, desc:"ok" (succeeded). It is required when an error occurred. "activeNumMax" (the max. number of
device allowed for License access is reached), "licenseExpired" (License expired)*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "test",
    /*ro, req, string, task ID of using picture to search picture*/
    "responseStatusStrg":  "OK",
    /*ro, req, string, searching status description*/
    "numOfMatches":  1,
    /*ro, req, int, number of matched results returned this time*/
    "totalMatches":  1,
    /*ro, req, int, total number of matched results by this search*/
    "targets": [
    /*ro, opt, array, list of matched records, subType:object*/
        {
            "id":  1,
            /*ro, req, int, No.*/
            "captureTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, string, capture time*/
            "isSummerTime":  false,
            /*ro, opt, bool, whether the time returned currently is in DST (Daylight Saving Time) system*/
            "deviceType":  "Blade",
            /*ro, req, enum, device type, subType:string, desc:"Blade" (vehicle analysis server), "BladePS" (security vehicle analysis server), "CVR","DM"
(full analysis server), "DMbehavior" (abnormal event detection device), "DVR","DVS", "DockStation" (dock station), "FA" (real-time face picture comparison
and alarm server), "FD" (capture detection server), "FS" (face picture static data server), "HAWK" (video structural analysis server), "HMCP" (model
comparison server), "HybirdNVR" (hybrid DVR), "IPCamera" (network camera), "IPDome" (network speed dome), "IPZoom" (integrated camera), "InfoReleaseSys"
(information release system), "InfoTerminal" (information release terminal), "MCU", "NVR", "PHA" (security control panel), "PURE" (pure analysis server),
"RFID", "SipServer" (SIP server), "Switch", "lightFace" (lightweight facial recognition server)*/
            "deviceId":  "test",
            /*ro, opt, string, camera ID*/
            "captureSite":  "3#大门处",
            /*ro, opt, string, capture site (camera name)*/
```

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```
            "monitorId":  "1",
            /*ro, opt, string, stream ID (encoding stream ID in CVR). This node will not be returned if the device does not support this function*/
            "ageGroup":  "any",
            /*ro, req, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime",
"teenager", "unknown", "young"*/
            "gender":  "any",
            /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
            "glasses":  "no",
            /*ro, req, enum, whether the person wears glasses, subType:string, desc:"no" (not wear glasses), "sunglasses" (wears sunglasses), "unknown",
"yes"*/
            "smile":  "no",
            /*ro, req, enum, whether the person wears a smile, subType:string, desc:smile or not: "unknown", "yes", "no"*/
            "picUrl":  "xxx",
            /*ro, opt, string, background picture URL*/
            "subpicUrl":  "xxx",
            /*ro, req, string, face picture thumbnail URL*/
            "rect": {
            /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
                "height":  1.000,
                /*ro, opt, float, height, range:[0.000,1.000]*/
                "width":  1.000,
                /*ro, opt, float, width, range:[0.000,1.000]*/
                "x":  0.000,
                /*ro, opt, float, x-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*ro, opt, float, y-coordinate, range:[0.000,1.000]*/
            },
            "face_pose": {
            /*ro, opt, object, face angle*/
                "pitch":  0.514,
                /*ro, opt, float, face pitch angle, face pitching up is regarded as positive*/
                "yaw":  0.2909,
                /*ro, opt, float, face yaw angle, face yawing left is regarded as positive*/
                "roll":  0.3669
                /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive*/
            },
            "face_rect": {
            /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                "height":  1.000,
                /*ro, opt, float, height, range:[0.000,1.000]*/
                "width":  1.000,
                /*ro, opt, float, width, range:[0.000,1.000]*/
                "x":  0.000,
                /*ro, opt, float, x-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*ro, opt, float, y-coordinate, range:[0.000,1.000]*/
            },
            "similarity":  1.00,
            /*ro, opt, float, similarity*/
            "errInfo": {
            /*ro, opt, object, error description, desc:object, error information. This node is valid when the value of the node subpicUrl is empty*/
                "errMsg":  "test",
                /*ro, req, string, error description, desc:the reason when the value of the node subpicUrl is empty*/
                "errType":  0
                /*ro, req, int, error type, desc:error type, which is the reason type when the value of the node subpicUrl is empty*/
            },
            "channelID":  1,
            /*ro, opt, int, channel No.*/
            "URLType":  "cloudURL",
            /*ro, opt, string, type of returned picture URL, desc:type of returned picture URL*/
            "faceExpression":  "angry",
            /*ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-faced",
"sad", "surprised", "unknown"*/
            "mask":  "no",
            /*ro, req, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"*/
            "linkFaceBodyId":  "XXX",
            /*ro, opt, string, linkage ID between face and human body picture*/
            "monitorPointID":  "test",
            /*ro, opt, string, camera ID*/
            "thermometryUnit":  "celsius",
            /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
            "currTemperature":  38.0,
            /*ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃*/
            "temperatureCondition":  "normal"
            /*ro, opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
        }
    ]
}
```

15.5.9.4 Submit the task of asynchronously searching by picture

Request URL

POST /ISAPI/SDT/Face/searchByPic?format=json

Query Parameter

None

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Request Message

```
{
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*opt, string, start time of search*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*opt, string, end time of search*/
    "choiceChannel": [
    /*opt, array, list of selected cameras, subType:object*/
        {
            "channelID":  "test",
            /*opt, string, camera ID*/
            "channelName":  "channel1"
            /*opt, string, camera name, range:[0,64]*/
        }
    ],
    "similarityMin":  0.3,
    /*req, float, minimum threshold for comparison, range:[0.00,1.00], desc:minimum threshold for comparison*/
    "similarityMax":  1.0,
    /*req, float, maximum threshold for comparison, range:[0.00,1.00], desc:maximum threshold for comparison*/
    "age":  23,
    /*opt, int, age*/
    "ageGroup":  "any",
    /*req, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime", "teenager",
"unknown", "young"*/
    "gender":  "any",
    /*req, enum, gender, subType:string, desc:"any" (unlimited condition which is used for search), "male", "female", "unknown"*/
    "glasses":  "no",
    /*req, enum, whether the person wears glasses, subType:string, desc:"no" (not wear glasses), "sunglasses" (wears sunglasses), "unknown", "yes"*/
    "smile":  "no",
    /*req, enum, whether the person is smiling, subType:string, desc:"no", "unknown", "yes"*/
    "mask":  "no",
    /*req, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"*/
    "modelMaxNum":  100,
    /*req, int, maximum number of model comparison results (the maximum number of returned results), desc:maximum number of model comparison results (the
maximum number of returned results)*/
    "targetModelData":  "test",
    /*req, string, target model data*/
    "sortType":  "similarity",
    /*opt, enum, sorting type, subType:string, desc:"similarity" (sort by similarity), "time" (sort by time)*/
    "URLType":  "cloudURL",
    /*opt, string, type of returned picture URL, desc:the URL type of searched picture*/
    "pictureMerge":  false,
    /*opt, bool, whether to merge multiple pictures of one person, desc:the value of this node is false by default*/
    "thermometryUnit":  "celsius",
    /*opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
    "currTemperature":  38.0,
    /*opt, float, face temperature, range:[-20.0,150.0], unit:℃*/
    "temperatureCondition":  "normal"
    /*opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceInfo	[Message content]	application/json	--	--	--
faceImage	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

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Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "ffc98d57-c156-40a8-ae64-84cb98dcee10"
    /*ro, req, string, task ID of searching by picture*/
}
```

15.5.9.5 Get the task information

Request URL

GET /ISAPI/SDT/Face/searchByPic?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, req, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskInfo": [
    /*ro, opt, array, task information, subType:object*/
        {
            "taskID":  "abc",
            /*ro, req, string, task ID of searching by picture, range:[0,64]*/
            "taskCreateTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, datetime, task creation time*/
            "timeUsed":  100,
            /*ro, opt, int, task execution duration, unit:s, desc:unit: second*/
            "progress":  100,
            /*ro, opt, int, task progress*/
            "TaskDetails": {
            /*ro, opt, object, task details*/
                "startTime":  "2004-05-03T17:30:08+08:00",
                /*ro, opt, string, start time of search*/
                "endTime":  "2004-05-03T17:30:08+08:00",
                /*ro, opt, string, end time of search*/
                "choiceChannel": [
                /*ro, opt, array, list of selected cameras, subType:object*/
                    {
                        "channelID":  "test",
                        /*ro, opt, string, camera ID*/
                        "channelName":  "channel1"
                        /*ro, opt, string, camera name*/
                    }
                ],
                "similarityMin":  0.3,
                /*ro, req, float, minimum threshold for comparison, range:[0.00,1.00], desc:it should be accurate to two decimal place, the default value is
0*/
                "similarityMax":  1.0,
                /*ro, opt, float, maximum threshold for comparison, range:[0.00,1.00], desc:it should be accurate to two decimal place, the default value is
0*/
                "age":  23,
                /*ro, opt, int, age*/
                "ageGroup":  "any",
                /*ro, opt, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime",
"teenager", "unknown", "young"*/
                "gender":  "any",
                /*ro, opt, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
                "glasses":  "no",
                /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no" (not wear glasses), "sunglasses" (wears sunglasses),
"unknown"*/
                "smile":  "no",
                /*ro, opt, enum, whether the person wears a smile, subType:string, desc:"no", "unknown", "yes"*/
                "mask":  "no",
                /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"*/
                "modelMaxNum":  100,
                /*ro, opt, int, maximum number of model comparison results, desc:the maximum number of returned results*/
                "targetModelData":  "test",
                /*ro, opt, string, target model data*/
                "sortType":  "similarity",
                /*ro, opt, enum, sorting type, subType:string, desc:"similarity" (sort by similarity), "time" (sort by time)*/
                "URLType":  "cloudURL",
                /*ro, opt, string, type of returned picture URL, desc:the URL type of searched picture*/
                "pictureMerge":  false,
                /*ro, opt, bool, whether to merge multiple pictures of one person, desc:the value of this node is false by default*/
                "thermometryUnit":  "celsius",
                /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
                "currTemperature":  38.0,
                /*ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃*/
                "temperatureCondition":  "normal"
                /*ro, opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
            }
        }
    ]
}
```

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15.5.9.6 Delete the task of searching picture by picture asynchronously

Request URL

DELETE /ISAPI/SDT/Face/searchByPic?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

15.5.9.7 search for captured pictures via picture comparison in synchronous mode

Request URL

POST /ISAPI/SDT/Face/searchByPic?supportSync=<supportSync>

Query Parameter

Parameter Name	Parameter Type	Description
supportSync	string	Supports setting as true only.

Request Message

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```
{
    "searchResultPosition": 0,
    /*req, int, the start position of the search result in the result list*/
    "maxResults": 0,
    /*req, int, the maximum number of search results this time*/
    "taskID": "test",
    /*opt, string, task ID, desc:task ID, which is returned from the device. The same task ID represents the same search conditions. When the task ID is
transmitted, the value of required fields for search does not need to be transmitted*/
    "isSearchOver": false,
    /*opt, bool*/
    "startTime": "2004-05-03T08:30:08+08:00",
    /*opt, datetime, search start time*/
    "endTime": "2004-05-03T17:30:08+08:00",
    /*opt, datetime, search end time*/
    "choiceChannel": [
    /*opt, array, list of selected cameras, subType:object, desc:all channels are included when this node is not configured*/
        {
            "channelID": "test"
            /*opt, string, camera ID*/
        }
    ],
    "similarityMin": 0.3,
    /*req, float, minimum threshold for comparison*/
    "similarityMax": 1.0,
    /*opt, float, maximum threshold for comparison, desc:maximum threshold for comparison*/
    "dataType": "URL",
    /*req, enum, picture data type, subType:string, desc:"URL" (picture URL), "binaryData" (binary picture data), "modelData" (model data)*/
    "pictureMerge": false,
    /*opt, bool, whether to merge multiple pictures of one person, desc:the value of this node is false by default*/
    "ageGroup": "any",
    /*opt, enum, age group, subType:string, desc:“any”, “child”, “infant”, “kid”, “middle”, “middleAged” (the middle age), “old”, “prime”, “teenager”,
“unknown”, “young”*/
    "gender": "any",
    /*opt, enum, gender, subType:string, desc:"any” (unlimited condition which is used in search), "male”, “female”, “unknown"*/
    "glasses": "no",
    /*opt, enum, whether the person wears glasses, subType:string, desc:“no” (not wear glasses), “sunglasses” (wears sunglasses), “unknown"*/
    "smile": "no",
    /*opt, enum, whether the person wears a smile, subType:string, desc:"no", "unknown", "yes”*/
    "modelMaxNum": 100,
    /*opt, int, maximum number of model comparison results (the maximum number of returned results)*/
    "targetModelData": "test",
    /*opt, string, target modeling data, dep:and,{$.dataType,eq,modelData}, desc:<targetModelData> is required when <dataType> is modelData*/
    "faceURL": "test",
    /*opt, string, picture storage URL for uploading the face picture by URL*/
    "sortType": "similarity",
    /*opt, enum, sorting type, subType:string, desc:“similarity”, “time”*/
    "orderType": "sequence",
    /*opt, enum, subType:string*/
    "mask": "no",
    /*opt, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes”*/
    "faceExpression": "angry",
    /*opt, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-faced", "sad",
"surprised", "unknown"*/
    "thermometryUnit": "celsius",
    /*opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
    "currTemperature": 38.0,
    /*opt, float, face temperature, range:[-20.0,150.0], unit:℃*/
    "temperatureCondition": "normal",
    /*opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
    "organizationID": ["test"],
    /*opt, array, subType:string, range:[1,300000]*/
    "whetherLinkHumanBody": true,
    /*opt, bool*/
    "whetherDriverExists": true,
    /*opt, bool*/
    "whetherFrontPassengerExists": true,
    /*opt, bool*/
    "whetherLinkNonmotorVehicle": true,
    /*opt, bool*/
    "whetherLinkVehicle": true,
    /*opt, bool*/
    "rect": {
    /*ro, opt, object*/
        "height": 1.000,
        /*ro, opt, float, range:[0.000,1.000]*/
        "width": 1.000,
        /*ro, opt, float, range:[0.000,1.000]*/
        "x": 0.000,
        /*ro, opt, float, range:[0.000,1.000]*/
        "y": 0.000
        /*ro, opt, float, range:[0.000,1.000]*/
    },
    "collectEnabled": "yes",
    /*opt, enum, subType:string*/
    "filterEnabled": false
    /*opt, bool*/
}
```

Response Message

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```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, see the description of this node and the above nodes in the message of JSON_ResponseStatus, desc:this node is required when the value of statusCode is not 1*/
    "responseStatusStrg": "OK",
    /*ro, opt, string, searching status description*/
    "taskID": "ffc98d57-c156-40a8-ae64-84cb98dcee10",
    /*ro, opt, string, task ID, desc:task ID, which is returned from the device for the first search. The same task ID represents the same search conditions and can be transmitted for subsequent searches*/
    "numOfMatches": 1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches": 1,
    /*ro, opt, int, total number of matched results*/
    "targets": [
    /*ro, opt, array, array type,matched result details, subType:object*/
        {
            "id": 1,
            /*ro, req, int, No.*/
            "captureTime": "2004-05-03T17:30:08+08:00",
            /*ro, opt, datetime, capture time*/
            "isSummerTime": false,
            /*ro, opt, bool, whether the time returned currently is in DST (Daylight Saving Time) system*/
            "organizationID": "test123456789",
            /*ro, opt, string, range:[1,32]*/
            "captureSite": "3#",
            /*ro, opt, string, capture location*/
            "ageGroup": "child",
            /*ro, opt, string, age group*/
            "gender": "male",
            /*ro, opt, string, gender*/
            "glasses": "yes",
            /*ro, opt, string, whether the person wears glasses*/
            "smile": "yes",
            /*ro, opt, string, whether the person wears a smile*/
            "picUrl": "xxx",
            /*ro, opt, string, background picture URL*/
            "subpicUrl": "xxx",
            /*ro, req, string, face picture thumbnail URL*/
            "subPicUUID": 1,
            /*ro, opt, int*/
            "rect": {
            /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
                "height": 1.000,
                /*ro, opt, float, height, range:[0.000,1.000]*/
                "width": 1.000,
                /*ro, opt, float, float type,width, range:[0.000,1.000]*/
                "x": 0.000,
                /*ro, opt, float, x-coordinate, range:[0.000,1.000]*/
                "y": 0.000
                /*ro, opt, float, y-coordinate, range:[0.000,1.000]*/
            },
            "face_pose": {
            /*ro, opt, object, face angle*/
                "pitch": 0.514,
                /*ro, opt, float, face pitch angle, face pitching up is regarded as positive, desc:float type,face raising or bowing angle,face raising is regarded as positive. This node does not exists if it is invalid.*/
                "yaw": 0.2909,
                /*ro, opt, float, face yaw angle, face yawing left is regarded as positive, desc:float type,face siding left or right angle,face siding left is regarded as positive. This node does not exists if it is invalid.*/
                "roll": 0.3669
                /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive, desc:float type,face tilting clockwise or anticlockwise angle,face tilting clockwise is regarded as positive. This node does not exists if it is invalid.*/
            },
            "face_rect": {
            /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                "height": 1.000,
                /*ro, opt, float, height, range:[0.000,1.000]*/
                "width": 1.000,
                /*ro, opt, float, width, range:[0.000,1.000]*/
                "x": 0.000,
                /*ro, opt, float, x-coordinate, range:[0.000,1.000]*/
                "y": 0.000
                /*ro, opt, float, y-coordinate, range:[0.000,1.000]*/
            },
            "similarity": 1.00,
            /*ro, opt, float, similarity*/
            "rowKey": "test",
            /*ro, opt, string, main key of face picture library, desc:searching by rowKey can be more efficient*/
            "mask": "unknown",
            /*ro, opt, string, whether the person wears a mask*/
            "faceExpression": "unknown",
            /*ro, opt, string, facial expression*/
            "linkFaceBodyId": "XXX",
            /*ro, opt, string, linkage ID between face and human body picture*/
            "monitorPointID": "test"
```


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```
            "monitorPointID":  "test",
            /*ro, opt, string, camera ID*/
            "thermometryUnit":  "celsius",
            /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
            "currTemperature":  38.0,
            /*ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃*/
            "temperatureCondition":  "normal",
            /*ro, opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
            "videoUrl":  "test",
            /*ro, opt, string, Alarm video URL.*/
            "traceUUID":  "fdb4465d7aa5495884845127a8bbbe3c",
            /*ro, opt, string, capture target UUID, range:[1,64], desc:the target person is assigned with a consistent UUID by the device capture algorithm during his/her consistent stay in the video image; if the target person leaves (or is blocked) and appears in the image again, he/she will be assigned with a new UUID*/
            "traceIdx":  1,
            /*ro, opt, int, target capture picture ID, range:[1,10], desc:during the target person's consistent stay in the video image where he/she is assigned with a consistent UUID by the device capture algorithm, the ID of pictures captured starts from 1; if the target person leaves (or is blocked) and appears in the image again when he/she is assigned with a new UUID, the ID of pictures captured starts from 1 again*/
            "collectEnabled":  "yes"
            /*ro, opt, enum, subType:string*/
        }
    ]
}
```


15.6 人体识别

15.6.1 以人搜人

15.6.1.1 Get progress of exporting results of searching by human body picture

Request URL

GET /ISAPI/Intelligent/humanRecognition/searchByPic/result/export/progress?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	UUID format

Request Message

None

Response Message


```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode, this field is required when statusCode is not 1*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "progress":  0,
    /*ro, opt, int, when errcode is 1 and errMsg is "ok",value range: [0,100], range:[0,100], desc:when errcode is 1 and errMsg is "ok",value range: [0,100]*/
    "URL":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile?id=00000001/temp001/212_14444186_17930&token=7a57a5a7ffffffffc1a0316369671314"
    /*ro, opt, string, URL for downloading the exported results, desc:this field is required when progress is 100*/
}
```


15.6.1.2 Stop exporting the results of human body search

Request URL

PUT /ISAPI/Intelligent/humanRecognition/searchByPic/result/export/stop?format=json

Query Parameter

None

Request Message

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```
{
    "taskID":  "5d811723-d165-11ea-9b07-005056c00008"
    /*req, string, task ID, desc:task ID*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

15.6.1.3 Get the capability of searching for the task of searching by human body picture

Request URL

GET /ISAPI/Intelligent/humanRecognition/searchByPic/taskInfo/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "searchID": {
    /*ro, req, object, search ID, desc:this node is used to check whether searches are the same one; due to device performance limitation, you can apply the
search conditions multiple times (the search ID should be the same) to complete one search and get the search progress*/
        "@min":  0,
        /*ro, opt, int, the minimum value, range:[0,32]*/
        "@max":  32
        /*ro, opt, int, the maximum value, range:[0,32]*/
    },
    "maxResultNumber": {
    /*ro, opt, object, the maximum number of records that need to be searched*/
        "@min":  0,
        /*ro, opt, int, the minimum value, range:[0,4294967295]*/
        "@max":  2000,
        /*ro, opt, int, the maximum value, range:[0,4294967295]*/
        "@def":  0
        /*ro, opt, int, the default value, range:[0,4294967295]*/
    },
    "maxResults": {
    /*ro, req, object, the maximum number of records that can be displayed on each page by this search*/
        "@min":  0,
        /*ro, opt, int, the minimum value, range:[0,100]*/
        "@max":  100
        /*ro, opt, int, the maximum value, range:[1,100]*/
    }
}
```

15.6.1.4 Set conditions of searching for human body by modeled picture data and submit the search task

Request URL

POST /ISAPI/Intelligent/humanRecognition/searchByPic?format=json

Query Parameter

None

Request Message

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```
{
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*req, string, start time, desc:ISO 8601 format*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*req, string, end time, desc:ISO 8601 format*/
    "choiceChannel": [
    /*opt, array, camera list, subType:object, desc:which specifies cameras where the human body pictures are captured. If this node is not configured, it
indicates no camera limit*/
        {
            "channelID":  "abc",
            /*opt, string, camera ID, desc:the maximum string size is 32*/
            "channelName":  "channel1"
            /*opt, string, camera name, range:[0,64]*/
        }
    ],
    "similarity":  0.99,
    /*opt, float, similarity, range:[0.00,1.00], desc:the larger the value is, the more accurate it will be*/
    "modelData":  "abc",
    /*req, string, target model data, desc:target model data*/
    "taskID":  "5d811723-d165-11ea-9b07-005056c00008",
    /*opt, string, task ID of searching by picture, desc:task ID of searching by picture*/
    "URLType":  "cloudURL",
    /*opt, string, the URL type of searched picture, desc:this node is for devices supporting local URL, and the local URL will be returned by default*/
    "maxResults":  100
    /*opt, int, the maximum number of search results this time, desc:the search will stop when the number of results reaches the configured value*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
searchByPic	[Message content]	application/json	--	--	--
humanImage	[Binary picture data]	image/jpeg	humanImage	humanImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "5d811723-d165-11ea-9b07-005056c00008"
    /*ro, req, string, task ID, desc:task ID*/
}
```

15.6.1.5 Delete a task of searching by human body picture

Request URL

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DELETE /ISAPI/Intelligent/humanRecognition/searchByPic?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	Task ID, which is returned by the URL (POST /ISAPI/Intelligent/humanRecognition/searchByPic?format=json).

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, Status Code, desc:this field is required when the status code is not "1" (success)*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:this field is required when the status code is not "OK" (success)*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:this field is required when the status code is not "ok" (success)*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error message, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string, error codes categorized by functional modules, desc:N/A*/
}
```

15.7 周界入侵检测

15.7.1 区域入侵

15.7.1.1 Intrusion detection

EventType:fielddetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"fielddetection" (intrusion detection)-->fielddetection
  </eventType>
```

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```
<eventState>
  <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
</eventState>
<eventDescription>
  <!--ro, req, string, event description-->test
</eventDescription>
<deviceID>
  <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
</deviceID>
<DetectionRegionList>
  <!--ro, opt, array, detection area list, subType:object-->
  <DetectionRegionEntry>
    <!--ro, opt, object, detection area-->
    <regionID>
      <!--ro, req, string, area ID-->test
    </regionID>
    <sensitivityLevel>
      <!--ro, opt, int, sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
    </detectionTarget>
    <TargetRect>
      <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
      <X>
        <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
      </X>
      <Y>
        <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
      </Y>
      <width>
        <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
      </width>
      <height>
        <!--ro, req, float, target height, range:[0.000,1.000]-->1.000
      </height>
    </TargetRect>
    <sceneID>
      <!--ro, opt, int, scene ID-->1
    </sceneID>
  </DetectionRegionEntry>
</DetectionRegionList>
<channelName>
  <!--ro, opt, string, channel name, range:[1,64]-->test
</channelName>
<Extensions>
  <!--ro, opt, object, extended information-->
  <serialNumber>
    <!--ro, opt, string, serial No.-->test
  </serialNumber>
  <eventPush>
    <!--ro, opt, string, event notification-->test
  </eventPush>
</Extensions>
<detectionPictureTransType>
  <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
</detectionPictureTransType>
<detectionPicturesNumber>
  <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
</detectionPicturesNumber>
<bkgUrl>
  <!--ro, opt, string, background picture URL, desc:background picture URL-->test
</bkgUrl>
<URLCertificationType>
  <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
</URLCertificationType>
<pId>
  <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
</pId>
<visibleLightURL>
  <!--ro, opt, string, visible light picture URL-->test
</visibleLightURL>
<thermalURL>
  <!--ro, opt, string, thermal imaging picture URL-->test
</thermalURL>
<targetAttrs>
  <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
</targetAttrs>
<TaskInfo>
```

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```
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <targetSpeed>
    <!--ro, opt, int-->0
  </targetSpeed>
  <targetDistance>
    <!--ro, opt, int, unit:m-->2
  </targetDistance>
  <triggerType>
    <!--ro, opt, enum, subType:string-->video
  </triggerType>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits-->test
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
  <nonTargetRate>
    <!--ro, opt, int, range:[1,100], unit:%-->1
  </nonTargetRate>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
```

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```
      <focus>
        <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
      </focus>
    </visibleLightAbsoluteHigh>
    <thermalAbsoluteHigh>
      <!--ro, opt, object-->
      <elevation>
        <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
      </elevation>
      <azimuth>
        <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
      </absoluteZoom>
      <focus>
        <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
      </focus>
    </thermalAbsoluteHigh>
    <deviceLocation>
      <!--ro, opt, object-->
      <longitudeType>
        <!--ro, req, enum, subType:string-->W
      </longitudeType>
      <latitudeType>
        <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
      </latitudeType>
      <longitude>
        <!--ro, req, object-->
        <degree>
          <!--ro, req, int, degree(s), range:[0,180]-->1
        </degree>
        <minute>
          <!--ro, req, float, minute(s), range:[0,60]-->0.0
        </minute>
        <sec>
          <!--ro, req, float, range:[0,60]-->0.0
        </sec>
      </longitude>
      <latitude>
        <!--ro, req, object-->
        <degree>
          <!--ro, req, int, range:[0,90]-->1
        </degree>
        <minute>
          <!--ro, req, float, minute(s), range:[0,60]-->0.0
        </minute>
        <sec>
          <!--ro, req, float, range:[0,60]-->0.0
        </sec>
      </latitude>
    </deviceLocation>
    <laserRanging>
      <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
    </laserRanging>
  </EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
fielddetection	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.7.2 越界侦测

15.7.2.1 Line crossing detection

EventType:linedetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"linedetection" (Line Crossing Detection)-->linedetection
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
```

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```
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
      <TargetRect>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <X>
          <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
        </X>
        <Y>
          <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
        </Y>
        <width>
          <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
        </width>
        <height>
          <!--ro, req, float, target height, range:[0.000,1.000]-->1.000
        </height>
      </TargetRect>
      <sceneID>
        <!--ro, opt, int, scene ID-->1
      </sceneID>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <visibleLightURL>
    <!--ro, opt, string, visible light picture URL-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
```

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```
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <targetSpeed>
    <!--ro, opt, datetime-->2019-12-30T15:29:23+08:00
  </targetSpeed>
  <targetDistance>
    <!--ro, opt, int, unit:m-->2
  </targetDistance>
  <triggerType>
    <!--ro, opt, enum, subType:string-->video
  </triggerType>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits-->test
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
  <applicationScene>
    <!--ro, opt, enum, Application Scene, subType:string-->indoor
  </applicationScene>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </visibleLightAbsoluteHigh>
  <thermalAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </thermalAbsoluteHigh>
  <deviceLocation>
    <!--ro, opt, object-->
    <longitudeType>
```

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```
      <!--ro, req, enum, subType:string-->W
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
    </latitudeType>
    <longitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, degree(s), range:[0,180]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </longitude>
    <latitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, range:[0,90]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </latitude>
  </deviceLocation>
  <laserRanging>
    <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
  </laserRanging>
</EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
linedetection	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.7.3 进入区域

15.7.3.1 Region entrance

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EventType:regionEntrance

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"regionEntrance" (region entrance)-->regionEntrance
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area points coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
      <TargetRect>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <X>
          <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
        </X>
        <Y>
          <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
        </Y>
        <width>
          <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
        </width>
        <height>
          <!--ro, req, float, target height, range:[0.000,1.000]-->1.000
        </height>
      </TargetRect>
      <sceneID>
```

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```
        <!--ro, opt, int, scene ID-->1
      </sceneID>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <visibleLightURL>
    <!--ro, opt, string, visible light picture URL-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
  <targetAttrs>
    <!--ro, opt, object, transparent transmission field-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选，设备ID，string类型，最大长度为
64*/ "deviceChannel": 1, /*可选，设备通道号，integer32类型*/ "deviceName": "shebei", /*可选，设备名称，string类型，最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
```

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```
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits--
>test
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:Rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:Rule Name-->test
  </ruleName>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </visibleLightAbsoluteHigh>
  <thermalAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </thermalAbsoluteHigh>
  <deviceLocation>
    <!--ro, opt, object-->
    <longitudeType>
      <!--ro, req, enum, subType:string-->W
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
    </latitudeType>
    <longitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, degree(s), range:[0,180]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </longitude>
    <latitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, range:[0,90]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </latitude>
  </deviceLocation>
  <laserRanging>
    <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
  </laserRanging>
</EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
regionEntrance	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.7.4 离开区域

15.7.4.1 Region exiting

EventType:regionExiting

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome".
The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP--
>HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
```

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```
</dateTime>
<activePostCount>
  <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
</activePostCount>
<eventType>
  <!--ro, req, enum, event type, subType:string, desc:"regionExiting" (region exiting)-->regionExiting
</eventType>
<eventState>
  <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
</eventState>
<eventDescription>
  <!--ro, req, string, event description-->test
</eventDescription>
<deviceID>
  <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
</deviceID>
<DetectionRegionList>
  <!--ro, opt, array, detection area list, subType:object-->
  <DetectionRegionEntry>
    <!--ro, opt, object, detection area-->
    <regionID>
      <!--ro, req, string, area ID-->test
    </regionID>
    <sensitivityLevel>
      <!--ro, opt, int, sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
    </detectionTarget>
    <TargetRect>
      <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
      <X>
        <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
      </X>
      <Y>
        <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
      </Y>
      <width>
        <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
      </width>
      <height>
        <!--ro, req, float, target height-->1.000
      </height>
    </TargetRect>
    <sceneID>
      <!--ro, opt, int, scene ID-->1
    </sceneID>
  </DetectionRegionEntry>
</DetectionRegionList>
<channelName>
  <!--ro, opt, string, channel name, range:[1,64]-->test
</channelName>
<Extensions>
  <!--ro, opt, object, extended information-->
  <serialNumber>
    <!--ro, opt, string, serial No.-->test
  </serialNumber>
  <eventPush>
    <!--ro, opt, string, event notification-->test
  </eventPush>
</Extensions>
<detectionPictureTransType>
  <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
</detectionPictureTransType>
<detectionPicturesNumber>
  <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
</detectionPicturesNumber>
<bkgUrl>
  <!--ro, opt, string, background picture URL, desc:background picture URL-->test
</bkgUrl>
<URLCertificationType>
  <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
</URLCertificationType>
<pId>
  <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
</pId>
<visibleLightURL>
  <!--ro, opt, string, visible light picture URL-->test
</visibleLightURL>
<thermalURL>
```

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```
    <thermalURL>
      <!--ro, opt, string, thermal imaging picture URL-->test
    </thermalURL>
    <targetAttrs>
      <!--ro, opt, object, transparent transmission field-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
    </targetAttrs>
    <TaskInfo>
      <!--ro, opt, object, task information submitted by data center products-->
      <taskID>
        <!--ro, req, string, task ID, range:[0,64]-->test
      </taskID>
      <algorithmType>
        <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
      </algorithmType>
      <taskName>
        <!--ro, opt, string, task name-->test
      </taskName>
      <algorithmID>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmID>
      <monitorPointID>
        <!--ro, opt, string, camera No.-->test
      </monitorPointID>
      <monitorPointName>
        <!--ro, opt, string, camera name-->test
      </monitorPointName>
      <streamType>
        <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
      </streamType>
      <ruleID>
        <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
      </ruleID>
      <ruleCustomName>
        <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
      </ruleCustomName>
      <algorithmInfo>
        <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
        <algorithmId>
          <!--ro, opt, string, algorithm package ID-->test
        </algorithmId>
        <algoVendorId>
          <!--ro, opt, string, manufacturer ID-->test
        </algoVendorId>
        <algoVersionId>
          <!--ro, opt, string, algorithm version No.-->test
        </algoVersionId>
        <algoChipType>
          <!--ro, opt, string, chip type-->test
        </algoChipType>
        <algoModelVersion>
          <!--ro, opt, string, model version-->test
        </algoModelVersion>
      </algorithmInfo>
    </TaskInfo>
    <timeStamp>
      <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
    </timeStamp>
    <uid>
      <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits-->test
    </uid>
    <isDataRetransmission>
      <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers.)-->true
    </isDataRetransmission>
    <ruleID>
      <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
    </ruleID>
    <ruleName>
      <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
    </ruleName>
    <visibleLightAbsoluteHigh>
      <!--ro, opt, object-->
      <elevation>
        <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
      </elevation>
      <azimuth>
        <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
      </absoluteZoom>
      <focus>
        <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
      </focus>
    </visibleLightAbsoluteHigh>
    <thermalAbsoluteHigh>
```

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```
      <!--ro, opt, object-->
      <elevation>
        <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
      </elevation>
      <azimuth>
        <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
      </absoluteZoom>
      <focus>
        <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
      </focus>
    </thermalAbsoluteHigh>
    <deviceLocation>
      <!--ro, opt, object-->
      <longitudeType>
        <!--ro, req, enum, subType:string-->W
      </longitudeType>
      <latitudeType>
        <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
      </latitudeType>
      <longitude>
        <!--ro, req, object-->
        <degree>
          <!--ro, req, int, degree(s), range:[0,180]-->1
        </degree>
        <minute>
          <!--ro, req, float, minute(s), range:[0,60]-->0.0
        </minute>
        <sec>
          <!--ro, req, float, range:[0,60]-->0.0
        </sec>
      </longitude>
      <latitude>
        <!--ro, req, object-->
        <degree>
          <!--ro, req, int, range:[0,90]-->1
        </degree>
        <minute>
          <!--ro, req, float, minute(s), range:[0,60]-->0.0
        </minute>
        <sec>
          <!--ro, req, float, range:[0,60]-->0.0
        </sec>
      </latitude>
    </deviceLocation>
    <laserRanging>
      <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
    </laserRanging>
  </EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
regionExiting	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

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- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.8 混合目标检测

15.8.1 混合目标检测

15.8.1.1 Get the rule parameters capability of frequently appeared person detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetDetectionCap": {
    /*ro, opt, object*/
        "enabled":  "true,false",
        /*ro, req, string, whether to enable multi-target-type detection*/
        "TypesDetectionCap": [
        /*ro, opt, array, subType:object*/
            {
                "ruleMode":  "human,face,vehice,nonMotor",
                /*ro, opt, string, rule mode*/
                "enabled":  "true,false",
                /*ro, opt, string, whether to enable the function*/
                "Property": [
                /*ro, opt, array, subType:object*/
                    {
                        "description":  "playMobilePhone",
                        /*ro, opt, string, attributes description*/
                        "enabled":  "true,false"
                        /*ro, opt, string, whether to enable multi-target-type detection*/
                    }
                ]
            }
        ],
        "RuleInfoCap": {
        /*ro, opt, object, rule information, desc:rule information,currently,only one rule is supported*/
            "maxSize":  1,
            /*ro, req, int, capability of maximum queue management rules*/
            "ruleID": {
            /*ro, req, object, rule ID*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "ruleMode": {
            /*ro, opt, object, rule mode, desc:rule type: face + human body + vehicle + non-motor vehicle,human body + vehicle + non-motor vehicle*/
                "@opt":  "face_human_vehice_nonMotor,human_vehice_nonMotor,face_human",
                /*ro, req, string, options*/
                "@def":  "face_human_vehice_nonMotor"
                /*ro, req, string, the default value*/
            },
            "RegionCap": {
            /*ro, req, object, region that configured with rule, desc:polygon with 3 to 10 edges*/
                "minSize":  3,
                /*ro, req, int, capability of minimum edges of rule region*/
                "maxSize":  10,
                /*ro, req, int, capability of maximum queue management rules*/
                "x": {
                /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
```

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```
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "y": {
        /*ro, req, object, Y-coordinate, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        }
    },
    "InterPupileRegionCap": {
    /*ro, req, object, minimum rectangle region to detect pupil distance*/
        "x": {
        /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "y": {
        /*ro, req, object, float,Y-coordinate,the value is between 0.000 and 1, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "width": {
        /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "height": {
        /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        }
    },
    "MaxInterPupilRegionCap": {
    /*ro, opt, object*/
        "x": {
        /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "y": {
        /*ro, req, object, Y-coordinate, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "width": {
        /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "height": {
        /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        }
    },
    "LaneLineInfoCap": {
```


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```
            /*ro, opt, object, lane information capability, desc:it is valid when ruleMode is "face_human_vehice"*/
                "LaneListCap": {
                /*ro, req, object, lane list capability*/
                    "maxSize":  4,
                    /*ro, req, int, capability of maximum left lane lines*/
                    "LeftLaneLineCap": {
                    /*ro, req, object, left lane line capability*/
                        "maxSize":  2,
                        /*ro, req, int, capability of maximum queue management rules*/
                        "x": {
                        /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
                            "@min":  0.000,
                            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                            "@max":  1.000,
                            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                            "#text":  0.120
                            /*ro, opt, float, range:[0.000,1.000]*/
                        },
                        "y": {
                        /*ro, req, object, Y-coordinate, desc:the origin is the lower-left corner of the screen*/
                            "@min":  0.000,
                            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                            "@max":  1.000,
                            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                            "#text":  0.120
                            /*ro, opt, float, range:[0.000,1.000]*/
                        }
                    },
                    "RightLaneLineCap": {
                    /*ro, req, object, right lane line capability*/
                        "maxSize":  2,
                        /*ro, req, int, capability of maximum queue management rules*/
                        "x": {
                        /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
                            "@min":  0.000,
                            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                            "@max":  1.000,
                            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                            "#text":  0.120
                            /*ro, opt, float, range:[0.000,1.000]*/
                        },
                        "y": {
                        /*ro, req, object, Y-coordinate, desc:the origin is the lower-left corner of the screen*/
                            "@min":  0.000,
                            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                            "@max":  1.000,
                            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                            "#text":  0.120
                            /*ro, opt, float, range:[0.000,1.000]*/
                        }
                    }
                },
                "alarmDisplayMode": {
                /*ro, opt, object*/
                    "@opt":  "face_human_vehice_nonMotor,human_vehice_nonMotor,face_human",
                    /*ro, opt, string*/
                    "@def":  "face_human_vehice_nonMotor"
                    /*ro, opt, string*/
                },
                "retrogradeDetectionList": {
                /*ro, opt, object*/
                    "@size":  2,
                    /*ro, opt, int*/
                    "retrogradeDetectionLine": {
                    /*ro, opt, object*/
                        "@size":  2,
                        /*ro, opt, int*/
                        "x": {
                        /*ro, req, object*/
                            "@min":  0.000,
                            /*ro, req, float, range:[0.000,1.000]*/
                            "@max":  1.000,
                            /*ro, req, float, range:[0.000,1.000]*/
                            "#text":  0.120
                            /*ro, opt, float, range:[0.000,1.000]*/
                        },
                        "y": {
                        /*ro, req, object*/
                            "@min":  0.000,
                            /*ro, req, float, range:[0.000,1.000]*/
                            "@max":  1.000,
                            /*ro, req, float, range:[0.000,1.000]*/
                            "#text":  0.120
                            /*ro, opt, float, range:[0.000,1.000]*/
                        }
                    },
                    "retrogradeDirectionLine": {
                    /*ro, opt, object*/
                        "startPoint": {
                        /*ro, req, object*/
                            "x": {
                            /*ro, req, object*/
                                "@min":  0.000,
```

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```
                    "@min":  0.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, req, object*/
                    "@min":  0.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                }
            },
            "stopPoint": {
            /*ro, req, object*/
                "x": {
                /*ro, req, object*/
                    "@min":  0.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, req, object*/
                    "@min":  0.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                }
            }
        }
    }
},
"KeyRegionInfo": {
/*ro, opt, object, key region information*/
    "enabled":  "true,false",
    /*ro, req, string, whether to enable the key region*/
    "RegionList": {
    /*ro, opt, object, key region array*/
        "maxSize":  8,
        /*ro, req, int, capability of maximum queue management rules*/
        "id": {
        /*ro, req, object, region ID*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  8
            /*ro, req, int, the maximum value*/
        },
        "Region": {
        /*ro, req, object, key region frame (rectangle)*/
            "x": {
            /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                "#text":  0.120
                /*ro, opt, float, range:[0.000,1.000]*/
            },
            "y": {
            /*ro, req, object, Y-coordinate, desc:the origin is the lower-left corner of the screen*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                "#text":  0.120
                /*ro, opt, float, range:[0.000,1.000]*/
            },
            "width": {
            /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                "#text":  0.120
                /*ro, opt, float, range:[0.000,1.000]*/
            },
            "height": {
            /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                "#text":  0.120
                /*ro, opt, float, range:[0.000,1.000]*/
            }
```


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```
            }
        }
    },
    "MultiScene": {
    /*ro, opt, object, multi-scene information*/
        "maxSceneNum":  8,
        /*ro, req, int, maximum number of scenes*/
        "sid": {
        /*ro, req, object, scene ID*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  8
            /*ro, req, int, the maximum value*/
        },
        "sceneName": {
        /*ro, opt, object, scene name*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  256
            /*ro, req, int, the minimum value*/
        },
        "parkTime": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  256
            /*ro, req, int, the maximum value*/
        }
    },
    "isSupportCapturePicture":  true,
    /*ro, opt, bool, whether supports setting capture parameters for multi-target-type detection, desc:whether supports setting capture parameters for
multi-target-type detection,if not supports,this node will not be returned*/
    "isSupportCapturePicOverlap":  true,
    /*ro, opt, bool, whether supports setting parameters of overlaying information on captured pictures, desc:whether supports setting parameters of
overlaying information on captured pictures,if not supports,this node will not be returned*/
    "isSupportAlgParam":  true,
    /*ro, opt, bool, whether supports setting advanced algorithm library parameters for multi-target-type detection, desc:whether supports setting
advanced algorithm library parameters for multi-target-type detection,if not supports,this node will not be returned*/
    "isSupportAlarmInfoGet":  true,
    /*ro, opt, bool, whether supports getting captured multi-target-type information, desc:whether supports getting captured multi-target-type
information,"true"-support,"false"-not support*/
    "province": {
    /*ro, opt, object, reserved, desc:reserved*/
        "@opt":  "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,255",
        /*ro, opt, string, options*/
        "@def":  33
        /*ro, opt, int, the default value*/
    },
    "ruleModeChangedPrompt": {
    /*ro, opt, object*/
        "opt":  "prompt1"
        /*ro, req, string, options, desc:options*/
    },
    "MultiSceneMode": {
    /*ro, opt, object*/
        "sceneMode": {
        /*ro, opt, object*/
            "@opt":  "normal,expert",
            /*ro, req, string, options*/
            "@def":  "normal"
            /*ro, req, string, the default value*/
        },
        "normalSceneNum": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max": 1
            /*ro, req, int, the maximum value*/
        },
        "expertSceneNum": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  8
            /*ro, req, int, the maximum value*/
        }
    },
    "isSupportOverlay":  true,
    /*ro, opt, bool, whether it supports OSD overlay of mixed targets, desc:whether it supports OSD overlay of mixed targets,"true" will be returned for
supporting; Otherwise,this node will not be returned*/
    "pictureQuality": {
    /*ro, opt, object, picture quality*/
        "@opt":  "high,middle,low",
        /*ro, req, string*/
        "@def":  "middle"
        /*ro, req, string, the default value*/
    },
    "isSupportBinaryPicUp":  true,
    /*ro, opt, bool*/
    "convertBinToBmpEnabled": {
    /*ro, opt, object*/
        "@opt": [true, false]
        /*ro, req, array, range, subType:bool*/
```

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```
        },
        "MEF": {
        /*ro, opt, object*/
            "enabled":  "true,false",
            /*ro, req, string, whether to enable multi-target-type detection, desc:whether to enable multi-target-type detection*/
            "offsetTimes": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, the minimum value, unit:min, desc:the minimum value*/
                "@max":  10,
                /*ro, req, int, maximum value, unit:min*/
                "@def":  1
                /*ro, req, int, the default value, unit:min*/
            },
            "leftOffsetTimes": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, the minimum value, unit:s, desc:the minimum value*/
                "@max":  300,
                /*ro, req, int, maximum value, unit:s*/
                "@def":  60
                /*ro, req, int, the default value, unit:s*/
            },
            "rightOffsetTimes": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, the minimum value, unit:s, desc:the minimum value*/
                "@max":  30,
                /*ro, req, int, maximum value, unit:s*/
                "@def":  30
                /*ro, req, int, the default value, unit:s*/
            }
        },
        "plateInfo": {
        /*ro, opt, object*/
            "region": {
            /*ro, opt, object*/
                "@opt": ["AFandAM"],
                /*ro, opt, array, subType:string*/
                "@def":  "APAC"
                /*ro, opt, string*/
            },
            "CRIndex": {
            /*ro, opt, object*/
                "@opt": ["0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10", "11", "12", "13", "14", "15", "16", "17", "18", "19", "20", "21", "22",
"23", "25", "26", "28", "29", "30", "31", "33", "34", "35", "36", "37", "38", "39", "40", "42", "43", "44", "45", "46", "47", "49", "51", "53", "55", "59",
"60", "62", "65", "68", "70", "71", "72", "73", "76", "77", "84", "87", "89", "90", "94", "95", "91", "104", "107", "108", "110", "114", "115", "130",
"137", "139", "140", "144", "145", "146", "151", "156", "157", "169", "174", "175", "177", "180", "181", "188", "206", "210", "215", "216", "217", "218",
"219", "220", "221", "222", "227", "228", "256"]
                /*ro, opt, array, subType:string*/
            },
            "relatedRegionAndCRList": [
            /*ro, opt, array, subType:object*/
                {
                    "region":  "AFandAM",
                    /*ro, req, enum, subType:string*/
                    "CRIndex": {
                    /*ro, opt, object*/
                        "@opt": ["0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10", "11", "12", "13", "14", "15", "16", "17", "18", "19", "20", "21",
"22", "23", "25", "26", "28", "29", "30", "31", "33", "34", "35", "36", "37", "38", "39", "40", "42", "43", "44", "45", "46", "47", "49", "51", "53", "55",
"59", "60", "62", "65", "68", "70", "71", "72", "73", "76", "77", "84", "87", "89", "90", "94", "95", "91", "104", "107", "108", "110", "114", "115", "130",
"137", "139", "140", "144", "145", "146", "151", "156", "157", "169", "174", "175", "177", "180", "181", "188", "206", "210", "215", "216", "217", "218",
"219", "220", "221", "222", "227", "228", "256"]
                        /*ro, opt, array, subType:string*/
                    }
                }
            ]
        },
        "nonMotorizedViolationUpload": {
        /*ro, opt, object*/
            "postEnabled": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, req, array, subType:bool*/
            },
            "illegalEnabled": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, req, array, subType:bool*/
            }
        }
    },
    "isSupportLineCrossingCapture":  true,
    /*ro, opt, bool*/
    "isSupportOneClickZoom":  true
    /*ro, opt, bool*/
}
```

15.8.1.2 Set the captured picture overlay parameters of multi-target-detection

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Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/capturePicOverlap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "MixedTargetCapturePicOverlap": {
    /*req, object*/
        "OverlapItemList": [
        /*req, array, subType:object*/
            {
                "overlapItem": {
                /*req, object*/
                    "overlapItemId":  1,
                    /*req, int*/
                    "itemType":  "cameraNo"
                    /*req, enum, subType:string*/
                }
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

15.8.1.3 Get OSD overlay configuration capability for detecting mixed targets

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/overlay/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
{
    "MixedTargetOverlayCap": {
    /*ro, req, object, OSD overlay configuration capability for mixed targets*/
        "enabled": {
        /*ro, opt, object, whether to enable OSD overlay capability for detecting mixed targets; OSD calculation data will be cleared if this node is
reset*/
            "@opt":  "true,false",
            /*ro, opt, string, options*/
            "@def":  false
            /*ro, opt, bool, the default value*/
        },
        "x": {
        /*ro, opt, object, x-coordinate; the value ranges from 0.000 to 1; the upper-left corner is set as the origin point*/
            "@min":  0.000,
            /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000
            /*ro, opt, float, the maximum value, range:[0.000,1.000]*/
        },
        "y": {
        /*ro, opt, object, y-coordinate; the value ranges from 0.000 t0 1; the upper-left corner is set as the origin point*/
            "@min":  0.000,
            /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000
            /*ro, opt, float, the maximum value, range:[0.000,1.000]*/
        },
        "fontSize": {
        /*ro, opt, object, font size of overlaid information*/
            "@opt": ["16*16", "32*32", "48*48", "64*64", "96*96", "adaptive"]
            /*ro, opt, array, "adaptive"-size adaptive (by default), subType:string*/
        },
        "targetDynamicMaskEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "targetDynamicMaskLevel": {
        /*ro, opt, object*/
            "@opt": ["high", "low", "medium"]
            /*ro, opt, array, subType:string*/
        },
        "isSupportOSDReset":  true,
        /*ro, opt, bool, if "true" is returned, it indicates that manually clearing OSD calculation data of mixed targets detection is supported (the URI is
available: /ISAPI/Intelligent/channels/<ID>/mixedTargetDetection/overlay/OSDReset?format=json*/
        "isSupportOSDAutoReset":  true
        /*ro, opt, bool, if "true" is returned, it indicates that auto clearing OSD calculation data of mixed targets detection is supported (the URI is
available: /ISAPI/Intelligent/channels/<ID>/mixedTargetDetection/overlay/OSDAutoReset?format=json*/
    }
}
```

15.8.1.4 Get OSD overlay configuration parameters for detecting mixed targets

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/overlay?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
{
    "MixedTargetOverlay": {
    /*ro, req, object, OSD overlay configuration parameters for detecting mixed targets*/
        "enabled":  false,
        /*ro, opt, bool, whether to enable OSD overlay capability for detecting mixed targets. By default, it is set to false*/
        "x":  0.120,
        /*ro, opt, float, X-coordinate, the value ranges from 0.000 to 1. The upper-left corner is set as the origin point, range:[0.000,1.000]*/
        "y":  0.120,
        /*ro, opt, float, Y-coordinate, the value ranges from 0.000 to 1. The upper-left corner is set as the origin point, range:[0.000,1.000]*/
        "fontSize":  "adaptive",
        /*ro, opt, enum, font size of overlaid information, subType:string, desc:"16*16", "32*32", "48*48", "64*64", "96*96", "adaptive" (size adaptive, by
default)*/
        "targetDynamicMaskEnabled":  true,
        /*ro, opt, bool*/
        "targetDynamicMaskLevel":  "high"
        /*ro, opt, enum, subType:string, dep:or,{$.MixedTargetOverlay.targetDynamicMaskEnabled,eq,true}*/
    }
}
```

15.8.1.5 Set single-scene mode parameters of multi-target-type detection

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "MixedTargetDetection": {
    /*opt, object, rule parameters of multi-target-type detection (by default, the first scene)*/
        "enabled":  true,
        /*req, bool, whether to enable globally*/
        "TypesDetection": [
        /*opt, array, subType:object*/
            {
                "ruleMode":  "human",
                /*opt, enum, rule type, subType:string, desc:human: human body; face; vehice; nonMotor: non-motor vehicle;*/
                "enabled":  true,
                /*opt, bool, whether to enable*/
                "Property": [
                /*opt, array, subType:object*/
                    {
                        "description":  "jacketColor",
                        /*opt, enum, subType:string*/
                        "enabled":  false
                        /*opt, bool, whether to enable*/
                    }
                ]
            }
        ],
        "RuleInfo": [
        /*opt, array, rule information, subType:object*/
            {
                "ruleID":  1,
                /*req, int, rule ID*/
                "ruleMode":  "face_human_vehice_nonMotor",
                /*req, enum, rule type, subType:string, desc:rule type: face + human body + vehicle + non-motor vehicle,human body + vehicle + non-motor
vehicle*/
                "Region": [
                /*req, array, region configured with rules, subType:object, range:[3,10], desc:the origin is the lower-left corner of the screen*/
                    {
                        "x":  0.120,
                        /*req, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.120
                        /*req, float, Y-coordinate, range:[0.000,1.000]*/
                    }
                ],
                "InterPupileRegion": {
                /*req, object, minimum rectangle region to detect pupil distance, desc:The origin is the lower-left corner of the screen.*/
                    "x":  0.120,
                    /*req, float, X-coordinate of the point in the top left corner, range:[0.000,1.000]*/
                    "y":  0.120,
                    /*req, float, Y-coordinate of the point in the top left corner, range:[0.000,1.000]*/
                    "width":  0.120,
                    /*req, float, rectangle width, range:[0.000,1.000]*/
                    "height":  0.120
                    /*req, float, rectangle height, range:[0.000,1.000]*/
                },
                "MaxInterPupilRegion": {
                /*opt, object, maximum rectangle region to detect pupil distance, desc:the origin is the lower-left corner of the screen.*/
```

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```
                /* opt, object, maximum rectangle region to detect pupil distance, desc:the origin is the lower-left corner of the screen. */
                    "x":  0.120,
                    /*req, float, X-coordinate of the point in the top left corner, range:[0.000,1.000]*/
                    "y":  0.120,
                    /*req, float, Y-coordinate of the point in the top left corner, range:[0.000,1.000]*/
                    "width":  0.120,
                    /*req, float, rectangle width, range:[0.000,1.000]*/
                    "height":  0.120
                    /*req, float, rectangle height, range:[0.000,1.000]*/
                },
                "LaneInfo": {
                /*opt, object, lane information, desc:lane information,it is valid when ruleMode is "face_human_vehice"*/
                    "LaneList": [
                    /*req, array, lane list, subType:object*/
                        {
                            "LeftLaneLine": [
                            /*req, array, left lane line capability, subType:object, range:[0,2], desc:the origin is the lower-left corner of the screen.*/
                                {
                                    "x":  0.120,
                                    /*req, float, Y-coordinate, range:[0.000,1.000]*/
                                    "y":  0.120
                                    /*req, float, Y-coordinate, range:[0.000,1.000]*/
                                }
                            ]
                        }
                    ],
                    "RightLaneLine": [
                    /*req, array, right lane line, subType:object, range:[0,2], desc:the origin is the lower-left corner of the screen.*/
                        {
                            "x":  0.120,
                            /*req, float, Y-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*req, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                }
            }
        ],
        "KeyRegionInfo": {
        /*opt, object, key region information*/
            "enabled":  true,
            /*req, bool, whether to enable key region*/
            "RegionList": [
            /*opt, array, key region array, subType:object*/
                {
                    "id":  1,
                    /*req, int, detection area ID*/
                    "Region": [
                    /*req, array, key region, subType:object, desc:The origin is the lower-left corner of the screen.*/
                        {
                            "x":  0.120,
                            /*req, float, X-coordinate, range:[0.000,1.000]*/
                            "y":  0.120,
                            /*req, float, Y-coordinate, range:[0.000,1.000]*/
                            "width":  0.120,
                            /*req, float, rectangle width, range:[0.000,1.000]*/
                            "height":  0.120
                            /*req, float, rectangle height, range:[0.000,1.000]*/
                        }
                    ]
                }
            ]
        },
        "province":  1,
        /*opt, enum, reserved, subType:int, desc:reserved*/
        "pictureQuality":  "high",
        /*opt, enum, picture quality, subType:string*/
        "isSupportBinaryPicUp":  true,
        /*opt, bool*/
        "convertBinToBmpEnabled":  true,
        /*opt, bool*/
        "MEF": {
        /*opt, object*/
            "enabled":  true,
            /*req, bool, whether to enable multi-target-type detection*/
            "offsetTimes":  1,
            /*opt, int, range:[0,10], unit:min*/
            "leftOffsetTimes":  60,
            /*opt, int, range:[0,300], unit:s*/
            "rightOffsetTimes":  30
            /*opt, int, range:[0,30], unit:s*/
        }
    }
}
```


Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

15.8.1.6 Get single-scene mode parameters of multi-target-type detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetDetection": {
    /*ro, opt, object*/
        "enabled":  true,
        /*ro, req, bool, whether to enable multi-target-type detection*/
        "TypesDetection": [
        /*ro, opt, array, subType:object*/
            {
                "ruleMode":  "human",
                /*ro, opt, enum, rule type, subType:string, desc:face + human body + vehicle + non-motor vehicle, human body + vehicle + non-motor vehicle*/
                "enabled":  true,
                /*ro, opt, bool, whether to enable multi-target-type detection*/
                "Property": [
                /*ro, opt, array, Attribute Information, subType:object*/
                    {
                        "description":  "jacketColor",
                        /*ro, opt, enum, subType:string*/
                        "enabled":  false
                        /*ro, opt, bool, whether to enable multi-target-type detection*/
                    }
                ]
            }
        ],
        "RuleInfo": [
        /*ro, opt, array, rule information, subType:object*/
            {
                "ruleID":  1,
                /*ro, req, int, rule ID*/
                "ruleMode":  "face_human_vehice_nonMotor",
                /*ro, opt, enum, rule type, subType:string, desc:rule type: face + human body + vehicle + non-motor vehicle,human body + vehicle + non-motor
vehicle*/
                "Region": [
                /*ro, req, array, rule area, subType:object, range:[3,10], desc:the origin is the lower-left corner of the screen*/
                    {
                        "x":  0.120,
                        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.120
                        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    }
                ],
                "InterPupileRegion": {
                /*ro, req, object, minimum rectangle region to detect pupil distance, desc:the origin is the lower-left corner of the screen*/
                    "x":  0.120,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.120,
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    "width":  0.120,
                    /*ro, req, float, rectangle height, range:[0.000,1.000]*/
                    "height":  0.120
                    /*ro, req, float, rectangle width, range:[0.000,1.000]*/
                },
                "MaxInterPupilRegion": {
                /*ro, opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen*/
                    "x":  0.120,
```

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```
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.120,
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    "width":  0.120,
                    /*ro, req, float, rectangle height, range:[0.000,1.000]*/
                    "height":  0.120
                    /*ro, req, float, rectangle width, range:[0.000,1.000]*/
                },
                "LaneInfo": {
                /*ro, opt, object, lane information, desc:it is valid when ruleMode is "face_human_vehice"*/
                    "LaneList": [
                    /*ro, req, array, lane list, subType:object*/
                        {
                            "LeftLaneLine": [
                            /*ro, req, array, left lane line capability, subType:object, range:[0,2], desc:the origin is the lower-left corner of the
screen*/
                                {
                                    "x":  0.120,
                                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                                    "y":  0.120
                                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                                }
                            ]
                        }
                    ],
                    "RightLaneLine": [
                    /*ro, req, array, right lane line, subType:object, range:[0,2], desc:the origin is the lower-left corner of the screen*/
                        {
                            "x":  0.120,
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                },
                "retrogradeDetectionList": [
                /*ro, opt, array, subType:object, range:[0,2]*/
                    {
                        "retrogradeDetectionLine": [
                        /*ro, opt, array, subType:object, range:[2,2]*/
                            {
                                "x":  0.120,
                                /*ro, req, float, range:[0.000,1.000]*/
                                "y":  0.120
                                /*ro, req, float, range:[0.000,1.000]*/
                            }
                        ],
                        "retrogradeDirectionLine": {
                        /*ro, opt, object*/
                            "startPoint": {
                            /*ro, req, object*/
                                "x":  0.120,
                                /*ro, req, float, range:[0.000,1.000]*/
                                "y":  0.120
                                /*ro, req, float, range:[0.000,1.000]*/
                            },
                            "stopPoint": {
                            /*ro, req, object*/
                                "x":  0.120,
                                /*ro, req, float, range:[0.000,1.000]*/
                                "y":  0.120
                                /*ro, req, float, range:[0.000,1.000]*/
                            }
                        }
                    }
                ],
                "nonMotorOverspeedThreshoid":  30,
                /*ro, opt, int, range:[10,50], step:1, unit:km/h*/
                "vehicleOverspeedThreshoid":  40
                /*ro, opt, int, range:[30,80], step:1, unit:km/h*/
            }
        ],
        "KeyRegionInfo": {
        /*ro, opt, object, key region information*/
            "enabled":  true,
            /*ro, req, bool, whether to enable multi-target-type detection*/
            "RegionList": [
            /*ro, opt, array, key region array, at least two key regions are required, subType:object*/
                {
                    "id":  1,
                    /*ro, req, int, detection area ID*/
                    "Region": [
                    /*ro, req, array, Key Area, subType:object, desc:the origin is the lower-left corner of the screen*/
                        {
                            "x":  0.120,
                            /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                            "y":  0.120,
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                            "width":  0.120,
                            /*ro, req, float, rectangle height, range:[0.000,1.000]*/
                            "height":  0.120
                            /*ro, req, float, rectangle width, range:[0.000,1.000]*/
                        }
                    ]
```

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```
                }
            ]
        },
        "province": 1,
        /*ro, opt, enum, reserved, subType:int, desc:reserved*/
        "pictureQuality": "high",
        /*ro, opt, enum, picture quality, subType:string, desc:picture quality*/
        "isSupportBinaryPicUp": true,
        /*ro, opt, bool*/
        "convertBinToBmpEnabled": true,
        /*ro, opt, bool*/
        "MEF": {
        /*ro, opt, object*/
            "enabled": true,
            /*ro, req, bool, whether to enable multi-target-type detection*/
            "offsetTimes": 1,
            /*ro, opt, int, range:[0,10], unit:min*/
            "leftOffsetTimes": 60,
            /*ro, opt, int, range:[0,300], unit:s*/
            "rightOffsetTimes": 30
            /*ro, opt, int, range:[0,30], unit:s*/
        },
        "plateInfo": {
        /*ro, opt, object*/
            "region": "AFandAM",
            /*ro, opt, enum, subType:string*/
            "CRIndex": "0"
            /*ro, opt, enum, subType:string*/
        },
        "nonMotorizedViolationUpload": {
        /*ro, opt, object*/
            "postEnabled": true,
            /*ro, opt, bool*/
            "illegalEnabled": true
            /*ro, opt, bool*/
        }
    }
}
```

15.8.1.7 Set the shielded area parameters of multi-target-detection event

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/Shield/mixedTargetDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Shield xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, enum, subType:string-->true
  </enabled>
  <RegionList>
    <!--opt, array, subType:object-->
    <Region>
      <!--opt, object-->
      <id>
        <!--req, string-->1
      </id>
      <enabled>
        <!--req, enum, subType:string-->true
      </enabled>
      <RegionCoordinatesList>
        <!--opt, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </Region>
  </RegionList>
</Shield>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

15.8.1.8 Multi-target-type detection

EventType:mixedTargetDetection

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "dateTimeMillisecond":  999,
    /*ro, opt, int, the milliseconds of the alarm trigger time, range:[0,999], desc:it should be used together with field dateTime*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:the event triggering frequency. The pulse event is the event which is triggered continuously according to the detection frequency of the device, e.g., the motion detection. The instant event is the event triggered when object recognition occurs, and the recognition of each object will trigger the event once, e.g., the facial recognition. When the pulse events are triggered, this node is used to tell whether the events are triggered by the same source. For example, if it is motion detection, the alarms will be uploaded continuously according to the device detection frequency; if it changes to another source, then the counting will start again*/
    "eventType":  "mixedTargetDetection",
    /*ro, req, enum, event type, subType:string, desc:"mixedTargetDetection" (multi-target-type detection event)*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "Mixed target detection",
    /*ro, req, enum, event description, subType:string, desc:"Mixed target detection" (multi-target-type detection event)*/
    "channelName":  "ABC",
    /*ro, opt, string, channel name (camera name), desc:it can be configured via URI (/ISAPI/Streaming/channels/<ID>)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID, desc:it can be configured via URI (/ISAPI/System/Network/Ehome)*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished, desc:whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers.)*/
    "customChanID":  "test",
    /*ro, opt, string, custom camera channel No., desc:the maximum string size is 64. The value of this node is that of the node <customChanID> configured by /ISAPI/ContentMgmt/InputProxy/channels/<ID>*/
    "faceBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"faceBackgroundImage" (face background picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "humanBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"humanBackgroundImage" (human body background picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "humanImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"humanImage" (human body thumbnail)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
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        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "nonMotorBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"nonMotorBackgroundImage" (background picture of the non-motor vehicle)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "vehicleBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"vehicleBackgroundImage" (vehicle background picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "vehicleImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"vehicleImage" (vehicle captured picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "CaptureResult": [
    /*ro, opt, array, captured results of detected target, subType:object*/
        {
            "targetID":  1,
            /*ro, req, int, captured target ID*/
            "Face": {
            /*ro, opt, object, face attribute*/
                "Rect": {
                /*ro, opt, object, face thumbnail coordinates, desc:rectangular frame coordinates of the face thumbnail, which is based on the background
picture (faceBackgroundImage). The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, face attribute details, subType:object*/
                    {
                        "description":  "age",
                        /*ro, opt, enum, attributes description, subType:string, desc:"age" (age), "ageGroup" (age group), "gender" (gender), "glass"
(whether the person wears glasses), "smile" (whether the person wears a smile), "mask" (whether the person wears a mask), "faceExpression" (face
expression), "hat" (whether the person wears a hat), "score" (face score)*/
                        "value":  "23"
                        /*ro, opt, string, it depends on the value of description, desc:the returned value type of this field is string*/
                    }
                ],
                "contentID1":  "faceImage",
                /*ro, opt, enum, face thumbnail, subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "nonMotorImage"
(non-motor vehicle thumbnail), "vehicleImage" (vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share
one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "faceBackgroundImage",
                /*ro, opt, enum, "faceBackgroundImage" (background picture of the face), subType:string, desc:"faceBackgroundImage" (face background
picture), "humanBackgroundImage" (human body background picture), "nonMotorBackgroundImage" (non-motor vehicle background picture), "plateImage" (vehicle
license plate thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and
non-motor vehicle are detected, these two properties share one picture and "nonMotorBackgroundImage" will be returned*/
                "FacePictureRect": {
                /*ro, opt, object, coordinates of face picture containing head and shoulder, desc:it is based on the face background picture matting and
equals to the face thumbnail uploaded by the device. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, face thumbnail (contains head and shoulder) ID, which indicates the uniqueness of the picture data, range:[0,32],
desc:both "contentID" and "pId" are available for the data transmission format Content-Type: multipart/form-data, and implement the relation between picture
data and structured data of corresponding contents. "contentID1" and "contentID2" correspond to "contentID", "pId1" and "pId2" correspond to "name" and
"filename". The advantage of "pId" is that it has better compatibility, it can match data via "name" and indicate the file name and format via "filename".
Both "contentID" and "pId" need to be returned, while "faceImageURL" and "faceBackgroundImageURL" are not required*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, face background picture ID*/
                "faceImageURL":  "test",
                /*ro, opt, string, face thumbnail URL*/
                "faceBackgroundImageURL":  "test",
                /*ro, opt, string, face background picture URL, desc:face background picture URL*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, face capture time*/
                "snapTimeMillisecond":  999
                /*ro, opt, int, the milliseconds of face capture time, range:[0,999], desc:it should be used together with the field snapTime*/
            },
            "Human": {
            /*ro, opt, object, human body attributes*/
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            /*ro, opt, object, human body attributes*/
                "Rect": {
                /*ro, req, object, target frame (based on the background picture), desc:coordinates of the human body target frame, which is based on the background picture matting. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, attributes data, subType:object*/
                    {
                        "description":  "jacketColor",
                        /*ro, opt, enum, attributes description, subType:string, desc:"jacketColor" (top color), "trousersColor" (bottoms color), "ageGroup" (age group), "gender", "glass" (whether the person wears glasses), "ride" (whether the person is riding a bike), "bag" (whether the person is carrying a bag), "things"(whether the person is carrying things), "hat" (whether the person wears a hat), "trousersType" (bottoms type), "mask" (whether the person wears a mask), "hairStyle" (hairstyle), "cyclingPersonNumber" (number of riders), "direction" (target direction), "score" (human body grading), "hatStyle" (hat style), "cyclingType" (riding type), "jacketType" (top type)*/
                        "value":  "23"
                        /*ro, opt, string, it depends on the value of description, desc:the returned value type of this field is string*/
                    }
                ],
                "modeldata":  "test",
                /*ro, opt, string, human body model data (encoded by base64)*/
                "contentID1":  "humanImage",
                /*ro, opt, enum, human body thumbnail, subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "nonMotorImage" (non-motor vehicle thumbnail), "vehicleImage" (vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "humanBackgroundImage",
                /*ro, opt, enum, human body background picture, subType:string, desc:"faceBackgroundImage" (face background picture), "humanBackgroundImage" (human body background picture), "nonMotorBackgroundImage" (non-motor vehicle background picture), "plateImage" (vehicle license plate thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorBackgroundImage" will be returned*/
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, human body thumbnail ID, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one thumbnail. The value of pId1 is same as that of pId1 in nonMotor, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle thumbnail*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, background picture ID of the human body, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one background picture. The value of pId2 is same as that of pId2 in nonMotor, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle background picture*/
                "humanImageURL":  "test",
                /*ro, opt, string, human body thumbnail URL, desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one picture. The value of humanImageURL is same as that of nonMotorImageURL, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle thumbnail*/
                "humanBackgroundImageURL":  "test",
                /*ro, opt, string, background picture URL of the human body, desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one picture. The value of humanBackgroundImageURL is same as that of nonMotorBackgroundImageURL*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, human body capture time*/
                "snapTimeMillisecond":  999,
                /*ro, opt, int, milliseconds of human body capture time, range:[0,999], desc:it should be used together with the field snapTime*/
                "contentID3":  "humanMergeImage",
                /*ro, opt, string, "humanMergeImage" (picture of human body merging details)*/
                "pId3":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, ID of the picture of human body merging details, range:[0,32]*/
                "contentID4":  "humanBackgroundImage2",
                /*ro, opt, enum, 2nd background picture of human body, subType:string, desc:"humanBackgroundImage2" (2nd background picture of human body)*/
                "contentID5":  "humanBackgroundImage3",
                /*ro, opt, enum, 3rd background picture of human body, subType:string, desc:"humanBackgroundImage3" (3rd background picture of human body)*/
                "pId4":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, ID of 2nd background picture of human body, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one background picture. The value of pId4 is same as that of pId2 in nonMotor, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle background picture*/
                "pId5":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, ID of 3rd background picture of human body, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one background picture. The value of pId5 is same as that of pId2 in nonMotor, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle background picture*/
                "humanBackgroundImage2URL":  "test",
                /*ro, opt, string, URL of 2nd human body background picture, desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one picture. The value of humanBackgroundImageURL is same as that of nonMotorBackgroundImageURL*/
                "humanBackgroundImage3URL":  "test",
                /*ro, opt, string, URL of 3rd human body background picture, desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one picture. The value of humanBackgroundImageURL is same as that of nonMotorBackgroundImageURL*/
                "HumanImageRect": {
                /*ro, opt, object, target rectangle frame based on the human body thumbnail, desc:the coordinates of the human body target frame, which is based on the human body thumbnail matting. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "smallImageRectInBackground": {
                /*ro, opt, object, rectangle frame of human body thumbnail (based on the background picture), desc:the rectangle frame is the area of the small human body picture based on humanBackgroundImage. The matted picture is the same as humanImage. The origin is the upper-left corner of the screen*/
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                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "relatedMixedTargetUid": ["test", "1"],
                /*ro, opt, array, UID array of linked multi-target-type detection events, subType:string, desc:corresponding to the "uid" of
mixedTargetDetection (multi-target-type detection event). when a device uploads human body properties, other related targets attributes will be linked, and
the client software can get the information about targets around the captured human body*/
                "contentID6":  "featurePicture",
                /*ro, opt, enum, close-up picture, subType:string, desc:"featurePicture" (close-up picture)*/
                "pId6":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, close-up picture ID, range:[0,32]*/
                "featurePictureURL":  "test"
                /*ro, opt, string, close-up picture URL*/
            },
            "NonMotor": {
            /*ro, opt, object, non-motor vehicle attributes*/
                "Rect": {
                /*ro, req, object, frame information of non-motor vehicle thumbnail, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, attributes data, subType:object*/
                    {
                        "description":  "nonMotorType",
                        /*ro, opt, enum, attributes description, subType:string, desc:"detectType" (device type), "nonMotorType" (non-motor vehicle type),
"plateColor" (license plate color of non-motor vehicle), "plateNo" (license plate number of non-motor vehicle), "directionIndex" (vehicle driving
direction), "laneNo" (lane number), "RFID" (linked RFID), "EPlateChipNo" (chip number of electronic license plate), "EPlateCardNo" (card number of
electronic license plate), "EPlateListenAuthCode" (licensing authority code of electronic license plate), "EPlateNo" (electronic license plate number),
"EPlateManufacturingDate" (data of production of electronic license plate), "EPlateInspectionValidity" (inspection validity of electronic license plate),
"EPlateMandaScrapPeriod" (compulsory scrapping period of electronic license plate), "EPlatePassengersCapacity" (authorized passenger capacity),
"EPlateCarryingCapacity" (tractive tonnage), "EPlateVehicleColor" (non-motor vehicle color), "EPlateNonMotorType" (non-motor vehicle type),
"EPlateNonMotorPlateType" (license plate type of non-motor vehicle), "EPlateColor" (license plate color of non-motor vehicle), "EPlateUseType" (function
properties of non-motor vehicle), "EPlateTriggerTime" (detection trigger time), "retrograde" (wrong-way driving of non-motor vehicle), "manned" (carrying
passengers on non-motor vehicle), "hatStyle" (hat style)*/
                        "value":  "23"
                        /*ro, opt, string, it depends on the value of description, desc:the returned value type of this field is string*/
                    }
                ],
                "contentID1":  "nonMotorImage",
                /*ro, opt, enum, face picture ID, subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "nonMotorImage"
(non-motor vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body
and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "nonMotorBackgroundImage",
                /*ro, opt, enum, background picture of the non-motor vehicle, subType:string, desc:"faceBackgroundImage" (face background picture),
"humanBackgroundImage" (human body background picture), "nonMotorBackgroundImage" (non-motor vehicle background picture). When there are multiple properties
for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one
picture and "nonMotorBackgroundImage" will be returned*/
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, ID of the non-motor vehicle thumbnail, range:[0,32]*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, background picture ID of the non-motor vehicle, range:[0,32]*/
                "nonMotorImageURL":  "test",
                /*ro, opt, string, URL of the non-motor vehicle thumbnail*/
                "nonMotorBackgroundImageURL":  "test",
                /*ro, opt, string, background picture URL of the non-motor vehicle*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, non-motor vehicle capture time*/
                "snapTimeMillisecond":  999,
                /*ro, opt, int, it should be used together with the field snapTime, range:[0,999], desc:it should be used together with the field snapTime*/
                "contentID3":  "nonMotorBackgroundImage2",
                /*ro, opt, enum, 2nd background picture of the non-motor vehicle, subType:string, desc:"nonMotorBackgroundImage2" (2nd background picture of
the non-motor vehicle)*/
                "contentID4":  "nonMotorBackgroundImage3",
                /*ro, opt, enum, 3rd background picture of the non-motor vehicle, subType:string, desc:"nonMotorBackgroundImage3" (3rd background picture of
the non-motor vehicle)*/
                "contentID5":  "faceImage",
                /*ro, opt, enum, face thumbnail, subType:string, desc:"faceImage" (face thumbnail)*/
                "contentID6":  "plateImage",
                /*ro, opt, enum, license plate thumbnail, subType:string, desc:"plateImage" (license plate thumbnail)*/
                "pId3":  "F4F665D6A18E41308CE9934DCDDD2223",
                /*ro, opt, string, 2nd background picture ID of the non-motor vehicle, range:[0,32]*/
                "pId4":  "F4F665D6A18E41308CE9934DCDDD2224",
                /*ro, opt, string, 3rd background picture ID of the non-motor vehicle, range:[0,32]*/
                "pId5":  "F4F665D6A18E41308CE9934DCDDD2225",
                /*ro, opt, string, face thumbnail ID, range:[0,32]*/
                "pId6":  "F4F665D6A18E41308CE9934DCDDD2226",
                /*ro, opt, string, license plate thumbnail ID, range:[0,32]*/
                "nonMotorBackgroundImage2URL":  "test",
                /*ro, opt, string, 2nd background picture URL of the non-motor vehicle*/
                "nonMotorBackgroundImage3URL":  "test",
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                /*ro, opt, string, 3rd background picture URL of the non-motor vehicle*/
                "faceImageURL":  "test",
                /*ro, opt, string, face thumbnail URL*/
                "plateImageURL":  "test",
                /*ro, opt, string, license plate thumbnail URL*/
                "illegalCode":  "12313",
                /*ro, opt, string, violation code*/
                "illegalName":  "test",
                /*ro, opt, string, violation name, range:[0,128]*/
                "illegalDescription":  "test",
                /*ro, opt, string, violation description, range:[0,256]*/
                "dwIllegalTime":  100,
                /*ro, opt, int, duration of the violation, unit:ms, desc:it is the difference between the capture time of the last picture and the capture
time of the first picture*/
                "relatedMixedTargetUid": ["test", "1"],
                /*ro, opt, array, UID array of linked multi-target-type detection events, subType:string, desc:corresponding to the "uid" of
mixedTargetDetection (multi-target-type detection event). when a device uploads non-motor vehicle properties, other related targets attributes will be
linked, and the client software can get the information about targets around the captured non-motor vehicle*/
                "contentID7":  "featurePicture",
                /*ro, opt, enum, close-up picture, subType:string, desc:"featurePicture" (close-up picture)*/
                "pId7":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, close-up picture ID, range:[0,32]*/
                "featurePictureURL":  "test",
                /*ro, opt, string, close-up picture URL*/
                "nonMotorSpeed":  23.4
                /*ro, opt, float, driving speed of non-motor vehicle, range:[0,1024], unit:km/h*/
            },
            "Vehicle": {
            /*ro, opt, object, vehicle attributes*/
                "VehicleRect": {
                /*ro, req, object, target frame information of vehicle picture (based on background picture), desc:the origin is the upper-left corner of
the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "PlateRect": {
                /*ro, req, object, frame information of license plate picture, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, attributes data, subType:object*/
                    {
                        "description":  "plateNo",
                        /*ro, opt, enum, attributes description, subType:string, desc:"plateNo" (license plate No.), "plateType" (license plate type),
"plateColor" (license plate color), "vehicleType"(vehicle type), "vehicleColor" (vehicle color), "vehicleLogo" (vehicle brand), "vehicleSublogo" (vehicle
sub-brand), "vehicleSize" (vehicle size)*/
                        "value":  "23",
                        /*ro, opt, string, it depends on the value of description, desc:it depends on the value of description*/
                        "confidence":  99.00
                        /*ro, opt, float, confidence, range:[0.00,100.00]*/
                    }
                ],
                "contentID1":  "vehicleImage",
                /*ro, opt, enum, vehicle thumbnail (on the basis of the target frame, a certain proportion is expanded), subType:string, desc:"faceImage"
(face thumbnail), "humanImage" (human body thumbnail), "vehicleImage" (vehicle thumbnail). When there are multiple properties for the detected target,
multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and
"nonMotorImage" will be returned*/
                "contentID2":  "plateImage",
                /*ro, opt, enum, license plate thumbnail, subType:string, desc:"plateImage" (license plate thumbnail)*/
                "contentID3":  "vehicleBackgroundImage",
                /*ro, opt, enum, vehicleBackgroundImage, subType:string, desc:"vehicleBackgroundImage" (vehicle background picture 1)*/
                "contentID4":  "vehicleBackgroundImage2",
                /*ro, opt, enum, vehicleBackgroundImage2, subType:string, desc:"vehicleBackgroundImage2" (vehicle background picture 2)*/
                "contentID5":  "vehicleBackgroundImage3",
                /*ro, opt, enum, vehicleBackgroundImage3, subType:string, desc:"vehicleBackgroundImage3" (vehicle background picture 3)*/
                "contentID6":  "vehicleBackgroundImage4",
                /*ro, opt, enum, vehicleBackgroundImage4, subType:string, desc:"vehicleBackgroundImage4" (vehicle background picture 4)*/
                "contentID8":  "vehicleBackgroundImage5",
                /*ro, opt, enum, vehicleBackgroundImage5, subType:string, desc:"vehicleBackgroundImage5" (vehicle background picture 5)*/
                "contentID9":  "pilotPicture",
                /*ro, opt, enum, pilotPicture, subType:string, desc:"pilotPicture" (driver's picture)*/
                "contentID10":  "copilotPicture",
                /*ro, opt, enum, copilotPicture, subType:string, desc:"copilotPicture" (co-driver's picture)*/
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, vehicle thumbnail ID, range:[0,32]*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, license plate thumbnail ID, range:[0,32]*/
                "pId3":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, vehicle background picture ID, range:[0,32]*/
                "pId4":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 2nd vehicle background picture, range:[0,32]*/
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                "pId5":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 3rd vehicle background picture, range:[0,32]*/
                "pId6":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 4th vehicle background picture, range:[0,32]*/
                "pId8":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 5th vehicle background picture, range:[0,32]*/
                "pId9":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, driver's picture, range:[0,32]*/
                "pId10":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, co-driver's picture, range:[0,32]*/
                "vehicleImageURL":  "test",
                /*ro, opt, string, vehicle thumbnail URL*/
                "plateImageURL":  "test",
                /*ro, opt, string, license plate thumbnail URL*/
                "vehicleBackgroundImageURL":  "test",
                /*ro, opt, string, vehicle background picture URL*/
                "vehicleBackgroundImage2URL":  "test",
                /*ro, opt, string, URL of the 2nd vehicle background picture*/
                "vehicleBackgroundImage3URL":  "test",
                /*ro, opt, string, URL of the 3rd vehicle background picture*/
                "vehicleBackgroundImage4URL":  "test",
                /*ro, opt, string, URL of the 4th vehicle background picture*/
                "vehicleBackgroundImage5URL":  "test",
                /*ro, opt, string, URL of the 5th vehicle background picture*/
                "pilotPictureURL":  "test",
                /*ro, opt, string, driver's picture URL*/
                "copilotPictureURL":  "test",
                /*ro, opt, string, co-driver's picture URL*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, vehicle capture time*/
                "snapTimeMillisecond":  999,
                /*ro, opt, int, it should be used together with the field snapTime, range:[0,999], desc:it should be used together with the field snapTime*/
                "humanMergeImageURL":  "test",
                /*ro, opt, string, URL of the picture of human body merging details*/
                "illegalCode":  "12313",
                /*ro, opt, string, violation code, range:[0,64]*/
                "illegalName":  "test",
                /*ro, opt, string, violation name, range:[0,128]*/
                "illegalDescription":  "test",
                /*ro, opt, string, violation description, range:[0,256]*/
                "dwIllegalTime":  100,
                /*ro, opt, int, duration of the violation, unit:ms, desc:it is the difference between the capture time of the last picture and the capture time of the first picture*/
                "VehicleImageRect": {
                /*ro, opt, object, rectangle target frame in the vehicle thumbnail, desc:the frame is the target area based on vehicleImage. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "smallImageRectInBackground": {
                /*ro, opt, object, vehicle thumbnail frame (based on the background picture), desc:the rectangle frame of vehicle thumbnail is based on vehicleBackgroundImage. The matted picture equals to vehicleImage. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "MultiPlateInfo": [
                /*ro, opt, array, information about multiple license plates, subType:object*/
                    {
                        "MultiPlateRect": {
                        /*ro, opt, object, license plate thumbnail frame, desc:based on the vehicle background picture*/
                            "height":  1.000,
                            /*ro, req, float, height, range:[0.000,1.000]*/
                            "width":  1.000,
                            /*ro, req, float, width, range:[0.000,1.000]*/
                            "x":  0.000,
                            /*ro, req, float, x-coordinate, range:[0.000,1.000]*/
                            "y":  0.000
                            /*ro, req, float, y-coordinate, range:[0.000,1.000]*/
                        },
                        "plateNo":  "noPlate",
                        /*ro, req, enum, license plate number, subType:string, desc:"noPlate" (vehicle without license plate), "unknown" (no license plate recognized), XXXXXXXX (license plate number. For motor vehicles, it is a 16-byte string; for non-motor vehicles, it is a 48-byte string*/
                        "plateColor":  "black",
                        /*ro, req, enum, license plate color, subType:string, desc:"black", "blue", "civilAviationBlack" (civil aviation black), "civilAviationGreen" (civil aviation green), "golden", "green", "mixedColor" (mixed color), "newEnergyGreen" (new energy green), "newEnergyYellowGreen" (new energy green and yellow), "orange", "other" (other color), "red", "unknown", "white", "yellow"*/
                        "confidence":  100.0,
                        /*ro, opt, float, confidence, range:[0,100.0]*/
                        "province":  0,
                        /*ro, req, enum, province index, subType:int*/
                        "contentID7":  "HongkongPlateImage",
```


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```
                        /*ro, opt, enum, license plate thumbnail, subType:string, desc:license plate thumbnail*/
                        "pId7":  "F4F665D6A18E41308CE9934DCDDD3333",
                        /*ro, opt, string, license plate thumbnail ID, range:[0,32]*/
                        "multiplateImageURL":  "test"
                        /*ro, opt, string, thumbnail URL of multiple license plates*/
                    }
                ],
                "carDriveDirect":  "forward",
                /*ro, opt, enum, vehicle driving direction, subType:string, desc:"back" (backward), "eastWest" (from east to west), "forward" (forward),
"northSouth" (from north to south), "northeastToSouthwest" (from northeast to southwest), "northwestToSoutheast" (from northwest to southeast), "other"
(other), "southNorth" (from south to north), "southeastToNorthwest" (from southeast to northwest), "southwestToNortheast" (from southwest to northeas),
"unknown" (unknown), "westEast" (from west to east)*/
                "contentID11":  "featurePicture",
                /*ro, opt, enum, close-up picture, subType:string, desc:"featurePicture" (close-up picture)*/
                "pId11":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, close-up picture ID, range:[0,32]*/
                "featurePictureURL":  "test",
                /*ro, opt, string, close-up picture URL*/
                "vehicleSpeed":  23.4
                /*ro, opt, float, vehicle driving speed, range:[0,1024], unit:km/h*/
            },
            "AbsoluteHigh": {
            /*ro, opt, object, PTZ position information*/
                "elevation":  11,
                /*ro, opt, int, tilting angle, desc:tilting angle*/
                "azimuth":  22,
                /*ro, opt, int, azimuth, desc:panning angle*/
                "absoluteZoom":  33
                /*ro, opt, int, zooming ratio, desc:zooming rate,value range: 1 to 1000*/
            },
            "GPS": {
            /*ro, opt, object, device GPS information*/
                "longitude":  -180.000000,
                /*ro, req, float, longitude*/
                "latitude":  -90.000000
                /*ro, req, float, latitude*/
            },
            "FaceContrastResult": [
            /*ro, opt, array, face picture comparison result, one result can contains comparison results of multiple persons, the captured picture with
highest comparison similarity is adopted, subType:object*/
                {
                    "errorCode":  1,
                    /*ro, opt, int, error code, desc:error information returned when analyzing face picture failed*/
                    "errorMsg":  "ok",
                    /*ro, opt, string, the modeling data of base picture in the face picture library, desc:when analyzing face picture failed, the
corresponding error information will be returned*/
                    "modelData":  "test",
                    /*ro, opt, string, the modeling data of base picture in the face picture library, range:[0,2048], desc:base64 encoded*/
                    "faces": [
                    /*ro, opt, array, matched face information, subType:object, desc:the information includes detection results, properties, modeling data,
and matched list. One picture can contains multiple faces*/
                        {
                            "faceId":  1,
                            /*ro, opt, int, face ID, desc:the face ID cannot be duplicated in the same picture*/
                            "identify": [
                            /*ro, opt, array, features, subType:object*/
                                {
                                    "relationId": "9887",
                                    /*ro, opt, string, arming linkage ID, range:[0,64], desc:arming linkage ID*/
                                    "maxsimilarity":  0.204000,
                                    /*ro, opt, float, the maximum similarity, range:[0.000000,1.000000], desc:the maximum similarity in the matched results
the similarity is between 0 and 1, and it corrects to two decimals*/
                                    "candidate": [
                                    /*ro, opt, array, person information list, subType:object*/
                                        {
                                            "blacklist_id":  "2323",
                                            /*ro, opt, string, alarm list ID, range:[0,64], desc:it corresponds to the FDID of face picture library. For
cameras, only the default FDID will be returned, the custom face picture library ID will be returned in customFaceLibID*/
                                            "human_data": [
                                            /*ro, opt, array, face picture information list, subType:object*/
                                                {
                                                    "face_id":  "123445",
                                                    /*ro, opt, string, face ID*/
                                                    "contentID":  "faceLibImage",
                                                    /*ro, opt, string, picture in the face picture library*/
                                                    "similarity":  0.204000,
                                                    /*ro, opt, float, face picture similarity, range:[0.000000,1.000000]*/
                                                    "isNoSaveFDPicture":  true,
                                                    /*ro, opt, bool, whether to not save the picture of face picture library, desc:true (not save). If the
function of not saving original pictures during picture importing or modeling is enabled, true should be returned, otherwise, this field will not be
returned*/
                                                    "pId":  "F4F665D6A18E41308CE9934DCDDD766F",
                                                    /*ro, opt, string, picture ID in face picture library, range:[0,32], desc:picture ID in face picture
library*/
                                                    "face_picurl":  "http://127.0.0.1:6120/face_picurl"
                                                    /*ro, opt, string, face thumbnail URL*/
                                                }
                                            ],
                                            "human_id":  "123",
                                            /*ro, opt, string, corresponds to the FPID in the face picture library, range:[0,64], desc:person ID in the
list, which corresponds to FPID in the face picture library*/
                                            "reserve_field": {
                                            /*ro, req, object, transmitted data*/
                                                "name":  "张三"
```

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```
                                        "name": "张三",
                                        /*ro, opt, string, name, range:[0,96]*/
                                        "gender": "any",
                                        /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition), "unknown", "male", "female"*/
                                        "bornTime": "1970-01-01+08:00",
                                        /*ro, opt, date, date of birth*/
                                        "city": "130100",
                                        /*ro, opt, string, city code of person birth, desc:see the data dictionary for details*/
                                        "certificateType": "officerID",
                                        /*ro, opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "passport" (passport), "other" (other certificate)*/
                                        "certificateNumber": "test",
                                        /*ro, opt, string, certificate No., range:[0,32]*/
                                        "ageGroup": "unknown",
                                        /*ro, opt, enum, age group, subType:string, desc:“any”, “child”, “infant”, “kid”, “middle”, “middleAged” (the middle age), “old”, “prime”, “teenager”, “unknown”, “young”*/
                                        "phoneNumber": "test",
                                        /*ro, opt, string, phone number, range:[0,64]*/
                                        "cityName": "石家庄市",
                                        /*ro, opt, string, city (corresponding to the field "city"), range:[0,32], desc:city (corresponding to the field "city)*/
                                        "provinceName": "河北省"
                                        /*ro, opt, string, province, range:[0,32], desc:province*/
                                    },
                                    "similarity": 0.780,
                                    /*ro, opt, float, the maximum similarity, range:[0.000,1.000], desc:the maximum similarity of all matched pictures for the person*/
                                    "listType": "blackList",
                                    /*ro, opt, enum, list type, subType:string, desc:"blacklist" (blocklist), "whitelist" (allowlist)*/
                                    "extendData": [
                                    /*ro, opt, array, person extended information, subType:object*/
                                        {
                                            "extendID": 1,
                                            /*ro, opt, int, ID of the extended person information*/
                                            "enable": true,
                                            /*ro, opt, bool, whether to enable the person extended information*/
                                            "name": "test",
                                            /*ro, opt, string, name of person tag extended information, range:[0,96]*/
                                            "value": "test"
                                            /*ro, opt, string, person tag extended information, range:[0,64]*/
                                        }
                                    ],
                                    "FDLibName": "faceLib1",
                                    /*ro, opt, string, detection threshold of face picture library*/
                                    "FDLibThreshold": 50,
                                    /*ro, opt, int, detection threshold of face picture library, range:[0,100]*/
                                    "customFaceLibID": "test",
                                    /*ro, opt, string, custom face picture library ID*/
                                    "customHumanID": "test"
                                    /*ro, opt, string, custom person ID*/
                                }
                            ]
                        }
                    ],
                    "stayDuration": 20
                    /*ro, opt, int, dwell time duration in the image, unit:ms*/
                }
            ]
        }
    ],
    "ModelingResult": {
    /*ro, opt, object, modeling result of captured face picture*/
        "modelingStatus": "succeeded",
        /*ro, req, enum, modeling status, subType:string, desc:"succeeded" (modeling succeeded), "resourceInsuf" (insufficient engine capacity), "faceDetectingFail" (face detection failed), "faceModelingFail" (face modeling failed), "faceGradingFail" (face quality grading), "featureExtractFail" (extracting features failed), "featureAnalyzeFail" (features analysis failed), "otherErrors" (other error of the engine), "DSPInternalError" (DSP internal error)*/
        "facePicQuality": "lower",
        /*ro, opt, enum, quality level of face thumbnail, subType:string, desc:"lower", "medium", "high"*/
        "stayDuration": 20,
        /*ro, opt, int, dwell time duration in the image, unit:ms*/
        "modeldata": "test"
        /*ro, opt, string, model data (encoded by base64)*/
    },
    "sid": "123",
    /*ro, opt, string, scene ID*/
    "sceneName": "场景1",
    /*ro, opt, string, scene name*/
    "uid": "test",
    /*ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits*/
    "behaviorLinkageUid": "test",
    /*ro, opt, string, which can consists of time (accurate to millisecond) and random digits,the maximum value size is 64 bytes, range:[0,64], desc:which can consists of time (accurate to millisecond) and random digits,the maximum value size is 64 bytes*/
    "position": [500, 500],
    /*ro, opt, array, offset position coordinates of target center point, subType:int, desc:an X-coordinate is positive if it is to the right of the Y-axis and negative if it is to the left of the Y-axis. A Y-coordinate is positive if it is above of the X-axis and negative values are not supported. The origin is the center point (the center point of all lane lines) of the horizontal line where the camera is located and which is perpendicular to the lane line*/
    "laneNo": 1,
    /*ro, opt, int, number of the lane where the target is located*/
    "speed": 23.4,
    /*ro, opt, float, vehicle speed*/
    "targetType": "human",
```


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```
            /*ro, opt, enum, vehicle type, subType:string, desc:"vehicle", "human", "nonMotor" (non-motor vehicles)*/
            "horizonSpeed":  33.4,
            /*ro, opt, float, horizontal speed of target*/
            "NormalizationPosition": [500, 500],
            /*ro, opt, array, normalized coordinates, subType:int, desc:normalized position coordinates,the value of x-coordinate is between 0 and 1000,and
the value of y-coordinate is between -500 and 1500,e.g.,[500,500]*/
            "targetMaxY":  340,
            /*ro, opt, int, the maximum Y-coordinate of target, desc:the value should be smaller than 1500*/
            "TargetGPSInfo": [
            /*ro, opt, array, GPS information of the target, subType:object*/
                {
                    "longitude":  -180.000000,
                    /*ro, req, float, longitude information, corrects to 6 decimal places, and the value range is [-180.000000,180.000000]*/
                    "latitude":  -90.000000
                    /*ro, req, float, latitude information, corrects to 6 decimal places, and the value range is [-180.000000,180.000000]*/
                }
            ],
            "perimeterLinkageUid":  "test",
            /*ro, opt, string, UID of linked perimeter events uploading, desc:64-byte unique identifier, which can be composed of time (accurate to the
millisecond) and random digits*/
            "MEFData": [
            /*ro, opt, array, mobile electronic fence data corresponding to the target, subType:object, desc:mobile electronic fence data of multiple users
is supported (the data needs to be encrypted)*/
                {
                    "time":  "2004-05-03T17:30:08+08:00",
                    /*ro, req, datetime, electronic fence data acquired time*/
                    "IMSI":  "460001357924680",
                    /*ro, req, string, SIM card number (international mobile subscriber identity), range:[1,15]*/
                    "IMEI":  "359355041886388",
                    /*ro, opt, string, international mobile equipment identity, range:[0,17]*/
                    "bandIdx":  11
                    /*ro, opt, enum, frequency band type used by the current user, subType:int, desc:0 (China Unicom B3), 1 (China Telecom B3), 2 (China
Mobile B39), 3 (China Mobile B38), 4 (China Mobile B40), 5 (China Mobile B41), 6 (China Unicom B1), 7 (China Telecom B1), 8 (China Mobile 2G), 9 (China
Unicom 2G), 10 (China Unicom WCDMA), 11 (China Telecom CDMA), 12 (China Telecom CDMA2000), 13 (China Mobile TD), 19 (China Mobile B1), 20 (China Mobile B3),
21 (China Mobile B8)*/
                }
            ]
        }
    ],
    "captureResultTargetNum":  10,
    /*ro, opt, int, number of uploaded alarm targets*/
    "URLCertificationType":  "digest",
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no"-no authentication, "digest"-digest authentication (it is used for local
picture URL returned by NVR or DVR)*/
    "triggerSourceType":  "linkedTrackingCapture"
    /*ro, opt, enum, alarm source type, subType:string, desc:"ruleTrigger" (rule parameters triggered alarm), "linkedTrackingCapture" (alarm triggered by
linked tracking capture). If this field is not returned or the returned value is "ruleTrigger", it indicates that the alarm is triggered by multi-target-
type detection rules (corresponding URL for rule configurations: /ISAPI/Intelligent/channels/<ID>/mixedTargetDetection?format=json). If the returned value
is "linkedTrackingCapture", it indicates that the alarm is triggered by linked tracking capture (corresponding URL for configurations:
/ISAPI/MasterSlaveTracking/linkedTracking?format=json)*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name
mixedTargetDetection	[Message content]	application/json	--	--
..value	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg
..value	[Binary picture data]	image/jpeg	faceBackgroundImage	faceBackgroundImage.jpg
..value	[Binary picture data]	image/jpeg	humanImage	humanImage.jpg
..value	[Binary picture data]	image/jpeg	humanBackgroundImage	humanBackgroundImage.jpg
..value	[Binary picture data]	image/jpeg	humanMergeImage	humanMergeImage.jpg

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..value	[Binary picture data]	image/jpeg	humanBackgroundImage2	humanBackgroundImage2.jpg
..value	[Binary picture data]	image/jpeg	humanBackgroundImage3	humanBackgroundImage3.jpg
..value	[Binary picture data]	image/jpeg	nonMotorImage	nonMotorImage.jpg
..value	[Binary picture data]	image/jpeg	nonMotorBackgroundImage	nonMotorBackgroundImage.jpg
..value	[Binary picture data]	image/jpeg	nonMotorBackgroundImage2	nonMotorBackgroundImage2.jpg
..value	[Binary picture data]	image/jpeg	nonMotorBackgroundImage3	nonMotorBackgroundImage3.jpg
..value	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg
..value	[Binary picture data]	image/jpeg	plateImage	plateImage.jpg
..value	[Binary picture data]	image/jpeg	vehicleImage	vehicleImage.jpg
..value	[Binary picture data]	image/jpeg	plateImage	plateImage.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage	vehicleBackgroundImage.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage2	vehicleBackgroundImage2.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage3	vehicleBackgroundImage3.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage4	vehicleBackgroundImage4.jpg
..value	[Binary picture data]	image/jpeg	HongkongPlateImage	HongkongPlateImage.jpg
	[Binary			

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..value	[Binary picture data]	image/jpeg	MacaoPlateImage	MacaoPlateImage.jpg
.....value	[Binary picture data]	image/jpeg	faceLibImage	faceLibImage.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage5	vehicleBackgroundImage.jpg5
..value	[Binary picture data]	image/jpeg	pilotPicture	pilotPicture.jpg
..value	[Binary picture data]	image/jpeg	copilotPicture	copilotPicture.jpg
..value	[Binary picture data]	image/jpeg	featurePicture	featurePicture
..value	[Binary picture data]	image/jpeg	featurePicture	featurePicture
..value	[Binary picture data]	image/jpeg	featurePicture	featurePicture
..value	[Binary picture data]	image/jpeg	compositePicture	compositePicture.jpg
..value	[Binary picture data]	image/jpeg	compositePicture	compositePicture.jpg
..value	[Binary picture data]	image/jpeg	compositePicture	compositePicture.jpg
..value	[Binary picture data]	image/jpeg	copilotPicture	copilotPicture.jpg

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Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.

- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.9 视频识别

15.9.1 专业智能异常行为检测(S)

15.9.1.1 Get the intelligent capability

Request URL

GET /ISAPI/Intelligent/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IntelliCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <isFaceSupport>
    <!--ro, opt, bool, whether it supports face detection, desc:whether it supports face detection-->true
  </isFaceSupport>
  <isBehaviorSupport>
    <!--ro, opt, bool, whether it supports abnormal event detection, desc:whether it supports abnormal event detection-->true
  </isBehaviorSupport>
  <isLineDetectionSupport>
    <!--ro, opt, bool, whether it supports line crossing detection, desc:whether it supports line crossing detection-->true
  </isLineDetectionSupport>
  <isFieldDetectionSupport>
    <!--ro, opt, bool, whether it supports intrusion detection, desc:whether it supports intrusion detection-->true
  </isFieldDetectionSupport>
  <isRegionEntranceSupport>
    <!--ro, opt, bool, whether it supports region entrance detection, desc:whether it supports region entrance detection-->true
  </isRegionEntranceSupport>
  <isRegionExitingSupport>
    <!--ro, opt, bool, whether it supports region exiting detection, desc:whether it supports region exiting detection-->true
  </isRegionExitingSupport>
  <isLoiteringSupport>
    <!--ro, opt, bool, whether it supports loitering detection, desc:whether it supports loitering detection-->true
  </isLoiteringSupport>
  <isGroupSupport>
    <!--ro, opt, bool, whether it supports people gathering detection, desc:whether it supports people gathering detection-->true
  </isGroupSupport>
  <isRapidMoveSupport>
    <!--ro, opt, bool, whether it supports fast moving detection, desc:whether it supports fast moving detection-->true
  </isRapidMoveSupport>
  <isParkingSupport>
    <!--ro, opt, bool, whether it supports parking detection, desc:whether it supports parking detection-->true
  </isParkingSupport>
  <isUnattendedBaggageSupport>
    <!--ro, opt, bool, whether it supports unattended baggage detection, desc:whether it supports unattended baggage detection-->true
  </isUnattendedBaggageSupport>
  <isAttendedBaggageSupport>
    <!--ro, opt, bool, whether it supports object removal detection, desc:whether it supports object removal detection-->true
  </isAttendedBaggageSupport>
  <isTeacherSupport>
    <!--ro, opt, bool, whether it supports teachers' behavior detection, desc:whether it supports teacher's behavior detection-->true
  </isTeacherSupport>
  <isStudentSupport>
    <!--ro, opt, bool, whether it supports students' behavior detection, desc:whether it supports student's behavior detection-->true
  </isStudentSupport>
  <HumanRecognitionModeSearchCap>
    <!--ro, opt, object, whether it is false human body recognition alarm: "true"-yes, "false"-no, desc:whether it is false human body recognition alarm:
"true"-yes,"false"-no-->
    <searchTargetsNumMax>
      <!--ro, opt, int, maximum number of sample pictures that can be imported for searching by picture-->1
    </searchTargetsNumMax>
    <HumanMode>
      <!--ro, opt, object, similarity,range: [0.0,100.0]-->
      <searchCond opt="age_group,gender,jacet_color,glass,bag,ride,unlimit">
        <!--ro, opt, string, attr:opt{req, string}-->test
      </searchCond>
      <similarity min="0.0" max="100.0">
        <!--ro, opt, float, range:[0.0,100.0], attr:min{req, int},max{req, int}-->0.000
      </similarity>
    </HumanMode>
    <HumanInfo>
```

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```
      <!--ro, opt, object, human body information, desc:person attribute information,it is required when searching for picture by attributes-->
      <age_group opt="child,infant,kid,middle,middleAged,old,prime,teenager,unknown,young">
        <!--ro, req, enum, age group, subType:string, attr:opt{req, string}, desc:“any”, “child”, “infant”, “kid”, “middle”, “middleAged” (the middle age), “old”, “prime”, “teenager”, “unknown”, “young”-->any
      </age_group>
      <gender opt="female,male,unknown">
        <!--ro, req, enum, gender, subType:string, attr:opt{req, string}, desc:"any" (unlimited condition which is used for search), "male", "female", "unknown"-->any
      </gender>
      <jacet_color opt="black,black,blue,brown,cyan,deepBlue,deepGray,gray,green,mixture,orange,pink,purple,red,silver,unknown,white,yellow">
        <!--ro, opt, string, tops color, attr:opt{req, string}-->test
      </jacet_color>
      <glass opt="yes,no,unknown,sunglasses">
        <!--ro, req, enum, whether the person is wearing glasses, subType:string, attr:opt{req, string}, desc:"any” (unlimited condition which is used in search), “no” (not wear glasses), “sunglasses” (wears sunglasses), “unknown", “yes”-->any
      </glass>
      <bag opt="yes,no,unknown">
        <!--ro, req, enum, whether the person is carrying a bag, subType:string, attr:opt{req, string}, desc:whether the person is carrying a bag-->no
      </bag>
      <ride opt="yes,no,unknown">
        <!--ro, req, enum, whether the person is riding a bike, subType:string, attr:opt{req, string}, desc:"any” (unlimited condition which is used in search), “no”, "unknown”, “yes"-->any
      </ride>
    </HumanInfo>
    <positive opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </positive>
    <eventType opt="unlimit,humanRecognition">
      <!--ro, req, enum, event type, subType:string, attr:opt{req, string}, desc:"unlimit"-no limit, "humanRecognition"-human body detection alarm-->unlimit
    </eventType>
    <isSupportMultiChannelSearch>
      <!--ro, opt, bool, whether it supports multi-channel search-->true
    </isSupportMultiChannelSearch>
    <isSupportTotalSearchResult>
      <!--ro, opt, bool, whether it supports limiting number of results that can be obtained after a single search-->true
    </isSupportTotalSearchResult>
  </HumanRecognitionModeSearchCap>
  <VehicleRecognitionModeSearchCap>
    <!--ro, opt, object, capability of searching for vehicles, desc:event type: "unlimit"-no limit,"vehicleBlackList"-vehicle blocklist,"vehicleWhiteList"-vehicle allowList-->
    <VehicleMode>
      <!--ro, opt, object, similarity,range: [0.0,100.0]-->
      <searchCond opt="licensePlate,vehicleLogo,vehicleSubLogoRecog,vehicleType,vehicleColor,unlimit">
        <!--ro, opt, string, attr:opt{req, string}-->test
      </searchCond>
    </VehicleMode>
    <isSupportMultiChannelSearch>
      <!--ro, opt, bool, whether it supports multi-channel search-->true
    </isSupportMultiChannelSearch>
    <isSupportTotalSearchResult>
      <!--ro, opt, bool, whether it supports limiting number of results that can be obtained after a single search-->true
    </isSupportTotalSearchResult>
  </VehicleRecognitionModeSearchCap>
  <isSupportUploadFacePictureByForm>
    <!--ro, opt, bool, whether it supports uploading face pictures by form, desc:whether it supports uploading face pictures by form-->true
  </isSupportUploadFacePictureByForm>
  <isSupportMisinfoFilterStatisticalMode>
    <!--ro, opt, bool, whether it supports statistics of filtering face comparison, desc:whether it supports statistics of filtering face comparison-->true
  </isSupportMisinfoFilterStatisticalMode>
  <isSupportSafetyHelmetDetection>
    <!--ro, opt, bool, whether it supports hard hat detection, desc:whether it supports hard hat detection-->true
  </isSupportSafetyHelmetDetection>
  <isSupportFaceScore>
    <!--ro, opt, bool, whether it supports face grading configuration (camera), desc:whether it supports face grading configuration (camera)-->true
  </isSupportFaceScore>
  <isSupportUploadFacePictureByUrl>
    <!--ro, opt, bool, whether it supports uploading face pictures by URL, desc:whether it supports uploading face pictures by URL-->true
  </isSupportUploadFacePictureByUrl>
  <isSupportFireEscapeDetectionSearch>
    <!--ro, opt, bool, whether the device supports searching for fire engine access detection events of multiple channels, desc:related URI: URI/ISAPI/Intelligent/fireEscapeDetection/search/capabilities?format=json-->true
  </isSupportFireEscapeDetectionSearch>
  <isSupportTakingElevatorDetectionSearch>
    <!--ro, opt, bool, whether the device supports searching for elevator detection events of multiple channels, desc:related URI: /ISAPI/Intelligent/takingElevatorDetection/search/capabilities?format=json-->true
  </isSupportTakingElevatorDetectionSearch>
  <isSupportCityManagementSearch>
    <!--ro, opt, bool, whether the device supports searching for city management events of multiple channels, desc:related URI: /ISAPI/Intelligent/cityManagement/search/capabilities?format=json-->true
  </isSupportCityManagementSearch>
  <AnalysisImageFaceCap>
    <!--ro, opt, object, face picture analysis type, desc:face picture analysis type (related URI:/ISAPI/Intelligent/analysisImage/face)-->
    <modeling opt="false">
      <!--ro, opt, bool, whether the device supports modeling, attr:opt{req, string}, desc:whether the device supports modeling: "true,false"-->true
    </modeling>
  </AnalysisImageFaceCap>
</IntelliCap>
```

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15.9.1.2 Get the intelligent channel list

Request URL

GET /ISAPI/Intelligent/intelliChannelList

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, intelligent channel list, subType:object, attr:version{req, string, protocolVersion}-->
  <IntelliChannel>
    <!--ro, opt, object, intelligent channel-->
    <id>
      <!--ro, req, string, channel ID-->test
    </id>
  </IntelliChannel>
</IntelliChannelList>
```

15.9.1.3 People number change detection

EventType:peopleNumChange

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"peopleNumChange" (people number change detection)-->peopleNumChange
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
```

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```
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <personCount>
    <!--ro, opt, int, the number of people-->1
  </personCount>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
```

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```
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

15.9.1.4 Overstay detection

EventType:retention

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:retention (overstay detection)-->retention
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
```

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```
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

15.9.1.5 Playing mobile phone detection

EventType:playCellphone

```
{
```

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```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:times that the same alarm has been uploaded*/
    "eventType":  "playCellphone",
    /*ro, req, enum, event type, subType:string, desc:"playCellphone" (playing mobile phone detection)*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "playCellphone",
    /*ro, req, string, event description*/
    "channelName":  "ABC",
    /*ro, opt, string, channel name (camera name)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID (PUID)*/
    "targetAttrs": {
    /*ro, opt, object, transmitted parameters*/
        "deviceId":  "1",
        /*ro, opt, string, device ID*/
        "deviceChannel":  1,
        /*ro, opt, int, device channel No.*/
        "deviceName":  "shebei"
        /*ro, opt, string, device name, range:[0,128]*/
    },
    "PlayCellphone": {
    /*ro, opt, object, Playing Mobile Phone Detection*/
        "TaskInfo": {
        /*ro, opt, object, task information submitted by data center products*/
            "taskID":  "test",
            /*ro, req, string, task ID, range:[0,64]*/
            "algorithmType":  1,
            /*ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)*/
            "taskName":  "test",
            /*ro, opt, string, task name*/
            "algorithmID":  "test",
            /*ro, opt, string, algorithm package ID*/
            "monitorPointID":  "test",
            /*ro, opt, string, camera No.*/
            "monitorPointName":  "test",
            /*ro, opt, string, camera name*/
            "streamType":  "realtime",
            /*ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)*/
            "ruleID":  "1",
            /*ro, opt, string, rule ID*/
            "ruleCustomName":  "ruleName",
            /*ro, req, string, user-defined event name, range:[0,128], desc:It is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed.*/
            "algorithmInfo": {
            /*ro, opt, object, algorithm version information, this field is required for secondary analysis*/
                "algorithmId":  "test",
                /*ro, opt, string, algorithm package ID*/
                "algoVendorId":  "test",
                /*ro, opt, string, manufacturer ID*/
                "algoVersionId":  "test",
                /*ro, opt, string, algorithm version No.*/
                "algoChipType":  "test",
                /*ro, opt, string, chip type*/
                "algoModelVersion":  "test"
                /*ro, opt, string, model version*/
            }
        },
        "BackgroundImage": {
        /*ro, opt, object, background picture when the event occurred*/
            "resourcesContentType":  "base64",
            /*ro, opt, enum, resource transmission type, subType:string, desc:"url", "binary" "base64"*/
            "resourcesContent":  "test"
            /*ro, opt, string, resource ID, desc:When the returned alarm message contains additional resources (e.g., pictures), this field should be returned. When the value of resourcesContentType is "binary", it must be the same as Content-ID of the resource. When the value of resourcesContentType is "url", its value is the resource URL. When the value of resourcesContentType is "base64", its value is the data encoded via base64.*/
        },
        "timeStamp":  "2019-12-30T15:29:23+08:00",
        /*ro, opt, datetime, ISO 8601 format*/
        "ruleID":  0,
        /*ro, opt, int, rule No., range:[0,7], desc:rule No.*/
```

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```
        "ruleName":  "test"
        /*ro, opt, string, rule name, range:[0,32], desc:rule name*/
    }
}
```

15.9.1.6 Getting up or falling down detection

EventType:failDown

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event uploading, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"failDown" (getting up or falling down detection)-->failDown
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
```

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```
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle, the origin is the upper-left corner of the screen-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:area origin coordinates-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:Rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:Rule Name-->test
  </ruleName>
  <isDataRetransmission>
    <!--ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be
uploaded again after the device recovers-->true
  </isDataRetransmission>
</EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
failDown	[Message content]	application/xml	--	--	--
video	[Video data]	video/mp4	随机值	video.mp4	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.9.1.7 Violent motion detection

EventType:violentMotion

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"violentMotion" (violent motion detection)-->violentMotion
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
```

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```
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (ENDME2.0, ENDME4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
```

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```
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
  <isDataRetransmission>
    <!--ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be
uploaded again after the device recovers-->true
  </isDataRetransmission>
</EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
violentMotion	[Message content]	application/xml	--	--	--
video	[Video data]	video/mp4	随机值	video.mp4	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.9.2 人脸库管理

15.9.2.1 Get the configuration capability of the armed face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/armingLibCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "ArmingLibCapacity": {
    /*ro, req, object, configuration capability of the armed face picture library*/
        "capacity": {
        /*ro, opt, object, capacity of the armed face picture library*/
            "@opt": ["100W", "300W"]
            /*ro, req, array, options, subType:string*/
        }
    }
}
```

15.9.2.2 Get the parameters of the armed face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/armingLibCfg?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "ArmingLibCfg": {
    /*ro, opt, object*/
        "capacity":  "100W"
        /*ro, req, string, capacity of the armed face picture library, desc:which is the maximum number of list libraries that can be loaded by the analysis unit (it can be configured by calling the URI /ISAPI/SDT/Management/listLoading?format=json)*/
    }
}
```

15.9.2.3 Set parameters of the armed face picture library

Request URL

PUT /ISAPI/Intelligent/FDLib/armingLibCfg?format=json

Query Parameter

None

Request Message

```
{
    "ArmingLibCfg": {
    /*opt, object*/
        "capacity":  "100W"
        /*req, string, capacity of the armed face picture library, desc:which is the maximum number of list libraries that can be loaded by the analysis unit (it can be configured by calling the URI /ISAPI/SDT/Management/listLoading?format=json)*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

15.9.2.4 Get face picture library capability

Request URL

GET /ISAPI/Intelligent/FDLib/capabilities?format=json

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Query Parameter

None

Request Message

None

Response Message

```
{
    "requestURL": "test",
    /*ro, opt, string, request URL*/
    "statusCode": 1,
    /*ro, req, int, status code*/
    "statusString": "test",
    /*ro, req, string, status description*/
    "subStatusCode": "test",
    /*ro, req, string, sub status code*/
    "errorCode": 1,
    /*ro, opt, int, error code, this field is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, detailed error description, this field is required when the value of statusCode is not 1*/
    "FDNameMaxLen": 64,
    /*ro, req, int, maximum length of face picture library name*/
    "customInfoMaxLen": 192,
    /*ro, req, int, maximum length of custom information*/
    "FDMaxNum": 3,
    /*ro, req, int, maximum number of face picture libraries*/
    "FDRecordDataMaxNum": 12345,
    /*ro, req, int, maximum face records supported by face picture library*/
    "supportFDFunction": "post,delete,put,get,setUp",
    /*ro, req, string, the supported operations on face picture library, desc:"post" (create), "delete" (delete), "put" (edit), "get" (search), "setUp"
(set)*/
    "isSuportFDSearch": true,
    /*ro, req, bool, whether supports searching in face picture library, desc:whether supports searching in face picture library*/
    "isDisplayCaptureNum": {
    /*ro, opt, object*/
        "@opt": [true, false]
        /*ro, opt, array, subType:bool*/
    },
    "isSupportFDSearchDataPackage": true,
    /*ro, req, bool, whether supports packaging the found data in the face picture library*/
    "isSuportFSsearchByPic": true,
    /*ro, req, bool, whether supports searching by picture in the face picture library, desc:whether supports searching by picture in the face picture
library*/
    "isSuportFSsearchByPicGenerate": true,
    /*ro, req, bool, whether supports exporting results of using picture to search picture from the face picture library*/
    "isSuportFDSearchDuplicate": true,
    /*ro, req, bool, whether supports duplication checking*/
    "isSuportFDSearchDuplicateGenerate": true,
    /*ro, req, bool, whether supports exporting the duplication checking results*/
    "isSuportFCSearch": true,
    /*ro, req, bool, whether supports searching face picture comparison alarms*/
    "isSupportFCSearchDataPackage": true,
    /*ro, req, bool, whether supports packaging the search results of face picture comparison alarms*/
    "isSupportFDExecuteControl": true,
    /*ro, req, bool, whether supports creating relation between face picture libraries and cameras*/
    "generateMaxNum": 1234,
    /*ro, opt, int, maximum face records can be exported from face picture library*/
    "faceLibType": "blackFD,staticFD,infraredFD",
    /*ro, opt, string, face picture library type*/
    "modelMaxNum": 1000,
    /*ro, opt, int, the maximum number of search results*/
    "isSupportModelData": true,
    /*ro, opt, bool, whether it supports applying model data*/
    "isSuportFDLibArmingType": true,
    /*ro, opt, bool, whether it supports face picture library arming type*/
    "isSuportFDLibSearch": true,
    /*ro, opt, bool, whether it supports searching face picture library*/
    "FDArmingRecordDataMaxNum": 12345,
    /*ro, opt, int, the supported maximum number of face records in the face picture arming library*/
    "isSupportControlPersonRecordByHumanId": true,
    /*ro, opt, bool, whether it supports modifying and deleting the face record by humanId*/
    "isSupportControlPersonRecordByRowKey": true,
    /*ro, opt, bool, whether it supports modifying and deleting the face record by rowKey*/
    "isSupportFaceLibRebuildCfg": true,
    /*ro, opt, bool, whether it supports recreating face picture library information and configuration*/
    "isSupportFDMove": true,
    /*ro, opt, bool, whether it supports moving face data in the face picture library in a batch, desc:the related URI is
/ISAPI/Intelligent/FDLib/FDMove/capabilities?format=json*/
    "faceURLLen": 1024,
    /*ro, opt, int, the maximum size of the face picture URL, desc:if this node is not returned, the default size of the face picture URL supported by the
device is 256 bytes; otherwise, the device should support that the value of this node is greater than or equal to 256*/
    "isSupportArmingLibCfg": true,
    /*ro, opt, bool, whether it supports configuring parameters of the armed face picture library, desc:related URI:
/ISAPI/Intelligent/FDLib/armingLibCfg/capabilities?format=json*/
    "isSupportModelTransformation": true,
    /*ro, opt, bool, whether it supports converting face picture models in the face picture list library, desc:related URI:
/ISAPI/Intelligent/FDLib/model/transformation/capabilities?format=json*/
    "featurePointTypeList": ["face", "leftEye", "rightEye", "leftMouthCorner", "rightMouthCorner", "nose"],
```

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```
    /*ro, opt, array, feature point types of face pictures supported by the device, subType:string*/
    "isSupportPresort":  true,
    /*ro, opt, bool, whether it supports configuring pre-categorization, desc:related URI: /ISAPI/Intelligent/FDLib/presort/capabilities?format=json*/
    "libAttribute": {
    /*ro, opt, object, library attribute type, desc:"general" (normal library), "blockList" (blocklist library), "VIP" (VIP library), "passerby" (allowlist
library, which cannot be deleted)*/
        "@opt": ["general", "blackList", "VIP", "passerby"]
        /*ro, opt, array, options, subType:string*/
    },
    "faceType": {
    /*ro, opt, object, face picture type*/
        "@opt": ["normalFace", "patrolFace", "hijackFace", "superFace"]
        /*ro, opt, array, options, subType:string*/
    },
    "saveFacePic": {
    /*ro, opt, object, whether to save the registered face picture*/
        "@opt": [true, false]
        /*ro, opt, array, options, subType:bool*/
    },
    "leaderPermission": {
    /*ro, opt, object, first authentication permission*/
        "@size":  4,
        /*ro, opt, int, the maximum number of array members, range:[1,4]*/
        "@min":  1,
        /*ro, opt, int, member value range, range:[1,4]*/
        "@max":  4
        /*ro, opt, int, member value range, range:[1,4]*/
    },
    "facePicFormat": {
    /*ro, opt, object*/
        "@opt": ["jpg", "png", "bmp"]
        /*ro, req, array, subType:string*/
    }
}
```

15.9.2.5 Search face picture library information by arming mode

Request URL

POST /ISAPI/Intelligent/FDLib/search?format=json

Query Parameter

None

Request Message

```
{
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults":  0,
    /*req, int, the maximum number of search results this time*/
    "libArmingType":  "armingLib"
    /*opt, enum, arming type, subType:string, desc:"armingLib" (armed library (default)), "nonArmingLib" (not armed library)*/
}
```

Response Message

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```
{
    "responseStatusStrg":  "OK",
    /*ro, opt, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched
data).*/
    "numOfMatches":  1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches":  1,
    /*ro, opt, int, total number of matched results*/
    "FDLib": [
    /*ro, opt, array, face picture library information, subType:object*/
        {
            "FDID":  "test",
            /*ro, opt, string, face picture library ID*/
            "faceLibType":  "blackFD",
            /*ro, opt, enum, face picture library type, desc:"blackFD" (list library), "staticFD" (static library)*/
            "name":  "test",
            /*ro, opt, string, name of the face picture library, range:[0,48]*/
            "customInfo":  "test",
            /*ro, opt, string, custom information, range:[0,192]*/
            "libArmingType":  "armingLib",
            /*ro, opt, enum, arming type, subType:string, desc:"armingLib" (armed library (default)), "nonArmingLib" (not armed library)*/
            "libAttribute":  "blackList"
            /*ro, opt, enum, library attribute type, subType:string, desc:"blackList" (blocklist), "VIP" (VIP library), "passerby" (passer-by library, which
cannot be deleted)*/
        }
    ]
}
```

15.9.2.6 Delete all face picture libraries

Request URL

DELETE /ISAPI/Intelligent/FDLib?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

15.9.2.7 Create a face picture library

Request URL

POST /ISAPI/Intelligent/FDLib?format=json

Query Parameter

None

Request Message

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```
{
    "faceLibType":  "blackFD,staticFD",
    /*req, string, face picture library type, desc:face picture library type*/
    "name":  "test",
    /*req, string, name of the face picture library, range:[0,48], desc:name of the face picture library*/
    "customInfo":  "test",
    /*opt, string, custom information, range:[0,192], desc:custom information*/
    "libArmingType":  "armingLib",
    /*opt, enum, subType:string*/
    "libAttribute":  "blackList",
    /*opt, enum, subType:string*/
    "FDID":  "test"
    /*opt, string, face picture library ID, desc:face picture library ID*/
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, req, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
    "FDID":  "test"
    /*ro, opt, string, face picture library ID, desc:face picture library ID*/
}
```

15.9.2.8 Delete a specified face picture library

Request URL

DELETE /ISAPI/Intelligent/FDLib?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	"blackFD" (list library), "staticFD" (static library)
terminalNo	string	Terminal No. It is an integer starting from 1.

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

15.9.2.9 Edit the information of a specific face picture library information, including name and custom information

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Request URL

PUT /ISAPI/Intelligent/FDLib?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	--
terminalNo	string	--

Request Message

```
{
    "name":  "test",
    /*opt, string, face picture library name, range:[0,48]*/
    "customInfo":  "test",
    /*opt, string, custom information, range:[0,192]*/
    "libArmingType":  "armingLib",
    /*opt, enum, arming type of the list library, subType:string, desc:“armingLib” (armed face picture library), “nonArmingLib” (not armed face picture library). The default value is "armingLib”*/
    "libAttribute":  "blackList"
    /*opt, enum, library type, subType:string, desc:“blackList” (blocklist library), “VIP” (VIP library), “passerby” (passerby library). The passerby library cannot be deleted*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode when statusCode is not 1*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

15.9.2.10 Get the information, including library ID, library type, name, and custom information, of all face picture libraries

Request URL

GET /ISAPI/Intelligent/FDLib?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	"blackFD” (list library), "staticFD” (static library), "infraredFD" (IR face picture library)
terminalNo	string	Terminal No. It is an integer starting from 1.

Request Message

None

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code, desc:status code*/
    "statusString":  "OK",
    /*ro, req, string, status description, range:[1,64], desc:status description*/
    "subStatusCode":  "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:sub status code*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:error description*/
    "faceLibType":  "blackFD",
    /*ro, opt, enum, face picture library type, subType:string, desc:"blackFD" (list library), "staticFD" (static library)*/
    "name":  "test",
    /*ro, opt, string, face picture library name, range:[1,48]*/
    "customInfo":  "test",
    /*ro, opt, string, custom information, range:[0,192]*/
    "libArmingType":  "armingLib",
    /*ro, opt, enum, library arming type, subType:string, desc:library arming type*/
    "libAttribute":  "blackList",
    /*ro, opt, enum, subType:string*/
    "personnelFileEnabled":  true
    /*ro, opt, bool*/
}
```

15.9.3 人脸库记录管理

15.9.3.1 Get the total number of face records in all face picture libraries

Request URL

GET /ISAPI/Intelligent/FDLib/Count?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "requestURL":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, see the description of this node and above nodes in the message of JSON_ResponseStatus, desc:this node is required when the value of
statusCode is not 1*/
    "FDRecordDataInfo": [
    /*ro, opt, array, information of face records in face picture library, subType:object, desc:this node is valid when errorCode is 1 and errorMsg is
"ok"*/
        {
            "FDID":  "test",
            /*ro, opt, string, face picture library ID, desc:the maximum size is 63 bytes*/
            "faceLibType":  "blackFD",
            /*ro, opt, enum, face picture library type, subType:string, desc:face picture library type "blackFD" list library,"staticFD" static library, the
maximum size is 32 bytes*/
            "name":  "test",
            /*ro, opt, string, face picture library name, desc:the maximum size is 48 bytes*/
            "recordDataNumber":  123,
            /*ro, opt, int, number of records*/
            "libArmingType":  "armingLib",
            /*ro, opt, enum, arming type, subType:string, desc:arming type*/
            "libAttribute":  "general",
            /*ro, opt, enum, subType:string*/
            "personnelFileEnabled":  true
            /*ro, opt, bool*/
        }
    ],
    "totalRecordDataNumber":  500000,
    /*ro, opt, int, total number of records, desc:total number of records*/
    "FDCapacity": {
    /*ro, opt, object*/
        "total":  0.000,
        /*ro, req, float*/
        "use":  0.000,
        /*ro, req, float*/
        "remain":  0.000,
        /*ro, req, float, remaining space, desc:remaining space*/
        "maxRecordDataNumber":  0,
        /*ro, opt, int*/
        "useRecordDataNumber":  0,
        /*ro, opt, int*/
        "remainRecordDataNumber":  0
        /*ro, opt, int*/
    }
}
```

15.9.3.2 Get the total number of face records in a face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/Count?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	"blackFD" (list library), "staticFD" (static library), "infraredFD" (IR face picture library)
terminalNo	string	--

Request Message

None

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "FDID":  "test",
    /*ro, opt, string, face picture library ID, range:[1,64]*/
    "faceLibType":  "blackFD",
    /*ro, opt, enum, face picture library type, subType:string, desc:face picture library type*/
    "name":  "test",
    /*ro, opt, string, name of the face picture library, desc:the maximum length is 48*/
    "recordDataNumber":  123
    /*ro, opt, int, total number of records*/
}
```

15.9.3.3 Add a face record to the face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/FaceDataRecord?format=json

Query Parameter

None

Request Message

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```
{
    "faceURL":  "test",
    /*opt, string, picture storage URL inputted when uploading the face picture by URL, desc:select this field or modelData*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:face picture library type*/
    "FDID":  "test",
    /*req, string, face picture library ID, range:[1,64]*/
    "FPID":  " test",
    /*opt, string, face record ID, range:[1,1000], desc:face record ID,it can be generated by device or inputted. If it is inputted,it should be the unique ID with the combination of letters and digits,and the maximum length is 63 bytes; if it is generated by the device automatically,it is the same as the employee No. (person ID)*/
    "name":  "张三",
    /*req, string, name of person in the face picture, range:[1,1000], desc:the maximum size is 96 bytes*/
    "gender":  "male",
    /*opt, enum, gender of person in the face picture, subType:string, desc:“male”, “female”, “unknown”; the maximum size is 32 bytes*/
    "bornTime":  "2004-05-03",
    /*req, string, birthday of person in the face picture, desc:the maximum size is 20 bytes*/
    "city":  "130100",
    /*opt, string, city code of birth for the person in the face picture, desc:the maximum size is 32 bytes*/
    "certificateType":  "ID",
    /*opt, enum, certificate type, subType:string, desc:"ID" (ID card), "officerID" (officer ID), "other" (other type), "LawyerCertificate" (lawyer qualification), "birthCertificate", "householdRegister", "judgeCertificate", "studentIDCard", "workCard" (work permit),  "policeID", "foreignPermanentResidentIDCard", "passport", "pressCard", "prosecutorCard", "temporaryIDCard", "temporaryResidencePermit"*/
    "certificateNumber":  "test",
    /*opt, string, certificate No., range:[1,64], desc:the maximum size is 32 bytes*/
    "caseInfo":  "test",
    /*opt, string, case information, desc:the maximum size is 192 bytes; it is valid when faceLibType is "blackFD”*/
    "tag":  "aa,bb,cc,dd",
    /*opt, string, custom tag, desc:up to 4 tags, which are separated by commas. The maximum size is 195 bytes. It is valid when faceLibType is "blackFD”*/
    "address":  "test",
    /*opt, string, person address, desc:the maximum size is 192 bytes. It is valid when faceLibType is "staticFD”*/
    "customInfo":  "test",
    /*opt, string, custom information, desc:the maximum size is 192 bytes. It is valid when faceLibType is "staticFD"*/
    "modelData":  "test",
    /*opt, string, target modeling data, desc:target model data,non-modeled binary data needs to be encrypted by Base64 during transmission*/
    "transfer":  true,
    /*opt, bool, whether to enable transfer, desc:whether to enable transfer*/
    "operateType":  "byTerminal",
    /*opt, enum, operation type, subType:string, desc:"byTerminal” (operate by terminal)*/
    "terminalNoList": [1],
    /*opt, array, terminal ID list, subType:int, desc:this node is required when operation type is "byTerminal"; currently, only one terminal is supported*/
    "PicFeaturePoints": [
    /*opt, array, feature points to be applied, subType:object*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, coordinates of the feature point, desc:object,coordinates of the feature point*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized X-coordinate which is between 0 and 1000*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized Y-coordinate which is between 0 and 1000*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:this node is required when featurePointType is "face"*/
                "height":  1
                /*opt, int, height, range:[0,1000], desc:this node is required when featurePointType is "face"*/
            }
        }
    ],
    "cutOutRect": {
    /*opt, object, picture cropping frame, desc:the origin is the upper-left corner of the screen, which is used for the target frame in matting*/
        "x":  0.000,
        /*req, float, x-coordinate, range:[0,1], desc:normalize coordinate*/
        "y":  0.000,
        /*req, float, y-coordinate, range:[0,1], desc:normalize coordinate*/
        "width":  0.000,
        /*req, float, width, range:[0,1]*/
        "height":  0.000
        /*req, float, height, range:[0,1]*/
    },
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string, desc:"normalFace" (normal face, default value), "patrolFace" (patrol person face), "hijackFace" (duress person face), "superFace" (super person face)*/
    "saveFacePic":  true,
    /*opt, bool, whether to save face pictures*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, first authentication permission, subType:int, range:[1,4], desc:if the array is empty, it indicates clearing first authentication permission of the person; if not, it indicates that the person have first authentication permission for the door 1, 2, 3, and 4*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceURL	[Message content]	application/json	--	--	--
img	[Binary picture data]	image/jpeg		facePic.jpg	--

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Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:this field is required when the value of statusCode is not 1*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:see the description of this node and above nodes in the message of JSON_ResponseStatus*/
    "FPID":  "test",
    /*ro, opt, string, face record ID, range:[1,64], desc:face record ID returned when the face record is added,it is unique,and the maximum size is 63
bytes. This node is valid when errorCode is "1" and errorMsg is "ok"*/
    "rowKey":  "test"
    /*ro, opt, string, rowKey of face record, desc:rowKey of face record is returned after face data is added. It is unique and in string format. The
maximum length is 64 bytes.*/
}
```

15.9.3.4 Get the capability of moving face data in the face picture library in a batch

Request URL

GET /ISAPI/Intelligent/FDLib/FDMove/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FDMove": [
    /*ro, opt, array, subType:object*/
        {
            "id": {
            /*ro, opt, object, number of data that can be moved each time*/
                "@min":  0,
                /*ro, opt, int*/
                "@max":  0
                /*ro, opt, int, the maximum value*/
            },
            "FDSrc": {
            /*ro, opt, object, source face picture library from which the face record will be moved out*/
                "FDID": {
                /*ro, opt, object, face picture library ID*/
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  0
                    /*ro, opt, int, the maximum value*/
                },
                "FPID": {
                /*ro, opt, object, face record ID*/
                    "@min":  0,
```

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```
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "humanId": {
        /*ro, opt, object, person ID*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        }
    },
    "FDDst": {
    /*ro, opt, object, destination face picture library which the face record will be moved to*/
        "FDID": {
        /*ro, opt, object, face picture library ID*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "FPID": {
        /*ro, opt, object, face record ID*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "humanId": {
        /*ro, opt, object, person ID*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "name": {
        /*ro, opt, object, name of the person in the face picture*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "gender": {
        /*ro, opt, object, gender of the person in the face picture*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "bornTime": {
        /*ro, opt, object, birth date of the person in the face picture*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "city": {
        /*ro, opt, object, city code of birth of the person in the face picture*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "certificateType": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int*/
        },
        "certificateNumber": {
        /*ro, opt, object, certificate No.*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "caseInfo": {
        /*ro, opt, object, remarks*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "tag": {
        /*ro, opt, object, custom tag*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "address": {
        /*ro, opt, object, person address*/
```

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```
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  0
                    /*ro, opt, int, the maximum value*/
                },
                "customInfo": {
                /*ro, opt, object, custom information*/
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  0
                    /*ro, opt, int, the maximum value*/
                },
                "rowKey": {
                /*ro, opt, object, main key of face picture library*/
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  0
                    /*ro, opt, int*/
                },
                "transfer": {
                /*ro, opt, object, whether to transfer*/
                    "@opt": [true, false]
                    /*ro, opt, array, subType:bool*/
                }
            }
        }
    ]
}
```

15.9.3.5 Move face data in the face picture library in a batch

Request URL

POST /ISAPI/Intelligent/FDLib/FDMove?format=json

Query Parameter

None

Request Message

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```
{
    "FDMove": [
    /*req, array, subType:object*/
        {
            "id":  1,
            /*req, int, serial No. of the element in the array,it starts from 1 and is used to match returned results*/
            "FDSrc": {
            /*opt, object, face records to be moved*/
                "FDID":  "test",
                /*req, string, face picture library ID*/
                "FPID":  "test",
                /*req, string, face record ID*/
                "humanId":  "humanIDXXX"
                /*opt, string, person ID*/
            },
            "FDDst": {
            /*opt, object, destination face picture library (face records will be moved to this face picture library)*/
                "FDID":  "test",
                /*req, string, face picture library ID, desc:the maximum length is 63 bytes. Multiple libraries are separated by commas*/
                "FPID":  " ",
                /*req, string, face record ID*/
                "humanId":  "humanIDXXX",
                /*opt, string, person ID*/
                "name":  "test",
                /*opt, string, name*/
                "gender":  "any",
                /*req, enum, gender, subType:string, desc:"any" (unlimited condition which is used for search), "male", "female", "unknown"*/
                "bornTime":  "2004-05-03",
                /*req, string, birth date of the person in the face picture, range:[0,20]*/
                "city":  "130100",
                /*opt, string, city code, desc:city of birth*/
                "certificateType":  "ID",
                /*req, enum, certificate type, subType:string, desc:certificate type*/
                "certificateNumber":  "test",
                /*opt, string, certificate No.*/
                "caseInfo":  "test",
                /*opt, string, remarks*/
                "tag":  "aa,bb,cc,dd",
                /*opt, string, custom tag*/
                "address":  "test",
                /*opt, string, person address*/
                "customInfo":  "test",
                /*opt, string, custom information*/
                "rowKey ":  "test",
                /*opt, string, key of the face record database,the maximum size is 64 bytes. Searching by rowKey can be efficient*/
                "transfer":  true
                /*opt, bool, whether to enable transferring, desc:whether to enable transferring*/
            }
        }
    ]
}
```

Response Message

```
{
    "FDMoveResult": [
    /*ro, req, array, result of batch moving, subType:object*/
        {
            "id":  1,
            /*ro, req, int, serial No. of the element in the array,it starts from 1 and is used to match returned results, desc:serial No. of the element in
the array,it starts from 1 and is used to match returned results*/
            "errorCode":  1,
            /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
            "errorMsg":  "ok"
            /*ro, req, string, error description, desc:this node is required when the value of statusCode is not 1*/
        }
    ]
}
```

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15.9.3.6 Delete the face record(s) in the face picture library

Request URL

PUT /ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&FDID=<FDID>&faceLibType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID, and the maximum length is 64 bytes. Create a face picture library: POST /ISAPI/Intelligent/FDLib?format=json, and it is returned for success.
FDType	enum	Face picture library type: "blackFD" (list library), "staticFD" (static library)

Request Message

```
{
    "FPID": [
    /*req, array, list of face record ID, subType:object, desc:list of face record ID*/
        {
            "value":  "test",
            /*req, string, face record ID, range:[0,64]*/
            "rowKey":  "test"
            /*opt, string, main key of face picture library, range:[0,64], desc:main key of face picture library*/
        }
    ],
    "operateType":  "byTerminal",
    /*opt, enum, operation type, subType:string, desc:"byTerminal" (by terminal)*/
    "terminalNoList": [1]
    /*opt, array, terminal ID list, subType:int, desc:terminal ID list*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

15.9.3.7 Search for the face records in the a face picture library or multiple face picture libraries

Request URL

POST /ISAPI/Intelligent/FDLib/FDSearch?format=json

Query Parameter

None

Request Message

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```
{
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults":  100,
    /*req, int, the maximum number of search results this time*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:face picture library type: "blackFD"-list library,"staticFD"-static library,the maximum
size is 32 bytes*/
    "FDID":  "test",
    /*req, string, face picture library ID, desc:face picture library ID,the maximum size is 63*/
    "FPID":  " ",
    /*opt, string, face record ID*/
    "startTime":  "2004-05-03",
    /*opt, string, start date of birth*/
    "endTime":  "2004-05-03",
    /*opt, string, end time of birth*/
    "name":  "test",
    /*opt, string, name*/
    "gender":  "any",
    /*req, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
    "city":  "130100",
    /*opt, string, city code, desc:city code of birth for the person in the face picture,the maximum size is 32 bytes*/
    "certificateType":  "ID",
    /*req, enum, certificate type, subType:string, desc:the maximum size is 10 bytes,certificate type: "officerID"-officer ID,"ID"-identify
card,passport,other*/
    "certificateNumber":  "test",
    /*opt, string, certificate No.*/
    "isInLibrary":  "yes",
    /*opt, enum, whether the picture is in library (whether modeling is successful), subType:string, desc:whether modeling is successful: unknown, no, yes*/
    "isDisplayCaptureNum":  true,
    /*opt, bool, whether to display number of captured pictures*/
    "rowKey":  "test",
    /*opt, string, main key of face picture library, desc:searching by rowKey can be more efficient*/
    "transfer":  true
    /*opt, bool, Transferring, desc:whether to enable transfer*/
}
```

Response Message

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```
{
    "requestURL": "test",
    /*ro, opt, string, request URL*/
    "statusCode": 1,
    /*ro, req, int, status code*/
    "statusString": "test",
    /*ro, req, string, status description*/
    "subStatusCode": "activeNumMax",
    /*ro, req, enum, sub status code, subType:string*/
    "errorCode": 1,
    /*ro, opt, int, detailed error description, this field is required when the value of statusCode is not 1*/
    "errorMsg": "ok",
    /*ro, opt, string, detailed error description, this field is required when the value of statusCode is not 1*/
    "responseStatusStrg": "OK",
    /*ro, opt, enum, status search, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data).*/
    "numOfMatches": 1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches": 1,
    /*ro, opt, int, total number of matched results*/
    "MatchList": [
    /*ro, opt, array, list of matched records, subType:object*/
        {
            "FDID": "test",
            /*ro, opt, string, face picture library ID*/
            "FDName": "名单库A",
            /*ro, opt, string*/
            "FPID": "test",
            /*ro, opt, string, face record ID*/
            "faceURL": "test",
            /*ro, opt, string, face picture URL*/
            "name": "test",
            /*ro, opt, string, name*/
            "gender": "any",
            /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
            "bornTime": "2004-05-03",
            /*ro, opt, string, birth date of the person in the face picture*/
            "city": "130100",
            /*ro, opt, string*/
            "certificateType": "ID",
            /*ro, req, enum, certificate type, subType:string*/
            "certificateNumber": "test",
            /*ro, opt, string, certificate No.*/
            "caseInfo": "test",
            /*ro, opt, string, remarks*/
            "tag": "aa,bb,cc,dd",
            /*ro, opt, string, custom tag*/
            "address": "test",
            /*ro, opt, string, person address*/
            "customInfo": "test",
            /*ro, opt, string, custom information*/
            "isInLibrary": "yes",
            /*ro, opt, enum, subType:string, desc:whether modeling is successful: unknown, no, yes*/
            "captureNum": 12,
            /*ro, opt, int, number of captured pictures*/
            "rowKey": "test",
            /*ro, opt, string, main key of face picture library, desc:searching by rowKey can be more efficient*/
            "modelData": "test",
            /*ro, opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64
method*/
            "faceType": "normalFace",
            /*ro, opt, enum, face picture library type, subType:string*/
            "saveFacePic": true,
            /*ro, opt, bool*/
            "leaderPermission": [1, 2, 3, 4],
            /*ro, opt, array, subType:int, range:[1,4]*/
            "numberOfArrivalsAtSiteLastDays": [
            /*ro, opt, array, subType:object*/
                {
                    "day": 1,
                    /*ro, opt, int, range:[1,7]*/
                    "numberOfArrivalsAtSite": 3
                    /*ro, opt, int, range:[1,99]*/
                }
            ],
            "dwellTimeOnPreviousDay": 4,
            /*ro, opt, int, range:[0,86400], unit:min*/
            "totalNumberOfArrivalsAtSite": 4
            /*ro, opt, int, range:[1,99]*/
        }
    ]
}
```

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15.9.3.8 Edit a face record in a specific face picture library

Request URL

PUT /ISAPI/Intelligent/FDLib/FDSearch?format=json&FDID=<FDID>&FPID=<FPID>&faceLibType=<faceLibType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID, and the maximum length is 63 bytes. Create a face picture library: POST /ISAPI/Intelligent/FDLib?format=json, and it is returned for success.
FPID	string	Face record ID, and the maximum length is 63 bytes. Add face record: POST /ISAPI/Intelligent/FDLib?format=json, and it is returned for success.
faceLibType	string	Face picture library type: "blackFD" (list library), "staticFD" (static library).

Request Message

```
{
    "faceURL":  "test",
    /*opt, string, face picture URL, range:[0,256]*/
    "name":  "张三",
    /*req, string, name of person in the face picture, range:[0,96]*/
    "gender":  "male",
    /*opt, enum, gender of person in the face picture, subType:string, desc:"male", "female", "unknown"*/
    "bornTime":  "1970-01-01+08:00",
    /*req, date, birthday of person in the face picture, ISO8601 time format, the maximum size is 20 bytes*/
    "city":  "130100",
    /*opt, string, city code of birth for the person in the face picture, the maximum size is 32 bytes*/
    "certificateType":  "officerID",
    /*opt, enum, certificate type, subType:string, desc:certificate type*/
    "certificateNumber":  "test",
    /*opt, string, certificate No., the maximum size is 32 bytes*/
    "caseInfo":  "test",
    /*opt, string, remarks, range:[0,192], desc:case information,the maximum size is 192 bytes,it is valid when faceLibType is blackFD*/
    "tag":  "aa,bb,cc,dd",
    /*opt, string, custom tag, range:[0,195], desc:up to 4 tags, which are separated by commas. It is valid when faceLibType is blackFD*/
    "address":  "test",
    /*opt, string, person address, range:[0,192], desc:person address,the maximum size is 192 bytes,it is valid when faceLibType is staticFD.*/
    "customInfo":  "test",
    /*opt, string, custom information, range:[0,192], desc:custom information,the maximum size is 192 bytes,it is valid when faceLibType is staticFD.*/
    "modelData":  "test",
    /*opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method*/
    "rowKey":  "test",
    /*opt, string, main key of face picture library, range:[0,64], desc:searching by rowKey can be more efficient*/
    "transfer":  true,
    /*opt, bool, whether to enable transfer, desc:whether to enable transfer*/
    "PicFeaturePoints": [
    /*opt, array, feature points to be applied., subType:object, desc:If the device only supports three types of feature points, when the platform applies
more than three types of feature points, the device will not return error information*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of
mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, region, desc:object,coordinates of the feature point*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized X-coordinate which is between 0 and 1000*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized Y-coordinate which is between 0 and 1000*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:width which is between 0 and 1000. This node is required when featurePointType is "face"*/
                "height":  1
                /*opt, int, height, range:[0,1000], desc:height which is between 0 and 1000. This node is required when featurePointType is "face"*/
            }
        }
    ],
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string, desc:face picture type*/
    "saveFacePic":  true,
    /*opt, bool, whether to save face pictures*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, subType:int, range:[1,4]*/
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

15.9.3.9 Search face pictures by picture in face picture library, and the picture can be the binary data, URL, or modeled data

Request URL

POST /ISAPI/Intelligent/FDLib/searchByPic?format=json

Query Parameter

None

Request Message

```
{
    "searchResultPosition":  0,
    /*req, int, the start position in the search result list, desc:in a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults":  0,
    /*req, int, the maximum number of search results this time*/
    "modelMaxNum":  100,
    /*req, int, the maximum records to be found, desc:the maximum records to be found,the default value is 100*/
    "taskID":  "4af36915-4aa3-47da-bddb-9b942d114869",
    /*opt, string, task ID, desc:this node is returned by device. For the first tome search, this node is not required; for the further search, this ID can help to improve the search efficiency without any other conditions*/
    "FDLib": [
    /*opt, array, face picture library list, subType:object*/
        {
            "FDID":  "test"
            /*opt, string, face picture library ID*/
        }
    ],
    "dataType":  "URL",
    /*req, enum, picture data types, subType:string, desc:"URL" (picture URL), "binaryData" (picture binary data (transmit in form format)), "modelData" (modeled data)*/
    "faceURL":  "test",
    /*opt, string, face picture URL, desc:it is required when dataType is "URL"*/
    "targetModelData":  "test",
    /*opt, string, target modeling data, desc:it is required when dataType is "modelData"*/
    "startTime":  "2004-05-03",
    /*opt, string, start time of search*/
    "endTime":  "2004-05-03",
    /*opt, string, end time of search*/
    "name":  "test",
    /*opt, string, name*/
    "gender":  "any",
    /*opt, enum, gender, subType:string, desc:"any" (unlimited condition (used in search)), "male", "female", "unknown"*/
    "city":  "130100",
    /*opt, string, city code, desc:city code of birth,the maximum string length is 32 bytes*/
    "certificateType":  "ID",
    /*opt, enum, certificate type, subType:string, desc:certificate type: "officerID"-officer ID,"ID"-identity card,the maximum string length is 10 bytes*/
    "certificateNumber":  "test",
    /*opt, string, certificate number*/
    "maxSimilarity":  0.999,
    /*req, float, the maximum similarity, range:[0.000,1.000], step:0.001, desc:it is accurate to two decimal places*/
    "minSimilarity":  1.000,
    /*req, float, the minimum similarity, range:[0.000,1.000], step:0.001, desc:it is accurate to two decimal places*/
    "isDisplayCaptureNum":  true,
    /*opt, bool, whether to display number of captured pictures, desc:whether to display the number of captured pictures: true,false (default)*/
    "faceLibType":  "blocklistFD"
    /*opt, enum, subType:string*/
}
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
searchByPic	[Message content]	application/json	--	--	--
FaceImage	[Binary picture data]	image/jpeg	FaceImage	FaceImage	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "activeNumMax",
    /*ro, req, enum, sub status code, subType:string*/
    "errorCode":  1,
    /*ro, opt, int, this node is required when the value of statusCode is not 1*/
    "errorMsg":  "ok",
    /*ro, opt, string, this node is required when the value of statusCode is not 1*/
    "responseStatusStrg":  "OK",
    /*ro, opt, string, "MORE" (searching)*/
    "taskID":  "4af36915-4aa3-47da-bddb-9b942d114869",
    /*ro, opt, string, task ID*/
    "numOfMatches":  1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches":  1,
    /*ro, opt, int, total number of matched results*/
    "MatchList": [
    /*ro, opt, array, the matched data information, subType:object*/
        {
            "FPID":  "test",
            /*ro, opt, string, face record ID*/
            "FDID":  "test",
            /*ro, opt, string, face picture library ID*/
            "FDName":  "test",
            /*ro, opt, string, face picture library name*/
            "faceURL":  "test",
            /*ro, opt, string, face picture URL*/
            "name":  "test",
            /*ro, opt, string, person name*/
            "gender":  "any",
            /*ro, req, enum, gender, subType:string, desc:"any"-unlimited condition (used in search), "male", "female", "unknown",*/
            "bornTime":  "2004-05-03",
            /*ro, opt, string, date of birth*/
            "city":  "130100",
            /*ro, opt, string, city of birth*/
            "certificateType":  "ID",
            /*ro, req, enum, certificate type, subType:string, desc:certificate type: "officerID"-officer ID,"ID"-identity card,the maximum string length is
10 bytes*/
            "certificateNumber":  "test",
            /*ro, opt, string, certificate No.*/
            "caseInfo":  "test",
            /*ro, opt, string, remarks, desc:it is valid only when type is "blackFD"*/
            "tag":  "aa,bb,cc,dd",
            /*ro, opt, string, custom tag, desc:it is valid only when faceLibType is "blackFD"*/
            "address":  "test",
            /*ro, opt, string, person's address*/
            "customInfo":  "test",
            /*ro, opt, string, custom information*/
            "similarity":  0.999,
            /*ro, req, float, similarity, desc:It should be accurate to three decimal place*/
            "captureNum":  12,
            /*ro, opt, int, number of captured pictures*/
            "rowKey":  "test",
            /*ro, opt, string, main key of face picture library, range:[0,64], desc:searching by rowKey can be more efficient*/
            "numberOfArrivalsAtSiteLastDays": [
            /*ro, opt, array, subType:object*/
                {
                    "day":  1,
                    /*ro, opt, int, range:[1,7]*/
                    "numberOfArrivalsAtSite":  3
                    /*ro, opt, int, range:[1,99]*/
                }
            ],
            "dwellTimeOnPreviousDay":  4,
            /*ro, opt, int, range:[0,86400], unit:min*/
            "totalNumberOfArrivalsAtSite":  4
            /*ro, opt, int, range:[1,99]*/
        }
    ]
}
```

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15.9.4 戴安全帽检测

15.9.4.1 Get the parameters of hard hat detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, hard hat detection parameters, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, opt, object, normalized value-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width, range:[1,1000]-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height, range:[1,1000]-->1000
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <enabled>
    <!--ro, opt, bool, whether to enable the rule-->true
  </enabled>
  <SafetyHelmetDetectionRegionList>
    <!--ro, opt, array, subType:object-->
    <SafetyHelmetDetectionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, opt, string, region ID-->test
      </id>
      <RegionCoordinatesList>
        <!--ro, opt, array, region coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </SafetyHelmetDetectionRegion>
  </SafetyHelmetDetectionRegionList>
  <stackTarget>
    <!--ro, req, bool, whether to overlay target frame on the alarm picture-->true
  </stackTarget>
  <stackRule>
    <!--ro, req, bool, whether to overlay rule frame on the alarm picture-->true
  </stackRule>
  <alarmLinkageDuration>
    <!--ro, req, enum, alarm linkage duration, subType:string, desc:alarm linkage duration-->5
  </alarmLinkageDuration>
  <ImageParam>
    <!--ro, opt, object-->
    <Resolution>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, picture ID-->test
      </id>
      <name>
        <!--ro, req, enum, pixels, subType:string, desc:pixels-->1080p
      </name>
      <value>
        <!--ro, req, enum, resolution, subType:string, desc:resolution-->1920*1080
      </value>
    </Resolution>
    <quality>
      <!--ro, req, enum, picture quality, subType:string, desc:picture quality-->best
    </quality>
    <alarmUpload>
      <!--ro, req, bool, whether to upload alarm picture-->true
    </alarmUpload>
    <localStorage>
      <!--ro, opt, bool, whether to store pictures locally-->true
    </localStorage>
  </ImageParam>
  <videoOverlay>
    <!--ro, opt, object, video overlay parameters-->
    <stackTarget>
      <!--ro, req, bool, whether to overlay alarm target frame-->true
    </stackTarget>
    <stackRule>
      <!--ro, req, bool, whether to overlay alarm rule frame-->true
    </stackRule>
  </videoOverlay>
  <wearHelmetEnabled>
```

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```
<wearHelmetEnabled>
  <!--ro, opt, bool-->true
</wearHelmetEnabled>
<analysisMode>
  <!--ro, opt, enum, analysis mode, subType:string, desc:the engine also has its analysis mode (refer to the <analysisMode> node in
/ISAPI/Intelligent/analysisEngines/<ID>). The device will be added to the engine with the same analysis mode via the load balancing method. If the engine is
not configured correspondingly,this node will be failed to e returned,and the error code of taskNotMatchEngine will be returned; "realTimeMode"-real time
mode,"autoSwitchMode"-auto-switch mode-->realTimeMode
</analysisMode>
<helmetFilterEnabled>
  <!--ro, opt, bool-->false
</helmetFilterEnabled>
<durationCapEnabled>
  <!--ro, opt, bool-->false
</durationCapEnabled>
<durationCapInterval>
  <!--ro, opt, int, range:[0,3600]-->30
</durationCapInterval>
<shieldRegionList>
  <!--ro, opt, array, subType:object-->
  <shieldRegion>
    <!--ro, opt, object-->
    <id>
      <!--ro, opt, string-->test
    </id>
    <regionCoordinatesList>
      <!--ro, opt, array, subType:object, range:[3,10]-->
      <regionCoordinates>
        <!--ro, opt, object-->
        <positionX>
          <!--ro, req, int, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, range:[0,1000]-->1
        </positionY>
      </regionCoordinates>
    </regionCoordinatesList>
  </shieldRegion>
</shieldRegionList>
<sensitivity>
  <!--ro, opt, int, range:[0,100]-->1
</sensitivity>
</SafetyHelmetDetection>
```

15.9.4.2 Set the parameters of hard hat detection

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, hard hat detection, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--opt, object-->
    <normalizedScreenWidth>
      <!--req, int, normalized width, range:[1,1000]-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height, range:[1,1000]-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <enabled>
    <!--opt, bool, whether to enable the hard hat detection rules-->true
  </enabled>
  <SafetyHelmetDetectionRegionList>
    <!--opt, array, subType:object-->
    <SafetyHelmetDetectionRegion>
      <!--opt, object-->
      <id>
        <!--opt, string, detection area ID-->test
      </id>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin (0, 0) is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </SafetyHelmetDetectionRegion>
  </SafetyHelmetDetectionRegionList>
  <stackTarget>
    <!--req, bool, whether to overlay the alarm target frame-->true
  </stackTarget>
  <stackRule>
    <!--req, bool, whether to overlay the alarm rule frame-->true
  </stackRule>
  <alarmLinkageDuration>
    <!--req, enum, alarm linkage duration time, subType:string, desc:alarm linkage duration time-->5
  </alarmLinkageDuration>
  <ImageParam>
    <!--opt, object, image parameters-->
    <Resolution>
      <!--opt, object-->
      <id>
        <!--req, string, picture ID-->test
      </id>
      <name>
        <!--req, enum, subType:string-->1080p
      </name>
      <value>
        <!--req, enum, resolution, subType:string, desc:resolution-->1920*1080
      </value>
    </Resolution>
    <quality>
      <!--req, enum, picture quality, subType:string, desc:picture quality-->best
    </quality>
    <alarmUpload>
      <!--req, bool-->true
    </alarmUpload>
    <localStorage>
      <!--opt, bool-->true
    </localStorage>
  </ImageParam>
  <videoOverlay>
    <!--opt, object-->
    <stackTarget>
      <!--req, bool, whether to overlay the target frame-->true
    </stackTarget>
    <stackRule>
      <!--req, bool, whether to overlay the rule frame-->true
    </stackRule>
  </videoOverlay>
  <wearHelmetEnabled>
    <!--opt, bool-->true
  </wearHelmetEnabled>
  <analysisMode>
    <!--opt, enum, analysis mode, subType:string, desc:analysis mode-->realTimeMode
  </analysisMode>
</SafetyHelmetDetection>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
SafetyHelmetDetection	[Message content]	application/xml	--	--	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

15.9.4.3 Get the configuration capability of hard hat detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetectionCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, opt, object, ro,normalized value-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width, range:[1,1000]-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
```

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```
      <!--ro, req, int, normalized height, range:[1,1000]-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <SafetyHelmetDetectionRegionList size="8">
    <!--ro, opt, array, detection area list, subType:object, attr:size{req, int}-->
    <SafetyHelmetDetectionRegion>
      <!--ro, opt, object, detection area-->
      <id>
        <!--ro, opt, string-->test
      </id>
      <RegionCoordinatesList size="10">
        <!--ro, opt, array, subType:object, range:[3,10], attr:size{req, int}-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:area coordinates-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </SafetyHelmetDetectionRegion>
  </SafetyHelmetDetectionRegionList>
  <stackTarget opt="true,false">
    <!--ro, req, bool, whether to overlay alarm target frame, attr:opt{req, string}-->true
  </stackTarget>
  <stackRule opt="true,false">
    <!--ro, req, bool, whether to overlay alarm rule frame, attr:opt{req, string}-->true
  </stackRule>
  <alarmLinkageDuration opt="5,10,30,60,120,300,600">
    <!--ro, req, enum, alarm linkage duration, subType:int, attr:opt{req, string}, desc:alarm linkage duration-->5
  </alarmLinkageDuration>
  <ImageParam>
    <!--ro, opt, object, picture parameters-->
    <ResolutionList size="3">
      <!--ro, opt, array, resolution, subType:object, attr:size{req, int}-->
      <Resolution>
        <!--ro, opt, object, resolution-->
        <id>
          <!--ro, req, string, ID-->test
        </id>
        <name>
          <!--ro, req, enum, resolution name, subType:string, desc:resolution name-->1080p
        </name>
        <value>
          <!--ro, req, enum, subType:string-->1920*1080
        </value>
      </Resolution>
    </ResolutionList>
    <quality opt="best,good,general">
      <!--ro, req, enum, picture quality, subType:string, attr:opt{req, string}, desc:"best","good","general"-normal-->best
    </quality>
    <alarmUpload opt="true,false">
      <!--ro, req, bool, whether to upload alarm picture, attr:opt{req, string}-->true
    </alarmUpload>
    <localStorage opt="true,false">
      <!--ro, opt, bool, whether to store pictures locally, attr:opt{req, string}-->true
    </localStorage>
  </ImageParam>
  <AdvanceConfiguration>
    <!--ro, opt, object-->
    <headSensitivity min="1" max="5" def="3">
      <!--ro, opt, int, head detection sensitivity, attr:min{req, int},max{req, int},def{req, int}, desc:larger value indicates more sensitive detection-->1
    </headSensitivity>
    <humanSensitivity min="1" max="5" def="3">
      <!--ro, opt, int, human body detection sensitivity, attr:min{req, int},max{req, int},def{req, int}, desc:larger value indicates more sensitive
detection-->1
    </humanSensitivity>
    <alarmAnalysisFrame min="1" max="50" def="12">
      <!--ro, opt, int, number of frames required for alarm analysis, attr:min{req, int},max{req, int},def{req, int}, desc:larger value indicates that more
frames are needed to be analyzed and more slowly the alarm will be triggered-->1
    </alarmAnalysisFrame>
    <FDLibList size="10">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <FDLib>
        <!--ro, opt, object, face picture library-->
        <id min="1" max="10">
          <!--ro, req, int, No., range:[1,10], attr:min{req, int},max{req, int}-->1
        </id>
        <FDID min="1" max="10">
          <!--ro, req, string, attr:min{req, int},max{req, int}-->test
        </FDID>
        <thresholdValue min="1" max="10">
          <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
        </thresholdValue>
        <customFaceLibID min="1" max="10">
          <!--ro, opt, string, custom face picture library ID, attr:min{req, int},max{req, int}, desc:custom face picture library ID-->test
        </customFaceLibID>
      </FDLib>
    </FDLibList>
```

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```
    </AdvanceConfiguration>
    <SearchDescriptionCap>
      <!--ro, opt, object, ability of searching for safety helmet event, desc:ability of searching for safety helmet event-->
      <timeSpanMaxNum>
        <!--ro, req, enum, the maximum number of time periods in which search can be conducted simultaneously, subType:string, desc:the maximum number of time periods in which search can be conducted simultaneously-->all
      </timeSpanMaxNum>
      <resultMaxNum>
        <!--ro, req, int, the maximum number of items that can be searched for at a time-->1
      </resultMaxNum>
      <faceContrastResult opt="all,success,failed">
        <!--ro, req, string, comparison result, attr:opt{req, string}-->test
      </faceContrastResult>
      <FDLibList size="10">
        <!--ro, opt, array, subType:object, attr:size{req, int}-->
        <FDLib>
          <!--ro, opt, object, face picture library-->
          <FDID min="1" max="10">
            <!--ro, req, string, attr:min{req, int},max{req, int}-->test
          </FDID>
        </FDLib>
      </FDLibList>
      <name min="1" max="10">
        <!--ro, opt, string, (search condition) name, attr:min{req, int},max{req, int}, desc:720p,D1"-->test
      </name>
    </SearchDescriptionCap>
    <videoOverlay>
      <!--ro, opt, object, video overlay parameters-->
      <stackTarget opt="true,false">
        <!--ro, req, bool, whether to overlay alarm target frame, attr:opt{req, string}-->true
      </stackTarget>
      <stackRule opt="true,false">
        <!--ro, req, bool, whether to overlay alarm rule frame, attr:opt{req, string}-->true
      </stackRule>
    </videoOverlay>
    <wearHelmetEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </wearHelmetEnabled>
    <analysisMode opt="realTimeMode,autoSwitchMode">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </analysisMode>
    <helmetFilterEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->false
    </helmetFilterEnabled>
    <durationCapEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->false
    </durationCapEnabled>
    <durationCapInterval min="0" max="3600">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->30
    </durationCapInterval>
    <shieldRegionList size="8">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <shieldRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, opt, string-->test
        </id>
        <regionCoordinatesList size="10">
          <!--ro, opt, array, subType:object, range:[3,10], attr:size{req, int}-->
          <regionCoordinates>
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->1
            </positionY>
          </regionCoordinates>
        </regionCoordinatesList>
      </shieldRegion>
    </shieldRegionList>
    <sensitivity min="0" max="100">
      <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
  </SafetyHelmetDetectionCap>
```

15.9.4.4 Hard hat detection results

EventType:safetyHelmetDetection

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The
```

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```
value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video
channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI
protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:times that the same alarm has been uploaded*/
    "eventType":  "safetyHelmetDetection",
    /*ro, req, string, event type, desc:"safetyHelmetDetection" (hard hat detection event)*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the event
ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "Safety Helmet Detection",
    /*ro, req, string, event description, desc:Hard Hat Detection*/
    "channelName":  "ABC",
    /*ro, req, string, channel name (camera name)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID which is also the PUID, desc:this field must be returned when ISAPI event information is transmitted via ISUP*/
    "safetyHelmetDetectionBackgroundImageResolution": {
    /*ro, opt, object*/
        "height":  1,
        /*ro, req, int, range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, range:[1,65535], unit:px*/
    },
    "Target": [
    /*ro, opt, array, target, subType:object, desc:multiple targets can be output after being recognized once*/
        {
            "recognitionType":  "safetyHelmet",
            /*ro, req, enum, recognition type, subType:string, desc:"safetyHelmet" (hard hat detection)*/
            "TargetInfo": {
            /*ro, opt, object, target information*/
                "recognition":  "safetyHelmet",
                /*ro, req, enum, recognition type, subType:string, desc:"safetyHelmet" (hard hat detection)*/
                "dataSource":  "timing"
                /*ro, req, enum, detection data source, subType:string, desc:"timing" (scheduled), "trigger" (when exceeded the configured level limit)*/
            },
            "rect": [
            /*ro, opt, array, rectangle frame of the hard hat, subType:object, desc:only one element will be returned. The origin is the upper-left corner
of the screen*/
                {
                    "height":  1.000,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                }
            ],
            "contentID":  "safetyHelmet1_image",
            /*ro, opt, string, thumbnail content ID, range:[0,256]*/
            "pId":  "F4F665D6A18E41308CE9934DCDDD2222",
            /*ro, opt, string, thumbnail ID of hard hat detection alarm, range:[0,64]*/
            "uid":  "test",
            /*ro, opt, string, unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits*/
            "safetyHelmetURL":  "test",
            /*ro, opt, string, thumbnail URL of hard hat detection alarm, range:[0,256]*/
            "faceContrastResult":  "success",
            /*ro, opt, enum, face picture comparison result, subType:string, desc:this field is returned only when the linkage between hard hat detection
and face picture (face comparison linkage) has been configured. "success" (comparison succeeded), "failed" (comparison failed)*/
            "Face": {
            /*ro, opt, object, face picture information*/
                "FDID":  "1123",
                /*ro, req, string, face picture library ID, range:[0,63]*/
                "FPID":  "test",
                /*ro, opt, string, face record ID, range:[0,63], desc:unique ID consisting of letters and digits*/
                "contentID":  "faceLibPic",
                /*ro, opt, string, contentID of picture, which matches the captured face picture, range:[0,256]*/
                "pId":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of picture, which matches the captured face picture, range:[0,64]*/
                "libFacePicURL":  "test"
                /*ro, opt, string, URL of picture, which matches the captured face picture, range:[0,256]*/
            },
            "FaceList": [
            /*ro, opt, array, face picture information list, subType:object, desc:face picture information list*/
                {
                    "FDID":  "1123",
                    /*ro, req, string, face picture library ID, range:[0,63]*/
                    "FPID":  "test",
                    /*ro, opt, string, face record ID, range:[0,63], desc:unique ID consisting of letters and digits*/
                    "contentID":  "faceLibPic",
                    /*ro, opt, string, contentID of picture, which matches the captured face picture, range:[0,256]*/
                    "pId":  "F4F665D6A18E41308CE9934DCDDD3333",
                    /*ro, opt, string, ID of picture, which matches the captured face picture, range:[0,64]*/
                    "libFacePicURL":  "test",
```

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dojojin@gmail.com

```
                    /*ro, opt, string, URL of picture, which matches the captured face picture, range:[0,256]*/
                    "name":  "test",
                    /*ro, opt, string, person name, range:[1,32], desc:the sensitive information should be encrypted*/
                    "certificateType":  "passportID",
                    /*ro, opt, enum, certificate type, subType:string, desc:the sensitive information should be encrypted. "officerID" (officer ID), "ID"
(ID card), "passportID" (passport), "other" (other type)*/
                    "certificateNumber":  "test"
                    /*ro, opt, string, certificate number, range:[8,32], desc:the sensitive information should be encrypted*/
                }
            ],
            "status":  "wear",
            /*ro, opt, enum, whether the person wears a hard hat, subType:string, desc:"unwear" (not wear, default value), "wear" (wear)*/
            "safetyHelmetColour":  "yellow",
            /*ro, opt, enum, subType:string, dep:and,{$.Target[*].status,eq,wear}*/
            "reflectiveVestStatus":  "wear",
            /*ro, opt, enum, subType:string*/
            "presetID":  0
            /*ro, opt, int*/
        }
    ],
    "contentID":  "background_image",
    /*ro, opt, string, thumbnail content ID, range:[0,256]*/
    "pId":  "F4F665D6A18E41308CE9934DCDDD1111",
    /*ro, opt, string, thumbnail ID of hard hat detection alarm, range:[0,64]*/
    "imageURL":  "test",
    /*ro, opt, string, background picture URL, range:[0,256]*/
    "URLCertificationType":  "digest"
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (it is for the cloud storage protocol), "digest" (digest authentication,
it is for the device local storage)*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
safetyHelmetDetection	[Message content]	application/json	--	--	--
.value	[Binary picture data]	image/jpeg	safetyHelmet1_image	safetyHelmet1_image.jpg	--
.value	[Binary picture data]	image/jpeg	faceLibPic	faceLibPic.jpg	--
.value	[Binary picture data]	image/jpeg	background_image	background_image.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

15.9.5 人脸低频检测

dojoin@gmail.com VISION AND SECURITY SYSTEM CO., LTD.

15.9.5.1 Set the condition of exporting low frequency person alarm and submit the task

Request URL

POST /ISAPI/SDT/LFPD/search/dataPackage?format=json

Query Parameter

None

Request Message

```
{
    "DataPackageCond": {
    /*req, object*/
        "searchResultPosition":  0,
        /*opt, int, the start position of the search result in the result list, desc:In a single search, if you cannot get all the records in the result
list, you can mark the end position and get the following records after the marked position in the next search*/
        "maxResults":  1000,
        /*opt, int, the maximum number of search results this time, desc:by default, it equals to maximum value which does not exceed 1000 terms*/
        "facelibDataIndex":  "test",
        /*opt, string, information index in the face picture library, desc:information index in the face picture library*/
        "Description": {
        /*opt, object, packaging search condition, desc:packaging search condition*/
            "startTime":  "1970-01-01T00:00:00+08:00",
            /*opt, datetime, start time of search*/
            "endTime":  "1970-01-01T00:00:00+08:00",
            /*opt, datetime, end time of search*/
            "channels": [1, 2],
            /*opt, array, array of the linked channel No., subType:int, desc:all channels are searched if channel information is not specified*/
            "highestOccurrences":  7
            /*opt, int, the highest frequency,default value:7,range: [0,30], range:[0,30], desc:default value: 7*/
        },
        "FDID":  "2323",
        /*opt, string, face picture library ID, range:[0,64], desc:face picture library ID*/
        "PID":  "123"
        /*opt, string, person ID of face picture library, range:[0,64], desc:person ID of face picture library*/
    }
}
```

Response Message

```
{
    "TaskInfo": {
    /*ro, req, object, task information*/
        "id":  "test"
        /*ro, opt, string, it is required for success*/
    }
}
```

15.9.5.2 Get the search results of rarely appeared person alarms

Request URL

POST /ISAPI/SDT/LFPD/searchTask/result/recordData?format=json

Query Parameter

None

Request Message

```
{
    "PagingCond": {
    /*req, object*/
        "facelibDataIndex":  "test",
        /*req, string, information index in the face picture library, range:[1,32], desc:information index in the face picture library*/
        "FDID":  "2323",
        /*req, string, face picture library ID, range:[1,64]*/
        "PID":  "123"
        /*req, string, person ID of face picture library, range:[1,64]*/
    }
}
```

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
{
    "RecordData": {
    /*ro, req, object*/
        "progress": 100,
        /*ro, opt, int, progress, range:[0,100], desc:you can get the detailed data when it reaches 100*/
        "targets": [
        /*ro, opt, array, list of matched records, subType:object*/
            {
                "id": 1,
                /*ro, req, int, No.*/
                "captureTime": "1970-01-01T00:00:00+08:00",
                /*ro, opt, datetime, capture time*/
                "isSummerTime": false,
                /*ro, opt, bool, whether the time returned currently is in DST (Daylight Saving Time) system*/
                "captureSite": "3#大门处",
                /*ro, opt, string, capture site (camera name)*/
                "channel": 1,
                /*ro, req, int, channel No., desc:channel No.*/
                "monitorPointIndexCode": "test",
                /*ro, opt, string, camera No., range:[0,32], desc:camera No.*/
                "ageGroup": "any",
                /*ro, opt, enum, age group, subType:string, desc:age group*/
                "range": 5,
                /*ro, opt, int, age deviation, desc:age deviation*/
                "value": 23,
                /*ro, opt, int, age*/
                "gender": "any",
                /*ro, req, enum, gender, subType:string, desc:gender*/
                "glasses": "no",
                /*ro, req, enum, whether the person wears glasses, subType:string, desc:whether the person wears glasses*/
                "smile": "no",
                /*ro, req, enum, whether the person wears a smile, subType:string, desc:"no", "unknown", "yes"”*/
                "mask": "no",
                /*ro, req, enum, whether the person wears a mask, subType:string, desc:whether the person wears a mask*/
                "picUrl": "http://127.0.0.1",
                /*ro, opt, string, picture URL*/
                "subpicUrl": "http://127.0.0.1",
                /*ro, opt, string, thumbnail URL*/
                "rect": {
                /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
                    "height": 0.514,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width": 0.290,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "y": 0.366,
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    "x": 0.183
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                },
                "face_rect": {
                /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                    "height": 0.514,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width": 0.290,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "y": 0.366,
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    "x": 0.183
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                },
                "similarity": 1.00,
                /*ro, opt, float, similarity*/
                "faceExpression": {
                /*ro, opt, object, facial expression*/
                    "confidence": 99.99990,
                    /*ro, opt, float, confidence*/
                    "value": "angry"
                    /*ro, req, enum, value, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-faced", "sad",
"surprised", "unknown"*/
                }
            }
        ]
    }
}
```


15.10 图像智能检索

15.10.1 以文搜图

15.10.1.1 Get capability of text-to-image search task

Request URL

GET /ISAPI/ContentMgmt/AsynSearchByTextTask/capabilities?format=json

Query Parameter

VISION AND SECURITY SYSTEM CO., LTD. dejijin@gmail.com

None

Request Message

None

Response Message

```
{
    "AsynSearchByTextTaskCap": {
    /*opt, object, capability of submitting the text-to-image search task, desc:related URL (POST /ISAPI/ContentMgmt/AsynSearchByTextTask?format=json)*/
        "taskID": {
        /*opt, object, range of task ID, desc:the platform specifies the task ID. Applied by platform by default; if not, generated by device*/
            "@def":  "fdb4465d7aa5495884845127a8bbbe3c",
            /*opt, string, the default value*/
            "@min":  1,
            /*req, int, minimum length*/
            "@max":  32
            /*req, int, maximum length*/
        },
        "startTime":  "2021-01-28T02:00:00.000+08:00",
        /*req, string, search start time example*/
        "endTime":  "2021-01-28T02:00:00.000+08:00",
        /*req, string, search end time example*/
        "channelList": {
        /*opt, object, channel list range, desc:according to the video channel No. connected to the device. If this field is absent, channelList configured
in API PicSearchByText takes effect*/
            "@size":  256,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  0,
            /*opt, int, the minimum number of array elements*/
            "@min":  1,
            /*req, int, the minimum value*/
            "@max":  256
            /*req, int, the maximum value*/
        },
        "streamIDList": {
        /*opt, object, stream ID list range, desc:stream ID returned by API (GET /ISAPI/ContentMgmt/InputProxy/channels)*/
            "@size":  1024,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  1,
            /*opt, int, the minimum number of array elements*/
            "@min":  1,
            /*req, int, minimum length*/
            "@max":  32
            /*req, int, the maximum length*/
        },
        "cameraIndexCodeList": {
        /*opt, object, camera ID list range, desc:the camera ID configured in video analysis tasks applied by the platform. If this field is absent,
cameraIndexCodeList configured in API PicSearchByText takes effect*/
            "@size":  256,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  1,
            /*opt, int, the minimum number of array elements*/
            "@min":  1,
            /*req, int, minimum length*/
            "@max":  64
            /*req, int, maximum length*/
        },
        "recordIDList": {
        /*opt, object, offline video file list range, desc:according to recordID returned in SearchRecordFolderResourceList. If the field is absent,
recordIDList configured in PicSearchByText takes effect*/
            "@size":  100,
            /*opt, int, the maximum number of array elements*/
            "@minSize": 1,
            /*opt, int, the minimum number of array elements*/
            "@min":  10000,
            /*req, int, the minimum value*/
            "@max": 10100
            /*req, int, the maximum value*/
        },
        "text": {
        /*req, object, value range of search text content, desc:supports Chinese, English, and their combination*/
            "@def":  "test",
            /*opt, string, the default value*/
            "@min":  0,
            /*req, int, minimum length*/
            "@max":  128
            /*req, int, maximum length*/
        },
        "language": {
        /*opt, object, language type range, desc:the language of text content (corresponding field "text")*/
            "@opt": [],
            /*opt, array, language type enumeration, subType:string, range:[1,50]*/
            "@def":  "SimChinese"
            /*opt, string, default value*/
        },
        "scenario": {
        /*opt, object, value range of search text content type, desc:it is used for multilingual scenarios and can be absent when the field "language" is
English*/
            "@opt": [],
```

WUHAN DOJOJIN SECURITY SYSTEM CO.,LTD dojojin@gmail.com

```
            /*opt, array, enumeration list of search text content type, subType:string, range:[1,3]*/
            "@def":  "exampleTerm"
            /*opt, string, default value*/
        },
        "key": {
        /*opt, object, value range of term corresponding key, desc:it can be obtained via field "key" returned by API (SearchExampleKeyword). It is used for multilingual scenarios and can be absent when the field "language" is English*/
            "@def":  1,
            /*opt, int, default value*/
            "@min":  1,
            /*req, int, the minimum value*/
            "@max":  128
            /*req, int, the maximum value*/
        },
        "englishText": {
        /*opt, object, the English content of search text, desc:it can be obtained via field "englishText" returned by API (GenerateAcuSeekText). It is used for multilingual scenarios and can be absent when the field "language" is English*/
            "@def":  "text",
            /*opt, string, default value*/
            "@min":  0,
            /*req, int, minimum length*/
            "@max":  128
            /*req, int, the maximum length*/
        },
        "similarity": {
        /*opt, object, similarity threshold range*/
            "@def":  "1",
            /*opt, number, the default value*/
            "@min":  "0",
            /*req, number, the minimum value*/
            "@max":  "100"
            /*req, number, the maximum value*/
        },
        "timeSortType": {
        /*opt, object, range of time sequence types, desc:ascending (from earlier to later) and descending (from later to earlier)*/
            "@opt": [],
            /*opt, array, time sequence type enumeration, subType:string, range:[1,2]*/
            "@def":  "reverse"
            /*opt, string, the default value*/
        },
        "similaritySortType": {
        /*opt, object, range of similarity sequence types, desc:ascending (from low to high) and descending (from high to low)*/
            "@opt": [],
            /*opt, array, similarity sequence type enumeration, subType:string, range:[1,2]*/
            "@def":  "reverse"
            /*opt, string, the default value*/
        },
        "picStorageType": {
        /*opt, object, value range of picture storage type, desc:the storage address of searched picture. If this field is absent, the value is "localURL" (device local storage) by default*/
            "@opt": [],
            /*opt, array, enumeration list of searched picture storage type, subType:string, range:[1,3]*/
            "@def":  "localURL"
            /*opt, string, the default value*/
        },
        "type": {
        /*opt, object, target type range*/
            "@size":  6,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  0,
            /*opt, int, the minimum number of array elements*/
            "@opt": []
            /*opt, array, array element value list, subType:string, range:[1,6]*/
        }
    }
}
```

15.10.1.2 Submit the text-to-image search task

Request URL

POST /ISAPI/ContentMgmt/AsynSearchByTextTask?format=json

Query Parameter

None

Request Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
{
    "taskID":  "fdb4465d7aa5495884845127a8bbbe3c",
    /*opt, string, task ID, range:[1,32], desc:the platform specifies the task ID. Applied by platform by default; if not, generated by device*/
    "startTime":  "2021-01-28T02:00:00.000+08:00",
    /*req, datetime, search start time*/
    "endTime":  "2021-01-28T02:00:00.000+08:00",
    /*req, datetime, search end time*/
    "channelList": [],
    /*opt, array, channel list, subType:integer, range:[0,256], desc:according to the video channel No. connected to the device. If this field is absent,
channelList configured in API PicSearchByText takes effect*/
    "streamIDList": [],
    /*opt, array, stream ID list, subType:string, range:[1,1024], desc:stream ID returned by API (GET /ISAPI/ContentMgmt/InputProxy/channels)*/
    "cameraIndexCodeList": [],
    /*opt, array, camera ID list, subType:string, range:[1,256], desc:the camera ID configured in video analysis tasks applied by the platform. If this
field is absent, cameraIndexCodeList configured in API PicSearchByText takes effect*/
    "recordIDList": [],
    /*opt, array, offline video file list, subType:integer, range:[1,100], desc:according to recordID returned in SearchRecordFolderResourceList. If the
field is absent, recordIDList configured in PicSearchByText takes effect*/
    "text":  "test",
    /*req, string, search content, range:[0,128], desc:search content*/
    "language":  "SimChinese",
    /*opt, enum, language, subType:string,
    desc:language of text:
    "SimChinese" (Simplified Chinese), "TraChinese" (Traditional Chinese), "English", "Russian", "Bulgarian", "Hungarian", "Greek", "German", "Italian",
"Czech", "Slovakia" (Slovak), "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish", "Japanese", "Danish", "Swedish", "Norwegian",
"Finnish", "Korean", "Thai", "Estonia" (Estonian), "Vietnamese", "Hebrew", "Latvian", "Arabic", "Slovenian", "Croatian", "Lithuanian", "Serbian",
"BrazilianPortuguese" (Brazilian Portuguese), "Indonesian", "Ukrainian", "EURSpanish" (Spanish (European)), "Sovenian", "Uzbek", "Kazak", "Kirghiz", "Farsi"
(Persian), "Azerbaidzhan" (Azerbaijani), "Burmese", "Mongolian", "Anglicism (British English)", "HongKongTraChinese" (Traditional Chinese (Hong Kong,
China)), "Albanian", "LatinAmericanSpanish" (Latin American Spanish)*/
    "scenario":  "exampleTerm",
    /*opt, enum, search text content type, subType:string, desc:"exampleTerm" (example term), "generateText" (generated text), "historicalTerm" (history
search term). It is used for multilingual scenarios and can be absent when the field "language" is English*/
    "key":  1,
    /*opt, int, term corresponding key, range:[1,128], desc:it can be obtained via field "key" returned by API (SearchExampleKeyword). It is used for
multilingual scenarios and can be absent when the field "language" is English*/
    "englishText":  "text",
    /*opt, string, the English content of search text, range:[0,128], dep:and,{scenario,eq,generateText}, desc:it can be obtained via field "englishText"
returned by API (GenerateAcuSeekText). It is used for multilingual scenarios and can be absent when the field "language" is English*/
    "similarity":  "1",
    /*opt, number, similarity threshold*/
    "timeSortType":  "reverse",
    /*opt, enum, sort by time, subType:string, desc:"sequence" (ascending order, from the earlier to later), "reverse" (descending order, from the later to
earlier)*/
    "similaritySortType":  "reverse",
    /*opt, enum, sort by similarity, subType:string, desc:"sequence" (ascending order, from the low to high), "reverse" (descending order, from the high to
low)*/
    "picStorageType":  "localURL",
    /*opt, enum, picture storage type, subType:string, desc:the storage address type of searched pictures: "localURL" (device local storage),
"simpleStorage" (simple storage protocol), "cloudURL" (cloud storage). If this field is absent, it is "localURL" by default*/
    "type": []
    /*opt, array, type of targets to be searched, subType:string, range:[0,6]*/
}
```

Response Message

```
{
    "taskID":  "fdb4465d7aa5495884845127a8bbbe3c"
    /*req, string, task ID, range:[1,32], desc:the platform specifies the task ID. Applied by platform by default; if not, generated by device*/
}
```

15.10.1.3 Get capability of text-to-image search

Request URL

GET /ISAPI/ContentMgmt/PicSearchByText/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO., LTD. doujin@gmail.com

```
{
    "PicSearchByTextCap": {
    /*opt, object, capability of text-to-image search parameters, desc:related URL (GET or PUT /ISAPI/ContentMgmt/PicSearchByText?format=json)*/
        "enabled": {
        /*req, object, whether to enable image modeling of text-to-image search, desc:if it is enabled, the device will model and store pictures of specified video sources for the further text-to-image search*/
            "@opt": [],
            /*req, array, whether to enable image modeling of text-to-image search, subType:boolean, range:[1,2]*/
            "@def":  "true"
            /*opt, boolean, the default value*/
        },
        "channelList": {
        /*opt, object, the range of the channel number list, desc:the video channel number accessed to devices*/
            "@size":  256,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  1,
            /*opt, int, the minimum number of array elements*/
            "@min":  1,
            /*req, int, the minimum value*/
            "@max":  256
            /*req, int, the maximum value*/
        },
        "cameraIndexCodeList": {
        /*opt, object, camera ID list range, desc:the camera ID configured in video analysis tasks applied by the platform*/
            "@size":  256,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  1,
            /*opt, int, the minimum number of array elements*/
            "@min":  1,
            /*req, int, minimum length*/
            "@max":  64
            /*req, int, maximum length*/
        },
        "recordIDList": {
        /*opt, object, offline video file list range, desc:according to recordID returned by API SearchRecordFolderResourceList*/
            "@size":  100,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  1,
            /*opt, int, the minimum number of array elements*/
            "@min":  10000,
            /*req, int, the minimum value*/
            "@max":  10100
            /*req, int, the maximum value*/
        }
    },
    "DualVCARemindCap": {
    /*opt, object, capability of camera configuration (intelligent information overlay) for AcuSeek, desc:corresponding API: GET or PUT /ISAPI/Intelligent/AcuSeek/DualVCARemind?format=json*/
        "isReminder": {
        /*req, object, value range (whether to prompt users of camera configuration for AcuSeek), desc:true (the Web page, control client, and platform prompt users of camera configuration). The camera configuration information is from API (/ISAPI/Streaming/channels/dualVCA?format=json)*/
            "@opt": [],
            /*req, array, enumeration list (whether to prompt users of camera configuration for AcuSeek), subType:boolean, range:[1,2]*/
            "@def":  "true"
            /*opt, boolean, default value*/
        }
    },
    "HistoricalTermsCollectCap": {
    /*opt, object, capability of collecting history search records, desc:corresponding API: GET or PUT /ISAPI/Intelligent/AcuSeek/HistoricalTermsCollect?format=json*/
        "enabled": {
        /*req, object, whether to enable collecting history search records, desc:once enabled, the device's search history will be collected. With your consent, the search history will be exported and used as algorithm training inputs to enhance the search accuracy. When the function is disabled, APIs for getting search history (GetHistoricalTerms) and exporting search history (HistoricalTermsExport) will be unavailable*/
            "@opt": [],
            /*req, array, enumeration list, subType:boolean, range:[1,2]*/
            "@def":  "true"
            /*opt, boolean, default value*/
        },
        "reportEnabled": {
        /*opt, object, value range of whether to upload the text-to-image search event, desc:true (enable uploading the text-to-image search event, i.e., agree to join the User Experience Program. And the device will upload 5 pieces of data per time), false (disable). Corresponding event: "HistoricalTermsEvent".*/
            "@opt": [],
            /*req, array, enumeration list of whether to enable uploading the text-to-image search event, subType:boolean, range:[1,2]*/
            "@def":  "true"
            /*opt, boolean, default value*/
        },
        "isReminder": {
        /*opt, object, whether to show the message of inviting to User Experience Program, desc:true (the Web page, control client, and platform show the message of inviting to User Experience Program)*/
            "@opt": [],
            /*req, array, enumeration list of whether to show the message of inviting to User Experience Program, subType:boolean, range:[1,2]*/
            "@def":  "true"
            /*opt, boolean, default value*/
        }
    }
}
```


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15.10.1.4 Get capability of searching for text-to-image search status

Request URL

GET /ISAPI/ContentMgmt/SearchTextTaskStatus/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
{
    "SearchTextTaskStatusCap": {
    /*opt, object, capability of searching for text-to-image search progress, desc:related URL (POST /ISAPI/ContentMgmt/SearchTextTaskStatus?format=json)*/
        "taskID": {
        /*req, object, range of task ID, desc:according to the taskID returned by API AsynSearchByTextTask*/
            "@def":  "fdb4465d7aa5495884845127a8bbbe3c",
            /*opt, string, the default value*/
            "@min":  0,
            /*req, int, minimum length*/
            "@max":  32
            /*req, int, maximum length*/
        },
        "searchResultPosition": {
        /*req, object, start position of the search result in the result list*/
            "@def":  1,
            /*opt, int, the default value*/
            "@min":  1,
            /*req, int, the minimum value*/
            "@max":  2147483647
            /*req, int, the maximum value*/
        },
        "maxResults": {
        /*req, object, range of maximum number of results in this search*/
            "@def":  100,
            /*opt, int, the default value*/
            "@min":  0,
            /*req, int, the minimum value*/
            "@max":  20000
            /*req, int, the maximum value*/
        },
        "channelList": {
        /*opt, object, channel list range, desc:filter search results by channel, which is according to the configured input parameters of
AsynSearchByTextTask*/
            "@size":  256,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  0,
            /*opt, int, the minimum number of array elements*/
            "@min":  1,
            /*req, int, the minimum value*/
            "@max":  256
            /*req, int, the maximum value*/
        },
        "streamIDList": {
        /*opt, object, stream ID list range, desc:filter search results by stream ID, which is according to the stream ID in the input parameters of
AsynSearchByTextTask*/
            "@size":  1024,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  1,
            /*opt, int, the minimum number of array elements*/
            "@min":  1,
            /*req, int, minimum length*/
            "@max":  32
            /*req, int, the maximum length*/
        },
        "similarity": {
        /*opt, object, similarity threshold range, desc:filter according to the search results of taskID*/
            "@def":  "1",
            /*opt, number, the default value*/
            "@min":  "0",
            /*req, number, the minimum value*/
            "@max":  "100"
            /*req, number, the maximum value*/
        },
        "timeSortType": {
        /*opt, object, range of time sequence types, desc:sort the search result by time: ascending (from earlier to later) and descending (from later to
earlier)*/
            "@opt": [],
            /*opt, array, time sequence type enumeration, subType:string, range:[1,2]*/
            "@def":  "reverse"
            /*opt, string, the default value*/
        },
        "similaritySortType": {
        /*opt, object, range of similarity sequence types, desc:sort the search result according to similarity: ascending (from low to high) and descending
(from high to low)*/
            "@opt": [],
            /*opt, array, similarity sequence type enumeration, subType:string, range:[1,2]*/
            "@def":  "reverse"
            /*opt, string, the default value*/
        },
        "targetType": {
        /*opt, object, target type range*/
            "@size":  5,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  0,
            /*opt, int, the minimum number of array elements*/
            "@opt": []
            /*opt, array, array element value list, subType:string, range:[1,5]*/
        }
    }
}
```

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15.10.1.5 Search for text-to-image search results

Request URL

POST /ISAPI/ContentMgmt/SearchTextTaskStatus?format=json

Query Parameter

None

Request Message

```
{
    "taskID":  "fdb4465d7aa5495884845127a8bbbe3c",
    /*req, string, task ID, range:[0,32], desc:according to the taskID returned by API AsynSearchByTextTask*/
    "searchResultPosition":  1,
    /*req, int, start position of the search result in the result list, range:[1,2147483647]*/
    "maxResults":  100,
    /*req, int, maximum number of results, range:[0,20000]*/
    "channelList": [],
    /*opt, array, channel list, subType:integer, range:[0,256], desc:filter search results by channel, which is according to the configured input parameters
of AsynSearchByTextTask*/
    "streamIDList": [],
    /*opt, array, stream ID list, subType:string, range:[1,1024], desc:filter search results by stream ID, which is according to the stream ID in the input
parameters of AsynSearchByTextTask*/
    "similarity":  "1",
    /*opt, number, similarity threshold, desc:filter search results*/
    "timeSortType":  "reverse",
    /*opt, enum, sort by time, subType:string, desc:"sequence" (ascending order, from the earlier to later), "reverse" (descending order, from the later to
earlier)*/
    "similaritySortType":  "reverse",
    /*opt, enum, sort by similarity, subType:string, desc:"sequence" (ascending order, from the low to high), "reverse" (descending order, from the high to
low)*/
    "targetType": []
    /*opt, array, target type, subType:string, range:[0,5]*/
}
```

Response Message

```
{
    "taskID":  "fdb4465d7aa5495884845127a8bbbe3c",
    /*req, string, task ID, range:[0,64]*/
    "runStatus":  "searching",
    /*req, enum, task status, subType:string, desc:"searching", "completed", "abnormal"*/
    "startTime":  "2021-01-28T02:00:00.000+08:00",
    /*opt, datetime, task start time*/
    "endTime":  "2021-01-28T02:00:00.000+08:00",
    /*opt, datetime, task end time*/
    "progress":  100,
    /*req, int, overall progress, range:[0,100], unit:%*/
    "numOfMatches":  100,
    /*req, int, number of results returned this time, range:[0,20000]*/
    "totalMatches":  1000,
    /*opt, int, total number of matched records, range:[0,2147483647], dep:and,{progress,eq,100}*/
    "matchResults": [
    /*opt, array, result list, subType:object, range:[0,256]*/
        {
            "id":  1,
            /*req, int, No., range:[1,256]*/
            "image": {
            /*opt, object, searched picture information, desc:background picture, which contains the target and background. The size is about 300 to 500
KB*/
                "filePathType":  "URL",
                /*req, enum, file path type, subType:string, desc:filePath is the file path generated by the device. When the value is "URL", the platform
gets the file from the device via the designated file transfer link. Authentication is required when the URL is used to download files from device, and the
method is the same with that in device login. Direct request is allowed if the URL parameter contains token(?token=). If the returned URL contains the IP
and port, the device is only accessible under a direct network connection. the platform should replace the IP and port in the URL based on the device's
network configuration*/
                "filePath":  "undefined"
                /*req, string, file path, range:[0,1024], desc:different filePathType values suggest different ways of file access*/
            },
            "targetImage": {
            /*opt, object, target picture information, desc:target thumbnail, which contains only the target. It is cropped from the background picture
according to the detected target area. The size is about 20 to 30 KB*/
                "filePathType":  "URL",
                /*req, enum, file storage path type, subType:string, desc:"URL" (file storage address), and the "filePath" is the path generated by device
The platform obtains files from the device through a special file transfer link. Authentication is required when the URL is used to download files from
device, and the method is the same with that in device login. Direct request is allowed if the URL parameter contains token(?token=). For other ways, the
authentication method is the same with that in device login. If the returned URL contains the IP and port, the device is only accessible under a direct
network connection, the platform should replace the IP and port in the URL based on the device's network configuration*/
                "filePath":  "undefined"
                /*req, string, file path, range:[0,1024], desc:different filePathType values suggest different ways of file access*/
            },
            "time":  "2021-01-28T02:00:00.000+08:00",
            /*opt, datetime, picture timestamp*/
            "imageUUID":  "fdb4465d7aa5495884845127a8bbbe3c",
            /*opt, string, target picture UUID, range:[1,64], desc:the device capture algorithm assigns a consistent UUID to the target during continuous
presence in the video. If the target leaves (or is tempered) and reappears, a new UUID is assigned*/
```

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```
            "modelData":  "",
            /*opt, string, picture modeling data, range:[0,2048], desc:the model data generated from pictures after HMS video structural modeling. It is
recommended to use this model data for picture search, as it has higher accuracy compared to searches using picture data. If this field is absent, picture
data can still be used for search.*/
            "similarity":  "1",
            /*opt, number, similarity*/
            "channelID":  1,
            /*opt, int, channel No., range:[1,256]*/
            "streamID":  "test",
            /*opt, string, stream ID, range:[1,32]*/
            "cameraIndexCode":  "test",
            /*opt, string, camera ID, range:[1,64]*/
            "traceUUID":  "fdb4465d7aa5495884845127a8bbbe3c",
            /*opt, string, target UUID, range:[1,64], dep:and,
            desc:1. It is returned when cameraIndexCode (camera ID) or recordID (offline video ID) exists.
                            2. The target ID is from the video analysis task result linked to cameraIndexCode or recordID. It is used for the platform to
link the searched picture with the picture information in the analysis task result.
                            3. For face picture analysis tasks, the traceUUID is the same as the traceUuid in captureResult.
            For vehicle analysis tasks, the traceUUID is the same as the passID in vehicleRcogResult.
            For video structural analysis tasks, the traceUUID is the same as the globalID in hawkResult.*/
            "recordID":  10000,
            /*opt, int, offline video ID, range:[10000,10100]*/
            "targetType":  "face",
            /*opt, enum, target type, subType:string, desc:"face", "human", "vehicle", "other"*/
            "targetRect": {
            /*opt, object, target frame, desc:the origin is the upper-left corner of the screen*/
                "height":  "1",
                /*req, number, height, desc:normalized coordinates*/
                "width":  "1",
                /*req, number, width, desc:normalized coordinates*/
                "x":  "0",
                /*req, number, x-coordinate, desc:normalized coordinates*/
                "y":  "0"
                /*opt, number, y-coordinate, desc:normalized coordinates*/
            },
            "targetCenterPoint": {
            /*opt, object, target center coordinates, desc:the origin is the upper-left corner of the screen*/
                "positionX":  "0.001",
                /*req, number, x-coordinate, desc:normalized coordinates*/
                "positionY":  "0.001"
                /*req, number, y-coordinate, desc:normalized coordinates*/
            },
            "targetID":  15883,
            /*opt, int, range:[0,2147483647]*/
            "customInfo":  "test"
            /*opt, string, platform-customized pass-through information for picture modeling, range:[0,2048], desc:custom information, which is from the
customInfo of API (AcuSeekImageModel)*/
        }
    ]
}
```

15.10.1.6 Search for AcuSeek searched picture attributes

Request URL

POST /ISAPI/Intelligent/AcuSeek/SearchImageAttributes?format=json

Query Parameter

None

Request Message

```
{
    "imageUUID":  "fdb4465d7aa5495884845127a8bbbe3c"
    /*req, string, target picture UUID, range:[1,64], desc:the value of field imageUUID returned by SearchTextTaskStatus*/
}
```

Response Message

```
{
    "faceAttributes": {
    /*opt, object, facial features*/
        "name":  "test",
        /*opt, string, name, range:[0,256]*/
        "gender":  "male",
        /*opt, enum, gender, subType:string, desc:"female", "male", "unknown"*/
        "age":  25,
        /*opt, int, age, range:[0,100]*/
        "certificateType":  "ID",
        /*opt, enum, certificate type, subType:string, desc:this field is considered sensitive and requires encrypted transmission (field value to be
encrypted when the URL parameter "security" is 1 or 2; whole message to be encrypted when "security" is 3 or 4) "officerID" (officer ID), "ID" (identify
card), "passport" (passport)*/
        "certificateNumber":  "test",
        /*opt, string, certificate number, range:[0,32], desc:this field is considered sensitive and requires encrypted transmission (field value to be
encrypted when the URL parameter "security" is 1 or 2; whole message to be encrypted when "security" is 3 or 4)*/
```

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```
        "phoneNumber":  "test",
        /*opt, string, phone number, range:[1,64]*/
        "faceModel":  "test"
        /*opt, string, face picture model data, range:[0,1024], desc:the model data can be used for the targetModelData field of the API (POST
/ISAPI/SDT/Face/searchByPic)*/
    },
    "humanAttributes": {
    /*opt, object, human body attributes*/
        "ageGroup":  "child",
        /*opt, enum, gender group, subType:string, desc:"child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime", "teenager",
"unknown", "young"*/
        "gender":  "female",
        /*opt, enum, gender, subType:string, desc:"female", "male", "unknown"*/
        "ride":  "unknown",
        /*opt, enum, whether the person is riding, subType:string, desc:"unknown", "yes", "no"*/
        "jacketColor":  "black",
        /*opt, enum, top's color, subType:string, desc:"black", "blue", "brown", "cyan", "deepBlue" (dark blue), "deepGray" (dark gray), "gray", "green",
"mixture", "orange", "pink", "purple", "red", "silver", "unknown", "white", "yellow"*/
        "trousersColor":  "black",
        /*opt, enum, bottom color, subType:string, desc:"black", "blue", "brown", "cyan", "deepBlue" (dark blue), "deepGray" (dark gray), "gray", "green",
"mixture", "orange", "pink", "purple", "red", "silver", "unknown", "white", "yellow"*/
        "cyclingPersonNumber":  25,
        /*opt, int, number of riders, range:[0,100], dep:and,{ride,eq,yes}*/
        "cyclingType":  "bicycle",
        /*opt, enum, riding type, subType:string, dep:and,{ride,eq,yes}, desc:"unknown", "bicycle", "motorcycle" (motorcycle / electric moped),
"threeWheelVehicle" (tricycle)*/
        "humanModel":  "test"
        /*opt, string, human body model data, range:[0,1024], desc:the model data can be used for the modelData field of the API (POST
/ISAPI/Intelligent/humanRecognition/searchByPic/sync?format=json)*/
    },
    "vehicleInfo": {
    /*opt, object, vehicle features*/
        "vehicleType":  "concreteMixer",
        /*req, enum, vehicle types, subType:string, desc:"SUVMPV" (SUV/MPV), "buggy" (small-sized truck), "bus", "concreteMixer" (concrete mixer),
"containerTruck" (container truck), "coupe", "crane", "hatchback", "largeBus" (large-sized bus), "lightTruck" (light truck), "mediumBus" (middle-sized bus),
"mediumHeavyTruck" (medium and heavy truck), "miniCar" (mini sedan (it will be converted to vehicle)), "minibus", "minitruck", "motorVehicle" (motor vehicle
(it will be converted to vehicle)), "nonmotorVehicle" (non-motor "vehicle" (it will be converted to threeWheelVehicle)), "oilTankTruck" (oil tank truck),
"pickupTruck" (pickup truck (it will be converted to buggy)), "platformTrailer" (platform trailer), "saloon", "slagTruck" (dump truck), "smallCar" (small
sedan (it will be converted to vehicle)), "threeWheelVehicle" (tricycle), "truck", "twoWheelVehicle" (two wheeler), "unknown", "van", "vehicle" (sedan)*/
        "vehicleColor":  "black",
        /*req, enum, vehicle color, subType:string, desc:"black", "blue", "brown", "cyan", "deepBlue" (dark blue), "deepGray" (dark gray), "gray", "green",
"orange", "pink", "purple", "red", "silver", "unknown", "white", "yellow"*/
        "vehicleLogo": {
        /*opt, object, vehicle brand information, desc:refer to the vehicle main and sub brand dictionary (VehicleLogoInfo)*/
            "vehicleLogoRecog":  1025,
            /*req, int, vehicle main brand, range:[1025,1801], desc:refer to the vehicle main brand and sub brand list*/
            "vehileSubLogoRecog":  0,
            /*opt, int, vehicle sub brand, range:[0,100], desc:refer to the vehicle main brand and sub brand list*/
            "vehileModel":  0
            /*opt, int, model year of vehicle sub brand, range:[0,100], desc:refer to the vehicle main brand and sub brand list*/
        },
        "plateInfo": [
        /*opt, array, license plate information, subType:object, range:[1,1]*/
            {
                "plateNo":  "浙A12345",
                /*req, string, license plate No., range:[1,64],
                desc:1. "NOPLATE" is returned when no license plate is detected.
                2. "unknown" is returned when no license plate recognized.*/
                "plateType":  "92TypeCivil",
                /*req, enum, license plate type, subType:string, desc:"02TypePersonalized" (type 02 custom vehicle), "04NewMilitay" (type 04 new military
vehicle), "92FarmVehicle" (two-line license plate civil vehicle), "92TypeCivil" (type 92 civil vehicle), "arm" (police vehicle), "civilAviation" (civil
aviation license plate), "coach" (driver training vehicle), "consulate" (consular vehicle), "embassy" (embassy vehicle), "emergency" (emergency license
plate), "green1325FarmVehicle" (agricultural vehicle (green 1325)), "hongKongMacao" (Hong Kong/Macao entrance vehicle), "leftRightMilitay" (military
vehicle, left-right type), "motorola" (motorcycle), "newEnergy" (new energy vehicle), "oneLineArm"(new armed police vehicle (one-line)),
"oneLineArmHeadquarters" (armed police headquarter license plate (one-line)), "tempEntry" (temporary vehicle), "tempTravl" (vehicle with temporary license
plate), "trailer" (trailer), "twoLineArm" (new armed police vehicle (two-line)), "twoLineArmHeadquarters" (armed police headquarter license plate (two-
line)), "unknown", "upDownMilitay" (military vehicle, top-bottom type), "yellow1225FarmVehicle" (agricultural vehicle (yellow 1225)),
"yellow1325FarmVehicle" (agricultural vehicle (yellow 1325)), "yellowTwoLine" (yellow two-line rear license plate)*/
                "plateColor":  "blue"
                /*req, enum, license plate color, subType:string, desc:"black", "blue", "brown", "cyan", "deepBlue" (dark blue), "deepGray" (dark gray),
"gray", "green", "mixture", "orange", "pink", "purple", "red", "silver", "unknown", "white", "yellow"*/
            }
        ],
        "vehicleModel":  "test"
        /*opt, string, vehicle model data, range:[0,1024], desc:the model data can be used for the modelData field in the API of vehicle picture search
(POST /ISAPI/Intelligent/vehicleRecognition/searchByPic/sync?format=json)*/
    }
}
```

15.11 音频通用

15.11.1 录音

15.11.1.1 Get monthly audio information of an audio channel

Request URL

POST /ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json

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Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--

Request Message

```
{
    "year":  2021,
    /*req, int, year*/
    "month":  1
    /*req, int, month, range:[1,12]*/
}
```

Response Message

```
{
    "AudioList": [
    /*ro, opt, array, monthly calendar list, subType:object*/
        {
            "dayID":  1,
            /*ro, req, int, day, range:[1,31]*/
            "record":  true
            /*ro, req, bool, whether there is audio*/
        }
    ]
}
```

15.11.1.2 Get the capability of audio recording schedule parameters

Request URL

GET /ISAPI/System/RecordAudio/channels/<audioID>/AudioRecordPlan/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--

Request Message

None

Response Message

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```
{
    "AudioRecordPlanCap": {
    /*opt, object, get the capability of audio recording schedule parameters, desc:related URL: GET/PUT
/ISAPI/System/RecordAudio/channels/<audioID>/AudioRecordPlan?format=json*/
        "enable": {
        /*req, object, whether to enable recording schedule range*/
            "@opt": [true, false],
            /*opt, array, whether to enable recording schedule enumeration list, subType:bool*/
            "@def":  true
            /*opt, bool, default value*/
        },
        "mode": {
        /*opt, object, recording type range*/
            "@opt": ["timing"],
            /*opt, array, recording type enumeration list, subType:string*/
            "@def":  "timing"
            /*opt, string, default value*/
        },
        "weekTimeList": {
        /*opt, object, week time*/
            "@size":  7,
            /*opt, int, the maximum number allowed*/
            "@minSize":  1,
            /*opt, int, the minimum number allowed*/
            "dayOfWeek": {
            /*opt, object, day of a week range*/
                "@opt": [1, 2, 3, 4, 5, 6, 7],
                /*opt, array, day of a week enumeration list, subType:int*/
                "@def":  1
                /*opt, int, default value*/
            },
            "timeRange": {
            /*req, object, time period*/
                "@size":  8,
                /*opt, int, the maximum number allowed*/
                "@minSize":  1,
                /*opt, int, the minimum number allowed*/
                "startTime":  "undefined",
                /*req, string, start time value example*/
                "endTime":  "08:00:00"
                /*req, string, end time value example*/
            }
        },
        "holidayTimeList": {
        /*opt, object, holiday time*/
            "@size":  32,
            /*opt, int, the maximum number allowed*/
            "startTime":  "undefined",
            /*req, string, start time value example, desc:format: 00:00*/
            "endTime":  "undefined"
            /*req, string, end time value example, desc:format: 00:00*/
        },
        "preRecordTimeSeconds": {
        /*opt, object, pre-record time range*/
            "@def":  5,
            /*opt, int, default value*/
            "@min":  0,
            /*opt, int, the minimum value*/
            "@max":  10
            /*opt, int, the maximum value*/
        },
        "postRecordTimeSeconds": {
        /*opt, object, post-record time range*/
            "@def":  5,
            /*opt, int, default value*/
            "@min":  0,
            /*opt, int, the minimum value*/
            "@max":  10
            /*opt, int, the maximum value*/
        }
    }
}
```

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15.11.1.3 Set audio recording schedule

Request URL

PUT /ISAPI/System/RecordAudio/channels/<audioID>/AudioRecordPlan?format=json

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	audio channel

Request Message

```
{
    "enable":  true,
    /*req, bool, whether to enable audio recording schedule*/
    "mode":  "timing",
    /*opt, enum, audio recording type, subType:string, desc:"timing" (scheduled audio recording)*/
    "weekTimeList": [
    /*opt, array, week time list, subType:object*/
        {
            "dayOfWeek":  1,
            /*opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)*/
            "timeRange": [
            /*req, array, time period, subType:object*/
                {
                    "startTime":  "undefined",
                    /*req, string, start time*/
                    "endTime":  "08:00:00"
                    /*req, string, end time, range:[0,64]*/
                }
            ]
        }
    ],
    "holidayTimeList": [
    /*opt, array, holiday time list, subType:object*/
        {
            "startTime":  "undefined",
            /*req, string, start time, desc:format: 00:00*/
            "endTime":  "undefined"
            /*req, string, end time, desc:format: 00:00*/
        }
    ],
    "preRecordTimeSeconds":  5,
    /*opt, int, pre-record, range:[0,10], unit:s*/
    "postRecordTimeSeconds":  5
    /*opt, int, post-record, range:[0,10], unit:s*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

16 How-To Video Guidance

If you need access to corresponding video guidance for device integration, please register on https://tpp.hikvision.com and visit our Training Center: https://tpp.hikvision.com/tpp/Training. The Training Center is specifically designed to provide technical training and guidance resources for our partners. On this platform, you can find integration video tutorials for various devices, enabling better understanding and learning of the integration process. To offer more personalized service, our Training Center also supports filtering by integration protocols, devices, and applications.

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